

BS-480 Chemistry Analyzer

Operator's Manual

Basic Volume



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- all installation operations, expansions, changes, modifications and repairs of this product are conducted by Mindray authorized personnel;
- the electrical installation of the relevant room complies with the applicable national and local requirements; and
- the product is used in accordance with the instructions for use.



Warning

It is important for the hospital or organization that employs this equipment to carry out a reasonable service/maintenance plan. Neglect of this may result in machine breakdown or personal injury.



Note

This equipment must be operated by skilled/trained clinical professionals.

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- Malfunction of the instrument or part whose serial number is not legible enough.
- Others not caused by instrument or part itself.

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Preface

This manual contains the instructions necessary to operate the product safely and in accordance with its function and intended use. Please read this manual thoroughly before using the product. This manual is based on the maximum configuration and therefore some contents may not apply to your product. If you have any questions, please contact us.

Observance of this manual is a prerequisite for proper performance and correct operation, and it ensures patient and operator safety. All graphics including screens and printouts in this manual are for illustration purpose only and must not be used for any other purposes. The screens and printouts on the product should prevail.

The following three parts are included in this manual:

I. Basic Volume:

- Copyright
- Preface
- Safety Information
- Chapter 1 System Description
- Chapter 2 General Operating Procedure
- Chapter 3 System Setup
- Chapter 4 Operation Theories

II. Advanced Volume:

- Chapter 5 Reagents
- Chapter 6 Calibration
- Chapter 7 Quality Control
- Chapter 8 Sample Programming and Processing
- Chapter 9 Result Printouts

Preface

- Chapter 10 Chemistries
- Chapter 11 System Commands and Setup Options
- Chapter 12 Use of ISE Module
- Chapter 13 Use of Bar Code
- Chapter 14 LIS and RMS

III. Maintenance Volume:

- Chapter 15 Diagnostics
- Chapter 16 Maintenance
- Chapter 17 Alarms and Troubleshooting
- Chapter 18 Template Modifying Software
- Vocabulary
- Index

Safety Information

This chapter provides you with safety symbols used in this manual and their meanings, summarizes the safety hazards and operating precautions that should be considered seriously when the instrument is being operated, and lists the labels and silkscreen that has been applied to the instrument and their indications.

Safety Symbols

Safety symbols are used in this manual in order to remind you of the instructions necessary to operate the product safely and in accordance with its function and intended use. A safety symbol and text constitutes a notice as shown in the table below:

Symbol	Text	Description
	WARNING	Read the statement following the symbol. The statement is alerting you to an operating hazard that can cause personal injury.
	BIOHAZARD	Read the statement following the symbol. The statement is alerting you to a potentially biohazardous condition.
	CAUTION	Read the statement following the symbol. The statement is alerting you to a possibility of system damage or unreliable results.
	NOTE	Read the statement following the symbol. The statement is alerting you to information that requires your attention.

Summary of Hazards

Introduction

Observe the following safety precautions when using the product. Ignoring any of these safety precautions may lead to personal injury or equipment damage.



WARNING

If the product is used in a manner not specified by our company, the protection provided by the product may be impaired.

Electric Shock Hazards

Observe the following instructions to prevent electric shock.



WARNING

- When the MAIN POWER is turned on, users other than the servicing personnel authorized by our company must not open the rear cover or side cover.
 - Spillage of reagent or sample on the product may cause equipment failure and even electric shock. Do not place sample and reagent on the product. In case of spillage, switch off the power immediately, remove the spillage and contact our Customer Service Department or your local distributor.
-

Moving Parts Hazards

Observe the following instructions to prevent personal injury caused by moving parts.



WARNING

- Do not touch such moving parts as sample probe, reagent probe, mixers , cuvette wash station, sample carousel, reagent carousel and reaction carousel,when the system is in operation.
 - Do not put your fingers or hands into any open part when the system is in operation.
-

Photometer Lamp Hazards

Observe the following instructions to prevent personal injury caused by photometer lamp.



WARNING

- Eye injury could occur from light emission from the photometer lamp. Do not stare into the lamp when the system is in operation.
 - If you want to replace the photometer lamp, first switch off the MAIN POWER and then wait at least 15 minutes for the lamp to cool down before touching it. Do not touch the lamp before it cools down, or you may get burned.
-

Laser Beam Hazards

Observe the following instructions to prevent personal injury caused by laser beam.



WARNING

Light emitted by the bar code reader may cause eye injury. Do not stare into the laser beam radiated from the bar code reader when the system is in operation.

Sample, Calibrator and Control Hazards

Observe the following instructions to protect against the biohazardous infection by samples, calibrators and control samples.



BIOHAZARD

- Inappropriately handling samples, controls and calibrators may lead to biohazardous infection. Do not touch samples, mixtures or waste with your bare hands. Wear gloves and lab coat and, if necessary, goggles.
 - In case your skin contacts the sample, control or calibrator, follow the standard laboratory safety procedure and consult a doctor.
-

Reagent and Wash Solution Hazards

Observe the following instructions to protect against the biohazardous infection by reagents and wash solution.



WARNING

Reagents and concentrated wash solution are corrosive to human skins. Exercise caution when using reagents and concentrated wash solution. In case your skin or clothes contact them, wash them off with soap and clean water. If reagents or wash solution spills into your eyes, rinse them with much water and consult an oculist.

Waste Hazards

Observe the following instructions to prevent environmental pollution and personal injury caused by waste.



BIOHAZARD

- Some substances contained in reagent, control, concentrated wash solution and waste are subject to regulations of contamination and disposal. Dispose of the waste in accordance with your local or national rule for biohazard waste disposal and consult the manufacturer or distributor of the reagents for details.
 - Wear gloves and lab coat and, if necessary, goggles.
-

System Disposal Hazards

Observe the following instructions to dispose of the waste analyzer.



WARNING

Materials of the analyzer are subject to contamination regulations. Dispose of the waste analyzer in accordance with your local or national rule for waste disposal.

Fire and Explosion Hazards

Observe the following instructions to prevent fire and explosion.



WARNING

Ethanol is flammable substance. Please exercise caution while using ethanol around the instrument in order to prevent fire and explosion.

Removal of Analyzer from Use for Repair or Disposal

To minimize or eliminate the hazards involved in repair, transportation, disposal process, please observe the following instruction.



WARNING

When the analyzer is not in use, for example, in repair, transportation or disposal process, please clean and sterilize the parts that may cause biohazards (sample probe, reagent probe and mixers, etc.) and remind the person who handles the device of the related hazards.

Precautions on Use

Introduction

To use the product safely and efficiently, pay attention to the following operating precautions.

Intended Use



WARNING

The instrument is an automated chemistry analyzer for in vitro diagnostic use in clinical laboratories and designed for in vitro quantitative determination of clinical chemistries in serum, plasma, urine and cerebrospinal fluid samples. Please consult us before you use the instrument for other purposes.

When drawing a clinical conclusion, please also refer to patients' clinical symptoms and other test results.

Environment Precautions



CAUTION

Evaluate the electromagnetic environment prior to operating the system.

Please install and operate the system in an environment specified by this manual. Installing and operating the system in other environment may lead to unreliable results and even equipment damage.

To relocate the system, please contact our Customer Service Department or your local distributor.

Installation Precautions



WARNING

The product is switched on and off via a switch or breaker. Before installing the system, ensure that the building in which the system will be located has been equipped with a switch or breaker that complies with IEC 61010-1:2001, is in close proximity to the equipment and within easy reach of you, and is marked as the disconnecting device for the equipment.

Electromagnetic Noise Precautions



CAUTION

Electromagnetic noise may interfere with operations of the system. Do not install devices generating excessive electromagnetic noise around the system. Do not use such devices as radio transmitters in the room housing the system. Do not use other CRT displays around the system.

Do not use other medical instruments around the system that may generate electromagnetic noise to interfere with their operations.

Do not use this device in close proximity to sources of strong electromagnetic radiation (e.g. mobile phones or radio transmitters), as these may interfere with the proper operation.

The electromagnetic environment should be evaluated prior to operation of the device.

This device has been designed and tested to CISPR 11 Class A, and in a domestic environment may cause radio interference, in which case, you may need to take measures to mitigate the interference.



NOTE

It is the manufacturer's responsibility to provide equipment electromagnetic compatibility information to the customer or user.

It is the user's responsibility to ensure that a compatible electromagnetic environment for the equipment can be maintained in order that it will perform as intended.

Operating Precautions



CAUTION

- Take the clinical symptoms or other test results of the patient into considerations when making a diagnosis based on the measuring results produced by the system.
 - Operate the system strictly as instructed by this manual. Inappropriate use of the system may lead to unreliable test results or even equipment damage or personal injury.
 - When using the system for the first time, first run calibrations, and then QC tests to make sure the system is in proper state.
 - Be sure to run QC tests every time when you use the system, otherwise the result may be unreliable.
 - Do not uncover the reagent carousel when the system is in operation. Keep the reagent carousel cover closed.
 - The RS-232 port on the analyzing unit is used for connection with the operation unit only. Do not use it for other connections. Use the cables provided by our company or your local distributor for the connection.
 - The operation unit is a personal computer with the operating software installed. Installing other software or hardware on the computer may interfere with the system operation. Do not run other software when the system is working.
 - Computer virus may destroy the operating software or test data. Do not use the computer for other purposes or connect it to the Internet. If the computer is infected by virus, please install anti-virus software to check for and clear virus.
 - Do not touch the display, mouse or keyboard with wet hands or hands with chemicals.
 - Do not place the MAIN POWER to ON again within 10 seconds since placing it to OFF; otherwise the system may enter the protection status. If it does so, place the MAIN POWER to OFF and place it to ON again.
-

Maintenance and Servicing Precautions



CAUTION

- Maintain the system strictly as instructed by this manual. Inappropriate maintenance may lead to unreliable results, equipment damage or personal injury.
 - To wipe off dust from the system surface, use a soft, clean and moistened cloth soaked with soap water rather than organic solvents such as ethanol. After cleaning, wipe the surface dry with dry cloth.
 - Shut down and turn off analyzer and disconnect the power plug before cleaning. Take necessary measures to prevent water ingress, otherwise equipment damage or personal injury may be caused.
 - Replacement of major parts as photometer lamp, sample probe, reagent probe, mixers and syringe plunger assembly must be followed by re-calibration.
 - Replacement of the photometer lamp should be done when the system power has been switched off for at least 15 minutes.
 - If the system is failed and needs servicing, contact our Customer Service Department or your local distributor. The system may need to be stopped or transported during servicing, which will probably cause biohazards, electric shock hazards and moving part hazards. Exercise caution when prepare the system for servicing.
-

Chemistry Parameter Configuration Precautions



CAUTION

To define such parameters as sample volume, reagent volume and wavelength, follow the instructions in this manual and the instructions of reagents.

ISE Module Precautions



CAUTION

To prevent ISE electrodes from being damaged due to water scarcity, if the system, when equipped with an ISE module, will be powered off for a long time, perform the electrode storage maintenance.

Sample Precautions



CAUTION

- Use samples that are completely free of insoluble substances like fibrin or suspended matter; otherwise the sample probe may be clogged.
 - Medicines, anticoagulants or preservative in the samples may lead to unreliable results.
 - Hemolysis, icterus or lipemia in the samples may lead to unreliable test results; running a sample blank, therefore, is recommended.
 - Store the samples properly. Improper storage may change the compositions of samples and lead to unreliable results.
 - Sample volatilization may lead to unreliable results. Do not leave the sample open for a long period.
 - The system has a specific requirement on the sample volume. Refer to this manual for proper sample volume.
 - Load samples to correct positions on the sample carousel before the analysis begins; otherwise reliable results may not be obtained.
-

Reagent, Calibrator and Control Precautions



CAUTION

- Use proper reagents, calibrators and controls on the system.
 - Select appropriate reagents according to the performance characteristics of the system. Consult the reagent suppliers, our company or our authorized distributor for details, if you are not sure about your reagent choice.
 - Store and use the reagents, calibrators and controls strictly as instructed by the suppliers; otherwise, reliable results or best performance of the system may not be obtained. Improper storage of reagents, calibrators and controls may lead to unreliable results and bad performance of the system even in validity period.
 - Perform calibration after changing the reagents, otherwise reliable results may not be obtained.
 - Contamination caused by carryover among reagents may lead to unreliable test results. Consult the reagent suppliers for details.
-

Data Archiving Precautions



NOTE

The system automatically stores the data to the built-in hard disk. Data loss, however, is still possible due to mis-deletion or physical damage of the hard disk. You are recommended to regularly archive the data to such medium as CDs.

To avoid the data loss caused by unexpected power failure, UPS (uninterrupted power supply) is recommended.

External Equipment Precautions



WARNING

For operating instructions and precautions of the computer and printer, please refer to their operation manuals.

External equipment connected to the analogue and digital interfaces must be authorized and complied with relevant safety and EMC standards (e.g., IEC 60950 Safety of Information Technology Equipment Standard and CISPR 22 EMC of Information Technology Equipment Standard (CLASS B)). Any person, who connects additional equipment to the signal input or output ports and configures an IVD system, is responsible for ensuring that the system works normally and complies with the safety and EMC requirements. If you have any questions, consult the technical services department of your local representative.

External Vacuum Pump Precautions



WARNING

Make sure the vacuum pump tubing is connected properly without any twists or sharp angles so that it can work normally.

Tubing and cables connected to the vacuum pump must be protected to prevent damage and breaks due to human or other causes.

Set the vacuum pump on a solid flat platform or ground.

Tube and Liquid Container Precautions



WARNING

When the tube or the part that contain liquid become aged or damaged, please stop its use immediately and contact our customer service department or your local distributor to check and replace it.

Labels and Silkscreen

Introduction

The following non-warning and warning labels and silkscreen are used on the product for system identification and operating instruction.

Check the labels regularly for cleanliness and integrity. If any of the labels becomes vague or peels off, contact our Customer Service Department or your local distributor for replacement.

Non-Warning Labels and Silkscreen

Serial number

This symbol, contained in the product label which is attached to the rear cover of the system, indicates the production serial number of the product.



Date of manufacture

This symbol, contained in the product label which is attached to the rear cover of the system, indicates the manufacture date of the product.



In vitro diagnostic equipment

This symbol, contained in the product label which is attached to the rear cover of the system, indicates that the product is in vitro diagnostic equipment.



European community representative

This symbol, contained in the product label which is attached to the rear cover of the system, indicates the name and address of the authorized representative in the European Community.



WEEE label

The following definition of the WEEE label applies to EU member states only.

The use of this symbol indicates that this product should not be treated as household waste. By ensuring that this product is disposed of correctly, you will help prevent bringing potential negative consequences to the environment and human health. For more detailed information with regard to returning and recycling this product, please consult the distributor from whom you purchased the product.



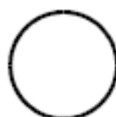
Main power switch: ON

This symbol located on the main power switch indicates that the system power is on when the switch is toggled upwards to expose this symbol and ON character and the green light is lightened.



Main power switch: OFF

This symbol located on the main power switch indicates that the system power is off when the switch is toggled downwards to expose this symbol and OFF character and the green light is extinguished. All components including the reagent refrigeration system will be stopped.



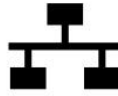
Analyzer power switch

This symbol located on the analyzer power switch indicates that the analyzer power is on when the switch is on the dotted-circle portion and off when it is on the blank-circle portion.



Network interface

This symbol located on the network interface indicates the connection between the analyzer and the operation unit.



Electrical ground

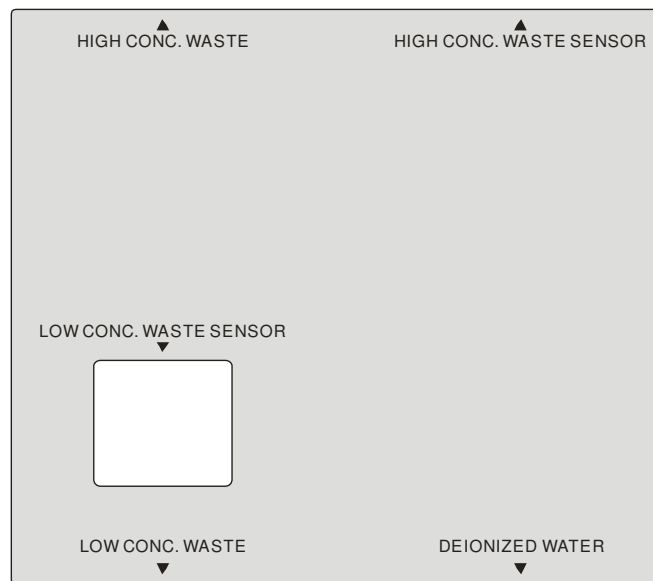
This symbol indicates an electrical ground.



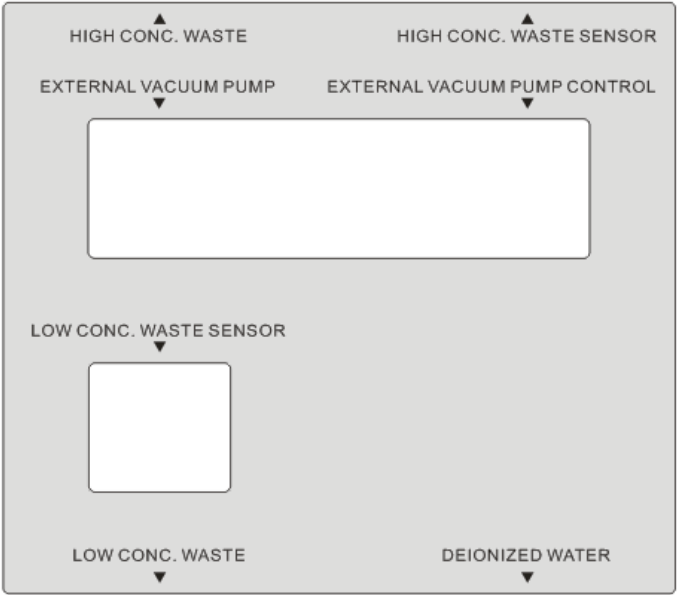
Interfaces for fluid connection

This symbol located on the fluid connection interfaces indicates the connection of fluid tubing.

The fluidic interfaces for standard configuration are shown as follows:



The fluidic interfaces for optional configuration are shown as follows:



Warning Labels

Biohazard warning

This label indicating the risk of biohazardous infection is located in the following positions:

- Sample probe
- Sample carousel
- Waste outlet
- Waste tank



Risk of electrical shock

This label is located on the power supply shielding box and vacuum pump. Be cautious of electrical shock!



Moving parts warning

This symbol and text indicating the hazardous moving parts is located in the following positions:

- Reagent probe and sample probe;
- Mixers
- Reagent carousel, reaction carousel and sample carousel
- Wash station



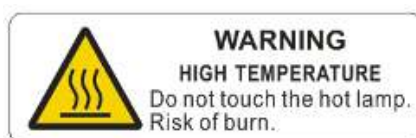
Laser warning

This symbol and text located near the sample bar code reader and the reagent bar code reader reminds you of not staring into the laser beam.



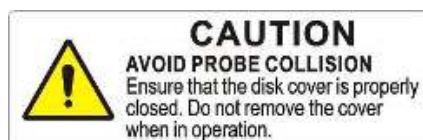
Photometer lamp warning

This symbol and text located on the lamp housing reminds you of not touching the lamp before it gets cool.



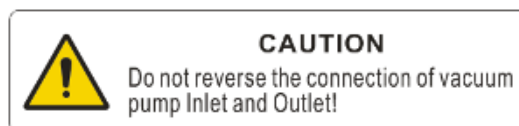
Probe collision warning

This symbol and text located near the sample carousel and reagent carousel reminds you of not opening the cover to prevent from damaging the probes.



Vacuum pump connection warning

This symbol and text located near the external vacuum pump and the built-in vacuum pump reminds you of connecting the inlet and outlet tubing correctly.



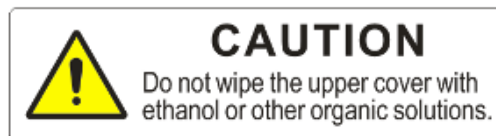
Water supply/drainage/external vacuum module warning

This symbol and text located on the water supply module, drainage module and external vacuum module reminds you of not pressing or placing heavy goods on these modules.



Upper cover

This symbol and text located on the transparent upper cover reminds you of not wipe the upper cover with ethanol or other organic solutions.



ISE module

This symbol and text located in the side cover of the ISE module. Please turn off the main power before opening the side door.



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1 System Description

This chapter describes the system from the installation, hardware, software and specifications perspectives, including:

- Installation requirements and methods of the instrument
- Hardware components
- Optional modules that can be configured with the instrument
- Introduction and operation of software screens
- Technical specifications

1.1 Installation Requirements and Procedure

1.1.1 Installation Requirements



CAUTION

Install the instrument in a place meeting the requirements presented in this section; otherwise, it will not perform as intended.

Installation environment

- The system is for indoor use only.
- The bearing platform (or ground) should be level (with gradient less than 1/200).
- The bearing platform (or ground) should be able to support at least 300Kg weight.
- The installation site should be well ventilated.
- The installation site should be free of dust.
- The installation side should not be in direct sun.
- The installation site should be kept away from a heat or draft source.
- The installation site should be free of corrosive gas and flammable gas.
- The bearing platform (of ground) should be free of vibration.
- The installation site should be kept away from large noise and power supply interference.
- Keep the system away from brush-type motors and electrical contact device that is frequently switched on and off.
- Do not use such devices as mobile phones and radio transmitter near the system.
- The system should be installed in a place with altitude height between -400 to 4000 meters. In places with an altitude height between 2000m to 4000m, an external vacuum pump should be employed.

Power supply

- Connect the system to a power supply meeting the requirements specified in this manual. For more information, refer to 1.5 System Specifications(page1-38).
- The system is provided with a three-wire power cord, which has good grounding performance.
- The system should be connected to a properly-grounded power socket.
- Grounding voltage must be configured.

**WARNING**

Make sure the power socket is grounded correctly. Improper grounding may lead to electric shock or equipment damage. Check if the power sockets outputs voltage meeting the specified requirements and has a proper fuse installed.

Temperature and humidity

- Ambient temperature: 15°C-30°C
 - Relative humidity: 35%-85%, without condensation.
-

**CAUTION**

Operating the system in an environment other than the specified may lead to unreliable test results. If the temperature or relative humidity does not meet the above-mentioned requirements, use air-conditioning equipment.

Water supply and drainage

- The supplied water must meet the requirements of CLSI type II.: Resistance is more than 1M Ω .CM.; Silicate is lower than 0.1 mg/L,
-

**CAUTION**

The supplied water must meet the requirements of CLSI type II; otherwise insufficiently purified water may result in misleading test results.

- Flow: no less than 38L/H for average flow, and 2L/M for transient peak flow.
 - If you use water supply equipment, make sure that the water supply pressure is within 95kPa-392kPa and the length of the inlet tubing is no longer than 10m.
 - Make sure that the outlet is no less than 50mm wide and no greater than 100mm high, and the length of the waste tubing does not exceed 5 meters.
-

**BIOHAZARD**

Dispose of the waste liquid according to the local regulations.

After installing the instrument, connect it with the fluidic components as instructed in the figure below.

**BIOHAZARD**

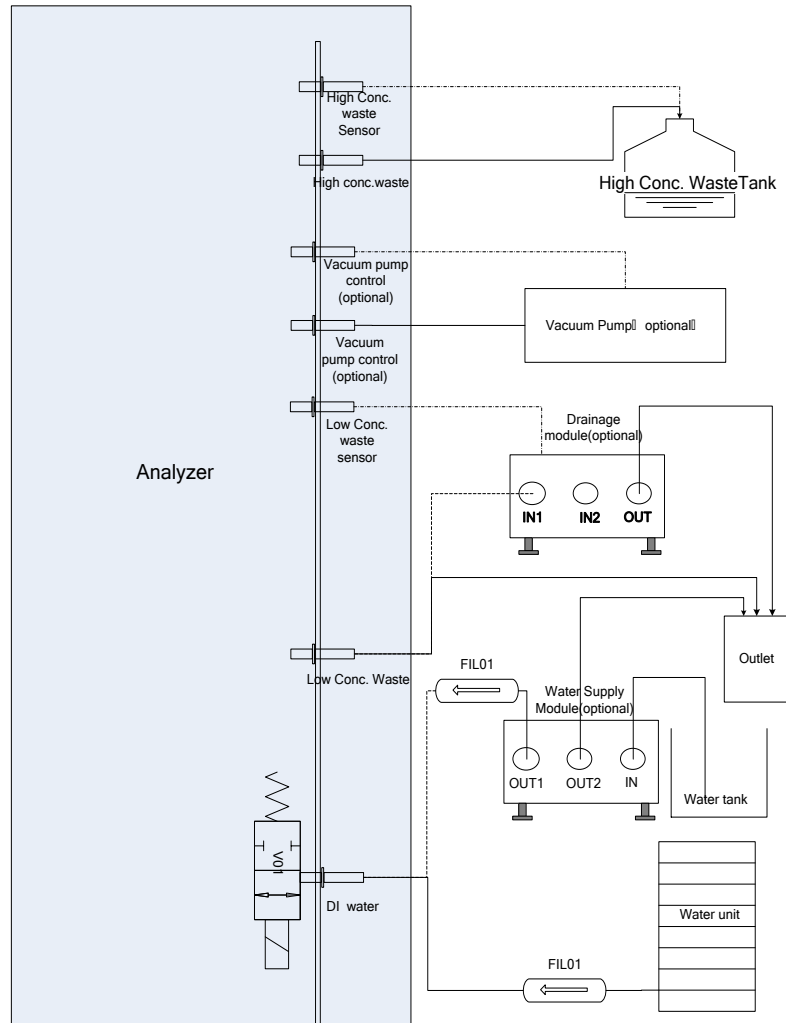
Wear gloves and lab coat, if necessary, goggles.

**CAUTION**

When connecting the tubes, exercise caution to avoid folding or pressing them.

1 System Description

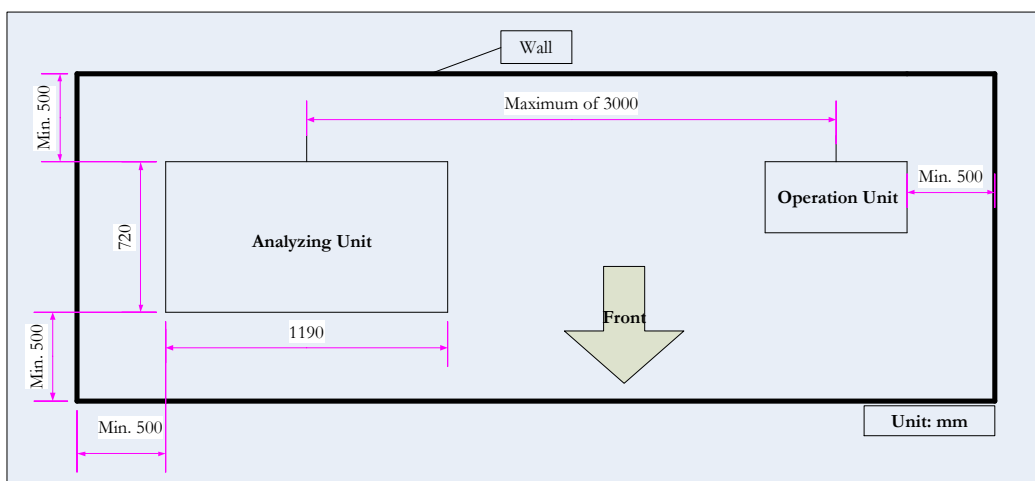
Figure 1.1 Fluidic connection diagram



Space and accessibility requirements

Install the instrument according to the clearance requirements as shown in the figure below.

Figure 1.2 System clearances



Recommended computer configuration

Table 1.1 Recommended computer configuration

Item	Description
CPU	At least P4 2.6GHz above
Random access memory (RAM)	At least 2GB for each RAM
Network adapter	The computer is connected to the chemistry analyzer through a network adapter. If you are going to connect the computer with the LIS or Internet, you should prepare another network adapter (Intel gigabit network adapter)
Serial port	The computer should provide an RS232 serial port, which is used to connect it with the chemistry analyzer.
Hard disk defragment	At least 160GB for hard disk. Install the operating system in the C drive and the operating software of the instrument in the D drive. Make sure that the C drive is over 30G and D drives over 100G, and the disk file system is of NTFS format. Deselect the two options at the bottom of the disk properties window: "Compress drive to save disk space" and "Allow Indexing Service to index this disk for fast file searching".
Operating system	The operating system installed on the computer must be an activated or free version Microsoft Windows 8(64 bit) or Windows 7(32 bit) .

1 System Description

Item	Description
Application software	Except for the operating system, other application software must not be installed or reserved on the computer. If an anti-virus application has been installed, then remove the automatic scheduled scanning and add the operating software and BSLOG to the trust list.
Screen saver and system standby	Turn off the screen saver and BS Special Power Policy power scheme, and then disable the hibernation option.
Screen display properties	Set the screen resolution as 1280*1024 pixels and color quality as Highest (32 bit).
Automatic synchronization with Internet time server	Disable the Automatically synchronize with an Internet time server option.
Automatic updates	Turn off the automatic updates.
Auto awake and shutdown setup	If you are going to use the auto awake/shutdown function, perform necessary settings for BIOS and network adapters while referring to their operation manuals.
Sound card and speaker	The system must be configured with sound card and speaker.

Recommended printer configuration

You are suggested to choose one of the following printers for use with the computer:

- Ink jet printer
- Laser printer
- Stylus printer

1.1.2 Installation Procedure



WARNING

The system should be installed only by technicians of or authorized by our company.

The system should be installed by technicians of or authorized by our company. Before the technicians arrive, prepare a proper site to install the system.

Before installation

When you receive the package, check it carefully. If you find any signs of mishandling or damage, file a claim immediately with our Customer Service Department or your local distributor.

After opening the package, check the delivered goods against the packing list, and then visually check the system appearance. If you find anything missing or damaged, alert our Customer Service Department or your local distributor immediately.

System relocation

If you want to relocate your system, contact our Customer Service Department or your local distributor.

1.2 Hardware Structure

1.2.1 System Overview

The chemistry analyzer consists of the analyzing unit (analyzer), operation unit (computer), output unit (printer), accessories and consumables.

The **analyzing unit**, the analyzer, determines various clinical chemistries in samples and displays the test results. It is composed of the following components:

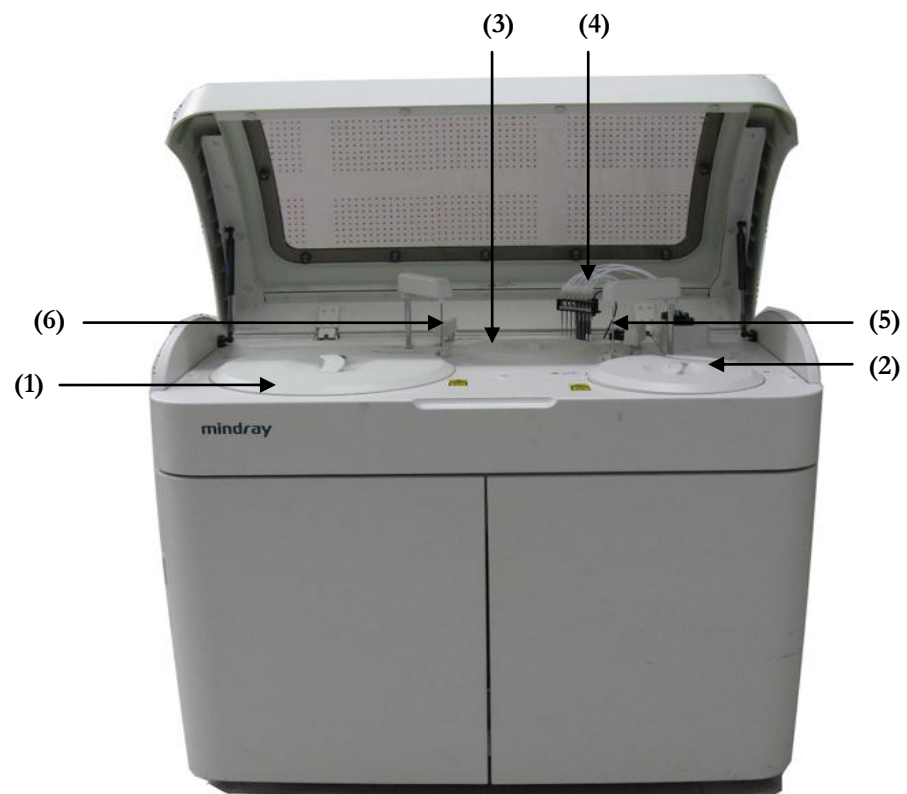
- Sample handling system
- Reagent handling system
- Reaction system
- Cuvette wash station
- Photometric system
- Mixer assembly

The **operation unit**, a computer configured with the operating software, controls the analyzing unit to finish tests and produce test results.

The **output unit** is a printer used to print out test results and other data.

Accessories and consumables are components that are required for sample processing and should be replenished regularly.

Figure 1.3 Front view



- 1. Reagent carousel
- 2. Sample Carousel
- 3. Reaction carousel

- 4. Auto wash station
- 5. Sample probe
- 6. Reagent probe

Figure 1.4 Rear view



- 1. Radiating fan
- 2. handle
- 3. Fluidic interfaces

- 4. Network port
- 5. Serial port
- 6. Power jack

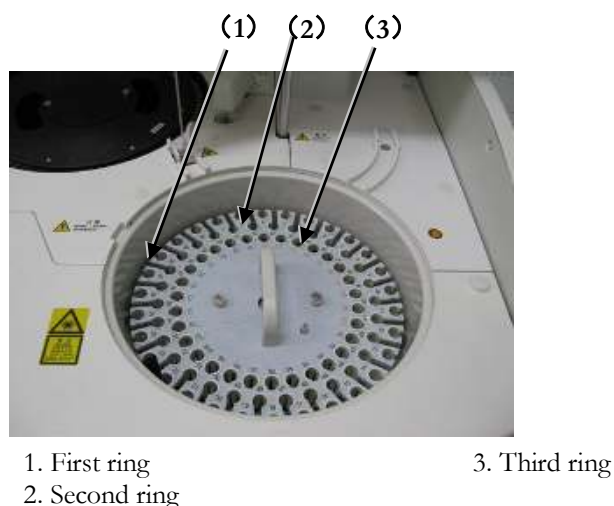
1.2.2 Sample Handling System

The sample handling system is used to hold samples and provides them for analysis. It consists of the following assemblies:

- Sample carousel assembly
- Sample bar code reader
- Sample dispenser assembly
- Sample containers

Sample carousel assembly

The sample carousel is a turntable located on right side of the analyzer panel. It holds sample tubes and carries each of them to the sample aspirate position for aspirating.

Figure 1.5 Sample carousel

The sample loading indicator tells you when you should load samples and when you must not. Do not operate on the movement path of sample probe, or collision might happen.

The sample loading indicator has the following states:

- Flash: indicates that the corresponding carousel is rotating or is going to rotate.
- ON: indicates that the corresponding carousel is stopped for sample aspirating or that the sample probe is aspirating on the reaction carousel during test for the diluted samples
- OFF: indicates that the corresponding carousel has no sample being aspirated and will not rotate in the next period.

The sample carousel includes three circles: inner, middle and outer, each of which provides 30 tube positions. 90 positions in total are available on the sample carousel and assigned as follows:

- Routine samples: 1-60
- Calibrators: S1-S10
- Controls: C1-C7
- STAT samples: E1-E10

Others:

- D1 (No.88) for ISE cleaning solution
- D2 (No.89) for wash solution
- W (No.90): for physiological saline

To remove the sample carousel,

- Loosen the two retaining screws on the sample carousel.

1 System Description

- Grab the handle and pull the sample carousel upward to remove it from the rotor.

To install the sample carousel,

- Align the positioning pins on the sample carousel to the counterparts on base.
- Set the sample carousel and tighten the two retaining screws on the carousel.

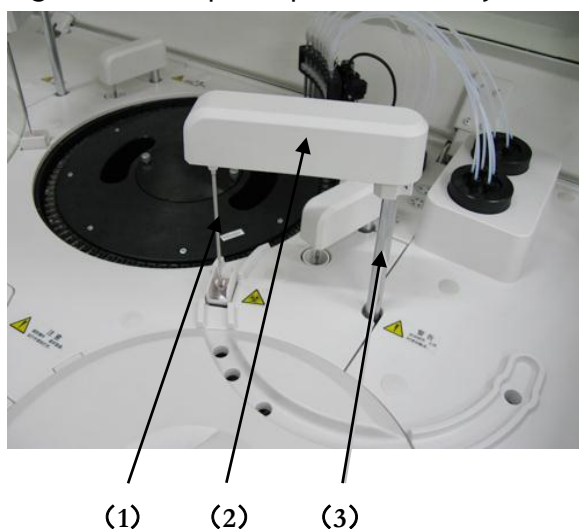
Sample bar code reader

The sample bar code reader is an optional module and used to obtain sample information through reading a sample bar code. For more information, refer to 1.3.3 Built-in Sample Bar Code Reader (page 1-23).

Sample dispenser assembly

The sample dispenser assembly located above the sample carousel is composed of the sample probe, probe arm, probe rotor, syringe and related tubing. It aspirates the specified amount of sample from a sample tube and then dispenses it into a cuvette for reaction and analysis.

Figure 1.6 Sample dispenser assembly



1. Sample probe
3. Sample probe rotor

2. Sample probe arm

Sample probe

The system has one sample probe, which aspirates the specified amount of sample for different type of chemistries:

- Biochemical chemistries: 1.5 μ L-45 μ L, with increment of 0.1 μ L.
- ISE chemistries: plasma/serum: 70 μ L; diluted urine: 140 μ L.

The sample probe is capable not only of aspirating sample but also of the following

functions:

- Clog detection: checks the sample probe for blockage. When detecting blockage, the system produces a warning and prompts you with the next step.
- Horizontal obstruct detection: detects obstacles in the horizontal direction. When the sample probe collides with an obstacle in the horizontal direction, the auto guard system is started to prevent the sample probe from being damaged.
- Vertical obstruct detection: detects obstacles in the vertical direction. When the sample probe collides with an obstacle in the vertical direction, the auto guard system is started to prevent the sample probe from being damaged.
- Level detection and tracking: detects the sample level and determines the depth of lowering down into the sample based on the specified aspirate volume.



WARNING

When the system is in operation, do not place any part of your body or any obstacle in the route where the sample probe arm moves; otherwise, personal injury or equipment damage may be caused.

Sample probe washing

The sample probe is cleaned in its wash well with water spraying its interior and exterior from two opposite directions.

Sample syringe

When the front left door of the analyzer is opened, you will see the sample syringe as shown below.

Figure 1.7 Sample syringe



(1)

1. Sample syringe

Sample containers

- Microtube: Mindray sample cup $\Phi 14 \times 25\text{mm}$, $\Phi 12 \times 37\text{mm}$

1 System Description

- Primary tube or plastic tube: $\Phi 12 \times 68.5\text{mm}$, $\Phi 12 \times 99\text{mm}$, $\Phi 12.7 \times 75\text{mm}$, $\Phi 12.7 \times 100\text{mm}$, $\Phi 13 \times 75\text{mm}$, $\Phi 13 \times 95\text{mm}$, $\Phi 13 \times 100\text{mm}$,

Sample tubes varying in specification requires different minimum sample volumes. Each sample tube must contain the minimum amount of sample; otherwise, correct aspirating cannot be ensured. The minimum sample volume is the sum of the minimum sample volume for analysis and the dead volume of the sample container.

The table below shows the dead volume of each type of sample container.

Table 1.2 Specification and dead volume of sample containers

Sample Container	Specification	Dead Volume
Sample tube	$\Phi 14 \times 25\text{mm}$, 0.5ml	70 μl
	$\Phi 14 \times 25\text{mm}$, 2ml	150 μl
	$\Phi 12 \times 37\text{mm}$, 2ml	100 μl
Primary tube or plastic tube	$\Phi 12 \times 68.5\text{ mm}$	8mm more over the unacceptable sample level height
	$\Phi 12 \times 99\text{ mm}$	
	$\Phi 12.7 \times 75\text{ mm}$	
	$\Phi 12.7 \times 100\text{ mm}$	
	$\Phi 13 \times 75\text{ mm}$	
	$\Phi 13 \times 95\text{ mm}$	
	$\Phi 13 \times 100\text{ mm}$	

1.2.3 Reagent Handling System

The reagent handling system is used to hold reagents and provides them for reacting with samples. It consists of the following assemblies:

- Reagent carousel assembly
- Reagent bar code reader
- Reagent dispenser assembly

Reagent carousel assembly

The reagent carousel assembly consists of a reagent carousel (including a cover) and a refrigeration unit.

The reagent carousel holds reagent bottles and carries the defined reagent bottle to the aspirate position for aspirating reagent.

The refrigeration unit keeps reagents in a low-temperature environment to keep them stable and minimize volatilization.

The reagent carousel provides a refrigerating environment which is constant within

2°C-10°C for 24 hours a day. The reagents stored in such environment can be kept stable with little volatilization.

Reagent positions: There are 80 bottle positions on the reagent carousel. D is for wash solution and W is for deionized water or physiological saline.



NOTE

The refrigeration unit has a power supply independent of that of the analyzing unit. The refrigeration unit is operational once the MAIN POWER is turned to the ON position.

Figure 1.8 Reagent carousel



(1)

1. Reagent carousel cover



CAUTION

Note: Every day before analysis, remove the plug on the reagent carousel in order to prevent mechanical reset failure and bending reagent probe.

Ensure that the reagent carousel is closed while the system is analyzing. Opening the reagent carousel cover during analyzing will abort the analysis and invalidate the tests that are running.

One virtual reagent carousel (80 positions) is allowed. You'll be reminded to change the reagent carousel on which all reagent aspiration is finished.

To remove the reagent carousel,

- Loosen the two retaining screws on the reagent carousel.
- Grab the handle and pull the reagent carousel upward to remove it from the rotor.

To install the reagent carousel,

1 System Description

- Align the positioning pins on the reagent carousel to the counterparts on base.
- Set the reagent carousel and tighten the two retaining screws on the carousel.

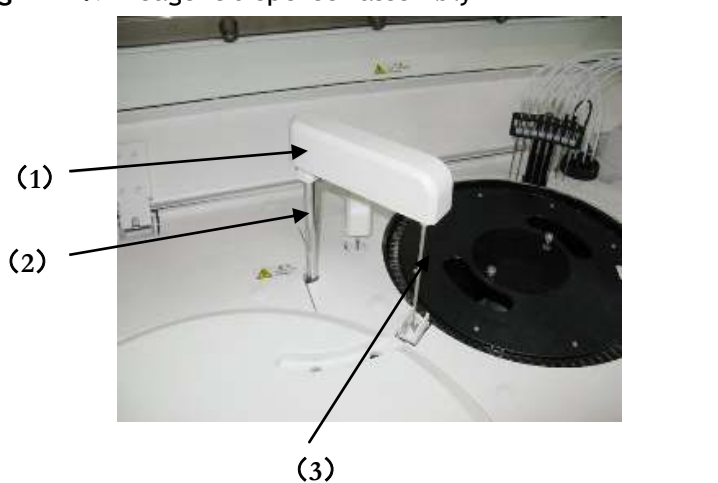
Reagent bar code reader

The reagent bar code reader is an optional module and used to obtain reagent information through reading a reagent bar code. For more information, refer to 1.3.4 Built-in Reagent Bar Code Reader (page 1-24).

Reagent dispenser assembly

The sample dispenser assembly located on the upper right of the reagent carousel consists of the reagent probe, probe arm, probe rotor, syringes and related tubing. It aspirates the specified amount of reagent from a reagent bottle and then dispenses it into a cuvette for reaction and analysis.

Figure 1.9 Reagent dispenser assembly



1. Probe arm
2. Probe rotor

3. Reagent Probe

Reagent probe

- The system has one probe. Reagent volume: 10 μ L ~350 μ L with 1 μ L increment.

The reagent probe is capable not only of aspirating reagent but also of the following functions:

- Horizontal obstruct detection: detects obstacles in the horizontal direction. When the reagent probe collides with an obstacle in the horizontal direction, the auto guard system is started to prevent the reagent probe from being damaged.
- Vertical obstruct detection: detects obstacles in the vertical direction. When the reagent probe collides with an obstacle in the vertical direction, the auto guard system is started to prevent the reagent probe from being damaged.

- Level detection and tracking: detects the reagent level and determines the depth of lowering down into the reagent based on the specified aspirate volume.
- Empty aspiration alarm: When the reagent probe cannot aspirate or aspirate insufficient reagent due to reagent insufficiency or air bubble, the system will give an alarm.



WARNING

When the system is in operation, do not place any part of your body or any obstacle in the route where the reagent probe arm moves; otherwise, personal injury or equipment damage may be caused.

Reagent probe washing

The reagent probe is cleaned in its wash well with water spraying its interior and exterior from two opposite directions.

Reagent syringe

When the left door of the analyzer is opened, you will see the reagent syringes as shown below.

Figure 1.10 Reagent syringe



(1)

1. Reagent syringe

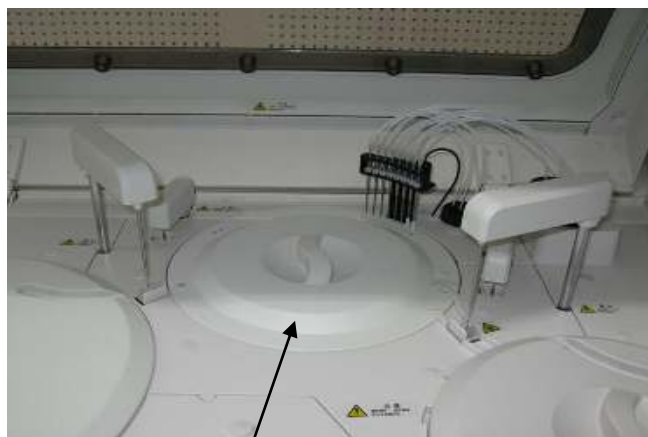
Reagent bottle

The reagent carousel is compatible with the following reagent bottles: Mindray outer ring reagent bottle 40ml or 20ml; Mindray inner ring reagent bottle 40 ml or 62 ml. Only 40 ml or 62 ml. Reagent bottle can be placed on the inner ring and only 40ml or 20ml reagent bottle can be placed on the outer ring.

1.2.4 Reaction System

The reaction carousel assembly consists of a reaction carousel, cuvettes and heater. See Figure below.

Figure 1.11 Reaction carousel



(1)

(1) Reaction carousel

The reaction carousel holds the cuvettes, in which the sample reacts with the reagent(s) and colorimetric readings are taken.

The semi-permanent plastic cuvettes are designed for reaction of sample and reagents, and also colorimetric measurement.

The heater provides a constant-temperature environment for reaction. Cuvettes on the reaction carousel are carried to the reagent dispensing position, sample dispensing position, mixing position and then washing position successively.

Reaction carousel

The reaction carousel rotates counter-clockwise, carrying the specified cuvette to reagent dispensing position, sample dispensing position, mixing position and then washing position successively.

The reaction carousel is circular and can hold 90 semi-permanent plastic cuvettes.

Reaction volume: 120 μ l-360 μ l.

The reaction carousel is capable of temperature control and provides a constant environment at $37\pm0.3^{\circ}\text{C}$ with fluctuation of $\pm0.1^{\circ}\text{C}$.

Reaction cuvette

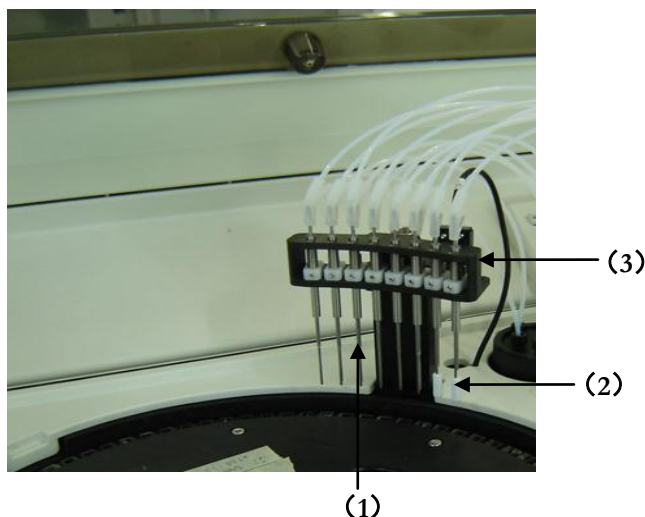
The plastic cuvette is provided by the factory. The light pathlength of the cuvette is $5\text{mm}\pm0.03\text{mm}$, and its inside dimension is 5mm (length)*5mm (depth)*29mm (height).

When finishing a test, the system washes and dries the cuvette automatically for later use.

1.2.5 Cuvette Wash Station

The system provides an eight-phase auto wash function, through which the cuvettes are washed via eight wash probes when a test is finished. The cuvette wash station consists of the wash probes, elevating motor and related tubing. The wash probes driven by the elevating motor to go up and down during each wash phase dispenses and aspirates wash solution in the cuvette to finish washing.

Figure 1.12 Cuvette wash station



- 1. Wash probe
- 2. Wipe block

- 3. Cuvette wash station

The cuvette wash station cleans the cuvettes with wash solution and Deionized water in eight phases, which are divided as follows:

- Phase 1 and 2: the cuvette is washed with diluted wash solution
- Phase 3 to 6: the cuvette is rinsed with deionized water
- Phase 7 and 8: the cuvette is dried and wiped

The cuvette is washed and rinsed with preheated diluted wash solution and deionized water in phase 1 to 6. After the washing, the waste fluid is discharged in two flows: high-concentration waste and low-concentration waste. The system is capable of detecting the waste fluid level and produces an alarm when detecting excessive waste.

1.2.6 Photometric System

The photometric system located inside of the analyzing unit measures the absorbance of the reaction mixture in the cuvette. The photometric system, composed of the photometer assembly and the signal detection assembly, measures the light transmitted through the reaction mixture and then converts the light change signal into electrical signal, which reflects the change of the light intensity.

The photometer assembly, which consists of the light source, colorimetric

1 System Description

component and optical component, provides sufficient monochromatic light and reliable colorimetric structure.

The signal detection assembly consists of the AD conversion component and the AD signal collection component. It converts the monochromatic light transmitted through the reaction mixture into an electrical signal, which is amplified and output as photometric data and then sent to the corresponding control unit for absorbance calculating.

The table below shows the main technical parameters of the photometric system.

Table 1.3 Specifications of photometric system

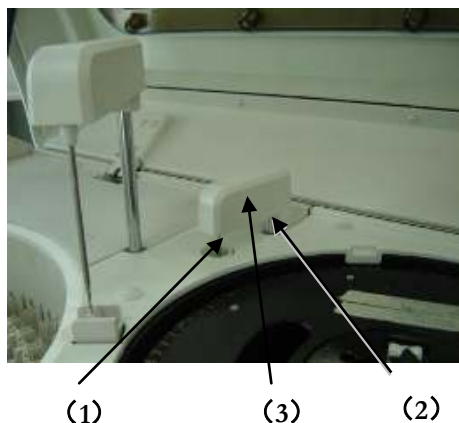
Name	Value
Light source	Tungsten-halogen lamp, 12V/20W
Colorimetric component	Reaction cuvette
Light transmission component	Holographic concave flat-field gratings
Light transmission mode	Reversed optics
Signal detector	Photodiode array
Measuring wavelength	12 wavelengths: 340nm, 380nm, 412nm, 450nm, 505nm, 546nm, 570nm, 605nm, 660nm, 700nm, 740nm and 800nm
Wavelength accuracy	±2nm
Measurement range	0-3.3A
Full width at half maximum (FWHM)	<10nm

1.2.7 Mixer Assembly

The mixer assembly, consisting of the sample mixer assembly and the reagent mixer assembly, is used to stir the reaction mixture when sample/R3 and R2/R4 are dispensed into the cuvette. There are two mixers: one is sample mixer, the other reagent probe.

Sample mixer assembly

Sample mixer assembly, located on the upper left side of the reaction carousel, mixes the sample in cuvette once it is dispensed and after R3 is dispensed.

Figure 1.13 Sample mixer assembly

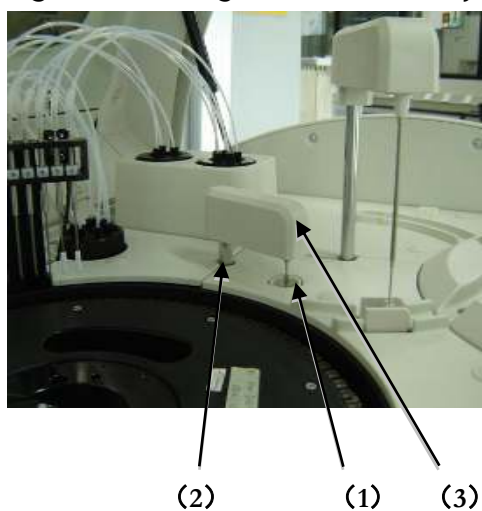
- 1. Sample mixer
- 2. Mixer rotor

- 3. Mixer arm

When stirring is finished, the mixer moves automatically to its wash well for cleaning.

Reagent mixer assembly

Reagent mixer assembly, located on the upper right side of the reaction carousel, mixes the sample in cuvette after R2 and R4 is dispensed.

Figure 1.14 Reagent mixer assembly

- 1. Reagent mixer
- 2. Mixer rotor

- 3. Mixer arm

1.2.8 Operation Unit

The operation unit is a computer configured with the operating software. It consists of the monitor, computer, keyboard and mouse.

1.2.9 Output Unit

The output unit is a printer used to print out test results and other data. The system supports three types of printer, which include inkjet printer, laser printer and stylus printer.

1.2.10 Accessories and Consumables

Accessories and consumables are replenishable components required to run tests and should be checked regularly for refilling and replacement.

Please use the accessories and consumables manufactured or recommended by our company in order to achieve the promised system performance. If needed, contact our customer service department or your local distributor.

For more information about accessories and consumables, refer to 16.1.3 Consumables (page 16-3).

1.3 Optional Modules

1.3.1 Introduction

Optional modules are not provided as standard configuration accompanying the instrument when it is delivered. They can be configured according to your requirements. The following modules are supplied:

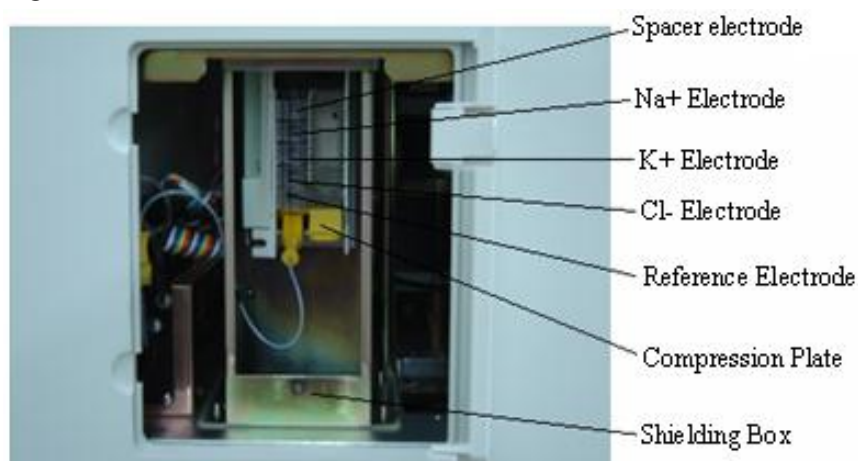
- ISE module
- Built-in sample bar code reader
- Built-in reagent bar code reader
- Water supply module
- Drainage module
- External vacuum pump

1.3.2 ISE Module

ISE (Ion-Selective Electrode) module consists of the Na⁺ electrode, K⁺ electrode, Cl⁻ electrode, reference electrode, pump, tubes and connectors. The ISE module measures the concentration of Na, K and Cl in serum, plasma and diluted urine.

The sample volume for measuring serum and plasma is 70µl; the sample volume for measuring diluted urine is 140µl. The theory of measurement is direct ion-selective electrode method.

Figure 1.15 ISE Module



1.3.3 Built-in Sample Bar Code Reader

The sample bar code reader is located on the left inside the sample carousel. The outer ring and middle ring of the sample carousel support bar code scanning. The

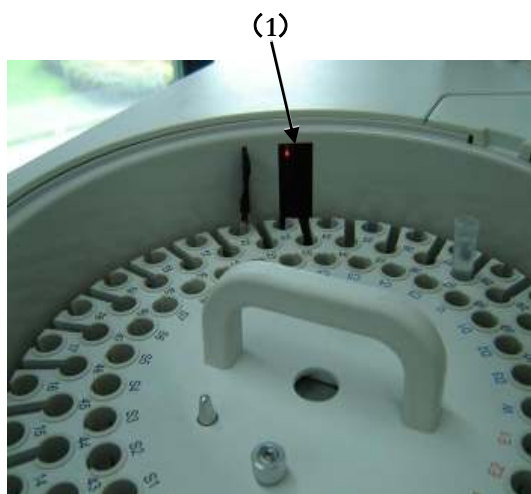
1 System Description

sample bar code reader assembly consists of the following components:

- Sample bar code reader
- Bar code label
- Hardware and software to control bar code scanning

When sample tubes are loaded to the sample carousel, the system scans the bar code label on the sample tubes to read the sample information and then display it on the screen.

Figure 1.16 Sample bar code scanning window



1. Sample bar code scanning window



WARNING

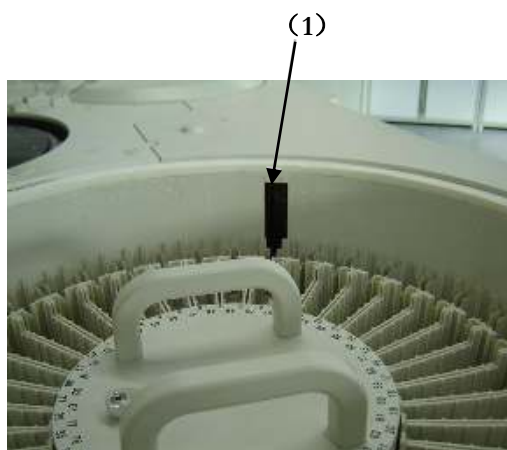
The light radiated from the sample bar code reader may hurt your eyes. Do not stare into the laser beam coming from the sample bar code reader.

1.3.4 Built-in Reagent Bar Code Reader

The reagent bar code reader located on the right inside the reagent carousel consists of the following components:

- Reagent bar code reader
- Bar code label
- Hardware and software to control bar code scanning

When the reagent carousel cover is closed after reagent bottles are loaded, select **End Load F2**, the system scans all reagents positions to reader reagent information and then displays it on the screen.

Figure 1.17 Reagent bar code scanning window

1. Reagent bar code scanning window

**WARNING**

The light emitted by the reagent bar code reader may cause eye injury. Do not stare into the laser beam coming from the reagent bar code reader.

1.3.5 Water Supply Module

The water supply module provides deionized water for the chemistry analyzer. When water is required during the measuring process, the water supply module turns on the internal inlet valve and transmits water while driven by the pneumatic pump; when water is not needed, the water supply module turns off the internal inlet valve and cuts off the power supply of the pneumatic pressure pump to stop supplying water.

1 System Description

Figure 1.18 Water supply module

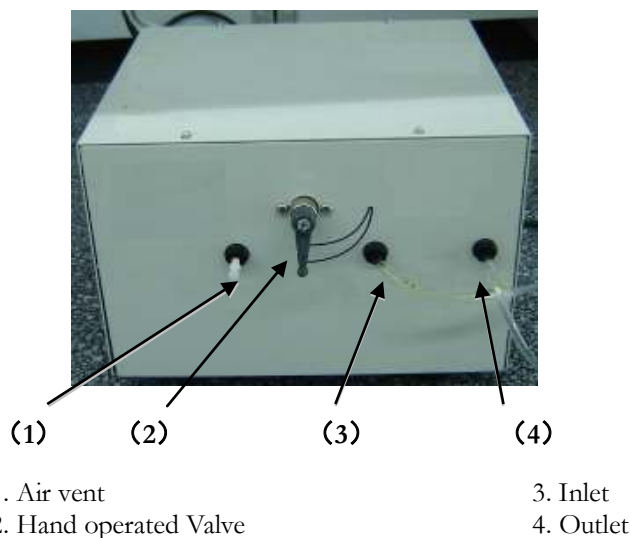
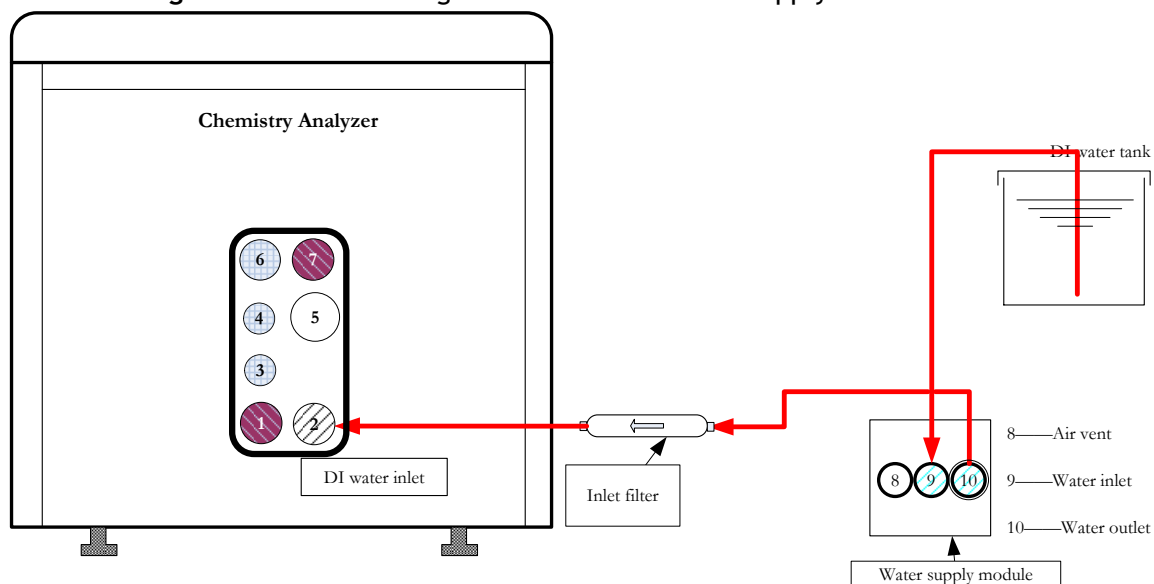


Figure 1.19 Connecting instrument with water supply module



Make sure that there is sufficient space between the water supply module and the wall so that it is convenient to connect or disconnect the power cord. Sufficient deionized water should be prepared in the water tank when using the water supply module. Make sure the water supply module is powered on before running. The module should be powered off if not used for a long time.

If there is something wrong with the water supply module, please consult our customer service department or your local distributor

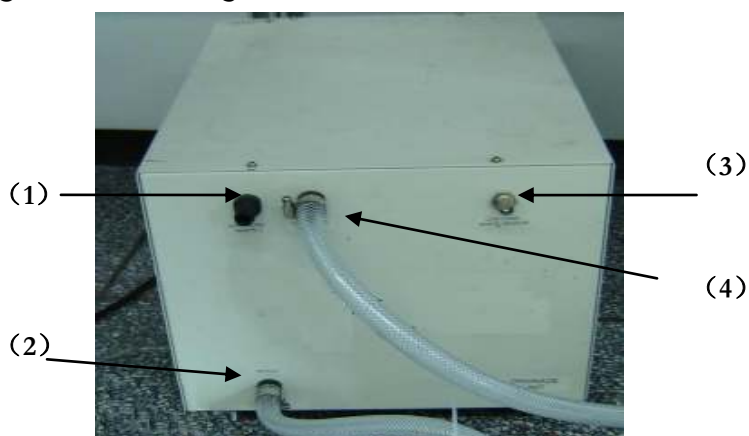
1.3.6 Drainage Module

The drainage module increases the pressure of waste liquids, which will be discharged more easily. When the outlet tubing of the analyzer exceeds 5m or the sewer is higher than 1.2m, a drainage module is required. Make sure that there is sufficient space between the drainage module and the wall so that it is convenient to connect or disconnect the power cord.

The drainage module collects and discharges to the waste buffer the low-concentration waste liquids from the outlet on rear panel of the analyzer. When the liquid level sensor detects that specified amount of waste liquids have been collected in the waste buffer, the waste pump of the drainage module starts running, discharging the waste to the sewer.

If there is something wrong with the drainage module, please consult our customer service department or your local distributor.

Figure 1.20 Drainage module



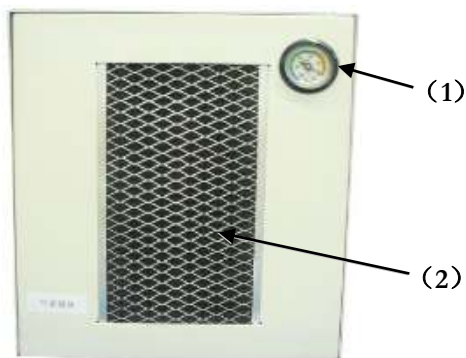
(1) Low conc. Waste 1
(2) Outlet

(3) Low conc. waste sensor
(4) Low conc. Waste 2

1.3.7 External Vacuum Pump

When operated in a place with the altitude above 2,000m, the system may be degraded in its liquid aspirating performance due to the decreased atmospheric pressure. In this situation, an external vacuum pump is required to assist the system with liquid aspiration.

Figure 1.21 Front view of the external vacuum pump



- 1. Pressure gauge
- 2. Dust screen

Figure 1.22 Rear view of the external vacuum pump



- 1. Gas connector
- 2. Control interface
- 3. Cooling fans
- 4. Power switch
- 5. Power jack

Make sure that there is sufficient space between the external vacuum pump and the wall so that it is convenient to connect or disconnect the power cord. Before using the vacuum pump, connect the gas connector and control interface with the counterpart connectors on the rear panel of the analyzer; connect the vacuum pump to the power supply with the three-wire power cord. The external vacuum pump will be controlled by the analyzer when powered on and requires no manual operations. When finishing all tests every day, you are recommended to power off the external vacuum pump. Before starting the tests every day, please make sure the external vacuum pump is powered on.

The pointer of the pressure gauge is deviated from the 0 point when the vacuum pump works normally. If the pointer stops at the 0 point while the vacuum pump is running, there must be something wrong with the external vacuum pump. Consult our customer service department or your local distributor.

The external vacuum pump should be installed and adjusted only by the technicians

of or authorized by our company.

1.3.8 Other Optional Modules

For more information about other optional modules, contact our customer service department or your local distributor.

1.4 Software Description

1.4.1 Main Screen

Figure 1.23 Main screen



1. Status display area
2. Function buttons area
3. Prompt message area
4. Function window
5. Shortcut icons area

Status display area

The status display area shows the status of analysis, LIS connection, printer and system date/time.

“None-test status” in this manual refers to Incubation, Standby, Stopped and Sleep; while “test status” refers to other status.

Table 1.4 Status display area

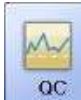
Status Indicator	Description
Biochemistry/ISE	This indicator appears on the left of the status display area. If Biochemistry/ISE appears intact, the biochemistry module and ISE module are enabled. The status of the biochemistry system includes: Initializing, Incubation, Standby, Running, Sample Load, Reagent Load, Inventory Check, Stop, Restoring, Sleep, Wake Up, Alignment, Maintenance, Diagnostics, SPT (System Performance Test) and Shutdown. The status of the ISE module includes: Standby, Running, Sample Load, Reagent Load, Pause, Maintenance, Stop, Restoring, Sleep, Wake Up and Alignment.
Analyzing time left	This indicator appears in the middle of the status display area. The time indicates how many minutes left the analysis will be finished.
Sample Stop/Reagent Stop	This indicator appears on the right of the status display area. The time indicates how many minutes left the dispensing of sample or reagent will be stopped.
	LIS status This indicator appears on the left of the status display area. The following information is indicated: - If  appears in blue, the LIS host is connected and online. - If  appears in grey, the LIS host is offline.
	Printer status This indicator appears on the left of the status display area. It indicates the status of the printer: not printing and printing. - If the icon appears in grey, the printer is not printing or unconnected. - If the icon appears in blue, the printer is printing.
Operator	This indicator appears in the middle of the status display area. It indicates the user who logs in the system.
Date and time	This indicator appears on the right of the status display area. It indicates the system date and time.

Function buttons area

The function buttons area contains the following buttons used to access various

1 System Description

function windows of the system:

- : used to program patient samples and control samples, and view sample carousel status.
- : used to recall test results of patient samples and controls and view the result statistics and test statistics.
- : used to load reagents, define/edit calibrators, request calibrations, recall calibration results and view reagent carousel status.
- : used to define/edit controls and rules, recall QC results and summary.
- : used to execute instrument commands, set up chemistry and system parameters, perform system maintenance and diagnostics, and view component status.
- : used to recall and handle error logs and deleting/editing logs.
- : used to exit the system by sleeping, logging off or shutting down.

Prompt message area

The prompt message area contains two lines, the upper line displaying operation prompts for screen controls and the lower line displaying error messages.

Function window

The function window contains options, buttons and other controls used to perform various functions of the system.

Shortcut icons area

The shortcut icons area contains the following icons used to quickly access certain function window or perform an operation:

- : Start icon. Select it to display the **Start Conditions** window, on which you are allowed to start new analysis or resume early testing.

- : Sample Stop icon. Select it to stop sample dispensing. And then you are allowed to load new samples or replenish samples.
- : Emergency stop icon. Select it to stop all tests. All tests that are running will be invalidated.
- : STAT icon. Select it to display the **STAT Sample Program** window, on which you are enabled to program emergency samples quickly.
- : Online help icon. Select it to display the online help of the current window, where you will find description of parameters and operations.

1.4.2 Using a Mouse

Move

The mouse is presented on the screen in the form of pointer. Place the mouse on a flat platform, and then move it to make the pointer lap over the object that you want to select or edit.

Select

Move the mouse to make the pointer lap over the object that you want to select or edit, and then press the left mouse button and release it quickly.

Double-click

Move the mouse to make the pointer lap over the object that you want to select or edit, and then quickly press the left mouse button twice and release it.

Drag

Dragging is used to move the slider on a screen in order to choose a scale. Move the mouse to make it stop over the slider, press and hold the left mouse button, move the mouse left and right to adjust the slider to the desired scale.

Using a mouse in conjunction with a keyboard

Some lists on the screen allow you to select more than one object at one time, and you can achieve this by using a mouse in conjunction with a keyboard. When selected, the objects will be highlighted for easy identification.

Perform the following operations to select more than one object:

- To select discontinuous objects, press the left mouse button to select the first object, press and hold the **Ctrl** key, use the mouse to select other desired objects, and then release the **Ctrl** key.

1 System Description

- To select continuous objects, press the left mouse button to select the first object, press and hold the **Shift** key, use the mouse to select the last object, and then release the **Shift** key.

1.4.3 Using Online Help

The system provides you with help information about the screens. If you do not understand a parameter or an operation on a screen, you can go to the online help for relevant information.

Accessing the online help

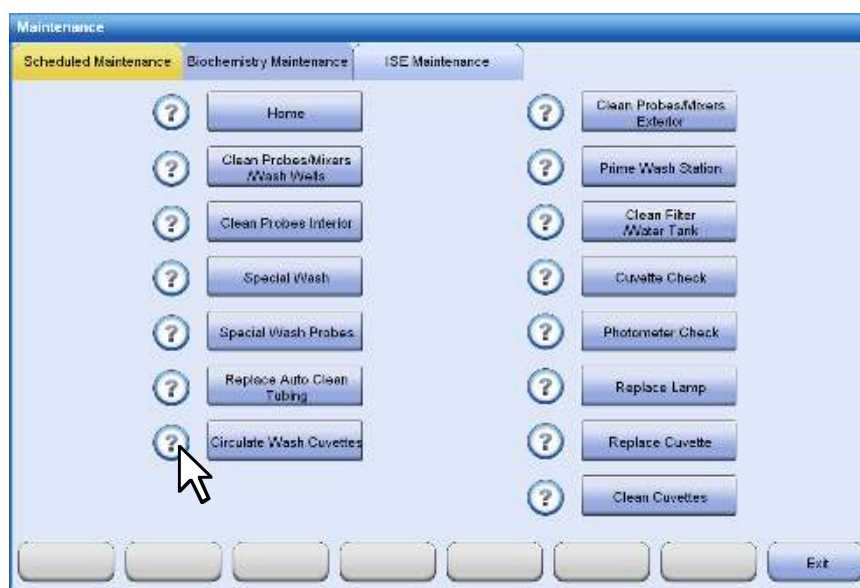
Access the online help from the following screens:

- Select the  icon on the upper right corner to display the help topic related to the current screen.

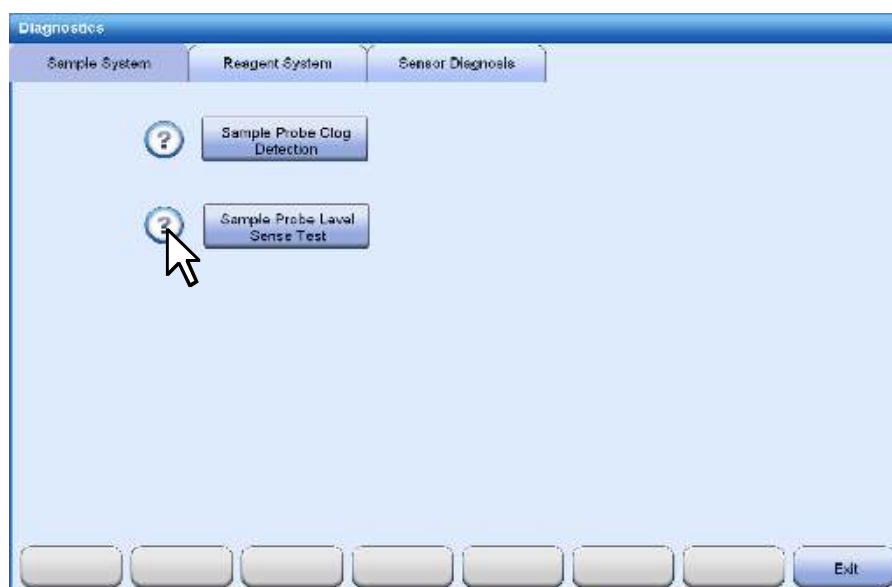
Figure 1.24 Accessing the online help from the main screen



- Select the  icon in front of each maintenance instruction or item to display the relevant operating instructions.

Figure 1.25 Accessing the online help from the Maintenance window

- Select the  icon in front of each diagnostic test to display the corresponding topic.

Figure 1.26 Accessing the online help from the Diagnostics screen

- Select the  icon in front of each error log to display the corresponding topic.

Figure 1.27 Accessing the online help from the Error Log screen



- Select the  icon on a warning message window to display the corresponding descriptions and solutions.
- Select the  icon on an error message window to display the corresponding descriptions and solutions.
- Press the shortcut combination key **Alt+F1** to display the topics related to the current screen or window.

Viewing screen information

The online help document contains descriptions of parameters, operations, maintenance and troubleshooting of the operating software. To view the information related to the current screen or window, perform the following steps.

1 Access the online help in the following ways:

- Select the  button on the upper right corner of the main screen, or press the shortcut combination key **Alt+F1**.
- To perform maintenance operations, select the  icon in front of the desired maintenance procedure.
- To view details of an error log, select the  icon in front of the error log.

- To view details of an alarm message, select the  icon on a warning or error message window.
- 2 Read the help topics. Move the scroll bar on the right side of the help window to view more information.
 - 3 Select  to close the help window.

Viewing other information

To view other information in the online help,

- 1 Select the  icon on the upper right corner of the main screen, or press the shortcut combination key **Alt+F1**.
- 2 Select the following tabs to view relevant information:
 - **Contents:** to navigate through all topics of the online help.
 - **Index:** to view topics related to the input keywords.
 - **Search:** to view topics containing the input keywords.
 - **Favourites:** to view your favorite topics.
- 3 Read the help topics. Move the scroll bar on the right side of the help window to view more information.
- 4 Select  to close the help window.

1.5 System Specifications

1.5.1 Technical Parameters

Throughput and reaction type

Table 1.5 Throughput and reaction type

Parameter	Description
Throughput for biochemistries	400 tests/hour for single-/double-reagent chemistries.
Throughput for ISE chemistries	Serum/plasma:240 samples/hour, and urine:171 tests/hour (including K, Na, Cl)
Biochemistries and ISE chemistries	560 tests/hour
Maximum number of tests run simultaneously	84 tests, which include 78 biochemistries, 3 ISE chemistries and 3 serum index chemistries.
Principles of analysis	Colorimetry, turbidity, and ISE method
Reaction types	Endpoint, fixed-time, and Kinetic
Reagent mode	Supporting single-/double-/triple-/quadruple-reagent tests
Wavelength	Supporting single/double-wavelength mode

Sample handling system

Table 1.6 Specifications of the sample handling system

Parameter	Description
Sample carousel	One carousel, including three rings. Each provides 30 positions, 90 positions in total.
Sample volume for routine chemistry	1.5µl-45µl, with increment of 0.1µl
Sample volume for ISE chemistry	Serum/plasma 70µL;diluted urine:140µl
Sample probe	One sample probe available, featuring level detection, horizontal/vertical obstruct detection, clog detection and level tracking.
Sample probe washing	The sample probe is cleaned in its wash well with water spraying its interior and exterior
Emergent samples	Emergent samples can be analyzed at any time with highest priority.

Parameter	Description
Rerunning mode	Supporting auto dilution, standard volume, increment, decrement, decrement dilution, increment dilution.

Reagent handling system

Table 1.7 Specifications of the reagent handling system

Parameter	Description
Reagent carousel	Reagent carousel provides 80 positions of which D is used for wash solution and W for physiological saline water.
Reagent volume	10 μ l~350 μ l with increment of 1 μ l.
Reagent probe	One reagent probe, featuring level detection, horizontal/vertical obstruct detection, empty aspiration alarm and level tracking.
Reagent probe washing	The reagent probe is cleaned in its wash well with water spraying its interior and exterior.

Mixer assembly

Table 1.8 Specifications of the mixer assembly

Parameter	Description
Mixer assembly	Composed of mixer, probe arm and probe rotor
Mixer	Two mixers available, one sample mixer and one reagent mixer

Reaction system

Table 1.9 Specifications of the reaction system

Parameter	Description
Reaction carousel	90 positions available
Reaction temperature	37°C $\pm$ 0.3°C
Reaction cuvette	Plastic cuvette 5mm $\times$ 5mm $\times$ 29mm (length $\times$ depth $\times$ height), light pathlength of 5mm $\pm$ 0.03mm
Reaction mixture volume	120 μ l-360 μ l

Photometric system

Table 1.10 Specifications of the photometric system

Parameter	Description
Light transmission mode	Holographic concave flat-field gratings
Light source	12V/20W tungsten-halogen lamp
Measuring wavelength	12 wavelengths: 340nm, 380nm, 412nm, 450nm, 505nm, 546nm, 570nm, 605nm, 660nm, 700nm, 740nm and 800nm
Measuring period	9 seconds

Water consumption

Less than 20L/H

Water supply module

Table 1.11 Specifications of water supply module

Parameter	Description
Power supply	100V-240V~, 50Hz/60Hz
Voltage fluctuation	±10%
Rated input power	50VA
Flux	0.6LPM
Tube length and connecting method	4*6mm PU tubes Connecting the water tank and the analyzer,<10m IN and the water tank ,<5m OUT2 and the waste outlet,<10m
Weight	9.7Kg(±1)
Size(length*width*height)	321.8mm×303.5mm×241.2mm(±5mm)
Maintenance requirement	No need to perform the maintenance procedure

Drainage module

Table 1.12 Specifications of the drainage module

Parameter	Description
Power supply	100V-240V~, 50Hz/60Hz
Voltage fluctuation	±10%

Parameter	Description
Rated input power	50VA
Flux	1LPM
Tube length and connecting method	12*18 mm braided tubes Connecting the analyzer and IN1/IN2,<5m OUT and the discharge outlet,<10m
Weight	12.5Kg(±1)
Size(length*width*height)	436.5mm×312.8mm×287.7mm(±5mm)
Maintenance requirement	No need to perform the maintenance procedure

External vacuum pump

Table 1.13 Specifications of the external vacuum pump

Parameter	Description
Power supply	110V: 110V/115V~, 60Hz 220V: 220V-240V~, 50Hz 220V/230V~, 60Hz
Voltage fluctuation	±10%
Rated input power	500VA
Vacuum flux	>10SLPM under the atmospheric pressure of -33KPa
Tube	PU tube, 7*10mm, <3m
Weight	29.7±1.2kg
Size	478mm×425mm×466mm
Maintenance requirement	Clean the dust screen monthly according to the operation guide

1.5.2 Power supply

Table 1.14 Power supply

Parameter	Description
Power supply	110V: 110V/115V~, 60Hz 220V: 220V-240V~, 50Hz 220V/230V~, 60Hz
Rated power consumption	1500VA
Voltage fluctuation	±10%
Frequency fluctuation	±1Hz

1.5.3 Environmental Requirements

Operating environment

- Temperature: 15°C-30°C
- Relative humidity: 35%-85%, without condensation
- Altitude height: -400m-2000m (An external vacuum pump is required for areas with altitude height between 2000-4000m.)

Storage environment

- Temperature: 0°C-40°C
- Relative humidity: 30%-85%, without condensation
- Altitude height: 50kPa-106kPa

1.5.4 Dimensions and Weight

- Dimension: $\leq 1190\text{mm}(\text{length}) \times 720\text{mm}(\text{depth}) \times 1155\text{mm}(\text{height})$
- Weight: $\leq 300\text{Kg}$

1.5.5 Input Device

- Keyboard (prepared by user)
- Mouse (prepared by user)
- Display monitor (prepared by user)
- Bar code reader

- LIS: HL7 and ASTM1394 (communicating through serial port or net port of the TCP/IP interface of static IP address)

1.5.6 Output Device

- Printer (prepared by user)
- Display monitor (prepared by user)
- LIS: HL7 and ASTM1394 (communicating through serial port or net port of the TCP/IP interface of static IP address)

1.5.7 Noise and Fuse

Table 1.15 Noise and fuse

Parameter	Description
Noise	≤ 65dBA
Fuse	For 110V: 250V 16A For 220V: 250V 13A

1.5.8 Communication Interfaces

Table 1.16 Communication interfaces

Communication Interfaces	Description
RS232 serial port	• Used for communication between the analyzing unit and the operation unit • Used for communication between the LIS and the operation unit • Used for connecting the operation unit with a printer
Network interface	• Used for communication between the analyzing unit and the operation unit • Used for communication between the LIS and the operation unit
USB interface	• Used for connecting the operation unit with a printer • Used for connecting the operation unit with an external storage device

1.5.9 Safety Classification

Table 1.17 Safety classification

Parameter	Description
Overvoltage type	Class II
Pollution degree	2
Device type	Fixed device
Work type	Continuous
Degree of IP(Ingress Protection)	Common device ,IPX0 (no protection against liquids)

1.5.10 EMC Requirements

This equipment complies with the emission and immunity requirements described in EN 61326-1:2006 and EN 61326-2-6:2006.

2 General Operating Procedure

This chapter illustrates the methods of using the instrument and the routine

- Check before powering on
- Powering on
- Checking system status
- Loading reagents
- Calibration
- Quality control
- Programming routine samples
- Programming STAT samples
- Test status and emergency stop
- Daily maintenance
- Powering off
- Check after powering off

2.1 General Operating Procedure

Table 2.1 General operating procedure

Procedures	Description	Page
1. Check before powering on	Check if the following components are ready for analysis: water supply, power supply, printing paper, low-/high-concentration waste connection, probes/mixers, concentrated wash solution inventory.	Page 2-3
2. Powering on	Turn on the water inlet valve, switch on the water supply module, drainage module and analyzing unit, and open the operating software.	Page 2-6
3. Checking system status	Check the status of the system, alarms, reagent/calibration, maintenance and subsystems.	Page 2-9
4. Preparing reagents	Prepare the biochemical reagents, ISE reagents and wash solutions.	Page 2-15
5. Calibration	Request calibrations, prepare calibrators and run calibration tests.	Page 2-27
6. Quality Control	Program, prepare and run control samples.	Page 2-33
7. Programming routine samples	Program, prepare and run routine samples.	Page 2-38
8. Programming STAT samples	Run emergent and STAT samples	Page 2-47
9. Test status and analysis control	View reagent status and reagent carousel status as well as the running status of calibrators, control samples, routine samples and STAT samples, pause and stop the analysis.	Page 2-53
10. Daily maintenance	Clean the ISE electrodes, sample and reagent compartments, analyzer panel, etc.	Page 2-59
11. Powering off	Switch off the water supply and power supply	Page 2-59
12. Check after powering off	Restore the reagent carousel cover, take out the calibrators, controls and samples from the sample carousel and store them properly, clean the analyzer panels, and empty the waste tank.	Page 2-61

2.2 Check before Powering On

2.2.1 Checking Water Supply

- 1 Check the deionized water tank or other water reservoirs, and make sure that water can be supplied continuously.
- 2 Check if the connections between the water supply, water supply module and analyzer are correct and tight.
- 3 Check if the water tubes are free of twists and leaks.

2.2.2 Checking Power Supply

- 1 Check if the power supply is available and can provide correct voltage:
- 2 Check the connections among the analyzing unit, operation unit and printer. Make sure the connections are correct and secure. Check the power cords of the analyzing unit, operation unit and printer and make sure they are well connected to the power sockets.

2.2.3 Checking Printing Paper

Check if sufficient printing paper is prepared in the printer. If not, refill the printing paper.

2.2.4 Checking Waste Tanks and Tubing

The waste fluid of the system is discharged in two flows: high-concentration waste and low-concentration waste. The former is drained through the waste tank and then disposed according to relevant regulations, or drained to the sewer; the latter is directly drained to the sewer.



BIOHAZARD

While checking the waste tanks and tubing, wear gloves and lab coat, if necessary, goggles.

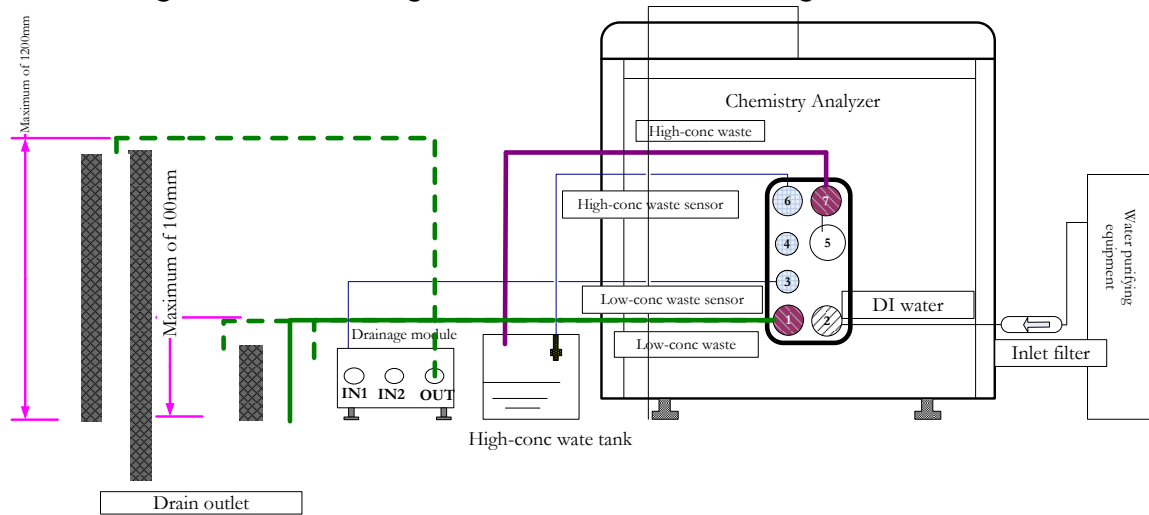
-
- 1 Check if the high-concentration waste tank has been emptied. If not, empty it.

High-concentration waste output: 1.5L/H (including ISE waste).

2 General Operating Procedure

- 2 Check if the low-concentration waste tubing is not bent and the sewer opening is lower than the waste outlet of the system.

Figure 2.1 Connecting instrument with waste drainage facilities



2.2.5 Checking Probes and Mixers

The sample probe, reagent probe and mixers are easy to be polluted or damaged. Check them carefully for dirt and bend before powering on the system.

- 1 Check the sample probe for dirt and bend.
 - If it is polluted, clean it.
 - If it is bent, replace it.
- 2 Check the reagent probe for dirt and bend.
 - If it is polluted, clean it.
 - If it is bent, replace it.
- 3 Check the sample mixer for dirt and bend.
 - If it is polluted, clean it.
 - If it is bent, replace it.
- 4 Check the reagent mixer for dirt and bend.
 - If it is polluted, clean it.
 - If it is bent, replace it.

2.2.6 Checking Concentrated Wash Solution

Insufficient concentrated wash solution may terminate the measurements. A tank of concentrated wash solution is 2L and can be used for analysis for 6 days on condition that plastic cuvettes are used and 2000 tests are done every day. Please check and refill the concentrated wash solution according to the consumption and tank volume.

- 1 Check the wash solution placed on sample and reagent carousels. If necessary, fill more or replace the wash solution.
- 2 Open the front door of the analyzer and check the concentrated wash solution. If necessary, fill more or replace the wash solution.

2.3 Powering On

2.3.1 Turning On Water Supply, Water Supply Module and Drainage Module

Turn on the water supply and power on the water supply module and drainage module and make sure the water entering the system is within 95kPa-392kPa.

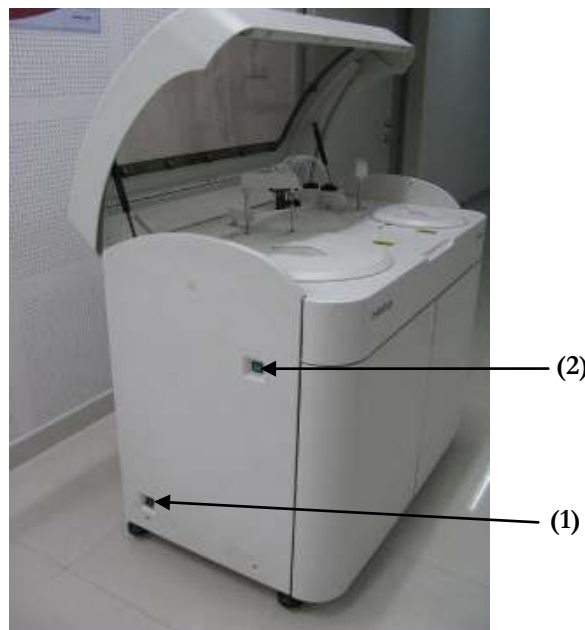
The water supply module and drainage module are optional equipments. For instructions of operating and maintaining these equipments, refer to their operation manual.

2.3.2 Powering On the System

After connecting correctly the system to the power sockets, switch on the power in the sequence presented below:

- 1 Turn on the main power switch of the analyzer.

Figure 2.2 Main power switch of the analyzer



(1) Main power switch of analyzer

(2) Analyzing unit power switch

- Place the switch to the  position to turn it on.
- Place the switch to the  position to turn it off.

- 2 Turn on the analyzing unit power switch (see the above figure).

- 3 Press the power button on the monitor of the operation unit.
- 4 Press the power button on the computer of the operation unit.
- 5 Press the power button of the printer.

2.3.3 Starting the Operating Software



Note

If virtual reagent carousel is used, please make sure that the loaded one is the NO.1 reagent carousel before starting the operating software each time.

- 1 When the operation unit (computer) is turned on, the operating software will run automatically.

If the system detects that the hardware and software environments of the computer do not meet the requirements, a prompt message will appear to ask for your confirmation to convert the screen resolution. If you cancel the conversion or the conversion is failed, you are allowed to abort the startup or reboot the system.

- 2 Enter the username and password in the **Login** window, and then select **OK**.
-



NOTE

The default username and password for administrator is Admin. Please note that the password is case sensitive. You are recommended to change the password when logging on the system for the first time in order to prevent others from abusing the privileges of the administrator.

If an operator forgets his password, he may ask the administrator to log on the system and delete the username and then redefine a username; or he may contact our customer service department or your local distributor. If the administrator forgets his password, contact our customer service department or your local distributor.

- 3 When the startup check is passed, the main screen shows. The startup procedure is finished.

The system will display prompt message when detecting unsatisfied environment during the startup process. Please take actions according to the instructions in the message box.



CAUTION

To ensure accurate test results, do not start measurement until the system status turns to Standby and the system has been turned on for about 20 minutes, so that the light source and reaction temperature gets steady.

2.4 Checking System Status

After the startup procedure is finished, check the system status, such as system status, alarm status, reagent/calibration status, maintenance status and sub system status. If the status is not satisfied for measurement, troubleshoot and maintain the system as instructed by 17 Alarms and Troubleshooting (page 17-1) and 16 Maintenance (page 16-1).

2.4.1 Checking System Status

Printer status

Check the printer status indication in the system status area of the main screen:

- If the  icon appears in blue, the printer is printing.
- If the  icon appears in grey, the printer is not printing.

ISE module status

Check the ISE module status indication in the system status area of the main screen:

- If Initializing is displayed, it indicates that the ISE module is performing the startup procedure. Do not start measurements until the initialization is finished.
- If Standby is displayed, it indicates that an ISE module is steady and ready for measurement.
- If Running is displayed, it indicates that the ISE module is performing measurements.
- If Stop is displayed, it indicates that the ISE module goes wrong or is stopped. Troubleshoot the ISE module and take relevant corrective solutions.

LIS status

Check the LIS status indication in the system status area of the main screen:

- If  appears in blue, the LIS host is connected and online.
- If  appears in grey, the LIS host is offline.

2.4.2 Checking Alarm Status

1 Check the **Alarm** button on the left of the main screen.

- If it appears in yellow, it indicates that a warning occurs. Proceed to the next step.

2 General Operating Procedure

- If it appears in red, it indicates that an error occurs, or both warning and error occur. Proceed to the next step.

2 Select the **Alarm** button. The **Error Log** screen is displayed.

Figure 2.3 Error Log screen



- 3 New alarm messages are indicated by corresponding colors. Select the help button in front of a new alarm message to view relevant description and solutions.
- 4 Take actions according to the recommended solutions.

2.4.3 Checking Reagent/Calibration Status

1 Check the **Reagent** button on the left of the main screen.

- If it appears in yellow, it indicates that a warning occurs. Proceed to the next step.
- If it appears in red, it indicates that an error occurs, or both warning and error occur. Proceed to the next step.

2 Select the **Reagent** button. The **Reagent/Calibration** screen is displayed.

Figure 2.4 Reagent/Calibration screen

Pos	Chem	Chems Left	Rpt Type	Tests Left	Days Left	Lot No.	Cal Status	Time Left
64	ALT	0	R1	0	<58d	2016	Cal Required	
27			R2		<59d	1017		
85	ALT-		IR2	208	<59d		Calibrated	
51	CREA-J		IR1	0	<28d	2016	Cal Required	
52	CREA-E		IR1	0	<50d	2006	Cal Required	
30	GLHK		IR2	214	<59d		Calibrated	
63	TP	0	R1	0	<59d	2017	Cal Required	
70			R1	0	<59d	2017		
2	TP-		R1	103	<59d		Calibrated	
3 M			R1		20d			
41	UA	0	R1	0	<58d	2012	Cal Required	
42			R1	0	<58d	2012		
44			R1	0	<59d	2013		
53			R2	0	<59d	2012		

- 3 View the reagent status. When a reagent is insufficient or exhausted, the corresponding chemistry name and chemistries left will be indicated as follows:
 - Yellow: indicates that the reagent is insufficient or expired, and the analysis will continue. Refill or replace the reagent.
 - Red: indicates that the reagent is exhausted or at least one reagent type is not loaded, and the analysis is stopped. Refill or replace the reagent.
- 4 View the calibration status. When the calibration is succeeded or failed, the **Cal Status** column of the chemistry shows the calibration status in corresponding color.
 - Yellow: indicates that the calibration factors of the chemistry have been calculated, or extended, edited or overridden.
 - Red: indicates that the calibration of the chemistry is failed or expired, or the chemistry needs to be calibrated.
- 5 Check the calibration time left.
- 6 Take actions according to the calibration status.

For more information about calibration, refer to 2.6 Calibration (page 2-27).

2.4.4 Checking Maintenance Status

When the system is started up, it is necessary to check the maintenance status. If a maintenance procedure is expired, perform it immediately to make sure that the system will run normally. When a maintenance procedure is expired, the following buttons and options will be indicated by corresponding color:

- **Utility** button on the left of the main screen
- **Maintenance** tab
- **Maintenance** button
- **Scheduled Maintenance** tab
- Maintenance frequency tab
- Maintenance procedure

- 1 Check the **Utility** button on the left of the main screen. If it appears in yellow, it indicates that a maintenance procedure is expired.
- 2 Select **Utility-Maintenance-Maintenance**.
- 3 Check if the **Scheduled Maintenance** tab and maintenance frequency tabs appear in yellow. If they do, it indicates that at least one maintenance procedure is expired.
- 4 Select the maintenance frequency tab appearing in yellow, find the expired maintenance procedure, and then perform the maintenance.
- 5 Repeat steps 3 and 4 until the maintenance frequency tabs and maintenance procedures are displayed in normal color.

2.4.5 Checking Subsystems

The subsystem status indicates the current working status of each subsystem and hardware component, which includes the status summary, cycle count, temperature, fans, Hydropneumatic subsystem, and control modules.

Checking subsystems

- 1 Select **Utility-Status**.
- 2 Choose a subsystem tab;
- 3 Check the subsystem status. When abnormality occurs, troubleshoot errors with the following methods:

- If the cycle count of a component reaches certain limit and an alarm occurs, replace the component or contact out customer service department or your local distributor for replacement of the component.
- If a component's temperature is beyond the valid range or abnormal and an alarm occurs, exit the operating software and switch off the analyzing unit power. After that, switch on the analyzing unit power again and run the operating software. If the error remains, contact out customer service department or your local distributor for replacement of the component.
- If the status of the fans is abnormal, exit the operating software and switch off the analyzing unit power. After that, switch on the analyzing unit power again and run the operating software. If the error remains, contact out customer service department or your local distributor for replacement of the fan.
- If a Hydropneumatic component is beyond the valid range or abnormal and an alarm occurs, exit the operating software and switch off the analyzing unit power. After that, switch on the analyzing unit power again and run the operating software. If the error remains, contact out customer service department or your local distributor for replacement of the component.
- If a smart module is abnormal and an alarm occurs, exit the operating software and switch off the analyzing unit power. After that, switch on the analyzing unit power again and run the operating software. If the error remains, contact out customer service department or your local distributor for replacement of the component.
- If the control unit is abnormal and an alarm occurs, exit the operating software and switch off the analyzing unit power. After that, switch on the analyzing unit power again and run the operating software. If the error remains, contact out customer service department or your local distributor for replacement of the component.

Description of subsystem status

Status summary

The status summary provides a high-level summary of the status of the system temperatures, fans, Hydropneumatic, and control modules.

Cycle count

The cycle count provides an approximation of a component's usage, which can be useful for estimating the maintenance frequencies or anticipating component failure.

Temperatures

The actual temperature and valid range of the deionized water, reagent carousel, reaction carousel, and wash station are displayed.

Fans

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The actual status of the reagent refrigeration fans, lamp housing fan, vacuum pump cooling fan is displayed

Hydropneumatic subsystem

Status for the Hydropneumatic subsystem shows: working status of various tanks.

The actual air pressure and valid range for air pressure equipment

Smart modules

Smart module status monitors the working status of each smart module, which includes probes, mixers, carousels, cuvette wash station, ISE unit, etc.

2.5 Preparing Reagents

After confirming the system status and performing the daily checks, prepare the reagents for measurement. Chemistries without reagents loaded can be requested but will not be included in measurements. Loading reagents is allowed when the system status is Standby and Incubation. In the case of Running, you must request Reagent stop before you load the reagents. In the case of Sleep, reagents cannot be loaded until the instrument is woken up. After assigning reagent positions, print out the reagent list and then manually load reagents according to it. When all reagents are loaded, the system will check the reagent inventory during measurement and then display it on the **Reagent/Calibration** screen. You are recommended to perform inventory check manually after loading reagents; otherwise, the tests left will not be displayed on the **Reagent/Calibration** screen.



WARNING

The probe tip is sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the probes.



BIOHAZARD

Wear gloves and lab coat, if necessary, goggles.

Do not touch the reagent directly with your body; otherwise, skin wound or inflammation may be caused.

2.5.1 Loading Biochemical Reagents

The system supports manual and auto load of biochemical reagents. Each chemistry can have more than one bottle of reagent loaded; however the reagent of same chemistry must be loaded on the same reagent carousel. If your system is not equipped with a reagent bar code reader, you need to enter the reagent information manually when loading reagents; if a reagent bar code reader is configured, the system will scan all reagents automatically and read reagent information from the bar code.

If no bar code is scanned on automatically loaded reagents, they will be unloaded automatically. To reload these reagents, input the bar code manually. No matter if reagents are loaded manually or automatically, the newly scanned bar code will substitute for the previous one if they are not the same.

When one or more reagents of a multi-reagent chemistry are not loaded, the “!” sign will appear near the chemistry’s reagent types that have been loaded.

Reagents can be loaded manually or via bar code scanning. For more information about loading bar-coded reagents, refer to 13.2.3 Loading Bar-Coded Reagents (page 13-19).

**NOTE**

Before loading biochemistry reagent, ensure that there are no air bubbles inside the reagent bottle so as to avoid inaccurate test results.

Manual load

When loading reagents manually, you need to enter the reagent information, which is the only information source of the loaded reagents. You are allowed to input reagent information before, during or after loading reagents to the reagent carousel. If loaded reagents are bar-coded, the reagent information cannot be edited; otherwise, all reagent information except for position, chemistry and reagent type can be edited. Manually loaded reagents have the letter “M” (Manual) appearing near them.

Figure 2.5 Flag for manually loaded reagents



(1) Flag “M” for manually loaded reagents

1 Check the system status and operate accordingly.

- Standby: Proceed to the next step.
- Running: Select **Reagent-Reagent/Calibration**. Select **Load F1** to stop reagent aspirating and dispensing. Meanwhile **Load F1** becomes **No load F1**. If you want to abort load, select **No load F1**. When the countdown for

reagent stop becomes 0 and the system status is Reagent Load, a message box pops up. Select **OK**, and then proceed to the next step.

- Incubation: Proceed to the next step.
- Sleep: Select **Utility-Commands-Wake up**, and then proceed to step 2.

- 2 Select **Reagent-Reagent/Calibration**.
- 3 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 4 Select the down-arrow button on the right side of the screen to display the biochemical reagents.
- 5 Choose a position to which you want to load a reagent.
- 6 Select **Load F1**. The **Load Reagent** window is displayed.

Figure 2.6 Load reagent window

The screenshot shows the 'Load Reagent' window with the following fields and values:

Field	Value
Position	4
Bar Code	
Chem	Glu-H
Bottle Type	40ml
Serial No.	
Lot No.	
Rgt Type	R1
Exp Date	10/26/2013

At the bottom, there are five buttons: Prev F1, Next F2, Load F3, Unload F4, and Exit F5.

- 7 Enter the following reagent information:

- Bar code
- Chemistry name (required)
- Reagent type (required)
- Lot number

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- Serial number
- Bottle type (required)
- Expiration date

- 8 Select **Load F3** to save the input information.
- 9 Select **Prev F1** and **Next F2** to load reagents for other chemistries.
- 10 Select **Print F7** to print out the biochemical reagent list.
- 11 Remove the reagent carousel cover.



CAUTION

If the system is running tests, after requesting reagent stop, do not remove the reagent carousel cover until the countdown for reagent stop is 0, the system status is Reagent Load, and the popup message is confirmed; otherwise, probe collision or other error may occur.

- 12 Load reagents according to the reagent load list. Place the reagents in position 1-78 on the reagent carousel, and then uncap the reagent bottles.
- 13 Restore the reagent carousel cover.
- 14 Select **End Load F2**.

Auto load

Auto load is to load bar-coded reagents to the reagent carousel, which are identified by bar code scanning.

- 1 Check the system status and operate accordingly.
 - Standby: Proceed to the next step.
 - Running: Select **Reagent-Reagent/Calibration**. Select **Load F1** to stop reagent aspirating and dispensing. Meanwhile **Load F1** becomes **No load F1**. If you want to abort load, select **No load F1**. When the countdown for reagent stop becomes 0 and the system status is Reagent Load, a message box pops up. Select **OK**, and then proceed to the next step.
 - Incubation: Proceed to the next step.
 - Sleep: Select **Utility-Commands-Wake Up** to awake the system, and then start loading reagents.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 3 Remove the reagent carousel cover.

**CAUTION**

If the system is running tests, after requesting reagent stop, do not remove the reagent carousel cover until the countdown for reagent stop is 0, the system status is Reagent Load, and the popup message is confirmed; reagent probe collision or other error may occur.

- 4 Place the reagents in position 1-78 of the reagent carousel and then uncap the reagent bottles.
- 5 Restore the reagent carousel cover.
- 6 Select **End Load F2**.

If reagent barcode reader has been configured, the system scans all reagent positions and read the following reagent information from the bar code:

- Chemistry name
- Reagent type
- Days left
- Lot number
- Serial number and bottle type

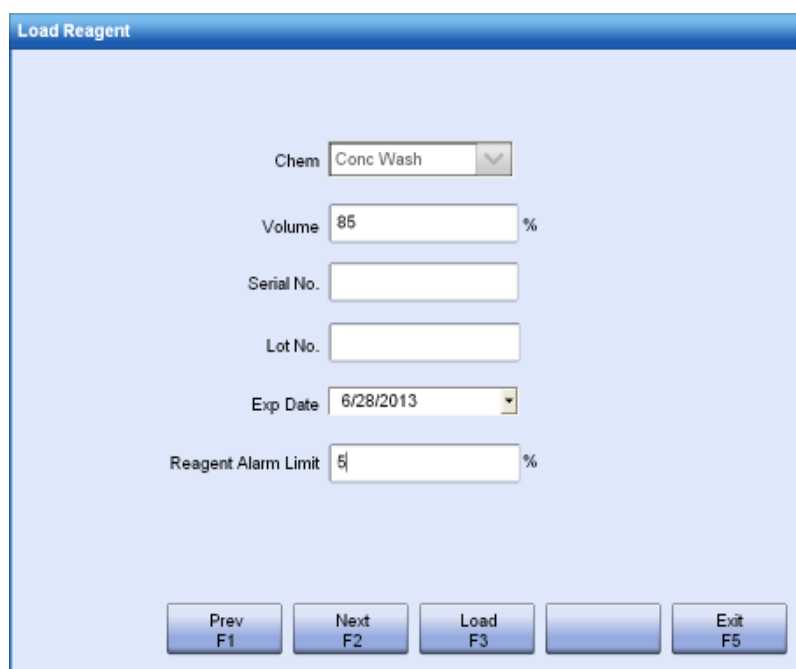
2.5.2 Loading Concentrated Wash Solution

Concentrated wash solution is used to clean reaction cuvettes and can only be loaded manually. The lot number, serial number, expiration date, volume, reagent alarm limit and other information of the loaded wash solution must be entered. Before loading concentrated wash solution, ensure that the diluted wash solution is enough for the tests in progress.

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select **Conc Wash** in the lower reagent list.
- 3 Select **Load F1**. The **Load Reagent** window is displayed.

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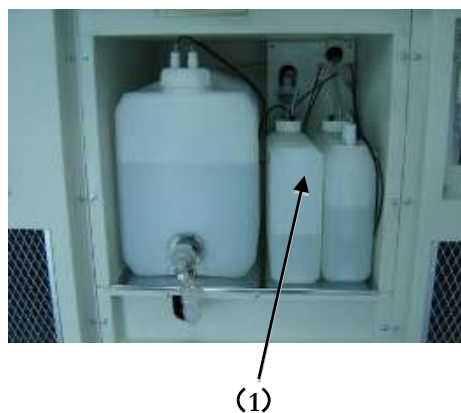
Figure 2.7 Load reagent window(Concentrated wash solution)



4 Open the front door of the analyzer.

5 Load the concentrated wash solution.

Figure 2.8 Positions for concentrated wash solution



(1) Concentrated wash solution

6 Close the front door of the analyzer.

7 Enter the following information:

- Volume % (required)
- Serial number

- Expiration date
- Lot number
- Reagent alarm limit

8 Select **Load F3**.

9 Select **Exit F5** to close the window.

2.5.3 Loading Reagent Probe Wash Solution

Reagent probe wash solution can only be loaded manually. The volume, lot number, serial number, expiration date, bottle type and other information of the loaded wash solution must be entered.

You are recommended to check the reagent probe wash solution every day to ensure its sufficiency.



NOTE

Before loading wash solution, ensure that there are no air bubbles inside the reagent bottle so as to avoid affecting washing effects.

1 Check the system status and operate accordingly.

- Standby: Proceed to the next step.
- Running: Select **Reagent-Reagent/Calibration**. Select **Load F1** to stop reagent aspirating and dispensing. Meanwhile **Load F1** becomes **No load F2**. If you want to abort load, select **No load F2**. When the countdown for reagent stop becomes 0 and the system status is Reagent Load, a message box pops up. Select **OK**, and then proceed to the next step. Incubation: Proceed to the next step.
- Sleep: Select **Utility-Commands-Wake up**, and then proceed to step 2.

2 Select **Reagent-Reagent/Calibration**.

3 Select **Wash D** in the lower reagent list.

4 Select **Load F1**. The **Load Reagent** window is displayed.

Figure 2.9 Load reagent window(Reagent probe wash solution)

The screenshot shows a software window titled "Load Reagent". It contains the following fields and controls:

- Position:** A dropdown menu with "79" selected.
- Bar Code:** An empty text input field.
- Chem:** A dropdown menu with "Wash D" selected.
- Bottle Type:** A dropdown menu with "40ml" selected.
- Serial No.:** An empty text input field.
- Lot No.:** An empty text input field.
- Volume:** A text input field with "100" and a "%" symbol to its right.
- Exp. Date:** A date picker showing "2015- 9-13".
- Reagent Alarm Limit:** A text input field with "5" and a "%" symbol to its right.

At the bottom of the window, there are five buttons labeled "Prev F1", "Next F2", "Load F3", "Unload F4", and "Exit F5".

- 5 Remove the reagent carousel cover.
- 6 Place the wash in position D (No.79) of the reagent carousel.
- 7 Restore the reagent carousel cover.
- 8 Enter the following information:
 - Volume (%)
 - Serial number
 - Expiration date
 - Lot number
 - Bottle type (required)
 - Reagent alarm limit
- 9 Select **Load F3**.
- 10 Select **Exit F5** to close the window.
- 11 If a reagent bar code reader has been configured, select **End Load F2**, the system will scan the reagent carousel.

2.5.4 Loading Sample Probe Wash Solution

Sample probe wash solution is used to clean the sample probe, and can only be loaded manually. When the sample probe wash solution is expired or exhausted, fill more sample probe wash solution.

Three special washes will be conducted for the sample probe when every batch of tests is finished, and about 90µl wash solution is consumed for each wash. The amount of concentrated wash solution for weekly cleaning of reaction cuvettes is 1ml. You are recommended to check and replace the sample probe wash solution every day to ensure its sufficiency.



NOTE

Before loading wash solution, ensure that there are no air bubbles inside the test tube so as to avoid affecting washing effects.

- 1 Check the system status and operate accordingly.
 - Standby: Proceed to the next step.
 - Running: Select the  button on the upper-right corner of the main screen to stop sample aspirating and dispensing. When the countdown for sample stop becomes 0 and the system status is Sample Load, proceed to the next step.
 - Incubation: Proceed to the next step.
 - Sleep: Proceed to the next step.
- 2 Remove the sample carousel cover;
- 3 Place sample probe wash in position D2 of the sample carousel.
- 4 Restore the sample carousel cover;

2.5.5 Loading Physiological Saline

Physiological saline is used to run sample blanks, reagent blanks and calibrations, and dilute samples, and it can only be loaded manually. The bottle type and volume of the loaded saline must be entered. Physiological saline used for running sample blanks and diluting samples should be loaded to the position W on the reagent carousel; and that for running reagent blanks and calibrations should be loaded manually to the position W on the sample carousel.

Loading physiological saline on the sample carousel

- 1 Check the system status and operate accordingly.
 - Standby: Proceed to the next step.

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- Running: Select the  button on the upper-right corner of the main screen to stop sample aspirating and dispensing. When the countdown for sample stop becomes 0 and the system status is Sample Load, proceed to the next step.
- Incubation: Proceed to the next step.
- Sleep: Proceed to the next step.

2 Remove the sample carousel cover.

3 Place physiological saline in position W of the sample carousel.

4 Restore the sample carousel cover;

Loading physiological saline on the reagent carousel

1 Check the system status and operate accordingly.

- Standby: Proceed to the next step.
- Running: Select **Reagent-Reagent/Calibration**. Select **Load F1** to stop reagent aspirating and dispensing. Meanwhile **Load F1** becomes **No load F1**. If you want to abort load, select **No load F1**. When the countdown for reagent stop becomes 0 and the system status is Reagent Load, a message box pops up. Select **OK**, and then proceed to the next step.
- Incubation: Proceed to the next step.
- Sleep: Select **Utility-Commands-Wake Up** to awake the system, and then proceed to step 2.

2 Select **Reagent-Reagent/Calibration**.

3 Select **Saline W** in the lower reagent list.

4 Select **Load F1**. The **Load Reagent** window is displayed.

Figure 2.10 Load reagent window(Physiological saline)

Load Reagent

Position: 80

Bar Code:

Chem: Saline W

Bottle Type: 40ml

Serial No.:

Lot No.:

Volume: 42 %

Exp Date: 10/ 4/2012

Reagent Alarm Limit: %

Prev F1 Next F2 Load F3 Unload F4 Exit F5

- 5 Remove the reagent carousel cover.



CAUTION

If the system is running tests, after requesting reagent stop, do not remove the reagent carousel cover until the countdown for reagent stop is 0, the system status is Reagent Load, and the popup message is confirmed; otherwise, Probe collision or other error may occur.

- 6 Place the physiological saline for sample blanks and sample dilution in position W (No.80) of the outer ring of the reagent carousel.
- 7 Restore the reagent carousel cover.
- 8 Enter the following information of physiological saline for sample blanks and sample dilution:
 - Volume %
 - Bottle type
 - Reagent alarm limit
- 9 Select **Load F3**.
- 10 Select **Exit F5** to close the window.

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11 Select **End Load F2**.

2.6 Calibration

Running calibration is to calculate calibration factors for sample result calculation. Generally, calibration is required when one of the following conditions occurs:

- A new chemistry is configured.
- QC alarms are given while the reagent, calibrator and control sample are within the expiration date.
- Reagent lot or bottle is changed.
- The calibration factors of a chemistry are expired.
- The ISE electrodes are adjusted or the ISE module is maintained.
- The calibration rules are changed, such as calibration method, replicates, concentration and calibrator.
- The chemistry parameters are changed, such as primary wavelength, secondary wavelength, blank time, reaction time, reagent volume (R1/R2/R3/R4), sample volume, sample dilution parameters, reaction type, reaction direction, sample blank and result unit.
- The lamp, syringe or sample probe is replaced.

If any of the following chemistry parameters are changed, a calibration is required:

- Primary wavelength
- Secondary wavelength
- Blank time
- Reaction time
- Reagent volume(R1/R2/R3/R4)
- Sample volume
- Reaction type
- Reaction direction
- Sample blank and result unit
- Twin chemistries

For more information about calibration setup, refer to 3.3 Calibration Setup (page 3-33).

2.6.1 Requesting Calibrations

General calibration request

When one of the above-mentioned conditions is happened, request a calibration

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according to the steps stated below.

Before requesting a calibration, make sure that the calibrator has been loaded to correct position.

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 3 Select the down-arrow button on the right side of the screen to display the biochemical reagents.

Figure 2.11 Reagent/Calibration screen



- 4 Select chemistries you want to calibrate.

Select the up-/down-arrow buttons to select more chemistries.

- 5 Select **Cal F5**.
- 6 Select **Calibration**.
- 7 Select **OK**.

Reagent lot calibration

After selecting the **Manage Reagents by Lot** option on the **System Setup** screen, you are allowed to calibrate each reagent lot of a chemistry and view all calibration results on the **Biochemistry Calibration** screen.

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 3 Select the down-arrow button on the right side of the screen to display the biochemical reagents.

Figure 2.12 Reagent/Calibration screen



- 4 Select reagent lots you want to calibrate.
Select the up-/down-arrow buttons to select more chemistries.
- 5 Select **Cal F5**.
- 6 Select **Calibration**.
- 7 Select **OK**.
- 8 Select the **Biochemistry Calibration** tab to view calibration results.

Requesting a calibration based on calibration status

When a chemistry has the calibration status of Cal Required, Cal Failed or Cal Time Out, the system will give an alarm. Perform the following steps to request a calibration based on the calibration status:

- 1 Check the **Reagent** button on the left of the main screen.
 - Yellow: indicates that a warning occurs.
 - Red: indicates that a serious error occurs.
- 2 If the **Reagent** button is highlighted, select **Reagent-Reagent/Calibration**.
- 3 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 4 Select the down-arrow button on the right side of the screen to display the biochemical reagents.
- 5 Check the biochemical chemistries of which the calibration status is highlighted.
- 6 Select chemistries that you want to calibrate.
- 7 Select **Cal F5**.
- 8 Select **Calibration**.
- 9 Select **OK**.

Auto calibration

The system provides the auto calibration option. When the conditions are satisfied, the system displays a message indicating calibration required and then stops running the corresponding chemistry. The conditions for auto calibration include:

- Calibration factors are expired
- Reagent lot is changed
- Reagent bottle is changed

For more information about auto calibration, refer to 6.5 Auto Calibration (page 6-14).

2.6.2 Loading Calibrators



BIOHAZARD

Inappropriate handling of calibrators may lead to biohazardous infection. Do not touch the calibrators directly with your hands. Wear gloves and lab coat, if necessary,

goggles. In case your skin contacts the calibrators, follow standard laboratory safety procedure and consult a doctor.

**CAUTION**

Do not use expired calibrators; otherwise, unreliable test results may be caused.

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 3 Select the down-arrow button on the right side of the screen to display the biochemical reagents.
- 4 Select **Load List F4**.

The calibrator list shows all requested chemistries as well as calibrators, positions, concentration, lot number and expiration date.

- 5 Select **Print F7**.
 - 6 Select **Close F8**.
 - 7 Load calibrators to the sample carousel according to the calibrator list.
-

**NOTE**

Calibrators of a chemistry must be placed and analyzed on the same sample carousel.

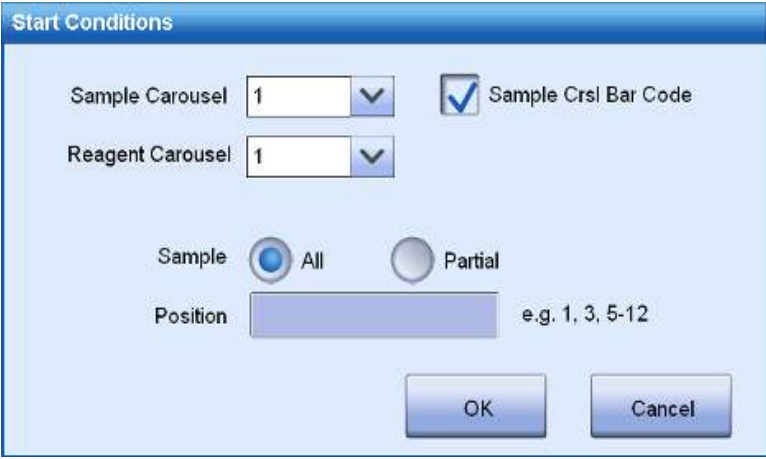
2.6.3 Running Calibrations

After requesting calibrations and load calibrators to the sample carousel, you can start the calibration test.

- 1 Select  on upper right corner of the main screen. The **Start Conditions** window is displayed.

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Figure 2.13 Start Conditions window



The 'Start Conditions' window is a dialog box with a blue title bar. It contains the following controls:

- Sample Carousel:** A dropdown menu showing '1'.
- Reagent Carousel:** A dropdown menu showing '1'.
- Sample Crsl Bar Code:** A checkbox that is checked.
- Sample:** Two radio buttons, 'All' (selected) and 'Partial'.
- Position:** A text input field with a blue background, followed by the text 'e.g. 1, 3, 5-12'.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

- 2 Select a sample carousel to which the calibrators are loaded.
- 3 Select a reagent carousel to which the reagents are loaded.
- 4 Select **OK** to start analysis.

2.7 Quality Control

QC results are tools used to monitor the system performance. To check if the system is running normally and steadily, you are recommended to run control samples every day. The system provides two modes to run control samples, auto and manual. New chemistries can be added no matter in which status the control samples are. The control programs can be edited when the control status is Programmed rather than In Progress.

2.7.1 Programming Control Samples

QC runs are requested by programming control samples. You are allowed to choose a control, control position and sample cup type as well as chemistries and panels for measurement. At least one chemistry must be selected for control programming. If Manage reagent by lot is selected on the System setup screen, the system also allows you to run QC samples by reagent lot. If a chemistry has no QC parameters set up, such as mean concentration and standard deviation, the chemistry cannot be used to programming controls.

- 1 Select **Program-Quality Control**.

Figure 2.14 Quality Control screen



- 2 Select a control from the **Control** pull-down list.

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The chemistries assigned for the control are selected automatically.

- If the control has never been programmed, all chemistries assigned for it will be selected.
- If the control has been programmed, the chemistries in recent programming will be selected automatically.

3 Select a position from the **Pos** pull-down list.

The options include all positions defined for the control. The default is the position on the first defined sample carousel in ascending numerical order. For more information about control position assignment, refer to 3.4.2 Defining/Editing a Control (page 3-42).

4 Choose a sample cup type to be used by the selected control.

The options include Standard and Microtube.

5 Choose desired chemistries and panels in the chemistry list.

If the chemistries included in a panel are set up for QC parameters, they will be selected automatically; otherwise, the panel can be selected but will not be programmed for quality control.

If you want to run the QC test by the reagent lot number of the chemistry:

- First select **Options F2**.
- Then select reagent lot number for the chemistry.
- Select **Save**.

6 Select **Save F8**

7 To program other controls, select **Prev F4** or **Next F5**, and then repeat steps 3 and 6.

2.7.2 Loading Control Samples



BIOHAZARD

Inappropriate handling of control samples may lead to biohazardous infection. Do not touch the control samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the control samples, follow standard laboratory safety procedure and consult a doctor.

**CAUTION**

Do not use expired control samples; otherwise, unreliable test results may be caused.

1 Select **Program-Sample**.

2 Select **List F5**.

The sample list shows all programmed patient samples, control samples and chemistries, including the following information:

- Program date and time
- Sample ID or control name
- Position
- Patient name (of patient samples)
- Chemistry
- Sample status

3 Select **Print F7**.

Samples and controls are printed out respectively.

4 Select **Exit F8**.

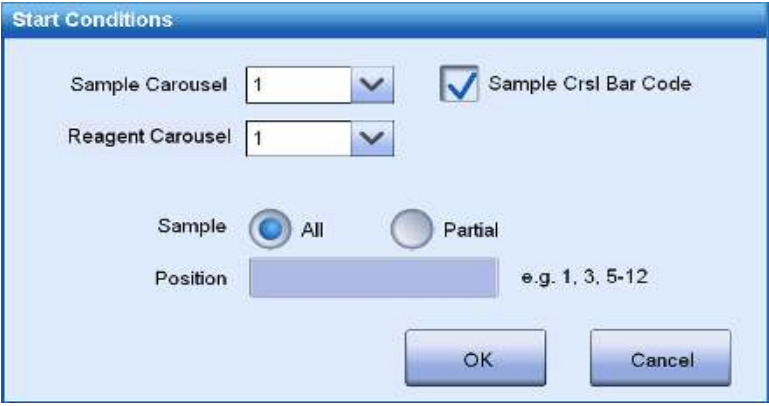
5 Load control samples to the sample carousel according to the printed list.

2.7.3 Running Control Samples

After programming and load the control samples, you can start the QC test.

- 1 Select  on upper right corner of the main screen. The **Start Conditions** window is displayed.

Figure 2.15 Start Conditions window

The image shows a software window titled "Start Conditions". It contains several input fields and controls. At the top, there are two dropdown menus: "Sample Carousel" and "Reagent Carousel", both set to "1". To the right of these is a checkbox labeled "Sample Crsl Bar Code" which is checked. Below the dropdowns, there are two radio buttons: "All" (which is selected) and "Partial". Underneath the radio buttons is a text input field labeled "Position" with a blue background, followed by the text "e.g. 1, 3, 5-12". At the bottom right of the window are two buttons: "OK" and "Cancel".

- 2 Select a sample carousel to which the control samples are loaded.
- 3 Select a reagent carousel to which the reagents are loaded.
- 4 Select **OK** to start analysis.

2.7.4 Auto quality control

Controls can be run automatically based on specified samples and calibration. When auto QC is enabled, the system will automatically run all chemistries of the selected controls once the conditions are met.

- 1 Select **Utility-System Setup**, and then select **Instrument F1**.
- 2 Select **9 QC Evaluation**.
- 3 Select **Auto QC on Carousel**, and then select controls for auto QC in the control list.
- 4 Set up the conditions for auto quality control:
 - Number of samples
 - When calibrated

For more information about auto QC setup, refer to 7.3 Auto Quality Control (page 7-8).

- 5 Select  on upper right corner of the main screen. The **Start Conditions** window is displayed.
- 6 Select **OK**.

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When conditions for auto quality control on sample carousel are satisfied, the system will run controls automatically for relevant chemistries through the sample carousel.

2.8 Programming Routine Samples

After running quality controls, if the test results indicate that the system is in control, you can start programming patient samples. This section describes how to program and run routine samples. For information about bar-coded samples, refer to 13.1.3 Programming Bar-Coded Routine Samples (page 13-4).

2.8.1 Programming Routine Samples

You are allowed to program samples one by one or in batch. Batch program is not allowed when the sample status is In Progress, Incomplete, Rerun or Complete. If the sample status is Programmed, the new program information will overwrite the previous program information.

Programming a sample

- 1 Select **Program-Sample**.

Figure 2.16 Sample screen

The screenshot shows the 'Sample' screen in a laboratory information system. The top bar includes 'Standby/Standby', 'HOST', 'Admin', and a timestamp '11:28 17:50'. Below the top bar are icons for Play, Pause, Stop, and a question mark. The main area is divided into sections: 'Sample' (ID 1, Crea 1, Pos 1, STAT, Sample Type: Setting, Status), 'Bar Code', 'Patient ID', and 'Panels'. A large grid of buttons contains various test names (e.g., SI, Mg, Glu-H, Glu-G, T-Bil-V, D-Bil-V, D-M-D, T-M-D, ADA, PA, TBA, ALT, AST, ALP, γ-GT, Lp(a), TP, CHE, ALB, LIP, α-AMY, ApoA1, ApoB, TG, LDL-C, HDL-C, TC, CREA-S, CREA-J, UA, CysC, UREA, CO2, CK-MB, CK, LDH, α-HBDH, FUN, Hb-CRP, ASO, RF, C3, C4, IgA). Below the grid are buttons for 'Panel 3', 'Panel 2', and a row of buttons: Deneg (F1), Options (F2), Batch (F3), Clear (F4), List (F5), Prev (F6), Next (F7), and Save (F8). The bottom status bar shows 'Select a sample type' and 'Setting up succeeded!'.

- 2 Enter the sample ID in the **ID** field.

Sample ID is composed of numbers, or letters and numbers. Up to 10 digits can be entered. The default sample ID ranges from 1 to 9000. The first sample on

each day is numbered as 1. Duplicate sample IDs are not allowed before the next time the samples are released.

3 Enter the sample position.

A sample position consists of **CrsI** and **Pos**. Routine samples can be programmed with virtual sample carousel. Up to 10 virtual sample carousels are provided, and the programming on each day starts from position No.1 of sample carousel 1. Occupied positions must not be used for programming before being released.

4 Select a sample type from the **Sample Type** pull-down list.

The options include serum, plasma, urine, CSF and other.

5 Enter sample barcode in Barcode field or scan the barcode with a handheld barcode reader

6 Enter the number of the panel or numbers of the panels or the number range of the panels in the Panels field. Its or their corresponding chemistries are selected automatically. Or Choose desired panels. When selected, the panels will appear in a blue frame.

7 Choose desired chemistries.

Chemistries in various statuses are indicated by symbols and color.

Table 2.2 Description of chemistry statuses

Symbol or Color	Chemistry	Description
▲	Chemistry for increment test	The chemistry will be run with sample volume increased.
▼	Chemistry for decrement test	The chemistry will be run with sample volume decreased.
	Masked chemistry	The chemistry is masked. It can be requested but cannot be run.
Chemistry name in black	Available chemistry	The chemistry can be requested for analysis.
Chemistry name in red	Unavailable chemistry	The chemistry can be requested but not allowed for analysis due to the following reasons: The reagent is not loaded or inventory is 0.

2 General Operating Procedure

Symbol or Color	Chemistry	Description
		The calibration status of the chemistry is Cal Required, Cal Failed or Cal Time Out.
Chemistry frame active	Available chemistry	The chemistry can be requested for analysis.
Chemistry frame inactive and appearing in grey	Unavailable chemistry	The chemistry cannot be requested for analysis due to the following reasons: Serum index is not applicable to samples other than serum and plasma. Requested chemistries cannot be chosen again for samples that are in progress, rerun, complete or incomplete.
Chemistry frame in normal color	Unselected chemistry	The chemistry is not selected.
Chemistry frame in blue	Selected chemistry	The chemistry is selected.
Chemistry frame in dark blue	Auto selected serum index chemistry	The serum index chemistry is automatically selected for serum and plasma samples. When deselected and requested again, the chemistry appears in a blue frame.

8 According to your actual needs, select **Options F2** to set up the following parameters:

- Sample volume
- Sample tube
- Off-line dilution factor
- Number of replicates.
- Predilution factor

Figure 2.17 Options window

- 9 Enable or disable sample blank for the sample.

Only when the **Set Sample Blank Individually** checkbox is selected on the **Factory Settings** screen, the **Sample Blank** option will appear. If you need the settings, contact our customer service department or your local distributor.

- 10 If you want to run a chemistry with different parameter, enter the values in the chemistry option area:

- Sample Vol
- Replicates
- Predilution
- Sample blank

- 11 Select **OK**.

- 12 Select **Save F8**.

Batch programming

For batch-programmed samples, all program information such as sample information, chemistries and patient demographics other than position, ID and bar code are the same.

- 1 Select **Program-Sample**.
- 2 Enter the sample ID of the first sample.

2 General Operating Procedure

- 3 Enter the start position to place the samples.
- 4 Select a sample type from the **Sample Type** pull-down list.
- 5 Enter the number of the panel or numbers of the panels or the number range of the panels in the Panels field. Its or their corresponding chemistries are selected automatically. Or Choose desired panels. When selected, the panels will appear in a blue frame.
- 6 Choose desired chemistries.
- 7 According to your actual needs, select **Options F2** to set up the following parameters:
 - Sample volume
 - Sample tube
 - Off-line dilution factor
 - Number of replicates.
 - Predilution factor
 - Sample blank

Figure 2.18 Options window

Chemistry	Sample Vol	Replicates	Predilution	Sample Blank	Chemistry	Sample Vol	Replicates	Predilution	Sample Blank
TP	Standard	1							

- 8 If you want to run a chemistry with different parameter, enter the values in the chemistry option area:

- Sample Vol
- Replicates
- Predilution
- Sample blank

9 Select **OK**.

10 Select **Batch F3**.

Figure 2.19 Program Batch window



Program Batch

End Sample ID

OK Cancel

11 Enter the sample ID of the last sample.

12 Select **OK**.

Editing patient information

You can enter the patient information at any time. Patient information can be customized through **System setup** screen. For more details, please refer to 8.9 Customizing Patient Demographics. When sample analysis is finished, you can view and edit the sample information on the **Current Results** and **History Results** screens.

1 Select **Program-Sample**.

2 Enter the sample ID in the **ID** field.

3 Select **Demog F1**.

Figure 2.20 Demographics window

- 4 To change the priority of the sample, select or deselect the **STAT** checkbox.
- 5 Enter the relevant patient information.
- 6 Select **Save F7** to save your input.
- 7 To edit demographics of other patients, select **Prev F4** or **Next F5**.
- 8 Select **Exit F8** to close the window.

Editing and confirming program information

If the programmed sample is not in progress, you are allowed to edit the program information and add more chemistries. Samples that are being analyzed, rerun, incomplete or complete must not be edited. New chemistries can be added to samples of any status. All program information of samples in Programmed status can be edited.

- 1 Select **Program-Sample**.
- 2 Enter the sample ID in the **Sample ID** field, or enter the sample position in the **Crsl** and **Pos** fields.

The program information of the sample is displayed.

- 3 Edit the following information:
 - STAT property
 - Sample type
 - Comment
 - Chemistries

- Panels
 - Patient demographics
 - Sample options and chemistry options
- 4 Confirm the program information.
 - 5 Select **Save F8**.
 - 6 Select **Prev F6** or **Next F7** to view other samples.

2.8.2 Loading Routine Samples



BIOHAZARD

Inappropriate handling of samples may lead to biohazardous infection. Do not touch the samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the samples, follow standard laboratory safety procedure and consult a doctor.



CAUTION

Do not use expired samples; otherwise, unreliable test results may be caused.



NOTE

Before loading sample, ensure that there are no air bubbles inside the sample cup so as to avoid inaccurate test results.

- 1 Select **Program-Sample**.
- 2 Select **List F5**.

The sample list shows all programmed samples, controls and chemistries, including the following information:

- Program date and time
- Sample ID or control name
- Position
- Patient name of patient samples
- Chemistry
- Sample status

- 3 Select **Print F7**.

Samples and controls are printed out respectively.

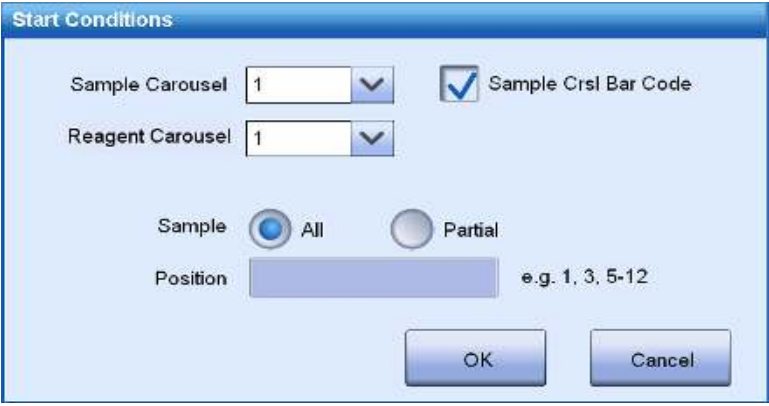
- 4 Select **Exit F8**.
- 5 Load samples to the sample carousel according to the printed list.

2.8.3 Running Routine Samples

After programming and loading the samples, you can start the analysis. To view sample results, refer to 8.12 Results Recall (8-43).

- 1 Select  on upper right corner of the main screen. The **Start Conditions** window is displayed.

Figure 2.21 Start Conditions window



The screenshot shows the 'Start Conditions' dialog box. It has a title bar 'Start Conditions'. Inside, there are two dropdown menus: 'Sample Carousel' and 'Reagent Carousel', both set to '1'. To the right of these is a checked checkbox labeled 'Sample Crsl Bar Code'. Below the dropdowns are two radio buttons: 'All' (selected) and 'Partial'. Below the radio buttons is a text input field labeled 'Position' with the example text 'e.g. 1, 3, 5-12'. At the bottom right are 'OK' and 'Cancel' buttons.

- 2 Select a sample carousel to which the samples are loaded.
- 3 Select a reagent carousel to which the reagents are loaded.
- 4 Select a patient sample range: All or Partial. When you select Partial, you should specify a sample position range for analysis.
- 5 Select **OK**.

2.9 Programming STAT Samples

STAT sample program allows emergent samples to be programmed and analyzed with high priority. The system provides common STAT and quick STAT program. Common STAT program is used in daytime to run emergent samples with higher priority than routine samples. Quick STAT program is mainly used in nighttime and at weekends to program emergent samples quickly with higher priority than routine and common STAT samples.

2.9.1 Programming STAT Samples

Programming single STAT Sample

- 1 Select **Program-Sample**.
- 2 Enter the sample ID in the **Sample ID** field.

Sample ID is composed of numbers, or letters and numbers. Up to 10 digits can be entered. The first sample on each day is numbered as 1. Duplicate sample IDs are not allowed before the next time the samples are released.

- 3 Mark the **STAT** checkbox.

When the **STAT** checkbox is marked, the default sample position is the STAT position available on the sample carousel.

- 4 If you do not want to use the default position, then enter the sample position.
- 5 Select a sample type from the **Sample Type** pull-down list.
- 6 Enter barcode in barcode field or scan the barcode with a handheld barcode reader.
- 7 Enter the number of the panel or numbers of the panels or the number range of the panels in the Panels field. Its or their corresponding chemistries are selected automatically. Or Choose desired panels. When selected, the panels will appear in a blue frame.
- 8 Choose desired chemistries.
- 9 According to your actual needs, select **Options F2** to set up the following parameters:

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- Sample volume
- Sample tube
- Off-line dilution factor
- Number of replicates.
- Predilution factor
- Sample blank

Only when the **Set Sample Blank Individually** checkbox is selected on the **Factory Settings** screen, the **Sample Blank** option will appear. If you need the settings, contact our customer service department or your local distributor.

10 If you want to run a chemistry with different parameter, enter the values in the chemistry option area:

- Sample Vol
- Replicates
- Predilution
- Sample blank

11 Select **OK**.

12 Select **Save F8**.

Batch programming STAT Samples

For batch-programmed samples, all program information such as sample information, chemistries and patient demographics other than position, ID and bar code are the same.

1 Select **Program-Sample**.

2 Enter the sample ID of the first sample.

3 Mark the **STAT** checkbox.

When the **STAT** checkbox is marked, the default sample position is the STAT position available on the sample carousel.

4 If you do not want to use the default position, then enter the sample position.

5 Select a sample type from the **Sample Type** pull-down list.

6 Enter the number of the panel or numbers of the panels or the number range of the panels in the Panels field. Its or their corresponding chemistries are

selected automatically. Or Choose desired panels. When selected, the panels will appear in a blue frame.

- 7 Choose desired chemistries.
- 8 According to your actual needs, select **Options F2** to set up the following parameters:
 - Sample volume
 - Sample tube
 - Off-line dilution factor
 - Number of replicates.
 - Predilution factor
 - Sample blank
- 9 If you want to run a chemistry with different parameter, enter the values in the chemistry option area:
 - Sample Vol
 - Replicates
 - Predilution
 - Sample blank

10 Select **OK**.

11 Select **Batch F3**.

Figure 2.22 Program Batch window



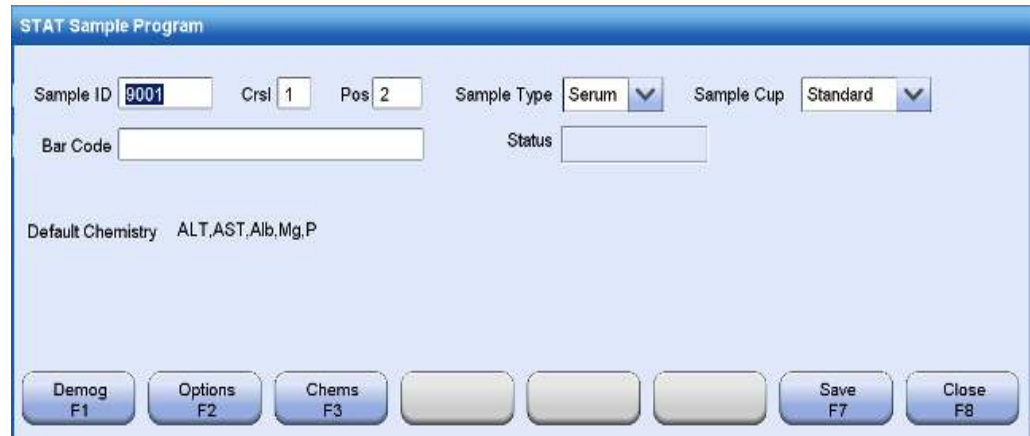
12 Enter the sample ID of the last sample.

13 Select **OK**.

Quickly programming STAT Samples

- 1 Select  on upper right corner of the main screen. The **STAT Sample Program** window is displayed.

Figure 2.23 STAT Sample Program window



STAT Sample Program

Sample ID Crsl Pos Sample Type Sample Cup

Bar Code Status

Default Chemistry ALT,AST,Alb,Mg,P

Demog F1 Options F2 Chems F3 Save F7 Close F8

- 2 Enter the sample ID. The first emergent sample on each day is numbered as 9001.

Sample ID is composed of numbers, or letters and numbers. Up to 10 digits can be entered. Duplicate sample IDs are not allowed before the next time the samples are released.

- 3 If you do not want to use the default position, then enter the sample position.
- 4 Select a sample type from the **Sample Type** pull-down list.
- 5 Select a sample tube type. The options include micro and standard.
- 6 Confirm the default chemistries.
- 7 To select more chemistries, perform the following steps:
 - Select **Chems F3**.
 - Choose chemistries and panels to be run for emergent samples.
 - Select **Save F7**.
- 8 Select **Demog F1** to enter patient demographics.
- 9 According to your actual needs, select **Options F2** to set up the following parameters:

- Sample volume
 - Sample blank
 - Sample tube
 - Off-line dilution factor
 - Number of replicates.
 - Predilution factor
- 10** If you want to run a chemistry with different parameter, enter the values in the chemistry option area:
- Sample Vol
 - Replicates
 - Predilution
 - Sample blank
- 11** Select **OK**.
- 12** Select **Save F7**;
- 13** Select **Close F8** to close the window.

2.9.2 Starting Analysis



BIOHAZARD

Inappropriate handling of samples may lead to biohazardous infection. Do not touch the samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the samples, follow standard laboratory safety procedure and consult a doctor.

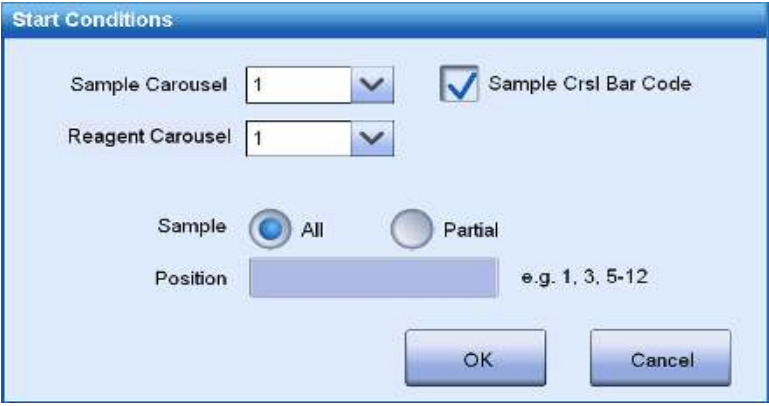


CAUTION

Do not use expired samples; otherwise, unreliable test results may be caused.

- 1** Load emergent samples to the sample carousel.
- 2** Select  on upper right corner of the main screen. The **Start Conditions** window is displayed.

Figure 2.24 Start Conditions window



The screenshot shows a 'Start Conditions' dialog box with a blue title bar. Inside, there are two dropdown menus for 'Sample Carousel' and 'Reagent Carousel', both set to '1'. To the right of these is a checked checkbox labeled 'Sample Crsl Bar Code'. Below the dropdowns are two radio buttons: 'All' (selected) and 'Partial'. Under the 'Partial' radio button is a text input field labeled 'Position' with a blue background, followed by the text 'e.g. 1, 3, 5-12'. At the bottom right are 'OK' and 'Cancel' buttons.

- 3** Select a sample carousel to which the samples are loaded.
- 4** Select a reagent carousel to which the reagents are loaded.
- 5** Select a patient sample range:

All or Partial. When you select Partial, you should specify a sample position range for analysis.

- 6** Select **OK**.

2.10 Test Status and Emergency Stop

During the analysis, you can check reagent inventory on the **Reagent/Calibration** screen, and view test status of calibrators, controls, routine and emergent samples on the **Program-Status** screen. View the reagent carousel status through **Reagent-Status** screen. To pause or stop analysis, select  icons on upper right corner of the main screen.

2.10.1 Checking Reagent Status

- 1 Select **Reagent-Reagent/Calibration**.

The screen displays by default the inventory of ISE reagent and ISE calibration status, as well as inventory and days left of wash solutions and physiological saline. When the inventory is less than the alarm limit, the system will give an alarm and mark the chemistry or wash solution name with different colors.

- Yellow: Warning. The reagent is insufficient or has been expired.
- Red: Serious. The reagent is exhausted.

Figure 2.25 ISE reagent/calibration screen



- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.

2 General Operating Procedure

- 3 Select the up-/down-arrow buttons to show the biochemistry screen.

Figure 2.26 Biochemistry reagent/calibration screen

Pos	Chem	Chems Left	Rpt Type	Tests Left	Days Left	Lot No.	Cal Status	Time Left
64	ALT	0	R1	0	< 58d	2015	Cal Required	
27			R2		< 58d	1017		
85	ALT		R2	208	< 58d		Calibrated	
51	CREA-J		R1	0	< 28d	2010	Cal Required	
52	CREA-S		R1	0	< 20d	2005	Cal Required	
30	GLUHK		R2	214	< 59d		Calibrated	
63	TP	0	R1	0	< 59d	2017	Cal Required	
70			R1	0	< 59d	2017		
2	TP		R1	103	< 58d		Calibrated	
3 M			R1		23d			
41	UA	0	R1	0	< 58d	2012	Cal Required	
42			R1	0	< 58d	2012		
44			R1	0	< 58d	2013		
53			R2	0	< 59d	2012		

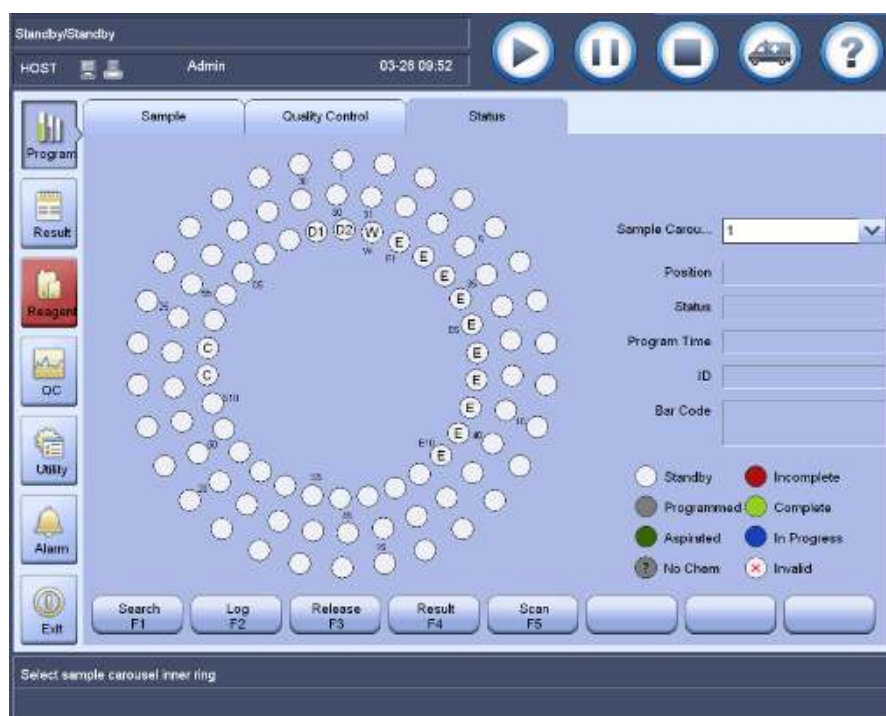
The screen displays the inventory and calibration status of the biochemistry reagents. When the reagent inventory is less than the alarm limit, the system will give an alarm and mark the chemistry name and chemistries left with different colors.

- Yellow: Warning. The number of chemistries left is lower than the alarm limit, or the calibration status of the reagent is Cal Time Extended, Calculated, Edited or Cal Overridden.
- Red: Serious. The number of chemistries left is 0, or the calibration status of the reagent is Cal Failed, Cal Timed Out, or Cal Required. The chemistry can still be requested but will not be run. The ongoing tests containing the chemistry will be invalidated.

2.10.2 Viewing Test Status

- 1 Select **Program-Status**.

Figure 2.27 Status screen



2 View the status of calibrators, controls and samples on the sample carousel graph.

- White: The position is not being used for analysis or has been released manually.
- Grey: The sample is programmed but not started for analysis.
- Dark green: The sample is dispensed into a reaction cuvette.
- Red: All chemistries of the sample are run, but one or more of them have no results.
- Green: All chemistries of the sample are run and have test results.
- Blue: The sample is being analyzed.
- : Indicate invalid sample.
 - The sign appears when sample bar code conflicts, or positions of controls and calibrators are occupied by patient samples, or invalid bar code is detected. The conflicting positions of samples rather than controls and calibrators can be released manually.
 - A bar code is deemed invalid if it contains invalid characters or exceeds the length limit, or is detected in an idle position but has no corresponding sample information or default panel for analysis.

- Select **Log F2** to find the specific causes.
 - : The sample does not have programmed chemistries.
- 3** Choose a sample on the sample carousel graph.

The detailed information of the selected sample is displayed on the right side of the screen:

- Sample position
 - Sample status
 - Program date and time
 - Sample ID (patient sample)
 - Bar code (patient sample)
 - Calibrator name and lot number (calibrator)
 - Name and lot number (control sample)
- 4** Choose the following buttons as needed:
- **Search F1**: used to search for desired calibrator, control or patient sample.
 - **Log F2**: used to recall controls and patient samples which are not complete due to some reasons within the recent 24 hours.
 - **Release F3**: used to release the specified or all positions on the current sample carousel.
 - **Result F4**: used to display the **Current Results** screen, on which you can recall all controls and patient samples that are programmed and analyzed since the system is started up.
 - **Scan F5**: used to scan the specified position or all positions on the selected sample carousel.

2.10.3 Viewing Reagent Carousel Status

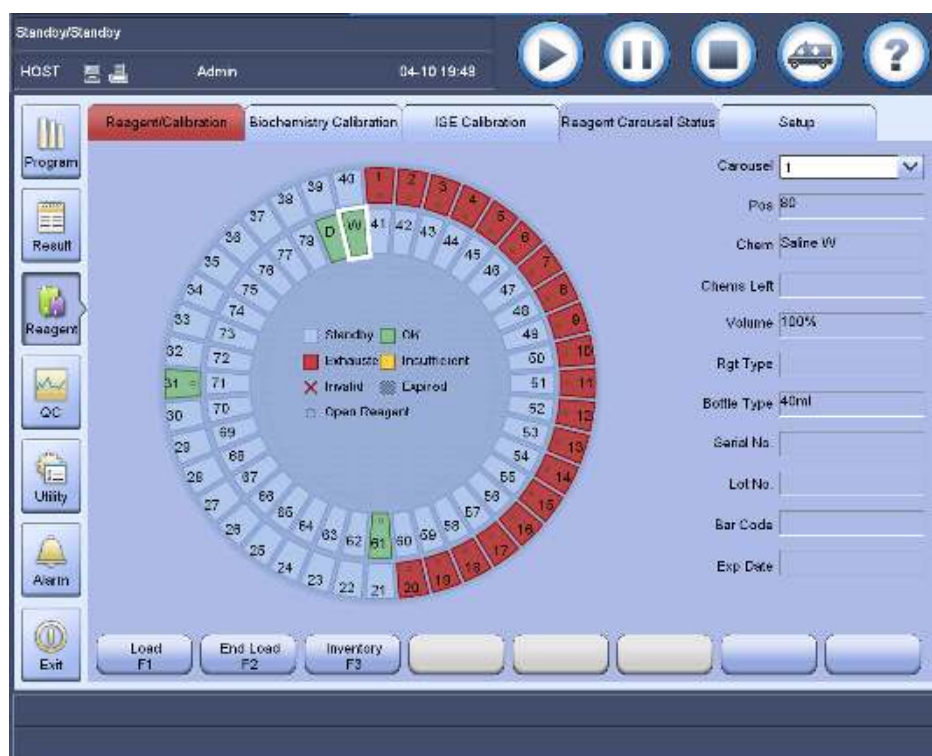
- 1** Select **Reagent-Reagent Carousel Status**.

The **Reagent Carousel** screen is displayed. You can view the status of the reagent carousel, each reagent and configured reagents.

- 2** Select a reagent carousel from the dropdown box of **reagent carousel**.

The simulated reagent carousel graph is shown in the middle of the screen. Reagent carousel status includes vacant, sufficient reagent, insufficient reagent, expired, and invalid reagent, marked by different colors.

Figure 2.28 Reagent carousel status



- 3 Select a position to display its reagent information on the right.
- 4 Select the following buttons to perform respective operations:
 - **Load F1**: select this button to load the reagent.
 - **End Load F2**: If reagent bar code reader is configured and the reagents have been loaded, select this button to scan the reagent carousel; if **Auto Refresh Reagent Inventory** is selected in system setup, the reagents whose inventory is 0 can be refreshed as available when **End Load F2** is selected.
 - **Inventory F3**: select this button to check reagent inventory.

2.10.4 Emergency Stop

Emergency stop will terminate all measurements on the instrument, and all tests that are not finished yet will be invalidated. Do not use emergent stop unless it is really needed, for example, system failure. Emergency stop can be performed in any system status.

Select the  icon on upper right corner of the screen, and then select **OK**. All unfinished actions of the system are cancelled, all pumps and valves are turned off,

2 General Operating Procedure

and the system enters the failure status.

To restore system failure, select **Utility-Commands**, and then select **Home**. To resume the analysis, select the  icon.

2.11 Daily Maintenance

After finishing all tests every day, you are required to perform the daily maintenance procedures and those maintenance procedures indicated in yellow.

Daily maintenance procedures include:

- Check probes/mixers/wash wells
- Check sample/reagent syringes
- Check deionized water connection
- Check waste tube connection
- Check concentrated wash solution
- Check sample probe wash solution
- Clean electrode tubes

2.12 Powering Off

- 1 Make sure that the system is in Standby status.
- 2 Select **Exit-Shut Down** on the left of the main screen. The Windows operating system will quit automatically.
- 3 Switch off the power in the following order:
 - Printer
 - Monitor display of the operation unit
 - Analyzing unit power switch
 - Water supply module (optional)
 - Drainage module(optional)

When the analyzing unit power is switched off, the refrigeration system is still running. If you are going to store the system for over 7 days, switch off the main power.

2.13 Check after Powering Off

- 1** Remove the sample carousel cover, and then remove the calibrators, controls and patient samples.
- 2** Check the analyzer panel for stains and wipe them off with clean gauze if any.
- 3** Check the high-concentration waste tank. Clear it if necessary.

3 System Setup

- System options
- Chemistries
- Calibration
- Quality control

3.1 System Setup Options

3.1.1 Introduction

This section summarizes the setup options on the **System Setup** screen as shown in the figure below.

Figure 3.1 System Setup screen



3.1.2 Sample Options and Reagent Alarm Limits

The sample options and reagent alarm limits allow you to:

- Set up default sample type, default sample tube and expiration date of samples
- Set up the interval of sample probe cleaning
- Enable/Disable auto serum index
- Set up alarm for exhaustion of reagent
- Manage reagents by lot
- Enable/Disable reaction temperature monitoring before analysis
- Result display settings
- Define result flags for reference range

- Set up inventory alarm limit for ISE reagent
- Set up the alarm volume and beep volume
- Set up ISE startup primes

Default sample type

The system supports a couple of sample types, which include serum, plasma, urine, cerebrospinal fluid samples (CSF) and other. The default is serum. When the default sample type is set up, it will be selected by default for programmed samples on the **Sample** screen.

Default sample cup type

The system supports the standard sample cup and Microtube. The default is the standard sample cup. When the default sample cup type is set up, it will be selected by default for programmed samples on the **Sample** screen.

Valid period of samples

Valid period of samples refers to the time interval that a patient sample is first loaded to the sample carousel and then expired. When the valid period of samples is set up, only samples within this period are allowed for analysis. If the valid period is not set up, the samples are valid all the time.

The valid period ranges from 1 to 99 in hour or day. The default is day.

Valid period is applicable to patient samples rather than calibrators and controls. Once the collection time is entered, the system will calculate the valid period from the time when the sample is collected; otherwise, the time when the sample is programmed will be used for calculating the valid period.

Special wash sample probe

The sample probe special wash function is used to execute additional cleaning procedure for the sample probe during measurement in order to avoid clogging. Enter the number of tests in the **Number of Tests** field. The input range is 100-10,000, and the default is 400. When the number of tests is finished, the system will clean the sample probe with wash solution in an additional cleaning procedure.

Auto serum index

When the auto serum index function is enabled, the SI chemistry on the **Sample** screen will be selected by default for programmed serum or plasma samples, and the system will measure the degree of Hemolysis, icterus and lipemia contained in these samples. If the **Qualitative Analysis** checkbox on the **Auto Serum Index** window is marked, the system will display qualitative flags of serum index on patient reports.

Serum index is only used to evaluate the integrity of samples rather than making a

diagnosis for patients.

Reaction temperature monitoring

The reaction temperature can be monitored before analysis begins.

- When the **Start Analysis When Temperature is Steady** checkbox is selected, the system will check before analysis begins if the reaction temperature is normal. If the temperature is normal, you are allowed to select  to start analysis; otherwise, a message will appear indicating analysis is forbidden in current condition.
- When the **Start Analysis When Temperature is Steady** checkbox is not selected, the system will still check before analysis begins if the reaction temperature is normal and within $37\pm 2.0^{\circ}\text{C}$. If the temperature is normal, you are allowed to select  to start analysis; otherwise, the system will remind you that the results may be influenced if you continue to start analysis. You may continue or abort the analysis.

Alarm when reagent exhausted

Each chemistry can have more than one bottle of reagent loaded. You can set up alarms for the case that the reagent is running out.

Select the **Alarm when reagent exhausted** option. When the reagent is exhausted, the system will give an alarm. If the option is not selected, the system will not give an alarm.

Manage reagents by lot

This option is used to monitor the calibration status and time of each reagent lot, supports reagent lot calibration, and displays calibration results of each reagent lot.

When this option is enabled, special attentions should be paid for the following operations:

- Loading reagents: You are required to input the lot number when loading reagents manually. The lot number of bar-coded reagents cannot be left blank; otherwise, reagent load will fail.
- Viewing calibration status and requesting calibration: You can view calibration status and time of each reagent lot, and request calibration accordingly. For more information of reagent lot calibration, refer to Reagent lot calibration (Page 2-29).
- Recalling calibration results: You can recall calibration results of each reagent lot on the **Biochemistry Calibration** screen.
- Auto calibration: Auto calibration by reagent bottle or lot is forbidden. When a different reagent lot is used, the system will request and run calibration

automatically. Reagent lots with valid calibration factors will not be calibrated again when used for measurement.

Result display settings

This option is used to set up flags and color for results less than or greater than the reference range, as well as color for results less than or greater than the critical range.

Click the relevant color setup button, choose desired color, and then select **OK**. The system will display flags in the **Flag** column of the **Current Results** and **History Results** screens and on patient reports if the test result is less than or greater than the reference range. The flags can be composed of numbers, letters and symbols for no more than 10 digits. The default flags for reference range are $\wedge$ and $\vee$. If a result is greater than the high limit, $\wedge$ will appear near the result; if a result is less than the low limit, $\vee$ will appear near the result.

If test results are beyond the critical range, they will appear in the set color.

Reagent alarm limit

Reagent alarm limit is only applicable to ISE reagent. If the inventory alarm limit is set up, the system will give an alarm and mark the reagent with colors when the reagent inventory is less than the alarm limit.

For more information about reagent inventory alarm limit, refer to 5.4 Reagent Inventory Alarm Limits Setup (page 5-8).

Alarm Sound volume

This option is to adjust the volume of alarm tone and beep. Alarm tone is the sound of a system alarm and beep is given when mis-input or mis-operation occurs. Volume of both sounds can be adjusted manually according to the practical conditions of the environment. Drag the slider in the **Alarm Volume** and **Beep Volume** fields horizontally. The scale is ascending from left to right. When the slider is moved to the leftmost position, the alarm buzzer is silenced.

Since the Windows 8 does not support alarming through buzzer, you should install an audio card on your computer in order to ensure the alarm and beep sound can be adjusted and given.

ISE prime cycle

For details of ISE primes setup, refer to 12.9 ISE Prime (page 12-23).

3.1.3 Auto Rerun Setup

The system provides a couple of conditions for auto rerun. When selected conditions are satisfied, chemistries for which auto rerun has been enabled will be rerun automatically with the specified sample volume type.

Select the checkbox to the left of a condition, and then select a sample volume type from the pull-down list to the right of the condition. The sample volume type options include: increased, standard, decreased, and last volume. When the condition is satisfied, chemistries with auto rerun enabled will be rerun automatically with the selected sample volume type.

Only users who have the permissions of system setup are allowed to set up auto rerun conditions.

Above Critical High

Select a rerun mode from the pull-down list box. It means that the system will rerun the tests with the selected mode when the test result exceeds the critical range high limit.

Unselection means this item will not be checked.

Below Critical Low

Select a rerun mode from the pull-down list box. It means that the system will rerun the tests with the selected mode when the test result is lower than the critical range low limit.

Unselection means this item will not be checked.

Above Linearity High

Select a rerun mode from the pull-down list box. It means that the system will rerun the tests with the selected mode when the test result exceeds the linearity high limit.

Unselection means this item will not be checked.

Below Linearity Low

Select a rerun mode from the pull-down list box. It means that the system will rerun the tests with the selected mode when the test result is lower than the linearity low limit.

Unselection means this item will not be checked.

Above Highest Calib.

Select a rerun mode from the pull-down list box. When selected, it means the analyzer will rerun the sample with the selected mode automatically if its response is beyond that of the highest-concentration calibrator.

Unselection means this item will not be checked.

Below Lowest Calib.

Select a rerun mode from the pull-down list box. When selected, it means the analyzer will rerun the sample with the selected mode automatically if its response is beyond that of the lowest-concentration calibrator.

Unselection means this item will not be checked.

Substrate Depletion

Select a rerun mode from the pull-down list box. When selected, it means the analyzer will rerun the tests with the selected mode automatically if the substrate ran out during running.

Unselection means this item will not be checked.

Prozone Check Error

Select a rerun mode from the pull-down list box. It means that the system will rerun the tests with the selected mode when prozone occurs during reaction process.

Unselection means this item will not be checked.

Nonlinear

Select a rerun mode from the pull-down list box. If the calculated linearity is greater than the defined linearity limit, the system will rerun the tests with the selected mode.

Unselection means this item will not be checked.

No Linear Interval

Select a rerun mode from the pull-down list box. It means that the system will rerun the tests with the selected mode when the number of measuring points within substrate limit is less than or equal to 3. This option applies to Kinetic method only.

Unselection means this item will not be checked.

No Calculation Interval

Select a rerun mode from the pull-down list box. If the number of measuring points within linearity range is less than 2 during high-activity enzyme measurement, the linearity range will be expanded. If the number of measuring points is less than 2 even when the lag time is included, the system will rerun the tests with the selected mode. This option applies to Kinetic method only.

Unselection means this item will not be checked.

3.1.4 Instrument Setup Options

In the **Instrument Setup** window, you are allowed to:

- Set up time for system auto sleep and auto startup
- Mask/Unmask chemistries
- Dictionary setup
- Set up system communication options
- Select language for the operating software

3 System Setup

- Upgrade the operating software, control software and ISE module software
- View software versions
- System date and time
- Set up control run length and auto QC
- Voice tone setup
- Set up auto release time of samples
- Optimizing result display
- Customize sample information
- Customize patient demographics
- Reagent/calibration setup

Sleep and awake setup

The **Sleep/Awake** setup option is used to set up the auto sleep time interval, and the auto awake time of the system.

If the auto sleep time interval is set up, the counter will start counting down once the system enters Standby status and begins to sleep when the countdown is finished.

The system allows you to choose a weekday and specific time that the system will be started up automatically. When the awake time is reached, the system will be woken up automatically no matter if it is off or sleeping. For more information, refer to 11.5 Sleep and Awake Setup (page 11-9).

Masking/Unmasking Chemistries

The Masking/Unmasking Chemistries option is used to disable chemistries, which will still be displayed on the **Sample, Quality Control** and **Reagent/Calibration** screens. Masked chemistries can be requested but will not be run for sample analysis.

For details of chemistry masking/unmasking, refer to 10.9 Masking/Unmasking Chemistries (page 10-25).

Dictionary setup

The Dictionary option is provided for setting up and managing frequent data information, including: result unit, sample type, sample comment, and QC comment.

For more information, refer to 11.6 Dictionary Setup (page 11-13).

System communication options

The Com Setup option is used to set up the IP address for connections between the PC and LIS. For more details, please refer to 14.2.2 Connection between PC and LIS Host.

Select language

The operating software is displayed by default in the same language as the current operating software. You are allowed to change the language of the operating software.

Select **System Setup-Instrument F1-5 Language**, and then choose a language from the following options: Chinese, English, Turkish, Russian, French, Portuguese, Italian, Spanish, and Polish. Select **OK** to save the settings. The language you select will take effect only when you reboot the operating software.

Software upgrading

By running the upgrade program, you are allowed to upgrade the operating software, control software and ISE module software. For more information, refer to 11.7 Software Upgrade (page 11-15).

Viewing software versions

The **Version Info** window shows the versions of the operating software, control software, ISE software and database. For more information, refer to 11.8 Software Version (page 11-16).

Date and time

The Date and Time option allows you to set the current date and time, select the date/time formats to be displayed on software screens and printed reports, and restore default date and time formats.

When adjusted, the date and time will influence the time left of reagents and calibration, shelf life of samples, and run length of two-control evaluation. The date and time cannot be edited when the system status is Running. Modification of the date and time will not affect samples on the Current Results screen or QC evaluation and Twin-Plot chart.

Follow this procedure to change system date and time:

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **8 Date/Time**.

Figure 3.2 Date/Time window



4 Select date in the **Date** area.

5 Set the time in the **Time** area.

Manually enter the hour, minute and second, or move the cursor to hour, minute and second, and then click the up/down arrows to adjust the time.

6 Choose a date format from the **Order** pull-down list.

- yyyy-mm-dd: e.g. 2010-07-28
- dd-mm-yyyy: e.g. 28-07-2010
- mm-dd-yyyy: e.g. 07-28-2010

7 Choose a time format from the **Time Format** pull-down list.

- 24-hour: e.g. 14:33:27
- 12-hour: e.g. 02:33:27

8 Select **OK** to save your input information.

9 To restore the date and time defaults, select **Restore Defaults**.

10 Select **Exit** to close the window.

QC run length and auto QC

By choosing the QC Evaluation, you are allowed to set up the QC run length and auto QC conditions.

For more information, refer to 7 Quality Control (page 7-1).

Auto release of samples

The system allows setting of daily release time of samples. When the set time is reached, samples that are currently in Complete status will be released automatically.

For more information, refer to 8.6.3 Auto Release of Samples (page 8-32).

Voice tone setup

This option is used to customize the alarm sound and beep sound.

For more information refer to “11.9 Voice Tone Setup” (Page 11-18).

Customize reagent display

This option is used to set up reagent information displayed on the biochemistry reagent/calibration screen.

For more information, refer to 5.2 Customizing Reagent Display (Page 5-6).

3.1.5 Print Setup

The **Print Setup** window allows you to set up paper size and default printer. For more information, refer to 9.2 Print Setup (page 9-8).

3.1.6 Bar Code Setup

The **Bar Code** option is used to set up sample and reagent bar code options. You are allowed to set up the bar code options only when you equip your system with the bar code module. For more information, refer to 13 Use of Bar Code (page 13-1).

3.1.7 Host Communication Setup

The **Host** option allows you to set up the host communication options and the transmission methods of test results. For more information, refer to 14.2 Host Communication.

3.1.8 User Accounts and Permissions

The **User** option allows you to define and edit user accounts, passwords and permissions. For more information, refer to 11.4 User and Password Setup (page 11-5).

3.1.9 Customizing Sample Information

Customizing sample information allows you to specify sample information to be displayed on the **Sample** screen. For more information, refer to 8.8.2 Customizing Sample Information.

3.1.10 Customizing Patient Demographics

You can specify patient demographics to be displayed, its default and its display order on the **Patient Demographics** screen. For more information, refer to 8.9 Customizing Patient Demographics.

3.1.11 Reagent/Calibration Setup

Via **Reagent/Calibration** option on instrument setup screen, you can configure whether to automatically refresh the reagent with 0 inventory as available for test, when reagent has been loaded and **End Load F2** button is selected. For more information, refer to 5.4.3 Auto refreshing reagent inventory.

3.2 Chemistries Setup

3.2.1 Introduction

Up to 200 chemistries can be defined and configured. The **Chemistries** screen is as shown below:

Figure 3.3 Chemistries screen



Definition and setup of user-defined chemistries will be described in detail in the following sections.

3.2.2 User-defined Chemistries Setup

Defining a chemistry

- 1 Select **Utility-Chemistries**.
- 2 Choose a blank frame in the chemistry list.
- 3 Select **Define F1**.
- 4 Enter the processing parameters and error detection limits of the chemistry.
- 5 Select **Next F5** to save your input information and define more chemistries. Or

- Select **Discard F6** to restore the default parameter settings.
- Select **Save F7** to save your input information.
- Select **Print F1** to print out the chemistry parameters.

6 Select **Close F8** to exit the window.

Editing user-defined chemistries

You are allowed to edit user-defined chemistries if:

- You have sufficient permissions, and
For user permission setup, refer to 11.4 User and Password Setup (page 11-5).
- The system is not running tests.

Editing user-defined chemistries is similar to defining a chemistry. Refer to other sections in this chapter for details.

Viewing user-defined chemistries

You are allowed to view the following information in any system status:

- Processing parameters
- Error detection limits
- Slope and offset
- Reference/Critical range

Perform the following steps to view chemistries you have defined:

- 1** Select **Utility-Chemistries**.
- 2** Choose a chemistry from the chemistry list.
- 3** Select **Define F1** to view the processing parameters, error detection limits and dilution factors.
- 4** Select **Close F8** to close the **Define/Edit Chemistries** window.
- 5** Select **Ref Range F4** to view the reference range and critical range.
- 6** Select **Slope/Offset F5** to view the slope and offset values.

Deleting a user-defined chemistry

Make sure that you have sufficient permission to delete a chemistry you have defined. For user permission setup, refer to 11.4 User and Password Setup (page 11-5). SI and ISE chemistries are not allowed to be deleted.

- 1 Remove the reagent from the reagent carousel.
- 2 Select **Utility-Chemistries**.
- 3 Select the chemistry in the chemistry list.
- 4 Check if the following conditions are satisfied:
 - The system is not running tests.
 - The selected chemistry is not requested or run for samples, calibrators and controls.
 - The selected chemistry is disabled.
 - The corresponding reagent has been unloaded from the reagent carousel.
- 5 Select **Delete F2**.

All test results, data and parameters related to the chemistry are cleared.

3.2.3 Processing Parameters

This section introduces the processing parameters for user-defined chemistries.

The processing parameters setup window is as shown below:

Figure 3.4 Processing parameters setup window

The screenshot shows the 'Define/Edit Chemistries' window with the following fields and controls:

- Chem:** GLU-H
- No.:** 14
- Sample Type:** Serum (dropdown)
- Chemistry:** GLU-H
- Print Name:** GLU
- Reaction Type:** Endpoint (dropdown)
- Reaction Direction:** Positive (dropdown)
- Pri Wave:** 340nm (dropdown)
- Sec Wave:** 700nm (dropdown)
- Unit:** mmol/L (dropdown)
- Decimal:** 0.01 (dropdown)
- Blank Time:** 47 and 49 (input fields)
- Reaction Time:** 76 and 78 (input fields)
- Sample Vol:** Standard 1.5 uL, Decreased, Increased (input fields)
- Aspirated:** (input field)
- Diluent:** (input field)
- Reagent Vol:** R1 120 uL, R2 30 uL, R3, R4 (input fields)
- Diluent:** (input field)
- Sample Blank:** ☐ (checkbox)
- Auto Rerun:** ☐ (checkbox)
- Buttons:** Print F1, Reflex F2, Qualitative F3, Prev F4, Next F5, Discard F6, Save F7, Close F8

Chem

Chemistry name is the only identity of a chemistry and must not be duplicate. A chemistry name can be composed of up to 10 characters.

No.

No. is a unique number for chemistry. It can be left blank but must not be duplicate. Chemistry number is composed of numbers and ranges from 0-9999.

Sample type

Sample type refers to the samples to which the chemistry is applicable. The options include serum, plasma, urine, CSF and other. The options available in the **Sample Type** pull-down list are those supported by the chemistry, and the default is the default sample type.

The system allows definition of chemistry parameters for more than one sample type, including the processing parameters and error detection limits. During definition of chemistries, the parameters should be firstly defined for serum sample, and then other sample types. Such chemistries will be calibrated with serum sample parameters by default.

Chemistry

Chemistry is the complete form of chemistry name. It can be composed of up to 36 characters. The input is not case sensitive. The **Chemistry** field can be left blank or duplicate.

A chemistry is only represented by its print name on patient reports and appears on other reports in the form of short name.

Print Name

Print name is displayed on patient reports representing a chemistry. It can be composed of up to 15 characters. The print name can be edited and duplicate. When this field is left blank, the short form of the chemistry name will appear on reports. A chemistry is represented by its short name on all reports other than patient reports.

Reaction Type

Reaction type is a measurement theory based on which chemistries are run for samples and then calculated. The system supports three reaction types, which are Endpoint, Fixed-time and Kinetic.

Table 3.1 Reaction types

Reaction Type	Description
Endpoint	Qualitative analysis is performed based on the absorption spectrum and absorbed light intensity of the reactant when the reaction becomes equilibrrious.
Fixed-time	For this reaction type, the reaction velocity is directly proportional to the substrate concentration. As the substrate is

Reaction Type	Description
	consumed continuously, the reaction velocity is decreasing gradually, and so is the absorbance change rate. It will take a long time for such reaction to become equilibrium, and the reaction can get steady only after a delay.
Kinetic	Kinetic, also called continuous monitoring method, is used to continuously measure the multiple change points of a reactant or substrate's concentration which varies with the enzymatic reaction, thus calculating the initial velocity of the enzymatic reaction and then the enzyme activity. This reaction type is mainly used for measurement of enzyme activity.

Reaction Direction

Reaction direction refers to the change trend of absorbance during the reaction process, and includes two options:

- Positive: indicates increasing absorbance with time.
- Negative: indicates decreasing absorbance with time.

Primary Wavelength

The primary wavelength is chosen based on the light absorption features of the reactant and used to measure the absorbed light intensity.

Options for primary wavelength include: 340nm, 380nm, 412nm, 450nm, 505nm, 546nm, 570nm, 605nm, 660nm, 700nm, 740nm and 800nm

Secondary Wavelength

The secondary wavelength is used to correct the absorbance measured at the primary wavelength and eliminate the influence of noise, such as light flash and drift, and scratches on cuvettes, etc. The two wavelengths cannot be equal.

Options for secondary wavelength include: blank, 340nm, 380nm, 412nm, 450nm, 505nm, 546nm, 570nm, 605nm, 660nm, 700nm, 740nm and 800nm

Unit

Changing the result units of the chemistries are allowed.

The result unit is blank by default. After changing the unit, you are required to update calibrator concentrations, control concentrations and standard deviations (SDs), reference ranges and offsets. Those test results calculated with the old unit will remain unchanged.

The following table summarizes the result units available for chemistries.

Table 3.2 Result units

Unit	Full name
1. mg/dl	Milligrams per decilitre
2. mg/L	Milligrams per liter
3. g/dl	Grams per decilitre
4. g/L	Grams per litre
5. mmol/L	Micromole per litre
6. μ mol/L	Millimole per litre
7. IU/L	International unit per litre
8. μ g/ml	Microgram per millilitre
9. μ g/dl	Microgram per decilitre
10. μ g/L	Microgram per liter
11. U/L	Unit per liter
12. %	Percent
13. IU/ml	International unit per millilitre
14. U/ml	Unit per millilitre
15. mg/ml	Milligram per millilitre
16. %GHB	Glycohemoglobin
17. %Alc	Hemoglobin A1C
18. nkat/L	Nanokatal per liter
19. μ kat/L	Microkatal per liter
20. ng/L	Nanograms per liter
21. nmol/L	Nanomoles per liter
22. KU/L	Kilo units per liter
23. N/A	/

Decimal

Decimal specifies the number of decimal places for test results. The decimal is allowed to be edited. Up to 3 decimal places can be set up and respectively correspond to 0, 0.1, 0.01 and 0.001.

Blank Time and Reaction Time

Blank time refers to the period between dispensing of the second reactant (reagent or sample) in reversed order and of the last reactant (reagent or sample).

For endpoint analysis, the reaction time refers to the time span from the start point of the reaction to the end point; for fixed-time and Kinetic analysis, it refers to the period from reaction equilibrium to the end of monitoring.

The blank time and reaction time are counted in measuring points.

Suppose the blank time range is N-P and the reaction time range is L-M. The start point is the first measuring point after dispensing of R1.

Table 3.3 Blank time and reaction time input ranges for endpoint analysis

Endpoint	Blank Time	Reaction Time
When the blank absorbance is read before the reaction begins,		
Single-reagent	$5 \leq N \leq P \leq 12$	$14 \leq L \leq M \leq 82$
Double-reagent	$14 \leq N \leq P \leq 49$	$51 \leq L \leq M \leq 82$
Triple-reagent	$51 \leq N \leq P \leq 90$	$104 \leq L \leq M \leq 172$
Quadruple-reagent	$104 \leq N \leq P \leq 139$	$141 \leq L \leq M \leq 172$
When the blank absorbance is read after the reaction begins,		
Single-reagent	$14 \leq N \leq P$	$P < L \leq M \leq 82$
Double-reagent	$51 \leq N \leq P$	$P < L \leq M \leq 82$
Triple-reagent	$104 \leq N \leq P$	$P < L \leq M \leq 172$
Quadruple-reagent	$141 \leq N \leq P$	$P < L \leq M \leq 172$
When the blank absorbance is not subtracted,		
Single-reagent	$N = P = 0$	$14 \leq L \leq M \leq 82$
Double-reagent	$N = P = 0$	$51 \leq L \leq M \leq 82$
Triple-reagent	$N = P = 0$	$104 \leq L \leq M \leq 172$
Quadruple-reagent	$N = P = 0$	$141 \leq L \leq M \leq 172$

Table 3.4 Blank time and reaction time input ranges for fixed-time and Kinetic analysis

Fixed-time and Kinetic	Blank Time	Reaction Time
When the blank absorbance is read before the reaction begins,		
Single-reagent	$5 \leq N < P \leq 12$	$14 \leq L < M \leq 82$
Double-reagent	$14 \leq N < P \leq 49$	$51 \leq L < M \leq 82$
Triple-reagent	$51 \leq N < P \leq 90$	$104 \leq L < M \leq 172$
Quadruple-reagent	$104 \leq N < P \leq 139$	$141 \leq L < M \leq 172$
When the blank absorbance is not subtracted,		
Single-reagent	$N = P = 0$	$14 \leq L < M \leq 82$
Double-reagent	$N = P = 0$	$51 \leq L < M \leq 82$
Triple-reagent	$N = P = 0$	$104 \leq L < M \leq 172$
Quadruple-reagent	$N = P = 0$	$141 \leq L < M \leq 172$

The blank time and reaction time are almost the same for both fixed-time and Kinetic analysis, except that $M - L \geq 2$ is required for Kinetic analysis, that is, the

reaction time should include at least 3 measuring points.

Sample Volume, Standard, Aspirated, Diluent, Increased, and Decreased

Sample volume is the standard sample amount, which should be dispensed in a normal test. It ranges from 1.5µl to 45µl with an increment of 0.1µl. The default is 1.5µl. A maximum of one decimal is allowed.

Aspirated volume refers to the amount of sample used for dilution at the specified ratio. It ranges from 2µl to 45µl with an increment of 0.1µl. The default is blank. A maximum of one decimal is allowed.

Diluent volume refers to the amount of diluent used for sample dilution. It ranges from 120µl to 350µl with an increment of 1µl. The default is blank. A maximum of one decimal is allowed.



NOTE

If aspirated volume for dilution and diluent volume are defined, ensure the total sum of them is within 120µl~360µl; otherwise, the settings cannot be saved.

The diluent volume for standard, increased and decreased analysis can be defined in the same way.

Decreased sample volume indicates the sample amount required for a decrement test. It ranges from 1.5µl to 45µl with an increment of 0.1µl. The default is blank. A maximum of one decimal is allowed.

Increased sample volume indicates the sample amount required for an increment test. It ranges from 1.5µl to 45µl with an increment of 0.1µl. The default is blank. A maximum of one decimal is allowed.



NOTE

If aspirated volume for dilution and diluent volume are defined, standard, decreased and increased analysis will be performed with diluted sample; otherwise, it will be done based on standard, decreased or increased sample volume.

Sample Blank

Sample blank is similar to sample analysis except for use of equivalent amount of physiological saline. Sample blank is used for removal of non-chromogenesis reaction, such as influence of sample interference (Hemolysis, icterus and lipemia) on absorbance readings.

The sample blank reaction curve is almost a straight line with slope of 0 during the reaction period, and therefore means nothing for fixed-time and Kinetic analysis. For double, triple and quadruple reagent endpoint analysis, the sample blank absorbance can be subtracted through parameter settings. Therefore, sample blank is only

effective for single-reagent endpoint chemistries.

Mark the **Sample Blank** checkbox with a tick. The chemistry will be sample blanked before the reaction begins.

Auto Rerun

The Auto Rerun option is used to rerun the chemistries when the auto rerun conditions are satisfied.

Mark the **Auto Rerun** checkbox means enabling the auto rerun option.

For more information about auto rerun, refer to 8.2.4 Rerunning Samples (page 8-4).

Reagent Volume and Diluent

Reagent volume specifies the reagent amount, which should be dispensed for measurement. The system allows the dispensing of four reagents: R1,R2,R3 and R4.

Reagent	Reagent volume	Diluent volume	Reagent volume+ Diluent volume
Non-concentrated reagent			
R1	120µl~350µl, with an increment of 1µl	none	120µl~350µl
R2	10µl~350µl, with an increment of 1µl	none	10µl~350µl
R3	10µl~350µl, with an increment of 1µl	none	10µl~350µl
R4	10µl~350µl, with an increment of 1µl	none	10µl~350µl
Concentrated reagent			
R1	10µl~340µl, with an increment of 1µl	10µl~340µl,with an increment of 1µl	120µl~350µl
R2	10µl~340µl, with an increment of 1µl	10µl~340µl,with an increment of 1µl	10µl~350µl
R3	10µl~340µl, with an increment of 1µl	10µl~340µl,with an increment of 1µl	10µl~350µl
R4	10µl~340µl, with an increment of 1µl	10µl~340µl,with an increment of 1µl	10µl~350µl

3 System Setup

The second, third and fourth reagents are allowed only when the reagent(s) prior to them are configured. For example, R2 can be set up with the prerequisite of R1; R3 with R1 and R2; R4 with R1, R2 and R3. If one of R2, R3 and R4 is removed, the remaining reagents behind it will also be removed and appear in grey.

Diluent volume refers to the amount of diluent used for reagent dilution. The combined volume of all reagents, reagent diluent and sample must be within 120 μ l and 360 μ l. If your input does not satisfy the requirements of reaction mixture volume, the system will display an error message. Check the sample volume, reagent diluent and reagent volumes you have entered, and change them if necessary.

3.2.4 Error Detection Limits

This section introduces the error detection limits for user-defined chemistries. For more information about error detection limits, refer to 4 Operation Theories (4-1).

The error detection limits setup window is as shown below:

Figure 3.5 Error detection limits setup window

The screenshot shows the 'Define/Edit Chemistries' window. At the top, 'Chem' is set to 'GLU-H', 'No.' is '14', 'Sample Type' is 'Serum', 'Chemistry' is 'GLU-H', and 'Print Name' is 'GLU'. The main area contains several input fields: 'linearity Range(Standard)' with values 0.6 and 30; 'linearity Range(Decreased)' and 'linearity Range(Increased)' are empty; 'R1 Blank Abs' has values -33000 and 33000; 'Blank Response' has values -33000 and 33000; 'Twin Chemistry' is empty. On the right, 'Linearity Limit' is empty; 'Substrate Depletion' is empty; 'Mixed Blank Abs' has values -33000 and 33000; 'Uncapping Time' is 28 Day(s); 'Reagent Alarm Limit' is 10. There is an 'Enzyme Linear Extension' checkbox. At the bottom, there are three radio buttons: 'Prozone Check' (unchecked), 'Rate Check' (checked), and 'Antigen Addition' (checked). Below these are four input fields: 'Q1', 'Q2', 'Q3', and 'Q4'. At the very bottom, there are two more input fields: 'PC' and 'ABS'. At the bottom of the window, there are eight buttons: 'Print F1', 'Reflex F2', 'Qualitative F3', 'Prev F4', 'Next F5', 'Discard F6', 'Save F7', and 'Close F8'.

Linearity Range

The linearity range indicates the measurable range of the system, during which the test result is linear to the response R. Determine the linearity range according to the reagent package insert.

Linearity range(standard):-99999 to 999999. The lower limit less than or equal to the

higher limit. The default is blank.

Linearity range(increased):-99999 to 999999. The lower limit less than or equal to the higher limit. The default is blank.

Linearity range(decreased):-99999 to 999999. The lower limit less than or equal to the higher limit. The default is blank.

The system compares the calculated sample concentration with the linearity range. When the high limit is exceeded, the > sign will appear near the result; when the low limit is exceeded, the < sign will appear. For more information of result flags, refer to 17.4 Data Alarm (page 17-14).

The default is blank, which means not performing this check.

Linearity Limit

Linearity limit is only applicable to Kinetic analysis, in which the absorbance change is linear to the reaction time. If the reagent undergoes substrate depletion, or the photometer fluctuates, or the reaction mixture is not stirred evenly, the test results may be unreliable. Therefore, the linearity of the measuring period is calculated and then compared with the set linearity limit.

If the reaction data within the linearity range does not satisfy the linearity limit, the system will flag the test result with “LIN” on the patient report. For more information of result flags, refer to 17.4 Data Alarm (page 17-14).

The linearity limit can be any number between 0 and 1 with a maximum of 2 decimals. The default is blank, which means not performing this check.

Substrate Depletion

The Substrate Depletion option is only applicable to Kinetic and fixed-time analysis. It can be obtained through the following formula:

$$\text{Substrate depletion limit} = \text{Input substrate depletion limit} + K(L1-Lb)$$

Where,

- L1: refers to the absorbance of primary wavelength measured at the first measuring point when sample is dispensed and stirred in sample analysis.
- Lb: refers to the absorbance of primary wavelength measured at the first measuring point when sample is dispensed and stirred in a reagent blank test or calibration with 0-concentration calibrator.
- K: correction factor of liquid volume

Results will not be adjusted when $L1-Lb \leq 0$ or the measurement is not a reagent blank or 0-concentration calibration. Substrate depletion is not applicable for calibrations.

We deem that substrate depletion occurs if the primary wavelength absorbance of the first measuring point is greater than the substrate depletion limit in ascending

reactions or lower than the substrate depletion limit in descending reactions. When substrate depletion occurs, the system will flag the test result with “BOE” in the patient report. For more information of result flags, refer to 17.4 Data Alarm (page 17-14).

The substrate depletion limit can be any number within -33,000-33,000. The default is blank, which means not performing this check.

R1 Blank Absorbance Range

The R1 Blank Abs indicates the allowable range of the maximum absorbance in the previous period prior to sample dispensing. The input range must be within -33,000-33,000, and the low limit lower than the high limit.

If the maximum absorbance in the previous period prior to sample dispensing is beyond the set range, the system will flag the test result with “RBK”.

The default is -33,000-33,000; the field can be left blank.

Mixed Blank Absorbance Range

The Mixed Blank Abs indicates the allowable range of the absorbance measured at the end point of a zero-concentration calibrator reaction or a reagent blank reaction. The input range must be within -33,000-33,000, and the low limit lower than the high limit.

If the absorbance measured at the reaction end point is beyond the set range, the system will flag the test result with “MBK”.

The default is -33,000-33,000; the field can be left blank.

Blank Response

The Blank Response specifies the allowable range of the response in a zero-concentration calibrator analysis or a reagent blank test. The input range can be any number within -33,000-33,000, and the low limit lower than the high limit.

If the response is beyond the set range, the system will flag the test result with “BLK”.

The default is -33,000-33,000; the field can be left blank.

Uncapping Time

The Uncapping Time refers to the number of days that the reagent can be kept valid since uncapped at the first time.

The input range must be within 1-999 days. The default is blank.

Twin Chemistry

Twin Chemistry is associated with the current chemistry, and the two chemistries are run with the same reagent. Results of two twin chemistries are calculated in the same

test.

The chemistry whose result will be firstly calculated should be defined prior to the associated chemistry. Volume of the shared reagent and sample volume must be the same for the two chemistries. Only the two chemistries that have had no reagents loaded can be configured as twins.

For more information about twin chemistries, refer to “10.1 Twin Chemistries” (Page 10-2).

Reagent Alarm Limit

Set up the reagent alarm limit for the chemistry. The input range is 1-100, and the default is 10. It can be left blank. When the number of chemistries left is lower than the limit, an alarm will occur; if no alarm limit is defined, the system will not give an alarm.

Only when sample type is Serum can reagent alarm limit be defined.

Enzyme Linear Extension

Linearity limit is only applicable to Kinetic analysis. Only when the substrate depletion has been set up properly, this option is activated.

Select this option to enable enzyme linear extension function. For more details, please refer to 4.5.6 Enzyme Linearity Range Extension.

Prozone Check

In the reaction of antigen and antibody, the amount of generated insoluble compound is closely related to the proportion of antigen and antibody. The maximum amount of compound will be generated at a proper proportion of antigen and antibody, at this point least light is passed and the greatest absorbance is obtained. For other proportions, the amount of insoluble compound will decrease with more light passed and lower absorbance calculated. Therefore, samples with quite different concentrations may generate the equivalent amount of insoluble antigen/antibody compound, and can have the same test results without a Prozone check.

The Prozone check can be performed in two ways: rate check and antigen addition.

- The rate check is based on the condition that the antibody excess reaction rather than the antigen excess reaction can reach equilibrium within the same specified period. This method is used for all chemistries.
- With the antigen addition method, more antigens are added to the finished reaction, and if the reaction does not continue, it indicates antigen excess. This method is only applicable to single and double reagent chemistries.

Rate check:

You are required to set up the following six parameters for the rate check method,

3 System Setup

which are Q1, Q2, Q3, Q4, PC and ABS. The unit is the same as the reaction time and blank time.

Enter the six parameters as follows:

- Single-reagent chemistries: $14 \leq q1 < q2 < q3 < q4 \leq 82$, “14” is the first measuring point after the sample is dispensed and stirred.
- Double-reagent chemistries: $51 \leq q1 < q2 < q3 < q4 \leq 82$ “51” is the first measuring point after R2 is dispensed and stirred.
- Triple-reagent chemistries: $104 \leq q1 < q2 < q3 < q4 \leq 172$, “104” is the first measuring point after R3 is dispensed and stirred.
- Quadruple-reagent chemistries: $141 \leq q1 < q2 < q3 < q4 \leq 172$, “141” is the first measuring point after R4 is dispensed and stirred.
- PC: a number between -99999999 and 99999999, with four decimals.
- ABS: any integer between -99999999 and 99999999.

Antigen addition:

For the antigen addition method, you need to enter the parameters, which are PCM, Q1, Q2, Q3 and Q4.

When $Q3=Q4=0$, absloqlimit can not be entered.

- $172 \geq q2 \geq 104$, $102 \geq q1 \geq \text{reaction end point}$.
- If one parameter among PCM, Q1 and Q2 is not entered, the antigen addition method is not applied.

3.2.5 Flag Qualitative Result

When the analyzer is in the status of standby, incubation, hibernation or stop, you can flag the result of the chemistries qualitatively and the results will be represented by a qualitative flag.

- 1 Select **Utility-Chemistries**.
- 2 Select a chemistry.
- 3 Select **Define F1**.

Figure 3.6 Define/edit chemistries

4 Select **Qualitative F3**.

Figure 3.7 Qualitative result window

5 Select **Use Qualitative Result**.

6 Enter the qualitative range and flag.

For instance, type in “10” in the first edit box of the **Range** field, and then enter “+” in the **Flag** field of the same row. If the chemistry result (L1)

contained in a sample is lower than 10, the “+” sign will be added to the result in the patient report. Type in “20” in the second edit box below the **Range** icon and “+” in the second edit box below the **Flag** icon. If the chemistry result (L2) is greater than 10 and lower than 20, the result will be flagged with the “+” sign. The cycle continues. If the result is greater than L5, the six flag will appear on the patient report.

- 7 Select **OK** to save the setup.

Refer to these steps to flag the qualitative result of ISE chemistries and calculations.

3.2.6 Slope and Offset Adjustment

The slope and offset are calculation factors that are used to compensate the test results of a chemistry when the QC result of the chemistry is slightly deviating.

When the measurement is finished, the system adjusts the test result with the following equation:

$$y=kx+b$$

Where, x is the test result before adjustment, y is the result after adjustment, k is the slope, and b is the offset.

Before setting up the calculating factors, make sure that you have sufficient permissions and the system is not running tests.

- 1 Select **Utility-Chemistries**.
- 2 Select **Slope/Offset F5**.

Figure 3.8 Slope/Offset Adjustment window

Chem	Slope	Offset	Unit
L	1	0	
H	1	0	
I	1	0	
Na(Serum)	1	0	mmol/L
K(Serum)	1	0	mmol/L
CL(Serum)	1	0	mmol/L
Na(Urine)	1	0	mmol/L
K(Urine)	1	0	mmol/L
CL(Urine)	1	0	mmol/L
ALT	1	0	U/L

Restore Defaults Save Discard Close

3 Choose a chemistry.

4 Double click the **Slope** field and then input the slope.

Positive, negative and decimal numbers (-99999999~99999999) can be entered.
The maximum input length is 8 digits.

5 Double click the **Offset** field and then input the offset.

Positive, negative and decimal numbers (-99999999~99999999) can be entered.
The maximum input length is 8 digits.

6 Repeat step 3 to 5 to set up the slope and offset for other chemistries.

7 Select **Save** to save your input information.

8 To restore the factory settings of slope and offset, select **Restore Defaults**.

9 Select **Close** to exit the window.

3.2.7 Reference/Critical Range Setup

The system allows the setup of reference/critical ranges for each chemistry.

- Reference range indicates the allowable concentration range of a normal sample.

- Critical range is the allowable result range from the perspective of clinical diagnosis.

If a result is greater than the high limit of the reference range, ↑ will appear near the result; if a result is less than the low limit of the reference range, ↓ will appear near the result. If a result is greater than the high limit of the critical range, ↑! will appear near the result; if a result is less than the low limit of the critical range, ↓! will appear near the result. You may enable the auto rerun function for a chemistry, which will be rerun automatically once the test result is beyond the critical range.

Prior to defining the reference/critical range, ensure that you have sufficient permissions and the system status is not Running.

Defining/Editing reference/critical range

- 1 Select **Utility-Chemistries**.
- 2 Select **Ref Range F4**.

Figure 3.9 Reference/Critical Range Setup window

- 3 Choose a chemistry from the **Chemistry** pull-down list.
- 4 Choose a sample type for the reference and critical range.
- 5 Choose patient gender for the reference and critical range.
- 6 Enter the age range in the **Age Range** field.
 - Enter the age low limit in the first edit box.
 - Enter the age high limit in the second edit box.

- Choose an age unit from year, month, day and hour.
- 7 Enter the reference range.
 - Enter the reference range low limit in the first edit box.
 - Enter the reference range high limit in the second edit box.
 - The maximum input length is 8 digits.
 - 8 Enter the critical range.
 - Enter the critical range low limit in the first edit box.
 - Enter the critical range high limit in the second edit box.
 - The maximum input length is 8 digits.
 - 9 To rerun the ISE chemistry when its test result is beyond the critical range, mark the **Auto Rerun** checkbox with a tick.

Biochemistries and user-defined calculations cannot be rerun. The **Auto Rerun** checkbox will not appear if the current chemistry is a biochemistry or a calculation.

For more information about auto rerun, refer to 8.2.4 Rerunning Samples (page 8-4).
 - 10 Select **Save F7**. The reference/critical range are displayed in the middle list.
 - Select **Discard F6** to abort the input information, or
 - Select **Set Defaults F1** to set the reference/critical range as the default for the chemistry.
 - 11 Select **Prev F4** or **Next F5** to set up reference/critical range for more chemistries.
 - 12 Select **Exit F8** to close the window.

Setting up default reference/critical range

You are allowed to select a default reference/critical range for a sample type and gender. The default range appears in red. Only one default reference/critical range is allowed for the same sample type and gender of each chemistry.

- 1 Select **Utility-Chemistries**.
- 2 Select **Ref Range F4**.
- 3 Choose the chemistry name, sample type, gender and age range.

- 4 Choose a reference/critical range in the middle list.
- 5 Select **Set Defaults F1**.

The selected reference/critical range is set as the default of the chemistry. The system will check the test result, and if necessary, flag and rerun the chemistry. For details of reference range flags, refer to Result display settings (page 3-5).

- 6 Select **Exit F8** to close the window.

Deleting a reference/critical range

- 1 Select **Utility-Chemistries**.
- 2 Select **Ref Range F4**.
- 3 Choose the chemistry name, sample type, gender and age range.
- 4 Choose a reference/critical range you want to remove.
- 5 Select **Delete F2**.
- 6 Select **OK**.
- 7 To clear all ranges of the chemistry, select **Del All F3**.



NOTE

The reference/critical range cannot be recovered once deleted. Think twice before the deletion.

- 8 Select **OK**.
- 9 Select **Exit F8** to close the window.

3.3 Calibration Setup

3.3.1 Introduction

Perform calibration settings in the following order:

- Define a calibrator
- Set up calibrator concentrations
- Set up calibration rules
- Set up calibrator acceptance limits

You are allowed to add, edit and delete calibrators only when the system status is not Running.

3.3.2 Defining a Calibrator

The system allows the definition of up to 99 calibrators. You are required to input the name and position for each defined calibrator.

1 Select **Reagent-Setup**.

2 Select **Define F1**.

Figure 3.10 Calibrator Definition window

Carousel	Pos
Sample Carousel 1	S6
Sample Carousel 2	
Sample Carousel 3	
Sample Carousel 4	
Sample Carousel 5	

3 Enter the calibrator name with 1-10 characters.

4 Enter the expiration date of the calibrator. The default is the current day in the next year.

3 System Setup

Calibrators beyond the expiration date cannot be used.

- 5 Enter the lot number.

The input range is 0-18 and accepts numbers and letters. Calibrators with the same name must not have the same lot number.

- 6 Assign positions for the calibrator.

You are allowed to assign one position of each sample carousel for the calibrator. The fourth ring (center) of the sample carousel is used to carry calibrators and controls. You may also place the calibrator on other idle positions of the sample carousel.

- 7 Select **Save** to save your input information.
- 8 To define more calibrators, repeat step 3 to 7.
- 9 Select **Close** to exit the window.

3.3.3 Importing a Calibrator

Calibrator parameters such as calibrator name, lot number, expiration date, concentration of each chemistry and dilution parameter can be imported.

- 1 Select **Reagent-Setup**.
- 2 Select **Define F1**.

Figure 3.11 Calibrator Definition window

Carousel	Pos
Sample Carousel 1	S6
Sample Carousel 2	
Sample Carousel 3	
Sample Carousel 4	
Sample Carousel 5	

- 3 Select **Import** and insert USB drive.
- 4 Select the path of .cif file.

Only .cif file can be imported; each .cif file stores information for one calibrator and each time only one calibrator can be imported. When the system reads the calibrator information, the following window is displayed:

Figure 3.12 Calibrator Definition window

Chem	Lot No.
Ca	
Tp	

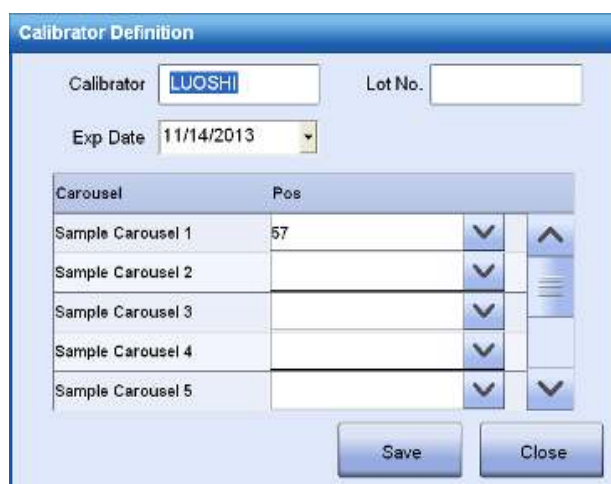
- 5 Enter Lot No.
- 6 Select OK.
- 7 Assign positions for the calibrator.
- 8 Select **Save** to save your input information
- 9 Select **Close** to exit the window.

3.3.4 Editing a Calibrator

You are allowed to edit calibrators only when the system is not running any tests. The default calibrator WATER (physiological saline) is placed in position W (No.90) of the sample carousel. It is used for reagent blank measurement and cannot be edited or deleted.

- 1 Select **Reagent-Setup**.
- 2 Choose a calibrator to edit and select **Edit F2**.

Figure 3.13 Calibrator Definition window



The screenshot shows the 'Calibrator Definition' window. It has a title bar with the text 'Calibrator Definition'. Inside, there are three input fields: 'Calibrator' with the text 'LUOSHI', 'Lot No.' which is empty, and 'Exp Date' with a dropdown menu showing '11/14/2013'. Below these is a table with two columns: 'Carousel' and 'Pos'. The table has five rows, each labeled 'Sample Carousel 1' through 'Sample Carousel 5'. The 'Pos' column for 'Sample Carousel 1' contains the number '57'. To the right of each row is a dropdown arrow. To the right of the table is a vertical stack of four buttons: an up arrow, a down arrow, a list icon, and a close icon. At the bottom right of the window are two buttons: 'Save' and 'Close'.

Carousel	Pos
Sample Carousel 1	57
Sample Carousel 2	
Sample Carousel 3	
Sample Carousel 4	
Sample Carousel 5	

- 3 Edit the following calibrator information:
 - Calibrator name
 - Expiration date
 - Lot number
 - Position
- 4 Select **Save** to save your input information.
- 5 Select **Close** to exit the window.

3.3.5 Setting up Calibrator Concentrations

You are required to set up calibrator concentrations for each chemistry after defining the calibrator. Only the calibrator with positions assigned and concentrations determined can be used for programming. You are allowed to change the calibrator concentrations when the system is not running any tests.

The default calibrator WATER has concentration of 0 for all chemistries. It has no lot number and expiration date and must not be edited or removed.

- 1 Select **Reagent-Setup**.

Figure 3.14 Calibrator concentration setup window

Stopped/Stopped

HOST Admin 09-26 15:20

Reagent/Calibration Biochemistry Calibration ISE Calibration Reagent Carousel Status Setup

Calibrator	Pos	Lot No.	Exp Date	Chem	Conc	Unit
WATER	W			B2-MG	9.20	mg/L
B2-MG	1-83		2013-05-28	B2-MG	9.20	mg/L
H2O	1-56		2013-11-14			
LUO9HI	1-57		2013-11-14			
ubc	1-54	20120110	2013-07-05			
water 1	1-59	1245	2013-12-05			

Define F1 Edit F2 Chem F3 Rules F4 Dilute F5 Delete F6 Discard F7 Save F8

Successfully saved!

- 2 Choose a calibrator in the left list.

The chemistries configured for the calibrator are displayed in the right list.

- 3 Choose chemistries to which the calibrator is applicable, and then select the corresponding **Conc** column and type in the calibrator concentration for it.

The concentration must be above 0.

- 4 Select the **Unit** from the dropdown box.
- 5 Select **Save F8** to save your input information.

A message box pops up indicating that parameters are changed and calibration is required.

3.3.6 Setting up Calibration Rules

You should set up the calibration rules after defining a calibrator and determining concentrations for it. You are allowed to set up or edit the calibration rules, replicates, K factor and auto calibration only when the system is not running any tests.

- 1 Select **Reagent-Setup**.

2 Select **Rules F4**.

Figure 3.15 Calibration Setup window

3 Choose a chemistry from the **Chem** pull-down list.

4 Choose a calibration method in the **Math Model** field.

The options include:

- K factor
- Two-point linear
- Multi-point linear
- Logit-Log 4P
- Logit-Log 5P
- Exponential 5P
- Polynomial 5P
- Parabola
- Spline

5 If you choose K Factor, type in the K factor in the **Factor** field.

This field is activated only when the one-point linear math model is chosen. When the K factor is determined, the calibration results will be calculated with the equation $Y=K \cdot X$. Where, Y is the calibration result, K is the factor, and X is the response. The K factor can be used to calculate sample results without running a calibration.

- 6 Choose the number of replicates.

The input range is 1-5, and the default is 1.

- 7 If you want to run auto calibration, choose the conditions.

- When reagent bottle is changed;
- When reagent lot is change; or
- When the calibration time is exceeded.

For more information about auto calibration, refer to 6.5 Auto Calibration (page 6-14).

- 8 Choose calibrators in the right list for the chemistry.

Up to 10 calibrators, including WATER, are allowed for each chemistry. The correspondence between the number of calibrators and calibration math model is shown in the table below.

Table 3.5 Correspondence between number of calibrators and calibration math model

Calibration Math Model	Number of Calibrators
K Factor	N=0 or 1
Two-point linear	N=2
Multi-point linear	$2 < N \leq 10$
Logit-Log 4P	$4 \leq N \leq 10$
Logit-Log 5P	$5 \leq N \leq 10$
Exponential 5P	$5 \leq N \leq 10$
Polynomial 5P	$5 \leq N \leq 10$
Parabola	$3 \leq N \leq 10$
Spline	$3 \leq N \leq 10$

- 9 Select **Save F7** to save your input information.
- 10 Select **Prev F4** or **Next F5** to set up the QC rules for other chemistries.
- 11 Select **Close F8** to close the window.

3.3.7 Calibrator Acceptance Limits

The calibration results are compared with the determined acceptance limits. If the calibration results exceed the acceptance limits, the system will give an alarm and flag the results on calibration reports.

- 1 Select **Reagent-Setup**.
- 2 Select **Rules F4**.
- 3 Enter the following acceptance limits in the **Acceptance Limits** area.

Table 3.6 Calibration acceptance limits

Acceptance Limits	Description
Calibration time	The validity period indicates the number of days that the calibration factors can be used. If the validity period is exceeded, the system will give an alarm. The input range must be within 1-9999 hours. The default is blank, which means the calibration factors of the chemistry can be used all the time.
Slope difference	The slope difference is applicable to linear calibration only. It is the K factor (slope) difference between two adjacent calibrations. The system will give the flag "FAC" and an alarm when the slope difference is exceeded. Type in the percentage within 0-100. The default is blank, which means not performing this check. The slope difference is not applicable to K factor calibration.
Standard deviation (SD)	The standard deviation is used for multi-point linear and non-linear calibrations. The system will give the flag "CSD" and an alarm when the SD value is exceeded. The input range must be within 0-999. The default is blank, which means not performing this check.
Sensitivity	The sensitivity is the absolute response difference between two calibrators with the maximum and minimum concentration. The system will give the flag "SEN" and an alarm when the sensitivity is exceeded. The input range must be within 0-33,000. The default is blank, which means not performing this check.
Repeatability	The repeatability is the difference of the maximum and minimum response of each calibrator. If the calculated calibrator response difference is lower than the set limit, the system will give the flag "DUP" and an alarm. The input range must be within 0-33,000. The default is blank, which means not performing this

Acceptance Limits	Description
	check.
Determination coefficient	The determination coefficient is the fit degree of the calibration curve. The system will give the flag "DET" and an alarm when the calibration fit degree is exceeded. The input range must be within 0-1. The default is blank, which means not performing this check.

- 4 Select **Save F7** to save your input information.
- 5 Select **Close F8** to close the window.

3.3.8 Deleting a Calibrator

You are allowed to remove the calibrators other than WATER. When a calibrator is deleted, all calibration settings and its position are cleared, and it cannot be used for programming. The stored test results of the calibrator can be recalled according to the chemistry name. only calibrators that are not requested or run can be deleted.

- 1 Select **Reagent-Setup**.
- 2 Choose a calibrator you want to remove.
- 3 Select **Delete F6**.
- 4 Select **OK**. The selected calibrator is deleted.

3.4 QC Setup

3.4.1 Introduction

Perform QC settings in the following order:

- Define a control
- Select chemistries
- Set up control concentrations
- Set up QC rules

3.4.2 Defining/Editing a Control

The system allows the definition of up to 99 controls. You are required to enter the control name and sample type. The combination of control name and lot number must not be duplicate and should be unique. If a control has no lot number, you are not allowed to define another control with the same name.

1 Select **QC-Setup**.

2 Select **Define F1**.

Figure 3.16 Define/Edit Controls window

Sample Carousel	Pos
Sample Carousel 1	1
Sample Carousel 2	
Sample Carousel 3	
Sample Carousel 4	
Sample Carousel 5	

3 Type in the control name.

4 Enter the control number.

The input range is 1-99.

- 5 Enter the lot number.

The lot number can be composed of characters or numbers. The combination of control name and lot number must not be duplicate.

- 6 Select a sample type from the **Sample Type** pull-down list.

The options include serum, plasma, urine, CSF and other. The default is serum.

- 7 Select expiration date for the control.

When the expiration date is exceeded, the control can still be programmed and analyzed, while the system flags the test result in the **Flag** column to remind you of replacing the expired control.

- 8 Assign positions for the control.

You are allowed to assign one position of each sample carousel for the control. The fourth ring (center) of the sample carousel is used to carry calibrators and controls. You may also place the control on other idle positions of the sample carousel.

- 9 Select **OK** to save your input information.

- 10 To define more controls, select **New** and repeat step 3 to 9.

- 11 Select **Exit** to exit the window.

3.4.3 Selection of Chemistries

After defining a control, you need to select chemistries for which the control will be used. When selecting chemistries, make sure that the system status is Incubation, Standby, Stop or Sleep, and the control status is not Programmed or Incomplete.

- 1 Select **QC-Setup**.
- 2 Choose a control in the left list.
- 3 Select **Chems F2**.

Figure 3.17 Chemistries window



- 4 Choose chemistries for the control. Use the right-arrow button to display more chemistries.
- 5 To choose all chemistries in the list, select **Select All**.
- 6 To deselect the chemistries, select **Clear**.
- 7 Select **OK**.

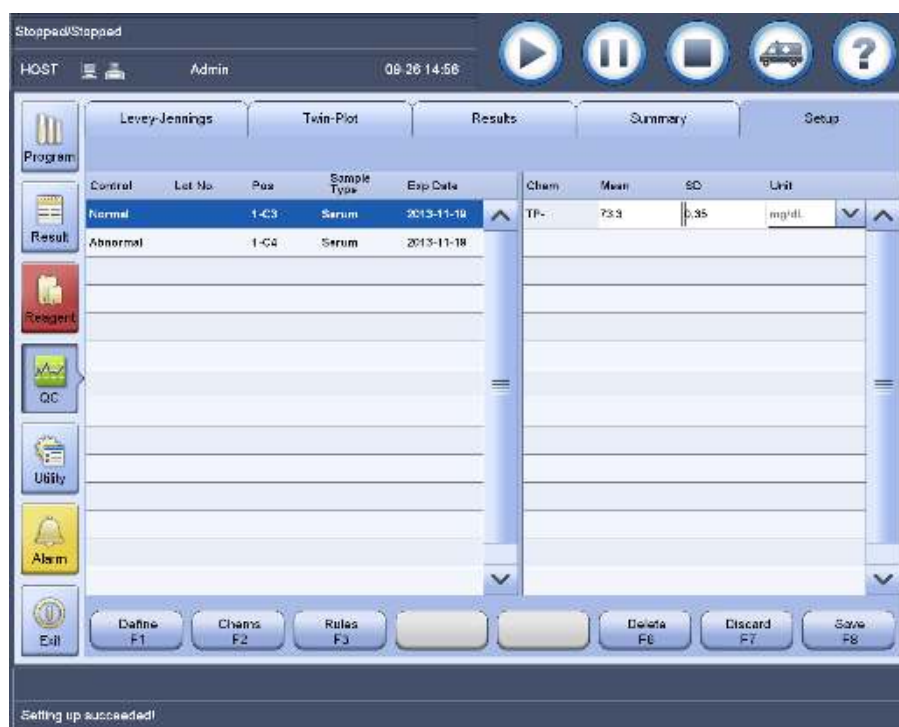
3.4.4 Setting up Control Concentrations

You are required to set up the average concentrations and SDs of a control for each chemistry after defining the control and choosing chemistries. Only the control with positions assigned and concentrations determined can be used for programming.

To run quality control for special calibrations, you must define the mean value and SD; otherwise, no control results will be calculated. If the sub chemistries of a special calculation have no mean value and SD, QC evaluation will not be done and QC plot cannot be recalled.

- 1 Select **QC-Setup**.

Figure 3.18 Setup screen



- 2 Choose a control in the left list.

The chemistries configured for the control are displayed in the right list.

- 3 Select the **Mean** column of a chemistry and type in the average concentration for it.

The concentration must be above 0 with no more than 8 digits.

- 4 Select the **SD** column of a chemistry and type in the standard deviation for it.

The SD must be above 0 with no more than 8 digits.

- 5 Select **Unit** from the dropdown box.
- 6 Select **Save F8** to save your input information.

3.4.5 Setting up QC Rules

You should set up the control rules after defining a control and determining concentrations for it. The controls without QC rule can still be programmed and analyzed but will not be monitored for error detection. You are allowed to change the QC rules when the system is not running any tests.

- 1 Select **QC-Setup**.
- 2 Select **Rules F3**. The **QC Rules Setup** window is displayed.

Figure 3.19 QC Rules Setup window

- 3 Choose a chemistry from the **Chem** list.
- 4 Choose QC rules in the **Westgard Rules** area.
- 5 If you assign a couple of controls for the chemistry, you are allowed to enable the Two-Control Evaluation option.

Those controls not contained in the two-control evaluation will be monitored according the Westgard rules.

- 6 Select the first control in the **Control(X)** field.
- 7 Select the second control in the **Control(Y)** field.
- 8 Select **OK** to save your input information.
- 9 Repeat step 3 to 8 to set up quality control rules for other chemistries.
- 10 Select **Exit** to exit the window.

3.4.6 Deleting a Control

You are allowed to change the control concentrations when the system is not running any tests. When a control is deleted, the control information, concentration parameters and QC results as well as the control position are cleared. If the deleted

control is included in the two-control evaluation, the relevant two-control evaluation will be disabled. Those controls programmed for analysis cannot be deleted.

- 1** Select **QC-Setup**.
- 2** Choose a control in the left list.
- 3** Select **Delete F6**.

4 Operation Theories

This chapter gives brief introduction of the operation theories of the instrument, which include:

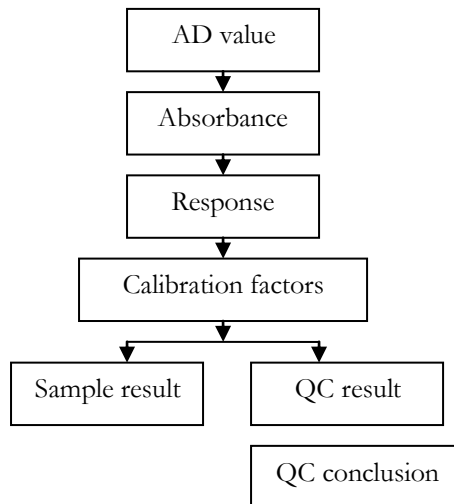
- Principles of measurement
- Calibration math model and calculation of factors
- Prozone check

4.1 Overview

The system is a fully automated computer-controlled clinical chemistry analyzer allowing the random selection of chemistries. It is capable of running a variety of chemistries based on the operation theories and measurement principles.

The system performs measurement and generates the test results in the following procedure:

Figure 4.1 Measurement workflow



The system measures the light intensity through photoelectric conversion, linear amplification and AD conversion, and then calculates the reaction mixture's absorbance and the absorbance change rate, that is, the response, based on which the calibration factors are obtained. The system performance is evaluated according to the test results of the control samples. If the system is working normally, you may start the analysis of patient samples and the system will calculate the sample results with the calibration factors.

4.2 Principles of Measurement

4.2.1 Introduction

The system performs measurement with the following principles:

- Endpoint
- Fixed-time
- Kinetic

In the description of the following sections, N and P indicate the blank read time range, L and M indicate the reaction read time range. In double-wavelength measurements, absorbance A is the absorbance difference between the primary and secondary wavelengths; in single-wavelength measurements, absorbance A is the absorbance measured at the primary wavelength.

4.3 Endpoint Measurements

4.3.1 Introduction

In endpoint measurements, the reaction reaches equilibrium after a period of time. Since the equilibrium constant is quite high, it can be considered that all substrates (analytes) have changed into products, and the absorbance of the reactant will not change any more. The absorbance change is directly proportional to the analytes' concentration. The endpoint method, also called equilibrium method, is most ideal for measurements.

The endpoint reaction is insensitive to minor changes in such conditions as the enzyme volume, pH value and temperature, provided the changes are not significant enough to affect the reaction time.

4.3.2 Calculation of Reaction Absorbance

Set up the reaction time range by understanding the following instructions:

- If $L=M$, that is, $[M]$ and $[M]$ are entered for the reaction time range, one measuring point will be used for absorbance calculation, and the reaction absorbance will be the absorbance measured at point M , i.e. $A_i=A_M$.
- If $L=M-1$, that is, $[M-1]$ and $[M]$ are entered for the reaction time range, two measuring points will be used for absorbance calculation, and the reaction absorbance will be the average of the absorbance measured at the two points, i.e.
$$A_i = \frac{A_M + A_{M-1}}{2}.$$
- If $L=M-2$, that is, $[M-2]$ and $[M]$ are entered for the reaction time range, three measuring points will be used for absorbance calculation, and the reaction absorbance will be the mediate absorbance measured at the three points, while the maximum and minimum absorbance is removed.
- If $M>L+2$, the reaction absorbance will be the average of the remaining absorbance when the maximum and minimum absorbance is removed.

4.3.3 Calculation of Blank Absorbance

The blank absorbance A_b is calculated in the same way as the reaction absorbance A_i .

When $N=P=0$, the blank absorbance A_b will not be calculated.

4.3.4 Calculation of K Factor

The system provides four K factors for result calculation, which are expressed through the following equations:

- $k1 = \frac{V_{R1}}{V_{R1} + V_S}$
- $k2 = \frac{V_{R1} + V_S}{V_{R1} + V_S + V_{R2}}$
- $k3 = \frac{V_{R1} + V_S + V_{R2}}{V_{R1} + V_S + V_{R2} + V_{R3}}$
- $k4 = \frac{V_{R1} + V_S + V_{R2} + V_{R3}}{V_{R1} + V_S + V_{R2} + V_{R3} + V_{R4}}$

Where, V_{R1} , V_{R2} , V_{R3} and V_{R4} are the volumes of R1, R2, R3 and R4; V_s is the actual volume of sample dispensed for reaction.

4.3.5 Calculation of Response

The response in endpoint measurements is calculated as follows:

$$R = A_i - k \cdot A_b$$

k is the calculation factor and varies with the chemistry parameters.

Table 4.1 Calculation of response for endpoint measurements

Endpoint	Blank Time	Reaction Time	K Factor
When the blank absorbance is read before the reaction begins,			
Single-reagent	$5 \leq N \leq 12$	$14 \leq L \leq M \leq 82$	K1
Double-reagent	$14 \leq N \leq P \leq 49$	$51 \leq L \leq M \leq 82$	K2
Triple-reagent	$51 \leq N \leq P \leq 90$	$104 \leq L \leq M \leq 172$	K3
Quadruple-reagent	$104 \leq N \leq P \leq 139$	$141 \leq L \leq M \leq 172$	K4
When the blank absorbance is read after the reaction begins,			
Single-reagent	$14 \leq N \leq P$	$P < L \leq M \leq 82$	1
Double-reagent	$51 \leq N \leq P$	$P < L \leq M \leq 82$	1
Triple-reagent	$104 \leq N \leq P$	$P < L \leq M \leq 172$	1
Quadruple-reagent	$141 \leq N \leq P$	$P < L \leq M \leq 172$	1
When the blank absorbance is not subtracted,			
Single-reagent	$N = P = 0$	$14 \leq L \leq M \leq 82$	0
Double-reagent	$N = P = 0$	$51 \leq L \leq M \leq 82$	0
Triple-reagent	$N = P = 0$	$104 \leq L \leq M \leq 172$	0
Quadruple-reagent	$N = P = 0$	$141 \leq L \leq M \leq 172$	0

4.3.6 Sample Blanked Response

Sample blank is used for removal of non-chromogenesis reaction, such as influence of sample interference (Hemolysis, icterus and lipemia) on absorbance readings. The sample blank reaction curve is almost a straight line with slope of 0 during the reaction period, and therefore means nothing for fixed-time and Kinetic analysis.

In single-reagent endpoint measurements, the response of the sample blank test is $R_{sb} = A_i - k \cdot A_b$, and the sample blanked response is $R' = R - R_{sb}$.

4.4 Fixed-time Measurements

4.4.1 Introduction

In fixed-time measurements, namely, rate measurements, the reaction velocity (v) is directly proportional to the substrate concentration $[S]$ within a specific period, that is, $v=k[S]$. As the substrate is consumed continuously, the reaction velocity is decreasing gradually, and so is the absorbance change rate. It takes a long time for the reaction to reach equilibrium. Theoretically, the absorbance reading can be taken at any time. The reaction can, however, become steady only after a lag because it is complicated at the beginning and there are miscellaneous reactions due to complex serum compositions.

For any rate measurements, the substrate concentration $[S]$ at a given point t since the reaction begins is obtained through the following formula:

$$[S] = [S_0] \times e^{-kt}$$

Where,

- S_0 : the initial substrate concentration
- e : base of the natural log
- k : velocity constant

The change of substrate concentration $\Delta[S]$ over a fixed time interval, t_1 to t_2 , is related to $[S_0]$ by the following equation:

$$[S_0] = \frac{-\Delta[S]}{e^{-kt_1} - e^{-kt_2}}$$

That is, the change in substrate concentration is directly proportional to its initial concentration within a fixed time interval. This is the common feature of rate measurements. Within this interval, the absorbance change is directly proportional to the analytes concentration. The fixed-time reaction is also called, rate reaction, first-order Kinetic reaction and two-point Kinetic reaction.

It is available in single-interval and double-interval according to the input mode of measuring points. In the double-interval reaction, the sample blank, which is the absorbance change at two points within the incubation time, is subtracted from the reaction absorbance.

The fixed-time measurements allow the check of substrate depletion at the two measuring points. When detecting substrate depletion, the system will flag the test result with “BOE” and give an alarm.

4.4.2 Calculation of Response

The response in fixed-time measurements is calculated as follows:

$$R = 60 * \left(\frac{A_M - A_L}{t_M - t_L} - k \cdot \frac{A_P - A_N}{t_P - t_N} \right)$$

k is the calculation factor and varies with the chemistry parameters.

Table 4.2 Calculation of response for fixed-time measurements

Fixed-time	Blank Time	Reaction Time	K Factor
When the blank absorbance is read before the reaction begins,			
Single-reagent	5 ≤ N < P ≤ 12	14 ≤ L < M ≤ 82	K1
Double-reagent	14 ≤ N < P ≤ 49	51 ≤ L < M ≤ 82	K2
Triple-reagent	51 ≤ N < P ≤ 90	104 ≤ L < M ≤ 172	K3
Quadruple-reagent	104 ≤ N < P ≤ 139	141 ≤ L < M ≤ 172	K4
When the blank absorbance is not subtracted,			
Single-reagent	N=P=0	14 ≤ L < M ≤ 82	0
Double-reagent	N=P=0	51 ≤ L < M ≤ 82	0
Triple-reagent	N=P=0	104 ≤ L < M ≤ 172	0
Quadruple-reagent	N=P=0	141 ≤ L < M ≤ 172	0

4.5 Kinetic Measurements

4.5.1 Introduction

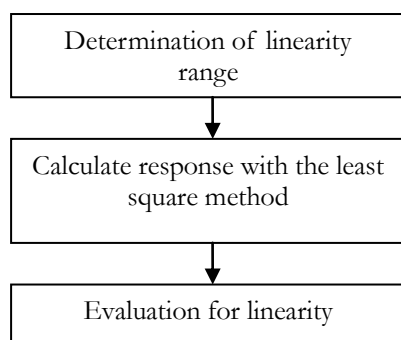
In Kinetic measurements, namely, zero-order Kinetic measurements or continuous-monitoring measurements, the reaction velocity is not related to substrate concentration and remains constant during the reaction process. As a result, the analytes absorbance changes evenly at a given wavelength, and the change rate ($\Delta A/\text{min}$) is directly proportional to the activity or concentration of the analytes. The Kinetic method is usually used to measure enzyme activity.

In fact, it is impossible for the substrate concentration to be absolutely high, and the reaction will be no longer a zero-order reaction when the substrate is consumed to certain degree. Therefore, the reaction type only stands within certain reaction period. In addition, the reaction can become steady only after a period of time, because the reaction is complicated at the beginning and there are miscellaneous reactions due to complex serum compositions.

In Kinetic reaction, the concentration or activity is obtained according to the absorbance change among specified measuring points.

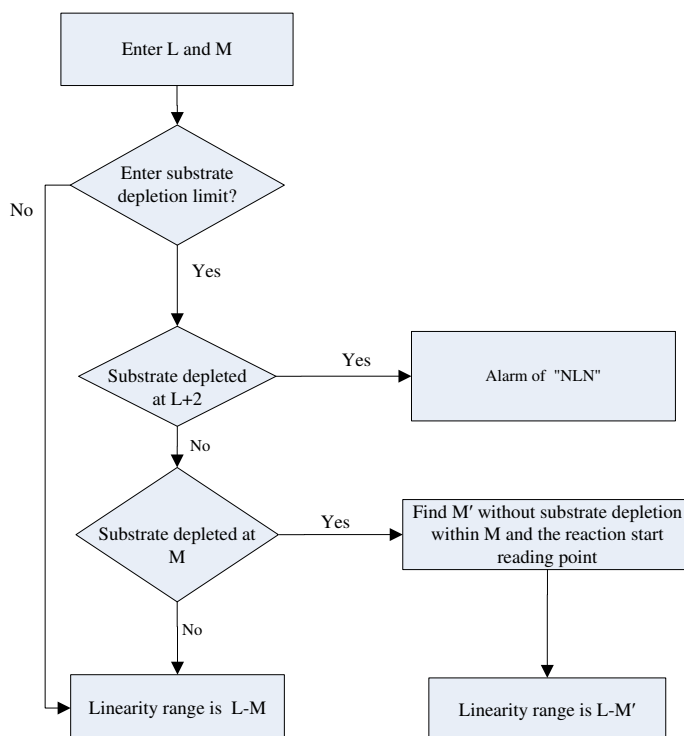
4.5.2 Data Calculation in Kinetic Measurements

Figure 4.2 Data calculation flow of Kinetic measurements



4.5.3 Determination of Linearity Range

The absorbance linearity range is determined based on the substrate depletion limit, and checked within the reaction time rather than the blank time.

Figure 4.3 Determination of linearity range for Kinetic measurements

The number (N) of measuring points within the substrate depletion limit is monitored for different operations:

- If $N \geq 3$, the linearity range includes all measuring points from the reaction start point to the substrate depletion limit;
- If $N = 2$, the system will give the flag “NLN” while using two measuring points for calculating the response.
- If $N = 0$ or 1 , when Enzyme Linear Extension option is selected on the chemistry parameter screen, enzyme linear extension will be enabled and the system gives the flag “NLN”; when Enzyme Linear Extension option is not selected on the chemistry parameter screen, enzyme linear extension will not be enabled and the system gives the flag “NLN” too.

4.5.4 Calculation of Response

Absorbance change rate $\Delta A_{LM'}$ within the reaction time

The response $\Delta A_{LM'}$ within L-M' is calculated with the least square method.

$$\Delta A_{LM'} = 60 * \frac{\sum_{i=L}^{M'} (T_i - \bar{T}) \cdot (A_i - \bar{A})}{\sum_{i=L}^{M'} (T_i - \bar{T})^2}$$

Where,

- L: start point of the linearity range
- M': end point of the linearity range
- A_i: absorbance measured at measuring point i
- $\bar{A}$: average absorbance within L-M'
- T_i: actual measuring time (second) at measuring point i
- $\bar{T}$: average measuring time within L-M

If there are less than two measuring points without substrate depletion within the reaction time, the system will calculate the absorbance change rate by extending the enzyme linearity range.

Absorbance change rate ΔA_{NP} within the blank time

The absorbance change rate ΔA_{NP} within the blank time is calculated with the same equation as $\Delta A_{LM'}$.

If N=P=0, the absorbance change rate within the blank time is 0.

Calculation of Response

The response in Kinetic measurements is calculated as follows:

$$R = \Delta A_{LM'} - K \cdot \Delta A_{NP}$$

k is the calculation factor and varies with the chemistry parameters.

Table 4.3 Calculation of response for Kinetic measurements

Kinetic	Blank Time	Reaction Time	K
When the blank absorbance is read before the reaction begins,			
Single-reagent	$5 \leq N < P \leq 12$	$14 \leq L < M \leq 82$	K1
Double-reagent	$14 \leq N < P \leq 49$	$51 \leq L < M \leq 82$	K2
Triple-reagent	$51 \leq N < P \leq 90$	$104 \leq L < M \leq 172$	K3
Quadruple-reagent	$104 \leq N < P \leq 139$	$141 \leq L < M \leq 172$	K4
When the blank absorbance is not subtracted,			
Single-reagent	N=P=0	$14 \leq L < M \leq 82$	0
Double-reagent	N=P=0	$51 \leq L < M \leq 82$	0
Triple-reagent	N=P=0	$104 \leq L < M \leq 172$	0
Quadruple-reagent	N=P=0	$141 \leq L < M \leq 172$	0

Note: $M-L \geq 2$ indicates that at least 3 measuring points should be included within the reaction time.

4.5.5 Evaluation for Linearity

$$\text{Linearity} = \frac{|\Delta A_f - \Delta A_b|}{|\Delta A_{u,v}|} \times 100 < \text{Linearity Limit}$$

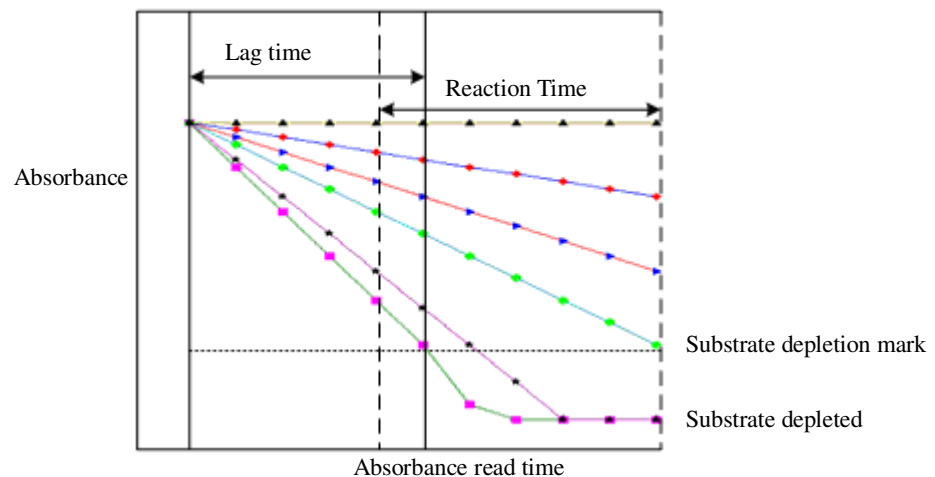
Where, ΔA_f , ΔA_b and $\Delta A_{u,v}$ are the absorbance change rates in the front part, back part and at all measuring points of the reaction. These three values are calculated based on the number of measuring points within the linearity range.

- When $N > 8$, ΔA_f is the absorbance change rate of the first 6 measuring points, ΔA_b of the last 6 measuring points, and $\Delta A_{u,v}$ of all measuring points.
- When $4 \leq N \leq 8$, ΔA_f is the absorbance change rate of the first 3 measuring points, ΔA_b of the last 3 measuring points, and $\Delta A_{u,v}$ of all measuring points.
- When $N \leq 3$, the system will not check the test results for linearity.
- When $|\Delta A_f - \Delta A_b| \leq 60$ or $|\Delta A_{u,v}| \leq 60$ (unit: A/10000/minute), the system will not check the test results for linearity.

The system will compare the calculated linearity with that defined for the chemistry, and will flag the test result with “LIN” and given an alarm if the configured linearity is exceeded.

4.5.6 Enzyme Linearity Range Extension

Figure 4.4 Reaction curve with extended enzyme linearity range



In high-activity enzyme measurements, the substrate may be depleted quickly and the reaction curve will appear obviously nonlinear (as a smooth curve). If the measurement is performed based on the general procedure, the system will flag the test result with “NLN” (no linearity interval), reminding the user to rerun the test after diluting the sample. This will more or less bring troubles to the user.

Extending enzyme linearity range:

Suppose the reaction start time is t_1 and the reaction time is t_L - t_M , then t_1 - t_L is the lag time.

If the number (N) of valid measuring points within t_L - t_M is less than 2 and too few to calculate the response, the sample response can be obtained by extending the enzyme linearity range.

Calculation of $\Delta A_{\max}$:

The linearity range t_1 - t_L without substrate depletion is found within the lag time t_1 - t_L .

If the number (N) of valid measuring points within t_L - t_M is less than 2, the system will not calculate the response but flag the test result with “ENC” (no calculation interval) and give an alarm;

or the system calculates the reaction rate $\Delta A = 60 \cdot (A_{i+1} - A_i) / (t_{i+1} - t_i)$, $i=1, 2 \dots L$ with the lag time t_1 - t_L . The maximum ΔA is taken as the response of the sample. Therefore, the enzyme linearity range is extended via the lag time. The results calculated by extending the enzyme linearity range will be flagged with “EXP” and “NLN”.

4.6 Calibration Math Model and Factors

The system provides linear and non-linear math models. The former is used for Colorimetry chemistries and the later for turbidity chemistries.

In this section,

- R: calibrator response
- C: calibrator concentration (or internal converting concentration in non-linear calibrations)
- K, R_0 , a, b, c and d: calibration factors

4.6.1 Linear Calibrations

Single-point linear calibration

The single-point linear calibration is also called the K factor method. Calculation formula: $C = K \times (R - R_0)$

Where, K is the user-defined K factor, R_0 is the reagent blank response of the first calibrator. If the chemistry is not reagent blanked, $R_0=0$.

Please note that the R and R_0 must be divided by 10,000.

Two-point linear calibration

Calculation formula: $C = K \times (R - R_0)$

The formula contains two factors, K and R_0 , where $K = \frac{C_2 - C_1}{R_2 - R_1}$, and

$$R_0 = R_1 - \frac{C_1}{K}.$$

The calibration math model requires two calibrators. C_1 and C_2 are the concentrations of calibrator 1 and 2; R_1 and R_2 are the responses of calibrator 1 and 2.

Multi-point linear calibration

Calculation formula: $C = K \times (R - R_0)$

The formula contains two factors, K and R_0 . The calibration math model requires $n(n \geq 3)$ calibrators. C_i is the concentration of calibrator i. R_i is the response of calibrator i. K and R_0 can be calculated with the least square method:

$$K = \frac{\sum_{i=1}^n C_i R_i - (\sum_{i=1}^n C_i)(\sum_{i=1}^n R_i) / n}{\sum_{i=1}^n R_i^2 - (\sum_{i=1}^n R_i)^2 / n}$$

$$R_0 = (\sum_{i=1}^n R_i) / n - \frac{(\sum_{i=1}^n C_i) / n}{K}$$

4.6.2 Non-Linear Calibrations

Logit-Log 4P

Calculation formula: $R = R_0 + K \frac{1}{1 + \exp[-(a + b \ln C)]}$

The formula contains four factors, which are R_0 , K , a and b .

The calibration math model requires at least four calibrators. The four factors can be calculated with the L-M method.

This calibration type is applied to the chemistries which have a calibration curve with the response reversely proportional to the concentration.

Logit-Log 5P

Calculation formula: $R = R_0 + K \frac{1}{1 + \exp[-(a + b \ln C + cC)]}$

The formula contains five factors, which are R_0 , K , a , b and c . The calibration math model requires at least five calibrators, and calculates the five factors with the L-M method.

This math model has the same application with the Logit-Log 4P except for a higher fitting.

Exponential 5P

Calculation formula: $R = R_0 + K \exp[a \ln C + b(\ln C)^2 + c(\ln C)^3]$

The formula contains five factors, which are R_0 , K , a , b and c . The calibration math model requires at least five calibrators, and calculates the five factors with the L-M method.

This calibration type is applied to the chemistries which have a calibration curve with the response directly proportional to the concentration.

Polynomial 5P

Calculation formula:
$$\ln C = a + b\left(\frac{R - R_0}{100}\right) + c\left(\frac{R - R_0}{100}\right)^2 + d\left(\frac{R - R_0}{100}\right)^3$$

The formula contains five factors, which are R_0 , a , b , c and d . The calibration math model requires at least five calibrators. The response (R) of the first calibrator (with internal converting concentration of 0) is R_0 , which is given.

Suppose, $y = \ln C$ and $x = \frac{R - R_0}{100}$.

Then, $y = a + bx + cx^2 + dx^3$ can be calculated with the least square method for polynomial expressions.

Parabola

Calculation formula:
$$R = aC^2 + bC + R_0$$

The formula contains three factors, which are a , b and R_0 . The calibration math model requires at least three calibrators. The three factors can be calculated with the least square method.

Spline

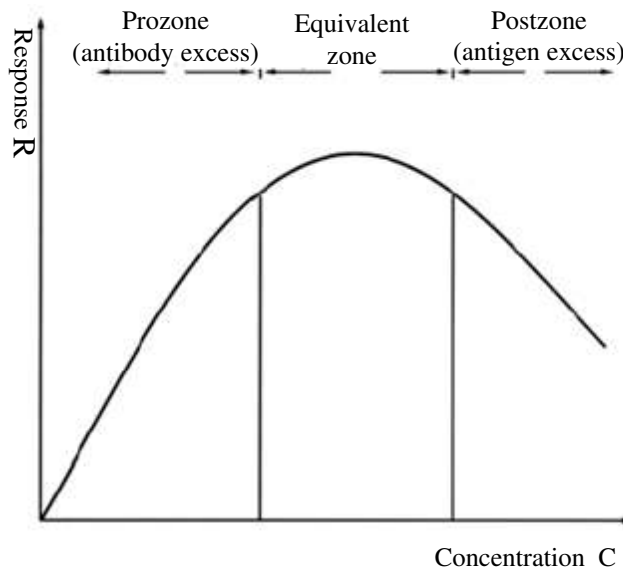
Calculation formula:
$$R = R_{0i} + a_i(C - C_i) + b_i(C - C_i)^2 + c_i(C - C_i)^3$$

The calibration math model requires 2-9 calibrators. Suppose the number of calibrators is n , then the calculation formula contains $4(n-1)$ factors, which are R_{0i} , a_i , b_i , and c_i . Due to the subsection fitting, this math model has the best fit curves than other math models.

4.7 Prozone Check

4.7.1 Introduction

Figure 4.5 Reaction curve of antigen and antibody



In the reaction of antigen and antibody, the amount of generated insoluble compound is closely related to the proportion of antigen and antibody. The maximum amount of compound will be generated at a proper proportion of antigen and antibody, at this point least light is passed and the greatest absorbance is obtained. For other proportions, the amount of insoluble compound will decrease with more light passed and lower absorbance calculated. Therefore, samples with quite different concentrations may generate the equivalent amount of insoluble antigen/antibody compound, and can have the same test results without a Prozone check. The Prozone check, therefore, is necessary for antigen-antibody reactions.

The Prozone limit is the allowable maximum or minimum PC when antigen excess does not happen.

The Prozone check factors include:

- PC_M (Prozone check limit), q1, q2, q3 and q4.
- Absorbance low limit: **ABS**

The Prozone check can be performed in two ways: rate check and antigen addition, which are described in detail in the following sections.

4.7.2 Antigen Addition Method

Antigen excess can be detected by further addition of antigen. When enough antibodies are provided, the antigen reacts with them in reaction medium and forms into stable compound particles, thus producing dispersed light, which increases dynamically with compound amount increased and reaction time extended (antibody excess). If the antibody keeps excess in specified period, it will continue to react with further added antigen, and the reaction will increase accordingly. If the antigen is excessive before further addition, the reaction will decrease. The antigen addition method is applicable to both single-/double-reagent chemistries.

Enter the Prozone check factors as follows:

- PC_M (Prozone check limit), q_1 and q_2 .
- If the absorbance low limit **ABS** appears in grey, that is $q_3=q_4=0$, it cannot be set up.
- $172 \geq q_2 \geq 104$, $102 \geq q_1 \geq$ Reaction end point.

If one of PC_M , q_1 and q_2 is not input, the system will not check the antigen.

- Sample $PC = A_{q_2} - k \times A_{q_1}$.
 - k is the calculation factor.
 - For single-reagent chemistries: $k = (VR1 + VS) / (VR1 + 2VS)$.
 - For double-reagent chemistries: $k = (VR1 + VS + VR2) / (VR1 + 2VS + VR2)$.

The system will flag the test result with “PRO” (Prozone check abnormal) and give an alarm if $PC < PC_M$ in positive reactions or $PC > PC_M$ in negative reactions.

4.7.3 Reaction Rate Method

The rate check is based on the condition that the antibody excess reaction rather than the antigen excess reaction can reach equilibrium within the same specified period. Enter the Prozone check factors as follows:

- PC_M (Prozone check limit), q_1 , q_2 , q_3 and q_4 .
- Absorbance low limit: **ABS**

- Sample PC : $PC = \frac{\frac{A_{q_4} - A_{q_3}}{q_4 - q_3}}{\frac{A_{q_2} - A_{q_1}}{q_2 - q_1}}$. If $PC > PC_M$, the system will flag the test result with “PRO” and give an alarm.

Enter the measuring points as follows:

- Single-reagent chemistries: $14 \leq q_1 < q_2 < q_3 < q_4 \leq 82$, “14” is the first measuring point after the sample is dispensed and stirred.

- Double-reagent chemistries: $51 \leq q1 < q2 < q3 < q4 \leq 82$. “51” is the first measuring point after R2 is dispensed and stirred.
- Triple-reagent chemistries: $104 \leq q1 < q2 < q3 < q4 \leq 172$. “104” is the first measuring point after R3 is dispensed and stirred.
- Quadruple-reagent chemistries: $141 \leq q1 < q2 < q3 < q4 \leq 172$. “141” is the first measuring point after R4 is dispensed and stirred.

If one of PC_M , $q1$, $q2$, $q3$ and $q4$ is not input, the system will not check the reaction rate.

Prozone check will be disabled if:

- (Reaction end point absorbance – Reaction start point absorbance) $< \mathbf{ABS}$
- The sample response is not within the calibrator response range for sample and control analysis of non-linear chemistries.

BS-480 Chemistry Analyzer

Operator's Manual

Advanced Volume

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5 Reagents

This chapter provides you with functions and operating instructions associated with reagent.

5.1 Overview

5.1.1 Introduction

This chapter introduces the advanced application of the reagent module. Perform the following operations according to the practical conditions in your laboratory:

- Sorting reagents
- Setting up reagent inventory alarm limits
- Checking reagent inventory
- Loading bar-coded reagents
- On-line load of reagents
- Off-line load of reagents
- On-line replacement of reagents
- Off-line replacement of reagents
- Unloading reagents

5.1.2 Reagent/Calibration Screen Overview

Select **Reagent** in the function button area of the main screen. The **Reagent/Calibration** screen is displayed. The screen is composed of the biochemistry reagent/calibration page and ISE reagent/calibration page. The latter is displayed by default.

Figure 5.1 ISE reagent/calibration screen

Standby/Standby

HOST Admin 11-27 10:48

Reagent/Calibration Biochemistry Calibration ISE Calibration Reagent Carousel Status Setup

Program Result Reagent QC Utility Alarm Exit

Carousel 1

Chem	Cal Status	Cal Date/Time	Time Left
Na	Calibrated	2012-11-26 16:38:37	5h
K	Calibrated	2012-11-26 16:38:37	5h
Cl	Calibrated	2012-11-26 16:38:37	5h

Reagent	Volume	Load Date	Days Left	Exp Date	Lot No.	Serial No.
ISE Reagent	41	%	2012-01-25	3d	2012-11-30	00009E8DE4
Conc Wash	85	%	2012-11-14	>99d	2013-06-28	
Wash D						
Saline W	42	%	2012-11-22			

1/7

Load F1 End Load F2 Inventory F3 Load List F4 Calibrate F5 Print F7 Cal Options F8

Unmatched software version.

The ISE reagent/calibration screen is divided into three areas. The upper list shows the ISE chemistries, calibration status, calibration date and calibration time left; the lower list shows the volume, load date, expiration date, lot number and serial number of all wash solutions and physiological saline; the function buttons at the bottom are used to access relevant functions.

Select the down-arrow button on the right side of the screen to display the biochemical reagents.

Figure 5.2 Biochemistry reagent/calibration screen



The screen shows all configured biochemistry reagents, including the following information:

- Position: position of the reagent on the reagent carousel.
- Chemistry: name of the chemistry.
- Chemistries left: It refers to the minimum tests left of R1, R2, R3 and R4. When the number of chemistries left is 0, the chemistry is still allowed for programming and measurement.
- Reagent type: reagent type of a multi-reagent chemistry. It includes R1, R2, R3 and R4.
- Tests left: It refers to the remaining tests of each reagent bottle.
- Days left: the difference of reagent expiration date and current date and the uncapping time, whichever the less. When a negative value is displayed, it indicates that the reagent is expired and should be replaced immediately.
- Lot number: lot number of the reagent. It can be input manually during reagent load.
- Calibration status: calibration status of the chemistry, including, Cal Required, Requested, Calibrated, Cal Failed, Cal Time Out, Cal Time Extended, Calculated, Edited, Cal Overridden and N/A.
- Time left: the time left when the calibration factors are expired. It will be displayed only when the calibration status is Calibrated, Cal Time Out or Cal

Time Extended. When the time left is less than 30 minutes, the system displays a message indicating calibration time out; when the calibration time is exceeded, the calibration factors can no longer be used, and you are allowed to recalibrate the chemistry or extend the calibration time.

5.2 Customizing Reagent Display

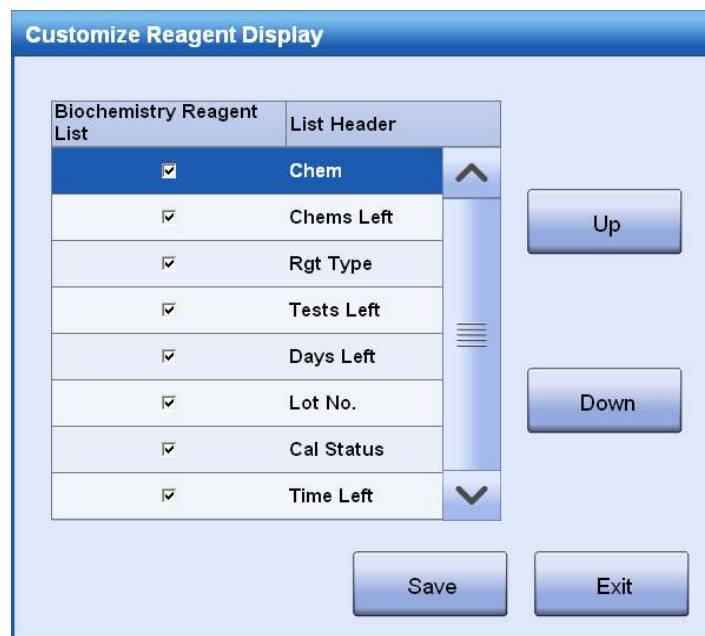
5.2.1 Introduction

The reagent information on the biochemistry reagent/calibration screen can be tailored and displayed in desired order.

5.2.2 Customizing Reagent Display

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **16 Customize Reagent Display**.

Figure 5.3 Customize Reagent Display window



- 4 Select checkbox of reagent information to display it on the reagent/calibration screen, deselect checkbox to cancel the display.
- 5 Select **Up** and **Down** to adjust the display order of reagent information.
- 6 Select **Save** to save the settings.
- 7 Select **Exit** to close the window.

5.3 Sort Reagents

5.3.1 Introduction

Reagents on the biochemistry reagent/calibration screen can be sorted by name, position, chemistries left, days left and calibration time left, and a V-type symbol appears to the right of the sort criteria. Prior to loading reagents or running calibrations, sort the reagents to display the desired ones in the front.

5.3.2 Sort Reagents

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 3 Select the down-arrow button on the right side of the screen to display the biochemical reagents.
- 4 Choose a sorting criterion, and then click on the corresponding list head to rearrange the reagents.

To view or load reagents, choose the following standards:

- Reagent position
- Chemistry name
- Chemistries left
- Days left

To view calibration status or run calibrations, choose the following standard:

- Calibration time left
- Calibration status

5.4 Reagent Inventory Alarm Limits Setup

5.4.1 Introduction

When the reagent inventory is lower than the alarm limit during or before the analysis, the system will give an alarm and display the reagents or wash solution name in yellow on the **Reagent/Calibration** screen, and the volume of ISE reagent and wash solutions become 0.

For Concentrated wash solution, reagent probe wash solution and physiological saline, please set up the reagent alarm limit on the **Load Reagent** window. Please refer to 2.5.2 Loading Concentrated Wash Solution, 2.5.3 Loading Reagent Probe Wash Solution 2.5.3 and 2.5.5 Loading Physiological Saline for details.

5.4.2 Setting up Reagent Inventory Alarm Limit for Biochemistry Reagent

- 1 Select **Utility-Chemistries**.
- 2 Select a Chemistry.
- 3 Select **Define F1**.
- 4 Select down arrow to get to the second page.

Figure 5.4 Define/Edit Chemistries screen

The screenshot displays the 'Define/Edit Chemistries' window. At the top, there are fields for 'Chem' (GLU-H), 'No.' (14), 'Sample Type' (Serum), and 'Chemistry' (GLU-H). Below these are 'Print Name' (GLU) and 'Linearity Limit' (empty). The main section contains several input fields for linearity ranges (Standard, Decreased, Increased), R1 Blank Abs, Blank Response, and Mixed Blank Abs. There are also fields for 'Uncapping Time' (28 Day(s)) and 'Reagent Alarm Limit' (10). A 'Twin Chemistry' dropdown is present. At the bottom, there are checkboxes for 'Prozone Check', 'Rate Check', and 'Antigen Addition', along with four 'Q' (Q1, Q2, Q3, Q4) and 'PC' (PC) input fields. The bottom of the window features a row of buttons: Print F1, Reflex F2, Qualitative F3, Prev F4, Next F5, Discard F6, Save F7, and Close F8.

- 5 Type in the reagent alarm limit.

Enter an integer between 1 and 100. The default is 10.

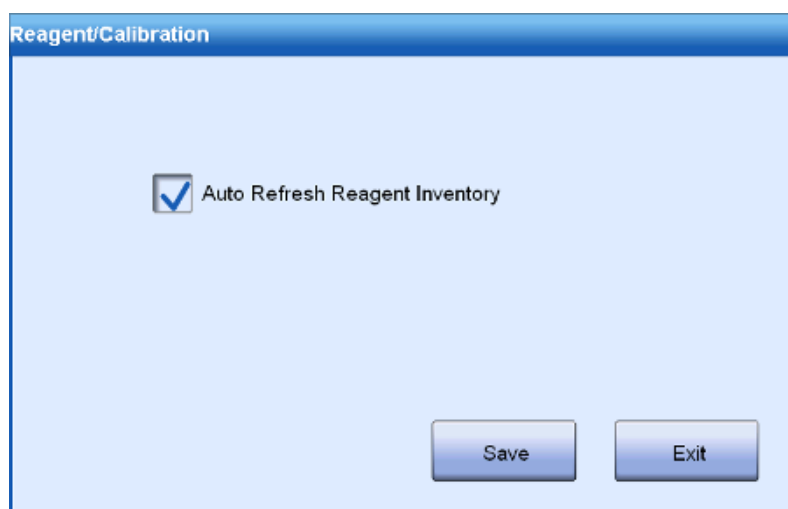
- 6 Select **Save F7**.

5.4.3 Auto refreshing reagent inventory

Via **Reagent/Calibration** option on instrument setup screen, you can configure whether to automatically refresh the reagent with 0 inventory as available for test, when reagent has been loaded and **End Load** button is selected.

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **15 Reagent/Calibration Setup**

Figure 5.5 Reagent/calibration window



- 4 Select or deselect the option **Auto refresh reagent inventory** which is unselected by default.
- 5 Click **Save**.

5.5 Reagent Inventory Check

5.5.1 Introduction

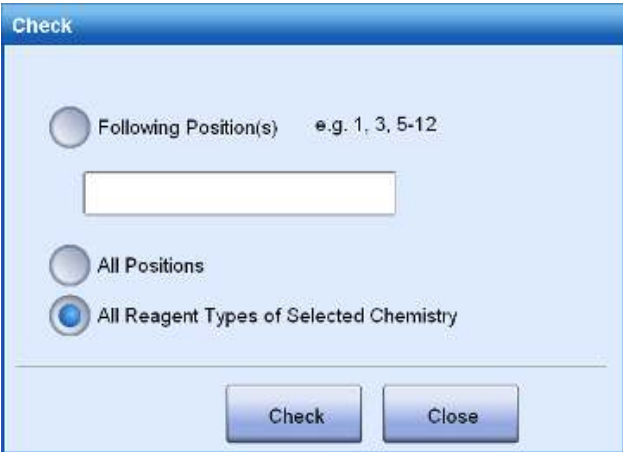
The system provides the manual and auto check of inventory of biochemical reagents. During the test, the system automatically checks the reagent inventory and displays it on the **Reagent/Calibration** screen. After the reagents are loaded, it is necessary to perform the inventory check in order to ensure that sufficient reagents are available for analysis.

Reagent inventory check is allowed only when the system status is Incubation or Standby. While the system is checking reagent inventory, loading or unloading reagents on the current module is not permitted, and the **Utility** button is disabled.

5.5.2 Checking Reagent Inventory

- 1 Select **Reagent- Reagent/Calibration**.
- 2 Select **Inventory F3**.

Figure 5.6 Check window



- 3 Choose reagent positions:
 - Following position(s): check the reagents on specified positions. Enter reagent positions and separate them with a comma. Enter single reagent positions like 1, 2, 3, or position range like 2-15, 20-25.
 - All positions: check all reagent positions of the reagent carousel.
 - All reagents of selected chemistry: check the inventory of all reagent types of the selected chemistry.
- 4 Select **Check**.

- The **Inventory F3** button on the **Reagent/Calibration** screen changes to **No Invent. F3**.
- The reagent carousel graph refreshes the reagent status automatically.
- The **Reagent/Calibration** screen refreshes the **Tests Left** of the selected chemistry, **Chems Left**, and the **Volume** of the selected wash solution.

5.5.3 Canceling Reagent Inventory Check

To cancel reagent inventory check,

- Select **Close** on the **Check** window, and then select **No Invent. F3** on the **Reagent/Calibration** screen. Or
- Select the  icon on the upper-right corner of the main screen, and then select **OK** to start analysis.

5.6 Bar-Coded Reagents Load

5.6.1 Loading Bar-Coded Reagents

If your system is equipped with a reagent bar code reader, you may put the bar-coded reagents on the reagent carousel, and the system will scan all reagent positions automatically and obtain reagent information from the bar code label.

The bar code scanning is only applied to biochemical reagents. The reagent probe wash solution and physiological saline can only be loaded manually rather than bar code scanning. If a bar code is scanned in the fixed positions (D and W), the wash solution or physiological saline in the positions will be unloaded automatically, but the scanned reagents will not be loaded.

For details of loading bar-coded reagents, refer to 13.2.3 Loading Bar-Coded Reagents (page 13-19).



WARNING

The probe tip is sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the probes.



BIOHAZARD

Wear gloves and lab coat, if necessary, goggles.

Do not touch the reagent directly with your body; otherwise, skin wound or inflammation may be caused.

5.7 On-line Load of Reagents

5.7.1 Introduction

The on-line load of reagents is performed while the system is running tests. Before starting an on-line load, request for reagent stop, do not load reagents until all started tests are finished for reagent dispensing. If the system is running calibrations, STAT samples or diluted samples, you are not allowed to start loading reagents unless all tests finish reagents dispensing.



WARNING

The probe tip is sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the probes.



BIOHAZARD

Wear gloves and lab coat, if necessary, goggles.

Do not touch the reagent directly with your body; otherwise, skin wound or inflammation may be caused.

5.7.2 On-Line Load of Reagents

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 3 Select **Load F1** to request for reagent stop. Meanwhile **Load F1** becomes **No load F1**. If you want to abort load, select **No load F1**.

The system status area shows a countdown for reagent stop, and a message box will be displayed when the countdown is finished.



CAUTION

Do not open the reagent carousel cover before the countdown is finished; otherwise, probe collision or other error may occur.

- 4 To load non-bar-coded reagents, select **OK** and then **Load F1**, and remove the reagent carousel cover; to load bar-coded reagents, just remove the reagent carousel cover.
- 5 Place the reagents in correct positions:

5 Reagents

Place the reagents in positions 1-78 of the reagent carousel.

6 Restore the reagent carousel cover.

- For load of non-bar-coded reagents, enter the reagent information on the **Load Reagent** window and then select **End Load F2**.
- For load of bar-coded reagents, the system scans all reagents positions automatically and read reagent information from the bar code, when **End Load F2** button is selected.

The system will resume the tests or select the  icon to start new test.

5.8 Off-line Load of Reagents

5.8.1 Introduction

The off-line load of reagents is performed while the system is not running any tests. You are allowed to directly place the reagents on the reagent carousel, sample carousel or other positions.



WARNING

The probe tip is sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the probes.



BIOHAZARD

Wear gloves and lab coat, if necessary, goggles.

Do not touch the reagent directly with your body; otherwise, skin wound or inflammation may be caused.

5.8.2 Off-line Load of Reagents

- 1 Remove the reagent carousel cover.
- 2 Place the reagents in correct positions:

Place the reagents in positions 1-78 of the reagent carousel.

- 3 Restore the reagent carousel cover.
 - For load of non-bar-coded reagents, enter the reagent information on the **Load Reagent** window.
 - For load of bar-coded reagents, when the reagents have been loaded and the reagent carousel cover has been restored, select **End Load F2** button, the system scans all reagent positions automatically and read reagent information from the bar code.

5.9 On-Line Replacement of Reagents

5.9.1 Introduction

When a reagent is insufficient or exhausted or going to be expired while the system is running tests, you should request for reagent stop and replace the reagent immediately to ensure that the following measurements will be done smoothly.

5.9.2 On-Line Replacement of Reagents

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 3 Confirm the reagent to be replaced and select the reagent position.
- 4 Select **Load F1** to request for reagent stop.

The system status area shows a countdown for reagent stop, and a message box will be displayed when the countdown is finished.



CAUTION

Do not open the reagent carousel cover before the countdown is finished; otherwise, probe collision or other error may occur.

- 5 To load non-bar-coded reagents, select **OK** and then **Load F1**, and remove the reagent carousel cover; to load bar-coded reagents, just remove the reagent carousel cover.
- 6 Remove the reagent.
- 7 Place the new reagent.
- 8 Restore the reagent carousel cover.

If Auto Refresh Reagent Inventory is enabled on the System Setup-Reagent/Calibration window(refer to 5.4.3 Auto refreshing reagent inventory), please follow the steps below:

- For load of non-bar-coded reagents, select the reagent position and enter the reagent information on the **Load Reagent** window and then select **End Load F2**.

- For load of bar-coded reagents, select **End Load F2** and the system scans all reagent positions automatically and read reagent information from the bar code.

If Auto Refresh Reagent Inventory is not enabled on the System Setup-Reagent/Calibration window(refer to 5.4.3 Auto refreshing reagent inventory), please follow the steps below:

- For load of non-bar-coded reagents, select the reagent position and select Load F1 and then select Unload F4. Enter the reagent information again on the **Load Reagent** window and then select **End Load F2**.
- For load of bar-coded reagents, select **End Load F2** and the system scans all reagent positions automatically and read reagent information from the bar code.

The system will resume the tests or select the  icon to start new test.

5.10 Off-Line Replacement of Reagents

5.10.1 Introduction

When a reagent is insufficient or exhausted or going to be expired while the system is not running any tests, you should replace the reagent immediately to ensure that the following measurements will be done smoothly.

5.10.2 Off-Line Replacement of Reagents

- 1 Remove the reagent carousel cover.
- 2 Remove the reagent.
- 3 Place the new reagent.
- 4 Restore the reagent carousel cover.
 - For load of non-bar-coded reagents, enter the reagent information on the **Load Reagent** window.
 - For load of bar-coded reagents, when the reagents have been loaded and the reagent carousel cover has been restored, select **End Load F2** button, the system scans all reagent positions automatically and read reagent information from the bar code.

If Auto Refresh Reagent Inventory is not enabled on the System Setup-Reagent/Calibration window(refer to 5.4.3 Auto refreshing reagent inventory), Before running the tests, please use the Inventory Check function to refresh the reagent inventory of the newly loaded reagent, otherwise they cannot be used. For more details about inventory check, refer to 5.5 Reagent Inventory Check(Page 5-10).

5.11 Unloading Reagents

5.11.1 Introduction

If some chemistries will not be used, you are allowed to clear the chemistry parameters and unload the relevant reagents. The Unload option is also used to remove reagents that are going to be exchanged. All reagents other than ISE reagent, sample wash solution, reagent wash solution, physiological saline and concentrated wash solution can be removed through the Unload function. When a chemistry is requested for quality control, sample analysis or calibration, all reagents of the chemistry still can be unloaded.

When a reagent is unloaded, all relevant information and its position are cleared. The reagents that are being used for analysis cannot be unloaded.

5.11.2 Unloading Biochemical Reagents

The following procedures are only applicable to unloading the reagents without bar code; for those reagents with barcode, when the reagents are taken away from the reagent carousel, they are unloaded automatically.

- 1 Make sure that the reagent to be unloaded is not being used for analysis.
- 2 Select **Reagent-Reagent/Calibration**.
- 3 Select the up and down arrow buttons to display the biochemical reagent/calibration screen.
- 4 Select the reagent position to unload reagent.
- 5 Select **Load F1**.
- 6 Select **Unload F4**.
- 7 Remove the reagent carousel cover.
- 8 Take out the reagent from the reagent carousel.
- 9 Restore the reagent carousel cover.
- 10 Select **Exit F5** to close the window.
- 11 Select **End Load F2**.

5 Reagents

6 Calibration

- Calibration status and alarm indication
- Calibrator dilution setup
- Reagent blank
- Auto calibration
- Extending calibration time
- Overriding a calibration
- Calibration results recall

6.1 Overview

In a calibration, the system measures the response of the calibrator with given concentration, and then calculates the factors in the concentration-response equation. In this way, a math equation about concentration and response is determined. The concentration of a patient sample can be calculated based on the math equation and the measured sample response.

When the calibration status is abnormal, the system will give an alarm and display the calibration status with specific color. The system allows multiple concentrations of a calibrator for multi-point calibration. The calibration factors can be adjusted through a reagent blank test. When you set up the auto calibration conditions, the system will automatically remind you of calibrating chemistries. Expired calibration factors can be used again by extending the calibration time. You are allowed to override a failed calibration and obtain results based on the failed calibration factors.

6.2 Calibration Status and Alarm

On the **Reagent/Calibration** screen, the chemistries are indicated with various texts and colors for different calibration status. Chemistries in Cal Required, Cal Failed or Cal Time Out status can be requested but will not be run.

Check the chemistries' calibration status frequently and take relevant actions according to the following table.

Table 6.1 Calibration status

Calibration Status	Description	Severity	Color
Cal Required	Indicates that the chemistry needs to be calibrated. This status appears when the chemistry is not calibrated or has no calibration factors calculated due to un-monotonic or in-convergent calibration curve.	Serious	Red
Requested	Indicates that the chemistry has been requested for calibration but the test has not begun.	Normal	No color indication
Calibrated	Indicates that the chemistry has been calibrated and has not exceeded the calibration period.	Normal	No color indication
Cal Failed	Indicates that the test has finished but cannot calculate the final result, or the calculated result exceeds the acceptance limits.	Serious	Red
Cal Time Out	Appears when the chemistry exceeds the calibration period or the reagent of different serial number and lot number is used.	Serious	Red
Cal Time Extended	Indicates that the calibration period has been extended and the current calibration factors can be used without time limit.	Warning	Yellow
Calculated	Indicates that the calibration factors of the chemistry have been calculated and can be used without time limit.	Warning	Yellow
Edited	Indicates that the calibration factors of the chemistry have been edited and can be used without time limit.	Warning	Yellow
Cal	Indicates that the test results of the	Warning	Yellow

6 Calibration

Calibration Status	Description	Severity	Color
Overridden	chemistry are based on a failed calibration, and flagged accordingly.		
N/A	Indicates the reagent has no calibration status.	Normal	No color indication

6.3 Calibrator Dilution Setup

6.3.1 Introduction

The system supports diluted calibrator(s) and allows one calibrator to have 9 concentrations for the same chemistry. You are only required to enter the final concentration of the diluted calibrator and the diluted calibrator volume aspirated by the sample probe during calibration. The system will automatically calculate the diluent volume and the sample volume for diluting. When you set up the dilution factors for a chemistry, its original calibrator concentration will be removed.

Diluted calibrator is only applied to biochemical chemistries rather than ISE chemistries.

You are allowed to edit or delete the calibrator dilution factors when the system is not running any tests.

6.3.2 Setting up Calibrator Dilution Factors

- 1 Select **Reagent-Setup**.
- 2 Choose a calibrator in the left list.
- 3 Choose a chemistry in the right list.
- 4 Select **Dilute F5**.

Figure 6.1 Calibrator Dilution Setup window

	Conc	Aspirated Vol	Neat Vol	Diluent Vol
1				
2				
3				
4				
5				
6				
7				
8				
9				

Buttons: Save, Edit, Delete, Close

- 5 Select the unit from the **Unit** dropdown box for the calibrator.

If the selected unit is different from the result unit of the chemistry, the configured parameters will be converted automatically to the values corresponding to the current result unit.

- 6 Enter the final concentration of the diluted calibrator in the **Conc** field.
- 7 Enter the calibrator volume dispensed by the sample probe during calibration in the **Aspirated Vol** field.

The input must be an integer multiple of 0.1 within 1.5µl-45µl. This field is required.

- 8 Enter the calibrator volume used for diluting in the **Neat Vol** field.

The input must be an integer multiple of 0.1 within 2µl-45µl. This field can be left blank.

- 9 Enter the diluent volume used for diluting in the **Diluent Vol** field.

The input must be an integer multiple of 1 within 120µl-350µl. This field can be left blank.



NOTE

If the neat sample volume and diluent volume are defined, ensure that the sum of the two volumes is within 120µl-360µl.

The two volumes must be defined or left blank simultaneously.

- 10 Select **Save**.
- 11 To define more concentrations for the calibrator, repeat step 5 to 9.
- 12 To set up dilution factors for other calibrators, repeat step 5 to 10.
- 13 Select **Close** to exit the window.

6.3.3 Editing Calibrator Dilution Factors

- 1 Select **Reagent-Setup**.
- 2 Choose a calibrator in the left list.
- 3 Choose the chemistries.
- 4 Select **Dilute F5**.

- 5 Choose a concentration line to edit.
- 6 Select **Edit**.

The selected concentration line is editable.

- 7 Change the concentration, sample volume, neat sample volume and diluent volume.

**NOTE**

If the neat sample volume and diluent volume are defined, ensure that the sum of the two volumes is within 120µl-360µl.

The two volumes must be defined or left blank simultaneously.

- 8 Select **Save**.
- 9 Select **Close** to exit the window.

6.3.4 Deleting Calibrator Dilution Factors

- 1 Select **Reagent-Setup**.
- 2 Choose a calibrator in the left list.
- 3 Choose a chemistry in the concentration list.
- 4 Select **Dilute F5**.
- 5 Choose a concentration line to delete.
- 6 Select **Delete**.
- 7 Select **Close** to exit the window.

6.4 Reagent Blank

6.4.1 Introduction

In a reagent blank test, the reagents react with the physiological saline or a calibrator with concentration of 0, and then the blank absorbance is calculated. When a reagent is uncapped for a long period, the reagent absorbance may be changed. At this time, you are allowed to run a reagent blank instead of calibration to calculate the reagent blank absorbance, which will be used to adjust the calibration factors of the reagent in order to ensure reliable sample results.

The reagent blank is allowed only in the following calibration status: Calibrated which means the calibration is successfully performed.

If the reagent blank results, including the mixed blank absorbance and blank response, are within the acceptance range, the system will update the calibration factors and the remaining calibration time based on the results. If the results exceed the acceptant limits, the system will give an alarm and remind you to rerun the reagent blank. The **Biochemistry Calibration** screen shows the calculated reagent blank response, absorbance and run date.

Reagent blank is applied to biochemical chemistries only.

6.4.2 Mixed Blank Absorbance and Response

When defining a chemistry, you need to set up the mixed blank absorbance and blank response to check the reagent blank results.

The mixed blank absorbance indicates the allowable range of the absorbance measured at the end point of a zero-concentration calibrator reaction or a reagent blank reaction. If the absorbance measured at the reaction end point is beyond the set range, the system will flag the test result.

The blank response specifies the allowable range of the response in a zero-concentration calibrator analysis or a reagent blank test. If the response is beyond the set range, the system will flag the test result.

- 1 Select **Utility-Chemistries**.
- 2 Choose a biochemical chemistry, or enter the chemistry name in the **Chemistry Name** field.
- 3 Select **Define F1**.
- 4 Select the down-arrow button to show the error detection parameters setup page.

- 5 Enter the mixed blank absorbance range in the **Mixed Blank Abs** field.

Both the low and high limits must be an integer within -33,000-33,000. The default is -33,000-33,000, and it can be left blank.

- 6 Enter the blank response range in the **Blank Response** field.

Both the low and high limits must be an integer within -33,000-33,000. The default is -33,000-33,000, and it can be left blank.

- 7 Select **Save F7**.

6.4.3 Requesting a Reagent Blank

Please note that reagent blank can only be run in following conditions:

- Chemistries with all calibration math models rather than two-point linear and K factor must have the 0-concentration calibrator set up.
- K factor chemistries must have calibrators set up.
- Chemistries of all calibration math models rather than K factor must have been successfully calibrated.

- 1 Select **Reagent-Reagent/Calibration**,
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 3 Select the up and down arrow buttons to display the biochemical reagent/calibration screen.
- 4 Choose the chemistries.
- 5 Check if the desired chemistries' calibration status is Calibrated, Cal Time Out or Cal Required.
- 6 Select **Cal F5**.
- 7 Choose **Rgt Blk**.
- 8 Select **OK**.
- 9 Select the  icon to start the analysis.

6.4.4 Recalling Reagent Blank Results

If the reagent blank results are within the acceptance limit range, they will be used to update the current calibration parameters. You are allowed to recall the reagent blank

response, absorbance and run date on the **Biochemistry Calibration** screen. Calibration curve of reagent blank cannot be recalled.

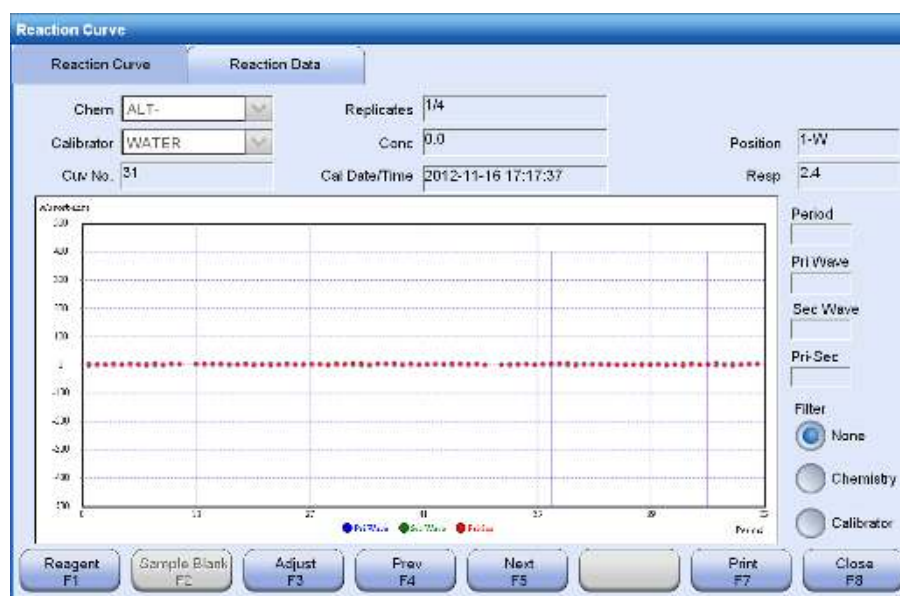
Recalling reagent blank response

- 1 Select **Reagent-Biochemistry Calibration**.
- 2 Choose a chemistry in the **Chem** pull-down list.
- 3 Select **Search F1**.

The calibration results and reagent blank results of the chemistry are displayed in the result list.

- 4 Choose a calibration result.
- 5 Select **Reac Curve F3**.

Figure 6.2 Reagent blank reaction curve



The response value current displayed is the updated reagent blank response.

- 6 Select the reaction data table to view the reagent blank reaction data.

Figure 6.3 Reagent blank reaction data



7 Choose the following buttons as needed:

- **Prev F4:** to view reaction curve and data of the previous calibrator of the chemistry.
- **Next F5:** to view reaction curve and data of the next calibrator of the chemistry.
- **Print F7:** to print the current reaction curve or data.

8 Select **Close F8**.

Recalling reagent blank trends

- 1 Select **Reagent-Biochemistry Calibration**.
- 2 Choose a chemistry in the **Chem** pull-down list.
- 3 Select **Search F1**.

The calibration results and reagent blank results of the chemistry are displayed in the result list.

- 4 Choose a calibration result.
- 5 Select **Trend F6**.

6 Calibration

Figure 6.4 Graphic Trend tab page



- 6 Choose a trend type you want to recall.

The options include:

- R1 blank absorbance
- Mixed blank absorbance
- Calibrator response
- K factor (for linear calibrations only)

- 7 Select the calibration time range.

- 8 Select **Search F1**.

The graphical trend of the selected chemistry within the specific period is displayed.

- 9 Select the **Tabular Trend** tab to view the trend data.

Figure 6.5 Tabular Trend tab page

Calibration Trends

Graphic Trend **Tabular Trend**

Chem: Cal Date: -

Trend Type:

Run Date/Time	Absorbance
2012-11-18 11:21:48	2.21127544523534
2012-11-18 11:21:52	-5.52805660960673
2012-11-18 11:21:57	-0.55287913953861
2012-11-18 11:22:01	-3.59351719562336
2012-11-19 14:29:38	-1.38184922226489
2012-11-19 14:29:41	-0.829373424383081
2012-11-19 14:29:45	1.38191958979448
2012-11-19 14:29:50	-1.65854651046113
2012-11-19 14:29:55	-4.97513214386564
2012-11-19 14:29:59	-1.10567280703619
2012-11-19 14:30:04	-5.63389507704376

Search F1 Read Curve F3 Prev F4 Next F5 Print F7 Close F8

10 Choose the following buttons as needed:

- **Prev F4:** to view the calibration trends and data of the previous chemistry.
- **Next F5:** to view the calibration trends and data of the next chemistry.
- **Print F7:** to print the current graphic trend or data.

11 Select **Close F8** to close the window.

6.5 Auto Calibration

6.5.1 Introduction

Based on the auto calibration conditions, the system can determine chemistries that need to be calibrated and remind you through calibration status and color indication. Auto calibration conditions include:

- Calibration factors' validity period
- Reagent lot changed
- Reagent bottle changed

When the lot number or serial number of R1, R2, R3 or R4 is changed, calibration is required. If no lot number or serial number is set for the reagents, the chemistries will not be calibrated automatically even though the conditions are met. When the calibration time is exceeded, the system will remind you of running calibrations.

6.5.2 Auto Calibration Setup

- 1 Select **Reagent-Setup**.
- 2 Select **Rules F4**.
- 3 Choose a chemistry from the **Chem** pull-down list.

Figure 6.6 Calibration Setup window

Calibrator	Conc	Pos	Lot No.
WATER	0.00	W	

- 4 Choose auto calibration conditions:

- **Bottle changed:** The system will remind you to run calibration when you use a different bottle of reagents.
- **Lot changed:** The system will remind you to run calibration when you use reagents of a different lot. If the reagents have no lot number, they will be considered as the same lot and different from other lots.
- **Calibration time:** The system will remind you in 30 minutes before the calibration is timed out and display the chemistry's calibration status with yellow.

**NOTE**

If the **Manage Reagents by Lot** option on the **System Setup** screen is enabled, **Bottle Changed** and **Lot Changed** will not appear. When a different reagent lot is used, the system will request and run calibration automatically.

- 5 Select **Save F7**.
- 6 Select **Prev F4** or **Next F5** to set up the calibration rules for other chemistries.
- 7 Select **Close F8** to close the window.

6.5.3 Auto Calibration Reminding

When the auto calibration conditions are satisfied, the system will remind you through the calibration status, prompt message and color indication.

- If you choose the **Bottle Changed** option, the system will display the calibration status as Cal Required and shows a message indicating calibration is required when you use a different bottle of reagents.
- If you choose the **Lot Changed** option, the system will display the calibration status as Cal Required and shows a message indicating calibration is required when you use reagents of a different lot.
- If you choose the **Cal Time** option, the system will remind you in 30 minutes before the calibration is timed out and display the chemistry name and calibration status with yellow.

6.5.4 Removing Auto Calibration

To disable the auto calibration, perform the following steps:

- 1 Select **Reagent-Setup**.
- 2 Select **Rules F4**.
- 3 Choose a chemistry from the **Chem** pull-down list.
- 4 Deselect all auto calibration conditions.

6 Calibration

- 5 Select **Save F7**.
- 6 Select **Close F8** to close the window.

6.6 Extending Calibration Time

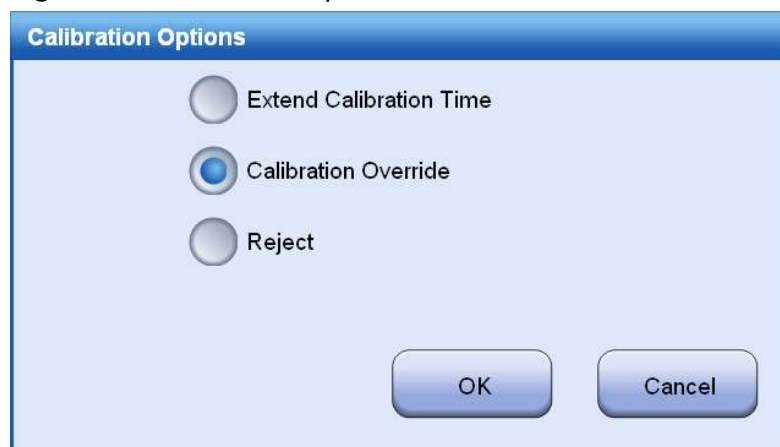
6.6.1 Introduction

Calibration factors that exceed the calibration period cannot be used for result calculation. The calibration status becomes Cal Required and the chemistry can no longer be run. The system will display a warning message in 30 minutes before the calibration is timed out, and you are allowed to recalibrate the chemistry or extend its calibration time. If you are certain that the calibration factors are correct and valid, you may prolong their validity period by using the calibration time extension function. A calibration time can be extended only if the current calibration of the chemistry is timed out. The results calculated based on extended calibration factors will be flagged.

6.6.2 Extending Calibration Time

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**;
- 3 Select the up and down arrow buttons to display the biochemical reagent/calibration screen.
- 4 Choose a chemistry you want to extend.
- 5 Select **Cal Options F8**.
- 6 Select **Extend Calibration Time** from the **Calibration Options** window.

Figure 6.7 Calibration Options window



- 7 Select **OK**. The calibration factors of the selected chemistry can be used without

6 Calibration

time limit.

6.6.3 Removing an Extended Status

Calibration extension is not absolutely definite. Recalibrate the chemistry to remove the extended status.

6.7 Calibration Override

6.7.1 Introduction

The Calibration Override option allows the system to override a failed calibration and calculate results based on the failed calibration factors. Calibration override is only applied to failed calibrations. Results that are obtained based on failed calibration factors will be flagged.



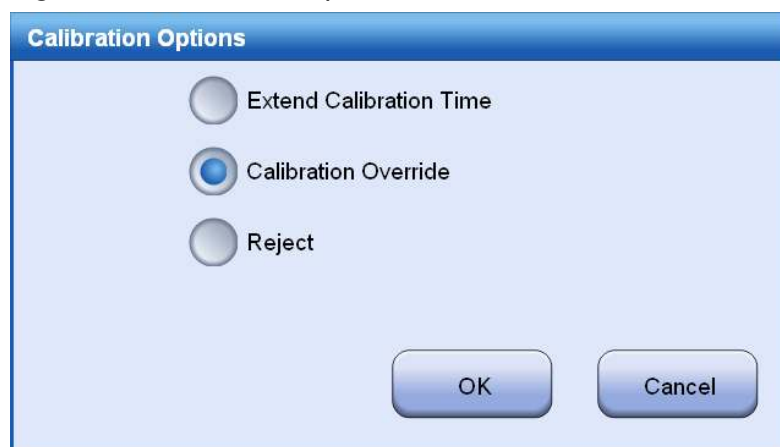
CAUTION

Before overriding a calibration, make sure that the calibration factors are within the acceptance limits of your laboratory. The magnitude of the error should be totally under the control of your laboratory. Use of overridden calibration factors may lead to unreliable results and influence the doctor's diagnosis. Think twice before overriding a failed calibration.

6.7.2 Overriding a Calibration

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Choose a chemistry you want to override.
- 3 Select **Cal Options F8**.
- 4 Select **Calibration Override** from the **Calibration Options** window.

Figure 6.8 Calibration Options window



- 5 Select **OK**. The failed calibration factors of the selected chemistry can be used for result calculation.

6.7.3 Removing Cal Overridden Status

Recalibrate the chemistry to remove its Cal Overridden status.

6.8 Reject

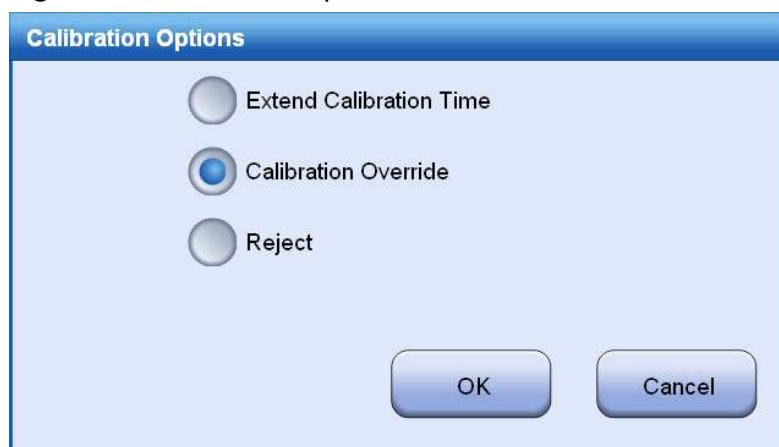
6.8.1 Introduction

If the current calibration fails but sample analysis needs to be performed immediately, you may use the Reject function to reject the current calibration factors, and use the latest valid ones for calculating sample results. Calibration factors of status other than Requested and Cal Required can be rejected. Rejected calibration factors cannot be rejected again.

6.8.2 Rejecting a Calibration

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Choose a chemistry you want to reject.
- 3 Select **Cal Options F8**.
- 4 Select **Reject** from the **Calibration Options** window.

Figure 6.9 Calibration Options window



- 5 Select **OK**. Calibration factors of the selected chemistry are rejected.

6.8.3 Removing Reject Status

Recalibrate the chemistry to remove its Reject status.

6.9 Recalling Calibration Results

On the **Biochemistry Calibration** screen you are enabled to recall the current and stored calibration factors of a chemistry. If the **Manage Reagents by Lot** option on the **System Setup** screen is enabled, you can recall calibration results of each reagent lot. The **Current** calibration factors are obtained in the recent calibration and are being used for result calculation. You are allowed to recall the calibration curve, calibration reaction curve and calibration trends during the specified period, and edit or recalculate the calibration factors.

6.9.1 Recalling Current Calibration Factors

- 1 Select **Reagent-Biochemistry Calibration**.

The screen shows all the calibrations requested on the day, including the following information:

- Chemistry name
- Lot number
- Result flag
- Calibration status
- R_0 : reagent blank response
- K: K factor
- A, B, C and D: factors a, b, c and d in nonlinear calibration equations
- Run date and time

- 2 Choose a chemistry from the **Chem** pull-down list.

- 3 Select **Search F1**.

The current calibration factors of the chemistry are displayed in the result list.

- 4 To print the calibration report, select **Print F7**.

6.9.2 Recalling History Calibration Factors

- 1 Select **Reagent-Biochemistry Calibration**.
- 2 Choose the **History** option button.
- 3 Choose a chemistry from the **Chem** pull-down list.
- 4 Select the date range in the **Cal Date** field.

5 Select **Search F1**.

The calibration factors used within the specified period are displayed on the screen.

6 Choose the following buttons as needed:

- **Cal Curve F2**
- **Reac Curve F3**
- **Edit F4**
- **Archive F5**
- **Trend F6**
- **Print F7**

6.9.3 Calibration Curve

A calibration curve reflects the mathematical relation between calibrator concentration and response. It is drawn based on the obtained response and the multiple values between the minimum and maximum concentrations of the calibrator. The calibration curve is a straight line in linear calibrations and a curve in nonlinear calibrations.

- 1** Search for desired calibration results on the **Biochemistry Calibration** screen.
- 2** Choose a chemistry in the result list.
- 3** Select **Cal Curve F2**. The **Calibration Curve** window is displayed.

Figure 6.10 Calibration Curve window



- 4 Choose the following buttons as needed:
 - **Prev F4:** to view the calibration curve of the previous chemistry.
 - **Next F5:** to view the calibration curve of the next chemistry.
 - **Recalculate F6:** to recalculate the calibration factors based on the specified math model.
 - **Print F7:** to print the current calibration curve.
- 5 Select **Close F8** to close the window.

Recalculating calibration factors

When a calibration is finished, the system allows recalculating of the calibration factors based on the new math model. A flag indicating that the calibration result is recalculated will appear on the **Biochemistry Calibration** screen.

Recalculating calibration factors is not applicable to K factor calibrations. Calibration factors that have been recalculated cannot be calculated again.

- 1 Select **Reagent-Biochemistry Calibration**.
- 2 Search for desired calibration results to recalculate.
- 3 Choose a chemistry in the result list.
- 4 Select **Cal Curve F2**.
- 5 Select **Recalculate F6**. The **Recalculate** window shows.

Figure 6.11 Recalculate window

Chem: AST Math Model: Two-Point Linear $C = R * (S - R_0)$

Calibrator	Conc	Resp
CalS	129.0000	475.9038
CalS	129.0000	479.4022
CalS	129.0000	481.3670
WATER	0.0000	1.4633
WATER	0.0000	1.8714
WATER	0.0000	0.3259

Para	Value	Para	Value
R0	1.4633	K	0.2698

Buttons: Recalc Curve F1, Discard F6, Save F7, Close F8

- 6 Choose a math model from the **Math Model** pull-down list.

The corresponding calculation formula is displayed in the text box to the right of the **Math Model** field.

- 7 Choose calibrators to recalculate in the left list. Move the scroll bar to view more calibrators.

Choose the correct number of calibrators corresponding to the math model. For more information, refer to 3.3.6 Setting up Calibration Rules (page 3-37).

- 8 Select **Save F7**.

The system will recalculate the calibration factors with the selected math model and calibrators.

- If the recalculation is succeeded, the new calibration factors will be displayed on the **Biochemistry Calibration** screen with the calibration status shown as Recalculated, and “CALR” will appear in the corresponding **Flag** column.
- If the recalculation is failed, the system will show a message box indicating the old calibration factors will remain to be used.

- 9 To view the reaction curve of the selected calibrator, select **Reac Curve F1**.

- 10 Select **Close F8** to close the window.

6.9.4 Calibration Reaction Curve

A calibration reaction curve reflects the relationship of the absorbance measured at the primary wavelength, secondary wavelength and primary-secondary wavelength. It is drawn based on the absorbance of the calibrator-reagent mixture measured within the reaction period.

Observing reaction curve:

- 1 Search for desired calibration results on the **Biochemistry Calibration** screen.
- 2 Choose a chemistry in the result list.
- 3 Select **Reac Curve F3**. The **Reaction Curve** window is displayed.

Figure 6.12 Reaction Curve window



- 4 Select a point on the curve. Relevant measuring period and absorbance are displayed on the right of the window.
- 5 Select a filter condition from the following options:
 - None: observe reaction curve and data in the default mode.
 - Chemistry: observe reaction curve of the results for the selected test.
 - Calibrator: observe reaction curve of the results for the selected calibrator.
- 6 Choose the **Reaction Data** tab to view the reaction data.

Figure 6.13 Reaction Data tab page

Reaction Curve

Reaction Data

Chem: ALT- Replicates: 14

Calibrator: Cal3 Conc: 114.0

Cuv No.: 45 Cal Date/Time: 2012-10-15 11:34:46

Position: 1-52 Resp: 239.7

Period	Pri Wave	Sec Wave	Pri-Sec	Period	Pri Wave	Sec Wave	Pri-Sec
1	17971.27	78.47	17892.80	2	17964.81	74.40	17890.41
3	17914.92	69.70	17846.21	4	17902.09	66.24	17835.84
5	17886.07	61.58	17804.49	6	17876.48	58.91	17819.57
7	17879.68	57.72	17821.96	8	17871.09	53.86	17817.83
9	17868.50	54.47	17814.02	10	17855.73	48.38	17807.35
11	17847.77	53.65	17794.11	12	17841.40	50.21	17791.19
14	16803.16	2390.02	16525.14	15	16866.78	2363.31	16525.47
16	16878.02	2348.22	16525.80	17	16869.06	2343.46	16525.60
18	16863.68	2338.96	16524.72	19	16867.26	2335.53	16531.73
20	16861.16	2331.04	16520.12	21	16861.16	2326.03	16525.13

Reagent F1 Sample Blank F2 Adjust F3 Prev F4 Next F5 Print F7 Close F8

7 Choose the following buttons as needed:

- **Reagent F1:** to view the calibrators and reagents used in calibration, and reagents for reagent blank test.
- **Sample Blank F2:** to view the sample blank reaction curve and reaction data of the calibrator.
- **Adjust F3:** to adjust the absorbance display range of current reaction curve. Refer to 8.12.8 Reaction Curve (Page 8-49) for details.
- **Prev F4:** to view reaction curve and data of the previous calibrator of the chemistry.
- **Next F5:** to view reaction curve and data of the next calibrator of the chemistry.
- **Print F7:** to print the current reaction curve or data.

8 Select **Close F8** to close the window.

Viewing reagent information:

On the calibration reaction curve window, you are allowed to view the calibrators and reagents used in calibration, and reagents for reagent blank test.

- 1 Search for desired calibration results on the **Biochemistry Calibration** screen.

- 2 Choose a chemistry in the result list.
- 3 Select **Reac Curve F3**. The **Reaction Curve** window is displayed.
- 4 Select **Reagent F1**.

Figure 6.14 Reagent window

The screenshot shows a software window titled "Reagent". At the top, there is a "Chem" dropdown menu set to "ALT-" and a "Cal Date/Time" text field showing "2012-10-15 11:34:46". Below this, there are two main sections. The first section is a table with columns "Calibrator" and "Lot No.". It contains one row with "Cal3" in the "Calibrator" column. The second section is a table with columns "Reagent", "Lot No.", and "Serial No.". It contains two rows with "R1" and "R2" in the "Reagent" column. Both tables have vertical scroll bars on their right sides. At the bottom right of the window is a "Close" button.

The window shows the calibrators and reagents used in calibration, and reagents for reagent blank test.

- 5 Select **Close** to exit the window.

6.9.5 Editing Calibration Factors

If the calibration factors of linear calibration are higher or lower than the expected values or than those obtained on other instruments, you are allowed to edit them to keep them consistent with the expected ones or those on other instruments. A flag will appear for results calculated based on edited calibration factors, and the calibration curve and reaction curve of edited calibration factors cannot be recalled.

Prior to editing calibration factors, ensure that you have sufficient permissions and the system status is not Running.

- 1 Select **Reagent-Biochemistry Calibration**.
- 2 Search for desired calibration results to edit.
- 3 Choose a desired chemistry.

- 4 Select **Edit F4**. The **Edit** window shows.

Figure 6.15 Edit window

- 5 Type in slope K and offset R_0 .

- 6 Select **Save**.

The system will refresh the calibration results and curves with the input slope and offset, and take the edited calibration factors as the defaults.

- 7 Select **Close** to exit the window.

6.9.6 Archiving Calibration Results

The system allows you to archive all searched calibration results to a designated hard disk directory or to a storage device, such as U disk. Archived calibration results are displayed in the same format as on the software screen. The archived content includes: chemistry name, flag, calibration status, R_0 , K factor, calibration coefficients A/B/C/D, and calibration time. The archiving file is of .csv format and named by date and time.

- 1 Select **Reagent-Biochemistry Calibration**.
- 2 Search for desired calibration results.
- 3 Select **Archive F5**.
- 4 Specify archiving path and file name.
- 5 Select **Save**.

6.9.7 Calibration Trends

Calibration graphical trends summarize a chemistry's calibrations during a period of time and reflect the trends of the calibrations. The calibration graphical trends show the chemistry's R1 blank absorbance, mixed blank absorbance and calibrator response.

R1 blank absorbance and mixed blank absorbance are available only for chemistries with 0-concentration calibrators. The K factor trends can be recalled for linear chemistries.

The following results can be recalled in a trend:

- R1 blank absorbance: displays calculated and valid calibration results with the calibration status of Calibrated, Cal Failed, Cal Time Out, Cal Time Extended and Cal Overridden.
- Mixed blank absorbance: displays calculated and valid calibration results with the calibration status of Calibrated, Cal Failed, Cal Time Out and Cal Time Extended.
- Calibrator response: displays calculated and valid calibration results with the calibration status of Calibrated, Cal Failed, Cal Time Out and Cal Time Extended.
- K factor: displays calculated and valid calibration results with the calibration status of Calibrated, Cal Failed, Cal Time Out and Cal Time Extended.

Follow this procedure to observe calibration trends:

- 1 Search for desired calibration results on the **Biochemistry Calibration** screen.
- 2 Choose a chemistry in the result list.
- 3 Select **Trend F6**. The **Calibration Trends** window is displayed.

Figure 6.16 Calibration Trends window



- 4 Choose a trend type you want to recall.

The options include:

- R1 blank absorbance
- Mixed blank absorbance
- Calibrator response
- K factor (for linear calibrations only)

- 5 Select the date range in the **Cal Date** field.

- 6 Select **Search F1**.

The trend within the specified period is displayed on the screen.

- 7 Choose the **Tabular Trend** tab to view the trend data.

Figure 6.17 Tabular Trend window

Calibration Trends

Graphic Trend **Tabular Trend**

Chem: CREA-J Cal Date: 2012- 3-13 2012-12-14

Trend Type: R1 Blank Abs Lot No.:

Run Date/Time	Absorbance
2012-06-28 10:19:57	-1.17
2012-06-28 10:20:05	-4.64
2012-06-28 10:20:15	1.01
2012-06-28 10:20:24	-1.23
2012-07-03 14:38:48	-1.01
2012-07-03 14:38:57	-0.87
2012-07-03 14:39:06	0.64
2012-07-03 14:39:15	-3.92

Search F1 Reac Curve F3 Prev F4 Next F5 Print F7 Close F8

- 8 Choose the following buttons as needed:
- **Reac Curve F3:** to view the reaction curve and data of the current calibration test.
 - **Prev F4:** to view the calibration trends and data of the previous chemistry.
 - **Next F5:** to view the calibration trends and data of the next chemistry.
 - **Print F7:** to print the current graphic trend or data.
- 9 Select **Close F8** to close the window.

7 Quality Control

This chapter describes applications of quality control, which include:

- Daily and monthly QC procedure
- QC alarm indications
- QC result flags
- Control status
- QC evaluation
- Auto QC
- Control results recall

7.1 Overview

7.1.1 Introduction

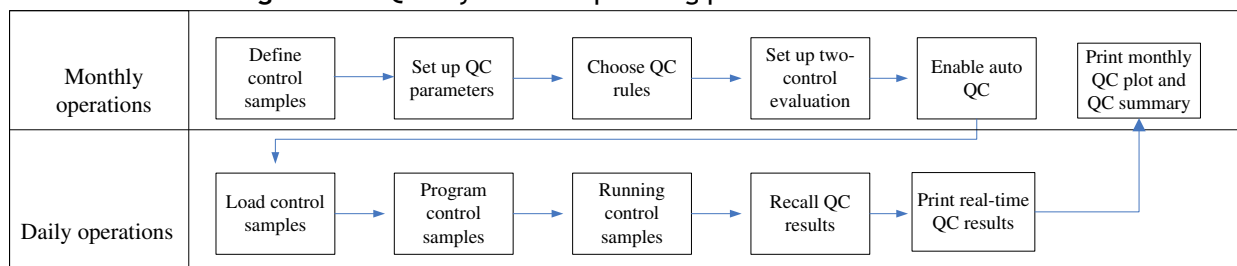
A QC run may require more than one control samples. You are recommended to use two control samples, one with normal values (within the reference range) and the other with abnormal values (beyond the reference range).

To ensure the system performance, run control samples every time after you perform a calibration, or change the reagent lot, or maintain and troubleshoot the instrument.

7.1.2 Quality Control Operating Procedure

After you define a control, chemistry and QC rules, there is no need to edit them frequently, and you are only required to run control samples every day to make sure that the system works well. Run control samples according to the following procedure:

Figure 7.1 Quality control operating procedure



7.1.3 QC Alarms

The system provides the real-time monitoring of quality controls, and check if the QC results are under control when a QC run is finished. If the results exceed the reference range, the system will give an audible alarm and shows an alarm message indicating the chemistry name, control name and control rules. For instance, “Chemistry ALT control C1 1-3s out of control!”. In this situation, you should stop the analysis and find the causes of the failure, and resume the analysis after solving the problem.

For QC alarms and corrective actions, refer to 17.5 Error Messages and Corrective Actions (page 17-29).

7.1.4 QC Result Flags

When a QC result is failed, the system will give an audible alarm and show alarm message to remind you of the failure. Moreover, the following flags will appear for failed results in the **Flag** column of the QC reports.

- 1-3s
- 2-2s
- R-4s
- 2-2s
- 4-1s
- 10-x

The system checks the failed QC results for system error or random error and then flag them accordingly. A “#” sign indicates a systematic error, and an asterisk “*” indicates a random error. For more information about QC result flags, refer to 17.4 Data Alarm (page 17-14).

7.1.5 Control Status

When you choose a control on the **Quality Control** screen, the current status of the control is displayed in the **Sample Status** field. It is necessary to understand the control statuses. The table below shows the various statuses of control samples.

Table 7.1 Descriptions of control status

Control Status	Description
N/A	Indicates that the control is not programmed for analysis.
Requested	Indicates that the control sample has been programmed but not analyzed yet.
In Progress	Indicates that the control sample is being analyzed.
Incomplete	Indicates that all chemistries of the control sample have been finished but one or more of them have no results.
Complete	Indicates that all chemistries of the control sample have been finished with results.

7.2 QC Evaluation

7.2.1 Introduction

The system provides the Westgard rules for evaluating QC results of the chemistries, and give alarms and flags when the obtained QC results are beyond the reference range. Since every chemistry may have one or more control samples, the QC results can be evaluated with different rules accordingly. Those controls that are not included in any lots will be evaluated as single controls.

7.2.2 Evaluation of Single Controls

The Westgard rules for evaluation of single controls are listed in the table below:

Table 7.2 Westgard rules for single controls

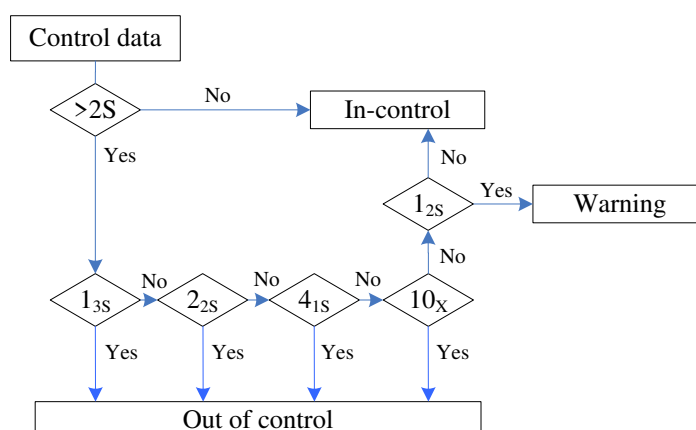
Rules	Description	Flag	Error Type
12s	One result is between ± 2 and ± 3 standard deviations from the assigned mean concentration.	N/A	N/A
13s	One result is greater than ± 3 standard deviations from the assigned mean concentration.	1-3s	*(1)
22s	Two continuous results are greater than +2 or -2 standard deviations from the assigned mean concentration, e.g. (X_n , X_{n-1})	2-2s	#(2)
41s	Four continuous results are greater than +1 or -1 standard deviation from the assigned mean concentration, e.g. (X_n , X_{n-1} , X_{n-2} , X_{n-3})	4-1s	#
10x	Ten results being compared are on the same side, e.g. (X_n , X_{n-1} , X_{n-2} , X_{n-3} .. X_{n-9})	10-x	#

(1) An asterisk “*” indicates a random error, which requires no special action but must not be ignored.

(2) A “#” symbol indicates a systematic error, which requires special consideration.

The evaluation procedure of single controls is shown in the figure below:

Figure 7.2 Evaluation procedure of single controls



7.2.3 Two-Control Evaluation

What is a run

A QC run is based on two control samples: C1 and C2, and at most one QC run is performed for each chemistry. The system allows the definition of QC run interval on the **System Setup** screen. The maximum QC run interval is 24 hours.

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Choose **9 QC Evaluation**.
- 4 Type in the QC run length in the **Run Length** field.
Enter an integer between 1 and 24. The default is 24.
- 5 Select **OK**.

Two-control evaluation rules

In every QC run, two results are obtained: X_n and Y_n , which are used to define a point on the Twin-plot chart. In this way, a complete twin-plot chart is drawn based on all the QC results and used for detecting systematic errors and random errors.

The Westgard rules for two-control evaluation are listed in the table below:

Table 7.3 Two-control evaluation rules

Rules	Description	Flag	Error Type
12s	One result is between ± 2 and ± 3 standard deviations from the assigned mean concentration.	N/A	N/A
13s	One result is greater than ± 3 standard deviations from the assigned mean concentration.	1-3s	*(1)
22SA	Two results (X_n , Y_n) of a run are simultaneously greater than +2 or -2 standard deviations from the assigned mean.	2-2s	#(2)
R4s	One result of a run is greater than +2 standard deviations from the assigned mean and the other greater than -2SDs.	R-4s	*
22SW	Two continuous results of a control are greater than +2 or -2 standard deviations from the assigned mean concentration, e.g. (X_n , X_{n-1}), (Y_n , Y_{n-1}).	2-2s	#
41SA	Results of two continuous runs are greater than +1 or -1 standard deviation from the assigned mean, e.g. (X_n , Y_n , X_{n-1} , Y_{n-1}).	4-1s	#
41SW	Four continuous results of a control are greater than +1 or -1 standard deviations from the assigned mean concentration, e.g. (X_n , X_{n-1} , X_{n-2} , X_{n-3}), (Y_n , Y_{n-1} , Y_{n-2} , Y_{n-3}).	4-1s	#
10XA	Results of five continuous runs (10 results) compared are on the same side, e.g. (X_n , Y_n , X_{n-1} , Y_{n-1} , X_{n-2} , Y_{n-2} , X_{n-3} , Y_{n-3} , X_{n-4} , Y_{n-4}).	10-x	#
10XW	Ten continuous results (10 results) of a control are on the same side, e.g. (X_n , X_{n-1} , X_{n-2} , X_{n-3} .. X_{n-9}), (Y_n , Y_{n-1} , Y_{n-2} , Y_{n-3} .. Y_{n-9}).	10-x	#

(1) An asterisk “*” indicates a random error, which requires no special action but must not be ignored.

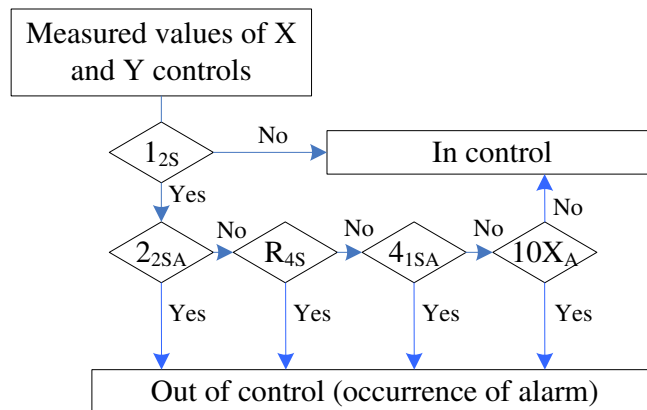
(2) A “#” symbol indicates a systematic error, which requires special consideration.

The random errors in two-control evaluation correspond to those in single-control evaluation as follows:

- $2_{2SA} \setminus 2_{2SW}$ corresponding to 2_{2s} .
- $4_{1SA} \setminus 4_{1SW}$ corresponding to 4_{1s} .
- $10_{XA} \setminus 10_{XW}$ corresponding to 10_x .

The procedure of two-control evaluation is shown in the figure below:

Figure 7.3 Two-control evaluation workflow



7.3 Auto Quality Control

7.3.1 Introduction

The system provides the auto quality control function. When the conditions for a QC run are satisfied, the system will request and run the specified controls automatically. The control samples automatically run can be selected on the **QC Parameters** window.

The conditions for auto quality control include:

- Number of samples: indicates the number of patient samples. After the given number of samples is finished, the system will run the selected control(s) automatically.
- When calibrated: The system will automatically run the chemistry for the selected control(s) every time when the chemistry is calibrated. Auto QC is not applicable to non-measurement calibrations, such as recalculation and editing.

When the control samples automatically run are selected, all chemistries configured for the control samples will be run.

7.3.2 Auto QC Setup

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Choose **9 QC Evaluation**.

Figure 7.4 QC Parameters window

Control	Lot No.
Mg ctrl	
TP control	
Normal	
Abnormal	
QC1	

- 4 Select **Auto QC on carousel**.
- 5 Set up the conditions for auto quality control:
 - **Number of Samples:** enter the number of samples for auto QC run. The input range is 10-500, 0 means auto QC is disabled.
 - **When Calibrated:** select the checkbox to allow the system to run controls when a chemistry is calibrated.
- 6 Choose controls to be run automatically.

One or more controls can be selected.

- 7 Select **OK**.

7.3.3 Auto Quality Control

After setting up auto QC conditions and selecting the **Auto QC** checkbox on the **Start Conditions** window, the system will insert QC runs automatically once the conditions are met.

- 1 After setting up auto QC, place controls on the sample carousel.
- 2 Select  on upper right corner of the main screen. The **Start Conditions** window is displayed.

7 Quality Control

- 3 Select **OK**. The system will insert a QC run in the current test queue.

7.3.4 Removing Auto QC Status

To remove an auto QC status, clear the auto QC settings on the **QC Parameters** window.

7.4 Recalling Control Results

The Recalling Control Results option allows you to view control sample results, L-J chart, twin-plot chart, analysis data and data summary.

Patient demographics and rerunning are not applicable to controls.

7.4.1 Control Sample Results

- 1 Select **Result-Current Results** or **History Results**.
 - The **Current Results** screen displays all incomplete patient samples and control samples, as well as those programmed on the current day.
 - The **History Results** screen displays all patient samples and control samples programmed before the current day.
- 2 Choose a result recall mode:
 - By sample
 - By chemistry
- 3 When recalling results by sample, choose a control in the left list. The right list displays all results of the control. When recalling results by chemistry, choose a chemistry in the left list. The right list displays all results of the chemistry.
- 4 Choose the following buttons as needed:
 - **Search F1**: to recall control results.
 - **Options F2**: to delete, edit, rerun or print control samples.
 - **Reac Curve F4**: to view the reaction curve of the selected control.
 - **Print F7**: to print control results.
 - **Host F8**: to transmit the selected control results to the LIS host.

Viewing control reaction curve

- 1 Search for desired control results on the **Current Results** or **History Results** screen.
- 2 Choose a chemistry in the result list.
- 3 Select **Reac Curve F4**. The **Reaction Curve** window is displayed.

Figure 7.5 Reaction Curve screen



4 Choose the **Reaction Data** tab to view the reaction data.

Figure 7.6 Reaction Data screen



5 Choose the following buttons as needed:

- **Sample Blank F2:** to view the sample blank reaction curve and reaction data of the selected control.

- **Prev F4:** to view the reaction curve and data of the previous chemistry.
- **Next F5:** to view the reaction curve and data of the next chemistry.
- **Print F7:** to print the current reaction curve or data.

6 Select **Close F8** to close the window.

Printing control results

You are allowed to print the selected or all control results on the **Current Results** or **History Results** screen.

- 1 Search for desired control results on the **Current Results** or **History Results** screen.
- 2 To print the selected controls, select them in the sample list.
- 3 Select **Print F7**.
- 4 Select **Print Sample Report**.
- 5 Choose the print range:
 - Selected Sample(s)
 - All Sample(s)
- 6 If you print all samples, you are allowed to skip those that are already printed out. Mark the **Bypass Printed Sample(s)** checkbox.
- 7 Select **OK**.

7.4.2 Recalling L-J Chart

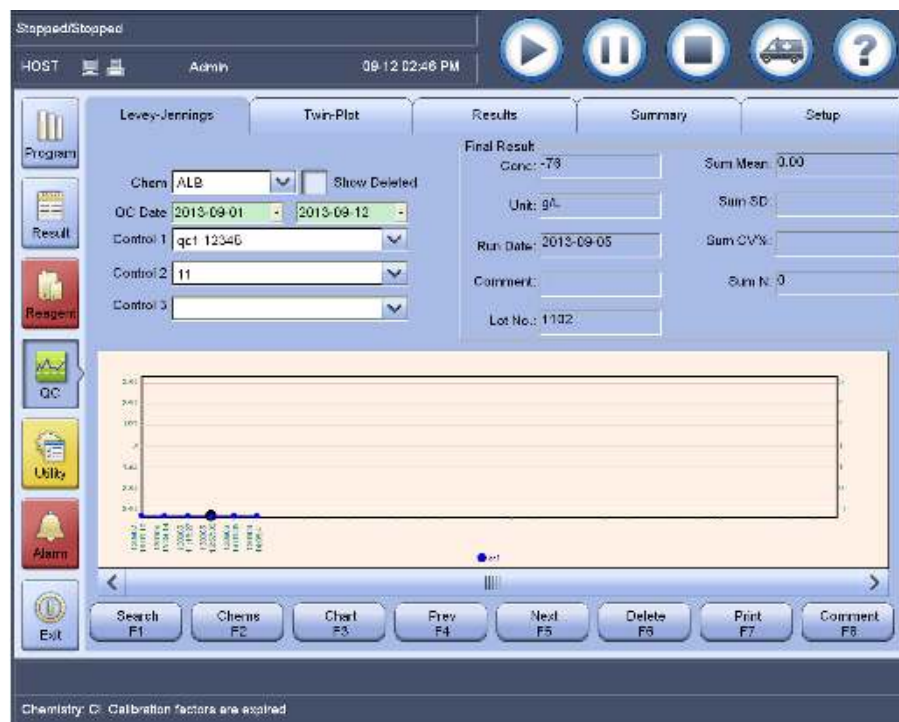
A Levey-Jennings (L-J) chart, drawn based on the QC date (X) and test results (Y), shows the QC result trend of a chemistry during the specified period. The graphical trends of up to 3 controls can be displayed on one L-J chart and distinguished with different colors. Each page can display 31 QC points. The query date must not be longer than 1 year.

Recalling L-J chart

- 1 Select **QC-Levey-Jennings**.
- 2 Choose a chemistry to recall in the **Chem** pull-down list box, or select **Chems F2** and then choose a chemistry.
- 3 Select the date range in the **QC Date** field.

- 4 Choose controls you desire to view. Up to 3 controls can be selected.
- 5 Select **Search F1**. The L-J chart area shows the QC result trends of the selected chemistry during the specified period.

Figure 7.7 Levey-Jennings screen



- 6 Choose the following buttons as needed:
 - **Prev F4**: to view the L-J chart of the previous chemistry.
 - **Next F5**: to view the L-J chart of the next chemistry.
 - **Delete F6**: to delete the selected point on the L-J chart. If you want to display the removed points on the L-J chart, mark the **Show Deleted Points** checkbox.
 - **Print F7**: to print the current L-J chart.
 - **Comment F8**: to add, modify and delete comments of a QC point.

Adding/Modifying comments

- 1 Select **QC-Levey-Jennings**.
- 2 Select a chemistry, QC date and controls, and then select **Search F1** to query the corresponding L-J chart.

- 3 Choose a QC point on the chart.
- 4 Select **Comment F8**, and then input comments for the QC point.
- 5 Select **OK**.

Select the QC point on the chart. The comments of this QC point are displayed in the **Comment** area at the upper-right corner of the screen.

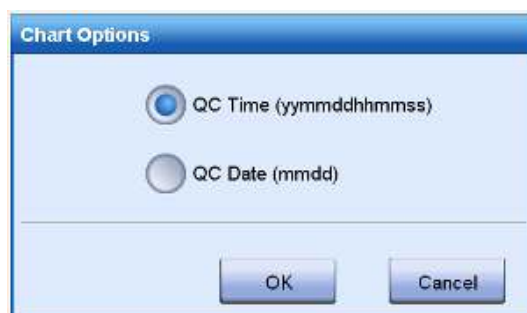
To delete the comments of a QC point, select the QC point on the chart, clear the comments, and then select **OK**.

Selecting chart option

The L-J chart can be drawn by QC date or QC time. Either of the two options can be selected to display the L-J chart. The default standard is QC time.

- 1 Select **QC-Levey-Jennings**.
- 2 Select **Chart F3**.

Figure 7.8 Chart Options window



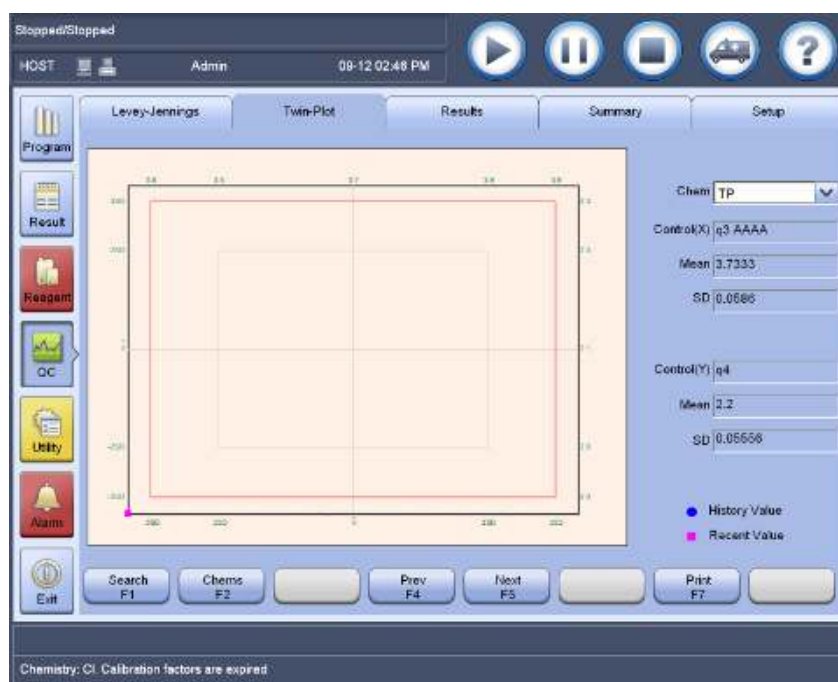
- 3 Choose an option to draw the L-J chart:
 - QC Time: The X coordinate of the L-J chart is displayed in the format of “YYMMDDHHMMSS”.
 - QC Date: The X coordinate of the L-J chart is displayed in the format of “MMDD”.
- 4 Select **OK**. The L-J chart is refreshed automatically and displayed in the selected format.

7.4.3 Recalling Twin-Plot Chart

A twin-plot chart, drawn based on the results of control X and control Y in the same run, is used to detect systematic errors and random errors. It shows the recent 10 QC results of a chemistry and excludes those that have been deleted.

- 1 Select **QC-Twin-Plot**.
- 2 Choose a chemistry to recall in the **Chem** pull-down list box, or select **Chems F2** and then choose a chemistry.
- 3 Select **Search F1**. The twin-plot chart area displays the recent 10 results of control X and control Y for the chemistry.

Figure 7.9 Twin-Plot screen



- 4 Choose the following buttons as needed:
 - **Prev F4**: to view the twin-plot chart of the previous chemistry.
 - **Next F5**: to view the twin-plot chart of the next chemistry.
 - **Print F7**: to print the current twin-plot chart.

7.4.4 Recalling QC Data

QC data includes QC results, and the set mean and standard deviation, and can be recalled based on control name, chemistry name and run date.

- 1 Select **QC-Results**.
- 2 Select **Chems F2**.
- 3 Choose a chemistry to recall, and then select **OK**.

- 4 Select the date range in the **QC Date** field.
- 5 Choose a control in the **Control** pull-down list.
- 6 Select **Search F1**.

The result list shows all results of the control for the chemistry during the specified period, as well as the set means and standard deviations.

Figure 7.10 Results screen



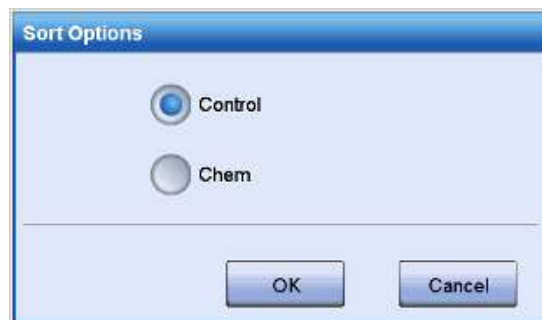
- 7 Choose the following buttons as needed:
 - **Sort F3**: to sequence the QC results by control or chemistry.
 - **Reac Curve F4**: to view the reaction curve and data of the selected QC result.
 - **Comment F5**: to add comments to the selected QC result.
 - **Archive F6**: to archive the currently displayed QC results to an external storage device.
 - **Print F7**: to print the QC results currently displayed in the result list.

Sort QC results

The searched QC results can be rearranged by control or chemistry.

- 1 Search for desired QC results on the **Results** screen.
- 2 Select **Sort F3**.

Figure 7.11 Sort window



- 3 Select a sorting criterion.
 - Control: control number + chemistry + run date/time
 - Chemistry: chemistry order + control + run date/time
- 4 Select **OK**.

The QC results on the **Results** screen are rearranged ascending based on the selected criterion.

Viewing control reaction curve

- 1 Search for desired QC results on the **Results** screen.
- 2 Choose a QC result to recall.
- 3 Select **Reac Curve F4**. The **Reaction Curve** window is displayed.

Figure 7.12 Control reaction curve



- 4 Select a point on the curve. Relevant measuring period and absorbance are displayed on the right of the window.
- 5 Select a filter condition from the following options:
 - None: observe reaction curve and data in the default mode.
 - Chemistry: observe reaction curve of the results for the selected test.
 - Control: observe reaction curve of the results for the selected control.
- 6 Select the **Reaction Data** tab to view the reaction data.

Figure 7.13 Control reaction data



7 Choose the following buttons as needed:

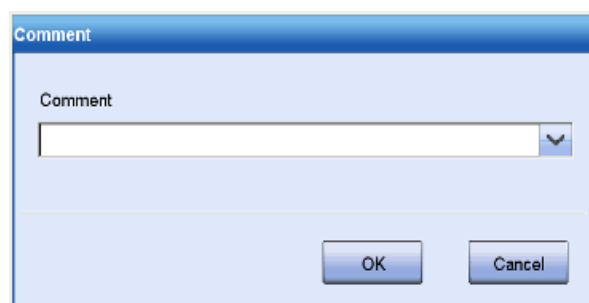
- **Reagent F1:** to view the reagents used for quality control, calibrators and reagents used in calibration, and reagents for reagent blank test. Refer to 8.12.8 Reaction Curve (Page 8-49) for details.
- **Sample Blank F2:** to view the sample blank reaction curve and reaction data of the selected control.
- **Adjust F3:** to adjust the absorbance display range of current reaction curve. Refer to 8.12.8 Reaction Curve (Page 8-49) for details.
- **Prev F5:** to view the reaction curve and data of the previous control.
- **Next F6:** to view the reaction curve and data of the next control.
- **Print F7:** to print the current reaction curve or data.

8 Select **Close F8** to close the window.

Add QC comments

Comments can be added to specific QC result for special notice.

- 1 Search for desired QC results on the **Results** screen.
- 2 Choose a QC result in the result list.
- 3 Select **Comment F5**.

Figure 7.14 Comment window

- 4 Type in comments for the selected QC result.

Up to 100 characters can be entered.

- 5 Select **OK**.

Archive QC data

The system allows archiving of QC results to a storage device. The file format is CSV and the default file name is QCData.csv, which cannot be edited. QC results must not be archived to the hard disk.

The archived QC results and data include the following information:

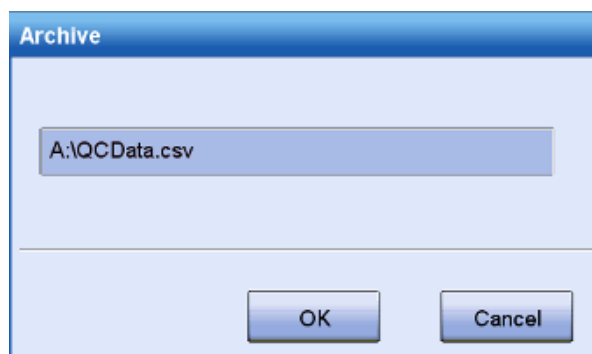
Table 7.4 Archived QC data types

ID	Control Result Information
1	Chemistry type
2	Chemistry number
3	Chemistry name
4	Control number
5	Control name
6	Lot number
7	Control concentration
8	Standard deviation
9	Measured results
10	Flag
11	Run date

Perform the following steps to archive QC results and data:

- 1 Search for desired QC results on the **Results** screen.
- 2 Select **Archive F6**.

Figure 7.15 Archive window



- 3 Select **OK**.

7.4.5 Recalling QC Summary

The QC summary reports the measurements of a control for the selected chemistry during the specified period. It presents you the means, standard deviations and coefficients of variation in this period, and compares them with the set mean and SD, enabling you to check if the system is working normally.

- 1 Select **QC-Summary**.
- 2 Select **Chems F2**.
- 3 Choose a chemistry to recall, and then select **OK**.
- 4 Select the date range in the **QC Date** field.
- 5 Choose a control in the **Control** pull-down list.
- 6 Select **Search F1**.

The result summary of the control for the chemistry is displayed on the screen.

Figure 7.16 Summary screen

Stopped/Stopped

HOST Admin 11-12-17:16

Levey-Jennings Twin-Plot Results Summary Setup

Control ALL QC Date 2012-3-1 2012-12-12

Control	Lot No.	Chem	Sample Type	Mean	SD	Unit	Summary Mean	Summary SD	Summary CV%	Summary Number
Normal		TP-		73.3	0.36	mg/dL	74.1073			1
Abnormal		TP-		65.6	0.3	mg/dL	62.034			1

Program Result Reagent QC Utility Alarm Exit

Search F1 Chems F2 Print F7

Select a control

Chemistry: Cl. Calibration factors are expired

7 To print the QC summary report, select **Print F7**.

8

Sample Programming and Processing

This chapter provides description of functions and operating instructions about sample analysis, which include:

- Modifying/Adding samples and chemistries
- Rerunning samples
- Programming samples with increased or decreased volume
- Programming diluted samples
- Sample blank
- Loading/Unloading samples
- Viewing unpositioned samples and assign positions for them
- Releasing finished samples
- Viewing sample logs
- Customizing sample information
- Viewing sample and chemistry lists
- Optimizing result display
- Sample results recall
- Workload statistics and result statistics
- Customizing patient demographics

8.1 Overview

Sample programming can be performed in manual and auto modes, in batch or by single, by rerunning or adding chemistry and samples, in common or quick STAT mode, and via virtual sample carousels. Chemistries selected for samples include biochemical chemistries, ISE chemistries, serum index, calculations, off-system chemistries and panels. If a chemistry will not be used in your laboratory, you are allowed to mask it and remove it from the chemistry list. Samples can be programmed and analyzed based on the running options. Patient demographics should be entered before or during the measurements. You may view the sample analyzing status through the **Status** screen. The system allows the deletion of programmed and complete samples.

These functions and operations will be described in detail in the following sections.

8.2 Sample Programming and Processing

8.2.1 Introduction

Except for analysis of routine samples, you often need to add samples or chemistries to the programming or rerun an abnormal sample. Samples can be diluted manually or prediluted automatically before being analyzed.

8.2.2 Adding Samples



BIOHAZARD

Inappropriate handling of samples may lead to biohazardous infection. Do not touch the samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the samples, follow standard laboratory safety procedure and consult a doctor.



CAUTION

Do not use expired samples; otherwise, unreliable test results may be caused.

- 1 Add samples to the existing programming according to 2.8.1 Programming Routine Samples (page 2-38).
- 2 Confirm the program information.
- 3 Select the  icon on the upper-right corner of the main screen to request for sample stop.
- 4 Check the sample stop countdown in the system status area and wait until it comes to 0.
- 5 Check the sample carousel indicators, and proceed to the next step when the indicators are extinguished.
 - Flash: indicates that the corresponding carousel is rotating or is going to rotate.
 - ON: indicates that the corresponding carousel is stopped for sample aspirating or that the sample probe is aspirating on the reaction carousel during test for the diluted samples
 - OFF: indicates that the corresponding carousel has no sample being aspirated and will not rotate in the next period.

- 6 Place the added samples on the assigned positions of the sample carousel, and then select  to start the analysis.
 - If the sample is within the specified position range of the selected sample carousel, it is analyzed automatically.
 - Otherwise, you should specify the sample carousel and position to start the analysis.

8.2.3 Adding/Modifying Chemistries

No matter in which status a sample is, new chemistries can be added, and dilution factors and replicates can be defined for them. For samples that are programmed but not analyzed yet, editing the sample information (except for STAT feature), patient demographics and chemistries is allowed; for samples in the status of In Progress, Rerun, Incomplete or Complete, the sample information and chemistries must not be edited, while patient demographics can be edited and new chemistries can be added.

- 1 Select **Program-Sample**.

- 2 Type in the sample ID.

The programming information of the sample is displayed on the screen.

- 3 Deselect chemistries you won't run, and then select chemistries you desire to run.
- 4 Deselect panels you won't run, and then select panels you desire to run.
- 5 Choose chemistries and panels to add to the sample.
- 6 Select **Save F8**.
 - If the system is running tests, it will analyze the added chemistries and panels automatically.
 - If the system is in Standby status, select the  icon on the upper-right corner of the main screen.

8.2.4 Rerunning Samples

Finished samples can be rerun in manual or auto mode. Only chemistries that have been finished can be rerun. If a chemistry is run for more than one replicate, it cannot be rerun only when all replicates are finished. Manual rerun is performed on the **List** screen, **Current Results** screen and **History Results** screen; auto rerun is performed when a result is beyond the set critical range or linearity range. Samples

in all status are allowed for rerunning.

Manual rerun on List screen

The **Rerun** window of the **List** screen allows you to manually rerun single or multiple samples that are in Complete, Incomplete, Rerun or In Progress status. When rerunning samples, you are allowed to edit the sample cup type, sample position, STAT feature and chemistries. If a chemistry is finished, it can be rerun with edited sample volume, replicates and predilution factor. Sample ID, bar code, sample type and collection time of rerunning samples must not be edited. If certain chemistries of a sample are not finished yet before the sample is rerun, the chemistries for rerunning cannot be modified.

Rerunning single sample:

A single sample can be rerun by specifying the bar code or sample ID.

- 1 Select **Program-Sample**.
- 2 Select **List F5**.
- 3 Select **Rerun F4**.

Figure 8.1 Rerun window



The screenshot shows a software window titled "Rerun". It contains two input fields: "Sample ID" with the value "1" and a hint "e.g. 1, 3, 5-12", and "Bar Code" which is empty. At the bottom, there are three buttons: "Select", "Batch", and "Close".

- 4 Type in the ID or bar code of the sample you desire to rerun or enter the barcode of the sample.
- 5 To edit sample information, select **Select**.

Figure 8.2 Rerun Samples window

- 6 Edit the following information:
 - Position: change the carousel number and position of the sample.
 - STAT: select or deselect the **STAT** checkbox.
 - Comment: choose or enter a sample comment.
 - Chemistry and panel: change chemistries and panels.
 - Options: edit the number of replicates and predilution factors for the sample or for a chemistry, and then modify the sample cup type.
- 7 Select **Save F8**.
- 8 Select **Exit F7**.
- 9 After confirming all rerun information, load samples to the assigned positions, and select  to start the analysis.
 - If the sample is within the specified position range of the selected sample carousel, it is analyzed automatically.
 - Otherwise, you should specify the sample carousel and position to start the analysis.

Rerunning batch samples:

Batch samples can be rerun by specifying the sample ID range and with same chemistries.

- 1 Select **Program-Sample**.
- 2 Select **List F5**.
- 3 Select **Rerun F4**.

Figure 8.3 Rerun window

- 4 Type in the sample ID or range you desire to rerun.

Separate single samples with comma, e.g. 5, 7, 9; and connect multiple continuous samples with a dash, e.g. 1-3.

- 5 To edit sample information, select **Select**.

Figure 8.4 Rerun Samples window

- 6 Edit the following information:
 - Position: change the carousel number and position of the sample.

8 Sample Programming and Processing

- **STAT**: select or deselect the **STAT** checkbox.
- **Comment**: choose or enter a sample comment.
- **Options**: edit the number of replicates and predilution factors for the sample or for a chemistry, and then modify the sample cup type.

7 Select **Save F8**.

8 Select **Exit F7**.

9 Select **Batch**.

Figure 8.5 Rerun Batch window



10 Choose chemistries for rerunning the samples.

The list includes all chemistries that have been enabled and configured. The selected chemistries will be requested for rerunning the samples.

11 Select **OK**.

12 After confirming all rerun information, load samples to the assigned positions, and select  to start the analysis.

- If the samples are within the specified position range of the selected sample carousel, they are analyzed automatically.
- Otherwise, you should specify the sample carousel and position to start the analysis.

Manual rerun on Current Results or History Results screen

1 Select **Result-Current Results** or **History Results**.

2 Choose the **By Sample** option.

3 Search for desired sample results.

- 4 Check the **Flag** column for flags indicating abnormalities.
- 5 Choose results you desire to rerun.
- 6 Select **Rerun F5**.

Figure 8.6 Rerun window

- 7 Change the carousel number and position of the sample.
- 8 Select a sample volume type to rerun the sample.

The sample volume is the same as that defined for the chemistry. If increased and decreased volumes are defined for the chemistry, Increased and Decreased are available here for selection.

- 9 Enable or disable sample blank for the sample.

Only when the **Set Sample Blank Individually** checkbox is selected on the **Factory Settings** screen, the **Sample Blank** option will appear. If you need the settings, contact our customer service department or your local distributor.

- 10 Choose a sample tube type. The options include micro and standard.
- 11 Enter the off-line dilution factor.

The input range is 2-9999, and the default is blank.

- 12 Enter the predilution factor.

8 Sample Programming and Processing

The input range is 4-176, and the default is blank. When dilution factors for both normal run and rerun are defined, the product of the two factors and the auto dilution factor must not be greater than 176.

13 If you want to run a chemistry with different sample volume, replicates and predilution factor, enter the values in the chemistry option area:

- **Sample Vol:** sample volume required to run the chemistry. The sample volume is the same as that defined for the chemistry. If increased and decreased volumes are defined for the chemistry, Increased and Decreased are available here for selection.
- **Predilution:** ratio at which samples containing the chemistry will be prediluted before being analyzed. When dilution factors for both normal run and rerun are defined, the product between the two factors and the auto dilution factor must not be greater than 176.
- **Sample blank:** set up sample blank for chemistries.

14 Select **Save**.

15 Load samples to the assigned positions, and select  to start the analysis.

- If the sample is within the specified position range of the selected sample carousel, it is analyzed automatically.
- Otherwise, you should specify the sample carousel and position to start the analysis.

Batch rerun on Current Results or History Results screen

When recalling results by chemistry on the **Current Results** or **History Results** screen, you are allowed to rerun multiple samples of a chemistry that are *Complete* or *Incomplete*.

- 1** Select **Result-Current Results** or **History Results**.
- 2** Choose the **By Chemistry** option.
- 3** Search for desired sample results.
- 4** Choose a chemistry and samples you desire to rerun.
- 5** Select **Rerun F5**.

Figure 8.7 Rerun window

Sample ID	Bar Code	Sample Vol	Predilution	Off-line Dilution	Sample Blank
1		Standard	▼		
1		Standard	▼		
1		Standard	▼		

The window shows the selected chemistry and samples, as well as sample ID, bar code, sample volume in previous test, predilution factor, sample blank and off-line dilution factor.

- 6 Enable or disable sample blank for the sample.
- 7 Modify the sample volume, predilution factor, and off-line dilution factor.
 - Predilution factor: The input range is 4-176, and the default is blank.
 - Off-line dilution factor: The input range is 2-9999, and the default is blank.
 - Sample blank: set up sample blank for chemistries.
- 8 Select **OK**.
- 9 Load samples to the original positions, and select  to start the analysis.

Auto rerun based on critical range

The auto rerun function can be enabled on the **Reference/Critical Range** window. Once the auto rerun is enabled, the system will check if the result is beyond the critical range, and if it is, will rerun the sample.

- 1 Select **Utility-Chemistries**.
- 2 Select **Ref Range F4**.
- 3 Choose a chemistry from the **Chemistry** pull-down list.
- 4 Set up the reference range and critical range.

- 5 Mark the **Auto Rerun** checkbox with a tick.

The system will rerun the sample if the chemistry result is beyond the critical range.

- 6 Select **Save F7** to save the settings.
- 7 Select **Exit F8** to close the window.

Rerun when meeting auto rerun conditions

The auto rerun function can be also enabled on the Define/Edit Chemistries window. Once the auto rerun is enabled, the system will check if the rerun conditions are met, and if they are, will rerun the sample.

For more information about auto rerun setup, refer to 3.1.3 Auto Rerun Setup. (Page 3-5).

- 1 Select **Utility-Chemistries**.
- 2 Choose a chemistry.
- 3 Select **Define F1**.
- 4 Select the down-arrow button to show the error detection parameters setup page.
- 5 Mark the **Auto Rerun** checkbox with a tick.
- 6 Select **Utility-System Setup**.
- 7 Set up auto rerun conditions.

The system will rerun the sample if the rerun conditions are met.

- 8 Select **Save F7** to save the settings.
- 9 Select **Close F8** to close the window.

Recalling rerun results

The rerun results of a sample are presented on the **Recall Rerun Results** window, through which you are allowed to recall all rerun results. Users with sufficient permissions are allowed to delete the rerun results of a sample.

- 1 Select **Result-Current Results** or **History Results**.
- 2 Search for desired sample results.

- 3 Choose a sample and then a chemistry you desire to recall.
- 4 Select **Cal Options F2**.
- 5 Select **Recall Rerun Results**. The **Recall Rerun Results** window is displayed.

The screen shows the sample information and all reruns results of the chemistry.

Figure 8.8 Recall Rerun Results window

Chemistry	Final Result	Run Date/Time	Default
ALT-	57.7	2012-10-15 11:43:10	Y

- 6 The latest rerun result is the default one. To change the default result, choose a result, and then select **Set Defaults**.

The **Default** column of the result shows **Y**, which stands for Yes.

- 7 Select **Exit** to exit the window.

8.2.5 Programming Samples with Increased or Decreased Volume

In common tests, chemistries are run with standard sample volume. Owing to the specificity of certain sample, the result may be high or low. To ensure accurate results, the system allows the processing of samples with increased or decreased volume. When a sample is analyzed with standard volume and a result is beyond the reference range or deemed abnormal, you are allowed to rerun the corresponding chemistry manually with the increased or decreased sample volume.

8 Sample Programming and Processing

- 1 Select **Utility-Chemistries**.
- 2 Choose a chemistry.
- 3 Select **Define F1**.

Figure 8.9 Define/Edit Chemistries window

The screenshot shows the 'Define/Edit Chemistries' window with the following fields and values:

- Chem: GLU--H, No.: 14, Sample Type: Serum
- Chemistry: GLU--H, Print Name: GLU
- Reaction Type: Endpoint, Reaction Direction: Positive
- Pri Wave: 340nm, Sec Wave: 700nm
- Unit: mmol/L, Decimal: 0.01
- Blank Time: 47, 49, Reaction Time: 78, 78
- Sample Vol: Standard 1.5 uL, Decreased, Increased
- Aspirated: uL, uL, uL
- Diluent: uL, uL, uL
- Reagent Vol: R1 120 uL, R2 30 uL, R3, R4
- Diluent: uL, uL, uL, uL
- Sample Blank: ☐, Auto Run: ☐

Buttons at the bottom: Print F1, Reflex F2, Qualitative F3, Prev F4, Next F5, Discard F6, Save F7, Close F8.

- 4 Type in the decreased and increased sample volume.
- 5 Type in the aspirated volume and diluent volume for decreased sample volume analysis.
- 6 Type in the aspirated volume and diluent volume for increased sample volume analysis.
- 7 Select **Save F7**.
- 8 Select **Close F8**.
- 9 Select **Program-Sample**.
- 10 Enter the following information:
 - ID
 - Sample position
 - STAT status

- Sample type
- Comment
- Chemistries and panels

11 Set the chemistry options:

- Select **Options F2**.
- Choose a sample volume type and sample cup type.
- Enter the replicates, off-line dilution factor and predilution factor for the sample.
- Choose a sample volume for the chemistry.
- Enter the replicates and predilution factor for the chemistry.
- Set up sample blank.
- Select **OK**.

12 Select **OK F8**.

13 Select the  icon to start the analysis.

8.2.6 Programming Diluted Samples

Due to patient specificity, certain results of a sample may be relatively high. In this condition, you are allowed to rerun the corresponding chemistries by manually or automatically diluting the sample at certain ratio for part or all of the chemistries. When a sample is analyzed and a result is beyond the reference range or deemed abnormal, you are allowed to rerun the corresponding chemistry manually with the sample diluted.

You are allowed to set the sample dilution factors when defining a chemistry or requesting the chemistry for sample analysis. When you set both the off-line dilution factor and predilution factor when requesting a chemistry, the result will be multiplied automatically by the two dilution factors. For chemistries with dilution factors,

- If the dilution factors are not set, the result will be directly multiplied by the predilution factor.
- If the dilution factors are set, the result will be directly multiplied by the product between the predilution factor and the ratio calculated based on the set dilution factors.

If the sample volume, replicates and predilution factor are set for both the sample and the chemistry, the chemistry will be run based on its own settings instead of those of the sample.

8 Sample Programming and Processing

Perform the following steps to run diluted samples.

- 1 Select **Utility-Chemistries**.
- 2 Choose a chemistry.
- 3 Select **Define F1**.
- 4 Type in the aspirated sample volume and diluent volume for standard sample volume analysis.
- 5 Type in the aspirated sample volume and diluent volume for decreased and increased sample volume analysis.
- 6 Select **Save**.
- 7 Select **Close**.
- 8 Select **Program-Sample**.
- 9 Enter the following information:
 - ID
 - Sample position
 - STAT status
 - Sample type
 - Comment
 - Chemistries and panels
- 10 Select **OK F8**.
- 11 Select **Options F2**.

Figure 8.10 Options window

12 Select a sample volume type to run the sample.

The options include:

- Standard
- Increased
- Decreased

13 Enable or disable sample blank for the sample.

14 Select a sample tube type from the **Sample Cup** pull-down list.

The options include:

- Microtube
- Standard

15 Enter the off-line dilution factor.

The input range is 2-9999, and the default is blank.

16 Enter the number of replicates.

The input range is 1-90, and the default is 1.

17 Enter the predilution factor.

The input range is 4-176, and the default is blank.

- 18 Choose a chemistry.
- 19 Select a sample volume type in the **Sample Vol** column for the chemistry.
- 20 Enter the number of replicates.
- 21 Enter the predilution factor.

When dilution factors for both normal run and rerun are defined, the product between the two factors and the predilution factor must not be greater than 176.

- 22 Move the scroll bar to view more chemistries.
- 23 Select **Save**.
- 24 Select the  icon to start the analysis.

8.2.7 Sample Blank

Sample blank is similar to sample analysis except for use of equivalent amount of physiological saline. Sample blank is used for removal of non-chromogenesis reaction, such as influence of sample interference (Hemolysis, icterus and lipemia) on absorbance readings.

The sample blank reaction curve is almost a straight line with slope of 0 during the reaction period, and therefore means nothing for fixed-time and Kinetic analysis. For double, triple and quadruple reagent endpoint analysis, the sample blank absorbance can be subtracted through parameter settings. Therefore, sample blank is only effective for single-reagent endpoint chemistries.

Running a sample blank

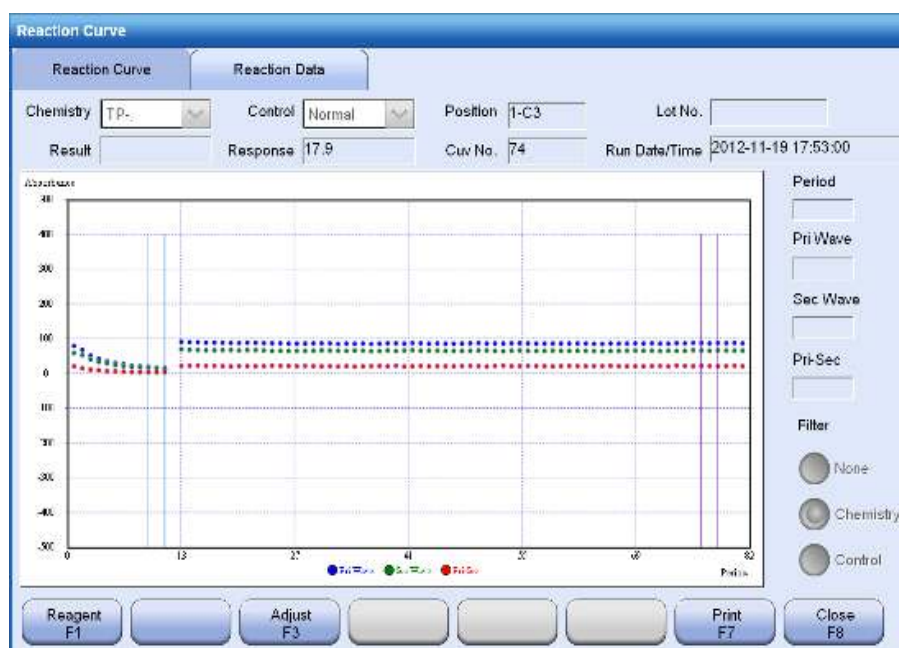
- 1 Select **Utility-Chemistries**.
- 2 Choose a chemistry.
- 3 Select **Define F1**.
- 4 Mark the **Sample Blank** checkbox with a tick.
- 5 Select **Save F7**.
- 6 Select **Close F8**.

The system will run a sample blank when running calibrators, controls and samples for the chemistry.

Recalling sample blank results

- 1 Select **Result-Current Results** or **History Results**.
- 2 Search for desired sample results.
- 3 Choose a sample and then a chemistry you desire to recall.
- 4 Select **Reac Curve F4**.
- 5 Select **Sample Blank F2**.

Figure 8.11 Sample blank reaction curve



- 6 Choose the **Reaction Data** tab to view the reaction data.

Figure 8.12 Sample blank reaction data

Reaction Curve

Chemistry: Control: Position: Lot No.:

Result: Response: Cuv No.: Run Date/Time:

Period	Pri Wave	Sec Wave	Pri-Sec	Period	Pri Wave	Sec Wave	Pri-Sec
1	79.25	58.70	20.55	2	67.36	52.48	14.88
3	51.00	40.25	10.75	4	42.25	33.27	8.98
5	35.26	29.01	6.25	6	30.22	23.57	6.24
7	26.73	21.26	5.48	8	22.65	19.75	4.10
9	20.71	15.81	3.90	10	19.35	16.04	3.32
11	17.03	13.51	3.12	12	15.48	12.17	3.31
14	90.77	69.40	21.36	15	99.80	67.64	21.66
16	89.80	67.25	22.64	17	88.23	66.87	21.37
18	88.43	66.67	21.76	19	88.04	67.26	20.78
20	87.45	67.45	20.00	21	87.26	66.28	20.97
22	88.04	67.25	20.78	23	87.84	67.06	20.78
24	87.06	66.09	20.87	25	86.67	64.53	22.14

Buttons:

7 To print the reaction curve or reaction data, select **Print F7**.

8 Select **Close F8** to close the window.

8.2.8 Sample Management

Before programming samples, it is necessary to understand the sample cups, and sample volume of the system, as well as how to load and unload samples.

Sample cup types

The sample carousel supports blood collecting tube, centrifugal tube, plastic tube and Microtube, which are available in the following specifications:

- Microtube: Mindray sample cup, $\Phi 14 \times 25$ mm, $\Phi 12 \times 37$ mm
- Blood collecting tube or plastic tube: $\Phi 12 \times 68.5$ mm, $\Phi 12 \times 99$ mm, $\Phi 12.7 \times 75$ mm, $\Phi 12.7 \times 100$ mm, $\Phi 13 \times 75$ mm, $\Phi 13 \times 95$ mm, $\Phi 13 \times 100$ mm.

Sample volume

The amount of sample required for a common measurement is 1.5-45 μ l, with an increment of 0.1 μ l. Analysis with insufficient samples may lead to inaccurate results.

If a sample is exhausted during the analysis, the system will automatically invalidate all incomplete chemistry of the sample. Before running samples, make sure that they are sufficient in volume for analysis.

Loading samples



BIOHAZARD

Wear gloves and lab coat, if necessary, goggles.

- 1 Check if the sample inside the sample tube is sufficient for analysis and the bar code label is applied correctly.
- 2 Check the system status.
 - If the system status is Running, select  to request for sample stop.
 - If the system status is Standby, proceed to the next step.
 - If the system status is Incubation, wait until the system gets steady, and then proceed to the next step.
- 3 Check if the sample carousel and the sample probe have stopped moving.
- 4 To load samples, remove the sample carousel cover.
- 5 Insert the sample tube into the tube holder until the tube bottom contacts the groove of the tube rack.
- 6 Restore the sample carousel.

Unloading samples



BIOHAZARD

Wear gloves and lab coat, if necessary, goggles.



Caution

When unloading $\Phi 16.5 \times 92$ mm sample cups, remove the sample carousel, or press the sample carousel with one hand and take out the sample cups with the other hand.

- 1 Check if the sample carousel and the sample probe have stopped moving.
- 2 If the system status is Running, select  to request for sample stop.
- 3 Remove the sample carousel cover.
- 4 Grab the sample tube and pull it upward to remove it from the tube holder.

8 Sample Programming and Processing

- 5** Repeat step 4 and 5 to unload more samples.
- 6** Restore the sample carousel.

8.3 Serum Index

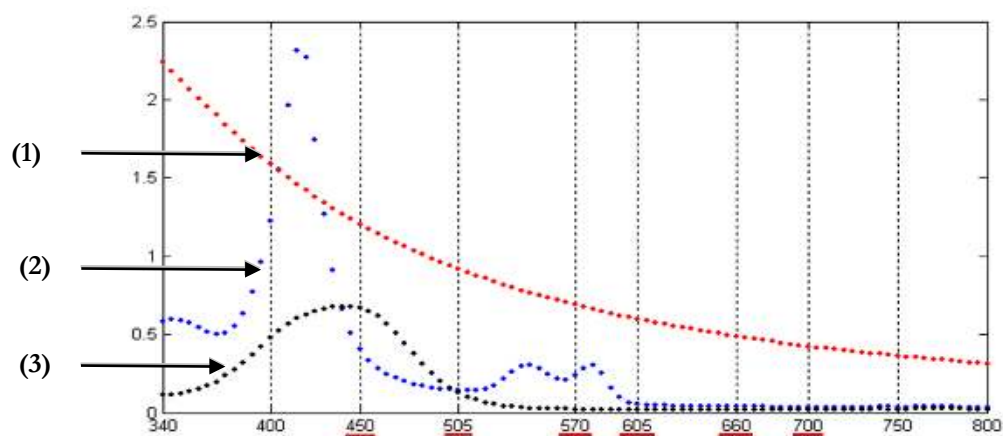
8.3.1 Introduction

Serum index is the degree of hemolysis, icterus and lipemia contained in serum sample. They are usually seen in serums and can influence the test results in physical or chemical way.

The serum index function is used to analyze the interferents in samples, helping clinical professionals to evaluate the test results.

8.3.2 Theory of Serum Index

Figure 8.13 Absorption spectrum of interferents in serum samples



The figure above shows the absorption spectrum of interferents in serum samples. (1) refers to lipemia, (2) refers to hemolysis, and (3) refers to icterus.

The three interferents are selective to wavelength and have complex absorption spectrums. They cannot be removed completely by means of double-wavelength measurements. The serum index option can be used to analyze the interferents contained in samples, helping clinical professionals to evaluate the test results.

Six wavelengths are chosen to determine the serum index. The equations of serum index are as follows:

- Lipemia: primary wavelength of 660, secondary wavelength of 700.

$$A_L = A_{660} - A_{700}$$
, lipemia index: $L = 1/C \times A_L$
- Hemolysis: primary wavelength of 570, secondary wavelength of 605.

$$A_H = A_{570} - A_{605}$$
, hemolysis index: $H = 1/A \times (A_H - B \times A_L)$
- Icterus: primary wavelength of 450, secondary wavelength of 505.

$$A_I = A_{450} - A_{505}, \text{ icterus index: } I = 1/D \times [A_I - E \times (A_H - B \times A_L) - F \times A_L]$$

Where,

- B and F: determined by the absorption spectrum of lipemia and not adjustable.
- E: determined by the absorption spectrum of hemolysis and not adjustable.
- C: determined by single lipemia, and can be user-defined.
- A: determined by single hemolysis, and can be user-defined.
- D: determined by single icterus, and can be user-defined.

8.3.3 Serum Index Setup

The serum index includes lipemia (L), hemolysis (H) and icterus (I), and has the common name of SI. The chemistry name, sample volume and reagent volume of SI are defined by the manufacturer and cannot be modified by users. The SI chemistry cannot be deleted. You are allowed to define the print names and qualitative result flags for the chemistry.

Defining print name

- 1 Select **Utility-Chemistries**.
- 2 Choose the **SI** chemistry.
- 3 Select **Define F1**.

Figure 8.14 Serum Index window

The screenshot shows the 'Serum Index' window with the following fields and controls:

- Chem:** SI
- Sample Vol:** 10 uL
- Reagent Vol:** 200 uL
- Lipemia Section:**
 - Print Name:** [Empty text box]
 - ☐ Use Qualitative Result
 - Range:** [5 empty input boxes]
 - Flag:** [5 empty input boxes]
- Hemolysis Section:**
 - Print Name:** [Empty text box]
 - ☐ Use Qualitative Result
 - Range:** [5 empty input boxes]
 - Flag:** [5 empty input boxes]
- Icterus Section:**
 - Print Name:** [Empty text box]
 - ☐ Use Qualitative Result
 - Range:** [5 empty input boxes]
 - Flag:** [5 empty input boxes]
- Buttons at the bottom:** Print F1, [Empty], [Empty], [Empty], [Empty], Discard F6, Save F7, Close F8

- 4 Type in the print name of lipemia in the **Print Name** of the **Lipemia** area. Up to 15 characters can be entered.

The lipemia index will appear as the print name on patient reports and as “SI” on other reports.

- 5 Repeat step 4 to define print names for hemolysis and icterus.
- 6 Select **Save F7**.
- 7 Select **Close F8**.

Defining qualitative result flags

The Qualitative Analysis option, when enabled, analyzes every sample for the detection of lipemia, hemolysis and icterus and calculates the numeric values of the index. If the volume of the interferents contained in a sample is beyond the set range, a flag will be added to the patient report.

The system allows 6 ranges and flags for each interferent.

- 1 Select **Utility-Chemistries**.
- 2 Choose the **SI** chemistry.
- 3 Select **Define F1**.
- 4 Mark the **Use Qualitative Result** checkbox in the **Lipemia** area.

The **Range** and **Flag** fields below are activated for editing.

- 5 Type in the detection range in the first edit box of the **Range** field, and then enter a flag in the first edit box of the **Flag** field.

For instance, type in “10” in the first edit box of the **Range** field in the **Lipemia** area, and then enter “+” in the **Flag** field of the same row. If the lipemia volume (L1) contained in a sample is lower than 10, the “+” sign will be added to the result in the patient report. Type in “20” in the second edit box below the **Range** icon and “+” in the second edit box below the **Flag** icon. If the lipemia volume (L2) is greater than 10 and lower than 20, the result will be flagged with the “+” sign. The cycle continues. If the result is greater than L5, the six flag will appear on the patient report.

- 6 Repeat step 4-5 to define ranges and flags for hemolysis and icterus.
- 7 Select **Save F7**.
- 8 Select **Close F8**.

8.3.4 Auto Serum Index

When the Auto Serum Index function is enabled, the system will select the SI chemistry automatically for serum or plasma samples. The SI chemistry will also be requested automatically when you program routine samples manually or by using the LIS host, or program STAT samples, or program routine samples with the default panels.

When programming samples with auto serum index, you are required to select at least one chemistry other than SI.

- 1 Select **Utility-System Setup**.
- 2 Mark the **Auto Serum Index** checkbox with a tick.
- 3 Select **Save F8**.

8.3.5 Running SI Chemistry

The SI chemistry is only applicable to serum and plasma samples (routine and STAT) rather than controls and calibrators.

To run the SI chemistry, choose the SI chemistry while programming samples. It will be analyzed along with other chemistries.

8.4 Clear Samples

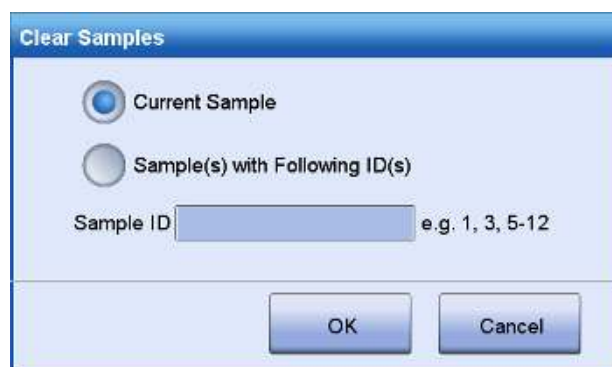
8.4.1 Introduction

The Clear Samples function is used to delete programmed samples that have not been analyzed. One or more samples can be cleared at one time. When samples are cleared, the sample information will be removed completely; the sample ID, position and bar code can be used for programming other samples. The action of clearing samples will be recorded in the edit logs.

8.4.2 Clearing Samples

- 1 Select **Program-Sample**.
- 2 Select **Clear F4**. The **Clear Samples** window appears.

Figure 8.15 Clear Samples window



- 3 Select samples you desire to clear.
 - Current sample: type in the sample ID on the **Sample** screen.
 - Sample(s) with following ID(s): type in the sample ID range in the **Sample ID** field. Single sample ID and sample range are acceptable.
- 4 Select **OK**.

The selected samples are cleared along with their programming information.

8.5 Unpositioned Samples

8.5.1 Introduction

Unpositioned samples are those:

- downloaded from the LIS host and not positioned yet. Such samples cannot be programmed for analysis until they have positions assigned. If your system is equipped with a sample bar code reader, the samples can be analyzed immediately without assigning positions for them.
- that are in Incomplete status when their positions are used for programming new samples.
- that are incomplete when their positions are released.

Once positioned, the samples will be removed from the unpositioned samples list.

8.5.2 Viewing Unpositioned Samples

- 1 Select **Program-Sample**.
- 2 Select **List F5**.
- 3 Select **Unpositioned F2**.

Figure 8.16 Unpositioned Samples window

Unpositioned Samples		
Program Date/Time	Sample ID	Bar Code
2010/08/03 19:26	A1	
2010/08/03 19:26	A2	
2010/08/03 19:26	A3	
2010/08/03 19:26	A4	
2010/08/03 19:26	A5	
2010/08/03 19:26	A6	

- 4 Move the scroll bar to view more samples.
- 5 Select **Exit** to exit the window.

8.5.3 Assigning Positions

- 1 Select **Program-Sample**.
- 2 Select **List F5**.
- 3 Select **Unpositioned F2**.
- 4 Select **Assign**.

Figure 8.17 Assign positions

The 'Assign' dialog box is shown with a blue title bar. Inside, there are four main input areas: 'Program Date' with a dropdown menu showing '2011-10-18'; 'ID' with a text box and a hint 'e.g. 1, 3, 5-12'; 'CrsI' with a dropdown menu showing '1'; and 'Pos' with two adjacent text boxes. At the bottom right, there are two buttons labeled 'OK' and 'Cancel'.

- 5 Select the program date of sample(s) to assign position.
- 6 Type in the sample ID or range in the **ID** field.
- 7 Choose a sample carousel on which you will place the sample.
- 8 Enter the positions in the **Pos** field.
 - To assign position for single sample, input the position number in the first edit box.
 - To assign positions for multiple samples, enter the start position number in the first edit box, and then the end position number in the second edit box. The system will assign positions for the samples ascending according to the sample ID.
 - If the available positions among the specified range are more than or equal to the number of samples, the extra positions will be neglected.
 - If the available positions among the specified range are less than the number of samples, the system will display a message indicating insufficient positions. Assign the positions again.
- 9 Select **OK**.

8 Sample Programming and Processing

- 10 To run the samples, select the  icon on the upper-right corner of the main screen.

8.6 Release Sample Position

8.6.1 Introduction

When a sample is analyzed, the position cannot be used for programming new sample until it is released. The **Status** screen provides the Release Sample Position function, which allows you to release the selected position or all positions on the current sample carousel that are not running any tests. When a sample is released, its results and programming information can be still recalled.

Sample positions can be released automatically at specified time every day. When the set time is reached,

- If the system is shut down, the sample positions in the status of Complete will be released next time when the system is started up.
- If the system is not running any tests, the sample positions in the status of Complete will be released.
- If the system status is Running, the sample positions in the status of Complete will be released when the system status becomes Standby or Failure at the first time.

When a sample is released, its results and programming information can be still recalled.

8.6.2 Releasing Sample Positions

Only patient samples rather than controls, calibrators, ISE wash solution and physiological saline can be released.

- 1 Select **Program-Status**.
- 2 Choose a sample carousel to release samples.
- 3 Select **Release F3**.

Figure 8.18 Release Positions window



- 4 Choose the sample range:
 - Following position(s): type in the sample ID or range to release in the edit box.
 - All positions: to release all positions of the selected sample carousel.
- 5 Select **OK**.

8.6.3 Auto Release of Samples

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **10 Auto Release Sample**.
- 4 Select the auto release time of patient samples in the **Auto Release Time** field.

Select an integer between 00 and 23. The default is 00.

- 5 Select **OK**.

When the time is reached, the system will release automatically all sample positions in the status of Complete.

8.7 Sample Logs

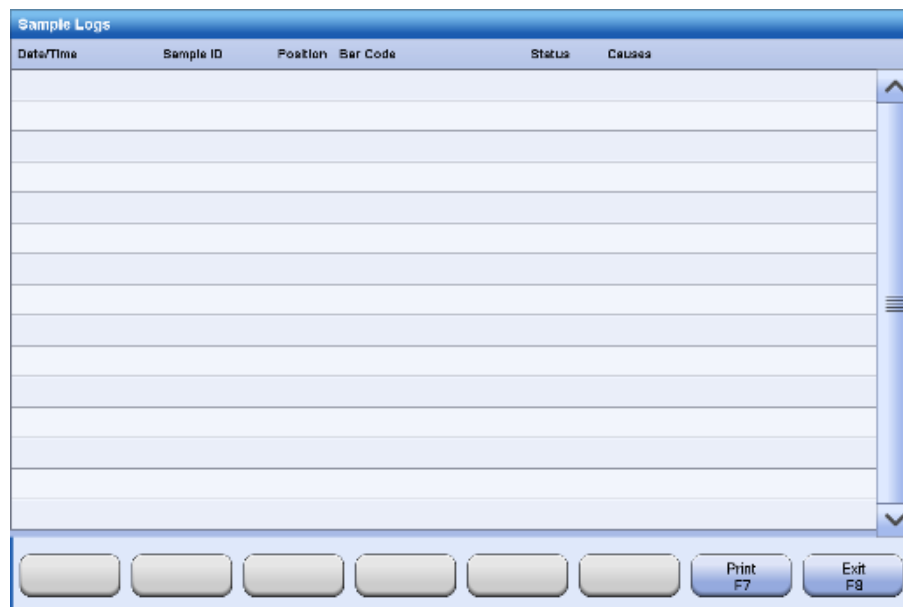
8.7.1 Introduction

The **Sample Logs** screen provides the controls and patient samples that are not complete within the recent 24 hours due to certain reasons. You are to rerun the samples or take other actions for the controls and samples. The sample logs refresh automatically as the analysis is performed, and can be printed out for archiving.

8.7.2 Viewing Sample Logs

- 1 Select **Program-Status**.
- 2 Select **Log F2**.

Figure 8.19 Sample Logs window



- 3 The screen shows the controls and patient samples that are not complete within the recent 24 hours due to certain reasons.

The table below summaries the sample status and relevant descriptions.

Table 8.1 Description and probable causes to sample statuses

Sample status	Description	Probable Causes
Incomplete	The sample is being analyzed and still has some tests which have no results.	
Programmed	The position is assigned to a sample but now occupied by another sample.	
Unprogrammed	The sample is identified, but no corresponding program is obtained from the LIS host.	Inquiring the LIS host is timed out. The sample has no program information on the LIS host and is programmed with the default panel.
Rerun	The sample is programmed automatically for rerun but cannot be run due to scanning error; or the sample is rerun automatically for a result is beyond the critical range.	The result is beyond the critical range. The Status column will display Rerun, and the Causes column will show the rerun results.
Removed	The sample has been unloaded but some tests are still being processed.	Samples that are being analyzed are unloaded.
Duplicate	The sample bar code has been assigned to a sample that already exists.	The positions for control and calibrator are occupied by patient samples. A sample is detected in an idle position but has no corresponding programming information or default panel.

- 4 To print the sample logs, select **Print F7**.
- 5 Select **Exit F8** to close the window.

8.8 Customizing Sample Information

8.8.1 Introduction

The system provides the Cust. Sample Info. option for specifying sample information to be displayed on the **Sample** screen. It includes: bar code, sample comment, patient ID, panels, sample status, and all editable information on the **Demographics** window like date of birth, gender, collection time, diagnosis, etc.

8.8.2 Customizing Sample Information

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **13 Cust. Sample Info**.

Figure 8.20 “Customize Sample Information” window

Sample Information	Customize
Status	☒
Bar Code	☒
Comment	☒
Patient ID	☒
Patient Name	☐
Gender	☐
Age	☐
Date of Birth	☐
Patient Comment	☐
Ordering Dept	☐

Save Cancel

- 4 Find desired sample information and mark the corresponding **Customize** checkbox.

Click the checkbox again to deselect it.

8 Sample Programming and Processing

- 5 Select **Save**.
- 6 Select **Cancel** to close the window.

8.9 Customizing Patient Demographics

You can specify patient demographics to be displayed, its default and its display order on the **Patient Demographics** screen.

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **14 Patient Demographics**

Figure 8.21 Patient Demographics

The screenshot shows the 'Patient Demographics' configuration window. It features two main columns of fields. The left column has a 'Default' column next to it. The right column has a list of fields with up/down arrows for reordering. Between the columns are 'Add' and 'Delete' buttons. To the right of the right column are 'Home', 'Up', 'Down', and 'End' buttons. At the bottom are 'OK', 'Cancel', and 'Exit' buttons.

Demographics	Default
Sample Type	
Sample Comment	
Patient ID	
Patient Name	
Gender	
Age	
Date of Birth	
P Comm	
Ordering Dept	
Ordering Physician	
Diagnosis	
Reviewer	
Tester	

Demographics
Patient ID
Patient Name
Gender
Age
Collection Date
Collection Time
Ordering Date
Ordering Dept
Ordering Physician
Test Date
Tester
Diagnosis
Reviewer

- 4 Select the desired information and the default value and then click **Add**.
- 5 Select the desired information and click **Delete** to delete it from the demographics list.
- 6 Select **Up**, **Down**, **Home** and **End** button to adjust the displayed order of patient demographics.
- 7 Select **OK** to save the setup.

8.10 Sample and Chemistry Lists

8.10.1 Introduction

The List option allows you to view, inquire and print all unfinished samples, and assign positions for unpositioned samples. You are also allowed to view the requested chemistries' calibration status, reagent status, tests left, and number of requests.

8.10.2 Sample List

Viewing programmed samples

The sample list shows all samples that have been programmed but not analyzed yet. Samples can be inquired by program date, sample status, ID, or bar code.

- 1 Select **Program-Sample**.
- 2 Select **List F5**.

Figure 8.22 Sample List tab page

[illegible]

- 3 Move the scroll bar to view more samples.
- 4 To print the sample list, select **Print F7**.
- 5 Select **Exit F8** to close the window.

Inquiring samples by program date, sample status or ID

- 1 Select **Search F1** on the **Sample List** tab page.

Figure 8.23 Search window

- 2 Enter the conditions:
 - Select the program date of samples you desire to inquire; and/or
 - Select a sample status, which is available in All, Programmed, In Progress, Incomplete, Complete, and Rerun; and/or
 - Type in the single sample ID or ID range in the **Sample ID** field.
- 3 Select **OK**. All samples that satisfy the conditions are displayed on the screen.

Inquiring a bar-coded sample

- 1 Select **Search F1** on the **Sample List** tab page.
- 2 Type in the sample bar code you desire to inquire.
- 3 Select **OK**. The corresponding sample is displayed on the screen.

8.10.3 Chemistry List

To view the summary of chemistries that are requested on the current day or requested before but not finished yet, perform the following steps:

- 1 Select **Program-Sample**.
- 2 Select **List F5**.
- 3 Select the **Chemistry List** tab.

8-40

Figure 8.24 Chemistry List tab page

[illegible]

The screen shows all requested chemistries, including the name, calibration status, number of requests, and tests left.

- 4 Move the scroll bar to view more chemistries.
- 5 To print the chemistry list, select **Print F7**.
- 6 Select **Exit F8** to close the window.

8.11 Optimizing Result Display

8.11.1 Introduction

Due to low sensitivity of certain reagents, samples with low concentration may have 0 or negative results, or cannot be represented accurately by results out of linearity range. To express sample concentration accurately, the system provides the Optimize Result Display option to customize such results. When less than the low limit of linearity range, results will show as "< Low limit of linearity range"; when greater than the high limit of linearity range, they will show as "> High limit of linearity range"; when less than both the low limit of linearity range and concentration of the lowest-concentration calibrator, they will show as "< Maximum of the two values"; when greater than the high limit of linearity range and concentration of the highest-concentration calibrator, they will show as "> Minimum of the two values".

Result optimizing will not affect storage, transmission and archiving of results. The calculations will be calculated with the actual results of relevant chemistries.

Only users who have the permissions of system setup are allowed to optimize result display.

8.11.2 Optimizing Result Display

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **12 Optimize Result Display**.

Figure 8.25 “Optimize Result Display” window

Chemistry	Low	High
5'-NT	☐	☐
ADA	☐	☐
AFU	☐	☐
ALB	☐	☐
ALP	☐	☐
ALT	☐	☐
ASO	☐	☐
AST	☐	☐
ApoA1	☐	☐
ApoB	☐	☐
C3	☐	☐
C4	☐	☐
CHE	☐	☐

Buttons: Select All, Clear, OK, Cancel

- 4 Find desired chemistry, and mark the corresponding **Low** and **High** checkboxes.

Click the checkboxes again to deselect them.

- Select **Low**. When a result is less than the low limit of linearity range or concentration of the lowest-concentration calibrator, it will show as “< Low limit of linearity range”, “< Concentration of the lowest-concentration calibrator”, or “< Maximum of the two values”.
- Select **High**. When a result is greater than the high limit of linearity range or concentration of the highest-concentration calibrator, it will show as “> High limit of linearity range”, “> Concentration of the highest-concentration calibrator”, or “> Minimum of the two values”.

- 5 Select **Save**.
- 6 Select **Cancel** to close the window.

8.12 Results Recall

8.12.1 Introduction

The Results Recall option allows routine samples, STAT samples and controls to be recalled and handled on the **Current Results** or **History Results** screen. The Current Results include those that are programmed and analyzed on the current day; the History Results are those programmed and analyzed before the current day. All results can be recalled by sample or by chemistry.

8.12.2 Displaying Current Results

1 Select **Result-Current Results**.

The screen shows all samples and controls that are programmed and analyzed on the current day. When certain test of a control sample or patient sample triggers a data alarm, the sample will appear in yellow.

Figure 8.26 Current Results screen



The sample type includes R, E and C. R stands for routine sample, E for STAT sample, and C for control.

The **Host** column indicates the transmission status of the sample. Y means that the sample has been sent to the LIS host, and N means the opposite.

The **Print** column indicates the print status of the sample. Y means that the sample has been printed, and N means the opposite.

Samples displayed in the sample list can be sorted by the type, ID, status, position, completion time, program date/time, host and print fields.

- 2 Choose a result recall mode:
 - By sample
 - By chemistry
- 3 When recalling results by sample, choose a sample in the left list. The right list displays all results of the sample. When recalling results by chemistry, choose a chemistry in the left list. The right list displays all results of the chemistry.
- 4 Choose the following buttons as needed:
 - **Search F1:** to inquire sample results.
 - **Options F2:** to delete, edit, rerun and print samples, recall rerun results, customize result display options, recalculate results, compensate results, and observe result trend.
 - **Demog F3:** to view patient demographics of the sample.
 - **Reac Curve F4:** to view the reaction curve of the sample.
 - **Rerun F5:** to rerun a finished sample.
 - **Review F6** to review the sample result.
 - **Print F7:** to print sample results.
 - **Host F8:** to transmit the selected sample results to the LIS host.

8.12.3 Recalling Current Results

Current results can be inquired by sample type, patient name, patient ID, sample ID or sample bar code, etc., along with the current date. Whichever status the system is, only one condition is required for inquiring desired results.

- 1 Select **Result-Current Results**.
- 2 Select **Search F1**.
- 3 Enter one or more search conditions.

Figure 8.27 Recall results window

The screenshot shows a software window titled "Recall Results". It contains a grid of input fields for searching sample records. The fields are organized as follows:

- Top Row:** Sample ID (text box with example "e.g. 1, 3, 5-12"), Sample Type (dropdown), Bar Code (text box), Sample Status (dropdown).
- Second Row:** Patient Name (text box), Patient ID (text box), MRN (text box).
- Third Row:** Sample Type (dropdown), Patient Zone (dropdown), Tester (dropdown).
- Fourth Row:** Pat. Type (dropdown), Blood Bag (text box), Reviewer (dropdown).
- Fifth Row:** Gender (dropdown), Bed No. (text box), Col Date (calendar icon showing 2012-3-28).
- Sixth Row:** Age (text box with a minus sign), Ordering Dept (dropdown), Col Time (time selection fields).
- Seventh Row:** (Empty text box), Ordering Physician (dropdown), Lot No. (text box).

At the bottom right of the window are two buttons: "OK" and "Cancel".

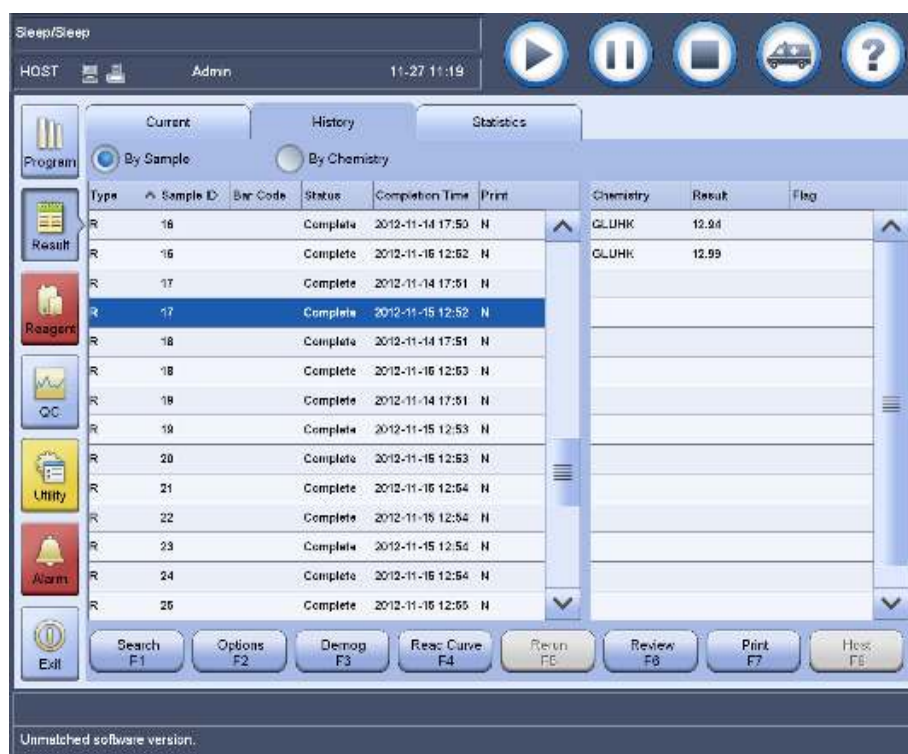
- 4 Select **OK**. The samples matching the condition are displayed on the screen.
- 5 Select a function button to perform relevant operations.

8.12.4 Displaying History Results

- 1 Select **Result-History Results**.

The screen shows all samples and controls that are programmed and analyzed before the current day.

Figure 8.28 History Results screen



The sample type includes R, E and C. R stands for routine sample, E for STAT sample, and C for control.

The **Host** column indicates the transmission status of the sample. Y means that the sample has been sent to the LIS host, and N means the opposite.

The **Print** column indicates the print status of the sample. Y means that the sample has been printed, and N means the opposite.

- 2 Choose a result recall mode:
 - By sample
 - By chemistry
- 3 Select **Search F1** to search for desired results.
- 4 When recalling results by sample, choose a sample in the left list. The right list displays all results of the sample. When recalling results by chemistry, choose a chemistry in the left list. The right list displays all results of the chemistry.
- 5 Choose the following buttons as needed:

- **Options F2:** to delete, edit, rerun and print samples, recall rerun results, customize result display options, recalculate results, compensate results, and observe result trend.
- **Demog F3:** to view patient demographics of the sample.
- **Reac Curve F4:** to view the reaction curve of the sample.
- **Rerun F5:** to rerun a finished sample.
- **Review F6** to review the sample results.
- **Print F7:** to print sample results.
- **Host F8:** to transmit the selected sample results to the LIS host.

8.12.5 Recalling History Results

Stored results can be inquired by sample type, patient name, patient ID, sample ID or sample bar code, along with the program date. Whichever status the system is, only one condition is required for inquiring desired results, while the **Program Date** field can be left blank. To quickly search for desired results from the tremendous amount of data, you are recommended to enter both the program date and any of the conditions.

- 1 Select **Result-History Results**.
- 2 Select **Search F1**.
- 3 Select the program date range you want to recall. Select the start date in the first box and the end date in the second box.
- 4 Enter one or more search conditions.

Figure 8.29 Recall Results window

The screenshot shows the 'Recall Results' window with the following fields and controls:

- Program Date:** Two date pickers showing '2012- 2-25' and '2012- 3-26'.
- Sample ID:** A text input field with a hint 'e.g. 1, 3, 5-12'.
- Sample Type:** A dropdown menu.
- Bar Code:** A text input field.
- Sample Status:** A dropdown menu.
- Patient Name:** A text input field.
- Patient ID:** A text input field.
- MRN:** A text input field.
- Sample Type (lower):** A dropdown menu.
- Patient Zone:** A dropdown menu.
- Tester:** A dropdown menu.
- Pat. Type:** A dropdown menu.
- Blood Bag:** A text input field.
- Reviewer:** A dropdown menu.
- Gender:** A dropdown menu.
- Bed No.:** A text input field.
- Col Date:** A date picker showing '2012- 3-26'.
- Age:** Two text input fields separated by a hyphen.
- Ordering Dept:** A dropdown menu.
- Col Time:** Four text input fields for hour, minute, second, and AM/PM.
- Ordering Physician:** A dropdown menu.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

- 5 Select **OK**. The samples matching the condition are displayed on the screen.
- 6 Select a function button to perform relevant operations.

8.12.6 Review Sample Results

Only when the sample status is complete, can the sample results be reviewed.

- 1 Select **Result-Current Results** or **History Results**.
- 2 Choose the **By Sample** option.
- 3 Choose a sample or more samples in the sample list.
- 4 Select **Review F6**.

The review status in the sample list turns from N to Y.

- 5 Select **No Review** to cancel the review operation.

8.12.7 Viewing/Editing Patient Demographics

Patient demographics can be viewed or edited in any system status.

- 1 Select **Result-Current Results** or **History Results**.
- 2 Choose the **By Sample** option.
- 3 Choose a sample in the sample list. Move the scroll bar to view more samples.
- 4 Select **Demog F3**.

Figure 8.30 Demographics window

The screenshot shows a software window titled "Demographics". At the top, there are input fields for "Crsl" (value 1), "Pos" (value 1), "Status", "ID" (value 1), and "Bar Code". Below these is a checkbox labeled "STAT". The main area contains several fields for patient information: "Patient ID", "Patient Name", "Gender" (dropdown), "Age" (with "Year" dropdown), "Collection Time" (with a colon separator), "Ordering Date" (value 2012- 3-26), "Ordering Dept" (dropdown), "Ordering Physi..." (dropdown), "Collection Date" (value 2012- 3-26), "Test Date" (value 2012- 3-26), "Tester" (dropdown), "Diagnosis" (dropdown), "Reviewer" (dropdown), "Conclusion" (dropdown), and "P Comm". At the bottom, there are seven function buttons: "Restore F3", "Prev F4", "Next F5", "Discard F6", "Save F7", and "Exit F8".

Patient demographics can be customized. For more information, please refer to customize patient demographics

- 5 Edit the related patient information:

- 6 Select **Save F7** to save your input.
- 7 Select **Prev F4** or **Next F5** to view demographics of the previous or next sample.
- 8 Select **Exit F8** to close the window.

8.12.8 Reaction Curve

A reaction curve reflects the relationship of the absorbance measured at the primary wavelength, secondary wavelength and primary-secondary wavelength. It is drawn based on the absorbance of the sample-reagent mixture measured within the reaction period.

Observing reaction curve:

- 1 Search for desired samples on the **Current Results** or **History Results** screen.
- 2 Choose a result recall mode:
 - By sample
 - By chemistry
- 3 Choose a chemistry or sample in the result list.
- 4 Select **Reac Curve F4**. The **Reaction Curve** window is displayed.

Figure 8.31 Sample reaction curve



- 5 Select a point on the curve. Relevant measuring period and absorbance are displayed on the right of the window.
- 6 Select a filter condition from the following options:
 - None: observe reaction curve and data in the default mode.
 - Chemistry: observe reaction curve of the results for the selected test.
 - Sample: observe reaction curve of the results for the selected sample.
- 7 Choose the **Reaction Data** tab to view the reaction data.

Figure 8.32 Sample reaction data

Reaction Curve

Reaction Curve Reaction Data

Chemistry: TP- Sample ID: 1 Position: Bar Code: Result: 68.0 Response: 1095.6 Cuv No.: 30 Run Date/Time: 2012-10-26 14:39:24

Period	Pri Wave	Sec Wave	Pri-Sec	Period	Pri Wave	Sec Wave	Pri-Sec
1	1350.56	2403.87	-1053.31	2	1346.72	2400.73	-1054.01
3	1348.00	2406.11	-1058.11	4	1348.85	2411.05	-1062.20
5	1344.16	2403.42	-1059.26	6	1347.15	2411.50	-1064.35
7	1352.27	2420.49	-1068.22	8	1348.00	2413.75	-1065.75
9	1345.44	2413.75	-1068.31	10	1348.43	2415.55	-1067.12
11	1342.88	2407.91	-1065.03	12	1348.43	2416.89	-1068.47
14	3737.23	4035.17	-297.94	15	4028.54	4226.79	-198.26
16	4200.42	4349.44	-149.02	17	4305.93	4423.31	-117.38
18	4390.30	4482.97	-92.67	19	4451.07	4526.98	-75.91
20	4498.77	4560.86	-62.09	21	4537.48	4587.94	-50.46
22	4565.86	4604.11	-38.25	23	4593.71	4623.21	-29.50
24	4621.65	4642.36	-20.70	25	4642.20	4655.72	-13.52

Reagent F1 Sample Blank F2 Adjust F3 Prev F4 Next F5 Print F7 Close F8

8 Choose the following buttons as needed:

- **Reagent F1:** to view the reagents used for sample analysis, calibrators and reagents used in calibration, and reagents for reagent blank test.
- **Sample Blank F2:** to view the sample blank reaction curve and reaction data of the selected sample.
- **Adjust F3:** to adjust the absorbance display range of current reaction curve. Refer to the following page for details.
- **Prev F4:** to view the reaction curve and data of the previous chemistry.
- **Next F5:** to view the reaction curve and data of the next chemistry.
- **Print F7:** to print the current reaction curve or data.

9 Select **Close F8** to close the window.

Viewing reagent information:

On the reaction curve window, you are allowed to view the reagents in sample measurement, the calibrators and reagents used in calibration, and reagents for reagent blank test.

- 1 Search for desired samples on the **Current Results** or **History Results** screen.
- 2 Choose a result recall mode:

8 Sample Programming and Processing

- By sample
 - By chemistry
- 3 Choose a chemistry or sample in the result list.
 - 4 Select **Reac Curve F3**. The **Reaction Curve** window is displayed.
 - 5 Select **Reagent F1**.

Figure 8.33 Reagent window

The screenshot shows a software window titled "Reagent". At the top, there is a "Chem" dropdown menu set to "TP-". Below this are two date/time fields: "Cal Date/Time" with the value "2012-10-15 11:35:49" and "Run Date/Time" with the value "2012-10-26 14:39:24". The main area contains three tables, each with columns for "Calibration", "Lot No.", and "Serial No.". The first table, labeled "Calibration", lists "WATER", "R1", "复合", and "R1". The second table, labeled "Rgt Blk", is currently empty. The third table, labeled "Reagent", lists "R1". Each table has a vertical scrollbar on its right side. A "Close" button is located at the bottom right of the window.

The window shows the calibration date and time, sample measurement date and time, calibrators, reagents for reagent blank test, and reagents for sample analysis.

- 6 Select **Close** to exit the window.

Adjusting display range:

The maximum absorbance display range of reaction curve can be adjusted automatically or manually. The adjustment is only applicable to the currently-displayed curve, which will restore the default display when opened next time.

- 1 Search for desired samples on the **Current Results** or **History Results** screen.

- 2 Choose a result recall mode:
 - By sample
 - By chemistry
- 3 Choose a chemistry or sample in the result list.
- 4 Select **Reac Curve F4**. The **Reaction Curve** window is displayed.
- 5 Select **Adjust F3**.

Figure 8.34 Adjust window



- 6 Choose an adjustment mode:
 - Auto: The system automatically determines the display range of X axis (measuring period) and Y axis (absorbance) according to the reaction data.
 - Manual: The system displays the reaction curve according to the specified absorbance range. Input the absorbance range (-40000~40000).
- 7 Select **OK**. The current reaction curve is refreshed accordingly.

8.12.9 Transmitting Results to LIS Host

Sample results and QC results can be sent to the LIS host in any system status if the LIS host is connected correctly. The Host option allows the transmission of single or multiple samples, or all samples to the LIS host.

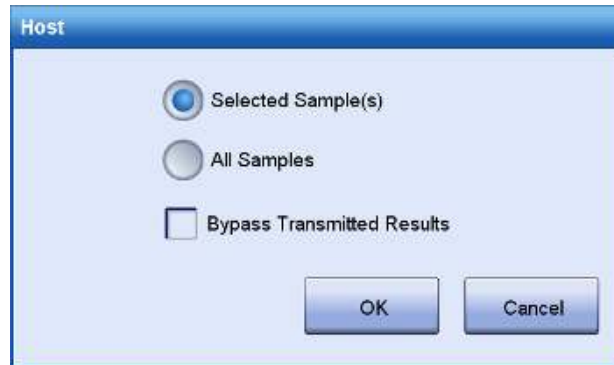
The results to be transmitted include:

- Patient demographics
- Sample results
- QC results

- 1 Search for desired samples on the **Current Results** or **History Results** screen.

- 2 Choose the **By Sample** option.
- 3 To transmit single or multiple samples, select them in the sample list.
- 4 Select **Host F8**.

Figure 8.35 Transmit Results window



- 5 Select the sample range you want to transmit:
 - Selected sample(s)
 - All samples
- 6 If you transmit all samples, you are allowed to skip those results that are already transmitted to the LIS host. Mark the **Bypass Transmitted Results** checkbox.
- 7 Select **OK**.

8.12.10 Printing Results

Samples can be printed manually on the **Current Results** and **History Results** screens. The system allows multiple samples to be printed on one report or one sample on one report. Before printing the recalled results, you should select a report template on the **System Setup** screen.

The Print option allows single or multiple samples, or all samples to be printed out.

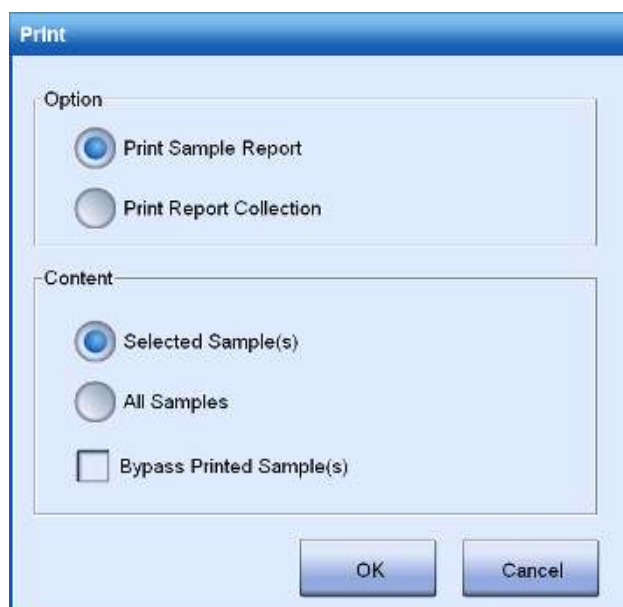
Print by Sample

Print by sample option allows you to print the test results of one or more samples when they are recalled by sample.

- 1 Search for desired samples on the **Current Results** or **History Results** screen.

- 2 Choose the **By Sample** option.
- 3 To print single or multiple samples, select them in the sample list.
 - To select current results, select the first sample, press and hold the **Shift** key, and then select the last sample; or select the first sample, press and hold the **Ctrl** key, and then select other samples; or press **Ctrl + A** to select all samples displayed on the current page. To display the first or last row in the current sample list, press **Ctrl + Home** or **End**.
 - To select history samples, directly click the type button of the samples to select them.
- 4 Select **Print F7**.

Figure 8.36 Print window



- 5 Select **Print Sample Report**.
- 6 Choose the print range:
 - Selected Sample(s)
 - All Sample(s)
- 7 If you print all samples, you are allowed to skip those that are already printed out. Mark the **Bypass Printed Sample(s)** checkbox.
- 8 Select **OK**.

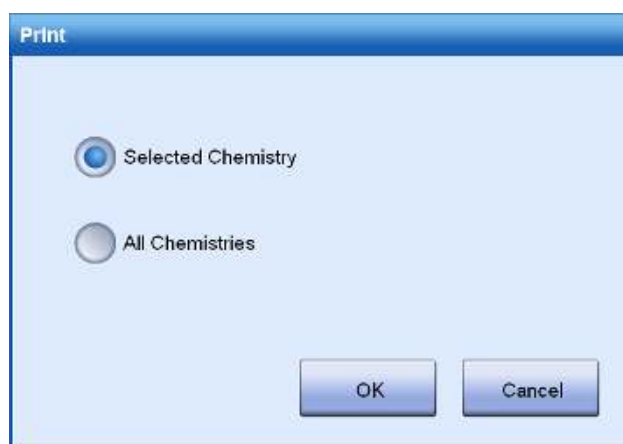
Print by Chemistry

Print by chemistry option allows you to print the test results of one or more chemistries when they are recalled by chemistries.

Laboratory Version

- 1 Search for desired samples on the **Current Results** or **History Results** screen.
- 2 Choose the **By Chemistry** option.
- 3 To print single chemistry in the chemistry list, select it; to print all chemistries, there is no need to select them.
- 4 Select **Print F7**.

Figure 8.37 Print window



- 5 Choose the print range:
 - Selected chemistry
 - All chemistries
- 6 Select **OK**.

Only the sample results that have been reviewed can be printed out.

8.12.11 Editing Results

The Edit Results option allows editing of results that slightly exceed the reference range or the linearity range but will not lead to mis-diagnosis of patients, or of results that are all on the high side or low side. This option is used for sample results only, exclusive of control results. Results of special calculations cannot be edited while results of off-system chemistry can be edited. Edited results will be flagged for

distinguishing from others.

Only the samples that have been analyzed and have results can be edited. For those tests that are run for over one time, result of each run can be edited. For rerun tests, only the default result can be edited.



CAUTION

Edit Results function gives doctors with freedom to modify the results, and therefore, must be used with cautions. Only users that have sufficient permissions are allowed to edit results.

- 1 Select **Result-Current Results** or **History Results**.
- 2 Choose a result recall mode:
 - By sample
 - By chemistry
- 3 Select **Search F1** to search for desired results.
- 4 Choose a sample or chemistry in the sample list which includes the off-system chemistries as well.
- 5 Select **Options F2**.
- 6 Select **Edit Results**.

The screen shows the samples or chemistry and all measured results.

Figure 8.38 Edit Results window - By sample(History results)

Chemistry	Final Result	Actual Result	Status
ALB	-2.6	-2.6	Complete

Figure 8.39 Edit Results window - By sample(Current results)

Chemistry	Final Result	Actual Result	Status
K	1.47	1.47	Complete

Figure 8.40 Edit Results window - By chemistry

Sample ID	Bar Code	Final Result	Actual Result	Status
28		-2.8	-2.8	Complete
29		-2.9	-2.9	Complete
30		-2.9	-2.9	Complete
31		-2.7	-2.7	Complete
32		-2.7	-2.7	Complete
33		-2.6	-2.6	Complete
34		-2.8	-2.8	Complete

Choose a chemistry to edit, and then input result in the **Final Result** column.

- On the **Edit Results** window- By sample (Current results), select **Prev** or **Next** to locate the results that require editing.
- For normal runs, only Complete chemistries can be edited.
- For reruns, only the default result can be edited.
- For off-system chemistries, the results can be edited at anytime.

8 Repeat step 7 to edit other results.

9 Select **Save** to save your editing.

10 Select **Cancel**.

8.12.12 Deleting Results

The system has a limited storage capacity and can store a maximum of 50,000 samples. The results with the earliest date will be overridden when the capacity is exceeded. The system allows deleting of routine samples, emergent samples and controls, while they are sent to the LIS host or printed out. When the system status is Running, samples in the status of Running cannot be deleted; when the system status is but Running, samples in any status can be removed. Deleted results cannot be restored. Make sure that you have archived them by sending them to the LIS host or printed out or in other ways.

Before deleting a result, check if you have sufficient permissions. Only users that have sufficient permissions are allowed to delete results. The deleting operation will be automatically recorded in event logs.

1 Select **Result-Current Results** or **History Results**.

2 Choose a result recall mode:

- By sample
- By chemistry

3 Select **Search F1** to search for desired results.

4 When recalling results by sample, choose samples in the sample list. When recalling results by chemistry, choose a chemistry in the left list.

All results of the selected sample or chemistry are displayed on the screen.

5 Select **Options F2**.

6 Select **Delete Results**.

All results of the samples are displayed on the screen.

Figure 8.41 Delete Results window



7 Choose the sample range:

- Selected result(s): to delete the results of the selected samples or chemistries.
- All results: to delete all results on the screen.

8 Select **OK**.

8.12.13 Customizing Result Display

The Customize Result Display option allows tailoring of sample and result display options on the **Current Results** and **History Results** screens. When recalling results by sample, the sample list and result list can be customized. When recalling results by chemistry, only the result list can be tailored.

1 Select **Result-Current Results** or **History Results**.

2 Choose a result recall mode:

- By sample
- By chemistry

3 Select **Options F2**.

4 Select **Customize Result Display**.

Figure 8.42 Customize Result Display window - By sample

The screenshot shows the 'Customize Result Display' window with two main sections: 'Sample List Setup' and 'Result List Setup'. Each section has a table with columns for 'Current', 'History', and 'List Header'. In the 'Sample List Setup' table, 'Type' is selected in the 'List Header' column, and 'Sample ID', 'Bar Code', 'Position', 'Status', 'Completion', 'Host', 'Patient Name', and 'Program Title' are listed in the 'List Header' column. In the 'Result List Setup' table, 'Chemistry' is selected in the 'List Header' column, and 'Result', 'Flag', 'Ref Range', and 'Unit' are listed in the 'List Header' column. There are 'Up' and 'Down' buttons between the two tables to adjust the order of the headers. At the bottom right, there are 'Save' and 'Cancel' buttons.

Figure 8.43 Customize Result Display window - By chemistry

The screenshot shows the 'Customize Result Display' window with the 'Result List Setup' section. The table has columns for 'Current', 'History', and 'List Header'. 'Type' is selected in the 'List Header' column, and 'Sample ID', 'Bar Code', 'Position', 'Status', 'Completion', 'Host', 'Program Title', and 'Print' are listed in the 'List Header' column. There are 'Up' and 'Down' buttons to the right of the table to adjust the order of the headers. At the bottom right, there are 'Save' and 'Cancel' buttons.

5 If recalling results by sample,

- Choose desired header names in the **Sample List Setup** area and screens where they are going to be displayed. Use the **Up** and **Down** buttons to adjust the display order of the header names.

To forbid display of a header name in the sample list, deselect the corresponding checkbox. Please note that the Type option for the **History Results** screen cannot be forbidden.

- Choose desired header names in the **Result List Setup** area. Use the **Up** and **Down** buttons to adjust the display order of the header names.

To forbid display of a header name in the result list, deselect the corresponding checkbox.

6 If recalling results by chemistry,

Choose desired header names in the **Result List Setup** area. Use the **Up** and **Down** buttons to adjust the display order of the header names.

To forbid display of a header name in the result list, deselect the corresponding checkbox.

7 Select **Save** to save the settings and close the window.

8.12.14 Recalculating Results

The Recalculate Results option is used to recalculate current sample results with the latest valid calibration factors of relevant chemistry. This option is often used when test result cannot be calculated due to incomplete or failed calibration.

Recalculate Results is only applicable to biochemistries. Result of samples in In Progress status cannot be recalculated. The recalculation will be automatically recorded in event logs.

- 1** Select **Result-Current Results**.
- 2** Select **Options F2**.
- 3** Select **Recalculate Results**.

Figure 8.44 Recalculate Results window

- 4 Choose a chemistry from the **Chem** pull-down list.
- 5 Input single sample ID or ID range you desire to recalculate results.
- 6 Select **Calculate**.

Results of the selected chemistry for the specified samples are recalculated automatically with the latest calibration factors and then displayed in the list at the bottom.

- 7 Select **Close** to exit the window.

8.12.15 Compensating Results

The Compensate Results option is used to recalculate multiple results of certain biochemistry through the linear formula $Y=K \cdot X+B$ with specified slope K and offset B .

Compensate Results is invalid for special calculations and off-system chemistries. A calculation will be recalculated automatically once its constituent chemistries are compensated. Only users that have sufficient permissions are allowed to compensate results. The compensation will be automatically recorded in event logs.

- 1 Select **Result - Current** or **History**.

- 2 Choose the **By Chemistry** option.
- 3 Choose the chemistry that you want to compensate in the left list.
- 4 Select **Options F2**.
- 5 Select **Compensate Results**.

Figure 8.45 “Compensate” window

The screenshot shows a software window titled "Compensate". At the top, there is a "Chemistry" dropdown menu set to "ALB". Below it are two input fields: "Slope" and "Offset". The main part of the window is a table with the following columns: "Sample ID", "Bar Code", "Status", "Actual Result", and "Final Result". The table contains 9 rows of data, all with a status of "Complete". To the right of the table is a vertical scrollbar. At the bottom right of the window are two buttons: "Save" and "Exit".

Sample ID	Bar Code	Status	Actual Result	Final Result
41		Complete	-2.4	-2.4
42		Complete	-2.6	-2.6
43		Complete	-2.2	-2.2
44		Complete	-2.5	-2.5
45		Complete	-2.5	-2.5
46		Complete	-2.6	-2.6
47		Complete	-2.5	-2.5
48		Complete	-2.4	-2.4
49		Complete	-2.4	-2.4

All results of the chemistry are displayed in the list at the bottom.

- 6 Input the slope K and offset B.
- 7 Select **Save**.

The system recalculates all results of the chemistry with the specified slope and offset. The final results are displayed in the list of the window.

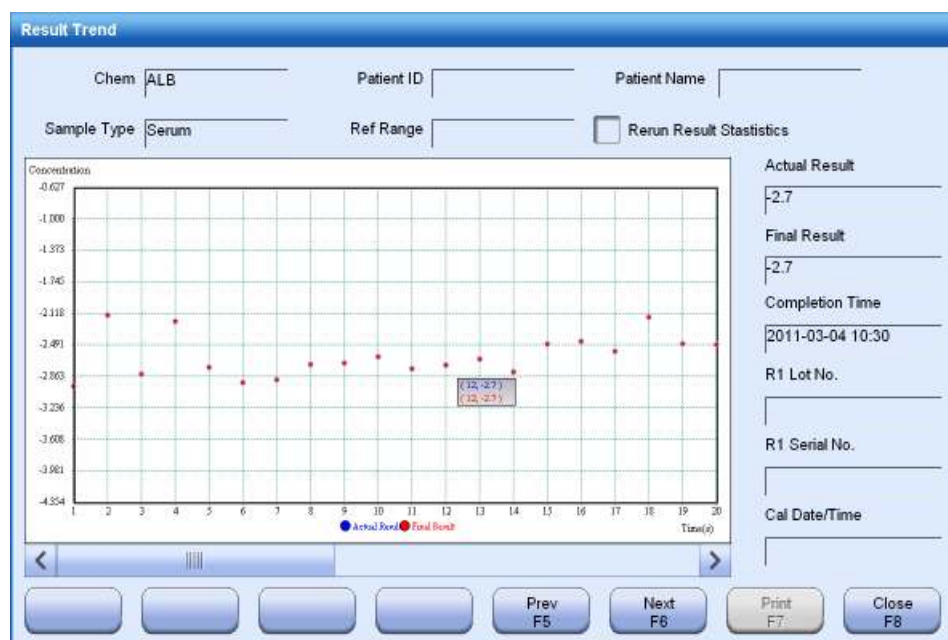
8.12.16 Recalling Result Trend

Result trend allows you to observe the result trend of the selected chemistry.

- 1 Select **Result-Current Results** or **History Results**.
- 2 Choose the result recall mode – By chemistry.
- 3 Select **Search F1** to search for desired results.
- 4 Choose a chemistry in the left list.

- 5 Select **Options F2**.
- 6 Select **Recall Result Trend**.

Figure 8.46 “Result Trend” window



The result trend curve of the selected chemistry is displayed on the window.

- 7 Move the cursor to certain point on the graphic trend. The actual result, final result, completion time, reagent lot number, serial number, and calibration time are displayed on the right of the window.
- 8 To show all results of repeated analysis or rerun tests, select the **Include Replicate Results** checkbox.
- 9 To observe result trend of other sample types, select **Prev F1** or **Next F2**.
- 10 Select **Print F7** to print the current trend curve.
- 11 Select **Exit F8** to close the window.

8.12.17 Archiving Results

The system allows archiving of sample results to a storage device. The file format is CSV and the default file name is SampleResultYYYYMMDD.csv. which cannot be edited. Sample results must not be archived to the hard disk.

The archived sample results include the following information:

Table 8.2 Archived sample result types

ID	Sample Result Information
1	Sample type (routine, STAT, control)
2	Sample ID
3	Sample bar code
4	Sample type (serum, plasma, etc)
5	Collecting date
6	Chemistry
7	Result
8	Response
9	Number of replicates
10	Unit
11	Result flag
12	Reference range
13	Run date and time
14	Module

Perform the following steps to archive sample results and data:

- 1 Search for desired sample results on the **Current Results** or **History Results** screen.

**NOTE**

It may take a long time to archive a large amount of results. You are recommended not to archive results over one week each time.

- 2 Select **Options F2**.
- 3 Select **Archive**.
- 4 Select **OK**.

8.13 Workload Statistics

On the Workload screen, you can view test requests and reagent application for each chemistry during a period, and you can sample requests and the quantity of its chemistries as well. Calibration test and QC test are not included in the statistics.

1 Select **Result-Statistics-Workload**.

2 Select **By Sample** or **By Chemistry**.

Figure 8.47 Workload screen-by sample



Figure 8.48 Workload screen-by chemistry



- By Sample: To view all requested samples and the quantity of its requested chemistries.
- By Test: To view test requisitions and reagent volume for the chemistries.

- 3 Select or enter the start date and end date in the **Date** field. the start date cannot be later than the end date.

4 Select **Search F1**.

All samples or tests requested during the period are displayed in the middle list of the **Workload** screen.

- 5 Select **Print** to print out the currently-displayed statistic information of measurements.

8.14 Result Statistics

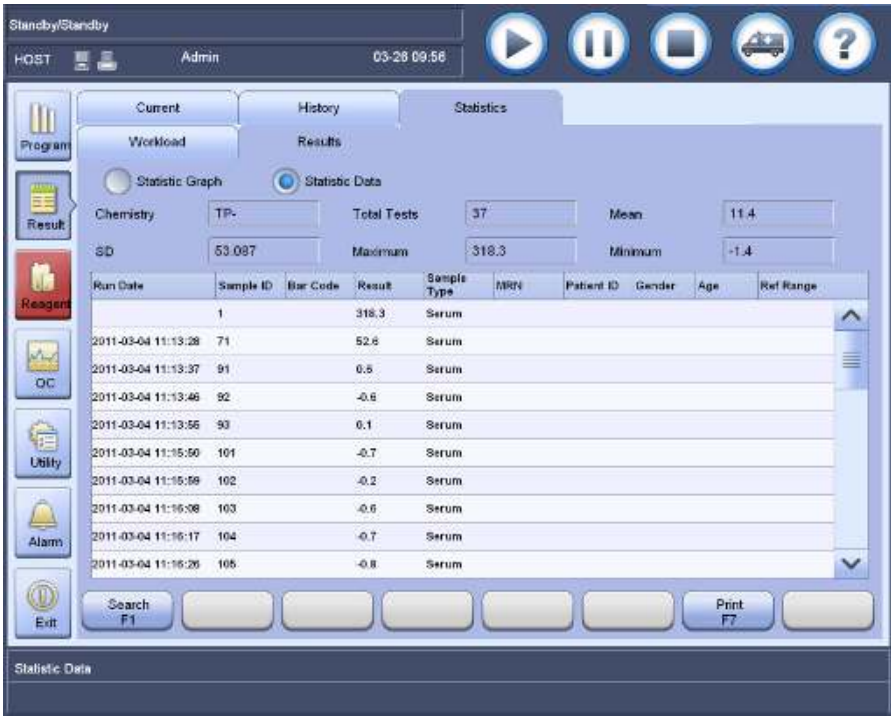
Result statistics option can summarize the total chemistries and the distribution trend of its results and provide the test data and graph. Calibration and control tests are not included in the statistics.

- 1 Select **Result-Statistics**.
- 2 Select the **Results**.
- 3 Select **Statistic Graph** or **Statistic Data**.

Figure 8.49 Result statistics screen -statistic graph



Figure 8.50 Result statistics screen -statistic data



4 Click **Search F1** and **Recall results** box pops up.

The 'Recall Results' dialog box is shown with the following fields and options:

- Program Date:** Two date pickers showing '2011- 3- 4' and '2012- 3-26'.
- Chemistry:** A dropdown menu with 'TP-' selected.
- Sample Type:** A dropdown menu.
- Gender:** A dropdown menu.
- Age:** Two input fields for age and a dropdown menu for 'Year'.
- MRN:** An input field.
- Patient ID:** An input field.
- Sample ID:** An input field with a hint 'e.g. 1, 3, 5-12'.
- Bar Code:** An input field.

At the bottom, there are 'OK' and 'Cancel' buttons.

5 Input one or more search conditions such as:

- Date
- Chemistry
- Sample type
- Gender
- Age
- Medical Record
- Patient ID
- Sample ID
- Bar code

6 Click **OK**.

The chemistry, number of tests, mean, standard deviation, maximum value, minimum value are displayed in the statistic results column. The statistic data include test time, result, sample ID, bar code, patient ID, medical number, sample type, reference range, sex and age.

7 Select **Print F7** to print out the statistic graph and statistic data.

9

Result Printouts

This chapter describes data archiving, print setup, auto print and manual print methods, and result printouts.

9.1 Data Import and Export

9.1.1 Introduction

The Data Import and Export function allows various data to be imported from or exported to an external storage device. Exporting data is allowed only when the system status is Standby, Incubation and Failure.

The following data can be imported:

- Open-/Closed-reagent chemistries, including biochemistries, Off-system chemistries, ISE chemistries, SI and calculations

The following data can be exported:

- Sample results (including results of all replicates): transmitted to LIS host
- Control results: transmitted to LIS host
- QC data: archived to external storage device
- Calibration results: archived to external storage device
- Open-reagent chemistries: exported to a .text file

9.1.2 Import/Export Chemistries

The system supports specified and default chemistries to be imported from an external file.

If an imported closed-reagent chemistry is no longer needed, it can be deleted with the **Delete F2** button on the **Chemistries** screen. Ensure the following conditions are met prior to deleting a closed-reagent chemistry:

- The system is not running tests.
- The selected chemistry is not requested or run for samples, calibrators and controls.
- The selected chemistry is disabled.
- The corresponding reagent has been unloaded from the reagent carousel.

Import default chemistry list

Those chemistries imported from the default parameter form can be run only based on reagents manufactured by our company. Only the print name, result unit, decimal places, error detection limits, auto rerun dilution factor and slope/offset can be modified and deleted, while the others can only be browsed.

- 1 Select **Utility-Chemistries**.

- 2 Select **Config F3**.
- 3 Select **Options**.
- 4 Select **Import**.
- 5 Select **Load Default**.

All chemistries contained in the default parameter form are displayed in the **Available Chemistries** list.

- 6 Use the following buttons to import desired chemistries:
 - **Add All>>**: add all chemistries in the **Available Chemistries** list to the **Imported Chemistries** list.
 - **Add ->**: add the selected chemistries in the **Available Chemistries** list to the **Imported Chemistries** list.
 - **<-Remove**: remove the selected chemistries from the **Imported Chemistries** list.
 - **<<Remove All**: remove all chemistries from the **Imported Chemistries** list.
- 7 Select **Import**.

All imported chemistries are enabled by default and can be used for measurement. If the result unit is changed, the corresponding chemistry must be recalibrated.

- 8 Select **Exit**.

Import specified chemistry list

Open-reagent chemistries can be imported from a .csv file, while closed-reagent chemistries can only be imported from an .item file. A maximum of 300 open-/closed-reagent chemistries can be imported, and the number of newly-imported chemistries must not exceed 200. The open-reagent chemistries include biochemistries, as well as the processing parameters, error detection limits, slop and offset, dilution factors, and sample type. The closed-reagent chemistries include biochemistries, ISE chemistries, SI and special calculations, as well as carryover pairs, reagent type, biochemistry calibration settings, ISE calibration settings, unit conversion rules, processing parameters, error detection limits, carryover settings and slop and offset.

When chemistries are imported, they are enabled by default if set up correctly. If the number of open-reagent chemistries imported exceeds the maximum limit, the excessive open-reagent chemistries will be disabled.

Only users with sufficient permission are allowed to import chemistries. Importing chemistries can be performed only when the system status is Standby, Incubation, Stop and Sleep.

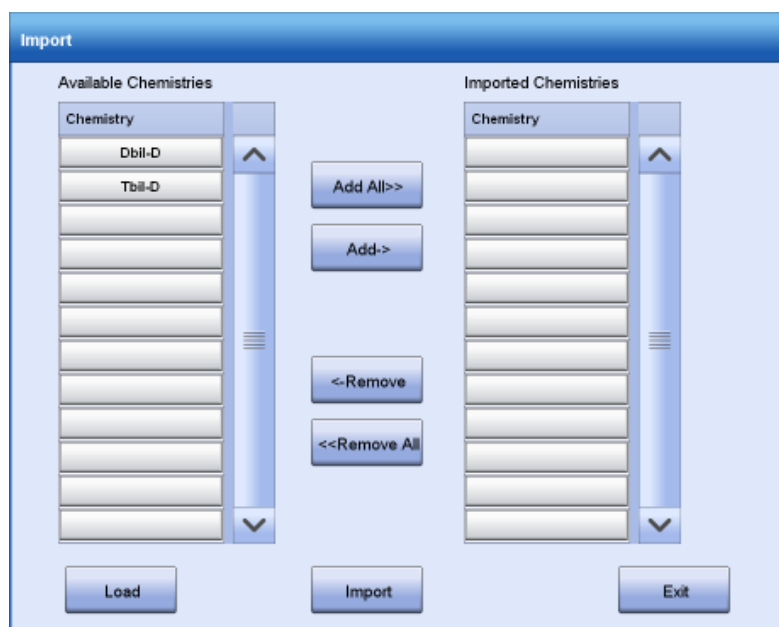


CAUTION

While importing chemistries, do not switch off the analyzing unit main power or exit the operating software.

- 1 Select **Utility-Chemistries**.
- 2 Select **Config F3**.
- 3 Select **Options**.
- 4 Select **Import**.

Figure 9.1 Import window



- 5 Select **Load**.
- 6 Locate the path of the parameter form, and then select it.
 - To import open-reagent chemistries, choose a .csv file.
 - To import closed-reagent chemistries, choose an .item file.
- 7 Select **Open**.

All chemistries contained in the parameter form are displayed in the **Available Chemistries** list.

8 Use the following buttons to import desired chemistries:

- **Add All>>**: add all chemistries in the **Available Chemistries** list to the **Imported Chemistries** list.
- **Add ->**: add the selected chemistries in the **Available Chemistries** list to the **Imported Chemistries** list.
- **<-Remove**: remove the selected chemistries from the **Imported Chemistries** list.
- **<<Remove All**: remove all chemistries from the **Imported Chemistries** list.

9 Select **Import**.

All imported chemistries with correct parameters are enabled by default and can be used for measurement. If you change any of the following parameters of an imported chemistry, recalibrate the chemistry:

- Reaction type
- Primary wavelength
- Secondary wavelength
- Reaction direction
- Reaction time
- Blank time
- Result unit
- Sample volume
- Reagent volume (R1, R2, R3 and R4)
- Sample dilution factors
- Sample blank

10 Select **Exit**.

Export chemistries

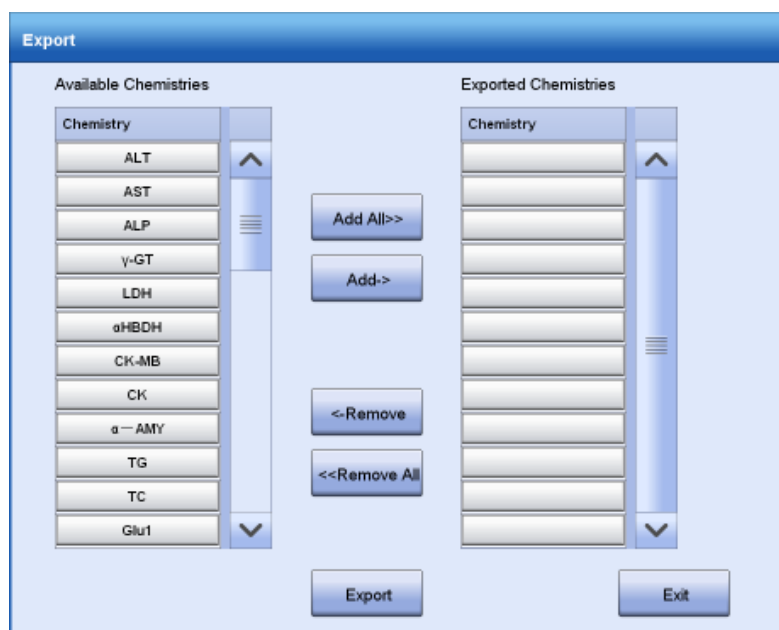
Open-reagent chemistries rather than closed-reagent chemistries can be exported, as well as the processing parameters, error detection limits and slop and offset. Only the open-reagent biochemistries can be exported from the system.

Only users with sufficient permission are allowed to export chemistries. Exporting chemistries can be performed when the system status is Standby, Incubation and

Failure.

- 1 Select **Utility-Chemistries**.
- 2 Select **Config F3**.
- 3 Select **Options**.
- 4 Select **Export**.

Figure 9.2 Export window



The **Available Chemistries** list shows all open-reagent chemistries other than those that have been masked or disabled.

- 5 Use the following buttons to export desired chemistries:
 - **Add All>>**: add all chemistries in the **Available Chemistries** list to the **Exported Chemistries** list.
 - **Add ->**: add the selected chemistries in the **Available Chemistries** list to the **Exported Chemistries** list.
 - **<-Remove**: remove the selected chemistries from the **Exported Chemistries** list.
 - **<<Remove All**: remove all chemistries from the **Exported Chemistries** list.
- 6 Select **Export**.
- 7 Select the path to export and input the file name.

The default file name is composed of the current date and time, such as 20100527_0951. The file format is .csv.

8 Select **Save**.

9 Select **Exit**.

9.1.3 Data Archive

You are recommended to regularly archive the ISE and biochemistry calibration results to the hard disk or a storage device, and archive QC data to an external storage device, such as U disk and floppy disk.

Archiving biochemistry calibration results

When archived, the biochemistry calibration results are displayed in the same format as those on software screens. The archived content includes: chemistry name, flag, calibration status, R0, K factor, calibration coefficients A/B/C/D and run date/time. The archiving file is of .csv format and named by date and time the results are archived.

For more information of archiving biochemistry calibration results, refer to 6.9.6 Archiving Calibration Results (page 6-29).

Archiving ISE calibration results

Both the current and early calibration factors of ISE chemistries can be archived. The archiving file is of .csv format and named by date and time the results are archived.

For more information of archiving ISE calibration results, refer to Archiving ISE calibration results (page 12-19).

Archiving QC data

The QC results and data can be archived to a storage device with the file name of QCData.csv, which cannot be edited.

For more information of archiving QC data, refer to Archive QC data (page 7-21).

9.1.4 Sending sample results and QC results to LIS

Sample results and QC results can be sent manually or in real-time mode to the LIS host for reviewing and storage. When a sample is analyzed with its all tests finished, the system can automatically send the test results to the LIS host; also you are allowed to search for desired results and then manually send them to LIS.

For more information about sending sample/QC results to LIS, refer to 14.4 Result Transmission (page 14-13).

9.2 Print Setup

9.2.1 Introduction

Results and data can be printed out with the specified template through the default printer. You are allowed not only to set up the printer type, default printer and printed hospital name, import print report but also define the print order of chemistries, edit print template and preview print template.

9.2.2 General Print Setup Options

1 Select **Utility-System Setup**.

2 Select **Print F3**.

Figure 9.3 Print setup screen

No.	Template Name	Paper Size	Default
01	Patient Report(1)	A4	Y
02	Patient Report(2)	B5	N
03	Patient Report(3)	Letter	N
04	Patient Report(4)	A42	N

3 Select a report type from the report list on the left of the window.

4 Select a template from the template list.

- The template list includes all the report templates of the selected report type.
- The template list has four columns. Only one template can be selected at the same time.

5 Select one option from the check box:

- Auto print patient report
 - Print after reviewal
 - Auto print QC report.
 - Auto print calibration report
- 6 Choose a printer type.

The system supports three types of printer, which include laser printer, inkjet printer and stylus printer.

- 7 Choose a default printer to print reports.
- 8 Choose a print mode between Paginal and Serial.
- 9 Select **OK**.

9.2.3 Set Up Default Template

- 1 Select **Utility-System Setup**.
- 2 Select **Print F3**.
- 3 Select a report type from the **Report** list on the left of the window.
- 4 Select a template from the template list.
- 5 Click **Set Defaults** to set the selected template in the template list as the default one.

9.2.4 Delete Template

- 1 Select **Utility-System Setup**.
- 2 Select **Print F3**.
- 3 Select a report type from the **Report** list on the left of the window.
- 4 Select a template from the template list.
- 5 Click **Delete** to delete the selected template.

If it is the default one or it has the print duty, it cannot be deleted.

9.2.5 Edit Print Template

- 1 Select **Utility-System Setup**.

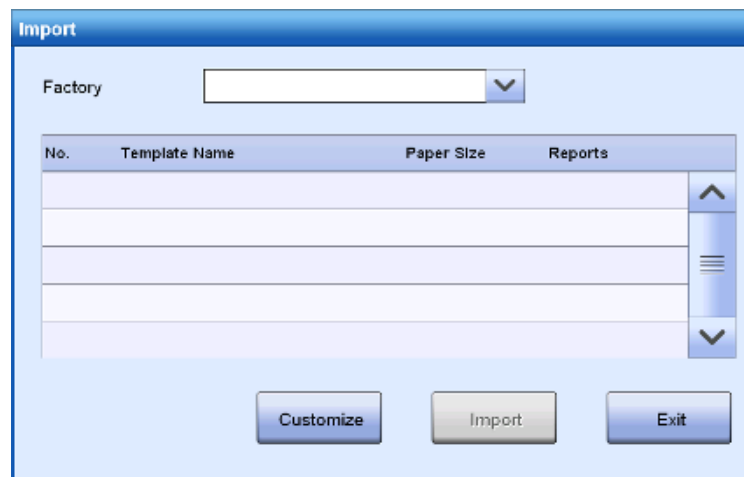
- 2 Select **Print F3**.
- 3 Select a report type from the **Report** list on the left of the window.
- 4 Select a template from the template list.
- 5 Click **Edit** to open the template modifying software

You can edit the report templates as needed. Refer to 18 Template Modifying Software for details.

9.2.6 Import Print Template

- 1 Select **Utility-System Setup**.
- 2 Select **Print F3**.
- 3 Click **Import** to import the print template.

Figure 9.4 Import print template window



- 4 Select a report type from the factory template dropdown box.
 - 5 Select one or more templates in the template list to be imported.
 - 6 Click **Import**.
- The selected template(s) can be imported
- 7 Click **Customize** to import the template you edited from a tmplt file.

The legal directory should be a portable storage device. The templates can be imported in batch.

- 8 Click **Import**

The imported templates will be displayed in the template list.

- 9 Click **Exit** to exit the window.

9.2.7 Defining Chemistry Print Order

- 1 Select **Utility-System Setup**.
- 2 Select **Print F3**.
- 3 Select **Print Order**.

Figure 9.5 Print Order window



- 4 Use the following buttons to adjust the chemistry print order:
 - **Home:** to move the chemistry to the first position.
 - **Up:** to move the chemistry to the previous position.
 - **Down:** to move the chemistry to the next position.
 - **End:** to move the chemistry to the last position.
- 5 Set up result print mode.
 - To print results on patient report, select the corresponding **Print** checkbox.
 - To forbid printing results on patient report, deselect the corresponding **Print** checkbox or leave it unselected.

9 Result Printouts

- 6 Select **OK** to save your settings.
- 7 To restore the factory settings, select **Restore Defaults**.

9.3 Sample Reports

9.3.1 Introduction

Sample reports are used to print sample results, sample list, reaction curve and data, as well as sample blank reaction curve and data.

The result reports include:

- Single sample report
- Multi-sample report
- Report collection
- Chemistry summary

The list reports include:

- Sample list report
- Control list report
- Chemistry list report

The above-mentioned reports and printing methods are described in detail in the following sections.

9.3.2 Single Sample Report

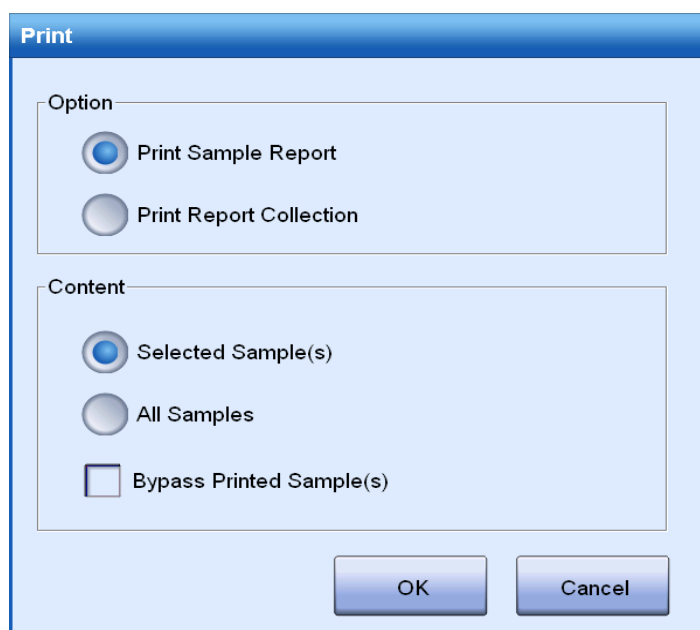
A single sample report contains all results of a sample, including emergent sample, routine sample and control sample. It can be printed out on:

- **Current Results** screen
- **History Results** screen

Print a single sample report by performing the following steps:

- 1 Select **Result-Current Results** or **History Results**.
- 2 Choose the **By Sample** option.
- 3 Search for desired results to print.
- 4 Choose a sample.
- 5 Select **Print F7**.

Figure 9.6 Print sample results window



- 6** Select **Print Sample Report**.
- 7** Select the **Selected Sample(s)** option button.
- 8** Select **OK**.

Figure 9.7 Single sample report example

Patient:	Sample ID:	1
Patient ID:	Sample Type:	Serum
Date of Birth:	Bar Code:	
Age:	Collection Date:	2012-08-09
Gender:	Collection Time:	
Doctor:	Department:	
Diagnosis:	Comment:	

Chemistry	Result	Unit	Flag	Ref Range
U/BIC	-21.9	µg/L		
U/BIC	-21.9	µg/L		
U/BIC	-24.8	µg/L		
U/BIC	-22.7	µg/L		
U/BIC	-22.6	µg/L		
U/BIC	-22.7	µg/L		
U/BIC	-22.7	µg/L		
U/BIC	-23.1	µg/L		

Ordering Date/Time:	2012-08-09	Test Date/Time:	2012-08-09	Print Date/Time:	2013-10-12 11:16:32
Tester:		Reviewer:			

The result are for this sample only

Page 2 of 2

9.3.3 Multi-Sample Report

A multi-sample report can print two or more samples of a patient on a report. If the patient demographics of the samples are not consistent, the demographics of the first will be printed by default. A multi-sample report can be printed out on:

- **Current Results** screen
- **History Results** screen

Print a multi-sample report by performing the following steps:

- 1 Select **Result-Current Results** or **History Results**.
- 2 Choose the **By Sample** option.
- 3 Search for desired results to print.
- 4 Choose results to print.
- 5 Select **Options F2**.
- 6 Select **Print Multi-Sample Report**.

Figure 9.8 Multi-sample report example

Patient:		Patient ID:	
Date of Birth:		Age:	
Department:		Gender:	
Doctor:		Diagnosis:	
Comment:			

Sample ID	Bar Code	Type	Chemistry	Result	Unit	Flag	Ref Range	Collection Date	Test Date/Time
1		Serum	CREA	-0.2	μmol/L				2011-03-04 09:56:20 AM
		Serum	UA	-7.0	μmol/L				2011-03-04 09:56:20 AM
		Serum	UREA	-0.64	mmol/L				2011-03-04 09:56:20 AM
2		Serum	CREA	-0.7	μmol/L				2011-03-04 09:56:33 AM
		Serum	UA	-8.0	μmol/L				2011-03-04 09:56:33 AM
		Serum	UREA	-0.96	mmol/L				2011-03-04 09:56:33 AM

Print Date/Time:	2011-10-18 04:58:27 AM	Reviewer:		Page	1 of 1
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9.3.4 Print Report Collection

Report collection contains the test results of all samples and QC samples for the purpose of archiving and internal audit of the clinical laboratories.

- 1 Select **Result-Current Results** or **History Results**.
- 2 Choose the **By Sample** option.
- 3 Search for desired results to print.
- 4 Select **Print F7**.

Figure 9.9 Print window

The 'Print' dialog box contains two main sections: 'Option' and 'Content'. In the 'Option' section, 'Print Sample Report' is selected with a radio button, while 'Print Report Collection' is unselected. In the 'Content' section, 'Selected Sample(s)' is selected with a radio button, 'All Samples' is unselected, and 'Bypass Printed Sample(s)' is unselected with a checkbox. At the bottom right, there are 'OK' and 'Cancel' buttons.

5 Select **Print Report Collection**.

6 Select **OK**.

All inquired sample results and the results of the QC samples will be printed out; however, only valid results can be printed. If a sample has been tested for a same chemistry for several times, the results of each time will be printed out.

Figure 9.10 Sample summary report

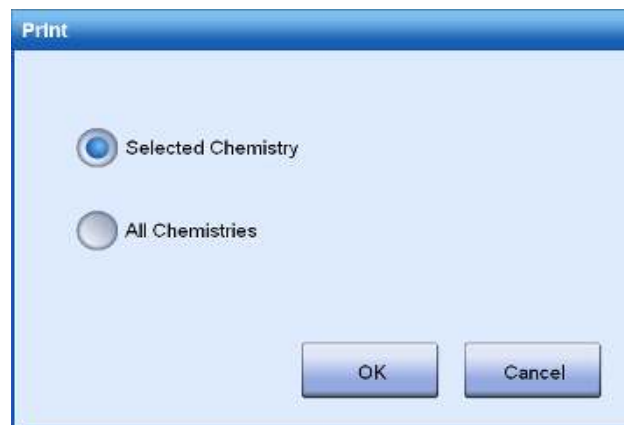
Sample Summary Report					
6				E	2012-09-14 09:52:15
AST	203.6	U/L			
AST	199.8	U/L			
AST	201.5	U/L			
AST	202.7	U/L			
AST	206.1	U/L			
7				E	2012-09-14 10:19:01
PA	256.8	mg/L			
PA	221.7	mg/L			
PA	258.6	mg/L			
PA	221.3	mg/L			
PA	275.8	mg/L			
PA	223.4	mg/L			
PA	259.0	mg/L			
PA	220.2	mg/L			
PA	255.9	mg/L			
PA	223.4	mg/L			
9				E	2012-09-14 11:38:18
CK	632.0	U/L			
CK	626.7	U/L			
CK	635.6	U/L			
CK	629.7	U/L			
37	521234567897			E	2012-09-12 14:53:30
ALP	93.6	U/L			
ALB	33.8	g/L			
38	521234567896			E	2012-09-12 14:53:34
v-GT	247.7	U/L			
CREA	283.1	μmol/L			
39	521234567895			E	2012-09-12 11:12:16
MG	-0.02	mmol/L			
TG	0.00	mmol/L			
40	521234567894			E	2012-09-12 11:12:20
MG	-0.02	mmol/L			
TG	0.00	mmol/L			
41	521234567893			E	2012-09-12 11:12:24
Glu	-0.10	mmol/L			
Print Date/Time: 2012-11-15 10:22:15				Page 1 of 371	

9.3.5 Chemistry Summary

Chemistry summary report contains the summary of a chemistry or all chemistries.

- 1 Select **Result-Current Results** or **History Results**.
- 2 Choose the **By Chemistry** option.
- 3 Search for desired results to print.
- 4 To print single chemistry in the chemistry list, select it; to print all chemistries, there is no need to select them.
- 5 Select **Print F7**.

Figure 9.11 Print window



- 6 Choose the print range:
 - Selected chemistry
 - All chemistries
- 7 Select **OK**.

9.3.6 Sample List Report

A sample list report contains all incomplete samples and patient demographics. It can be printed out on the **Sample List** screen.

- 1 Select **Program-Sample**.
- 2 Select **List F5**.
- 3 Select **Print F7**. All incomplete samples are printed with the specified template.

Figure 9.12 Sample list report example

Sample List					
Program Date/Time	Sample ID	Position	Patient	Chemistries	Status
2011-10-18 04:09:47 AM	1	1-1		ADA, ASO, C4, CHE, Cl, K, Na, TP-	Rerun
2011-10-18 05:00:03 AM	2	1-2		Cl, K, Na	Programmed
2011-10-18 05:00:04 AM	3	1-3		TP-, laohua	Programmed
2011-10-18 05:00:05 AM	4	1-4		CO2, CRP, w1, w2	Programmed
2011-10-18 05:00:07 AM	5	1-5		ASO, CRP, Ca, w2	Programmed
Print Date/Time:			2011-10-18 05:00:09 AM		
			Page 1 of 1		

9.3.7 Control List Report

A control list report contains all incomplete samples and control information. It can be printed out on the **Sample List** screen.

- 1 Select **Program-Sample**.
- 2 Select **List F5**.
- 3 Select **Print F7**.

All incomplete controls are printed with the specified template. If both patient samples and controls are shown on the screen, they will be printed out with different templates.

Figure 9.13 Control list report example

Control List					
Program Date/Time	Control	Position	Lot No.	Chemistries	Status
2011-10-18 05:01:34 AM	Randox2-M1	1-C1		ALB, ALP, ALT, AST, ApoB, CREA-S, Ca D-Bil-V, D-bil-D, Glu-G, Glu-H, HDL-C, LDL-C, T-Bil-V, T-bil-D, TC, TG, TP, TP- UA, UREA, γ-GT	Programmed
Print Date/Time:			2011-10-18 05:01:37 AM		
			Page 1 of 1		

9.3.8 Chemistry List Report

A chemistry list report contains all unfinished chemistries. It can be printed out on the **Chemistry List** screen.

- 1 Select **Program-Sample**.
- 2 Select **List F5**.
- 3 Select the **Chemistry List** tab.
- 4 Select **Print F7**. All incomplete chemistries are printed with the specified

template.

Figure 9.14 Chemistry list report example

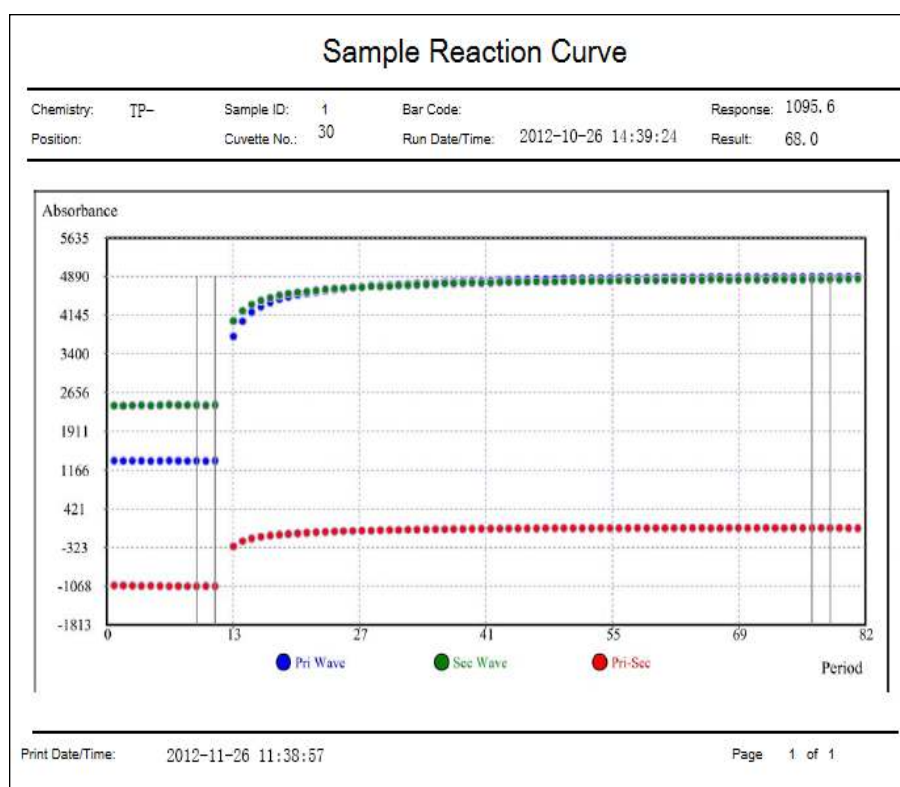
Chemistry List			
Chemistry	Cal Status	Chems Requested	Tests Left
ADA		1	
ALB	Cal Required	1	
ALP	Cal Required	1	
ALT	Cal Required	1	
ASO		1	
AST	Cal Required	1	
ApoB	Cal Time Out	1	
CK-MB	Cal Required	1	
CO2		1	
CREA-S	Cal Required	1	
CRP	Calibrated	1	
Ca	Cal Required	2	
D-Bil-V	Cal Required	1	
D-bil-D		1	
Glu-G	Cal Required	1	
Glu-H	Cal Required	1	
HDL-C	Cal Required	1	
LDL-C	Cal Required	1	
T-Bil-V	Cal Required	1	
T-bil-D		1	
TC	Cal Required	1	
TG	Cal Required	1	
TP	Cal Required	1	
TP-	Calibrated	1	245
UA	Cal Required	1	
UREA	Cal Required	1	
laohua	Extended	1	534
w1	Calibrated	1	
w2		1	
γ-GT	Cal Required	1	
Print Date/Time: 2011-10-18 05:02:16 AM Page 1 of 1			

9.3.9 Sample Reaction Curve and Data

Sample reaction curve and data can be printed out on the **Current Results** and **History Results** screens.

- 1** Select **Result-Current Results** or **History Results**.
- 2** Search for desired results to print.
- 3** Choose a sample.
- 4** Select **Reac Curve F4**.
- 5** Select **Print F7** to print the reaction curve.

Figure 9.15 Sample reaction curve example



6 Select the **Reaction Data** tab.

7 Select **Print F7** to print the reaction curve data.

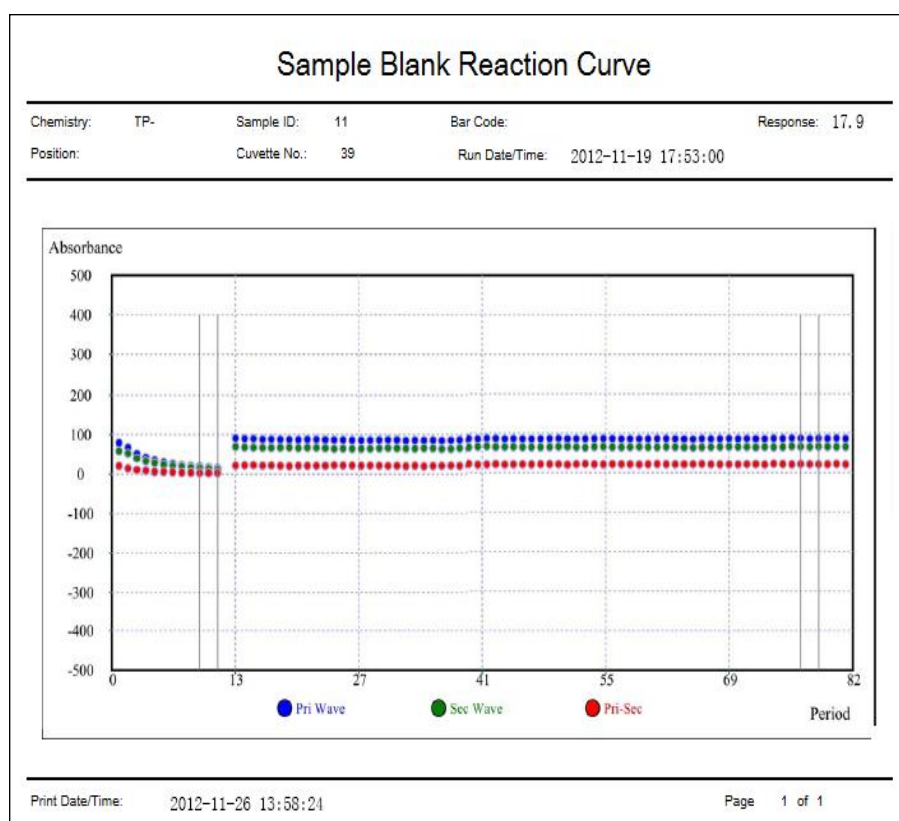
Figure 9.16 Sample reaction curve data example

Sample Reaction Data							
Chemistry:	TP-	Sample ID:	1	Bar Code:		Response:	1095.6
Position:		Cuvette No.:	30	Run Date/Time:	2012-10-26 14:39:24	Result:	68.0
Period	Pri Wave	Sec Wave	Pri-Sec	Period	Pri Wave	Sec Wave	Pri-Sec
1	1350.56	2403.87	-1053.31	2	1346.72	2400.73	-1054.01
3	1348.00	2406.11	-1058.11	4	1348.85	2411.05	-1062.20
5	1344.16	2403.42	-1059.26	6	1347.15	2411.50	-1064.35
7	1352.27	2420.49	-1068.22	8	1348.00	2413.75	-1065.75
9	1345.44	2413.75	-1068.31	10	1348.43	2415.55	-1067.12
11	1342.88	2407.91	-1065.03	12	1348.43	2416.89	-1068.47
14	3737.23	4035.17	-297.94	15	4028.54	4226.79	-198.26
16	4200.42	4349.44	-149.02	17	4305.93	4423.31	-117.38
18	4390.30	4482.97	-92.67	19	4451.07	4526.98	-75.91
20	4498.77	4560.86	-62.09	21	4537.48	4587.94	-50.46
22	4565.86	4604.11	-38.25	23	4593.71	4623.21	-29.50
24	4621.65	4642.36	-20.70	25	4642.20	4655.72	-13.52
26	4655.93	4663.87	-7.94	27	4677.18	4680.18	-3.00
28	4687.20	4684.85	2.35	29	4707.91	4705.89	2.02
30	4711.68	4701.79	9.89	31	4721.74	4707.06	14.68
32	4738.11	4723.45	14.66	33	4745.68	4724.62	21.05
34	4755.78	4734.01	21.77	35	4766.52	4740.47	26.05
36	4774.11	4744.58	29.53	37	4787.41	4759.87	27.54
38	4789.95	4758.10	31.85	39	4798.20	4765.76	32.44
Print Date/Time: 2012-11-26 11:38:57				Page 1 of 1			

9.3.10 Sample Blank Reaction Curve and Data

For endpoint measurement of single-reagent chemistries, the sample blank reaction curve and data can be printed out on the **Current Results** and **History Results** screens.

- 1 Select **Result-Current Results** or **History Results**.
- 2 Search for desired results to print.
- 3 Choose a sample.
- 4 Select **Reac Curve F4**.
- 5 Select **Sample Blank F2**.
- 6 Select **Print F7** to print the sample blank reaction curve.

Figure 9.17 Sample blank reaction curve example

- 7** Select the **Reaction Data** tab.
- 8** Select **Print F7** to print the sample blank reaction curve data.

Figure 9.18 Sample blank reaction curve data example

Sample Blank Reaction Data							
Chemistry:	TP-	Sample ID:	11	Bar Code:		Response:	17.9
Position:		Cuvette No.:	39	Run Date/Time:	2012-11-19 17:53:00		
Period	Pri Wave	Sec Wave	Pri-Sec	Period	Pri Wave	Sec Wave	Pri-Sec
1	79.25	58.70	20.55	2	67.36	52.48	14.88
3	51.00	40.25	10.75	4	42.26	33.27	8.98
5	35.26	29.01	6.25	6	30.22	23.97	6.24
7	26.73	21.26	5.46	8	22.85	18.75	4.10
9	20.71	16.81	3.90	10	19.36	16.04	3.32
11	17.03	13.91	3.12	12	15.48	12.17	3.31
14	90.77	69.40	21.38	15	89.80	67.84	21.96
16	89.80	67.26	22.54	17	88.23	66.87	21.37
18	88.43	66.67	21.76	19	88.04	67.26	20.78
20	87.45	67.45	20.00	21	87.26	66.28	20.97
22	88.04	67.26	20.78	23	87.84	67.06	20.78
24	87.06	66.09	20.97	25	86.67	64.53	22.14
26	86.67	65.12	21.55	27	86.09	64.53	21.55
28	85.69	64.73	20.97	29	86.09	64.73	21.36
30	86.48	65.51	20.97	31	86.87	66.28	20.58
32	86.67	65.89	20.78	33	84.91	65.12	19.80
34	85.89	64.92	20.97	35	85.30	65.89	19.41
36	86.09	65.70	20.38	37	84.91	64.14	20.77
38	85.69	64.34	21.36	39	86.67	66.09	20.58
40	86.48	64.53	21.94	41	85.69	65.89	19.80

Print Date/Time: 2012-11-26 14:07:20 Page 1 of 1

9.3.11 QC Reaction Curve and Data

The reaction curve and data of the QC samples can be printed out on QC reaction curve report and QC reaction data report.

- 1 Search for desired control results on the **Current Results** or **History Results** screen.
- 2 Choose a chemistry in the result list.
- 3 Select **Reac Curve F4**. The **Reaction Curve** window is displayed.
- 4 Choose the **Reaction Data** tab to view the reaction data.
- 5 Select **Print F7** to print the reaction curve or reaction data.

Figure 9.19 QC reaction curve report

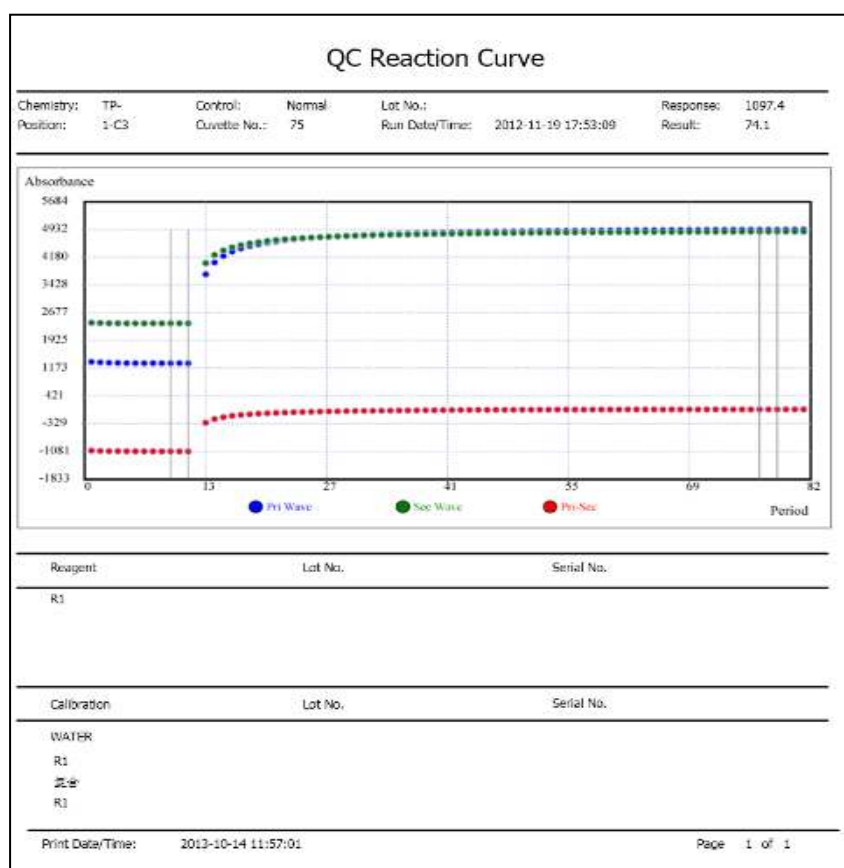


Figure 9.20 QC reaction data report

QC Reaction Data							
Chemistry:	TP-	Control:	Normal	Lot No.:		Response:	1097.4
Position:	1-C3	Cuvette No.:	75	Run Date/Time:	2012-11-19 17:53:09	Result:	74.1
Period	Pri Wave	Sec Wave	Pri-Sec	Period	Pri Wave	Sec Wave	Pri-Sec
1	1344.47	2403.58	-1059.11	2	1331.10	2398.94	-1067.83
3	1319.56	2392.23	-1072.66	4	1317.08	2390.42	-1073.35
5	1311.88	2388.10	-1076.22	6	1310.97	2388.36	-1077.38
7	1308.94	2388.10	-1079.16	8	1311.20	2390.42	-1079.22
9	1308.94	2389.13	-1080.19	10	1309.39	2390.16	-1080.77
11	1310.52	2390.68	-1080.16	12	1307.13	2388.87	-1081.74
14	3716.20	4019.86	-303.66	15	4038.25	4241.69	-203.43
16	4211.28	4364.79	-153.51	17	4330.71	4447.41	-116.69
18	4414.59	4507.09	-92.50	19	4480.07	4553.94	-73.87
20	4529.67	4588.75	-59.08	21	4571.99	4620.36	-48.38
22	4607.57	4645.74	-38.16	23	4638.00	4667.16	-29.16
24	4663.89	4683.60	-19.71	25	4686.51	4699.74	-13.22
26	4705.19	4712.19	-7.00	27	4721.57	4723.32	-1.75
28	4736.97	4733.44	3.53	29	4750.39	4743.92	6.47
30	4764.16	4752.72	11.45	31	4775.94	4760.51	15.43
32	4786.72	4768.98	17.74	33	4795.49	4775.43	20.06
34	4804.94	4782.22	22.72	35	4811.70	4787.31	24.38
36	4820.83	4792.75	28.08	37	4827.94	4798.19	29.74
38	4834.03	4802.28	31.76	39	4840.14	4806.71	33.43
40	4845.23	4810.45	34.77	41	4851.00	4814.88	36.11
42	4857.11	4817.61	39.50	43	4860.51	4820.68	39.83
44	4864.93	4823.75	41.18	45	4870.37	4826.82	43.55
46	4873.10	4829.90	43.20	47	4878.21	4832.29	45.92
48	4879.57	4835.36	44.20	49	4883.66	4838.10	45.56
50	4887.41	4839.81	47.60	51	4889.79	4842.54	47.25
52	4894.91	4845.28	49.63	53	4896.28	4846.65	49.63
54	4898.67	4847.33	51.33	55	4900.03	4849.04	50.99
56	4901.40	4851.10	50.30	57	4903.79	4853.49	50.30
58	4906.18	4854.18	52.00	59	4906.87	4854.86	52.00
60	4909.60	4856.23	53.37	61	4912.34	4859.32	53.02
62	4913.70	4860.00	53.70	63	4914.39	4860.35	54.04
64	4916.78	4863.09	53.69	65	4916.10	4862.40	53.70
66	4916.44	4864.12	52.32	67	4919.86	4864.80	55.06
68	4920.20	4864.80	55.40	69	4923.63	4867.89	55.74
70	4922.26	4868.58	53.68	71	4922.94	4868.58	54.37
72	4923.97	4868.92	55.05	73	4924.31	4869.60	54.71
74	4926.03	4870.98	55.05	75	4926.71	4870.98	55.73
76	4928.08	4872.69	55.39	77	4927.74	4872.01	55.73
78	4929.79	4873.04	56.76	79	4928.08	4873.38	54.70
80	4930.82	4874.75	56.07	81	4931.16	4875.10	56.07
82	4932.54	4875.78	56.75				
Print Date/Time: 2013-10-14 13:44:51				Page 1 of 1			

9.3.12 Sample Log Report

Sample log report contains the information of the samples and controls whose tests are not completed due to some reasons within 24 hours.

- 1 Select **Program-Status**.
- 2 Select **Log F2**.
- 3 Select **Print F7**.
- 4 Select **Exit F8**.

Figure 9.21 Sample log report

Sample Logs				
Date/Time	Sample ID	Position	Bar Code	Status Causes
2012-04-17 14:23:16	1			Incomplete
Print Date/Time: 2012-04-17 16:44:02		Page 1 of 1		

9.4 Reagent Reports

9.4.1 Introduction

Reagent reports are used to print the list of biochemistries and ISE chemistries, as well as reagent position, reagent type, tests and chemistries left, calibration status, etc. The reagent reports include:

- Biochemistry list report
- ISE chemistry list report

9.4.2 Biochemistry List Report

A biochemistry list report can be printed out on the biochemistry screen. The report contains the following information of all configured biochemistries:

- Reagent position
- Chemistry name
- Chemistries left
- Reagent type
- Tests left
- Days left
- Lot number
- Calibration status
- Calibration time left

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.
- 3 Select the down-arrow button to show the biochemical chemistry screen.
- 2 Select **Print F7**.
- 3 Select **By Chemistry** or **By Position**.
- 4 Select print range.
- 5 Select **OK**.

Figure 9.22 Biochemistry list report example - By chemistry

Reagent/Calibration List								
Carousel Position	1 Chemistry	Chems Left	Reagent Type	Tests Left	Days Left	Lot No.	Cal Status	Time Left
64	ALT	0	R1	0	<-99d	2015	Cal Required	
27			R2		<-99d	1017		
65	ALT-开放		R2	248	<-99d		Calibrated	
51	CREA-J		R1	0	-25d	2010	Cal Required	
52	CREA-S		R1	0	<-99d	2005	Cal Required	
30	GLUHK 罗氏		R2	214	<-99d		Calibrated	
63	TP	0	R1	0	<-99d	2017	Cal Required	
70			R1	0	<-99d	2017		
2	TP-开放		R1	103	<-99d		Calibrated	
3 M			R1		21d			
41	UA	0	R1	0	<-99d	2012	Cal Required	
42			R1	0	<-99d	2012		
44			R1	0	<-99d	2013		
53			R2	0	<-99d	2012		
56			R2	0	<-99d	2012		
61	UREA	0	R1	0	-87d	2018	Cal Required	
62			R2	0	-87d	2018		
43	α -AMY		R1	0	<-99d	2009	Cal Required	
54	β 2-mG		R1	0	<-99d	2008	Cal Required	
48			R1	0	<-99d	2010		
55			R1	0	<-99d	2010		
Print Date/Time: 09/26/2013 08:16:49 PM Page 1 of 1								

Figure 9.23 Biochemistry list report example - By position

Reagent/Calibration List								
Carousel Position	1 Chemistry	Chems Left	Reagent Type	Tests Left	Days Left	Lot No.	Cal Status	Time Left
2	TP-		R1	103	<-99d		Calibrated	
3 M			R1		21d			
27	ALT	0	R2		<-99d	1017	Cal Required	
30	GLUHK		R2	214	<-99d		Calibrated	
41	UA	0	R1	0	<-99d	2012	Cal Required	
42			R1	0	<-99d	2012		
43	α -AMY		R1	0	<-99d	2009	Cal Required	
44	UA	0	R1	0	<-99d	2013	Cal Required	
48	β 2-mG		R1	0	<-99d	2010	Cal Required	
51	CREA-J		R1	0	-25d	2010	Cal Required	
52	CREA-S		R1	0	<-99d	2005	Cal Required	
53	UA	0	R2	0	<-99d	2012	Cal Required	
54	β 2-mG		R1	0	<-99d	2008	Cal Required	
55			R1	0	<-99d	2010		
56	UA	0	R2	0	<-99d	2012	Cal Required	
61	UREA	0	R1	0	-87d	2018	Cal Required	
62			R2	0	-87d	2018		
63	TP	0	R1	0	<-99d	2017	Cal Required	
64	ALT	0	R1	0	<-99d	2015	Cal Required	
65	ALT-		R2	248	<-99d		Calibrated	
70	TP	0	R1	0	<-99d	2017	Cal Required	
Print Date/Time: 09/26/2013 08:20:22 PM Page 1 of 1								

9.4.3 ISE Chemistry List Report

An ISE chemistry list report can be printed out on the ISE chemistry screen. The report contains the following information of all ISE chemistries and wash solutions:

- Chemistry name
- Calibration status
- Calibration run date and time
- Calibration time left
- Reagent name
- Volume %

9 Result Printouts

- Load date
- Days left
- Expiration date
- Lot number
- Serial number

1 Select **Reagent-Reagent/Calibration**.

The screen shows the ISE chemistries and wash solutions of the system.

2 Select **Print F7**.

Figure 9.24 ISE chemistry list report example

Reagent/Calibration List						
Chemistry		Cal Status	Cal Date/Time		Time Left	
ISE	Na	Cal Time Out	2012-06-13 10:33:25		<-99d	
	K	Cal Time Out	2012-06-13 10:33:25		<-99d	
	Cl	Cal Time Out	2012-06-13 10:33:25		<-99d	
Carousel	1					
Reagent	Volume	Load Date	Days Left	Exp Date	Lot No.	Serial No.
ISE Reagent	43%	2012-01-25	4d	2012-11-30	00009E8DE4	
Conc Wash	85%	2012-11-14	>99d	2013-06-28		
Wash D						
Saline W	42%	2012-11-22				
Print Date/Time:		2012-11-26 14:42:07			Page 1 of 1	

9.5 Calibration Reports

9.5.1 Introduction

Calibration reports are used to print the calibration results of biochemistries and ISE chemistries.

9.5.2 Calibrator List Report

A calibrator list report contains the following information:

- Chemistry name
- Calibrator name
- Calibrator position
- Calibrator concentration
- Lot number
- Expiration date

The report can be printed out on the **Reagent/Calibration** screen.

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Select the down-arrow button to show the biochemical chemistry screen.
- 3 Select **Load List F4**.
- 4 Select **Print F7**.

Figure 9.25 Calibrator load list report example

Calibrator List					
Chemistry	Calibrator	Position	Conc	Lot No.	Exp Date
TP-	WATER	W2	0		
	Cal3	1-S6	45.5		2012-02-17
laohua	WATER	W2	0		
	laohua	1-S1	0.1		2012-03-01
Print Date/Time:		2011-10-18 05:08:48 AM		Page 1 of 1	

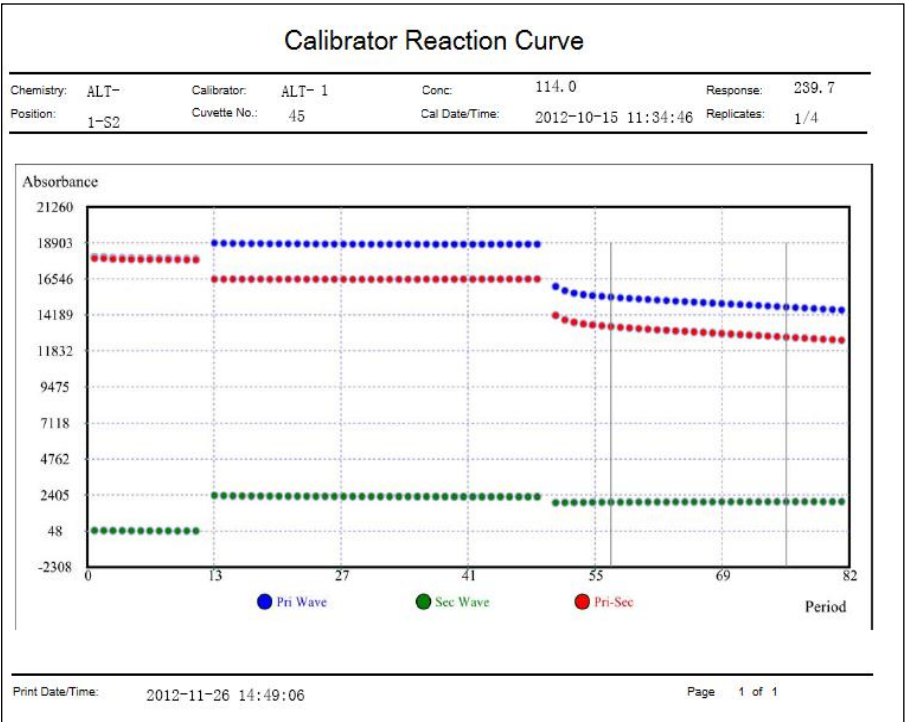
9.5.3 Calibrator Reaction Curve and Data

Calibrator reaction curve and data can be printed out on the **Biochemistry Calibration** screen.

- 1 Select **Reagent-Biochemistry Calibration**.

- 2 Select **Reac Curve F3**.
- 3 Select **Print F7**.

Figure 9.26 Calibrator reaction curve example



- 4 Select the **Reaction Data** tab.
- 5 Select **Print F7** to print the reaction curve data.

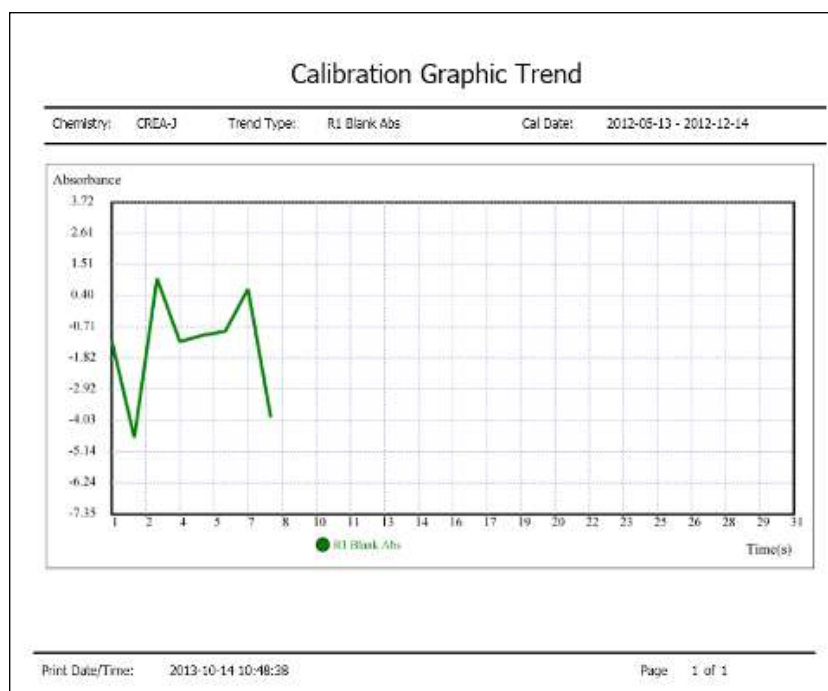
Figure 9.27 Calibrator reaction curve data example

Calibrator Reaction Data							
Chemistry:	ALT-	Calibrator:	ALT- 1	Conc:	114.0	Response:	239.7
Position:	1-S2	Cuvette No.:	45	Cal Date/Time:	2012-10-15 11:34:46	Replicates:	1/4
Period	Pri Wave	Sec Wave	Pri-Sec	Period	Pri Wave	Sec Wave	Pri-Sec
1	17971.27	78.47	17892.80	2	17964.81	74.40	17890.41
3	17914.92	68.70	17846.21	4	17902.09	65.24	17836.84
5	17896.07	61.58	17824.49	6	17876.48	56.91	17819.57
7	17879.68	57.72	17821.96	8	17871.69	53.86	17817.83
9	17868.50	54.47	17814.02	10	17855.73	48.38	17807.35
11	17847.77	53.66	17794.11	12	17841.40	50.21	17791.19
14	18903.16	2380.02	16523.14	15	18888.78	2363.31	16525.47
16	18878.02	2348.22	16529.80	17	18869.05	2343.46	16525.60
18	18863.68	2338.96	16524.72	19	18867.26	2335.53	16531.73
20	18851.16	2331.04	16520.12	21	18851.16	2326.03	16525.13
22	18851.16	2325.23	16525.93	23	18852.95	2324.44	16528.51
24	18840.44	2319.70	16520.75	25	18840.44	2316.53	16523.91
26	18835.09	2315.48	16519.61	27	18831.52	2311.79	16519.73
28	18831.52	2311.26	16520.26	29	18824.39	2310.74	16513.65
30	18824.39	2308.37	16516.02	31	18820.82	2307.05	16513.77
32	18820.82	2304.15	16516.67	33	18824.39	2304.42	16519.97
34	18822.60	2303.10	16519.50	35	18827.95	2304.15	16523.80
Print Date/Time: 2012-11-26 15:00:00				Page 1 of 1			

9.5.4 Calibration Trends and Data

Calibration trends and data can be printed out on the **Biochemistry Calibration** screen.

- 1 Select **Reagent-Biochemistry Calibration**.
- 2 Search for calibration result and select a chemistry.
- 3 Select **Trend F6**.
- 4 Select **Print F7**.

Figure 9.28 Calibration graphical trends example

5 Select the **Tabular Trend** tab.

6 Select **Print F7** to print the trends data.

Figure 9.29 Calibration tabular trends example

Calibration Tabular Trend			
Chemistry: ALT-		Trend Type: R1 Blank Abs	
		Cal Date: 2012-09-26 - 2012-11-26	
Run Date	Absorbance	Run Date	Absorbance
2012-10-15 11:34:19	17791.1923	2012-10-15 11:34:28	17835.8982
2012-10-15 11:34:37	17832.8795	2012-10-15 11:34:46	17788.4624
Print Date/Time: 2012-11-26 15:22:33		Page: 1 of 1	

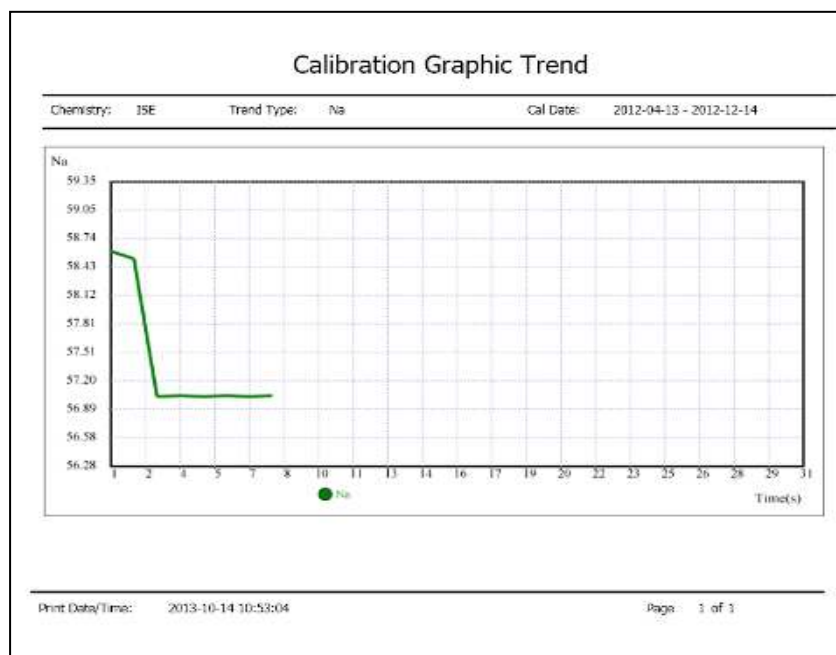
9.5.5 ISE Calibration Trends and Data

ISE calibration trends and data can be printed out on the **ISE Calibration** screen.

- 1** Select **Reagent-ISE Calibration**.
- 2** Search for ISE calibration result.
- 3** Select **Trend F6**.

4 Select **Print F7**.

Figure 9.30 Calibration graphical trends example



5 Select the **Tabular Trend** tab.

6 Select **Print F7** to print the trends data.

Figure 9.31 Calibration tabular trends example

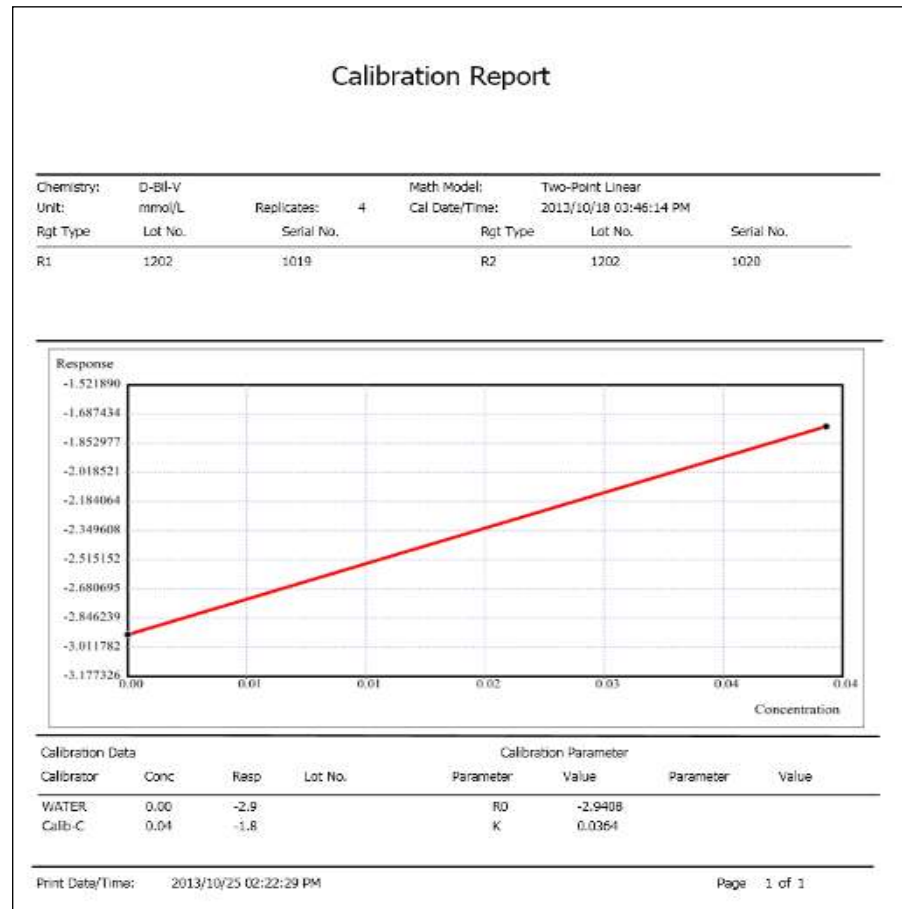
Chemistry: ISE		Trend Type: Na		Cal Date: 2012-04-13 - 2012-12-14	
Run Date/Time	Na	Run Date/Time	Na		
2012-06-13 10:33:25	58.6	2012-06-13 10:33:25	58.52		
2012-11-26 16:36:06	57.03	2012-11-26 16:36:06	57.04		
2012-11-26 16:38:37	57.03	2012-11-26 16:38:37	57.04		
2012-12-04 16:51:43	57.03	2012-12-04 16:51:43	57.04		
Print Date/Time: 2013-10-14 10:53:23		Page: 1 of 1			

9.5.6 Biochemistry Calibration Curve

A biochemistry calibration curve contains the calibration curve and factors of a biochemistry, and can be printed out on the **Biochemistry Calibration** screen.

1 Select **Reagent-Biochemistry Calibration**.

2 Select **Cal Curve F2**.

3 Select **Print F7**.**Figure 9.32** Biochemistry calibration curve example

9.5.7 Biochemistry Calibration Results Report

A biochemistry calibration results report can be printed out on the **Biochemistry Calibration** screen. It may contain the current calibration factors being used, or those in the specified period.

- 1 Select **Reagent-Biochemistry Calibration**.
- 2 Choose the **Current** or **History** option button.
- 3 Search for desired calibration results.
- 4 Select **Print F7**.

Figure 9.33 Biochemistry calibration results report example

Calibration Result List								
Chemistry	Cal Status	R0	K	A	B	C	D	Cal Date/Time
ApoB	Cal Time Out	76.600147		-2.183839		0.211068		
			40447.7564		0.671018			
C3	Cal Time Out	576.059120		-1.350882		0.053165		2011-02-17 03:35:45 PM
			24987.9477		1.001428			
CK	Cal Time Out	-1.128190						2011-02-17 03:36:52 PM
			1.051957					
CRP	Calibrated	614.576904		-3.773015		0.003391		2011-02-23 11:04:58 AM
			8821.46810		0.622558			
wf	Calibrated	0.000000						2011-02-17 02:08:44 PM
			10000.0000					2011-02-22 06:54:34 PM
Print Date/Time:		2011-10-18 05:13:29 AM						Page 1 of 1

9.5.8 ISE Calibration Results Report

The ISE calibration results report contains all ISE results during the specified period, and can be printed out on the **ISE Calibration** screen.

- 1 Select **Reagent-ISE Calibration**.
- 2 Search for desired calibration results.
- 3 Select **Print F7**.

Figure 9.34 ISE calibration results report example

Calibration Result List								
Chemistry	Cal Status	A Voltage1 mV	B Voltage1 mV	A Voltage2 mV	B Voltage2 mV	Slope1 mV/decade	Slope2 mV/decade	Ref Range mV/decade
Na	Cal Time Out	153.30	136.10	158.30	136.60	57.03	57.04	52.00~64.00
K	Cal Time Out	153.50	171.20	133.50	151.20	56.68	56.58	52.00~64.00
Cl	Cal Time Out	97.70	120.20	98.70	130.20	46.47	46.67	40.00~55.00
Lot No.:								
Cal Date/Time:		12/04/2012 04:51:43 PM						
Print Date/Time:		09/27/2013 02:12:05 PM						Page 1 of 1

9.6 QC Reports

9.6.1 Introduction

QC reports are used to print control results, such as actual results, L-J chart, twin-plot chart, QC data and QC summary. They can be printed out on:

- **Current Results** screen
- **History Results** screen
- **Levey-Jennings** screen
- **Twin-Plot** screen
- **Results** screen
- **Summary** screen

9.6.2 QC Results Report

The quality control results can be printed manually or automatically on the **Current Results** screen. Those QC results within a period can be printed on the **History Results** screen.

- 1 Select **Result-Current Results** or **History Results**.
- 2 Choose the **By Sample** option.
- 3 Search for desired results to print.
- 4 Choose controls to print.
- 5 Select **Print F7**.

Figure 9.35 QC results report example

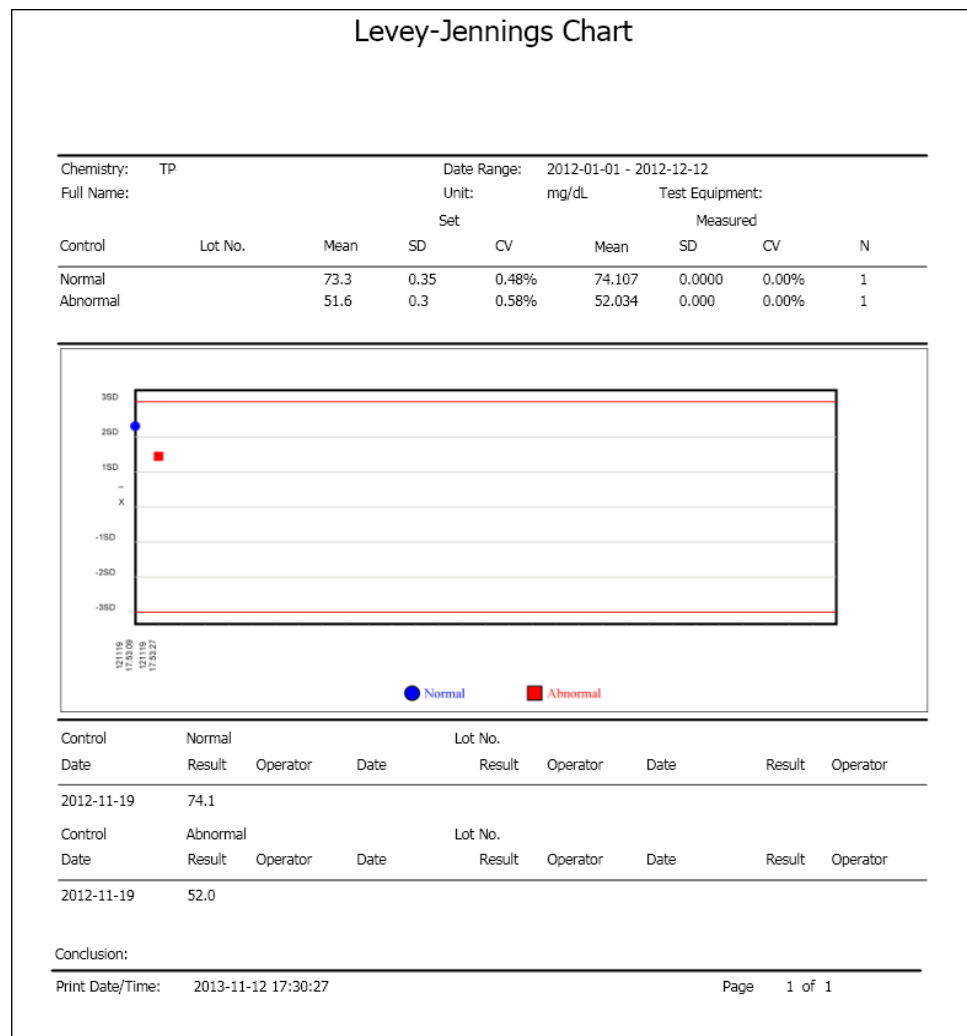
QC Result									
Control	Lot No.	Chemistry	Lot No.	Unit	Measured Result	Mean	SD	Flag	Run Date/Time
Normal		TP-开放		mg/dL	74.1	73.3	0.35		2012-11-19 17:53:09
Print Date/Time:					2013-11-12 17:26:35				
					Page 1 of 1				

9.6.3 Levey-Jennings Chart

The Levey-Jennings chart report contains the L-J chart and result values of a chemistry during the specified period.

- 1 Select **QC-Levey-Jennings**.
- 2 Search for desired results to print.
- 3 Select **Print F7**.

Figure 9.36 Levey-Jennings chart example

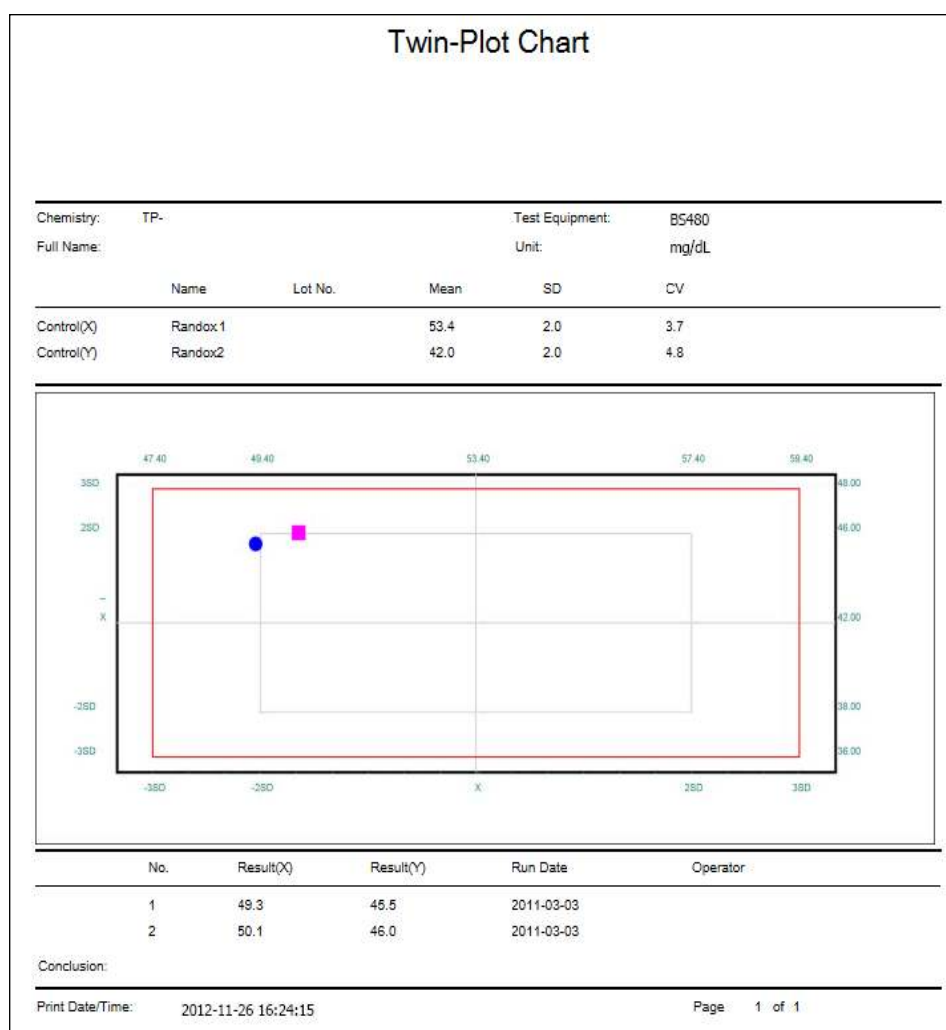


9.6.4 Twin-Plot Chart

The Twin-Plot chart report contains the Twin-Plot chart and result values of a two-control evaluation.

- 1 Select **QC-Twin-Plot**.
- 2 Select **Chems F2**, choose a chemistry from the list, and then select OK.
- 3 Select **Search F1**.
- 4 Select **Print F7**.

Figure 9.37 Twin-Plot chart example



9.6.5 QC Data Report

A QC data report contains all results of a control for a chemistry during the specified period, as well as the set means and standard deviations.

- 1 Select **QC-Results**.
- 2 Select **Chems F2**.
- 3 Choose chemistries to recall, select **OK**.
- 4 Select the date range in the **QC Data** field.
- 5 Select a control from the **Control** pull-down list.
- 6 Select **Search F1**.

The result list shows all results of the control for the chemistry during the specified period, as well as the set means and standard deviations.

- 7 Select **Print F7**.

Figure 9.38 QC data report example

QC Result									
Control	Lot No.	Chemistry	Lot No.	Unit	Measured Result	Mean	SD	Flag	Run Date/Time
Normal		TP-开放		mg/dL	74.1	73.3	0.35		2012-11-19 17:53:09
Print Date/Time: 2013-11-12 17:26:35					Page 1 of 1				

9.6.6 QC Summary Report

A QC summary report contains the mean values and standard deviations of controls analyzed within the specified period, as well as the set mean and SD value.

- 1 Select **QC-Summary**.
- 2 Select **Chems F2**.
- 3 Choose chemistries to recall, select **OK**.
- 4 Select the date range in the **QC Date** field.
- 5 Select a control from the **Control** pull-down list.
- 6 Select **Search F1**.

The result summary of the control for the chemistry is displayed on the screen.

- 7 Select **Print F7**.

Figure 9.39 QC summary report example

QC Summary								
Summary Date:		2011-01-01-2011-10-18						
Control	Lot No.	Chemistry	Mean	SD	Summary			
					Mean	SD	CV%	N
Randox2-M1		LDL-C	2.71	0.80	2.26	0.74	32.60	4
Randox2-M1		HDL-C	1.57	0.60	1.47	0.63	42.84	4
Randox2-M1		TC	4.30	0.40	4.48	0.36	8.06	3
Randox2-M1		CREA-S	128.9	2.0	121.7	0.0	0.00	1
Randox2-M1		UA	339.1	4.0	339.1	0.0	0.00	1
Randox2-M1		UREA	7.33	1.00	8.06	0.00	0.00	1
Randox2-M1		ALB	43.2	2.0	43.1	0.0	0.00	1
Randox2-M1		TP	53.3	4.0	56.4	13.2	23.32	2
Randox2-M1		ALP	191.1	3.0	0.0	0.0	0.00	0
Randox2-M1		ALT	36.0	2.0	33.0	0.0	0.00	1
Randox2-M1		AST	33.6	2.0	33.1	0.0	0.00	1
Randox2-M1		γ-GT	51.6	3.0	55.4	5.8	10.52	2
Randox2-M1		T-Bil-V	29.85	2.00	26.21	2.12	8.09	3
Randox2-M1		T-bil-D	26.64	2.00	0.00	0.00	0.00	0
Randox2-M1		D-Bil-V	16.58	2.00	16.17	0.00	0.00	1
Randox2-M1		D-bil-D	13.33	1.00	0.00	0.00	0.00	0
Randox2-M1		ApoB	0.40	0.01	0.00	0.00	0.00	0
Randox2-M1		Ca	1.91	0.50	3.22	0.90	28.03	2
Randox2-M1		Glu-G	6.29	2.00	8.78	3.89	44.27	2
Randox2-M1		Glu-H	6.09	1.00	5.76	0.00	0.00	1
Randox2-M1		TG	1.10	0.50	1.06	1.23	116.1	4
Randox2-M1		TP-	53.4	2.0	49.7	0.6	1.14	2
Randox2-M2		TP-	42.0	2.0	45.8	0.4	0.77	2
Print Date/Time:		2011-10-18 05:20:40 AM					Page 1 of 1	

9.7 Chemistry Parameter Reports

9.7.1 Introduction

Chemistry parameter reports are used to print out the parameters for the SI, ISE, off-system chemistries and common chemistries.

9.7.2 Chemistry Parameter Report

1 Select **Utility-Chemistries**.

2 Select a chemistry.

A chemistry may be an SI, ISE, Off-system chemistry or a common chemistry.

3 Select **Define F1**.

4 Select **Print F1**.

For ISE chemistry, select **Print**.

Figure 9.40 Biochemistry parameter report

Bio-Chemistry Parameters Report									
Chem	Glu-H			Chemistry					
Chemistry No.	414			Print Name	Glu				
Reaction Type	Endpoint			Reaction Direction	Positive				
Pri Wave	340nm			Sec Wave	700nm				
Decimal	0.01			Samp Type					
Blank Time	47	49		Reaction Time	80	82			
Unit	mmol/L								
	Aspirated Vol	Neat Vol	Diluent Vol			Reagent Vol	Diluent		
Standard	2 uL	uL	uL	R1	167 uL	uL	uL		
Decreased	uL	uL	uL	R2	84 uL	uL	uL		
Increased	uL	uL	uL	R3	uL	uL	uL		
				R4	uL	uL	uL		
linearity Range(Standard)	0.6	33	Linearity Limit						
linearity Range(Decreased)			Substrate Depletion						
linearity Range(Increased)			Mixed Blank Abs		-33000	33000			
R1 Blank Abs	-33000	33000	Uncapping Time		28	Day(s)			
Blank Response	-33000	33000	Reagent Alarm Limit		-1				
Twin Chemistry									
Prozone Check	---								
Q1	Q2			Q3					
Q4	PC			ABS					
Use Qualitative Result									
Range				Flag					
Slope and Offset									
Slope		Offset		Unit					
1		0		mmol/L					
Ref Range									
Sample Type	Gender	Age Range	Ref Range	Critical Range	Unit				
Print Date/Time: 09/27/2013 11:22:46 AM									
Page 1 / 1									

Figure 9.41 SI chemistry parameter report

SI Chemistry Parameters Report						
Chem	SI	Sample Vol	10	uL	Reagent Vol	200 uL
Lipemia		Hemolysis		Icterus		
Print Name						
Slope and Offset						
	Slope	Offset		Unit		
L	1	0		mg/dL		
H	1	0		mg/dL		
I	1	0		mg/dL		
Use Qualitative Result						
Range	Flag	Range	Flag	Range	Flag	
Print Date/Time: 09/27/2019 01:50:53 PM						
Page 1 / 1						

Figure 9.42 ISE parameter report

ISE Chemistry Parameters Report						
Print Name		Na	K		Cl	
Use Qualitative Result						
Range		Flag	Range		Flag	Range
Range		Flag	Range		Flag	Range
Unit		Decimal		Measur. Range		
Na(Serum)	mmol/L	0.1		100	200	
K(Serum)	mmol/L	0.01		1	8	
Cl(Serum)	mmol/L	0.1		50	150	
Na(Urine)	mmol/L	0		10	500	
K(Urine)	mmol/L	0		5	200	
Cl(Urine)	mmol/L	0		15	400	
Slope and Offset						
	Slope		Offset		Unit	
Na(Serum)	2		5		mmol/L	
K(Serum)	1.5		-1		mmol/L	
Cl(Serum)	5		-10		mmol/L	
Na(Urine)	100		-100		mmol/L	
K(Urine)	3		25		mmol/L	
Cl(Urine)	20		60		mmol/L	
Ref Range						
Chemistry	Samp Type	Gender	Age Range	Ref Range	Critical Range	Unit
Print Date/Time: 09/27/2013 01:54:14 PM						
Page 1 / 1						

Figure 9.43 Off-system chemistry parameter report

Off-System Chemistry Parameters Report					
Chem	Off 1		Chemistry	Off	
Chemistry No.	1		Print Name	Off 1	
Attribute	Quantitative		Reference		
Decimal			Unit	mmol/L	
Use Qualitative Result					
Range			Flag		
Ref Range					
Sample Type	Gender	Age Range	Ref Range	Critical Range	Unit
Print Date/Time: 09/27/2018 02:00:05 PM					
Page 1 / 1					

9.8 Chemistry Reports

9.8.1 Introduction

Chemistry reports are used to print sample/control panels and user-defined calculations. They can be printed out respectively on the **Panels** and **Calculations** screens.

9.8.2 Sample/Control Panels Report

Sample panels are used for patient sample analysis and only appear on the **Sample** screen; control panels are used for quality control and only displayed on the **Control** screen.

- 1 Select **Utility-Chemistries**.
- 2 Select **Panels F7**.
- 3 Select **Print F7**.

All user-defined sample panels and control panels are printed out with the default templates.

Figure 9.44 Sample panels report example

Sample Panel List	
Panels	Chemistries
1	ALB, TP, ALP, ALT, AST, γ-GT, T-Bil-V, D-Bil-V
2	LDL-C, HDL-C, TC, TG
3	CREA-S, UA, UREA
4	Na, K, Cl
5	ALP, ALT, AST, γ-GT, CHE
6	CREA-J, UA, UREA, CO2, CysC, Glu-G
7	CREA-J, UA, UREA, CO2, Glu-G
8	LDL-C, HDL-C, TC, ApoA1, ApoB, TG
9	CK, CK-MB
10	Na, K, Cl, Ca
11	C3, C4
12	ASO, CRP
13	Na, K, Cl, Ca
14	CREA-S, UA
Print Date/Time: 2011-10-18 04:59:19 AM Page 1 of 1	

Figure 9.45 Control panels report example

QC Panel List	
Panels	Chemistries
1	ALB, TP, ALP, ALT, AST, γ-GT, T-Bil-V, D-Bil-V
2	LDL-C, HDL-C, TC, TG
8	LDL-C, HDL-C, TC, ApoA1, ApoB, TG
9	CK, CK-MB
12	ASO, CRP
Print Date/Time: 2011-10-18 05:21:49 AM Page 1 of 1	

9.8.3 Calculations Report

The calculations report contains all user-defined calculations, including the name, chemistries, and enabling/disabling status.

- 1 Select **Utility-Chemistries**.
- 2 Select **Calculations F6**.
- 3 Select **Print F7**.

All defined calculations are printed out with the default template.

Figure 9.46 Calculations report example

Special Calculations		
Calculation	Formula	Enabled
Glo	[TP]-[ALB]	Yes
A/G	[ALB]/([TP]-[ALB])	Yes
AST/ALT	[AST]/[ALT]	Yes
IBIL-V	[T-Bil-V]-[D-Bil-V]	Yes
IBIL-D	[T-bil-D]-[D-bil-D]	Yes
Print Date/Time: 2011-10-18 05:22:29 AM Page 1 of 1		

9.9 Instrument Status Reports

9.9.1 Introduction

The instrument status reports contain the current working status of each subsystem and hardware component, which includes the status summary, cycle count, temperature, Hydropneumatic subsystem, and control modules. They can be printed out on the **Status** screen.

9.9.2 Status Summary Report

The status summary provides a high-level summary of the status of the system temperatures, hydropneumatic, and control modules. The status includes OK and Error.

- 1 Select **Utility-Status**.
- 2 Select the **Status Summary** tab.
- 3 Select **Print F7**.

Figure 9.47 Status summary report example

Status Summary	
Monitored Areas	Status
Temperature	OK
Hydro	OK
Smart Modules	OK
Main Control Unit	OK
Print Date/Time: 10/10/2013 04:29:47 PM Page 1 of 1	

9.9.3 Cycle Count Report

The cycle count report contains an approximation of a component's usage, which can be useful for estimating the maintenance frequencies or anticipating component failure. The report includes cycle count of the following components:

- Vacuum pump for cleaning reaction cuvettes
- Waste pump
- Sample syringe
- Reagent syringe
- ISE diluent syringe

- ISE tests

Print the cycle count report by performing the following steps:

- 1 Select **Utility-Status**.
- 2 Select the **Count** tab.
- 3 Select **Print F7**.

Figure 9.48 Cycle count report example

Count Status	
Name	Count
Vacuum Pump for Cleaning Cuvettes	16H
Vacuum pump	16H
Interior Wash Syringe	2312
Sample Syringe	1040
Reagent Syringe	1365
Phase 1-2 Wash Syringe	4309
Na	194
K	194
Cl	194
Reference	194
Print Date/Time: 10/10/2013 04:29:55 PM Page 1 of 1	

9.9.4 Temperature Report

The temperatures report contains the actual temperature and valid range of the deionized water, reagent carousel, reaction carousel, and wash station.

- 1 Select **Utility-Status**.
- 2 Select the **Temperature** tab.
- 3 Select **Print F7**.

Figure 9.49 Temperature report example

Temperature Status		
Name	Measured Value	Valid Range
Reaction Carousel Temperature	37.0 °C	36.7 °C - 37.3 °C
Cuvette Cleaning Fluid Temperature	44.0 °C	39.0 °C - 49.0 °C
Cuvette Wash Solution Temperature	42.5 °C	39.0 °C - 49.0 °C
Reagent Refrigeration Temperature	4.6 °C	2.0 °C - 10.0 °C
Internal Temperature of Whole Unit	26.5 °C	15.0 °C - 40.0 °C
Print Date/Time: 10/10/2013 04:29:55 PM Page 1 of 1		

9.9.5 Fan Report

The fan report displays the working status of the fans such as reagent refrigeration fan 1 and 2, lamp housing fan, vacuum pump cooling fan.

- 1 Select **Utility-Status**.
- 2 Select the **Fans** tab.
- 3 Select **Print F7**.

Figure 9.50 Fan report example

Fans Status		
Name	Measured Value	Valid Range
Reagent Refrigeration Fan 1	OK	OK / Error
Reagent Refrigeration Fan 2	OK	OK / Error
Lamp Housing Fan	OK	OK / Error
Vacuum Pump Cooling Fan	OK	OK / Error
Print Date/Time: 10/10/2013 04:30:10 PM Page 1 of 1		

9.9.6 Hydropneumatic Status Report

The Hydropneumatic status report contains the working status of various tanks and the actual air pressure and valid range for air pressure equipment.

- 1 Select **Utility-Status**.
- 2 Select the **Hydro** tab.
- 3 Select **Print F7**.

Figure 9.51 Hydropneumatic status report example

Hydropneumatic Status				
Name	Status	Name	Actual	Valid Range
Water Tank	OK	Primary Vacuum Pressure	-73.22 kPa	< -20 kPa
Diluted Wash Solution Tank	OK	Resistivity of Cleaning Water		≥ 1 MΩ.CM
Concentrated Wash Solution Tank	OK			
High-concentration Waste Collector	OK			
Low-concentration Waste Collector	OK			
Primary Vacuum Container	OK			
High Concentration Waste Tank	OK			
External Drain Canister	OK			
Print Date/Time: 10/10/2013 04:30:17 PM			Page 1 of 1	

9.9.7 Smart Module Status Report

The smart module report contains the working status of each smart module on the system. The status includes OK and Error.

- 1 Select **Utility-Status**.
- 2 Select the **Smart Modules** tab.
- 3 Select **Print F7**.

Figure 9.52 Smart modules status report example

Smart Module Status	
Name	Status
Reagent Probe Unit	OK
Sample Probe Unit	OK
Reagent Mixer Unit	OK
Sample Mixer Unit	OK
Probe Interior Wash Unit	OK
Rgt Carousel	OK
Sample Carousel	OK
Reaction Carousel Unit	OK
Temperature Unit	OK
Cuvette Wash Station	OK
Reagent Refrigeration Unit	OK
Sample/Reagent Mixer Unit	OK
ISE Unit	OK
Print Date/Time: 10/10/2013 04:30:24 PM	
Page 1 of 1	

9.9.8 Cuvette Status Report

The cuvette status report contains a cuvette's results and time in the recent two measurements at all wavelengths. It can be printed out while the **Cuvette Check** maintenance is performed.

- 1 Select **Utility-Maintenance**.
- 2 Select **Maintenance**.
- 3 Select **Biochemistry Maintenance**.
- 4 Select **Cuvette Check**.
- 5 Select **Continue**.
- 6 After cuvette check is finished, choose a cuvette to print.
- 7 Select **Print**.

Figure 9.53 Cuvette status report example

Cuvette Status			
Cuvette No.: 49			
Current Result		Last Result	
Test Time:	2013 - 09 - 18	Test Time:	2013 - 05 - 27
Wavelength(nm)	Absorbance	Wavelength(nm)	Absorbance
340	11707	340	11707
380	11707	380	11707
412	11707	412	11707
450	11707	450	11707
505	11707	505	11707
546	11707	546	11707
570	11707	570	11707
605	11707	605	11707
660	11707	660	11707
700	11707	700	11707
740	11707	740	11707
800	11707	800	11707
Print Date/Time:	09/26/2013 08:20:22 PM	Page	1 of 1

9.9.9 Cuvette Blank Result Report

This report shows the blank results of all cuvettes at all wavelengths. It can be printed out while the **Cuvette Check** maintenance is performed.

- 1 Select **Utility-Maintenance**.
- 2 Select **Maintenance**.
- 3 Select **Biochemistry Maintenance**.
- 4 Select **Cuvette Check**.

- 5 Select **Continue**.
- 6 After cuvette check is finished, choose **Result**.
- 7 Select **Print**.

Figure 9.54 Cuvette blank result report

Cuvette Blank Result												
Cuv	340	360	412	450	505	546	570	605	660	700	745	800

Print Date/Time: 2012-04-17 17:29:21 Page 1 of 1

9.9.10 Photometer Status Report

The photometer status report shows the results and time of the recent two photometer checks.

- 1 Select **Utility-Maintenance**.
- 2 Select **Maintenance**.
- 3 Select **Biochemistry Maintenance**.
- 4 Mark the **Select** checkbox of **Photometer Check**.
- 5 Select **Continue**.
- 6 When the photometer check is finished, select **Print**.

Figure 9.55 Photometer status report example

Photometer Check			
Current Result		Last Result	
Test Time:	2011-10-18	Test Time:	2011-08-08
Wavelength(nm)	Absorbance	Wavelength(nm)	Absorbance
340	11707	340	11707
380	11707	380	11707
412	11707	412	11707
450	11707	450	11707
505	11707	505	11707
546	11707	546	11707
570	11707	570	11707
605	11707	605	11707
660	11707	660	11707
700	11707	700	11707
740	11707	740	11707
800	11707	800	11707
Print Date/Time: 2011-10-18 05:30:54 AM		Page 1 of 1	

9.10 Statistics Report

The workload and result statistics of the samples and chemistries in a period of time can be printed out in the statistics reports.

9.10.1 Workload Statistics

The workload statistics can be printed out through the **Statistics-Workload** screen. Sample statistics report shows the sample information and the quantity of its requested chemistries; test statistics report shows the requested and complete quantity of the chemistries and the consumption of the reagents.

- 1 Select **Result-Statistics-Workload**.
- 2 Select **By Sample** or **By Chemistry**.
- 3 Select or enter the start date and end date in the **Date** field.
- 4 Select **Search F1**.
- 5 Select **Print F7**.

Figure 9.56 Sample statistics report

Sample Statistic List							
Program Time	Sample ID	Bar Code	Requested	Biochemistry	Off-system	Calculations	ISE
2012-03-09 17:03:23	1		0	0	1	0	0
2012-03-09 17:06:40	2		0	0	1	0	3
2012-03-09 17:19:09	3		0	0	1	0	0
2012-03-12 17:33:05	3		0	0	1	0	0
2012-03-12 17:57:16	5		1	0	0	0	0
2012-03-20 11:12:29	9001		3	0	1	0	0
2012-03-20 14:35:52	1		0	0	0	0	3
2012-03-26 15:00:57	1		1	0	0	0	0
Summary:			5	0	5	0	6
Print Date/Time: 03/28/2012 09:17:33 Page 1 / 1							

Figure 9.57 Test statistics report

Test Statistics List						
Chemistry	Request	Finished	R1	R2	R3	R4
Na	2	0				
K	2	0				
Cl	2	0				
ALT	1	0	0.0	0.0		
w1	1	0	0.0			
TP-	3	0	0.0			
Summary:	11	0	0.0	0.0		
Print Date/Time:			03/28/2012 09:18:05			
			Page 1 / 1			

9.10.2 Result Statistics

Result statistics option can summarize the total chemistries and the distribution trend of its results and provide the test data and graph.

- 1 Select **Result-Statistics-Results**.
- 2 Select **Statistic Graph** or **Statistic Data**.
- 3 Select **Search F1**.
- 4 Enter the search conditions.
- 5 Select **OK**.
- 6 Select **Print F7**.

Figure 9.58 Result statistic graph

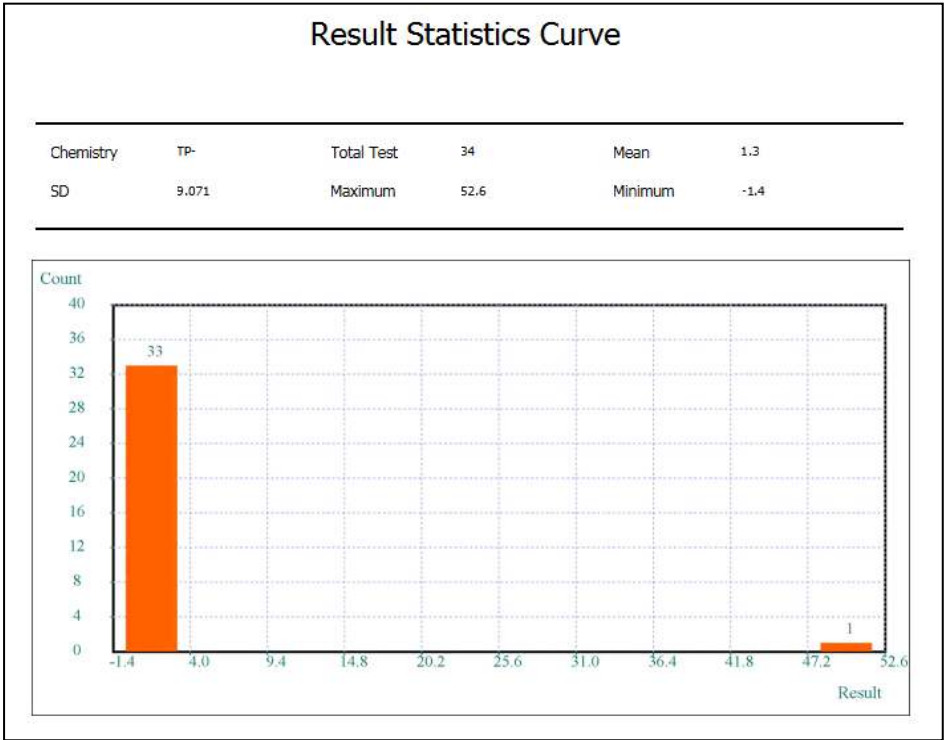


Figure 9.59 Result statistic data

Result Statistics List						
Chemistry	TP-	Test Total	34	Mean	1.3	
SD	9.071	Maximum	52.6	Minimum	-1.4	
Run Data	Sample ID	Bar Code	Result	Sample Type	Gender	Ref Range
2011-03-04 11:13:28	71		52.6	Serum		
2011-03-04 11:13:37	91		0.6	Serum		
2011-03-04 11:13:46	92		-0.6	Serum		
2011-03-04 11:13:55	93		0.1	Serum		
2011-03-04 11:15:50	101		-0.7	Serum		
2011-03-04 11:15:59	102		-0.2	Serum		
2011-03-04 11:16:08	103		-0.6	Serum		
2011-03-04 11:16:17	104		-0.7	Serum		
2011-03-04 11:16:26	105		-0.8	Serum		
2011-03-04 11:16:35	106		0.0	Serum		
2011-03-04 11:16:44	107		-0.1	Serum		
2011-03-04 11:16:53	108		-0.2	Serum		
2011-03-04 11:17:02	109		-0.2	Serum		
2011-03-04 11:17:11	110		-0.1	Serum		
2011-03-04 11:17:24	111		-1.4	Serum		
2011-03-04 11:17:33	112		-0.2	Serum		
2011-03-04 11:17:42	113		-0.2	Serum		
2011-03-04 11:17:51	114		-0.1	Serum		
2011-03-04 11:18:00	115		-0.1	Serum		
2011-03-04 11:18:09	116		-0.2	Serum		
2011-03-04 11:18:18	117		-0.2	Serum		
2011-03-04 11:18:27	118		-0.1	Serum		
2011-03-04 11:18:36	119		-0.2	Serum		
2011-03-04 11:18:45	120		-0.2	Serum		
2011-03-04 11:53:40	121		-0.1	Serum		
2011-03-04 11:53:49	122		-0.1	Serum		
2011-03-04 11:53:59	123		-0.4	Serum		
2011-03-04 11:54:07	124		-0.1	Serum		
2011-03-04 11:54:16	125		-0.3	Serum		
2011-03-04 11:54:25	126		-0.4	Serum		
2011-03-04 11:54:34	127		-0.3	Serum		
2011-03-04 11:54:43	128		-0.1	Serum		
2011-03-04 11:54:52	129		-0.1	Serum		
2011-03-04 11:55:01	130		-0.6	Serum		
Print Date/Time:			03/28/2012 09:19:25	Page 1 / 1		

9.11 Log Reports

9.11.1 Introduction

Log reports are used to print the inquired error logs, delete/edit logs and maintenance log, which can be respectively printed out on the **Error Log**, **Delete/Edit Log** screens and **Maintenance History** window.



NOTE

Printing logs will take a long time and requires great amount of paper. Think twice before printing logs.

9.11.2 Error Log Report

The error log report contains all error logs that are currently displayed on the **Error Log** screen.

- 1 Select **Alarm-Error Log**.
- 2 Search for desired error logs.
- 3 Select **Print F7**. All error logs are printed out with the default template.

Figure 9.60 Error log report example

Error Log				
Event ID	Date/Time	Description	Type	Component
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-17, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-8, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-7, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-6, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-5, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-4, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-20, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-19, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-18, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-9, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-16, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-15, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-14, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-13, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-12, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-11, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-2, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-10, has been expired	Warning	Other
C07028	2011-10-17 10:45:00 PM	Chemistry: w1, lot number: , position: 1-3, has been expired	Warning	Other
C07028	2011-10-17 05:54:38 PM	Chemistry: w1, lot number: , position: 1-8, has been expired	Warning	Other
C07028	2011-10-17 05:54:38 PM	Chemistry: w1, lot number: , position: 1-7, has been expired	Warning	Other
C07028	2011-10-17 05:54:38 PM	Chemistry: w1, lot number: , position: 1-6, has been expired	Warning	Other
C07028	2011-10-17 05:54:38 PM	Chemistry: w1, lot number: , position: 1-5, has been expired	Warning	Other
C07028	2011-10-17 05:54:38 PM	Chemistry: w1, lot number: , position: 1-4, has been expired	Warning	Other
Print Date/Time:		2011-10-18 05:31:31 AM		Page 1 of 85

9.11.3 Edit Log Report

The edit log report contains records of all deletions and editing performed by the operator, and can be printed out on the **Edit Log** screen.

- 1 Select **Alarm-Edit Log**.
- 2 Search for desired edit logs.
- 3 Select **Print F7**. All edit logs are printed out with the default template.

Figure 9.61 Edit log report example

Edit Log				
No.	Date/Time	Operator	Operation Type	Description
1	2011-10-18 05:21:20 AM	Admin	DEL	Delete chemistry: [13]
2	2011-10-18 05:21:19 AM	Admin	DEL	Delete chemistry: [14]
3	2011-10-18 05:21:16 AM	Admin	DEL	Delete chemistry: [10]
4	2011-10-18 05:21:15 AM	Admin	DEL	Delete chemistry: [11]
5	2011-10-18 05:21:13 AM	Admin	DEL	Delete chemistry: [3]
6	2011-10-18 05:21:12 AM	Admin	DEL	Delete chemistry: [5]
7	2011-10-18 05:21:12 AM	Admin	DEL	Delete chemistry: [4]
8	2011-10-18 05:21:11 AM	Admin	DEL	Delete chemistry: [6]
9	2011-10-18 05:21:10 AM	Admin	DEL	Delete chemistry: [7]
10	2011-10-18 05:04:45 AM	Admin	DEL	Delete control. Control: [Randox2-M1], lot number: [], program date: [2011-10-18 5:01:34 AM]
11	2011-10-18 05:04:36 AM	Admin	DEL	Delete sample. Sample ID: [7], bar code: [], program date/time: [2011-10-18 5:02:11 AM]
12	2011-10-18 05:04:36 AM	Admin	DEL	Delete sample. Sample ID: [8], bar code: [], program date/time: [2011-10-18 5:02:13 AM]
13	2011-10-18 05:04:35 AM	Admin	DEL	Delete sample. Sample ID: [6], bar code: [], program date/time: [2011-10-18 5:02:10 AM]
14	2011-10-18 05:01:30 AM	Admin	EDT	Control edited (Randox2-M1.)
15	2011-10-18 05:01:07 AM	Admin	DEL	Delete sample. Sample ID: [1], bar code: [], program date/time: [2011-10-18 4:09:47 AM]
16	2011-10-18 05:00:57 AM	Admin	DEL	Delete sample. Sample ID: [3], bar code: [], program date/time: [2011-10-18 5:00:04 AM]
17	2011-10-18 05:00:57 AM	Admin	DEL	Delete sample. Sample ID: [4], bar code: [], program date/time: [2011-10-18 5:00:05 AM]
18	2011-10-18 05:00:57 AM	Admin	DEL	Delete sample. Sample ID: [5], bar code: [], program date/time: [2011-10-18 5:00:07 AM]
19	2011-10-18 05:00:55 AM	Admin	DEL	Delete sample. Sample ID: [2], bar code: [], program date/time: [2011-10-18 5:00:03 AM]
Print Date/Time:		2011-10-18 05:32:41 AM		
		Page 1 of 1		

9.11.4 Maintenance Log Report

This report can be printed out through the **Maintenance History** window. It shows the maintenance history of the selected maintenance procedure.

- 1 Choose a maintenance procedure on the **Scheduled Maintenance** screen.
- 2 Select **History**. The **Maintenance Log** window is displayed.
- 3 View all performance records of the selected maintenance procedure.
- 4 To print maintenance log:
 - Mark the checkbox of one or more desired maintenance records.
 - Select **Print**.

Figure 9.62 Maintenance log report

Maintenance Log			
Procedure	Date	Logs	Operator
Cuvette Check:	2013-10-14 14:11:13		Admin
Cuvette Check:	2013-10-14 14:05:17	maintained	Admin
Print Date/Time:	2013-10-14 11:57:01		Page 1 of 1

10 Chemistries

This chapter introduces applications of chemistries, including:

- Definition and application of twin chemistries
- Definition and application of calculations
- Definition and application of panels
- Definition and application of off-system chemistries
- Configuration and application of serum index
- Masking and unmasking of chemistries
- Chemistry configuration
- Definition and application of default panels
- Carryover setup
- Reflex

10.1 Twin Chemistries

10.1.1 Introduction

Twin chemistries are run and calculated based on the same reagent. Results of the two twin chemistries are calculated through the same test. Take the reagent HbA1C as an example. It can be used for running two chemistries in the same test. The chemistry HB is measured during the former reaction period, while the chemistry HbA1c measured during the latter one. Finally, HbA1C (%) can be calculated based on results of the two chemistries.

Similar to normal chemistries, twin chemistries can be run only when the following settings are finished:

- defining chemistries
- assigning reagent position
- setting up calibrator and calibration rule
- setting up control and QC rule

Twin chemistries are run in the test order of the “latter chemistry” for which the twin is specified.

10.1.2 Chemistry Definition

Twin chemistries can be defined in the same way as normal chemistries. The following parameters, however, must be set up differently for two twin chemistries:

- Sample type
- Normal sample volume, increased sample volume, and decreased sample volume
- Volume of the same reagent type
- Prozone check

A chemistry that has been set as the twin of another chemistry must not have another twin. When twin chemistries are defined, both chemistries must be calibrated.

Figure 10.1 Chemistry definition window



For defining methods of chemistries, refer to “3.2 Chemistries Setup” (Page 3-13).

10.1.3 Removing Twin Relation

To remove the twin relation between two chemistries, cancel the selection of a twin chemistry. Only when reagents of the two chemistries are unloaded can the twin relation between them be removed.

10.1.4 Reagent Setup

Twin chemistries are run with the same reagent in the same position. The reagent can be loaded manually or through bar code scanning.

Manual load:

You are only required to manually set up reagents for one of the twin chemistries. The reagent of the same type for the other twin chemistry will be automatically set up with the same position.

Automatic load:

Place the bar-coded reagents of twin chemistries on the reagent carousel, the system will scan the reagent bar code and automatically assign the same position for the same reagent of the twin chemistries.

If reagent loading is failed for either of the twin chemistries, both chemistries cannot be run.

For details of reagent loading, refer to “2.5.1 Loading Biochemical Reagents” (Page 2-15).

10.1.5 Setting Up and Requesting Calibration

Calibration setup

The calibrator, number of replicates and auto calibration conditions must be the same for two twin chemistries.

For calibration settings, refer to “3.3 Calibration Setup” (Page 3-33).

Requesting calibration

Twin chemistries can be requested for calibration in the same way as normal chemistries. When either of the twin chemistries is requested, the other twin will be requested automatically, and finally both chemistries will be calibrated. You are allowed to recall the calibration results, calibration curves and reaction curves of the two chemistries.

10.1.6 Setting Up and Requesting Quality Control

QC setup

Twin chemistries must be defined with the same control, and the QC setup of twin chemistries is the same as that of normal chemistries. Refer to “3.4 QC Setup” (Page 3-42) for details.

Programming controls

Twin chemistries can be requested for quality control in the same way as normal chemistries. When either of the twin chemistries is requested, the other twin will be requested automatically, and finally both chemistries will be run for quality control. You are allowed to recall the QC results and QC reaction curves of the two chemistries.

10.1.7 Sample Programming and Processing

Twin chemistries can be requested for sample analysis in the same way as normal chemistries. When either of the twin chemistries is requested, the other twin will be requested automatically, and finally both chemistries will be run for sample analysis. You are allowed to recall the sample results and sample reaction curves of the two chemistries.

10.2 Special Calculations

10.2.1 Introduction

Calculation of certain chemistries can derive new chemistries of clinical purposes, such as A/G (ALB/ (TP-ALB)), I-BIL (T-Bil - D-Bil), etc.

A calculation is composed of chemistries, calculation operators and algorithm. Only users with sufficient permissions are allowed to define, modify and delete calculations. The system allows a maximum of 50 calculations to be defined.

For the print order of calculations, refer to 9.2.7 Defining Chemistry Print Order (Page 9-11).

10.2.2 Defining/Editing a Calculation

- 1 Select **Utility-Chemistries**.
- 2 Select **Calculations F6**.
- 3 Select **Define F1**.

Figure 10.2 Special Calculations window

- 4 Type in the calculation's short name in the **Chem** field.
- 5 If you are going to use the calculation for analysis, mark the **Enable** checkbox.

- 6 Choose a sample type to which the calculation will be applied.

The options include:

- Serum
- Plasma
- Urine
- CSF
- Other

- 7 Type in the calculation's full name in the **Chemistry** field.

- 8 Type in the print name of the calculation to appear on patient reports.

- 9 Choose a result unit from the **Unit** pull-down list.

- 10 Choose a result precision, that is, the number of decimal places.

The options include:

- 0
- 0.1
- 0.01
- 0.001

- 11 Edit the calculation formula:

- Choose chemistries in the **Chemistries** list. The chemistries are then displayed in the **Formula** field.
- Choose numbers and operators in the **Mathematical Symbols** area to constitute the calculation formula along with the chemistries.
- To remove a chemistry, number or operator, move the cursor behind them and select **BS**.
- To clear the entire formula, select **AC**.

- 12 Select **Flag qualitative results** if you want to flag the qualitative results.

For more information, please refer to 3.2.5 Flag Qualitative Result

- 13 Select **OK** to save the settings.

- 14 Select **Exit** to exit the window.

10.2.3 Enabling/Disabling Calculations

When a special calculation is defined, it is enabled by default and will be calculated for sample analysis. If a calculation is disabled, it will not be calculated for sample measurements. Before enabling or disabling a calculation, make sure that the system status is not Running.

Perform the following steps to enable or disable calculations:

- 1 Select **Utility-Chemistries**.
- 2 Select **Calculations F6**.
 - The calculation list shows all calculations and formulas.
 - When the **Enable** checkbox is marked, it indicates that the calculation will be included for result calculating.
 - When the **Enable** checkbox is not marked, it indicates that the calculation will not be included for result calculating.

Figure 10.3 Special Calculations window



- 3 To activate a calculation, mark the **Enable** checkbox.
- 4 To inactivate a calculation, deselect the **Enable** checkbox.

10.2.4 Deleting User-Defined Calculations

Calculations can be deleted by users with sufficient permissions while the system status is not Running. Only user-defined calculations rather than closed calculations can be deleted.

1 Select **Utility-Chemistries**.

2 Select **Calculations F6**.

Figure 10.4 Special Calculations window



3 Choose calculations to delete.

4 Select **Delete F2**.

10.2.5 Running Calculations

Calculations will not be run for calibration, but for quality control and sample analysis along with other chemistries.

If a chemistry contained in a calculation is run for more than one replicates, the final result of the chemistry will be used to calculate the result of the special calculation.

10.3 Panels

10.3.1 Introduction

A couple of chemistries combined together for certain clinical purposes can constitute a panel, such as liver function, kidney function, etc. Panels can help fast programming of samples.

Panels can be composed of biochemistries and ISE chemistries except for SI and calculations. The system allows a maximum of 100 panels to be defined. Only users with sufficient permissions are allowed to define, modify and delete panels.

10.3.2 Defining/Editing a Panel

- 1 Select **Utility-Chemistries**.
- 2 Select **Panels F7**.
- 3 Select **Define F1**.

Figure 10.5 Define/Edit Panels window

- 4 Type in the panel name.
- 5 Choose panel types.
 - Sample: indicates that the panel can be used for sample analysis.
 - QC: indicates that the panel can be used for quality control.

At least one panel type must be selected. A panel can be applied to both sample and control analysis.

- 6 Choose chemistries for the panel.

At least one biochemistry should be selected. The three ISE chemistries (Na, K and Cl) can be selected alone.

- 7 To remove a chemistry, click it again.

- 8 Select **Save F7**.

- 9 Select **Close F8** to close the window.

10.3.3 Adjusting Display Order of Panels

Display order of panels on the **Sample** and **Quality Control** screens can be adjusted manually for convenient test requisition.

- 1 Select **Utility-Chemistries**.
- 2 Select **Panels F7**.
- 3 Select **Up F3** to move the current panel to the previous position, or select **Down F4** to move it to the next position.
- 4 Select **Save F7** to save the settings.

10.3.4 Deleting Panels

Panels can be deleted by users with sufficient permissions while the system status is not Running. When a panel is removed, the chemistries contained in it will still remain and can constitute panels with other chemistries.

- 1 Select **Utility-Chemistries**.
- 2 Select **Panels F7**.
- 3 Choose panels to delete.
- 4 Select **Delete F2**.

10.3.5 Running Panels

Panels will not be run for calibration, but for sample and control analysis along with other chemistries.

10.4 Off-system Chemistries

10.4.1 Introduction

All the chemistries that are not run by the analyzer are referred to as the off-system chemistries. You can manually enter the off-system chemistry results into the system to print out them in the patient report.

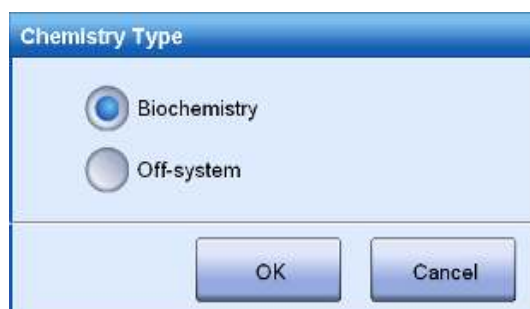
There are two types of off-system test: qualitative and quantitative.

- Qualitative chemistries: No numeric results are obtained but the flags you defined on the system. Reference value can be set for the qualitative chemistries
- Quantitative chemistries: Numeric results and defined flags are displayed and printed. Reference range can be set for the quantitative chemistries

10.4.2 Define/Edit Off-System Chemistries

- 1 Select **Utility-Chemistries**.
- 2 Select a blank chemistry box.
- 3 Select **Define F1**.

Figure 10.6 Chemistry type



- 4 Select **Off-system**.
- 5 Select **OK**

Figure 10.7 Define/Edit chemistries

- 6 Enter the chemistry parameters
- 7 Select **Use Qualitative Result** to flag the result of a quantitative off-system chemistry.
- 8 Enter **Range** and **Flag**.
- 9 Select **OK**.

10.4.3 Running the Off-system Chemistries

Off-system chemistries can participate in the panel setup and quantitative off-system chemistries can join the special calculations setup. You can program the off-system chemistries on the **Program- Sample** screen and the **STAT Sample Program** screen. The off-system chemistry can be added singly or in batch to the sample which needs rerun on the **Rerun Sample** window and **Rerun Batch** window. For details, please refer to 8.2.4 Rerunning Samples. After the off-system chemistries are programmed, you can edit their results on the **Result** screen. The results of the off-system chemistries can be edited in any status of the sample.

- 1 Program the off-system chemistries on the **Program- Sample** screen. Refer to 2.8.1 Programming Routine Samples
- 2 Select **Result-Current Results/History Results**.
- 3 Choose a result recall mode:
 - By sample

- By chemistry
- 4 Select **Search F1** to search for desired results.
 - 5 Select the sample or chemistry whose results you want to edit.
 - 6 Select **Option F2**.
 - 7 Select **Edit Results**.
 - 8 Edit the results.
 - 9 Click **Save**.

10.4.4 Deleting Off-system Chemistries

When the system status is not running, the off-system chemistries can be deleted.

- 1 Select **Utility-Chemistries**.
- 2 Select the off-system chemistry you want to delete.
- 3 Select **Delete F2**.

10.5 Serum Index

Serum index is the degree of lipemia, hemolysis and icterus contained in samples, and used to check if these interferents will influence the sample results.

For more information about serum index, refer to 8.3 Serum Index (page 8-23).

10.6 Chemistry Configuration

10.6.1 Introduction

The Chemistry Configuration function is used to enable/disable chemistries that have been defined correctly and customize their display order on the **Sample, STAT Sample Program** and **Quality Control** screens. When disabled, chemistries will no longer appear on the **Sample, Reagent/Calibration, Quality Control, Define/Edit Panels, Special Calculations, Current Results** and **History Results** screens. Only the enabled chemistries can be requested for measurements and recalled on results screens.

The system allows up to 200 chemistries to be enabled. The number of open-reagent chemistries can be adjusted according to the practical situations in your laboratory.

The **Chemistry Configuration** screen is as shown below:

Figure 10.8 Chemistry Configuration screen



10.6.2 Enabling Chemistries

All chemistries other than ISE chemistries and SI can be enabled or disabled. The closed-reagent chemistries are enabled by default after being imported from a chemistry file; while the open-reagent chemistries will be enabled only if the parameters are set up correctly. The SI is always enabled and cannot be disabled. If an ISE module is configured, the ISE chemistries will always be enabled.

To enable chemistries, perform the following steps:

- 1 Select **Utility-Chemistries**.
- 2 Select **Config F3**.
- 3 Choose one or more chemistries in the **Available Chemistries** list.
- 4 Select **Add->**.

The selected chemistries are enabled and appear in the **Configured Chemistries** list.

- 5 To enable all available chemistries, select **Add All>>**.

All chemistries in the **Available Chemistries** list are enabled and displayed in the **Configured Chemistries** list.

- 6 Select **OK**.

10.6.3 Disabling Chemistries

Some chemistries that will not be used for the moment can be disabled, and will no longer appear on request screens. ISE chemistries and SI are always available and cannot be disabled. Results of disabled chemistries cannot be recalled until the chemistries are enabled again.

A chemistry can be disabled only if:

- It is not an ISE chemistry.
- It is not SI.
- It has no reagent position.
- It has no calibrator position and has not been requested for calibration.
- It has no control position.
- It is not contained in samples and controls that are in Programmed, Incomplete or Rerun status.

Perform the following procedure to disable chemistries:

- 1 Select **Utility-Chemistries**.
- 2 Select **Config F3**.
- 3 Choose a chemistry in the **Configured Chemistries** list.

ISE chemistries and SI cannot be disabled.

- 4 Select **<-Remove**.

The selected chemistry is disabled and removed from the **Configured Chemistries** list.

- 5 To disable all chemistries, select **<<Remove All**.

All chemistries in the **Configured Chemistries** list that meet the requirements are disabled. The disabled open-reagent chemistries are indicated in red.

If one of the chemistries does not satisfy the requirements, the operation will be aborted and all the chemistries cannot be disabled.

- 6 Select **OK**.

10.6.4 Customizing Chemistry Display Order

Chemistries can be customized to match the test order of your laboratory and will be refreshed on the request screens. Chemistries on the **Chemistry Configuration** window are displayed alphabetically. In case an ISE module is configured, Na, K and Cl will appear on the first three positions after SI in the **Configured Chemistries** list. In the **Available Chemistries** and **Configured Chemistries** lists, click the **Chemistry** or **Module** header line to sort the chemistries by name or by module.

To adjust chemistry display order, perform the following steps:

- 1 Select **Utility-Chemistries**.
- 2 Select **Config F3**.
- 3 Choose a chemistry in the **Configured Chemistries** list.
- 4 Use the following buttons to adjust the chemistry's display order:
 - **Home**: to move the chemistry to the first position.
 - **Up**: to move the chemistry to the previous position.
 - **Down**: to move the chemistry to the next position.
 - **End**: to move the chemistry to the last position.
- 5 Select **OK**.

The chemistry list on the request screens are refreshed automatically.

10.6.5 Adjusting Test Order of Chemistries

Test order of configured biochemistries can be adjusted manually. During sample analysis, the chemistries are run in the order of ISE chemistries, SI, and then biochemistries. If multiple biochemistries are requested, they will be run in the

default order. If the test order is adjusted manually, the biochemistries will be run in the updated order.

Only users with corresponding permission are allowed to adjust the test order of biochemistries.

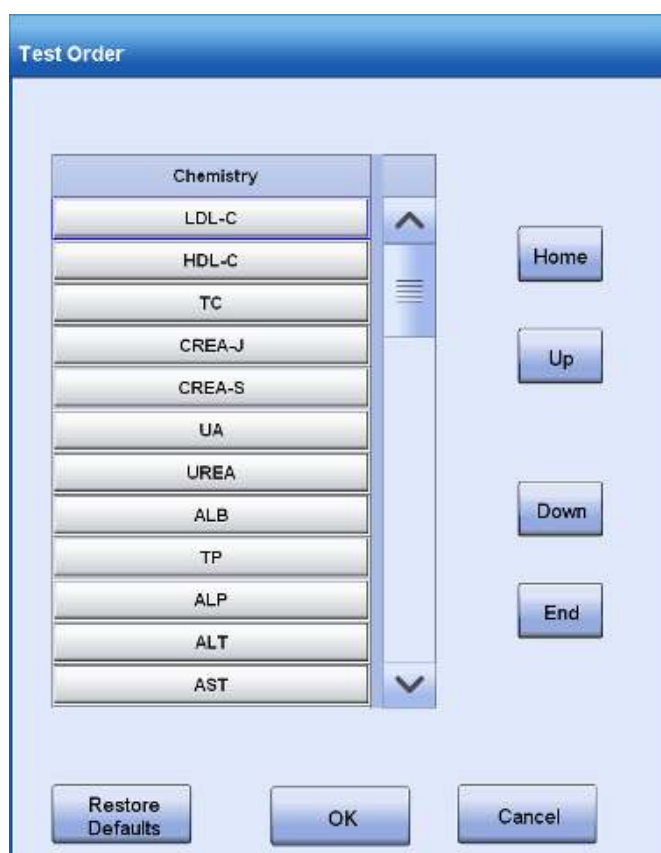
1 Select **Utility-Chemistries**.

2 Select **Config F3**.

3 Select **Options**.

4 Select **Test Order**.

Figure 10.9 Test Order window



5 Choose a chemistry in the **Configured Chemistries** list.

6 Use the following buttons to adjust the chemistry's test order:

- **Home:** to move the chemistry to the first position.
- **Up:** to move the chemistry to the previous position.
- **Down:** to move the chemistry to the next position.

- **End:** to move the chemistry to the last position.

7 Select **OK**.

8 To restore the default test order, select **Restore Defaults**.

10.7 Carryover Setup

10.7.1 Introduction

The Carryover Setup option is used to set up the carryover relations between open-reagent chemistries and between cuvettes. The system will insert a cleaning to reagent probe and cuvettes based on the carryover settings. The closed-reagent chemistries have been set up by the manufacturer and cannot be viewed or edited, while the open-reagent chemistries need to be set up on the **Carryover** window.

Note: When there exists carryover between the twin chemistries and a third chemistry, you should set up the carryover relationship of both the chemistries in the twin chemistries pair with the third chemistry. For example, HBA1C and HB are twin chemistries, if there exists the carryover between the twin chemistries and a third chemistry, then you should set up the carryover between HBA1C and the third chemistry and the carryover between HB and the third chemistry at the same time.

If carryover of one chemistry in the twin chemistries pair has been set up, edited or deleted, the carryover of the other chemistry should be set up, edited or deleted too.

Carryover setup can only be performed by users with sufficient permissions when the system status is not Running.

Figure 10.10 Carryover window

The screenshot shows the 'Carryover' window. On the left, there are radio buttons for 'Reagent' (selected) and 'Cuvette'. Below them are four dropdown menus labeled 'Contaminators', 'Contaminated', 'Contaminated', and 'Contaminated', followed by a checkbox for 'Special Wash'. The main area is a table titled 'Carryover Pairs' with columns: 'Contaminators', 'Contaminated', 'Reagent', 'Cuvette', 'Contaminators', 'Contaminated', and 'Special Wash'. The table contains two rows of data. At the bottom, there are buttons for 'Define F1', 'Delete F5', 'Discard F6', 'Save F7', and 'Close F8'.

Contaminators	Contaminated	Reagent	Cuvette	Contaminators	Contaminated	Special Wash
ALL	GLU-H			R1		
GLU-H	UA-1			R1	R1	

10.7.2 Defining/Editing Carryover Pair

- 1 Select **Utility-Chemistries**.

- 2 Select **Carryover F8**.
- 3 Mark the **Reagent** checkbox of the contaminated, if the contaminator may cause reagent cross-contamination with the contaminated. If the contaminator may cause cuvette cross-contamination, mark the **Cuvette** checkbox of the contaminated.
 - Only when the reagent checkbox of the contaminated is selected, all enabled chemistries are displayed in the **Contaminators** and **Contaminated** lists.
 - Once the reagent checkbox of the contaminated is selected, the special wash check box is selected as well and cannot be edited.
 - When the **Cuvette** checkbox of the contaminated is selected, the special wash check box is not selected by default and can be edited. When special wash check box is not selected, the system performs DI water wash.
- 4 Choose one or all contaminator chemistries that may contaminate other chemistries.
- 5 Choose one or all contaminated chemistries in the **Contaminated** list.

Note: All (the entire contaminator) to All (the contaminated) is not permitted to set up.

- 6 If different reagent type may cause reagent carryover, choose a reagent type in the contaminator reagent list and choose a reagent type in the contaminated reagent list.
- 7 Select **Save F7**.

The defined carryover pair appears in the **Carryover Pairs** list.
- 8 Select **Define F1** and follow the above steps to set up other carryover pairs.
- 9 Select **Close F8** to close the window.

10.7.3 Removing a Carryover Pair

- 1 Select **Utility-Chemistries**.
- 2 Select **Carryover F8**.
- 3 Choose desired carryover pair.

- 4 Select **Delete F5**.
- 5 Select **OK** to confirm the deletion.
- 6 Select **Close F8** to close the window.

10.8 Default Panel

10.8.1 Introduction

The system allows a maximum of one default panel to be defined. When a bar-coded sample has no relevant programming information on the LIS host or has not been programmed manually, it can be analyzed with the default panel. The default panel is only applicable to routine and emergent samples, and often used for a tremendous amount of samples that are analyzed with the same chemistries.

Only a sample panel rather than control panel can be set as the default.

10.8.2 Defining the Default Panel

- 1 Select **Utility-Chemistries**.
- 2 Select **Panels F7**.
- 3 Select **Define F1**.
- 4 Type in the number of the panel.
- 5 Type in the panel name.
- 6 Choose panel types.
 - Sample: indicates that the panel can be used for sample analysis.
 - QC: indicates that the panel can be used for quality control.

At least one panel type must be selected. A panel can be applied to both sample and control analysis.

- 7 Choose chemistries for the panel.

At least one biochemistry should be selected. The three ISE chemistries (Na, K and Cl) can be selected alone.

- 8 Select **Save F7**.
- 9 Select **Close F8**.
- 10 Select the defined panel in the panel list.
- 11 Mark the **Default** checkbox in the same row as the selected panel.
- 12 Select **Close F8** to close the window.

10.8.3 Running Default Panel for Routine Samples



BIOHAZARD

Inappropriate handling of samples may lead to biohazardous infection. Do not touch the samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the samples, follow standard laboratory safety procedure and consult a doctor.



CAUTION

Do not use expired samples; otherwise, unreliable test results may be caused.

- 1 Load samples to the sample carousel.
- 2 Select the  icon on the upper-right corner of the main screen.
- 3 Select a sample carousel to which the samples are loaded.
- 4 Select **OK**.

10.8.4 Running Default Panel for Emergent Samples



BIOHAZARD

Inappropriate handling of samples may lead to biohazardous infection. Do not touch the samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the samples, follow standard laboratory safety procedure and consult a doctor.



CAUTION

Do not use expired samples; otherwise, unreliable test results may be caused.

- 1 Load emergent samples to the sample carousel.
- 2 Select the  icon on the upper-right corner of the main screen.
- 3 Select a sample carousel to which the samples are loaded.
- 4 Select **OK**.

10.9 Masking/Unmasking Chemistries

10.9.1 Introduction

The chemistry masking function is used when a chemistry needs to be disabled temporarily due to abnormal result or reagent exhaustion. The masked chemistry will have a  symbol appearing on its upper-left corner, and will still be displayed on the **Sample**, **Quality Control** and **Reagent/Calibration** screens. Masked chemistries can be requested but cannot be run until they are unmasked.

In any system status chemistries can be masked or unmasked. Any users are allowed to mask or unmask chemistries.

If a sample contains masked chemistries, it will enter the Incomplete status when finished; if chemistries are unmasked while the sample status is Programmed, they will be run along with other chemistries; if chemistries are unmasked while the sample is being analyzed, they will be added automatically to the analysis; if chemistries are unmasked after the sample is analyzed, they will be run automatically when analysis begins next time.

10.9.2 Masking/Unmasking Chemistries

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **2 Mask/Unmask Chem.**

Figure 10.11 Mask/Unmask Chemistries window



10 Chemistries

- 4 Choose chemistries to mask, select **OK**.
- 5 To unmask chemistries, select them and then select **OK**.
- 6 Select **Exit** to close the window.

10.10 Reflex

10.10.1 Introduction

The Reflex option allows related chemistries to be requested and run automatically when the deciding biochemistry's result is within specified range. Each biochemistry may embrace multiple reflex conditions, and each condition may contain a maximum of 20 related chemistries.

Reflex conditions and related chemistries are open for observation, but only users with corresponding permission are allowed to set, modify or delete reflex relation.

10.10.2 Setting Reflex Relation

Before using the reflex function, it must be enabled with reflex conditions and related chemistries configured. Please note that the chemistries in a reflex condition must have existed. When the system status is running, the reflex function cannot be set up.

- 1 Select **Utility-Chemistries**.
- 2 Choose a chemistry for which you desire to configure reflex settings.
- 3 Select **Define F1**.
- 4 Select **Reflex F2**.

Figure 10.12 Reflex window

Reflex

Chemistry TP-

☒ Enable Reflex Function

No.	Conditions	Related Chemistry
1	>0.000 and <100.000	ADA,ASO,C4,CHE

Conditions:

☐ > 0 or < 100
☒ > 0.000 and < 100.000

Related Chemistry:

5'-NT	ADA	AFU	ALB	ALP
ALT	ASO	AST	ApoA1	ApoB
C3	C4	CHE	CK	CK-MB
CO2	CREA-J	CREA-S	CRP	Ca
CysC	D-Bil-V	D-bil-D	<	>

Buttons: Delete, Cancel, OK, Exit

5 Mark the **Enable Reflex Function** checkbox to activate this option.

6 Set up reflex conditions.

Two conditions are available: “or” and “and”:

- or: When the test result (concentration) is greater than certain value OR less than certain value, the related chemistries will be requested and run automatically.
- and: When the test result (concentration) is greater than certain value AND less than certain value, the related chemistries will be requested and run automatically.

Select an option and input the concentration range (0~9999.999).

7 Choose related chemistries in the chemistry list.

The options include all configured biochemistries.

8 Select **OK**.

The defined reflex relation is shown in the left list.

9 To define more relations for the current chemistry, repeat steps 6~8.

10 Select **Exit** to close the window.

- 11 To define reflex relations for other chemistries, repeat steps 1~8.

10.10.3 Editing Reflex Relation

Only users with corresponding permission are allowed to edit reflex relation.

- 1 Select **Utility-Chemistries**.
- 2 Choose a chemistry for which you desire to edit reflex settings.
- 3 Select **Define F1**.
- 4 Select **Reflex F2**.
- 5 Select a reflex relation in the left list.
- 6 Modify the condition and related chemistries.
- 7 Select **OK**.
- 8 Select **Exit** to close the window.

10.10.4 Deleting Reflex Relation

Only users with corresponding permission are allowed to delete reflex relation. If a chemistry is deleted, the corresponding reflex relation to which it is related will be removed automatically.

- 1 Select **Utility-Chemistries**.
- 2 Choose a chemistry for which you desire to delete reflex settings.
- 3 Select **Define F1**.
- 4 Select **Reflex F2**.
- 5 Select a reflex relation in the left list.
- 6 Select **Delete**.
- 7 Select **OK**.
- 8 Select **Exit** to close the window.

10.10.5 Measurement and Result Recall

Chemistries with reflex settings are run in the same way as routine biochemistries. When the test result meets the set condition, the related chemistries will be requested and run automatically while those that have been requested for the sample will be excluded. To view the results, select **Result-Current Results** or **History Results**.

11

System Commands and Setup Options

The system commands include:

- Home
- Stop print
- Wake up

The advanced setup options include:

- User and password setup
- System timers for auto sleep and auto startup
- Dictionary setup
- Software version upgrading
- Software version
- Voice tone setup

11.1 Home

11.1.1 Introduction

The Home command is used to initialize the biochemistry system and the ISE module, and to recover them from failures, making all components return to the home positions. When the Home command is executed, the system status becomes Standby.

11.1.2 Homing System

- 1 Select **Utility-Commands**.
- 2 Select **Home**.

11.2 Stop Print

11.2.1 Introduction

The Stop Print command will stop all print requests in the print queue and prevent them from being sent to the printer. This feature is used for stopping print requests of many pages, such as error logs, QC reports, multi-sample reports, etc. The print tasks that are Printing, Deleted, Canceling or Canceled in the print task window will not be deleted.

11.2.2 Stop Print

- 1 Select **Utility-Commands**.
- 2 Select **Stop Print**. All print requests in the print queue will be removed.

11.3 Sleep and Wake Up

11.3.1 Introduction

When hibernating manually through the sleep option, the system can be activated by the Wake Up command. Except for Wake Up, the system can also be activated based on the set auto awake time.

For more information about auto sleep and wake up, refer to 11.5 Sleep and Awake Setup (page 11-9).

11.3.2 System Hibernation

- 1 Select **Exit**.
- 2 Select **Sleep**.

The system starts hibernating and the system status changes into Sleep.

11.3.3 Waking up the System

- 1 Select **Utility-Commands**.
- 2 Select **Wake Up**.
- 3 The system is waking up, and the system status becomes Standby.

11.4 User and Password Setup

11.4.1 Introduction

Users can be defined, deleted or modified on the **User and Password** window. The system allows up to 100 users to be defined and belonged to two user groups: administrator and operator. Administrators are allowed to assign permissions for operators.

Figure 11.1 User and Password window

Username	User Group
Admin	Administrator



NOTE

The default username and password for administrator is Admin. Please note that the password is case sensitive. You are recommended to change the password when logging on the system for the first time in order to prevent others from abusing the privileges of the administrator.

If an operator forgets his password, he may ask the administrator to log on the system and delete the username and then redefine a username; or he may contact our customer service department or your local distributor. If the administrator forgets his password, contact our customer service department or your local distributor.

11.4.2 Defining a User

Only administrators are allowed to define users. Up to 100 users are allowed, including administrators. You should enter the username, password, confirm password and user group when defining a user.

- 1 Select **Utility-System Setup**.

- 2 Select **User F6**.
- 3 Enter the username.
- 4 Enter the password.

A maximum of 20 characters can be entered.

- 5 Enter the password again in the **Confirm** field.
- 6 Choose a user group in the **User Group** pull-down list.

The options include:

- Administrator
- Operator

- 7 Select a doctor from **Associated Doctor** dropdown box.

When the user and the associated doctor have been set up, the default tester in the patient demographics is the associated doctor of the current login user.

- 8 Select **New**. The defined user appears in the user list.
- 9 Select **Exit** to close the window.

11.4.3 Modifying a User

Only administrators are allowed to edit the user group of themselves and other users. Username and password can only be modified by the user himself rather than anyone else.

- 1 Select **Utility-System Setup**.
- 2 Select **User F6**.
- 3 Choose a user to edit in the user list.
- 4 Enter the new username.
- 5 Enter the new password.
- 6 Enter the new password again in the **Confirm Password** field.
- 7 Choose a user group in the **User Group** pull-down list.

The options include:

- Administrator
 - Operator
- 8 Select **Modify**.
 - 9 Select **Exit** to close the window.

11.4.4 Assigning/Modifying Permissions

Permissions are assigned to user groups, which include administrator and operator. Administrators are allowed to use, assign and modify all permissions that are assigned for operators; while operators are only allowed to use common functions, such as assigning reagent position; programming samples, controls and calibrators; recalling sample/QC/calibration results; and those assigned by the administrators.

- 1 Select **Utility-System Setup**.
- 2 Select **User F6**.
- 3 Choose a user you desire to setup permissions in the user list, and then select **Permission**.

Figure 11.2 Permission assignment

The screenshot shows a window titled "User and Password". Inside, the "Username" field is set to "aaa". Below this is a table with two columns: "Select" and "Function". The "Select" column contains buttons labeled "yes" or "no". The "Function" column lists various operations. The row for "Define, set up and modify calibrators" has the "yes" button selected. At the bottom of the window are two buttons: "Exit" and "Save".

Select	Function
yes	Delete result
no	Edit result
no	Export sample result
yes	Define, set up and modify calibrators
no	Load calibrator parameters
no	Recalculate calibration result
no	Edit calibration result
no	Export calibration result

- 4 Assign permissions for the selected user.
 - To assign new permissions, select the box in front of the relevant operation. The select button changes to Yes.

- To cancel permissions, deselect the box in front of the relevant operation.
The select button changes to No.

- 5 Select **Save** to save the settings.
- 6 Select **Exit** to close the window.

11.4.5 Deleting a User

The username that has been used to log on the system currently cannot be deleted. Only the administrators are allowed to delete users.

- 1 Select **Utility-System Setup**.
- 2 Select **User F6**.
- 3 Choose a username in the user list.
- 4 Select **Delete**.
- 5 Select **OK**.
- 6 Select **Exit** to close the window.

11.5 Sleep and Awake Setup

11.5.1 Introduction

The Sleep/Awake feature includes the Auto Sleep Setup and Auto Awake Setup options.

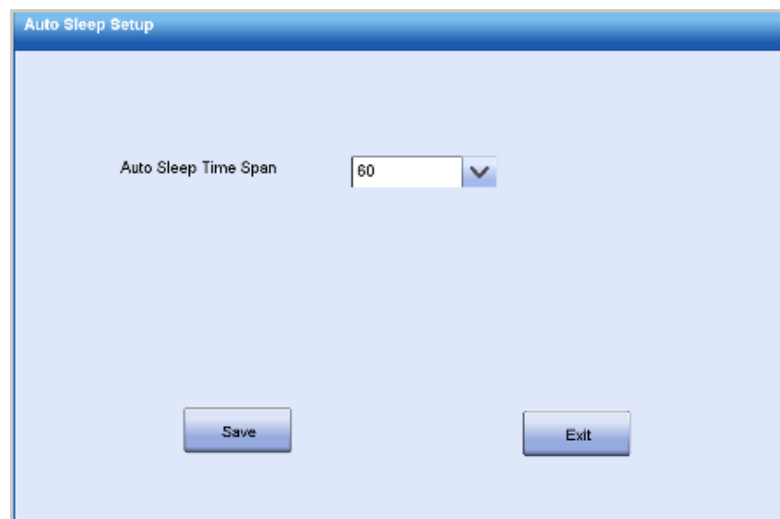
The Auto Sleep Setup option is used to set up the time interval of auto sleep time of the system. After the sleep time interval is set up, a countdown will begin from the moment that the system status becomes Standby. When the time interval is elapsed, the system will begin sleeping. Except for the auto sleep setting, the system can be woken up by means of the wake up command.

The Auto Awake Setup option allows to define date and time of starting up the system. When the time is reached, the system will be started up or woken automatically no matter if it is off or sleeping.

11.5.2 Auto Sleep Setup

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **1 Sleep/Awake**.
- 4 Select **1 Auto Sleep Setup**.

Figure 11.3 Auto Sleep Setup window



- 5 Type in the time interval for auto sleep.

The options include N/A, 30, 60, 90, 120, 150, 180, 210, 240, 270 and 300 and the default is 60 minutes. N/A means the auto sleep timer is disabled



NOTE

If auto sleep is not enabled, some components, such as lamp, may get aged quickly and degraded in performance. You are recommended to enable this option.

- 6 Select **Save**.

When the interval is elapsed, the system will start to sleep and the system status becomes Sleep.

- 7 Select **Exit**.

11.5.3 Auto Startup Setup

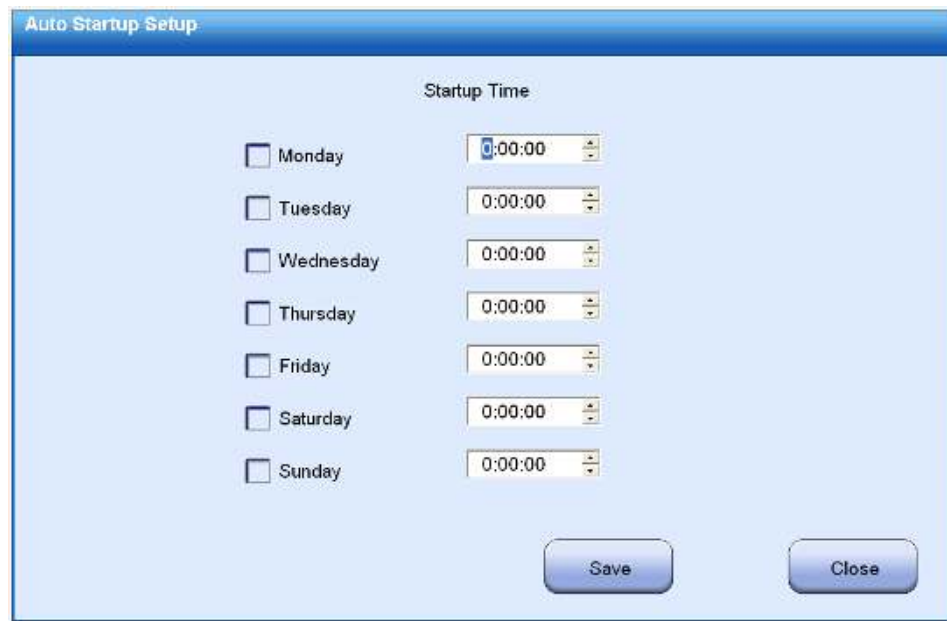


NOTE

After setting up the auto startup time, ensure that the operation unit and the analyzer are connected to power supply; otherwise, they cannot be started up automatically.

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **1 Sleep/Awake**.
- 4 Select **2 Auto Startup Setup**.

Figure 11.4 Auto Startup Setup



The image shows a software dialog box titled "Auto Startup Setup". Inside the dialog, there is a section labeled "Startup Time". Below this label, there are seven rows, each representing a day of the week: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, and Sunday. Each row consists of a checkbox on the left and a time selection field on the right. All checkboxes are currently unchecked, and all time fields are set to "0:00:00". At the bottom right of the dialog, there are two buttons: "Save" and "Close".

- 5 Choose the weekday for auto startup, and then set up the specific time.

Any time within a week(from Monday to Sunday) can be defined for the system to start up automatically.

- 6 Select **Save**.

When the date and time is reached, the system will be started up automatically no matter if it is off.

- 7 Select **Close**.

11.5.4 Auto Awake Setup



NOTE

After setting up the auto awake time, ensure that the operation unit and the analyzing unit are connected to power supply; otherwise, they cannot be woken up automatically.

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **1 Sleep/Awake**.
- 4 Select **2 Auto Awake Setup**.

Figure 11.5 Auto Awake Setup



- 5** Choose the weekday for auto startup, and then set up the specific time.

Any time within a week (from Monday to Sunday) can be defined for the system to start up automatically.

- 6** Select **Save**.

When the date and time is reached, the system will be started up or woken automatically no matter if it is off or sleeping.

- 7** Select **Exit**.

11.6 Dictionary Setup

11.6.1 Introduction

The Dictionary option is provided for setting up and managing frequent data information, including: result unit, sample type, sample comment, and QC comment. A maximum of 30 data options can be defined for each dictionary, and each dictionary must not contain duplicate data. Sample comment can be entered manually or selected from the **Comment** pull-down list on the **Sample** screen, **Levey-Jennings** screen, and (QC) **Results** screen.

Data options can be defined, edited or deleted in any system status. The default data options cannot be deleted or edited.

11.6.2 Defining, Editing and Deleting Data Option

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **3 Dictionary**.

Figure 11.6 Dictionary window

No.	Data	Description	Symbol
1			
2	mg/dL		
3	mg/L		
4	g/dL		
5	g/L		
6	mmol/L		
7	µmol/L		
8	mEq/L		
9	nKat/L		
10	µKat/L		
11	IU/L		
12	µg/mL		
13	ng/mL		

Data: Symbol:
 Description:

- 4 Choose desired dictionary in the Data list.

11 System Commands and Setup Options

To add a data option:

- Select **New**.
- Input the data description in the **Data** field.
- Select **Save**.

To modify a data option:

- Select desired data option in the data list.
- Modify the data description in the **Data** field.
- Select **Save**.

To delete a data option:

- Select desired data option in the data list.
- Select **Delete**.

5 Select **Close**.

11.7 Software Upgrade

11.7.1 Introduction

Software Upgrade is used to upgrade the operating software and ISE module software. When software versions is upgraded, the original data, including those in the database and saved in files, will not be destroyed and can be compatible with the new versions.

11.7.2 Software Upgrade

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **6 Version Upgrade**.
- 4 Insert the U disk containing the software into the USB interface of the computer.
- 5 Select **OK**, and then operate according to the screen prompts.

11.8 Software Version

11.8.1 Introduction

You are allowed to view the version number of the operating software and control software in any system status.

11.8.2 Software Version

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **7 Version Info**.

Figure 11.7 Software Version window



- 4 View the version number of the operating software, control software, database, and ISE module software.

If a new version is released, upgrade the operating software while referring to 11.7 Software Upgrade (page 11-15). If no ISE module is configured, the **ISE Software Version** area will be blank.

- 5 To view the version numbers of the smart module software, select **Details**.

Figure 11.8 Smart module software version window

- 6 Move the scroll bar to view more versions.
- 7 Select **OK**.

11.9 Voice Tone Setup

11.9.1 Introduction

The Voice Tone Setup option provides voice tone choices for system failures or user's mis-input or mis-operation. You are allowed to import audio files from an external storage device and set them as voice tone.

11.9.2 Importing Audio Files

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **11 Voice Tone Setup**.
- 4 Select **Import**.
- 5 Select the path and one or more favorite audio files.
- 6 Select **Open**.

The imported audio files are displayed in the **Alarm** and **Message Tip** pull-down lists.

11.9.3 Setting Up Voice Tone

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **11 Voice Tone Setup**.

Figure 11.9 Voice Tone Setup window

- 4 Choose a voice tone from the pull-down list box, and then select the corresponding **Test** button to test the voice effect until the proper one is found.
- 5 Select **Save** to save the settings.

12 Use of ISE Module

This chapter introduces the ISE module in the following aspects:

- Precautions on use
- Principles of measurement
- Chemistry parameters setup
- Preparing reagents for analysis
- Running ISE chemistries
- Results recall
- ISE primes and calibration factor expiration date

12.1 Precautions on Use

12.1.1 Introduction

Read the following precautions thoroughly prior to using the ISE module.

12.1.2 Precautions on Use

Operator Precautions



Warning

The ISE module must be operated by skilled/trained doctors, nurses or clinical professionals.

Driver parts precautions



Warning

Exercise caution while using the ISE module. Prevent your hair, legs or other parts of your body from being hurt by the driver parts.

Serum sample biohazards



BIOHAZARD

The serum samples remaining in the electrodes may contain a great number of viruses. Wear gloves to prevent infection while operating around the electrodes.

Calibration precautions



CAUTION

Calibrate the ISE chemistries for serum and urine before starting the measurement. If the result of a chemistry is based on the calibration factors of another chemistry, it may not be accurate enough.

After changing electrodes or other consumables, perform a calibration. You are recommended to perform calibration at least once every day to ensure accurate results.

Calibrator biohazards



BIOHAZARD

The calibrators contain preservatives. In case your skin contacts calibrators, wash them off with soap and water. In case the calibrators spill into your eyes, rinse them with water and consult an oculist. If you swallow them by mistake, see a doctor.



CAUTION

Use the calibrators specified by our company. Use of other reagents or calibrators may result in unreliable results, or damage the Hydropneumatic system, or even shorten the electrodes life span.

Prior to using the calibrators, check if they are within the expiration date.

Place them correctly; otherwise, it may cause unreliable results, or leak, or module damage.

ISE wash solution biohazards



BIOHAZARD

The ISE wash solution is sodium hypochlorite. Use the ISE wash solution carefully to prevent it from contacting your skins or eyes. If your skins or eyes contact the ISE wash solution, rinse them off with fresh water and consult a doctor.

12.2 Principles of Measurement

The ISE unit measures the concentration of Na⁺, K⁺ and Cl⁻ ions contained in serum and urine samples with the ion-selective electrode method. The relation between the electromotive force of ion-selective electrodes and the ion concentration is expressed in a Nernst formula. Serum is tested without dilution while urine should be diluted manually before test with buffer solution at the ratio of 1:9.

A single measurement of the ISE unit is conducted in the following order:

- Drainage: The calibrator in the ISE pipe is drained
- Sample analysis: The sample probe dispenses the sample (70μL for serum sample, 140μL for diluted urine) into the sample injection port of the ISE module and then the sample is absorbed into the flow cell for measurement. When the measurement is finished, the waste is drained from it.
- Cleaning pipework: 100μL calibrator A is dispensed into the ISE module for cleaning the ISE flow cell
- Single point calibration: 80μL calibrator is dispensed into the ISE module to perform single point calibration.

The table below lists the measurement range of the ISE module:

Table 12.1 Measurement range of ISE module

Chemistry	Serum	Urine
Na ⁺	100~200mmol/L	10~500mmol/L
K ⁺	1~8mmol/L	5~200mmol/L
Cl ⁻	50~150mmol/L	15~400mmol/L

12.3 ISE Chemistry Parameters

12.3.1 Introduction

The ISE module measures the concentration of K⁺, Na⁺ and Cl⁻ ions contained in human body fluid by means of electrodes, helping diagnosis of electrolyte disturbance, body fluid equilibrium, and other relevant diseases.

The ISE chemistries are applicable to serum and urine, and the default sample type is serum. If the sample is of a type other than serum and urine, it will be analyzed with the chemistry parameters for serum.

ISE chemistry parameters can be viewed but cannot be modified and reconfigured.

Figure 12.1 Define/Edit ISE Chemistries window

The screenshot shows a software window titled "Define/Edit ISE Chemistries". At the top, there are three input fields for "Print Name" with values "Na", "K", and "Cl". Below this, the window is organized into a table-like structure with rows for "Na(Serum)", "Na(Urine)", "K(Serum)", "K(Urine)", "Cl(Serum)", and "Cl(Urine)". Each row contains four columns: "Unit" (all set to "mmol/L"), "Decimal" (values: 0.1 for Na(Serum), 0 for Na(Urine), 0.01 for K(Serum), 0 for K(Urine), 0.1 for Cl(Serum), and 0 for Cl(Urine)), and "Measur. Range" (two input boxes per row: Na(Serum) 100-200, Na(Urine) 10-500, K(Serum) 1-8, K(Urine) 5-200, Cl(Serum) 50-150, Cl(Urine) 15-400). On the right side of the table, there are two vertical arrow buttons (up and down). At the bottom right, there are three buttons: "Print", "OK", and "Exit".

Print Name	Unit	Decimal	Measur. Range
Na Na	mmol/L	0.1	100 200
Na(Urine)	mmol/L	0	10 500
K K	mmol/L	0.01	1 8
K(Urine)	mmol/L	0	5 200
Cl Cl	mmol/L	0.1	50 150
Cl(Urine)	mmol/L	0	15 400

12.3.2 Viewing ISE Chemistry Parameters

The ISE chemistry parameters are opened to all users for viewing in any system status.

- 1 Select **Utility-Chemistries**.
- 2 Choose the **ISE** box.
- 3 Select **Define F1**.
- 4 View the parameters.

- 5 Click **Exit** to close the window.

12.3.3 Introduction to ISE Chemistry Parameters

ISE chemistry parameters and reference range are displayed on the Define/Edit ISE chemistries screen. ISE chemistry has two test mode: serum and urine. For urine, it has to be diluted manually before test.

In the following table, U stands for urine and S for serum.

Table 12.2 ISE chemistry parameters(cannot be edited)

Parameter/Chemistry	K ⁺	Na ⁺	Cl ⁻
Unit (S)	mmol/L	mmol/L	mmol/L
Unit (U)	mmol/L	mmol/L	mmol/L
Decimal (S)	0.01	0.1	0.1
Decimal (U)	1	1	1
Measurement Range (S)	1.00-8.00	100.0-200.0	50.0-150.0
Measurement Range (U)	5-200	10-500	15-400

Unit

The unit of K,Na and Cl is mmol/L which can be viewed but cannot be edited.

Decimal

The decimal of the result can be viewed but cannot be edited.

Measurement Range

The measurement range can be viewed but cannot be edited.

12.3.4 Flag ISE Qualitative Result

- 1 Select **Utility-Chemistries**.
- 2 Choose the **ISE** box.
- 3 Select **Define F1**.
- 4 Click the **Down** button.

Figure 12.2 Define/Edit ISE Chemistries window

Define/Edit ISE Chemistries

Print Name Na K Cl

Na

☐ Use Qualitative Result

Range	Flag

K

☐ Use Qualitative Result

Range	Flag

Cl

☐ Use Qualitative Result

Range	Flag

▲

▼

Print OK Exit

- 5 Select **Use Qualitative Result** under Na.
- 6 Enter **Range** and **Flag**.
- 7 Repeat 5-6 to flag the qualitative result for Na and Cl.
- 8 Click **OK** to save the setup.
- 9 Select **Exit** to close the window.

12.4 Preparing ISE Reagents for Measurement

12.4.1 Introduction

The ISE wash solution is placed on the sample carousel. When you cancel the configuration of the ISE module, the loaded reagents are deleted.

Please follow the steps below to load the ISE reagents:

12.4.2 Off-line Load ISE Reagent

Off-line load ISE reagent is performed when ISE is not running its tests.

- 1 Make sure the ISE module is not running its tests.
- 2 Select **Reagent-Reagent/Calibration-ISE Reagent**.
- 3 Select **Load F1**.

If it is the first time to load the ISE reagent, please enter Step 7; if not, unload the reagent first.

Figure 12.3 Unload Reagent

- 4 Select **Unload** to unload the reagent pack.
- 5 Disconnect the wand from the top of the Reagent Pack by pushing down the yellow button in the wand, which disconnects the wand from and Reagent Pack.

Set the wand on the table surface, which should not leak. Carefully remove the used Reagent Pack from the Chemistry Analyzer and dispose of it properly.

- 6 Click **OK** to load the reagent pack.
- 7 Remove the red caps from Reagent Pack first and push the wand just above the top of Reagent Pack. Make sure that the three pipe adapters at the bottom of the wand are opposite to those on the top of Reagent Package, then push down the wand to Reagent Package. The wand will only fit one way. Once the wand is connected to the reagent pack, place the reagent pack on the sliding tray

Figure 12.4 Installing Reagent Pack 1

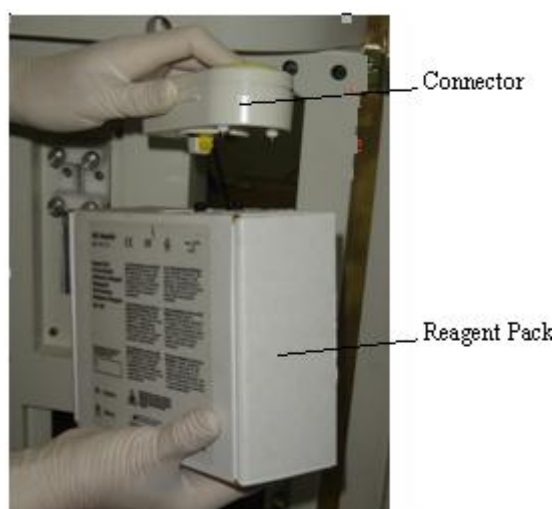


Figure 12.5 Installing Reagent Pack 2



NOTE

When installing ISE reagent pack, do not twist, press and squeeze the pipes of the ISE Module otherwise the ISE pipes may be clogged.

Wear gloves and lab coat and, if necessary, goggles.

Figure 12.6 Load Reagent

Rgt Load

Reagent Status OK	
Load Date 2012- 1-25	Exp Date 2012-11-30
A Volume 49 %	B Volume 88 %
Purge A Volume 400 ml	Purge B Volume 300 ml
Lot No. 00009E8DE4	Purge A/B 30

- 8** Enter the number in **Purge A/B** field which can be any integer within 1-50 and the default is 30.

- 9 Click **Prime**.
- 10 Select **OK** to calibrate ISE chemistry when Purge A/B is complete.

12.4.3 On-line Load ISE Reagent

On-line loading ISE reagent is performed when ISE module is running its tests.

- 1 Select **Reagent-Reagent/Calibration-ISE Reagent**.
- 2 Select **Load F1** to stop reagent aspirating and dispensing. Meanwhile **Load F1** becomes **No load F1**.
- 3 If you want to cancel the reagent stop, select **No Load F1**.
- 4 Click **OK** and then select **Load F1**. Here, the load steps are the same as off-system load ISE reagent.

12.4.4 Load ISE Wash Solution

ISE wash solution is used to wash the electrodes and can only be loaded manually.

- 1 Check the system status and operate accordingly.
 - Standby: Proceed to the next step.
 - Running: Select the  button on the upper-right corner of the main screen to stop sample aspirating and dispensing. When the countdown for sample stop becomes 0 and the system status is Sample Load, proceed to the next step.
 - Incubation: Proceed to the next step.
 - Sleep: Proceed to the next step.
- 2 Remove the sample carousel cover;
- 3 Place ISE wash solution in position D1 (No.88) of the sample carousel.
- 4 Restore the sample carousel cover;
- 5 Select the  icon to resume the test or start new test.

12.4.5 Replacing ISE Wash Solution

Replacing the wash solution can be performed when the ISE module status is Standby or Running. If the system status is Standby, you are allowed to directly replace the wash solution in the same way as it is loaded; if the system status is

Running, the wash solution can only be replaced after the current tests are finished; if the ISE module is running a calibration, you are not allowed to replace the wash solution until all tests of the calibration are finished.

- 1 Check the system status and operate accordingly.
 - Standby: proceed to the next step.
 - Running: Select the  button on the upper-right corner of the main screen to stop sample aspirating and dispensing. When the countdown for sample stop becomes 0 and the system status is Sample Load, proceed to the next step.
 - Incubation: proceed to the next step.
 - Sleep: Start replacing reagents.
- 2 Remove the sample carousel cover.
- 3 Remove the ISE wash solution from position D1 (No.88) of the sample carousel.
- 4 Place the new wash solution.
- 5 Restore the sample carousel cover.
- 6 Select the  icon to resume the test or start new test.

12.5 Calibration and Results Recall

12.5.1 Introduction

Current calibration factors and all intermediate data are provided on the **ISE Calibration** screen. Calibration results can be printed out or archived to an external storage device. The Trend option is provided to enable you to view the calibration trends of ISE chemistries during a period of time. When the calibration factors are expired, the Extend Calibration Time can help prolonging their validity period for measurement.

12.5.2 Calibration Setup

Set up ISE calibrators and the calibration time. When a calibrator is expired, it will be indicated in yellow and cannot be used for calibration.

Setting up ISE calibration time

- 1 Select **Reagent-Setup**.
- 2 Select **Rules F4**.
- 3 Choose a chemistry from the **Chemistry** pull-down list.

Figure 12.7 ISE calibration setup window

Calibrator	Cone	Pos	Lot No.

- 4 Enter the calibration time in the **Cal Time** field.

The input range is 1-9999, and the default is 8 hours. If the field is left blank, it indicates that the calibration factors can be always used.

5 Select **Auto Calibration** if you want to set up ISE auto calibration.

6 Enter the auto calibration time.

The range is 1-24 hour; the default is blank.

7 Select **Save F7** to save the setup.

8 Select **Prev F4** or **Next F5** to set up calibration rules for other chemistries.

9 Select **Close F8** to close the window.

12.5.3 Calibration Status and Alarm

On the **Reagent/Calibration** screen, the chemistries are indicated with various texts and colors for different calibration status. Chemistries in Cal Required, Cal Failed or Cal Time Out status can be requested but will not be run.

Check the chemistries' calibration status frequently and take relevant actions according to the following table.

Table 12.3 ISE calibration status

Calibration Status	Description	Severity	Color
Cal Required	Indicates that the chemistry needs to be calibrated. This status appears when the chemistry is not calibrated or the ISE reagent/electrode is replaced.	Serious	Red
Requested	Indicates that the chemistry has been requested for calibration but not finished yet.	Normal	No color indication
Calibrated	Indicates that the chemistry has been calibrated successfully and has not exceeded the calibration time.	Normal	No color indication
Cal Failed	Indicates that the chemistry has calibration factors calculated but they exceed the acceptance limits, or has no calibration factors calculated.	Serious	Red

Calibration Status	Description	Severity	Color
Cal Time Out	Appears when the chemistry exceeds the calibration period or the reagent of different serial number and lot number is used. Appears when the chemistry exceeds the calibration time.	Serious	Red
Cal Time Extended	Indicates that the calibration period has been extended and the current calibration factors can be used without time limit.	Warning	Yellow
Default	Appears when the result is calculated with the default calibration factors defined by the user rather than the factory.	Warning	Yellow
N/A	Indicates that the reagent is not loaded.	Normal	No color indication

12.5.4 ISE Calibration

ISE chemistries, which include Na, K and Cl, are calibrated without being divided into ISE serum and ISE urine

- 1 Select **Reagent-Reagent/Calibration**.

Figure 12.8 Reagent/Calibration screen



- 2 Choose **ISE Chemistry**.
- 3 Select **Calibrate F5**.
- 4 Select **OK** to start calibration.

12.5.5 Results Recall

The calibration data and trends of ISE chemistries are provided on the **ISE Calibration** screen. The system allows you to recall the current ISE calibration factors and results of recent 540 calibrations. If a calibration result is abnormal, a flag will be added on patient reports and on the **ISE Calibration** screen.

For more information about result flags, refer to 17.4 Data Alarm (page 17-14).

Recalling calibration results

- 1 Select **Reagent-ISE Calibration**.

Figure 12.9 ISE Calibration screen



The screen shows the calibration factors that are being used for calculating results.

- 2 Select the **History** option button, and then select date range that the chemistry is calibrated.
- 3 Select **Search F1**.

The calibration results of the chemistry are displayed in the result list.

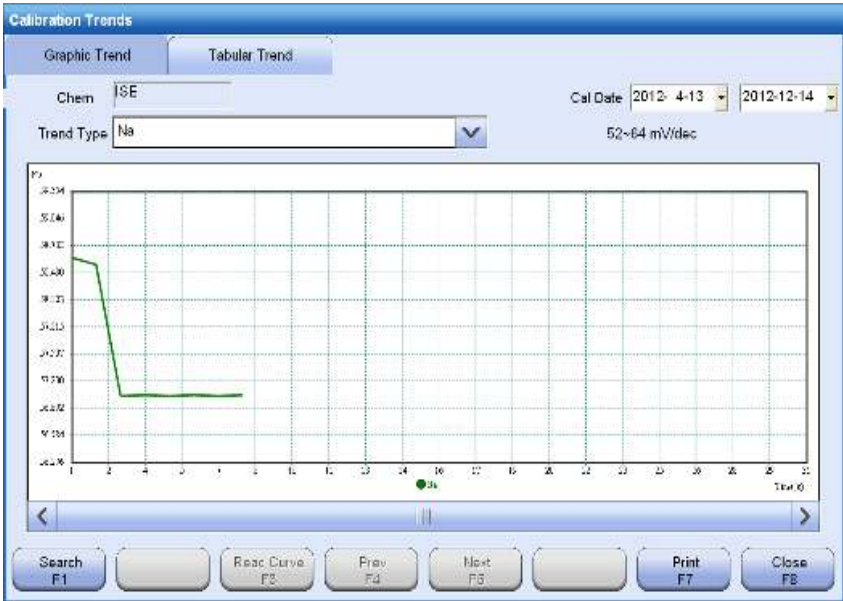
- 4 To print the calibration report, select **Print F7**.

Recalling calibration trends

- 1 Search for desired calibration results on the **ISE Calibration** screen.
- 2 Select **Trend F6**. The **Calibration Trends** window is displayed.
- 3 Choose desired trend type and calibration date range, and then select **Search F1**.

The trend of the chemistry within the specified time period is displayed on the screen. The trend type options will not include Reference Electrode when trends of ISE Urine are being recalled.

Figure 12.10 Calibration Trends window



4 Choose the **Tabular Trend** tab to view the trend data.

Figure 12.11 Tabular Trend window



5 Choose the following buttons as needed:

- **Prev F4:** to view the calibration trends and data of the previous chemistry.
 - **Next F5:** to view the calibration trends and data of the next chemistry.
 - **Print F7:** to print the current graphic trend or data.
- 6 Select **Close F8** to close the window.

Archiving ISE calibration results

Both the current and early calibration factors of ISE chemistries can be archived. The archiving file is of .csv format and named by the date and time the results are archived.

- 1 Select **Reagent-ISE Calibration**.
- 2 Search for desired calibration results.
- 3 Select **Archive F5**.
- 4 Specify archiving path and file name.
- 5 Select **Save**.

12.5.6 Extending ISE Calibration Time

When ISE calibration factors exceed the validity period, they cannot be used for measurement, and the calibration status changes to Cal Required. If you are certain that the calibration factors are correct and valid, you may prolong their validity period by using the calibration time extension function. A calibration time can be extended only if the current calibration of the chemistry is timed out or succeeded. The results calculated based on extended calibration factors will be flagged.

- 1 Select **Reagent-Reagent/Calibration**.
- 2 Choose a chemistry you want to extend.
- 3 Select **Cal Options F8**.
- 4 Select **Extend Calibration Time** from the **Calibration Options** window.
- 5 Select **OK**. The calibration factors of the selected chemistry can be used without time limit.
- 6 To remove the extended status, recalibrate the chemistry.

12.6 Quality Control and Results Recall

12.6.1 Quality Control and Results Recall

Control samples can be defined, run and recalled for the ISE chemistries in the same way as for biochemistries. The ISE chemistries are divided into the following based on the sample types:

- Na (serum)
- K (serum)
- Cl (serum)
- Na (urine)
- K (urine)
- Cl (urine)

For the operating procedure of quality control, refer to 2.7 Quality Control (page 2-33).

For details of QC evaluation and results recall, refer to 7 Quality Control (page 7-1).

12.7 Sample Programming and Results Recall

The ISE chemistries, like biochemistries, can be also used for analyzing routine samples, emergent samples, added samples and reruns, and requested along with the biochemistries. They are requested in the form of Na, K and Cl, and applicable to serum, plasma, urine, CSF and other sample types. The four sample types other than urine are programmed with the serum parameters, while urine sample is with the urine parameters.

Nevertheless, the ISE chemistries are slightly different from biochemistries for that they do not support the measurement with increased or decreased or prediluted samples.

For the operating procedure of sample analysis, refer to 2.8 Programming Routine Samples (page 2-38).

ISE test results have no reaction curves and can be recalled in the same way as other chemistries. Refer to 8 Sample Programming and Processing (page 8-1) for details.

12.8 Reagent Inventory Alarm Limit

12.8.1 Introduction

When the reagent inventory is lower than the alarm limit during or before the analysis, the system will give an alarm and display the volume of ISE reagent and wash solution as <x on the **Reagent/Calibration** screen and the ISE reagent is marked by yellow ;("x" stands for the defined alarm limit) When the ISE reagent inventory is 0, it is marked by red.

12.8.2 Setting up Reagent Inventory Alarm Limit

- 1 Select **Utility-System Setup**.

Figure 12.12 System Setup screen



- 2 Type in the inventory alarm limit of ISE reagent.

Enter an integer between 1 and 50. The default is 5.

- 3 Select **Save F8**.

12.9 ISE Prime Cycle

12.9.1 Introduction

While the analyzer is started up and new reagent pack is identified, the ISE module will prime automatically to replace the reagents inside of it with fresh reagents. The number of primes can be defined on the **System Setup** screen.

Only administrators are allowed to define or modify the startup prime times.

12.9.2 Defining/Modifying ISE Prime Times

- 1 Select **Utility-System Setup**.
- 2 Type in the number of primes in the **ISE Prime Cycle** field.

The input range is 1-50, and the default is 30.

- 3 Select **Save F8**.

12.10 Daily Maintenance

12.10.1 Daily Maintenance

To ensure the ISE module's life span and measurement performance, maintain it regularly as instructed in this manual. The system provides scheduled maintenance and maintenance instructions, in which the latter contains all of the scheduled maintenance procedures and some maintenance instructions that can be performed independently.

The table below is a summary of the scheduled maintenance procedures and maintenance instructions for the ISE module.

Table 12.4 Scheduled maintenance and instructions for ISE module

Schedule	Maintenance Procedures
Daily	Clean ISE electrode tubes
Monthly	Clean sample injection port
	Pump calibration
	Air bubble Detector calibration
Six month	Replace reference electrode
Other	Replace ISE electrode
	Store electrodes
	Remove reagent pack
	Two-point calibration
	Clean waste tube of ISE module
Maintenance Instructions	Clean electrode tubes
	Maintenance
	A purge
	B purge
	Pump calibration
	Air bubble Detector calibration
	Replace electrode
	Remove reagent pack
	Program check instruction
	Air bubble Detector calibration result
	Pump calibration result
	Reading and writing Dallas

For more information about ISE module maintenance, refer to 16 Maintenance (page 16-1).

12.11 Troubleshooting ISE Module

12.11.1 Troubleshooting ISE Module

The failures occurring on the ISE module may be related to the sample probe unit, sample carousel unit, Hydropneumatic unit, ISE module, reagent inventory, reference electrode and communication. The system provides the following processing methods for the failures:

- Alarm: displaying alarm messages and recording them in error logs without influencing the tests.
- Invalidating current ISE tests: invalidating the tests that are currently being run.
- ISE measurement stop: stopping the measurement after finishing the tests that have been started.
- ISE emergent stop: terminating all ISE tests immediately.

For troubleshooting of the ISE module, refer to 17 Alarms and Troubleshooting (page 17-1).

13

Use of Bar Code

The setup and operation instructions of the sample bar code reader and the reagent bar code reader are depicted in this chapter. The sample bar code reader is used to identify samples and obtain sample information by scanning the bar code label applied on sample tubes. The reagent bar code reader scans the bar code labels automatically when the reagents are loaded.

13.1 Sample Bar Code Reader

13.1.1 Introduction

The sample bar code reader obtains sample information from the bar code label. When bar-coded samples are loaded to the sample carousel, the system will make a full scan and locate samples through the bar code.

Sample bar code specifications

The sample bar code reader is in compliance with the Clinical and Laboratory Standards Institute (CLSI) and compatible with various application environments.

Table 13.1 Sample bar code specifications

Name	Description
Symbology	Codabar, ITF, Code128, Code39, UPC/EAN, and Code93
Minimum bar code density	0.19mm-0.5mm
Length	3-27 digits
Format and content	User-defined
Maximum width	55mm
Minimum height	10mm
Maximum inclination angle	±5°
Print quality	No less than Class C according to the <i>ANSI MH10.8M Print Quality Specification</i> .
Width and narrowness	2.5-3.0:1
Print paper	Coated paper or matte paper. Printing bar code on common paper may result in vague bar code or degraded bar code label. You are not suggested to print bar code on common print paper.
Characters	Meaningful characters, such as numbers (0-9) and upper-case letters (A-Z). You are recommended to print the check digit in order to check that a bar code is read accurately.

Information contained in a sample bar code

The system will obtain the following information from the LIS host based on sample bar code:

- Sample ID

- Sample type
- Requested chemistries

13.1.2 Sample Bar Code Setup

Before performing the setup procedure, check if your system is equipped with a sample bar code reader. If needed, contact our customer service department or your local distributor.

Perform the following steps to set up sample bar code:

- 1 Select **Utility-System Setup**.
- 2 Select **Bar Code F4**.
- 3 Choose **Sample Bar Code**.
- 4 Choose a bar code symbology and set up the check digit status.

The following symbologies are provided:

- Codabar
- Interleaved 2 of 5
- Code128
- Code39
- UPC/EAN
- Code93

Code 128, Code 93 and UPC/EAN requires a check digit by default, and other symbologies are not compulsive. The Code 128 is selected by default and cannot be modified.



CAUTION

You are recommended to enable the check function for all symbologies in order to prevent misreading of bar code.

- 5 Set up sample bar code applications according to actual demands.

The following three options are available:

- Enable/Disable sample carousel bar code
 - Mark the **Enable/Disable Sample Bar Code** checkbox to enable the sample bar code reader. **Bar Code** will appear on the **Current Results** and **History Results** screen in the place of **Sample ID**.
 - To disable the sample bar code reader, deselect the checkbox.

- Enable or disable auto numbering of bar-coded samples
 - When this option is enabled, the system will automatically number the bar-coded samples during bar code scanning. The start number will be the next available one since the last sample is programmed. The default start number for every day is 1. Sample barcode
- Extract sample information: if it is selected, the system will automatically extract the sample information according to the barcode. For example, if the barcode only contains the sample ID and Date, when being analyzed, the sample ID and Date are added to the information of the sample. Only when LIS communication mode is unidirectional, can the option be enabled.
- Define STAT sample positions on sample carousel
 - Samples placed in the specified STAT positions will be taken automatically as emergent samples. Input the start and end positions within the range of 1~115. The set positions will be indicated by E (Emergent) on the sample carousel status screen.

The **Sample Crsl Bar Code** and **Auto Number Scanned Samples** options are selected by default.

6 Select **OK** to save the setup.

7 Select **Format**

8 Define the bar code digits.

The system can scan a sample bar code of fixed length or within 3-27 digits. The Interleaved 2 of 5 only supports bar code of even number length and the digits of the barcode must be defined.

- To use a fixed-length bar code,
 - Mark the **Fixed Digits** checkbox of relevant symbology.
 - Type in the number of digits in the edit box to the right of the **Fixed Digits** field.
- To use a sample bar code within 3-27 digits, you have no need to define the fixed digits.

9 Select **OK** to save the setup.

13.1.3 Programming Bar-Coded Routine Samples

Program bar-coded routine samples by choosing an operating procedure according to the facilities in your laboratory.

**BIOHAZARD**

Inappropriate handling of samples may lead to biohazardous infection. Do not touch the samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the samples, follow standard laboratory safety procedure and consult a doctor.

**CAUTION**

Do not use expired samples; otherwise, unreliable test results may be caused.

**NOTE**

When manually entering sample program information in bar code mode, ensure that the input program information is consistent with the samples loaded to the sample carousel. After the manually programmed samples are analyzed, they must be released manually to leave space for other samples.

When a LIS is provided

- 1 Place the bar-coded samples in idle positions of the sample carousel.
- 2 Select the  icon on the upper-right corner of the main screen.
- 3 Select a sample carousel to which the samples are loaded, and mark the **Sample Crsl Bar Code** checkbox.
- 4 Select a patient sample range: All or Partial. When you select Partial, you should specify a sample position range for analysis.
- 5 Select **OK**.

The system starts scanning the samples on the sample carousel and then analyzes them according to the program information downloaded from the LIS host.

When no LIS is provided

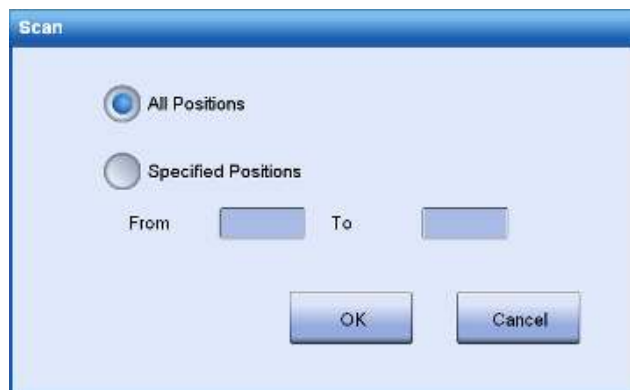
If your system is not equipped with a LIS host, you are allowed to program bar-coded samples with the default panel or to program them manually one by one or by batch. For more information, refer to 10.8 Default Panel (page 10-23).

- 1 Place bar-coded samples sequentially on the sample carousel.

If the auto numbering feature is enabled, the system will automatically number the samples according to the order in which they have been placed. The start number will be the next available one since the last sample is programmed.

- 2 Select **Program-Status**.
- 3 Select **Scan F5**. The **Scan** window is displayed.

Figure 13.1 Scan window



Prior to selecting the **Scan F5** button, ensure the following conditions have been satisfied:

- A sample bar code reader has been configured.
 - Sample bar code scanning is activated on the **Sample Bar Code** window.
 - The system status is *Standby* or *Sample Stop*.
- 4 Choose the scanning range.
 - All positions: to scan all positions on the sample carousel.
 - Specified positions: to scan the specified positions on the sample carousel. Input the start and end scanning positions.
 - 5 Select **OK**.

If a default panel has been defined, the scanned samples will be programmed with the default panel; otherwise, chemistries should be requested for the samples.

- 6 Select **Program-Sample**.
- 7 Enter the sample programming information.

To program a single sample,

- Input the sample ID, press **Enter**, and then choose chemistries for analysis.

- Select **Save F8**.

To batch-program samples,

- Input the start sample ID, press **Enter**, and then choose chemistries for analysis.
- Select **Batch F3**, input the end sample ID, and then select **OK**.

- 8 Select the  icon on the upper-right corner of the main screen.
- 9 Select a sample carousel to which the samples are loaded.
- 10 Select a patient sample range: All or Partial. When you select Partial, you should specify a sample position range for analysis.
- 11 Select **OK**.

13.1.4 Programming Bar-Coded STAT Samples

STAT sample program allows emergent samples to be programmed and analyzed with high priority. The system provides common STAT and quick STAT program. Common STAT program is of higher priority than routine samples. Quick STAT program is mainly used to program emergent samples quickly with higher priority than routine and common STAT samples.

Common STAT samples can be analyzed automatically by means of the sample bar code system and the LIS. Quick STAT samples, however, can only be analyzed by selecting the  icon at the upper-right corner of the main screen. Refer to Quickly programming STAT Samples (Page 2-50) for details.



BIOHAZARD

Inappropriate handling of samples may lead to biohazardous infection. Do not touch the samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the samples, follow standard laboratory safety procedure and consult a doctor.



CAUTION

Do not use expired samples; otherwise, unreliable test results may be caused.



NOTE

When manually entering sample program information in bar code mode, ensure that the input program information is consistent with the samples loaded to the sample carousel. After the manually programmed samples are analyzed, they must be released manually to leave space for other samples.

When a LIS is provided

- 1 Place the bar-coded samples in idle positions of the sample carousel.
- 2 Select the  icon on the upper-right corner of the main screen.
- 3 Select a sample carousel to which the samples are loaded, and mark the **Sample Crsl Bar Code** checkbox.
- 4 Select a patient sample range: All or Partial. When you select Partial, you should specify a sample position range for analysis.
- 5 Select **OK**.

The system starts scanning samples on the sample carousel and analyzes them automatically according to the program information obtained from the LIS host.

When no LIS is provided

If your system is not equipped with a LIS host, you are allowed to program bar-coded samples by using the default panel, or to program them one by one or by batch. For more information about analysis with default panel, refer to 10.8 Default Panel (page 10-23).

After locating a sample by entering the bar code, you are allowed to manually program the sample for analysis.

- 1 Place bar-coded STAT samples sequentially on the sample carousel.

If the auto numbering feature is enabled, the system will automatically number the samples according to the order in which they have been placed. The start number will be the next available one since the last sample is programmed.

- 2 Select **Program-Status**.
- 3 Select **Scan F5**. The **Scan** window is displayed.

Figure 13.2 Scan window



Prior to selecting the **Scan F5** button, ensure the following conditions have been satisfied:

- A sample bar code reader has been configured.
- Sample bar code scanning is activated on the **Sample Bar Code** window.
- The system status is *Standby* or *Sample Stop*.

4 Choose the scanning range.

- All positions: to scan all positions on the sample carousel.
- Specified positions: to scan the specified positions on the sample carousel. Input the start and end scanning positions.

5 Select **OK**.

If a default panel has been defined, the scanned samples will be programmed with the default panel; otherwise, chemistries should be requested for the samples.

6 Select **Program-Sample**.

7 Enter the sample programming information.

To program a single sample,

- Input the sample ID, press **Enter** or **Tab**, mark the **STAT** checkbox, and then choose chemistries for analysis.
- Select **Save F8**.

To batch-program samples,

- Input the start sample ID, press **Enter**, and then choose chemistries for analysis.

- Select **Batch F3**, input the end sample ID, and then select **OK**.
- 8 Select the  icon on the upper-right corner of the main screen.
 - 9 Select a sample carousel to which the samples are loaded.
 - 10 Select a patient sample range: All or Partial. When you select Partial, you should specify a sample position range for analysis.
 - 11 Select **OK**.

13.1.5 Adding New Samples or Chemistries



BIOHAZARD

Inappropriate handling of samples may lead to biohazardous infection. Do not touch the samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the samples, follow standard laboratory safety procedure and consult a doctor.



CAUTION

Do not use expired samples; otherwise, unreliable test results may be caused.

- 1 Select **Program-Sample**.
- 2 Select **List F5**.
- 3 Select **Download F7**.
- 4 Choose one of the following options:
 - All programmed samples: to download all samples programmed on the current day.
 - Latest samples: to download samples which are programmed on the current day but have not been downloaded.
 - Sample with the following IDs: to download samples with the specified program date and ID. Type in the single sample ID or ID range in the edit box.
 - Sample with the following bar code: to download the sample with the specified bar code. Enter the bar code of the desired sample.
- 5 Select **OK**.
- 6 Select **List F5**, and then select **Unpositioned F2** to view samples downloaded from the LIS host.

- 7 Select **Assign** and set positions for the downloaded samples.
 - Select the program date.
 - Input single sample ID or sample ID range.
 - Choose a sample carousel on which the sample(s) will be placed.
 - Input the start and end positions.
- 8 Type in the sample ID or bar code on the **Sample** screen, and then confirm the program information.
- 9 Select the  icon on the upper-right corner of the main screen to request for sample stop.
- 10 Check the sample stop countdown in the system status area and wait until it comes to 0.
- 11 Check the sample carousel indicators, and proceed to the next step when the indicators are extinguished.
 - Flash: indicates that the corresponding carousel is rotating or will start to rotate after 2 periods.
 - ON: indicates that the corresponding carousel is stopped for sample aspirating or that the sample probe is aspirating on the reaction carousel during test for the diluted samples
 - Off: indicates that the corresponding carousel has no sample being aspirated and will not rotate in the next period.
- 12 Place the bar-coded samples on the idle positions of the sample carousel.
 - If the system is running tests, it will analyze the added samples automatically.
 - If the system is not running any tests,
 - Select the  icon on the upper-right corner of the main screen.
 - Select a sample carousel to which the samples are loaded, and mark the **Sample Crsl Bar Code** checkbox.
 - Select a patient sample range: All or Partial. When you select Partial, you should specify a sample position range for analysis.
 - Select **OK**.

13.1.6 Rerunning Bar-Coded Samples



BIOHAZARD

Inappropriate handling of samples may lead to biohazardous infection. Do not touch the samples directly with your hands. Wear gloves and lab coat, if necessary, goggles. In case your skin contacts the samples, follow standard laboratory safety procedure and consult a doctor.



CAUTION

Do not use expired samples; otherwise, unreliable test results may be caused.

Programming rerun samples

- 1 Select **Result-Current Results**.
- 2 Choose results you desire to rerun.
- 3 Select **Rerun F5**.

Figure 13.3 Rerun window

- 4 Enter the sample position.
- 5 Select a sample volume type to rerun the sample.

The sample volume is the same as that defined for the chemistry. If increased and decreased volumes are defined for the chemistry, Increased and Decreased are available here for selection.

- 6 Type in the predilution factor for the sample.

The input range must be within 4-176. The default is blank, which indicates that the sample needs not to be prediluted before being analyzed.

- 7 Select **OK**.

Modifying and adding chemistries

- 1 Select **Program-Sample**.
- 2 Enter the sample ID in the **Sample ID** field.

Sample ID is composed of numbers, or letters and numbers. Up to 10 digits can be entered. The ID range for routine samples and common STAT samples is 1-9000; while quick STAT samples start from 9001 by default.

- 3 Modify the requested chemistries, or add new chemistries.
- 4 To change the sample volume and dilution factor, select **Options F2**, and then do necessary settings.
- 5 Select **Save F8**.

Starting analysis

- 1 Check that the rerun samples have been loaded to the sample carousel.
- 2 If the system is running tests, it will automatically scan all sample positions and then start the analysis.
- 3 If the system status is Standby, select the  icon on the upper-right corner of the main screen.
- 4 Select the sample carousel on which the samples are located.
- 5 Select a patient sample range: All or Partial. When you select Partial, you should specify a sample position range for analysis.
- 6 Select **OK**.

The system starts scanning the samples on the sample carousel and then analyzes them according to the program information downloaded from the LIS host.

Recalling rerun results and determining default result

Recalling rerun results and setting default results are the same with non-bar-coded samples. For more information refer to 8.2.4 Rerunning Samples (page 8-4).

13.1.7 Results Recall

Displaying current results

1 Select **Result-Current Results**.

The screen shows all incomplete samples and controls, as well as those programmed and analyzed on the current day.

Figure 13.4 Current Results screen



2 Choose a sample in the left list. The right list displays all results of the sample.

3 Choose the following buttons as needed:

- **Search F1:** to inquire sample results.
- **Options F2:** to delete, edit, rerun or print samples.
- **Demog F3:** to view patient demographics of the sample.
- **Reac Curve F4:** to view the reaction curve of the sample.

- **Rerun F5:** to rerun a finished sample.
- **Print F7:** to print sample results.
- **Host F8:** to transmit the selected sample results to the LIS host.

13.1.8 Recalling Current Results

Current results can be inquired by sample type, patient name, patient ID, sample ID or sample bar code, etc. along with the program date. Whichever status the system is, only one condition is required for inquiring desired results.

You are allowed to view patient demographics, reaction curve and data, to delete or edit results, to send results to the LIS host, and to print the results. For more information, refer to 8.12 Results Recall (page 8-43).

13.2 Reagent Bar Code Reader

13.2.1 Introduction

The reagent bar code reader obtains reagent information from the bar code label. When bar-coded reagents are loaded to the reagent carousel, the system will make a full scan and obtain reagent information from the bar code labels.

Reagent bar code specifications

The reagent bar code reader is compatible with various application environments. The code128 is selected by default with total bar code length of 13 digits. Users are allowed to set up the symbology and bar code compositions for open reagents. Open reagents are identified based on the symbology and bar code compositions defined by the user; while closed reagents are identified based on those defined by the manufacturer.

Table 13.2 Reagent bar code specifications

Name	Description
Symbology	Codabar, ITF, Code128, Code39, UPC/EAN, and Code93
Minimum bar code density	0.25mm-0.5mm
Length	13-30 digits
Format and content	User-defined
Maximum width	44mm
Minimum height	12mm
Maximum inclination angle	Less than 5°
Print quality	No less than Class C according to the ANSI MH10.8M Print Quality Specification.
Width and narrowness	2.5:1
Print paper	Coated paper or matte paper. Printing bar code on common paper may result in vague bar code or degraded bar code label. You are not suggested to print bar code on common print paper.
Characters	Meaningful characters, such as numbers (0-9) and upper-case letters (A-Z). You are recommended to print the check digit in order to check that a bar code is read accurately.

Information contained in a reagent bar code

The system will obtain the following information from a reagent bar code:

- Chemistry number
- Chemistry name
- Reagent type
- Bottle type
- Lot number
- Serial number
- Expiration date (YYMM)

The reagent information obtained from a bar code label cannot be modified.

13.2.2 Reagent Bar Code Setup

It is necessary to set up the reagent bar code symbologies, check digit and bar code information before using the reagent bar code scanning. Only open-reagent bar code rather than close-reagent bar code needs to be set up.

- 1 Select **Utility-System Setup**.
- 2 Select **Bar Code F4**.
- 3 Choose **Reagent Bar Code**.
- 4 Select or deselect **Analyze barcode of open reagent**.
 - If Reagent barcode system is configured , the option **Analyze barcode of open reagent** is not selected by default
 - When **Analyze barcode of open reagent** is selected, once the barcode of the open reagent is identified, its information is analyzed according to its setup.
 - While loading the reagent manually, you can enter the barcode of the open reagent on the condition that **Analyze barcode of open reagent** is not selected.
- 5 Choose a bar code symbology and set up the check digit status.

The following symbologies are provided:

- Codabar
- Interleaved 2 of 5
- Code128
- Code39

- UPC/EAN
- Code93

Code 128, Code 93 and UPC/EAN requires a check digit by default, and other symbologies are not compulsive. The Code 128 is selected by default and cannot be modified.



CAUTION

You are recommended to enable the check function for all symbologies in order to prevent misreading of bar code.

- 6 Define the total length of reagent bar code.
 - Type in the total length of the reagent bar code in the **T** field. The input range is 13-30 digits. The Interleaved 2 of 5 only supports bar code of even number length.
 - Type in the start digit of the reagent bar code in the **S** field.
 - Type in the end digit of the reagent bar code in the **E** field.
- 7 Determine reagent bar code compositions.
 - Type in the number of digits for reagent information in the **Digits** field.
 - Type in the start digit of the reagent information in the **S** field.
 - Type in the end digit of the reagent information in the **E** field.

Table 13.3 Reagent bar code compositions

Reagent Information	Number of Digits
Chemistry number	0-4 digits
Chemistry name	1-10 digits
Reagent type	1 digit ("1" stands for R1, "2" for R2, "3" for R3, and "4" for R4)
Serial number	0-5 digits
Bottle type	1-3 digits(one digit is recommended; "1" stands for Mindray 20ml outer ring reagent bottle; "2" for 40ml outer ring bottle; "3" for 40ml inner ring bottle; "4" for 62ml inner ring bottle.)
Lot number	0-18 digits
Expiration date	0, 4, 6 or 8 digits (4digits:yymm; 6digits:yyyymm; 8digits:yyyymmdd)

- 8 Select **OK**.

13.2.3 Loading Bar-Coded Reagents

Both open reagents and closed reagents can be loaded through bar code scanning.

If the system is equipped with a reagent bar code reader, you may put the bar-coded reagents on the reagent carousel, and the system will scan all reagent positions automatically and obtain reagent information from the bar code label. The information obtained from a reagent bar code include chemistry name, reagent type, expiration date, lot number, serial number and bottle type, which cannot be modified except for the reagent position.

The reagent bar code is enabled by default. Reagents are identified through bar code scanning with reagent information obtained, all of which but bar code can only be viewed and cannot be edited.

The bar code scanning is only applied to biochemical reagents. The sample probe wash solution, reagent probe wash solution, physiological saline ISE wash solution can only be loaded manually rather than bar code scanning.



Warning

The probe tip is sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the probe.



BIOHAZARD

Wear gloves and lab coat, if necessary, goggles.

Do not touch the reagent directly with your body; otherwise, skin wound or inflammation may be caused.

Perform the following steps to load bar-coded reagents:

- 1 Check the system status and operate accordingly.
 - Standby: proceed to the next step.
 - Running: Select **Reagent-Reagent/Calibration**. Select **Load F1** to stop reagent aspirating and dispensing. Meanwhile **Load F1** becomes **No load F1**. If you want to abort load, select **No load F1**. When the countdown for reagent stop becomes 0 and the system status is Reagent Load, a message box pops up. Select **OK**, and then proceed to the next step.
 - Incubation: proceed to the next step.
 - Sleep: Select **Utility-Commands-Wake Up** to awake the system, and then proceed to step 2.
- 2 Select a reagent carousel from the dropdown box of **reagent carousel**.

- 3 Remove the reagent carousel cover.



CAUTION

If the system is running tests, after requesting reagent stop do not remove the reagent carousel cover until the countdown for reagent stop is 0; otherwise, the tests currently run will be invalidated.

- 4 Place the bar-coded reagents on correct positions and then uncap the reagent bottles.
- 5 Restore the reagent carousel cover.
- 6 Select **End Load F2**.

The system scans all reagent positions automatically and read reagent information from the bar code.

13.3 Bar Code Reader Maintenance

13.3.1 Introduction

The sample and reagent bar code readers are located inside the analyzing unit and need not to be maintained. You are only required to regularly check the bar code scanning window, and clean them if dust or other stains, such as sample and reagent, accumulate.

13.3.2 Cleaning Sample and Reagent Bar Code Scanning Windows

For details about cleaning the sample and reagent bar code scanning windows, refer to 16.10.14 Bar Code Maintenance (Page 16-81).

13.4 Troubleshooting Bar Code Reader

For troubleshooting methods of the bar code reader, refer to 17 Alarms and Troubleshooting (page 17-1).

14 LIS and RMS

This chapter contains communication parameter setup of LIS and RMS, as well as sample analysis and result transmission when an LIS is connected.

14.1 Overview

The chapter provides detailed description of the LIS and RMS.

Laboratory Information System (LIS) is an external host computer connected with the chemistry analyzer through a fixed interface. The LIS is used to download sample program information to the analyzer and receives results sent from the analyzer.

You should set up the communication parameters and results transmission methods prior to using the LIS host.

Check that your analyzer is equipped with a LIS. If needed, contact our customer service department or your local distributor.

Remote Management System (RMS) provides a platform of remote diagnosis and maintenance based on the internet. The RMS allows transfer of data and files with the chemistry analyzers in hospitals, and helps the service engineers to find, collect, analyze, locate and solve the failures happening at the user end.

14.2 Host Communication

14.2.1 Introduction

The host communication parameters, such as transmission mode, IP address and port, should be set up prior to use of the LIS host. To download sample program information from or sent results to the host, you need to set up the chemistry code used for identification of chemistries on both the LIS host and the analyzer, which, otherwise, cannot identify the chemistries simultaneously.

14.2.2 Connection between PC and LIS Host

Follow the procedure below to set up the connection between the operation unit PC with the LIS host.

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **4 Com Setup**. The **System Communication** window is displayed.

Figure 14.1 System communication setup



- 4 Select **PC and LIS** (selected by default).
- 5 Choose a network connection in the **Network Adaptor** area.
- 6 Set up the connection between operation unit and LIS.
 - Auto Obtain IP Address(selected by default)
 - Or set using **Following IP Address**: type in the **IP Address**, **Subnet Mask** and **Default Gateway** for connecting the operation unit PC with the LIS host.

7 Select **Apply**.

A dialog box pops up: Check the network cable connection prior to applying new settings. Please check the connection of the network cable and then click **OK** to save the settings.

8 Select **Exit** to close the window.

14.2.3 Host Communication Parameters

1 Select **Utility-System Setup**.

2 Select **Host F5**. The **Host Communication Parameters** window shows.

Figure 14.2 Host communication Parameters window

3 Set up the following parameters:

Table 14.1 Host communication parameters

Parameter	Description
Transport	Choose a transport mode from the Transport Mode pull-down list. The options include Serial and TCP/IP. The default is Serial.
IP address	Enter the IP address of the LIS host. The connection between the analyzer and the LIS host is based on the network, i.e. TCP/IP protocol.
Port	Enter the interface number of the LIS host.

Parameter	Description
Serial communication parameters	If you choose Serial as the transport mode, set up the following parameters: • Serial port: The default is COM1. • Data bits: 7 or 8. The default is 8. • Stop bits: 1 or 2. The default is 1. • Parity: None, Odd, or Even. The default is None. • Baud rate: 300, 1200, 2400, 4800, 9600, or 19200. The default is 9600.
Protocol	Choose a protocol for connection between the analyzer and the LIS host from the Protocol pull-down list. The options include HL7 and ASTM 1394.
Mode	Choose a data transmission mode for the analyzer and LIS host. The available options are Unidirectional and Bidirectional. • Unidirectional: You are only allowed to send results and patient demographics to the host rather than downloading sample programs from it. • Bidirectional: You are allowed to send results and patient demographics to the host and downloading sample programs from it.
Timeout	Enter the time out limit for querying the LIS host. The input range is 30s-60s, and the default is 30s. If the time out limit is exceeded when you attempt to download sample programs from, or send results to, or connect the analyzer with the LIS host, the system will give an alarm indicating communication timed out.
Auto Connect to LIS	When the checkbox is selected, the system will connect to the LIS host automatically when started up.
Retry after Disconnection	When the checkbox is selected, the system will try to reconnect the LIS host for every set interval once the connection is interrupted.
Interval	Input the time interval for which the system will try to reconnect the LIS host for every set interval once the connection is interrupted. The default is 30 seconds.
Send Complete Samples	When the checkbox is selected, the system will automatically send results to the LIS host after a sample changes from <i>In Progress</i> to <i>Complete</i> . This function is only applicable to samples analyzed on the current day rather than those analyzed before.

Parameter	Description
Send Incomplete Samples	When the checkbox is selected, the system will automatically send results to the LIS host after a sample changes from <i>In Progress</i> to <i>Incomplete</i> . This function is only applicable to samples analyzed on the current day rather than those analyzed before.
Advanced options	Select Advanced. The Advanced window appears, providing the following options: • Send Programmed Samples: When the checkbox is selected, the system will automatically send the program information to the LIS host once a single or batch routine and STAT samples are programmed. • Rerun Finished Chemistries When Downloaded: When the checkbox is selected, chemistries that have been finished will be rerun if downloaded again. If this option is not selected, they will be neglected. • Send Actual Results and Rerun Results: When the checkbox is selected, all actual results and rerun results of each chemistry will be sent to the LIS. If this option is not selected, only the default result will be sent. • Bypass Results Beyond Linearity Range: When the checkbox is selected, those results that are beyond the linearity range will not be sent to the LIS. If this option is not selected, they will be sent. • Ignore Alarms for Unknown Chemistries: When the checkbox is selected, the system will not give an alarm if the samples downloaded from the LIS host contain unknown chemistries without identification code. If this option is not selected, an alarm will be given indicating sample programming failure.

- 4 Select **Save** to save your input information.
- 5 Select **Connect** to connect the analyzer with the LIS host.

14.2.4 Defining Chemistry Code

- 1 Select **Utility-System Setup**.
- 2 Select **Host F5**. The **Host Communication Parameters** window shows.
- 3 View the chemistry channel number list on the right of the window.

The screen shows the chemistries and code in two columns. The left column provides all chemistries that have been defined and set up correctly; the right column shows the code for identifying a chemistry on the LIS host.

- 4 Click on the **Channel No.** column of a chemistry, and then type in a code for it.
- 5 Repeat step 4 to define a code for other chemistries.
- 6 Select **Save**.

14.3 Programming Samples with LIS Host

14.3.1 Introduction

Sample programming information can be sent by or downloaded from the LIS host, and then the measured results are sent to it manually or in real-time mode.

Both bar-coded and non-bar-coded samples can be programmed with the LIS host. When a sample bar code module is configured, the system will automatically identify samples and match them with the programming information downloaded from the host. If there is no sample bar code module, you should manually assign positions for the downloaded samples.

14.3.2 Programming Functions

Samples can be downloaded manually or automatically from the LIS host. If the system status is Standby, you are allowed to download samples manually from LIS.

Sample programs downloaded from the LIS host can be edited. When programs are downloaded for samples that are in Programmed status, the requested chemistries in the programs will be used to overwrite the original chemistries; if the samples are in a status other than Programmed, the requested chemistries will be added to the original ones.

Sending sample programs from LIS

Sending bar-coded samples:

- 1 When samples are sent from the LIS host to the analyzer, select **Program-Sample**.
- 2 Select **List F5** to view downloaded samples.
- 3 On the **Sample** screen, type in the sample bar code, and then confirm the program information.
- 4 Select **Save F8**.
- 5 Load the samples to idle positions of the sample carousel.
- 6 Select the  icon on the upper-right corner of the main screen.

Sending non-bar-coded samples:

- 1 After program samples on the LIS host, send them to the analyzer, and then select **Program-Sample** on the analyzer.
- 2 Select **List F5** to view downloaded samples.
- 3 Select **Unpositioned F2**.
- 4 Select **Assign**.
- 5 Select the date the desired samples are programmed.
- 6 Type in the single sample ID or ID range in the **ID** field.
- 7 Choose a sample carousel on which you will place the sample.
- 8 Enter the sample position.

The options include all available positions of the selected sample carousel.

- To assign position for single sample, input the position number in the first edit box.
 - To assign positions for multiple samples, enter the start position number in the first edit box, and then the end position number in the second edit box. The system will assign positions for the samples ascending according to the sample ID.
 - If the available positions among the specified range are more than or equal to the number of samples, the extra positions will be neglected.
 - If the available positions among the specified range are less than the number of samples, the system will display a message indicating insufficient positions. Assign the positions again.
- 9 Select **OK**.
 - 10 Enter the sample ID on the **Sample** screen and edit the following information:
 - Position
 - STAT status
 - Sample type
 - Comment
 - Chemistries and panels
 - Sample options and chemistry options
 - 11 Select **Save F8**.

12 Load the samples to the assigned positions on the sample carousel.

13 Select the  icon on the upper-right corner of the main screen.

Obtaining samples automatically

When the system status is Standby or Sample Stop, load the samples to the sample carousel, and then select . The system will automatically scan the samples and then query the LIS host to download relevant program information. After matching the downloaded program information with the samples, the system will start the analysis.

The obtained sample program information includes:

- Patient demographics: patient name, gender, date of birth, etc.
- Requested chemistries: sample bar code, sample ID, sample type and chemistry code.

Downloading samples manually

Downloading bar-coded samples:

1 Select **Program-Sample**.

2 Select **Download F7**.

3 Choose one of the following options:

- All programmed samples: to download all samples programmed on the current day.
- Latest samples: to download samples that are programmed on the current day but have not been downloaded.
- Samples with the following IDs: to download samples with the specified program date and ID. Enter the sample IDs or ID range to download.
- Sample with the following bar code: to download the sample with the specified bar code. Enter the bar code of the desired sample.

4 Select **OK**.

5 Confirm the sample information and selected chemistries/panels.

6 Load the samples to idle positions of the sample carousel.

7 Select the  icon on the upper-right corner of the main screen.

8 Select **OK** to start analysis.

Downloading non-bar-coded samples:

- 1** Select **Program-Sample**.
- 2** Select **Download F7**.
- 3** Choose one of the following options:
 - All programmed samples: to download all samples programmed on the current day.
 - Latest samples: to download samples that are programmed on the current day but have not been downloaded.
 - Sample with the following IDs: to download samples with the specified program date and ID. Enter the sample IDs or ID range to download.
- 4** Select **OK**.
- 5** Select **List F5** to view downloaded samples.
- 6** Select **Unpositioned F2**.
- 7** Select **Assign**.
- 8** Select the date the desired samples are programmed.
- 9** Type in the single sample ID or ID range in the **ID** field.
- 10** Choose a sample carousel on which you will place the sample.
- 11** Enter the sample position.

The options include all available positions of the selected sample carousel.

- To assign position for single sample, input the position number in the first edit box.
 - To assign positions for multiple samples, enter the start position number in the first edit box, and then the end position number in the second edit box. The system will assign positions for the samples ascending according to the sample ID.
 - If the available positions among the specified range are more than or equal to the number of samples, the extra positions will be neglected.
 - If the available positions among the specified range are less than the number of samples, the system will display a message indicating insufficient positions. Assign the positions again.
- 12** Select **OK**.

14 LIS and RMS

- 13 Load the samples to the assigned positions on the sample carousel.
- 14 Select the  icon on the upper-right corner of the main screen.
- 15 Select **OK** to start analysis.

14.4 Result Transmission

14.4.1 Introduction

Sample results and QC results can be sent manually or in real-time mode to the LIS host for reviewing and storage. When a sample is analyzed with its all tests finished, the system can automatically send the test results to the LIS host; also you are allowed to search for desired results and then manually send them to LIS.

Patient demographics, sample results and QC results can be sent to the LIS host.

Patient demographics:

- Patient name
- Gender
- Age
- Patient ID
- Clinical diagnosis
- Ordering physician
- Ordering date
- Test date
- Tester
- Ordering department

Sample results:

- Sample type
- Sample bar code
- Sample ID
- Program date
- Completion date
- Sample status
- Chemistry name
- Result
- Unit
- Original result

Control results:

- Chemistry name
- Run date

- Control name
- Lot number
- Expiration date
- Concentration level
- Standard deviation
- Result

14.4.2 Result Transmission Setup

When all tests of a sample are finished and at least one of them has calculated a result, the result can be sent to the LIS host automatically. Duplicate results on the LIS host can be handled by means of adding, overwriting or ignoring. For rerun results, only the default result (that is currently displayed on the **Current Results** screen) is sent to the LIS host. The results of all replicates of a sample or chemistry will be sent to the LIS host.

- 1 Select **Utility-System Setup**.
- 2 Select **Host F5**. The **Host Communication Parameters** window shows.
- 3 Mark the **Send Complete Samples** or **Send Incomplete Samples** checkbox with a tick.

A sample will be sent to the LIS host automatically when it changes from *In Progress* to *Complete* or *Incomplete*.

If you won't send results, deselect the checkbox.

- 4 Select **Save**.

14.4.3 Manually Sending Results to LIS Host

- 1 Select **Result-Current Results** or **History Results**.
- 2 Search for control results or sample results to transmit.
- 3 Select desired samples in the sample list.
- 4 Select **Host F8**.
- 5 Select the sample range you want to transmit:
 - Selected sample(s)
 - All samples

- 6 If you transmit all results, you are allowed to skip those that are already transmitted to the LIS host. Mark the **Bypass Transmitted Results** checkbox.
- 7 Select **OK**.

14.5 Troubleshooting LIS

For troubleshooting methods of the LIS host, refer to 17 Alarms and Troubleshooting (page 17-1).

14.6 Use of RMS

14.6.1 Introduction

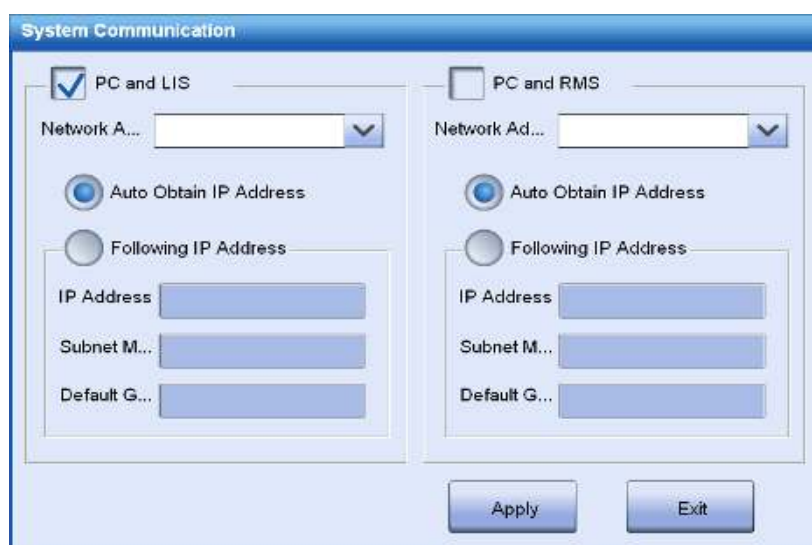
The RMS provides a platform of remote diagnosis and maintenance based on the internet. The RMS allows transfer of data and files with the chemistry analyzers in hospitals, and helps the service engineers to find, collect, analyze, locate and solve the failures happening at the user end. Before connecting the analyzer with the RMS, you should set up the IP address of the operation unit PC.

14.6.2 Connection between PC and RMS

Follow the procedure below to set up the IP address for connecting the operation unit PC with the RMS.

- 1 Select **Utility-System Setup**.
- 2 Select **Instrument F1**.
- 3 Select **4 Com Setup**. The **System Communication** window is displayed.

Figure 14.3 System communication setup



- 4 Select **PC and RMS**.
- 5 Choose a network connection in the **Network Adaptor** area.
- 6 Set up the connection between operation unit and RMS.
 - Auto Obtain IP Address(selected by default)

- Or set using **Following IP Address**: type in the **IP Address**, **Subnet Mask** and **Default Gateway** for connecting the operation unit PC with the RMS.

7 Select **Apply**.

A dialog box pops up: Check the network cable connection prior to applying new settings. Please check the connection of the network cable and then click **OK** to save the settings.

8 Select **Exit** to close the window.

14.6.3 Troubleshooting RMS

For troubleshooting methods of the RMS, refer to 17 Alarms and Troubleshooting (page 17-1).

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Operator's Manual

Maintenance Volume

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15

Diagnostics

This chapter provides test descriptions, test procedures, test results and corrective actions for diagnosis in Sample and Reagent systems.

15.1 Overview

Diagnostics consist of a series of tests and actions, which are used for troubleshooting errors. These tests and actions are made to detect failures, but cannot be used to confirm one specific failure. Users should make a judgment by integrating the information of diagnosis and warnings with the failure characteristics. Diagnostic tests available in two function modules are described in the table below.

Table 15.1 Categories of diagnostics

Function Module	Description
Sample System	Diagnostic tests here are used to detect failures of components in Sample system.
Reagent System	Diagnostic tests here are used to detect failures of components in Reagent system.
Sensor Diagnosis	Diagnostic tests here are used to detect failures of the sensors.

15.2 Diagnosis of Sample System

15.2.1 Introduction

The Sample System is responsible for delivering samples to the system for analysis. Tests include:

- Sample Probe Clog Detection
- Sample Probe Level Sense Test

15.2.2 Sample Probe Clog Detection

Test description

This test can help you find if the Sample Probe Clog Detection function works normally. Related data or text will be displayed after testing, which can be used to confirm the results.

Use this test when one of the following alarms is given:

- Clog detection board communication error.
- The sample probe is clogged while the sample is deemed OK.
- The sample probe is clogged during cleaning.
- Clog detection board working mode setting error.

Test procedure

- 1 Select **Utility -> Maintenance -> Diagnostics**.
- 2 Select **Sample System** tab.
- 3 Select **Sample Probe Clog Detection**.
- 4 Load one cuvette of water onto Position 1 on sample carousel, and click **Next** to open the **Sample Probe Clog Detection Diagnosis** window.

Figure 15.1 Sample Probe Clog Detection Diagnosis window

Sample Probe Clog Detection Diagnosis					
Test Item	SubItem	Reference	Final Result	Pass/Fail	Suggestions
Basic Check	PCBA Version	/			
	12V	10.8V-13.2V			
	5V	4.5V-5.5V			
	PCBA Pressure	9-16psia			
	Clog Signal	/			
Wash Check	Wash Pressure	30-60psia			
Clog Check	Final Result	/			
Sample Aspiration Check (1.5μl)	P0p	<7psi			
	Final Result	/			
Sample Aspiration Check (45μl)	P0p	<7psi			
	Final Result	/			
Start Exit					

5 Click **Start**.

The system starts to run each test for sample probe clog detection. Tests include:

- Basic Check
- Wash Check
- Clog Check
- Sample Aspiration Check (1.5μL)
- Sample Aspiration Check (45μL)

6 When tests are complete, the tested voltage and the level sense test data are displayed on the screen.**7** Click **Exit** to close the window.**Test results**

The testing result of each subitem is displayed on the screen. Judge if the result meets the requirements by comparing with the corresponding reference value. “PASS” in the PASS/FAIL column indicates the test is normal, while “FAIL” indicates the test is failed and it should be corrected based on the suggestions provided.

Corrective action

Table 15.2 Sample system obstruction detection reference range and corrective action

Test Type	Test Item	Reference Range	Corrective Action
Basic Check	Version of Clog Detection Board		Contact our customer service department or your local distributor.
	12V	10.8V-13.2V	
	5V	4.5V-5.5.V	
	Pressure of Clog Detection Board	9.0psia-16.0psia	
	Clog Signal	OK/Error/" /"	
Wash Check	Wash Pressure	30.0psia-60.0psia	
Clog Check	Final Result	OK/Error/" /"	
Sample Aspiration Check (1.5µL)	P0p	<7psi	
	Final Result	OK/Error/" /"	
Sample Aspiration Check (45µL)	P0p	<7psi	
	Final Result	OK/Error/" /"	

15.2.3 Sample Probe Level Sense Test

Test description

The Level Sense Test is used to diagnose the level detection performance of the sample system and gives related data that helps you locate the causes of an error.

Use this test when one of the following conditions happens:

- An alarm message appears indicating that the sample probe contacts no liquid in the aspiration positions (include sample carousel, reaction carousel and concentrated wash position) and the analysis is stopped.
- An alarm message appears indicating that the sample probe aspirates nothing in the aspiration positions and the analysis is stopped, and has confirmed that the failure is not caused by probe clog.
- An alarm message appears indicating that the sample probe contacts no liquid during dispensing samples into reaction carousel and the analysis is stopped, and has confirmed that the failure is not caused by neither reagent bubbles nor reagent probe level sensing.
- An alarm message appears indicating that problems related with level sensing occur during dispensing samples in ISE module, and has confirmed it is not the problems of ISE module itself.

- An alarm message appears indicating that the sample probe contacts no liquid during liquid dispensing (also called water testing), and the analysis is stopped.
- An alarm message appears indicating that the sample probe contacts no liquid in the wash well and the analysis is stopped, and has confirmed that it is not a hydropneumatic failure.

Test procedure

- 1 Select **Utility -> Maintenance -> Diagnostics**.
- 2 Select **Sample System** tab.
- 3 Select **Sample Probe Level Sense Test**.
- 4 Place a tube with its 2/3 full of water in test position, and click **Next** to open the **Sample Probe Level Sense Test Results** window.

Figure 15.2 Sample Probe Level Sense Test Results window

Sample Probe Level Sense Test Results

Test Sample Position

1#

Level Sense Board Voltage Check Results

Actual	Reference	Pass/Fail	Suggestions
2.8-4.8V			

Level Sense Test Data

Test Cycle	Status	Lowering Height
1	Error	0.00 mm

Test Result	Suggestions	Height Diff
Error	Contact manufacturer	0.000 mm

Start Exit

The default test position is position 1 on the sample carousel. To change the test position, click Change Pos and enter a new number within the range from 1 to 90, and then click **OK**.

- 5 Click **Start**.

The system will start to check the level sense board voltage of the sample probe, and continuously detect level in the test position for 20 times.

- 6 When tests complete, the tested voltage and the level sense test data will be

displayed on the screen.

- 7 Click **Exit** to close the window.

Test results

The results shown are described below:

Level Sense Board Voltage Check Results

If the Actual value falls in the Reference range, the result is PASS, indicating the voltage of the level detection board is normal; otherwise, the result is FAIL, indicating the voltage is abnormal. You should correct it based on the suggestions provided.

Level Sense Test Data

The system will continuously check the lowering height of the sample probe for 20 times, to judge if the lowering position is the vertical extreme position. If it is, abnormality exists. If the extreme difference of 20 lowering heights is greater than 1mm, then the result should be considered as abnormal, indicating that there are problems with connections of sample probe and Printed Circuit Board Assembly (PCBA), PCBA power, output voltage for level sense detection, or connections of level sense board and probe/mixer conversion board, and vice versa. You should correct it based on the suggestions provided.

Corrective action

If the operating voltage of the level detection board is beyond the reference range, contact our customer service department or your local distributor.

If the result of the level detection performance is abnormal, contact our customer service department or your local distributor.

15.3 Diagnosis of Reagent System

15.3.1 Introduction

The Reagent System is responsible for delivering reagents to the system for analysis. Test includes:

- Reagent Probe Level Sense Test

15.3.2 Reagent Probe Level Sense Test

Test description

The Level Sense Test is used to diagnose the level detection performance of the reagent probe and gives related data that helps you locate the causes of an error.

Use this test when one of the following conditions happens:

- An alarm message appears indicating that the reagent probe contacts no liquid on the reagent carousel, and the analysis is stopped. An alarm message appears indicating that probe aspirates nothing in the aspiration position and the analysis is stopped.
- An alarm message appears indicating that the reagent probe contacts no liquid during dispensing reagents and the analysis is stopped, and has confirmed that it is not caused by reagent bubbles.
- An alarm message appears indicating that the reagent probe contacts no liquid in the wash well and the analysis is stopped, and has confirmed that it is not a hydropneumatic failure.

Test procedure

- 1 Select **Utility -> Maintenance -> Diagnostics**.
- 2 Select **Reagent System** tab.
- 3 Select **Reagent Probe Level Sense Test**.
- 4 Place a tube with its 2/3 full of water in position 1 of reagent carousel, and click **Next** to open **Reagent Probe Level Sense Test Results** window.

Figure 15.3 Reagent Probe Level Sense Test Results window

Reagent Probe Level Sense Test Results

Test Position

1

Level Sense Board Voltage Check Results

Actual	Reference	Pass/Fail	Suggestions

Level Sense Test Data

Test Cycle	Status	Lowering Height
Test Result	Suggestions	Height Diff

Start

Exit

The default test position is position 1 on reagent carousel. To change the test position, click Change Pos and enter a new number within the range from 1 to 80, and then click **OK**.

- 5 Click Start.**

The system starts to check the level sense board voltage for reagent probe, and continuously detects level in the test position for 20 times.

- 6 When tests complete, the tested voltage and the level sense test data are displayed on the screen.
- 7 Click **Exit** to close the window.

Test results

The results shown are described below:

Level Sense Board Voltage Check Results

If the Actual value falls in the Reference range, the result is PASS, indicating the voltage of the level detection board is normal; otherwise, the result is FAIL, indicating the voltage is abnormal. You should correct it based on the suggestions provided.

Level Sense Test Data

The system will continuously check the lowering height of reagent probe for 20 times, to judge if the lowering position is the vertical extreme position. If it is, abnormality exists. If the difference of 20 lowering heights is greater than 1mm, then the result should be considered as abnormal, indicating that there are problems with connections of reagent probe and PCBA, PCBA power, output voltage for level sense detection, or connections of level sense board and probe/mixer conversion board, and vice versa. You should correct it based on the suggestions provided.

Corrective action

If the operating voltage of the level detection board is beyond the reference range of 2.8V-4.8V, contact our customer service department or your local distributor.

If the result of the level detection performance is abnormal, contact our customer service department or your local distributor.

15.4 Sensor Diagnosis

15.4.1 Introduction

Sensor diagnosis provides diagnosis results of the sensors of the analyzer to help locate the failure cause of the related parts.

15.4.2 Sensor Diagnosis

Use this test when one of the following conditions happens:

- Reaction carousel loses steps or positioning failed.
- Wash station loses steps or fails to find the mechanical zero position.
- Sample carousel loses steps or positioning failed.
- Sample probe loses steps in horizontal or vertical movement or fails to find the mechanical zero position.
- Sample syringe loses steps or fails to find the mechanical zero position.
- Reagent carousel loses steps or positioning failed.
- Reagent probe loses steps in horizontal or vertical movement or fails to find the mechanical zero position.
- Reagent syringe loses steps or fails to find the mechanical zero position.
- Sample mixer, reagent mixer fails to find the mechanical zero position.
- Interior wash syringe fails to find the mechanical zero position.
- Phase 1-2 wash syringe fails to find the mechanical zero position.

Test procedure

- 1 Select **Utility** -> **Maintenance** -> **Diagnostics**.
- 2 Select **Sensor Diagnosis** tab.
- 3 Select the optical coupler to be diagnosed.

Click **Select All** or **Cancel All** to select the sensors.

- 4 Select **Start** to inquire the status of the sensors.

To stop the test, select **Stop**.

Figure 15.4 Sensor diagnosis window



5 Click **Exit** to close the window.

Test result

Sensor high level signal is indicated by “unblocked” while low level signal is indicated by “blocked”.

Corrective action

Perform the operations: block or unblock the sensors and perform the diagnosis test. observe if the sensor signal is changed accordingly with your operations. If the displayed status is changed accordingly, it means the sensor works normally; If the displayed status is not changed with your operations, it means the sensor is abnormal and please contact our customer service department.

16 Maintenance

This chapter provides you with maintenance of the instrument, including frequently-used maintenance commands and scheduled maintenance procedures. The purpose, time, system status, precautions and steps of each maintenance procedure are described here.

16.1 Overview

16.1.1 Introduction

16.1.2 Introduction

Maintenance of the system should be performed regularly by trained personnel to ensure reliable performance and reduce unnecessary service calls. Even you are only an operator, it is important for you to read this chapter. Your thorough understanding will help you obtain the best performance of the system.

The Biochemistry Maintenance, ISE Maintenance and Scheduled Maintenance Log are provided. The Biochemistry Maintenance and ISE Maintenance features provide a list of the maintenance procedures that can be performed to optimize the system performance. The Scheduled Maintenance Log feature allows you to understand what maintenance is needed, when it is performed and who performed the procedure. It is capable of reminding you of the maintenance that is due and keeping track of what is happened during a maintenance procedure.

In the case of maintenance that is beyond your capability or not covered in this chapter, contact our customer service department or your local distributor.

The maintenance frequencies stated in this manual are based on working for 5 hours a day, that is $5 \times 400 = 2,000$ tests/day, and $5 \times 400 \times 25 = 50,000$ tests/month.



Warning

Do not perform any maintenance procedures that are not described in this chapter; otherwise, equipment damage or personal injury may be caused.

Do not touch the components other than those specified in this chapter.

Performing unauthorized maintenance procedures can damage the instrument and cause personal injury, or invalidate the applicable warranty provisions in the service contract.

After performing maintenance, make a verification to ensure that the system runs normally.

Do not spill water or reagent on mechanical or electrical components of the system.

If the system is to be stored for a long time (over 1 week) or transported, contact our customer service department or your local distributor to perform necessary maintenance in order to ensure the system's optimal performance in following use.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

16.1.3 Consumables

Please use the consumables manufactured or recommended by our company in order to achieve the promised system performance. If needed, contact our customer service department or your local distributor.

Table 16.1 Consumables

Item	Location	Description
20W tungsten-halogen lamp	Lamp housing	Regularly-replaced part. Replace it when it serves for over 2000 hours or the system shows a prompt message indicating light intensity too weak. You are recommended to replace the lamp when is it is used for no longer than 6 months.
Kloehn 100µl syringe plunger assembly	Sample syringe	Regularly-replaced part. Replace it when: it serves for 3 months; or it works for 100,000 tests; or it is obviously damaged.
Kloehn 1ml syringe plunger assembly	Reagent syringe	Regularly-replaced part. Replace it when: it serves for 3 months; or it works for 300,000 tests; or it is obviously damaged.
Sample syringe washer	Connection joint between the sample syringe and the T piece	Regularly-replaced part. Replace it when the sample syringe is reinstalled for 2 to 3 times.
Reagent syringe washer	Connection joint between the reagent syringe and the T piece	Regularly-replaced part. Replace it when the reagent syringe is reinstalled for 2 to 3 times.
Reagent probe assembly	Reagent probe arm	Irregularly-replaced part. Replace it when it is damaged or bent.
Sample probe assembly	Sample probe arm	Irregularly-replaced part. Replace it when it is damaged or bent.
Sample probe washer	Nut on the sample probe	Regularly-replaced part. Replace it when, the sample probe is reinstalled for 2 to 3 times; or the sample probe is replaced with a new one.

16 Maintenance

Item	Location	Description
Reagent probe washer	Nut on the reagent probe	Regularly-replaced part. Replace it when, the reagent probe is reinstalled for 2 to 3 times; or the reagent probe is replaced with a new one.
Mixer	Mixer arm	Irregularly-replaced part. Replace it when it is damaged.
A4 paper	Printer	Consumable
Reaction cuvette	Reaction carousel	Replace it as needed.
Deionized water filter assembly	Water supply	Replace it in every 6 months.
Filter core	Water inlet filter	Replace it in every 3 months.
Water inlet filter	Water supply module or water unit	Replace it in every 6 months.
Wash solution	Water tank bracket	Consumable
Reagent bottle for reagent carousel inner ring (40ml)	Reagent carousel	Consumable
Reagent bottle for reagent carousel inner ring (62ml)	Reagent carousel	Consumable
Reagent bottle for reagent carousel outer ring (20ml)	Reagent carousel	Consumable
Reagent bottle for reagent carousel outer ring (40ml)	Reagent carousel	Consumable
Na electrode	ISE module (optional)	Consumable
K electrode	ISE module (optional)	Consumable
Cl electrode	ISE module (optional)	Consumable
Reference electrode	ISE module (optional)	Consumable

Item	Location	Description
Spacer	ISE module (optional)	Consumable
Wash solution package	ISE module (optional)	Consumable
Reagent module	ISE module (optional)	Consumable
Urine diluent	ISE module (optional)	Consumable
ISE control	ISE module (optional)	Consumable

16.1.4 Tools Required for Maintenance

The following tools will be used for maintenance of the system.

Accompanying Tools

Table 16.2 Accompanying Tools

Item	Applicable Maintenance
Philips-head screwdriver $\phi 3.3 \times 100$	Removing the system enclosure and the cooling fans
Philips-head screwdriver $\phi 4.7 \times 100$	Installing the probes and lamp
Slot-head screwdriver $\phi 4.7 \times 100$	Installing/removing the probes, and installing the tube hoop.
Wash solution	Enhanced cleaning
Round-head needle, $0.25 \pm 0.01 \text{ mm} \times 125 \text{ mm}$	Unclogging the probes
Unclogging device for reagent probe	Unclogging the reagent probe
Unclogging device for sample probe	Unclogging the sample probe

Tools to be Prepared by User

Table 16.3 Tools to be Prepared by User

Item	Applicable Maintenance
Tube brush, ultrasound cleaner	Cleaning the filter core
Clean gauze	Cleaning the syringes, rotors, probes/mixers
Cotton swabs	Cleaning the wash well, sample compartment, etc.
Suction cleaner	Cleaning the fans and dust screens
Hair brush	Cleaning the dust screen
Tweezers	Removing/Installing probes and syringe washers

16 Maintenance

Item	Applicable Maintenance
Thread syringe	Unclogging the sample probe and reagent probe
Tube brush or ultrasound cleaner	Cleaning the filter core
Beaker	Cleaning the needle and unclogging device
Ethanol	Cleaning the photometer lens, probes, mixers and wash station
NaClO (0.5% sodium hypochlorite solution)	Cleaning the wash wells
Fiber-free gloves	Cleaning and replacing reaction cuvettes etc.
Large water container	Cleaning the deionized water tank
Screen and keyboard wash solution	Cleaning the screen and keyboard

16.2 Biochemistry Maintenance

16.2.1 Introduction

The Biochemistry Maintenance feature provides maintenance instructions for the biochemistry system. The following three types of maintenance are available.

Photometric system:

- Cuvette check
- Photometer check
- Replace lamp
- Replace cuvette

Hydropneumatics:

- Clean probes/mixers exterior
- Clean probes interior
- Prime wash station
- Clean filter and DI water tank
- Circulate wash cuvettes

Sample/Reagent handling system and mixer assembly:

- Home
- Clean probes/mixers/wash wells
- Special wash
- Special wash probes

The biochemistry maintenance is described in detail in the following pages.

16.2.2 Biochemistry Maintenance Screen Overview

Select **Utility-Maintenance-Maintenance-Biochemistry Maintenance**. The screen shows the biochemistry maintenance commands that are frequently used.

Figure 16.1 Biochemistry Maintenance screen**Maintenance procedures**

Provides frequently-used maintenance commands of the biochemistry system. Select a maintenance command button to start the maintenance procedure.

Online help

Online help information is provided for each biochemistry maintenance command. Select the  icon to the left of a maintenance command to show relevant instructions.

Exit

Select this button to close the **Maintenance** window.

16.3 ISE Maintenance

16.3.1 Introduction

The ISE Maintenance feature provides maintenance commands for the ISE module. The following maintenance procedures are included:

- Two-point calibration
- Clean electrode tubes
- Pump calibration
- Maintenance
- Air bubble detector calibration
- Purge A
- Purge B
- Replace electrode
- Remove reagent pack
- Program check instruction
- Air bubble detector calibration result
- Pump calibration result
- Read Dallas chip
- Write Dallas chip
- Store electrodes
- Clean sample injection port

The ISE maintenance is described in detail in the following pages.

16.3.2 ISE Maintenance Screen Overview

Select **Utility-Maintenance-Maintenance-ISE Maintenance**. The screen shows the ISE maintenance commands that are frequently used. Operate according to the screen prompts.

Figure 16.2 ISE Maintenance screen



Maintenance procedures

Provides frequently-used maintenance commands of the ISE module. Select a maintenance command button to start the maintenance procedure.

Online help

Online help information is provided for each ISE maintenance command. Select the  icon to the left of a maintenance command to show relevant instructions.

Exit

Select this button to close the **Maintenance** window.

16.4 Scheduled Maintenance Log

16.4.1 Introduction

Scheduled maintenance procedures are determined by use of the components and frequency of performance, and should be performed regularly by trained personnel to ensure reliable performance and reduce unnecessary service calls. Read this section carefully prior to doing the maintenance.

The Customize feature allows definition of maintenance procedures and configuration of manufacturer-/user-defined maintenance procedures for each maintenance frequency. The Electronic Maintenance Log is provided enabling you to record comments and other important information of maintenance.

Most of the scheduled maintenance procedures are performed by executing maintenance instructions, while the remaining part by manual operations. Perform the maintenance strictly as instructed in this manual.

16.4.2 Maintenance Schedule

The scheduled maintenance procedures are divided into the following periods:

- Daily: 1 day
- Weekly: 8 days
- Two-week: 15 days (No maintenance item for this model)
- Monthly: 31 days
- Three-month: 91 days
- Six-month: 181 days
- Other (As-needed/As-required)

The maintenance frequency is counted down from the date of performing. When the countdown becomes 0, the corresponding maintenance procedure is highlighted in yellow. To determine that a due maintenance procedure is due, check if the following items are displayed in yellow background:

- **Utility** button on the main screen
- **Maintenance** tab
- **Maintenance** button
- **Scheduled Maintenance** tab
- Maintenance frequency tab
- Maintenance procedure

The maintenance information will not be lost when the operating software version is upgraded. When new version software is installed to remove the system failure or fix the system, the maintenance counter returns to 0 and restarts a countdown.

16.4.3 Scheduled Maintenance Procedures

Maintenance procedures vary from different maintenance frequencies. The maintenance procedures described in this chapter are based on a complete configuration of the system. If some modules are not equipped on your system, you have no need to perform relevant maintenance.

Daily maintenance:

- Check probes/mixers/wash wells
- Check sample/reagent syringes
- Check deionized water connection
- Check waste tube connection
- Check concentrated wash solution
- Check sample probe wash solution
- Clean electrode tubes

Weekly maintenance:

- Clean sample/reagent probe exterior/ mixers
- Special wash
- Clean mixers
- Cuvette check
- Photometer check

Monthly maintenance:

- Clean wash wells
- Clean cuvette wash station
- Clean filter core
- Clean dust screens
- Clean sample injection port
- Pump calibration
- Air bubble detector calibration

Three-month maintenance:

- Clean DI water tank
- Replace filter core

Six-month maintenance:

- Replace lamp
- Replace water inlet filter
- Replace reference electrode

Other (As-needed/As-required):

- Clean analyzer panels
- Bar code maintenance
- Clean sample compartment
- Clean reagent compartment
- Clean sample probe interior
- Clean reagent probe interior
- Replace sample probe
- Replace reagent probe
- Replace sample mixer
- Replace reagent mixer
- Remove air bubbles in sample syringe
- Remove air bubbles in reagent syringe
- Special wash probes
- Replace cuvettes
- Replace ISE electrode
- Store electrodes
- Remove reagent pack
- Replace sample syringe
- Replace reagent syringe
- Clean rotors

Perform the scheduled maintenance according to the instruction in this chapter. Run calibration or quality control after performing the maintenance.

16.4.4 Maintenance Log Sheet

Refer to the following table for scheduled maintenance procedures you are supposed to perform. Please copy it every month and place a check mark in relevant day column every time after you performing maintenance.

16 Maintenance

Table 16.4 Maintenance Log Sheet

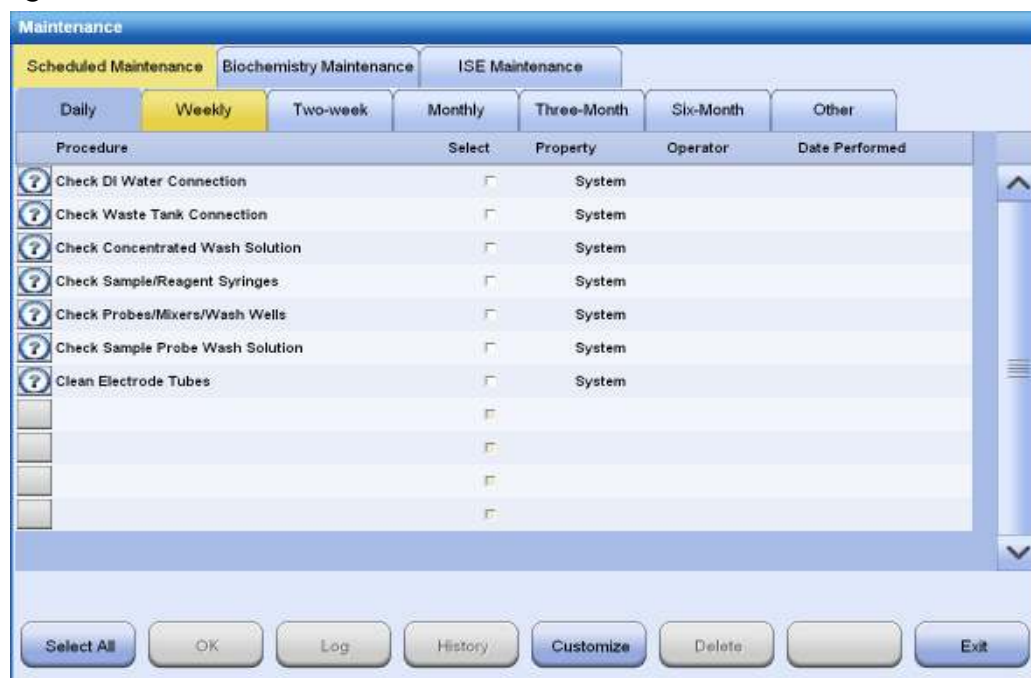
Maintenance Log Sheet																																
		Year																					Month				Page 1 of 2					
Daily Maintenance		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
1	Check Probes/Mixers/Wash Wells																															
2	Check Sample/Reagent Syringes																															
3	Check Deionized Water Connection																															
4	Check Waste Tube Connection																															
5	Check Concentrated Wash Solution																															
6	Check Sample Probe Wash Solution																															
7	Clean ISE Electrodes																															
Weekly Maintenance		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
1	Clean Sample/Reagent probe Exterior																															
2	Clean Mixers																															
3	Special Wash																															
4	Cuvette Check																															
5	Photometer Check																															
Monthly Maintenance		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
1	Clean Wash Wells																															
2	Clean Cuvette Wash Station																															
3	Clean Filter Core																															
4	Clean Dust Screens																															
5	Clean Sample Injection Port																															
6	Pump Calibration																															
7	Air Bubble Detector Calibration																															
8	Clean the Dust Screen of the External Vacuum Pump																															
Three-Month Maintenance		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
1	Clean DI Water Tank																															
2	Replace Filter Core																															

Six-Month Maintenance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
1 Replace Lamp																															
2 Replace Water Supply Filter																															
3 Replace Reference Electrode																															
As-Required/As-Needed Maintenance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
1 Clean Analyzer Panels																															
2 Bar Code Maintenance																															
3 Clean Sample Compartment																															
4 Clean Reagent Compartment																															
5 Clean Sample Probe Interior																															
6 Clean Reagent Probe Interior																															
7 Replace Sample Probe																															
8 Replace Reagent Probe																															
9 Replace sample mixer																															
10 Replace reagent mixer																															
11 Remove Air Bubbles In Sample Syringe																															
12 Remove Air Bubbles In Reagent Syringe																															
13 Special Wash Probes																															
14 Replace Cuvettes																															
15 Replace ISE Electrodes																															
16 Store Electrodes																															
17 Remove Reagent Pack																															
18 Replace sample syringe																															
19 Replace reagent syringe																															
20 Clean rotors																															
21 Clean waste tube of ISE module																															

16.4.5 Scheduled Maintenance Screen Overview

The **Scheduled Maintenance** screen contains maintenance frequency tabs, maintenance procedures, scroll bar, and function buttons. Select a tab to view the maintenance procedures to be performed in the period. Choose a maintenance procedure, and then select function buttons to access windows to execute an operation.

Figure 16.3 Scheduled Maintenance screen



Fields and buttons on the screen are introduced as follows.

Maintenance procedures

Shows the preset and user-defined maintenance procedures for the current maintenance frequency.

Select field

Choose a maintenance procedure and click on the corresponding **Select** checkbox. A tick appears in the middle of the checkbox, which indicates the maintenance procedure is chosen. Select the function buttons at the bottom of the screen to access a window or execute an operation. To deselect a maintenance procedure, click on the **Select** checkbox again. The tick inside the checkbox disappears, which indicates the maintenance procedure is deselected.

Property field

Shows how the maintenance procedure is defined. The Property includes two options: System and User. System indicates that the maintenance procedure is

defined by the manufacturer and cannot be configured; User indicates that the maintenance is defined by user and can be configured for each maintenance frequency.

Operator field

Shows who performs the maintenance procedure, that is, the user ID currently logging on the system.

Date Performed field

Shows the date confirmed by the operator on which the maintenance was performed. After performing a maintenance procedure, mark the **Select** checkbox and select **OK**. The date is refreshed and displayed as the current date. The system will restart the countdown of the maintenance frequency from the current date.

Scroll bar

If all maintenance procedures of a period are not shown on the current screen, move the scroll bar view more maintenance procedures.

Select All button

This function allows selection of all maintenance procedures currently available on the screen. When the **Select All** button is selected, a tick appears in all **Select** checkboxes to the right of the maintenance procedures. Choose the following buttons as needed:

- **OK**: allows the reviewal of the selected maintenance procedure and entering of the date performed.
- **Log**: allows recording of comments and other important information of maintenance.
- **History**: provides a stored history record of maintenance performance with date and operator for the procedure selected.

OK button

This function allows the reviewal of the selected maintenance procedure and entering of the date performed. When the approving a maintenance procedure, the date of performance will be displayed as the current date.

Log button

The electronic maintenance log function allows the recording of comments and other important information of maintenance. Choose one or more maintenance procedures, and then select the **Log** button. The **Maintenance Log** window shows. Input logs for the procedure selected, and then select **OK**. Your input information will be applied to the selected maintenance procedure.

History button

This feature provides a stored history record of maintenance performance with date and operator for the procedure selected. You are allowed to edit or delete a maintenance record. Please note that only one maintenance procedure can be recalled for history performance at one time.

- 1 Choose a maintenance procedure on the **Scheduled Maintenance** screen.
- 2 Select **History**. The **Maintenance Log** window is displayed.
- 3 View all performance records of the selected maintenance procedure.
- 4 To edit a maintenance record:
 - Mark the checkbox of the desired maintenance record.
 - Select **Edit**.
 - Modify the maintenance record.
 - Select **OK**.

Only one maintenance record can be edited at one time.

- 5 To delete maintenance records:
 - Mark the checkbox of one or more desired maintenance records.
 - Select **Delete**.
 - Select **OK**. The selected maintenance records are removed.
- 6 To print maintenance log:
 - Mark the checkbox of one or more desired maintenance records.
 - Select **Print**.
- 7 Select **Close** to exit the window.

Customize button

The Customize function allows definition of new maintenance procedures and configuration of manufactured-/user-defined maintenance procedures. User-defined maintenance procedures can be deleted.

Select **Customize** on the **Scheduled Maintenance** screen. The **Customize Maintenance Procedure** window is displayed.

To define a maintenance procedure:

- Select **New**.
- Enter the name of the new maintenance procedure.

- Select **OK**. The maintenance procedure is displayed in the **Available Procedures** list.
- Use **>>** and **<<** to configure or cancel user-defined maintenance procedures. The property of a user-defined maintenance procedure is User.
- Select **OK** to save the configuration, or select **Cancel** to abort it.

To configure a maintenance procedure:

- Choose a maintenance frequency in the **Frequency** pull-down list.
- Choose a maintenance procedure in the **Available Procedures** list. Move the vertical scroll bar to view more maintenance procedures.
- Select **>>**. The selected maintenance procedure appears in the **Enabled Procedures** list, and the relevant maintenance schedule screen will be refreshed automatically.

To remove a maintenance procedure:

- Choose a maintenance procedure in the **Enabled Procedures** list.
- Select **<<**. The selected maintenance procedure is removed from the **Enabled Procedures** list and appears in the **Available Procedures** list. The relevant maintenance schedule screen will be refreshed automatically.
- Select **OK** to save the configuration, or select **Cancel** to abort it.

Delete button

The system allows deleting of maintenance procedures that will no longer be used. Only user-defined rather than manufacturer-defined maintenance procedures can be deleted.

- 1 Choose a maintenance procedure on the **Scheduled Maintenance** screen.
- 2 Select **Delete**.
- 3 Select **OK**. The selected maintenance procedure is deleted. The **Available Procedures** list on the **Customize Maintenance Procedure** window is refreshed automatically.

Close

Select this button to close the **Maintenance** window.

16.5 Daily Maintenance

16.5.1 Introduction

The daily maintenance procedures should be performed every day prior to measurements. The maintenance for the biochemistry system is to check the sample probe, reagent probe, mixers, wash wells, syringes, deionized water connection, waste connection, and concentrated wash solution volume. The maintenance for the ISE module is to clean the electrodes in order to remove the proteins and lipid remaining on their surfaces.

16.5.2 Check Probes/Mixers/Wash Wells

Abnormal sample probe, reagent probe, wash wells or mixers may influence the measurement performance and result in inaccurate results. Prior to measurements every day, check the sample probe and reagent probe for stains and crystals, and check if the mixers cannot rotate normally or are lifted and the water flow in the wash wells is abnormal. If the above-mentioned abnormalities exist, clean or adjust the probes and mixers immediately.

Purpose

To check the sample probe and reagent probe for water dripping, stains and liquid flow abnormalities, and check if the mixers can rotate normally and the water flow in the wash wells is abnormal.

When to do

You are recommended to do this maintenance procedure every day before starting the analysis.

System status

Make sure that the system status is Standby.

Precautions



Warning

The probes and mixers are sharp and vulnerable. To prevent injury and equipment damage, exercise caution when working around the probes and mixers. Keep away from the probes and mixers to avoid collision with them.

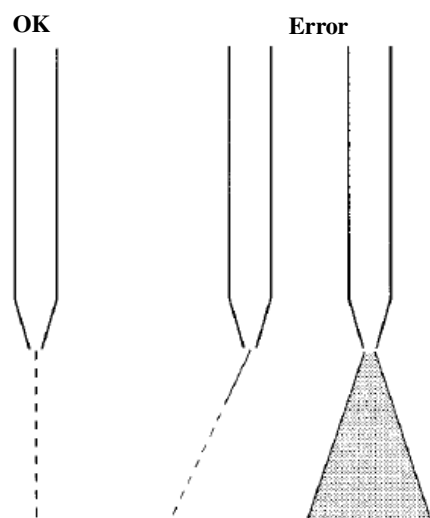


BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles.

How to do

- 1 Open the upper and rear protective shields of the analyzer.
- 2 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 3 Select **Clean Probes/Mixers/Wash Wells**.
- 4 Check the exterior of the probes/mixers for stains. If stains exist, perform the Clean Sample/Reagent Probe Exterior or Clean Mixers procedure.
- 5 Select **Continue** to clean the reagent probe and sample probe interiors.
- 6 Check the liquid flow of the sample probe and reagent probe. If the liquid flow is sprayed out or does not come out vertically, the probe may be clogged. Perform the Concentrated Wash Probes/Mixers procedure,, and then check them again. If the abnormality remains, perform the Clean Sample Probe Interior or Clean Reagent Probe Interior procedure. If the abnormality still remains, perform the Replace Sample Probe or Replace Reagent Probe procedure, or contact a service engineer.

Figure 16.4 Normal and abnormal liquid flows of sample probe and reagent probe

- 7 Select **Second Wash**. The probe interior wash can be performed again.
- 8 Select **Continue**.
- 9 Observe the water flow of the probe/mixer wash wells, and check if the water

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reaches to about 5mm of the probe/mixer from the tip. If it does, proceed to the next step; otherwise, contact a service engineer.

- 10 Select **Continue**.
- 11 Select **Done**.
- 12 Select **Scheduled Maintenance-Daily**.
- 13 Mark the **Select** checkbox to the right of **Check Probes/Mixers/Wash Wells**.
- 14 Select **OK** to refresh the current date as the performance date.
- 15 Select **Log**, and then record comments and other important information for the procedure.
- 16 Select **OK** to save your input information.
- 17 Restore the protective shields.

16.5.3 Check Sample/Reagent Syringes

The sample syringe and reagent syringes are precise devices used to aspirate/dispense small amount of sample and reagent. If the syringes leak, they cannot aspirate/dispense the correct amount of sample or reagent, and may even be damaged. Prior to measurements every day, check the sample/reagent syringes for leak.

Purpose

To check the sample/reagent syringes for leak and air bubbles.

When to do

You are recommended to do this maintenance procedure every day before starting the analysis.

Materials required

Clean gauze

System status

Make sure that the system status is Incubation or Standby.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles.

How to do

- 1 Open the front door of the analyzer. You will see two syringes on the two side of the water tank.
- 2 Check the T piece assembly and plunger guide cap for leak.
- 3 Use dry gauze to wipe the T piece, and then check if the gauze is moistened.
 - If it is not, proceed to the next step.
 - If it is, tighten the T piece.
 - Check the T piece and plunger guide cap again. If the leak remains, check if the washer inside the syringe connector is intact.
 - If the washer is damaged, replace it with a new one; otherwise, replace the syringe.
- 4 Check the syringe interior for air bubbles. If yes, remove the air bubbles.
- 5 Check if the retaining screws at the bottom of the syringe are tightened.
 - If not, tighten them and proceed to the next step.
 - If yes, proceed to the next step.
- 6 Close the front door of the analyzer.
- 7 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Daily**.
- 8 Mark the **Select** checkbox to the right of **Check Sample/Reagent Syringes**.
- 9 Select **OK** to refresh the current date as the performance date.
- 10 Select **Log**, and then record comments and other important information for the procedure.
- 11 Select **OK** to save your input information.

16.5.4 Check Deionized Water

If the deionized water tubes are not connected properly, deionized water cannot be supplied normally or leak may be caused, influencing the measurements.

Purpose

To check the DI water connection to ensure normal supply of DI water.

When to do

You are recommended to do this maintenance procedure every day before starting the analysis.

System status

Make sure that the system is powered off, or the system status is Incubation or Standby.

How to do

- 1 Check that the water tank or other water containers have sufficient deionized water.
- 2 Check that the tubes are not bent or folded or leaking.
- 3 Check that the water supply module is powered on.
- 4 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Daily**.
- 5 Mark the **Select** checkbox to the right of **Check DI Water Connection**.
- 6 Select **OK** to refresh the current date as the performance date.
- 7 Select **Log**, and then record comments and other important information for the procedure.
- 8 Select **OK** to save your input information.

16.5.5 Check Waste

If the waste tube is not connected properly or the high-concentration waste tank is full, overflow may be caused, resulted in environmental contamination or cross contamination, or even damaging the equipment. It is necessary to regularly check the waste tube connection and the high-concentration waste tank.

Purpose

To check the waste tube connection and the high-concentration waste tank to

prevent overflow.

When to do

You are recommended to do this maintenance procedure every day before starting the analysis.

System status

Make sure that the system is powered off, or the system status is Incubation or Standby.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

Dispose of the waste in accordance with your local or national guidelines for biohazard waste disposal.

How to do

- 1 Check if the waste drainage system works well, and make sure that the waste tube is not bent or folded and the high-/low-concentration waste is drained properly.
- 2 Check if the high-concentration waste tank has been emptied. If not, empty it.

High-concentration waste output: 1.5L/H, low-concentration waste output: no more than 18.5L/H, and water consumption: no more than 20L/H.
- 3 If leak remains after performing the above-stated steps, contact our customer service department or your local distributor.
- 4 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Daily**.
- 5 Mark the **Select** checkbox to the right of **Check Waste Tank Connection**.
- 6 Select **OK** to refresh the current date as the performance date.
- 7 Select **Log**, and then record comments and other important information for the procedure.
- 8 Select **OK** to save your input information.

16.5.6 Check Concentrated Wash Solution

Insufficient concentrated wash solution may terminate the measurements. Prior to measurements every day, check the concentrated wash solution volume, and fill more, if necessary.

A tank of concentrated wash solution is 2L and can be used for 6 days for plastic cuvettes on condition that 2000 tests are done every day. Please check and refill the concentrated wash solution according to the consumption and tank volume.

Purpose

To check the concentrated wash solution volume to prevent measurements from being terminated.

When to do

You are recommended to do this maintenance procedure everyday before starting the analysis.

System status

Make sure that the system is powered off, or the system status is Incubation or Standby.

Precautions



Warning

Concentrated wash solution is corrosive to human skins. Wear gloves and goggles while checking the concentrated wash solution. In case your hand or clothes contact the wash solution, wash them off with soap and water. If the wash solution spills into your eyes, rinse them with water and consult an oculist.



CAUTION

When the system is Initializing, it may be diluting the concentrated wash solution. Do not try to fill concentrated wash solution until the system status becomes Standby.

How to do

- 1 Open the front door of the analyzer and check the concentration wash solution.
If necessary, fill more or replace the wash solution.
- 2 Close the front door of the analyzer.
- 3 Power on the analyzer and run the operating software.
- 4 Make sure that the system status is Incubation or Standby.
- 5 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled**

Maintenance-Daily.

- 6 Mark the **Select** checkbox in the same row as **Check Concentrated Wash Solution**.
- 7 Select **OK** to refresh the current date as the performance date.
- 8 Select **Log**, and then record comments and other important information for the procedure.
- 9 Select **OK** to save your input information.

16.5.7 Check Sample Probe Wash Solution

Insufficient sample probe wash solution may cause probe clogging and cross contamination.

Three special washes will be conducted for the sample probe when every batch of tests is finished, and about 90µl wash solution is consumed for each wash. You are recommended to check and replace the sample probe wash solution every day to ensure its sufficiency.

Purpose

To check the sample probe wash solution volume to prevent measurements from being terminated.

When to do

You are recommended to do this maintenance procedure every day before starting the analysis.

System status

Make sure that the system is powered off, or the system status is Incubation or Standby.

Precautions**CAUTION**

You are recommended to replace the sample probe wash solution every day in order to prevent probe clogging and cross contamination.

While the system is running tests, do not try to fill sample probe wash solution until the system status becomes Standby.

How to do

- 1 Check the volume of the sample probe wash solution on the sample carousel position D2.

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- 2 If necessary, fill more or replace the wash solution.
- 3 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Daily**.
- 4 Mark the **Select** checkbox in the same row as **Check Sample Probe Wash Solution**.
- 5 Select **OK** to refresh the current date as the performance date.
- 6 Select **Log**, and then record comments and other important information for the procedure.
- 7 Select **OK** to save your input information.

16.5.8 Clean Electrode Tubes (For ISE Module)

When the ISE module finishes a great number of measurements, the proteins and lipid contained in samples may remain on surfaces of the electrodes, influencing their measurement performance. You should clean the electrodes regularly to ensure system performance. It will take about 2 minutes to perform this procedure.

Purpose

To remove the proteins and lipid remaining on the electrode surfaces.

When to do

You are recommended to perform this procedure after finishing all ISE tests of the day, or before shutting down the system, or 50 samples are analyzed.

Materials required

ISE wash solution, 2ml sample tube

System status

Make sure that the status of both the biochemistry system and ISE module is Standby.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles.

The wash solution may hurt your eyes and skins. Exercise caution while using the wash solution. If your eyes contact the wash solution, rinse them off with fresh water and consult a doctor.



CAUTION

Please use consumables recommended by our company. Use of other consumables may degrade the system performance.



NOTE

After performing this procedure, recalibrate the ISE electrodes prior to starting analysis.

How to do

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Clean Electrode Tubes**. The maintenance guide window shows.
- 3 Open the upper protective shield of the analyzer.
- 4 Fill a 2ml sample tube with 300µl ISE wash solution, and then load it to position D1 on the sample carousel.
- 5 Select **Continue**. The system starts cleaning the ISE electrodes.
- 6 Select **Done**.
- 7 Select **Scheduled Maintenance-Daily**.
- 8 Mark the **Select** checkbox in the same row as **Clean Electrode Tubes**.
- 9 Select **OK** to refresh the current date as the performance date.
- 10 Select **Log**, and then record comments and other important information for the procedure.
- 11 Select **OK** to save your input information.

16.6 Weekly Maintenance

16.6.1 Clean Sample/Reagent Probe Exterior

The sample probe and reagent probe are often dirty on their surfaces, causing carryover between samples or reagents and resulting in inaccurate results. You are recommended to perform this procedure every week.

Purpose

To clean the exterior of the sample probe and reagent probe to prevent cross contamination.

When to do

This procedure should be performed on weekly basis.

Materials required

2 pieces of clean gauze, ethanol, deionized water, tweezers

System status

Make sure that the system status is not Running.

Precautions**Warning**

The probe tip is sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the probes. If the probe is bent or damaged, replace it immediately; otherwise, unreliable results may be obtained.

**BIOHAZARD**

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Switch off the analyzing unit power.
- 2 Rotate the probe arm to move the probe to a position convenient for cleaning, and then use gauze soaked with ethanol to gently wipe the probe exterior. Clean the probe tip until it becomes clear without stain.

Do not pull the probe horizontally to prevent probe damage.

- 3 Use gauze moistened with deionized water to clear the ethanol on the probe.

- 4 After finishing the cleaning, turn on the analyzing unit power switch.
- 5 Select **Utility-Commands-Home** to reset the probe.
- 6 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Weekly**.
- 7 Mark the **Select** checkbox to the right of **Clean Sample/Reagent Probe Exterior**.
- 8 Select **OK** to refresh the current date as the performance date.
- 9 Select **Log**, and then record comments and other important information for the procedure.
- 10 Select **OK** to save your input information.

16.6.2 Clean Mixers

The mixers are often dirty on their surfaces, causing carryover between samples or reagents and resulting in inaccurate results. You are recommended to perform this procedure every week.

Purpose

To clean the sample mixer and reagent mixer to prevent cross contamination.

When to do

This procedure should be performed on weekly bases.

Materials required

2 pieces of clean gauze, ethanol, deionized water, tweezers

System status

Make sure that the system status is not Running.

Precautions



Warning

Exercise caution while working around the mixer. If it is bent or damaged, replace it immediately; otherwise, unreliable results may be obtained.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

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How to do

- 1 Switch off the analyzing unit power.
- 2 Rotate the mixer arm to move the mixer to a position convenient for cleaning, and then use gauze soaked with ethanol to gently wipe the mixer exterior until it becomes clear without stain.

Do not pull the mixer horizontally to prevent damage.
- 3 Use gauze moistened with deionized water to clear the ethanol on the mixer.
- 4 After finishing the cleaning, turn on the analyzing unit power switch.
- 5 Select **Utility-Commands-Home** to reset the mixer.
- 6 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Weekly**.
- 7 Mark the **Select** checkbox in the same row as **Clean Mixers**.
- 8 Select **OK** to refresh the current date as the performance date.
- 9 Select **Log**, and then record comments and other important information for the procedure.
- 10 Select **OK** to save your input information.

16.6.3 Special Wash

Special wash is to clean the sample probe, reagent probe, mixers, reaction cuvettes and wash station by using the concentrated wash solution, with the aim of eliminating carryover and preventing waste from leaving in the waste tubes. It will take about 30 minutes to perform this procedure.

Purpose

To eliminate cross contamination among the sample probe, reagent probe, mixers, cuvettes and wash station, and prevent waste from leaving in the waste tubes.

When to do

You are recommended to perform this procedure on weekly basis or when the equipment is to be stored for a long time.

Materials required

Concentrated wash solution manufactured by our company

System status

Make sure that the system status is Standby.

How to do

- 1 Open the upper protective shield of the analyzer.
- 2 Place more than 40ml concentrated wash solution in position D of the reagent carousel, and place more than 1ml concentrated wash solution in position D2 on the sample carousel.
- 3 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 4 Choose **Special Wash**.
- 5 Confirm if cuvette check is needed after the special wash. If it is, mark the checkbox in front of **Perform Cuvette Check**.
- 6 Select **Continue** to continue, or select **Exit** to abort the special wash.
- 7 The system starts cleaning the sample probe, reagent probe, mixers, cuvettes and wash station. To terminate the clean process, select **Stop**.
- 8 Perform the cuvette check procedure. Refer to 16.6.4 Cuvette Check (page 16-33) for details.
- 9 Select **Done**.
- 10 Restore the upper protective shield of the analyzer.
- 11 Select **Scheduled Maintenance-Weekly**.
- 12 Mark the **Select** checkbox in the same row as **Special Wash**.
- 13 Select **OK** to refresh the current date as the performance date.
- 14 Select **Log**, and then record comments and other important information for the procedure.
- 15 Select **OK** to save your input information.

16.6.4 Cuvette Check

After being used for a long time, the reaction cuvettes may have proteins or other stains left inside of them that are difficult to remove and will influence the light

transmittance of the cuvettes. If the cuvettes are polluted or scratched or damaged, the light transmittance will be affected, threatening the accuracy and stability of the results. Check the reaction cuvettes regularly to avoid unwanted results. It will take about 15~20 minutes to perform this procedure.

Purpose

To check if the reaction cuvettes are polluted and the light transmittance is decreased in order to prevent unreliable test results.

When to do

You are recommended to perform this procedure on weekly basis or after replacing the reaction cuvettes.

System status

Prior to performing the maintenance, make sure that the system has been power on for over 10 minutes and the system status is Standby. Check if the reaction carousel has a cuvette for each position. If not, load cuvettes.

Precautions



NOTE

When a cuvette is deemed dirty, clean or replace it immediately, and then perform the cuvette check again.

Stains inside cuvettes will influence the photometric measurement. You are recommended to perform the Cuvette Check after finishing the Concentrated Wash procedure.

How to do

- 1 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 2 Choose **Cuvette Check**.
- 3 Make sure that the lamp has been turned on for over 10 minutes. Select **Continue** and then select **Start**. When finishing the check, the system refreshes the cuvette status based on the check results. Record the cuvettes highlighted in yellow and please replace the cuvettes highlighted in yellow. To abort the cuvette check, select **Stop**.

The screen shows all cuvettes and highlights the dirty cuvettes with special color:

- No color indication: normal cuvette
- Yellow: dirty cuvette

- 4 Select **Result**. The **Cuvette Check Results** window appears and shows the

latest check result of the 90 cuvettes at all wavelengths.

- 5 Choose a cuvette in the result list. The **Cuvette Status** window pops up.

Choose the following buttons as needed:

- |<: to view the first cuvette.
- <: to view the previous cuvette.
- >: to view the next cuvette.
- >|: to view the last cuvette.
- **Print**: to print the results currently displayed on the screen.
- **Exit**: to close the **Cuvette Status** window.

- 6 Select **Exit** to close the **Cuvette Check** window.

- 7 Select **Scheduled Maintenance-Weekly**.

- 8 Mark the **Select** checkbox in the same row as **Cuvette Check**.

- 9 Select **OK** to refresh the current date as the performance date.

- 10 Select **Log**, and then record comments and other important information for the procedure.

- 11 Select **OK** to save your input information.

16.6.5 Photometer Check

Decreased light intensity and stability of the lamp will directly influence the accuracy and repeatability of the results. Check the lamp regularly, or if necessary, replace it. The Photometer Check procedure provides detection of too strong or too weak light intensity. The photometer status will be provided through an alarm message or prompt message.

Purpose

To check the light intensity by measuring absorbance of 5 cuvettes and help you determine whether to replace the lamp.

When to do

You are recommended to perform this procedure on weekly basis or after replacing the lamp.

System status

Prior to performing the maintenance, make sure that the system has been power on for over 10 minutes and the system status is Standby.

Precautions



NOTE

Before checking the lamp, perform the Cuvette Check procedure and replace or clean the dirty cuvettes; otherwise, the photometer check results are unreliable.

To ensure the photometer's measurement performance, replace the lamp in the case of weak light intensity.

How to do

- 1 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 2 Choose **Photometer Check**. The following window appears.
- 3 Make sure that the lamp has been turned on for over 10 minutes. Select **Continue** and then select **Start**. When finishing the check, the system displays the results and refreshes the photometer status. To abort the photometer check, select **Stop**.

On the left of the screen shows the absorbance at each wavelength in the current photometer check; on the right of the screen shows that of the previous photometer check. By checking the results of the previous and current photometer check, you may understand the status of the lamp.

- 4 If the **Current Status** field shows Normal, it indicates that the lamp's light intensity satisfies the requirements of measurement; if it shows "Light intensity is weak" in red, it indicates that the lamp has insufficient light intensity.
- 5 If an alarm occurs during the check, operate as follows:
 - If the alarm indicates the lamp is off, check if the lamp has been turn on. If not, manually turn on the lamp; if yes, contact our customer service department or your local distributor.
 - If the alarm indicates light intensity too strong, contact our customer service department or your local distributor.
 - If the alarm indicates light intensity weak, select **Replace** to replace the lamp. For more information, refer to 16.9.1 Replace Lamp (page 16-54).
- 6 Choose the following buttons as needed:
 - **Print**: to print the photometer check results currently available on the screen.
 - **Exit**: to close the window.

- 7 Select **Done** to close the **Photometer Check** window.
- 8 Select **Scheduled Maintenance-Weekly**.
- 9 Mark the **Select** checkbox in the same row as **Photometer Check**.
- 10 Select **OK** to refresh the current date as the performance date.
- 11 Select **Log**, and then record comments and other important information for the procedure.
- 12 Select **OK** to save your input information.

16.7 Monthly Maintenance

16.7.1 Clean Wash Wells

When the system is used for a long time, waste and dust may accumulate in the wash wells and block them. Clean the wash wells every month to keep them clean and smooth.

Purpose

To remove the waste and dust from the wash wells (of reagent probe, sample probe, sample mixer and reagent mixer).

When to do

This procedure should be performed on monthly basis.

Materials required

Cotton swabs and sodium hypochlorite solution (NaClO)

System status

Make sure that the system status is not Running.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Switch off the analyzing unit power.
- 2 Open the upper protective shield of the analyzer.
- 3 Rotate the same probe, reagent probe and mixers to keep them away from the wash wells.
- 4 Use clean cotton swabs moistened with NaClO to clean the wash wells.
- 5 After finishing the cleaning, turn on the analyzing unit power switch.
- 6 Select **Utility-Commands-Home** to reset the probes and mixers, check if the wash wells have a normal water flow.
- 7 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Monthly**.

- 8 Mark the **Select** checkbox in the same row as **Clean Wash Wells**.
- 9 Select **OK** to refresh the current date as the performance date.
- 10 Select **Log**, and then record comments and other important information for the procedure.
- 11 Select **OK** to save your input information.

16.7.2 Clean Wash Station

Clean the wash station regularly to prevent waste from accumulating on it.

Purpose

To clean the cuvette wash station in order to avoid waste buildup and cross contamination.

When to do

This procedure should be performed on monthly basis.

Materials required

Gauze, ethanol, deionized water, waste container (large beaker)

System status

Make sure that the system status is Standby.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process. Dispose of the used gauze in accordance with your local or national guidelines for biohazard waste disposal.

How to do

- 1 Open the upper protective shield of the analyzer.
- 2 Remove the cuvette wash station and use ethanol-moistened gauze to wipe the wash probes and wipe blocks.
- 3 Use gauze moistened with deionized water to clear the ethanol on the wash probes.
- 4 Restore the wash station.

- 5 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 6 Choose **Prime Wash Station**. The maintenance guide window shows. Select **Continue**.
- 7 Enter the wash cycle (1~100).
- 8 Select **Continue**.
- 9 When the cleaning and priming are finished, select **Done**.
- 10 Restore the upper protective shield of the analyzer.
- 11 Select **Scheduled Maintenance-Monthly**.
- 12 Mark the **Select** checkbox in the same row as **Clean Cuvette Wash Station**.
- 13 Select **OK** to refresh the current date as the performance date.
- 14 Select **Log**, and then record comments and other important information for the procedure.
- 15 Select **OK** to save your input information.
- 16 Select **Utility-Commands**, and then select **Home** to put the instrument into the Standby status.

16.7.3 Clean Filter Core

Clean the filter core every month to prevent accumulation of foreign matters and improve the water quality.

Purpose

To clean the filter core in order to prevent accumulation of foreign matters and improve the water quality.

When to do

This procedure should be performed on monthly basis.

Materials required

Tube brush or ultrasound cleaner

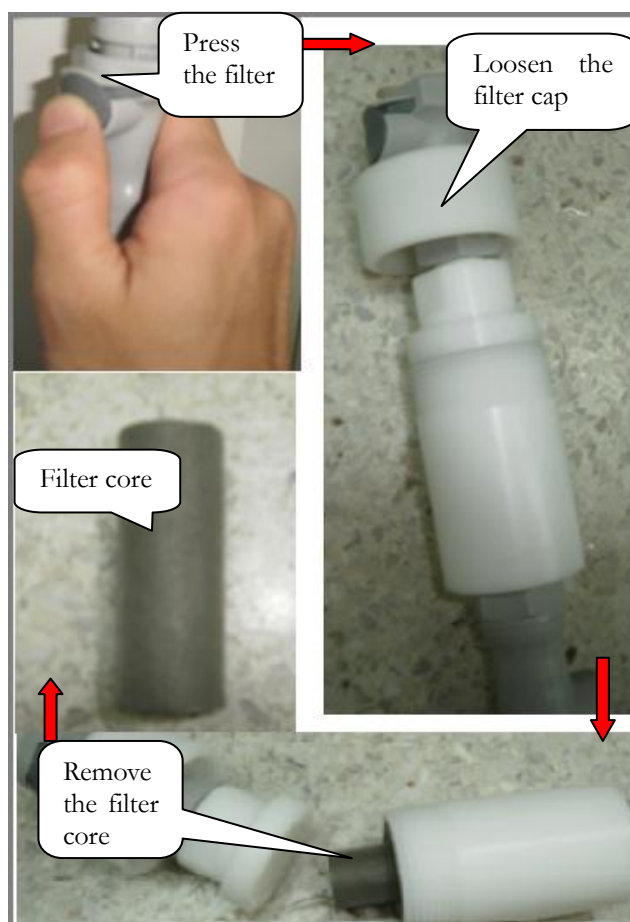
System status

Make sure that the system status is Standby.

How to do

- 1 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 2 Choose **Clean Filter/Water Tank**, and then select **Continue**.
- 3 Open the front door of the analyzer, unscrew the cap of the concentrated wash solution tank, and then remove the tank. The DI water filter appears in front of you.
- 4 Put a water container below the filter.
- 5 Remove the connectors from the DI water tank and the filter.
- 6 Tilt the filter assembly, loosen the filter cap and remove the filter core. Use a tube brush to clean the filter core's surface, or put it in an ultrasound cleaner for 10 minutes.

Figure 16.5 Remove deionized water filter core



- 7 Restore the filter core and filter cap according to the above-mentioned steps in

reversed order.

- 8 Select **Continue**. The system starts priming the tubes with deionized water.
- 9 When the priming is complete, select **Done**.
- 10 Close the front door of the analyzer.
- 11 Select **Scheduled Maintenance-Monthly**.
- 12 Mark the **Select** checkbox in the same row as **Clean Filter Core**.
- 13 Select **OK** to refresh the current date as the performance date.
- 14 Select **Log**, and then record comments and other important information for the procedure.
- 15 Select **OK** to save your input information.
- 16 Select **Utility-Commands**, and then select **Home** to put the instrument into the Standby status.

16.7.4 Clean Dust Screens

Dust may accumulate on the dust screens when the instrument is used for a long time, influencing the ventilation and heat elimination effects. It is necessary to clean the dust screens regularly.

Purpose

To clean the dust screens to ensure good ventilation.

When to do

This procedure should be performed on monthly basis.

Materials required

Suction cleaner, hair brush and fresh water

System status

Make sure that the analyzer main power is off.

Precautions



NOTE

Use a suction cleaner to clean the dust screens while keeping them uninstalled, or use a hair brush and fresh water to clean the dust screens after removing them from the analyzer.

Do not reinstall the dust screens until they are dry completely.

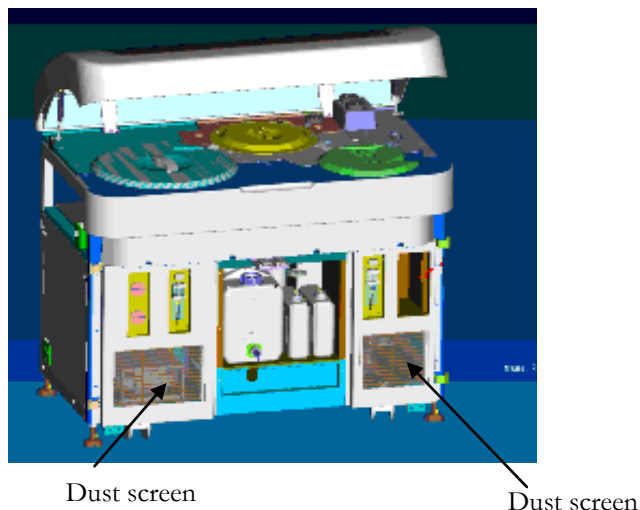
Install the dust screens correctly to avoid gaps.

To clean the dust screens by knocking them at solid ground, find an appropriate place, and then carefully knock them at the ground.

How to do

- 1 Switch off the analyzer's main power.
- 2 Open the front door of the analyzer and remove the dust screens.

Figure 16.6 Dust screens



- 3 Use the suction cleaner, or hair brush and fresh water to clean the dust screens, and then dry them in air.
- 4 Reinstall the dust screens when they are dry.
- 5 Close the front door of the analyzer.
- 6 Power on the analyzer and run the operating software.
- 7 Make sure that the system status is Incubation or Standby.
- 8 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Monthly**.
- 9 Mark the **Select** checkbox in the same row as **Clean Dust Screens**.
- 10 Select **OK** to refresh the current date as the performance date.
- 11 Select **Log**, and then record comments and other important information for the

procedure.

- 12 Select **OK** to save your input information.

16.7.5 Clean Sample Injection Port (For ISE Module)

When the ISE module is used for a period, stains may build up in the sample injection port and influence the measurement performance. Clean the sample injection port regularly to keep it clear.

Purpose

To remove the stains accumulating in the sample injection port.

When to do

You are recommended to perform this procedure every month.

Materials required

Deionized water, cotton swabs, and ethanol

System status

Make sure that the status of the ISE module is Standby or Failure.

How to do

- 1 Make sure that the system status is Incubation or Standby.
- 2 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 3 Choose the **Clean Sample Injection Port** option.
- 4 Open the upper protective shield of the analyzer.
- 5 Open the cover of the ISE module.
- 6 Use clean cotton swab soaked with ethanol to wipe the sample injection port (interior of the sample injection cup of the ISE module) until it is clean; then use a clean cotton swab soaked with DI water to wipe the interior and periphery of the sample injection port.
- 7 Select **Done**.
- 8 Select **Purge A** and **Purge B** to prime the ISE module.
- 9 Select **Scheduled Maintenance-Monthly**.
- 10 Mark the **Select** checkbox in the same row as **Clean Sample Injection Port**.

- 11 Select **OK** to refresh the current date as the performance date.
- 12 Select **Log**, and then record comments and other important information for the procedure.
- 13 Select **OK** to save your input information.
- 14 Restore the cover of the ISE module.
- 15 Restore the upper protective shield of the analyzer.

16.7.6 Pump Calibration (For ISE Module)

The peristaltic pump may get aging when used for a long time. It is necessary to calibrate it regularly.

Purpose

To calibrate the peristaltic pump to ensure accurate test result.

When to do

This procedure should be performed on monthly basis.

System status

Make sure that the status of the ISE module is Standby .

How to do

- 1 Make sure that the system status is Incubation or Standby.
- 2 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 3 Choose **Pump Calibration**.
- 4 Select **Start**. Place 500ul DI water in the place designated by the software.

When the calibration is complete, the calibration results are displayed on the screen.

- 5 Select **Done**.
- 6 To view the detailed results, select **Pump Calibration Result**. The detailed results are displayed in the data list.
- 7 Select **Scheduled Maintenance-Monthly**.
- 8 Mark the **Select** checkbox in the same row as **Pump Calibration**.
- 9 Select **OK** to refresh the current date as the performance date.

- 10 Select **Log**, and then record comments and other important information for the procedure.
- 11 Select **OK** to save your input information.

16.7.7 Air Bubble Detector Calibration (For ISE Module)

The air bubble detector may get aging when used for a long time. It is necessary to calibrate it regularly.

Purpose

To calibrate the air bubble detector to ensure accurate test result.

When to do

This procedure should be performed on monthly basis.

System status

Make sure that the status of the ISE module is Standby .

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Air Bubble Detector Calibration**.
- 3 Select **Start**.

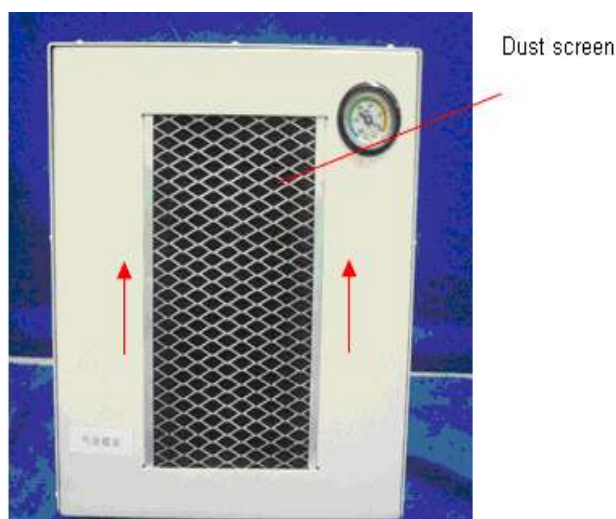
When the calibration is complete, the calibration results are displayed on the screen.
- 4 Select **Done**.
- 5 To view the detailed results, select **Air Bubble Detector Calibration Result**.
The detailed results are displayed in the data list.
- 6 Select **Scheduled Maintenance-Monthly**.
- 7 Mark the **Select** checkbox in the same row as **Air Bubble Detector Calibration**.
- 8 Select **OK** to refresh the current date as the performance date.

- 9 Select **Log**, and then record comments and other important information for the procedure.
- 10 Select **OK** to save your input information.

16.7.8 Clean the Dust Screen of the External Vacuum Pump

Dust may accumulate on the dust screens when the vacuum pump is used for a long time, influencing the ventilation and heat elimination effects. It is necessary to clean the dust screen monthly.

- 1 Remove the front dust screen of the external vacuum pump.
 - Push the dust screen upward gently along the groove and make its lower end leave the groove.



- Pull the dust screen out of the groove.

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2 Replace it with a clean one.

- Insert the upper end of the dust screen into the groove and push it upward.



- Press the dust screen and make it contact tightly with the groove.
- Push it downward gently and make the lower end inserted into the groove as well.



- 3 Wash the removed dust screen with water and dry it.

16.8 Three-Month Maintenance

16.8.1 Clean DI Water Tank

Stains will remain in the deionized water tank when it is used for a long time and may influence the cleaning effects of the system.

Purpose

To clean the deionized water tank to ensure good cleaning performance of the system.

When to do

You are recommended to perform this procedure every 3 months.

Materials required

Water container

System status

Make sure that the system status is Standby.

How to do

- 1 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 2 Choose **Clean Filter/Water Tank**, and then select **Continue**.
- 3 Open the front door of the analyzer. You will see the deionized water tank as shown in the figure below.

Figure 16.7 Deionized water tank

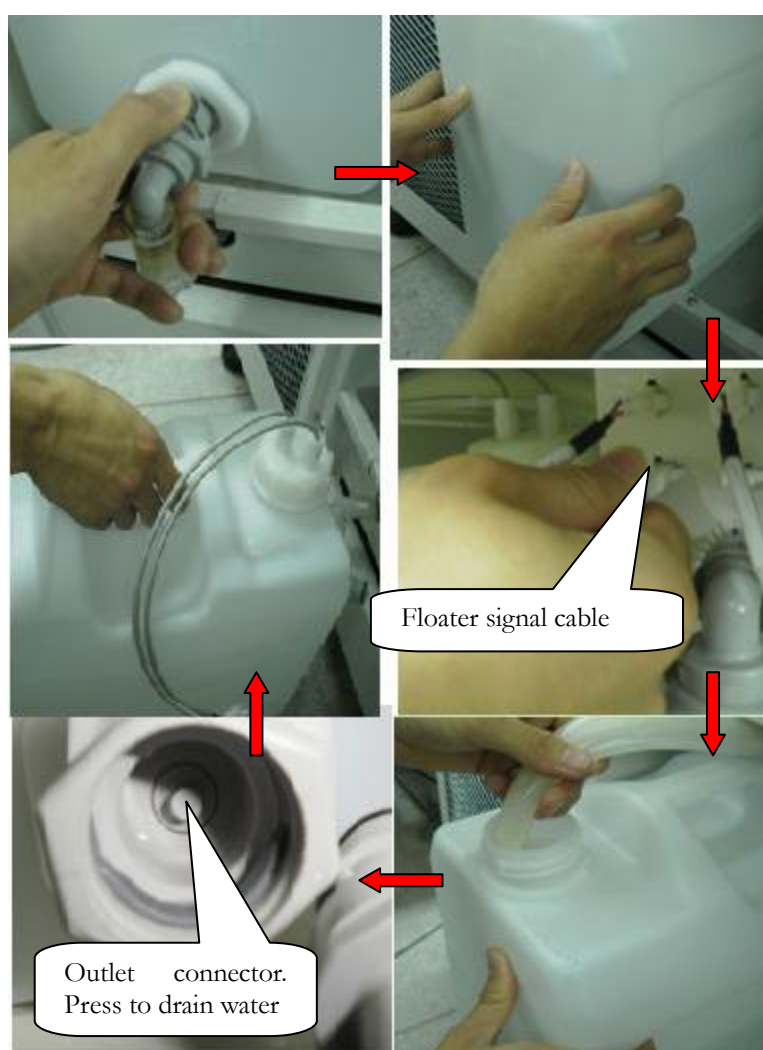


- 4 Remove the quick connector from the outlet of the water tank, and then pull the water tank outwards for a little to expose its opening.
- 5 Put a water container below the outlet of the DI water tank; insert another

normally open quick connector into the outlet to drain water into the water container. When the DI water tank is emptied, proceed to the next step. Or you may close the outlet with a solid plug, take out the water tank completely, and then empty it by inclining it. Choose this method if there is little water inside the DI water tank.

- 6 Remove the tubes from the tank inlet, disconnect the liquid level floater signal cable from the right panel of the water tank, take out the water tank completely, and then remove the liquid level floater. Perform this step according to the figure below.

Figure 16.8 Remove deionized water tank



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- 7 Clean the water tank repeatedly with deionized water.
- 8 Insert the floater into the connector on rear panel of the water tank, connect the backflow tube to the water tank, connect the floater signal cable and water supply tube to the water tank according to the labels on it, and then place the water tank in the cabinet of the analyzer.
- 9 Select **Continue**. The system automatically primes the deionized water tubes.
- 10 Take away the water container and close the front door of the analyzer.
- 11 Select **Scheduled Maintenance-Three-Month**.
- 12 Mark the **Select** checkbox in the same row as **Clean DI Water Tank**.
- 13 Select **OK** to refresh the current date as the performance date.
- 14 Select **Log**, and then record comments and other important information for the procedure.
- 15 Select **OK** to save your input information.
- 16 Select **Utility-Commands**, and then select **Home** to put the instrument into the Standby status.

16.8.2 Replace Filter Core

The filter may be blocked after being used for a long time. Replace the filter core every 3 months to ensure good filtering effects.

Purpose

To replace the filter core and ensure good filtering effects.

When to do

You are recommended to perform this procedure every 3 months.

Materials required

New filter core

System status

Make sure that the system status is Standby.

How to do

- 1 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.

- 2 Choose **Clean Filter/Water Tank**, and then select **Continue**.
- 3 Remove the deionized water filter core according to the Clean Filter Core procedure.
- 4 Put the new filter core in the filter and reinstall the filter.
- 5 Select **Continue**. The system starts priming the deionized water tubes.
- 6 When the replacement is complete, select **Done**.
- 7 Select **Scheduled Maintenance-Three-Month**.
- 8 Mark the **Select** checkbox in the same row as **Replace Filter Core**.
- 9 Select **OK** to refresh the current date as the performance date.
- 10 Select **Log**, and then record comments and other important information for the procedure.
- 11 Select **OK** to save your input information.
- 12 Select **Utility-Commands**, and then select **Home** to put the instrument into the Standby status.

16.9 Six-Month Maintenance

16.9.1 Replace Lamp

An aged lamp will have its energy decreased and influence the measurement accuracy. Failed lamp will make measurements impossible. To ensure the optimal performance of the system, replace the lamp regularly. Every time after you replacing the lamp, if the light intensity is insufficient, replace the lamp immediately. It will take about 10 minutes to perform this procedure.

Purpose

To ensure that the lamp works normally.

When to do

You are recommended to perform this procedure every 6 months or when you find that the lamp does not satisfy the requirements after performing the Photometer Check.

Materials required

New lamp, Philips-head screwdriver, cotton or antistatic gloves

System status

Make sure that the system status is Standby or Failure.

Precautions



NOTE

Too hot lamp may burn you. Do not replace the lamp until it gets cool.

Do not touch the light entrance on the lamp housing or the lens in front of the lamp. In case the light entrance is dirty, use cotton swabs moistened with absolute ethanol to clean it.



CAUTION

Please use consumables recommended by our company. Use of other consumables may degrade the system performance.

How to do

- 1 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 2 Choose **Replace Lamp**. The maintenance guide window pops up. Select **Continue**.

- 3 Make sure that the lamp has cooled down for 5 minutes, and then select **Continue**.
- 4 Remove the cover plate of the lamp.
- 5 Wear a pair of cotton or antistatic gloves, loosen the nuts on the cable terminals, and then remove the O-ring connectors from the terminals.
- 6 Loosen the retaining screw on the left side of the lamp.
- 7 Remove the lamp from the lamp housing.

**CAUTION**

Do not hold the lamp by its bulb to prevent contamination and damage.

- 8 Install the retaining screw, O-ring connectors, cable terminal nuts and lamp cover plate in the reversed order.
- 9 Select **Continue**.
- 10 When the lamp is incubated, select **Done**.

Perform the Photometer Check procedure to ensure the system power is normal. For more information, refer to 16.6.5 Photometer Check (page 16-35).
- 11 Select **Scheduled Maintenance-Six-Month**.
- 12 Mark the **Select** checkbox in the same row as **Replace Lamp**.
- 13 Select **OK** to refresh the current date as the performance date.
- 14 Select **Log**, and then record comments and other important information for the procedure.
- 15 Select **OK** to save your input information.

16.9.2 Replace Water Inlet Filter

When the water inlet filter is used for a long period, it may be blocked, influencing the filtering effects. Replace the water inlet filter every 6 months.

Purpose

To replace the water inlet filter to ensure the good filtering effects.

When to do

You are recommended to perform this procedure every 6 months.

Materials required

New water inlet filter

System status

Make sure that the system is powered off, or the system status is Incubation or Standby.

How to do

- 1 Check that the system is powered off, or the system status is Incubation or Standby.
- 2 Turn off the power switch of the water unit or other water supply equipment.
- 3 Prepare a new water inlet filter with connectors on its two ends.
- 4 Turn on the ball valve on the water supply module to release the remaining pressure. When the pressure gauge indicates 0, turn off the ball valve.
- 5 Press the tubing release button to remove the tubing from two ends of the old filter assembly.
- 6 Wash the tubing and insert them into the new filter. Make sure that the filter is installed in the same direction as the water flow.
- 7 Power on the water supply module, turn on the ball valve on it and wait for 5 minutes. When you see the water supply module is supplying water continuously which signifies the normal working of the module, turn off its ball valve. Ensure that the pressure gauge on the water supply module is about 0.25MPa.
- 8 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Six-Month**.
- 9 Mark the **Select** checkbox in the same row as **Replace Water Inlet Filter**.
- 10 Select **OK** to refresh the current date as the performance date.
- 11 Select **Log**, and then record comments and other important information for the procedure.

12 Select **OK** to save your input information.

16.9.3 Replace Reference Electrode(For ISE Module)

The reference electrode is a consumable and has a limited life span. When used for a long period or after measuring a large number of samples, it may have its performance degraded and should be replaced immediately. It will take about 5 minutes to perform this procedure.

Purpose

To replace the reference electrode to ensure the optimal measurement performance.

When to do

Replace the electrodes in the following conditions:

- when 10,000 ISE tests are performed, or the instrument is used for 6 months.
- when calibration is failed or quality control is abnormal as the result of degraded electrode performance.

Materials required

Reference electrode

System status

Make sure that the status of the ISE module is Standby or Failure.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles.

How to do

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Replace Electrode**.
- 3 Select reference electrode, and enter the lot number and expiration date. Select **Add** and then select **OK**.
- 4 Take out the insert from the reference electrode, and ensure no crystallized salt exists in and around it. If needed, clean the electrode with warm water. Meanwhile make sure the red ball of the reference electrode floats on the internal fluid. Make sure the O rings of all electrodes remain intact.
- 5 Select **Continue**.

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- 6 Open the ISE side door and remove the cover of the shielding box.
- 7 Open the electrode case, take out the electrode, remove the tapes around its inside, and then use clean tissue to wipe it.
- 8 Remove all electrodes from the ISE module.
- 9 Replace the old electrode and then press the compressor; place the reference electrode at the bottom of the ISE module and make the rear part of the electrode contact closely with the internal wall of the ISE module. Loosen the compressor and ensure the electrodes are fixed tightly.
- 10 Restore other electrodes in the order of Cl, K, Na and spacer from bottom to top.
- 11 Select **Continue**.



NOTE

The new electrode can be calibrated successfully only after certain time period (at least 30 minutes).

- 12 Select **Done**.
- 13 Restore the cover of the shielding box and close the side door of the ISE module.
- 14 Run ISE calibration.
- 15 Select **Scheduled Maintenance-Other**.
- 16 Mark the **Select** checkbox in the same row as **Replace Reference Electrode**.
- 17 Select **OK** to refresh the current date as the performance date.
- 18 Select **Log**, and then record comments and other important information for the procedure.
- 19 Select **OK** to save your input information.

16.10 As-Needed/As-Required Maintenance

16.10.1 Clean Analyzer Panels

The analyzer and computer are often accessed and easily get dirty. To keep a good operating environment and minimize the biohazards, clean the components that are often accessed, such as analyzer panel, carousel cover, screen, keyboard, etc.

Purpose

To clean the analyzer panels, carousel covers, screen and keyboard.

When to do

Perform this procedure when dust or other stains are found on the components.

Materials required

Clean gauze, screen wash solution, and deionized water

System status

Make sure that the system status is not Running.

Precautions



Warning

Do not spill liquid on the analyzer. Liquid ingress may cause equipment damage.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process. Dispose of the used gauze in accordance with your local or national guidelines for biohazard waste disposal.

How to do

- 1 Make sure that the system is not running tests, and then open the protective shield.
- 2 Use clean gauze moistened with ethanol to clean the analyzer panels and carousel covers.
- 3 Use wash solution to clean the screen and keyboard.
- 4 Restore the protective shield.
- 5 Select **Scheduled Maintenance-Other**.

- 6 Mark the **Select** checkbox in the same row as **Clean Analyzer Panels**.
- 7 Select **OK** to refresh the current date as the performance date.
- 8 Select **Log**, and then record comments and other important information for the procedure.
- 9 Select **OK** to save your input information.

16.10.2 Clean Sample Compartment

When samples are sprayed into the sample compartment, or dusts accumulate inside the compartment, clean them immediately in order to minimize the risks of cross contamination.

Purpose

To clean the sample carousel assembly to ensure clear operating environment and eliminate the risks of cross contamination.

When to do

Perform this procedure when samples are spilled into the sample compartment or dust is found inside of it.

Materials required

Clean gauze, deionized water, ethanol, and cotton swabs

System status

Make sure that the system status is Stopped or Standby.

Precautions



Warning

Do not spill water or ethanol into the sample compartment to prevent equipment damage.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process. Dispose of the used gauze in accordance with your local or national guidelines for biohazard waste disposal.

How to do

- 1 Make sure that the system is in Stopped or Standby status.
- 2 Remove the sample carousel cover and sample carousel, and then store them

properly.

- 3 Use clean gauze soaked with deionized water or ethanol to clean the interior of the sample compartment. If necessary, you can use gauze moistened with neutral wash solution.
- 4 Use clean gauze soaked with deionized water or ethanol to clean the sample carousel, and then use cotton swabs dipped with ethanol to clean the sample positions.
- 5 Install the sample carousel and the carousel cover.
- 6 Select **Scheduled Maintenance-Other**.
- 7 Mark the **Select** checkbox in the same row as **Clean Sample Compartment**.
- 8 Select **OK** to refresh the current date as the performance date.
- 9 Select **Log**, and then record comments and other important information for the procedure.
- 10 Select **OK** to save your input information.

16.10.3 Clean Reagent Compartment

When reagents are sprayed into the reagent compartment, or dusts accumulate inside the compartment, clean them immediately in order to minimize the risks of cross contamination.

Purpose

To clean the reagent carousel assembly to ensure clear operating environment and eliminate the risks of cross contamination.

When to do

Perform this procedure when reagents are spilled into the reagent compartment or dust is found inside of it.

Materials required

Clean gauze, deionized water, ethanol, and cotton swabs

System status

Make sure that the system status is Stopped or Standby.

Precautions



Warning

Do not spill water or ethanol into the reagent compartment to prevent equipment damage.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process. Dispose of the used gauze in accordance with your local or national guidelines for biohazard waste disposal.

How to do

- 1 Make sure that the system is in Stopped or Standby status.
- 2 Remove the reagent carousel cover and reagent carousel, and then store them properly.
- 3 Use clean gauze soaked with deionized water or ethanol to clean the interior of the reagent compartment. If necessary, you can use gauze moistened with neutral wash solution.
- 4 Use clean gauze soaked with deionized water or ethanol to clean the reagent carousel, and then use cotton swabs dipped with ethanol to clean the reagent positions.
- 5 Install the reagent carousel and the carousel cover.
- 6 Select **Scheduled Maintenance-Other**.
- 7 Mark the **Select** checkbox in the same row as **Clean Reagent Compartment**.
- 8 Select **OK** to refresh the current date as the performance date.
- 9 Select **Log**, and then record comments and other important information for the procedure.
- 10 Select **OK** to save your input information.

16.10.4 Clean Sample Probe Interior

The sample probe, once blocked, cannot aspirate or dispense sample correctly. When

you find that the sample probe is clogged and cannot aspirate or dispense sample, or when the sample probe is detected with abnormal liquid flow through the Check Probes/Mixers maintenance, perform this procedure to solve the problems.

Purpose

To clean the interior of the sample probe and avoid clogging.

When to do

Perform this procedure when you find that the sample probe is clogged and cannot aspirate or dispense sample, or when the sample probe is detected with abnormal liquid flow through the Check Probes/Mixers maintenance.

Materials required

Unclogging device, small slot-head screwdriver, small Philips-head screwdriver, beaker, deionized water, and thread syringe

System status

Make sure that the system status is Standby or Failure.

Precautions**BIOHAZARD**

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

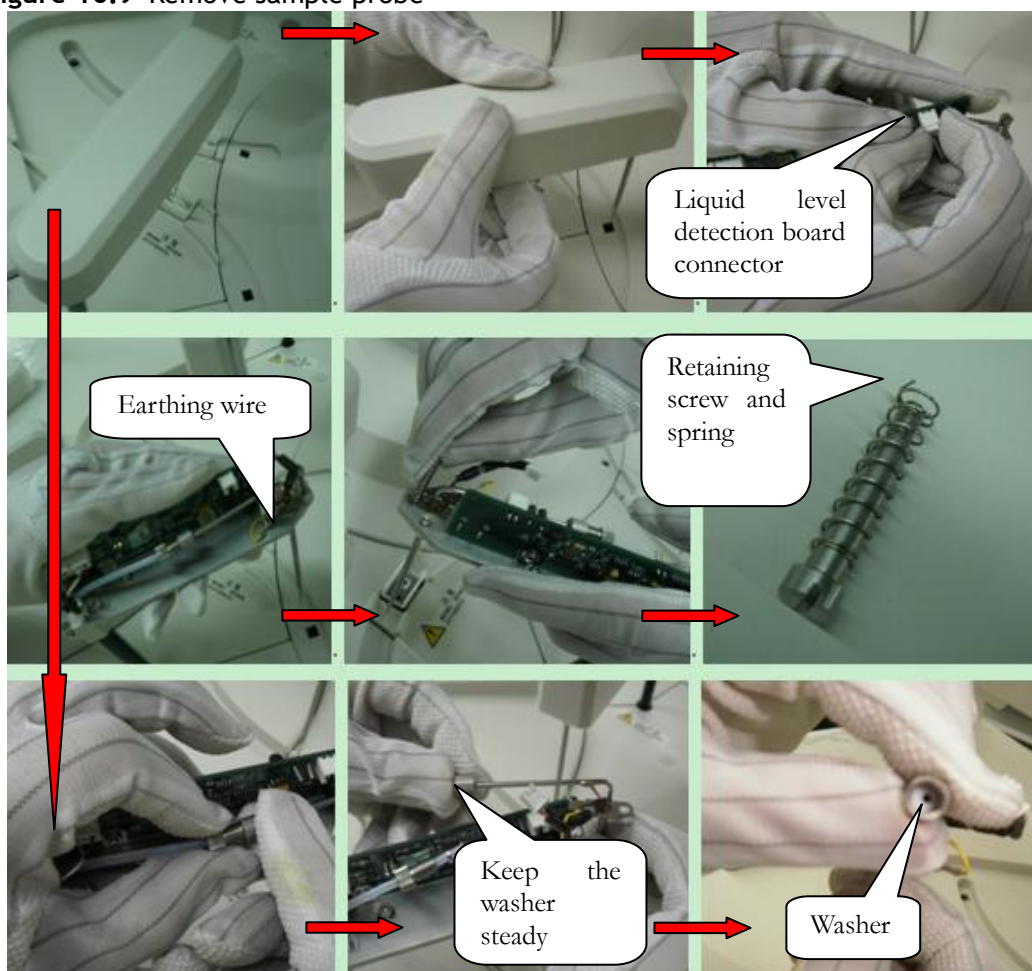
- 1** Recall the maintenance logs and check if the sample probe has been removed and reinstalled for 3 times. If it has, prepare a new washer and moisten it with deionized water. Store the washer properly to avoid being lost.
- 2** Switch off the analyzing unit power.
- 3** Grab the lower parts of the arm cover and pull them slightly from the opposite directions; remove the cover from the arm base.
- 4** Press the circuit board with one hand and unplug the tube connector with the other hand, and then use a small slot-head screwdriver to loosen the earthing wire on the sample probe.
- 5** Use a small screwdriver to remove the retaining screw from the sample probe and take out the spring.
- 6** While holding the connector on the sample probe with one hand, unscrew the

tube connector counterclockwise with the other hand until the tube connector is disconnected. Remove the tube from the sample probe.

Exercise caution to prevent the washer from dropping out. If the washer drops out, store it in a clear place for later installation. To replace the washer, take it out from the tube connector.

- 7 Remove the sample probe. See the figure below.

Figure 16.9 Remove sample probe



- 8 Connect the unclogging device to the sample probe, fill the syringe with deionized water and then connect it to the unclogging device. Put the sample probe inside the beaker while keeping the probe tip not contacting the beaker. Push the syringe to rinse the interior of the sample probe. Repeat this step for 10 times.

If the syringe plunger leaks and the sample probe cannot be unclogged due to serious blockage, replace the sample probe.

- 9 When continuous water flow comes out of the sample probe in the same direction with the probe, it indicates the cleaning procedure is finished successfully. Remove the unclogging device.
- 10 Insert the sample probe downwards into the hole on the probe arm while aligning the screw hole on the probe plate to the rod inside the arm.
- 11 To replace the washer, remove the old one from the tube connector and install the new one. Connect the tube connector to the sample probe and then tighten it.
- 12 Fix the earthing wire of the sample probe to the earthing terminal inside the arm; connect the probe connector to the liquid level detection board.
- 13 Sleeve the spring on the rod and tighten the retaining screw. Pay attention to the spring direction and make the thread opening face downwards.
- 14 Pinch the sample probe by the part near the probe arm. Push the sample probe upwards and then release it to check if the spring works well.
 - If it does, proceed to the next step.
 - If not, check if the spring is clamped or fixed too tightly.
- 15 Switch on the analyzing unit power, and then check if the No.D2 LED indicator on the circuit board inside the probe arm is lit.
 - If it is, the liquid level detection system is normal.
 - If not, contact our customer service department or your local distributor.
- 16 Install the probe arm cover properly until you hear a click.
- 17 Pinch the sample probe by the part near the probe arm. Push the sample probe upwards and then release it to check if the spring works well.
 - If it does, proceed to the next step.
 - If not, it indicates that the arm cover is not installed correctly. Reinstall the arm cover and check the spring until it can move freely.
- 18 Execute the Home maintenance command. Check if the water flow coming out

of the sample probe is continuous and in the same direction as the probe. If it is not, perform the Check Probes/Mixers procedure to troubleshoot the problems.

- 19 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 20 Mark the **Select** checkbox to the right of **Clean Sample Probe Interior**.
- 21 Select **OK** to refresh the current date as the performance date.
- 22 Select **Log**, and then record comments and other important information for the procedure.
- 23 Select **OK** to save your input information.
- 24 Select **Utility-Commands**, and then select **Home** to put the instrument into the Standby status.

16.10.5 Clean Reagent Probe Interior

The reagent probe, once blocked, cannot aspirate or dispense reagent correctly. It is necessary to clean the reagent probe interior at times.

Purpose

To clean the interior of the reagent probe and avoid clogging.

When to do

Perform this procedure when you find that a reagent probe is clogged and cannot aspirate or dispense sample, or when a reagent probe is detected with abnormal liquid flow through the Check Probes/Mixers maintenance.

Materials required

Unclogging device, small slot-head screwdriver, small Philips-head screwdriver, beaker, deionized water, and thread syringe

System status

Make sure that the system status is Standby or Failure.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Recall the maintenance logs and check if the reagent probe has been removed and reinstalled for 3 times. If it has, prepare a new washer and moisten it with deionized water. Store the washer properly to avoid being lost.
- 2 Switch off the analyzing unit power.
- 3 Clean the reagent probe by referring to step 3 to 18 in 16.10.4 Clean Sample Probe Interior.
- 4 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 5 Mark the **Select** checkbox to the right of **Clean Reagent Probe Interior**.
- 6 Select **OK** to refresh the current date as the performance date.
- 7 Select **Log**, and then record comments and other important information for the procedure.
- 8 Select **OK** to save your input information.
- 9 Select **Utility-Commands**, and then select **Home** to put the instrument into the Standby status.

16.10.6 Replace Sample Probe

Replace the sample probe when it is damaged and cannot be repaired, or blocked seriously, or bent.

Purpose

To replace the sample probe.

When to do

Perform this procedure when the sample probe is damaged and cannot be repaired due to the following causes, such as serious blockage, or bending.

Materials required

Small slot-head screwdriver, small Philips-head screwdriver, and new sample probe

System status

Make sure that the system status is Standby or Failure.

Precautions



Warning

The probe tip is sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the probes.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

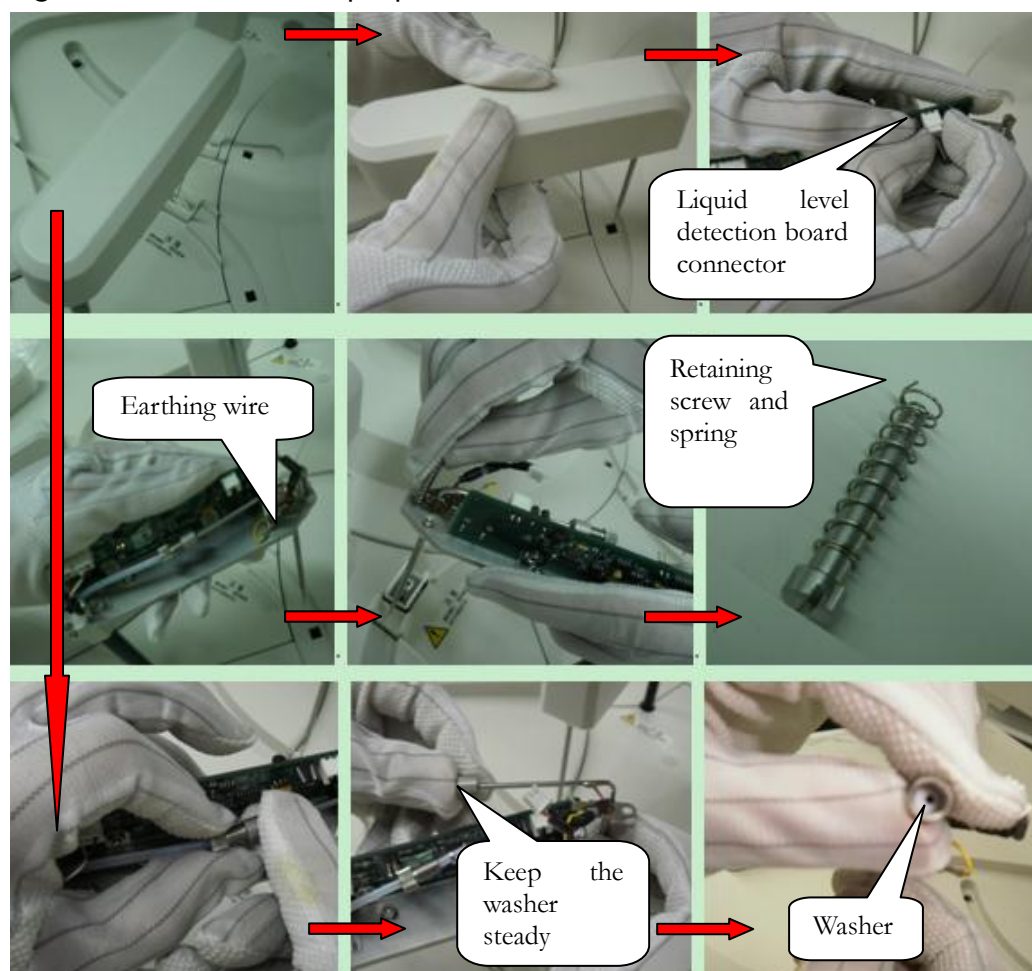
How to do

- 1** Prepare the new sample probe. Recall the maintenance logs and check if the sample probe has been removed and reinstalled for 3 times. If it has, prepare a new washer and moisten it with deionized water. Store the washer properly to avoid being lost.
- 2** Switch off the analyzing unit power.
- 3** Grab the lower parts of the arm cover and pull them slightly from the opposite directions; remove the cover from the arm base.
- 4** Press the circuit board with one hand and unplug the tube connector with the other hand, and then use a small slot-head screwdriver to loosen the earthing wire on the sample probe.
- 5** Use a small screwdriver to remove the retaining screw from the sample probe and take out the spring.
- 6** While holding the connector on the sample probe with one hand, unscrew the tube connector counterclockwise with the other hand until the tube connector is disconnected. Remove the tube from the sample probe.

Exercise caution to prevent the washer from dropping out. If the washer drops out, store it in a clear place for later installation. To replace the washer, take it out from the tube connector.

- 7** Remove the sample probe. See the figure below.

Figure 16.10 Remove sample probe



- 8 Insert the sample probe downwards into the hole on the probe arm while aligning the screw hole on the probe plate to the rod inside the arm.
- 9 To replace the washer, remove the old one from the tube connector and install the new one. Connect the tube connector to the sample probe and then tighten it.
- 10 Fix the earthing wire of the sample probe to the earthing terminal inside the arm; connect the probe connector to the liquid level detection board.
- 11 Sleeve the spring on the rod and tighten the retaining screw. Pay attention to the spring direction and make the thread opening face downwards.
- 12 Pinch the sample probe by the part near the probe arm. Push the sample probe

upwards and then release it to check if the spring works well.

- If it does, proceed to the next step.
- If not, check if the spring is clamped or fixed too tightly.

13 Switch on the analyzing unit power, and then check if the No.D2 LED indicator on the circuit board inside the probe arm is lit.

- If it is, the liquid level detection system is normal.
- If not, contact our customer service department or your local distributor.

14 Install the probe arm cover properly until you hear a click.

15 Execute the Home maintenance command. Check if the water flow coming out of the sample probe is continuous and in the same direction as the probe. If it is not, perform the Check Probes/Mixers procedure to troubleshoot the problems.

16 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.

17 Mark the **Select** checkbox in the same row as **Replace Sample Probe**.

18 Select **OK** to refresh the current date as the performance date.

19 Select **Log**, and then record comments and other important information for the procedure.

20 Select **OK** to save your input information.

21 Select **Utility-Commands**, and then select **Home** to put the instrument into the Standby status.

16.10.7 Replace Reagent Probe

Replace the reagent probe when they are damaged and cannot be repaired, or blocked seriously, or bent.

Purpose

To replace the reagent probe.

Materials required

Small slot-head screwdriver, small Philips-head screwdriver, and new reagent probe

System status

Make sure that the system status is Standby or Failure.

Precautions



Warning

The probe tip is sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the probes.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Prepare the new reagent probe. Recall the maintenance logs and check if the reagent probe has been removed and reinstalled for 3 times. If it has, prepare a new washer and moisten it with deionized water. Store the washer properly to avoid being lost.
- 2 Switch off the analyzing unit power.
- 3 Replace the reagent probe by referring to step 3 to 15 in 16.10.6 Replace Sample Probe (page 16-67).
- 4 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 5 Mark the **Select** checkbox to the right of **Replace Reagent Probe**.
- 6 Select **OK** to refresh the current date as the performance date.
- 7 Select **Log**, and then record comments and other important information for the procedure.
- 8 Select **OK** to save your input information.
- 9 Select **Utility-Commands**, and then select **Home** to put the instrument into the Standby status.

16.10.8 Replace Sample Mixer

Replace the sample mixer when they are bent or damaged and cannot be repaired.

Purpose

Replace the sample mixer.

When to do

Perform this procedure when the sample mixer are damaged and cannot be repaired.

Materials required

Ethanol, clean gauze, new sample mixer

System status

Make sure that the system status is Standby or Failure.

Precautions



Warning

The mixer tips are sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the mixers.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Switch off the analyzing unit power.
- 2 Gently pull the mixer to its highest point and rotate it to a position convenient to operate.
- 3 Pinch the mixer by the knurled part with one hand and unscrew (counter-clockwise) the retaining nut with the other hand until the mixer gets loose. Pull the mixer downward to remove it and remove the nut.



CAUTION

When trying to pull out the mixer, concentrate your force in the direction of the axis on the mixer arm. Biased force may damage the mixer and/or the axis.

- 4 Align the new mixer to the bigger hole of the retaining nut and gently screw it into the nut until the end of the mixer is in line with the smaller hole of the nut.
- 5 Pinch the mixer by the knurled part and align the hole of the nut to the axis on

the mixer and push the nut onto the mixer until it reaches the end of the mixer. Tighten the nut by screwing clockwise with the other hand.



CAUTION

When trying to push the mixer into the arm, concentrate your force in the direction of the axis on the mixer arm. Biased force may damage the mixer and/or the axis.

Ensure the mixer is all the way pushed to the end.

- 6 After replacing the bar, visually check whether the mixer is vertical to the bar arm.
 - If not, remove the mixer and reinstall it.
 - If so, proceed to the next step.
- 7 Pull the mixer arm to its highest point and rotate it back to a position above its wash well.
- 8 Turn on the analyzing unit power switch.
- 9 Select **Utility- Command**; Perform the **Home** maintenance procedure.
- 10 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 11 Mark the **Select** checkbox in the same row as **Replace Sample mixer**.
- 12 Select **OK** to refresh the current date as the performance date.
- 13 Select **Log**, and then record comments and other important information for the procedure.
- 14 Select **OK** to save your input information.

16.10.9 Replace Reagent Mixer

Replace the reagent mixer when they are bent or damaged and cannot be repaired.

Purpose

Replace the reagent mixer.

When to do

Perform this procedure when the reagent mixer are damaged and cannot be repaired.

Materials required

Ethanol, clean gauze, new reagent mixer

System status

Make sure that the system status is Standby or Failure.

Precautions



Warning

The mixer tips are sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the mixers.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Switch off the analyzing unit power.
- 2 Replace the reagent mixer by referring to step 2 to 9 in 16.10.9 Replace Reagent Mixer (page 16-73).
- 3 Select **Utility-Command-Home**.
- 4 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 5 Mark the **Select** checkbox in the same row as **Replace Reagent mixer**.
- 6 Select **OK** to refresh the current date as the performance date.
- 7 Select **Log**, and then record comments and other important information for the procedure.
- 8 Select **OK** to save your input information.

16.10.10 Remove Air Bubbles in Sample Syringe

Purpose

To remove the air bubbles possibly existing inside the tubes and clean/prime the probes, mixers and wash wells. It will take about 20 seconds to perform this procedure.

When to do

Perform this procedure when you find air bubbles inside the sample syringe.

Materials required

Concentrated wash solution

System status

Make sure that the system status is Standby.

Precautions**BIOHAZARD**

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1** Switch off the analyzing unit power, and open the front door of the analyzer.
- 2** Loosen counterclockwise the four retaining screws on top of the syringe, and then remove the screws and the fixing blocks.
- 3** Loosen counterclockwise the retaining screw at the bottom of the syringe and then remove it.
- 4** Hold the T piece with one hand and the syringe connector with the other hand. Loosen the syringe counterclockwise and then remove the washer.
- 5** Soak the syringe connector in the deionized water beaker, pull the plunger head to aspirate half syringe of deionized water, and then push the plunger head to remove the air. Repeat this pull-push operation until the air bubbles are removed from the syringe. Fill the syringe with half cylinder of deionized water to prevent new bubbles.
- 6** Put the washer in the T piece. Hold the T piece with one hand and the syringe connector with the other hand, and then screw the T piece clockwise.
- 7** Install the syringe on the bracket.
- 8** Install the fixing blocks and 4 retaining screws while having the retaining screws not tightened.
- 9** Align the plunger head to the retaining screw at the bottom of the syringe, and then tighten clockwise the retaining screw.

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- 10 Pinch the plunger guide cap to adjust the syringe height. For the sample syringe, make the syringe head over the upper fixing block for 7.5 scales; for the reagent syringes, make the syringe head over the upper fixing block for 15 scales.
- 11 Tighten the four retaining screws on the fixing blocks.
- 12 Turn on the analyzing unit power switch.
- 13 Perform the **Home** maintenance procedure. Check the new syringe for leak and bubbles, and if there is, perform the Check Sample/Reagent Syringes procedure.
- 14 Close the front door of the analyzer.
- 15 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 16 Mark the **Select** checkbox in the same row as **Remove Air Bubbles in Sample Syringe**.
- 17 Select **OK** to refresh the current date as the performance date.
- 18 Select **Log**, and then record comments and other important information for the procedure.
- 19 Select **OK** to save your input information.

16.10.11 Remove Air Bubbles in Reagent Syringe

Purpose

To remove the air bubbles possibly existing inside the tubes and clean/prime the probes, mixers and wash wells. It will take about 20 seconds to perform this procedure.

When to do

Perform this procedure when you find air bubbles inside the reagent syringe.

Materials required

Concentrated wash solution

System status

Make sure that the system status is Standby.

Precautions

**BIOHAZARD**

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Switch off the analyzing unit power, and open the front door of the analyzer.
- 2 Remove the air bubbles in reagent syringe by referring to step 2 to 14 in 16.10.10 Remove Air Bubbles in Sample Syringe (page 16-74).
- 3 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 4 Mark the **Select** checkbox in the same row as **Remove Air Bubbles in Reagent Syringe**.
- 5 Select **OK** to refresh the current date as the performance date.
- 6 Select **Log**, and then record comments and other important information for the procedure.
- 7 Select **OK** to save your input information.

16.10.12 Replace Cuvette

The reaction cuvettes, if contaminated by serum or other stains, or scratched or damaged, will result in inaccurate photometric measurement. Check the reaction cuvettes regularly, and if necessary, replace them immediately. It will take about half a minute to replace a cuvette.

Purpose

To ensure that the cuvettes are normal and not contaminated, scratched or damaged.

When to do

Replacing cuvettes is performed as needed or as required. Replace a cuvette if,

- it is detected abnormal through the Cuvette Check procedure; or
- scratches or cracks are found on the optical surface of the cuvette.

Materials required

Fiber-free gloves, dry cloth or gauze, reaction cuvettes, and concentrated wash solution manufactured by our company

System status

Make sure that the system status is Standby.

Precautions



Warning

While installing the reaction cuvettes, exercise caution to avoid scratching them. Do not touch the optical surface of the reaction cuvettes. If the optical surface is polluted, the obtained absorbance may be inaccurate.

While installing the reaction cuvettes, make sure that the optical surface is confronted with the outside of the reaction carousel.

Wear gloves free of fibre and powder to avoid polluting the optical surface of the reaction cuvettes.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.



CAUTION

Please use consumables recommended by our company. Use of other consumables may degrade the system performance.

**NOTE**

If a cuvette cannot be removed from the reaction carousel, remove 1 or 2 cuvettes to the right of the cuvette, use a knife to remove the metal plate next to it, and then use your hands or tweezers to take out the cuvette.

When serious problems occur such as overflow and require the reaction cuvettes to be maintained, contact our customer service department or your local distributor.

How to do

- 1 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 2 Choose **Replace Cuvette**.
- 3 Select **Continue**.
- 4 Remove the reaction carousel cover.
- 5 Type in the position number of the cuvette you want to replace.

The input range is 1-90. Only one position number can be entered each time.

- 6 Select **Replace**.
- 7 Wear a pair of gloves and remove the specified cuvette by pulling it outwards.
- 8 Install the provided or cleaned cuvette to the reaction carousel and make sure that the cuvette bottom can no longer proceed.
- 9 Restore the reaction carousel cover.
- 10 Select **Done**. The system resets mechanically.
- 11 Perform the Cuvette Check procedure to check if the new cuvettes meet the requirements.

For more information, refer to 16.6.4 Cuvette Check (page 16-33).

- 12 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 13 Mark the **Select** checkbox in the same row as **Replace Cuvette**.
- 14 Select **OK** to refresh the current date as the performance date.
- 15 Select **Log**, and then record comments and other important information for the

procedure.

- 16 Select **OK** to save your input information.

16.10.13 Special Wash Probes

Purpose

To eliminate cross contamination among the sample probe and reagent probe, and prevent waste from leaving in the waste tubes.

When to do

Perform this procedure when the probes are clogged or the carryover result exceeds the limit.

Materials required

Concentrated wash solution

System status

Make sure that the system status is Standby.

How to do

- 1 Open the upper protective shield of the analyzer.
- 2 Place more than 20ml concentrated wash solution in position D of the reagent carousel, and place more than 3ml concentrated wash solution in position D2 on the sample carousel.
- 3 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 4 Choose **Special Wash Probes**.
- 5 Select Special Wash Reagent Probe and Special Wash Sample Probe, and then select **Continue**.

The system resets and then clean the two probes.

- 6 When the cleaning is finished, select **Done**.
- 7 Restore the upper protective shield of the analyzer.
- 8 Select **Scheduled Maintenance-Other**.
- 9 Mark the **Select** checkbox in the same row as **Special Wash Probes**.
- 10 Select **OK** to refresh the current date as the performance date.

- 11 Select **Log**, and then record comments and other important information for the procedure.
- 12 Select **OK** to save your input information.

16.10.14 Bar Code Maintenance

This maintenance procedure is used to clean the sample and reagent bar code scanning windows in order to avoid influencing bar code scanning.

Purpose

To clean the glass of the sample and reagent bar code scanning windows in order to avoid influencing bar code scanning.

When to do

This maintenance should be performed if the glass of the sample or reagent bar code scanning window is contaminated and causes bar code scanning failure.

Materials required

Clean gauze, deionized water, ethanol, and cotton swabs

System status

Make sure that the system is not running any tests.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.



CAUTION

Exercise caution not to spray water or ethanol or other liquids on the glass of the bar code scanning window.

How to do

- 1 Remove the sample/reagent carousel covers and the carousels.
- 2 Use clean gauze to clean the bar code reader window inside the sample compartment and reagent compartment. If necessary, you can use gauze soaked with ethanol or deionized water. Make sure that there is no trace or dust left on the glass.
- 3 Install the carousels and carousel covers.

- 4 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 5 Mark the **Select** checkbox in the same row as **Bar Code Maint**.
- 6 Select **OK** to refresh the current date as the performance date.
- 7 Select **Log**, and then record comments and other important information for the procedure.
- 8 Select **OK** to save your input information.

16.10.15 Clean Probes/Mixers Exterior

It is necessary to clean the probes and mixers exterior after maintaining them as well as the wash wells, to remove air bubbles from the tubes.

Purpose

To clean the exterior of the probes and mixers to remove air bubbles.

When to do

This procedure should be performed after maintenance of probes, mixers and wash wells.

System status

Make sure that the system status is not Running.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Select **Utility-Maintenance-Maintenance- Biochemistry Maintenance**.
- 2 Choose **Clean Probes/Mixers Exterior**.
- 3 Input the wash cycle (1~100).
- 4 Select **Continue**.
- 5 Select **Done**.

16.10.16 Replace ISE Electrode

ISE electrodes are consumables and have a limited life span. When used for a long period or after measuring a large number of samples, the ISE electrodes may have their performance degraded and should be replaced immediately. It will take about 10 minutes to perform this procedure.

Purpose

To replace the ISE electrodes to ensure the optimal measurement performance.

When to do

Replace the electrodes in the following conditions:

- when 10,000 ISE tests are performed, or the instrument is used for 6 months.
- when calibration is failed or quality control is abnormal as the result of degraded electrode performance.

Materials required

ISE electrode

System status

Make sure that the status of the ISE module is Standby or Failure.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles.



NOTE

After performing this procedure, recalibrate the ISE electrodes prior to starting analysis.

How to do

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Replace Electrode**.
- 3 Select desired electrodes, and enter the lot number and expiration date. Select **Add** and then select **OK**.
- 4 Select **Continue**.
- 5 Open the ISE side door and remove the cover of the shielding box.
- 6 Open the electrode case, take out the electrode, remove the tapes around its

inside, and then use clean tissue to wipe it.



NOTE

Take out the insert from the reference electrode, and ensure no crystallized salt exists in and around it. If needed, clean the electrode with warm water. Meanwhile make sure the red ball of the reference electrode floats on the internal fluid. Make sure the O rings of all electrodes remain intact.

- 7 Remove all electrodes from the ISE module.
- 8 Replace the old electrode and then press the compressor; place the reference electrode at the bottom of the ISE module and make the rear part of the electrode contact closely with the internal wall of the ISE module. Loosen the compressor and ensure the electrodes are fixed tightly.
- 9 Restore other electrodes in the order of Cl, K, Na and spacer from bottom to top.
- 10 Select **Continue**.



NOTE

The new electrode can be calibrated successfully only after certain time period (at least 30 minutes). Some electrodes need to be activated by serum before calibration.

- 11 Select **Done**.



NOTE

The Na, K and Cl electrodes are of the same size and shape. Ensure that the electrodes are inserted in the correct order. If one of the electrodes cannot be easily pushed into the housing, check the electrode first and then repeat the installation process.

- 12 Restore the cover of the shielding box and close the side door of the ISE module.
- 13 Run ISE calibration.
- 14 Select **Scheduled Maintenance-Other**.
- 15 Mark the **Select** checkbox in the same row as **Replace ISE Electrode**.
- 16 Select **OK** to refresh the current date as the performance date.

- 17 Select **Log**, and then record comments and other important information for the procedure.
- 18 Select **OK** to save your input information.

16.10.17 Remove Reagent Pack (For ISE Module)

When powering off the analyzer for a long time, or storing the electrodes, or replacing the electrode tubes, remove the reagent pack first.

When to do

Perform this maintenance procedure when powering off the analyzer for a long time, or storing the electrodes, or replacing the electrode tubes.

System status

Make sure that the status of the ISE module is Standby or Failure.

How to do

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Remove Reagent Pack**.
- 3 Remove the tube of pump A and then reinstall the tube by switching the connectors of the tube. Handle pump B in the same way.
- 4 Select **Continue**.
- 5 Install the tubes of pump A and B in the reverse direction and then select **Continue**.

The system executes purge A and B each for 30 times.
- 6 Select **Continue**.
- 7 Select **Done**.
- 8 Select **Scheduled Maintenance-Other**.
- 9 Mark the **Select** checkbox in the same row as **Remove Reagent Pack**.
- 10 Select **OK** to refresh the current date as the performance date.
- 11 Select **Log**, and then record comments and other important information for the procedure.

12 Select **OK** to save your input information.

16.10.18 Store Electrodes (For ISE Module)

While the analyzer is powered off for a long time, the ISE electrodes cannot be moistened by regular prime, and may be damaged due to lack of water. It is necessary to store the electrodes properly before powering off the analyzer for a long period.

Purpose

To store the electrodes separately to prevent them from being damaged due to lack of water while the analyzer is powered off.

Materials required

Electrode cases and tapes

When to do

Perform this procedure when the analyzer is going to be powered off for over 3 days. If it will be powered off for no more than 3 days, prime the ISE electrodes to protect them from being damaged.

System status

Make sure that the status of the ISE module is Standby or Failure.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Remove the reagent pack according to 16.10.17 Remove Reagent Pack (Page 16-85).
- 2 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 3 Choose **Store Electrodes**.
- 4 Select **Continue**.
- 5 Open the ISE side door and remove the cover of the shielding box.
- 6 Remove all electrodes from the ISE module.

**NOTE**

Reference electrode: Put back the insert to the cell of the reference electrode and prevent the crystallized salt from clogging the cell. And then store the electrode in an electrode case at the room temperature in a sun-shielding place.

Na, K, Cl electrode: Take out a little calibrator A from the reagent pack, inject it into the cell of the electrode and seal it with tape. Make sure proper amount of calibrator is injected into the cell of the electrode. Store the capped electrodes in an electrode case at the room temperature in a sun-shielding place.

Note the storage temperature should be below 40°C.

- 7 Select **Continue**.
- 8 Restore the cover of the shielding box and close the side door of the ISE module.
- 9 Select **Done**.
- 10 Select **Scheduled Maintenance-Other**.
- 11 Mark the **Select** checkbox in the same row as **Store Electrodes**.
- 12 Select **OK** to refresh the current date as the performance date.
- 13 Select **Log**, and then record comments and other important information for the procedure.
- 14 Select **OK** to save your input information.

16.10.19 Two-Point Calibration (For ISE Module)

The Two-point Calibration option is used to calibrate the ISE module with calibrator A and B.

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Two-point Calibration**.

When the calibration is finished, the progress box is closed automatically, and the calibration results are displayed in the data list.

16.10.20 Maintenance (For ISE Module)

The Maintenance option is used during electrode replacement to discharge the calibrator from the inside of the electrodes. The auto prime for every 30 minutes will stop.

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Maintenance**.

The maintenance results are displayed in the data list.

16.10.21 Purge A (For ISE Module)

The Purge A option makes the peristaltic pump dispense 100µl calibrator A to the sample injection port and then to the inside of the electrodes.

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Purge A**.

The purge results are displayed in the data list.

16.10.22 Purge B (For ISE Module)

The Purge B option makes the peristaltic pump dispense 100µl calibrator B to the sample injection port and then to the inside of the electrodes.

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Purge B**.

The purge results are displayed in the data list.

16.10.23 Program Check Instruction (For ISE Module)

The Program Check Instruction option is used to query the software version of the ISE module.

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Program Check Instruction**.

The query results are displayed in the data list.

16.10.24 Read Dallas Chip (For ISE Module)

The Read Dallas Chip option is used to read information from the reagent pack chip.

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.

- 2 Choose **Read Dallas Chip**.

The read information is displayed in the data list.

16.10.25 Write Dallas Chip (For ISE Module)

The Read Dallas Chip option is used to write information to the reagent pack chip.

- 1 Select **Utility-Maintenance-Maintenance- ISE Maintenance**.
- 2 Choose **Write Dallas Chip**.

The written information is displayed in the data list.

16.10.26 Replace Sample Syringe

The sample syringe has a limited life span, and when due, may have leak or other phenomena causing inaccurate aspirating/dispensing and resulting in unreliable results.

Purpose

To replace the syringe plunger assembly to ensure optimal measuring performance.

When to do

Perform this procedure when the syringe is used for 100,000 times.

Materials required

Deionized water, beaker, and syringe plunger assembly

System status

Make sure that the system status is Standby.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Prepare a new syringe plunger assembly and washer, put the plunger head in the deionized water beaker to remove air from the syringe, and then moisten the washer in the deionized water.
- 2 Switch off the analyzing unit power.
- 3 Open the front door of the analyzer. You will see two syringes, reagent syringe

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to the left of the water tank and sample syringe to the right.

- 4 Loosen counterclockwise the four retaining screws on top of the syringe, and then remove the screws and the fixing blocks.
- 5 Loosen counterclockwise the retaining screw at the bottom of the syringe and then remove it.
- 6 Hold the T piece with one hand and the syringe connector with the other hand. Loosen the syringe counterclockwise and then remove the washer.
- 7 Loosen the plunger guide cap counterclockwise, hold the plunger head and pull it slightly to remove the plunger assembly from the syringe.
- 8 Insert the plunger head of the new plunger assembly into the bottom of the syringe, and then tighten the retaining screw to fix the plunger head.
- 9 Soak the new syringe connector in the deionized water beaker, pull the plunger head to aspirate half syringe of deionized water, and then push the plunger head to remove the air.
- 10 If there is no washer inside the T piece, put the new washer in the T piece. Hold the T piece with one hand and the syringe connector with the other hand, and then screw the T piece clockwise.
- 11 Install the syringe on the bracket.
- 12 Install the fixing blocks and 4 retaining screws while having the retaining screws not tightened.
- 13 Align the plunger head to the retaining screw at the bottom of the syringe, and then tighten clockwise the retaining screw.
- 14 Pinch the plunger guide cap to adjust the syringe height. For the sample syringe, make the syringe head over the upper fixing block for 7.5 scales; for the reagent syringes, make the syringe head over the upper fixing block for 15 scales.
- 15 Tighten the four retaining screws on the fixing blocks.
- 16 Turn on the analyzing unit power switch.

- 17 Perform the **Home** maintenance procedure. Check if the new syringe has leak. If it does, perform the Check Sample/Reagent Syringes procedure to check the syringe.
- 18 Close the front door of the analyzer.
- 19 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 20 Mark the **Select** checkbox to the right of **Replace Sample Syringe**.
- 21 Select **OK** to refresh the current date as the performance date.
- 22 Select **Log**, and then record comments and other important information for the procedure.
- 23 Select **OK** to save your input information.

16.10.27 Replace Reagent Syringe

The reagent syringe has a limited life span, and when due, may have leak or other phenomena causing inaccurate aspirating/dispensing and resulting in unreliable results.

Purpose

To replace the syringe plunger assembly to ensure optimal measuring performance.

When to do

Perform this procedure when the syringe is used for 300,000 times.

Materials required

Deionized water, beaker, and syringe plunger assembly

System status

Make sure that the system status is Standby.

Precautions



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.

How to do

- 1 Prepare a new syringe plunger assembly and washer, put the plunger head in the

deionized water beaker to remove air from the syringe, and then moisten the washer in the deionized water.

- 2 Switch off the analyzing unit power.
- 3 Replace the reagent syringe by referring to step 3 to 18 in 16.10.26 Replace Sample Syringe(page 16-89).
- 4 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 5 Mark the **Select** checkbox to the right of **Replace Reagent Syringe**.
- 6 Select **OK** to refresh the current date as the performance date.
- 7 Select **Log**, and then record comments and other important information for the procedure.
- 8 Select **OK** to save your input information.

16.10.28 Clean Rotors

Clean the rotors of the sample probe, reagent probe and mixers to eliminate noise and fraying.

Purpose

Clean the rotors of the probes and mixers to minimize noise and fraying due to movement in order to extend the service life.

When to do

This procedure should be performed when there are dirty substances or dust on the rotors.

Materials required

Clean gauze

System status

Make sure that the system status is Standby.

Precautions



Warning

The probe and mixer tip are sharp and may cause puncture wounds. To prevent injury, exercise caution when working around the probes and mixers.

**BIOHAZARD**

Wear gloves and lab coat, and if necessary, goggles during the maintenance process. Dispose of the used gauze in accordance with your local or national guidelines for biohazard waste disposal.

How to do

- 1 Switch off the analyzing unit power.
- 2 Pull the probe/mixer arm to the highest point, and then rotate the arm to move the probe/mixer to a position convenient to operate.
- 3 Wipe the rotor with clean gauze.
- 4 After finishing the cleaning, turn on the analyzing unit power switch.
- 5 Select **Utility-Commands-Home** to reset the probes and mixers.
- 6 Select **Utility-Maintenance-Maintenance**, and then select **Scheduled Maintenance-Other**.
- 7 Mark the **Select** checkbox in the same row as **Clean Rotors**.
- 8 Select **OK** to refresh the current date as the performance date.
- 9 Select **Log**, and then record comments and other important information for the procedure.
- 10 Select **OK** to save your input information.

16.10.29 Clean ISE Waste Tube

Samples containing insoluble substance like fibrin may accumulate in the ISE wand waste outlet after extended usage and clog the waste tube.

Purpose

Clean the waste tube of ISE module to prevent the sediment inside from clogging the tube.

When to do

When the alarm “No fluid in tube” or “Air in calibrator A” or “Air in calibrator B” occurs and the waste tube is clogged.

Materials required

16 Maintenance

Unclogging tool for the ISE waste tube, bleaching agent(Dilution Ratio 1:1) or 50% sodium hypochlorite and DI water

System status

Make sure that the system is not running test.

Precautions



Warning

Do not spill liquid on the analyzer. Liquid ingress may cause equipment damage.



BIOHAZARD

Wear gloves and lab coat, and if necessary, goggles during the maintenance process.



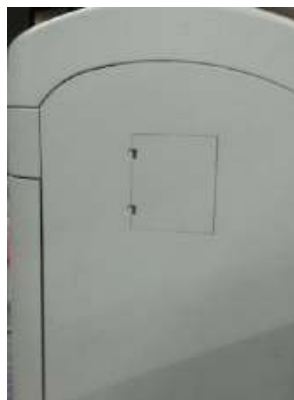
Note

Excessive bleach and DI water flushed into the ISE reagent pack waste bag may cause waste bag over expansion and clog the Cal A & Cal B reagent flow.

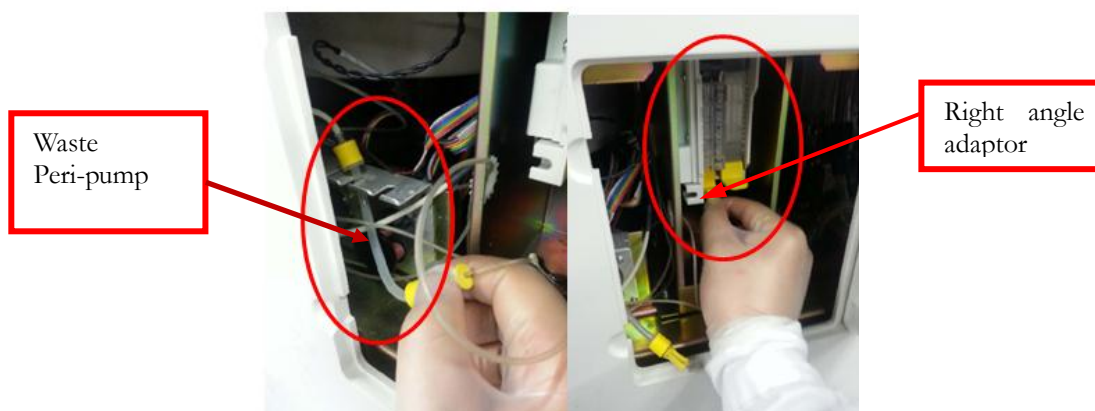
To prevent this problem, connect to an old used-up reagent pack and follow the procedures below.

How to do

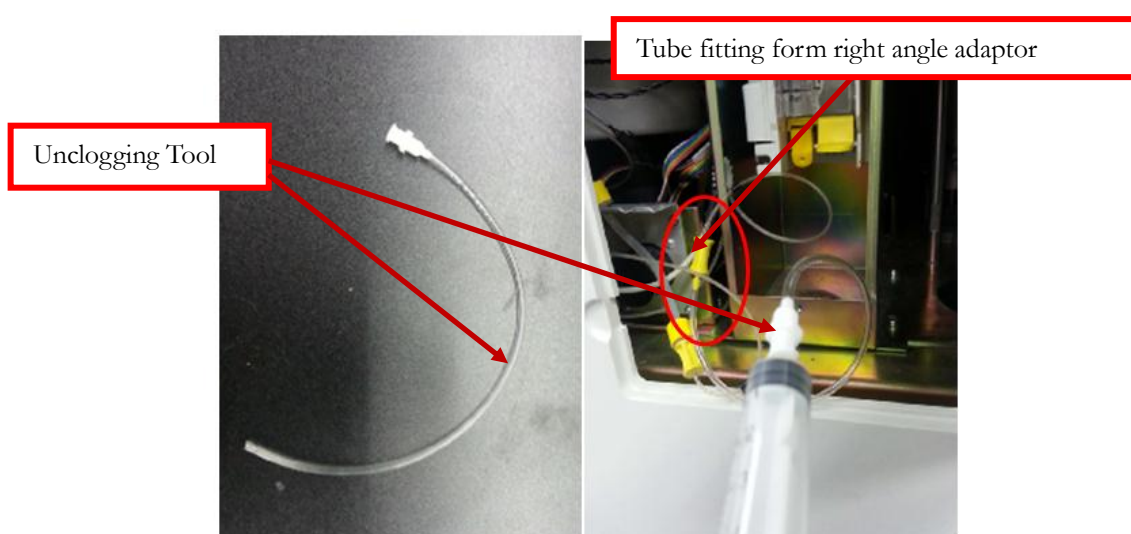
- 1 Ensure the analyzer is on idle (standby) condition. Open the ISE cover on the right side analyzer panel.



- 2 Remove the electrode housing cover. Remove the waste tube fitting from the bottom of the right angle adaptor. Remove waste peri-pump tube from the pump bracket. Refer to pictures below.



- 3 Connect the waste tube fitting to a syringe and unclogging tool with 5 mL of undiluted household bleach. Refer to pictures below.



- 4 Press the wand release button to remove the wand from the current in use ISE reagent pack and keep it in a safe place. Engage the wand to an old used-up reagent pack.
- 5 Inject bleach into the ISE waste tube and soak the tube for 5 minutes. Discharge the waste into the reagent pack.

Note: When the bleach cannot be injected into the ISE pack, remove the wand and push down to open the waste valve manually with a sharp object, and then inject again. If bleach can go through this time, the waste bag was clogged and cannot be used. If bleach still cannot be injected, replacing the ISE wand is recommended.

16 Maintenance



- 6** Repeat this step with 5 mL of DI water without the 5 minutes of soaking time.
- 7** Remove the wand from the old use-up pack and re-install it back to the current in use ISE pack. Re-install the waste tube fitting back to the ISE electrode housing right angle adaptor and waste peri-pump tube back to the pump bracket. Re-install the housing cover.
- 8** Calibrate ISE pump to ensure it passes, all reagents and waste flow are normal. Calibrate all electrodes and run QC to ensure they all pass within their given range.

17

Alarms and Troubleshooting

The following pages describe how to view and edit error logs and edit logs, and how to locate failure and determine relevant corrective actions. Read this chapter thoroughly to achieve the best performance of the instrument.

17.1 Classification of Logs

17.1.1 Introduction

The logs provided by the system are divided into:

- Error log
- Edit log

17.1.2 Error Logs

Error logs record all types of failures occurring on the components. The system stores the failures that occur within the latest 180 days and deletes those occurring beyond the latest 180 days.

Classification of failure

Failures are divided into the following types based on component, severity and processing method:

Table 17.1 Classification of failure based on component

No.	Failure by Component
1	Operating system
2	System communication
3	Database
4	Result calculation
5	Sample bar code
6	Reagent bar code
7	Host communication
8	Command execution
9	Sample probe unit
10	Reagent probe unit
11	Sample mixer unit
12	Reagent mixer unit
13	Reaction carousel unit
14	Sample carousel unit
15	Reagent carousel unit
16	Wash station
17	Temperature unit

No.	Failure by Component
18	ISE unit
19	Light source
20	Cuvette Wash Station
21	Other

Table 17.2 Classification of failure based on severity

ID	Failure by Severity	Description
1	Warning	Warning errors include: • Errors to warn user • Errors to invalidate tests • Errors to invalidate samples • Errors to invalidate reagents
2	Error	This type of errors includes serious errors other than warning errors.

Table 17.3 Classification of failure based on processing method

No.	Failure by Processing Method	Description
1	Errors to warn user	Such errors will not influence the system's running, and should be noticed.
2	Errors to invalidate tests	Such errors indicate that the current tests are invalidated due to the unqualified: • Tests • Reagents • Samples

No.	Failure by Processing Method	Description
3	Errors to pause	Such errors indicate that the failed component needs to be diagnosed and restored, while other components are not influenced. The errors may occur on the following components: • R1 aspirating and dispensing • R2 aspirating and dispensing • Sample carousel 1 • Sample carousel 2 • Sample probe • Sample mixer • Reagent mixer • Wash unit • Hydropneumatic assembly • ISE module
4	Errors to stop a component	Such errors indicate that the failed part cannot work normally and should be checked immediately. The components include: • ISE module • Reaction carousel
5	Errors to forbid a component	Such errors indicate that the failed unit is forbidden until it is restored. The components include: • LIS • Sample bar code module • Reagent bar code module
6	Errors to exit	When such error occurs, it indicates that the operating software cannot be started, or an error occurs during the startup. You are allowed to exit the operating software or restart it.

Error code

Each error has a unit code used for identification and locating probable causes and solutions. An error code consists of 6 letters and numbers, such as “C01001”, in which “C” indicates that the error occurs on the operation unit, “01” is the error description of instrument connection, and “001” is the serial number of the error. Therefore, “C01001” is described as “the first error of instrument connection on the operation unit”.

The following tables provide a summary of error codes for the operation unit and analyzing unit.

Table 17.4 Error code of the operation unit

Error Code	Description
C	Indicates that the error occurs on the operation unit.
00-99	Indicates the specific component on which the error occurs. • 00-Operating system • 01- System communication • 02-Database • 03-Result calculation • 04-Sample bar code • 05-Reagent bar code • 06-LIS host communication • 07-Other
000-999	Serial number of the error.

Table 17.5 Error code of the analyzing unit

Error Code	Description
A	Indicates that the error occurs on the analyzing unit.
00-99	Indicates the specific component on which the error occurs. • 00-Main control unit(Middle control unit) • 01-Sample probe unit • 02-Reagent probe unit • 04-Sample mixer • 05-Reagent mixer • 06-Reaction carousel unit • 07-Sample carousel unit (including sample bar code module) • 09-Reagent carousel unit (including reagent bar code module) • 11-Wash unit • 12-Temperature unit • 13-ISE module • 15-Other
000-999	Serial number of the error.

Help

Every error log is provided with online help information. Select the  icon in front of an error log. The descriptions, possible causes and solutions of the error are displayed.

17.1.3 Edit Logs

Edit logs record all deletions and part of editing actions performed by the user. The system stores the edit operations that are performed within the latest 1 year, and archives those occurring beyond the latest 1 year to an external file.

The delete logs record all deleting actions other than the error deletion.

The edit logs include editing of sample results and calibration factors.

17.2 Viewing and Handling Logs

All event logs are stored in folders named by the date when the logs are produced. The system automatically compresses the event logs on the previous day and then removes the relevant folder.

17.2.1 Description of Error Log Screen

Select **Alarm** in the function buttons area of the main screen. The **Error Log** screen is displayed by default and shows all errors occurring on the current day. On the **Error Log** screen you are allowed to view and handle all errors that occur within the latest 6 months.

Figure 17.1 Error Log screen



Every error log contains the event ID, date/time, error description (by processing method), event class (by subsystem) and symptom.

Choose the following buttons as needed:

- **Search F1:** to search for error logs by date, event ID, symptom, or event class.
- **Refresh F2:** to refresh the error logs based on the current search conditions.
- **Delete F3:** to remove specified error logs on the screen.
- **Print F7:** to print all error logs currently displayed on the screen.

17.2.2 Description of Edit Log Screen

Select **Alarm-Edit Log**. The **Edit Log** screen is displayed and shows all editing actions occurring on the current day. On the **Edit Log** screen you are allowed to view and handle all deleting/editing actions that occur within the latest 1 year.

Figure 17.2 Edit Log screen



Every edit log contains the serial number, date/time, operator, event type and description.

Choose the following buttons as needed:

- **Search F1:** to search for edit logs based on the occurring date.
- **Refresh F2:** to refresh the edit logs based on the current search conditions.
- **Delete F3:** to remove specified edit logs on the screen.
- **Print F7:** to print all edit logs currently displayed on the screen.

17.2.3 Recalling Logs

Error logs and edit logs can be recalled by all users in any system status. Error logs can be recalled by date, event ID, symptom and event class, while edit logs can only be recalled by occurring date.

Perform the following steps to recall desired event logs:

- 1 Select **Alarm-Error Log** or **Edit Log**.

- 2 Select **Search F1**.
- 3 Enter one or more of the following conditions:
 - Date
 - Event ID (available for error logs only)
 - Symptom (available for error logs only)
 - Event class (available for error logs only)
- 4 Select **OK**. The event logs satisfying the conditions are displayed on the screen.
- 5 Choose the following buttons as needed:
 - **Refresh F2**: to refresh the logs based on the current search conditions.
 - **Delete F3**: to remove specified logs on the screen.
 - **Print F7**: to print all logs currently displayed on the screen.

17.2.4 Refreshing Logs

To refresh the event logs, perform the following procedure:

- 1 Select **Alarm-Error Log** or **Edit Log**.
- 2 Select **Refresh F2**.
- 3 The system refreshes the logs based on the previous search conditions.
 - New error logs are displayed chronologically and highlighted by different colors. Yellow indicates a warning, and red indicates a serious error.
 - New edit logs are displayed chronologically on the front-most of the log list.
- 4 Choose the following buttons as needed:
 - **Delete F3**: to remove specified logs on the screen.
 - **Archive F4**: to archive all event logs occurring within a period of time.
 - **Print F7**: to print all logs currently displayed on the screen.

17.2.5 Clearing Logs

Since the system has a limited storage capacity, you should clear and manage the event logs regularly to ensure that the most-recent and important logs are kept.

Only users with sufficient permissions are allowed to delete event logs. For more information about user permissions, refer to 11.4.4 Assigning/Modifying Permissions (page 11-7).

17 Alarms and Troubleshooting

Perform the following steps to clear event logs:

- 1 Select **Alarm-Error Log** or **Edit Log**.
- 2 Select event logs you desire to delete.
- 3 Select **Delete F3**.
- 4 Select **OK**. To abort the deleting, select **Cancel**.

When you confirm the deleting, the system removes the selected event logs from the screen.

17.2.6 Printing Logs

After searching for desired logs on the **Error Log** or **Edit Log** screen, select **Print F7**. The event logs currently displayed are printed out in the same format as shown on the screen.

Printing logs will take a long time and requires a great number of papers. Think twice before printing logs.

To terminate the printing, select **Utility-Commands-Stop Print**.

17.3 Error Troubleshooting

17.3.1 Introduction

When an error occurs, it will be indicated in many ways. The following pages describe how to troubleshoot errors and help you determine solutions to such errors.

Generally, troubleshooting is divided into the following steps:

- An error occurs and is indicated in various ways.
- Check the error logs and component status.
- Identify the error and determine relevant solutions.
- Implement the solutions.
- Check and evaluate the implementation of the solutions.

17.3.2 Error Indications

Errors may occur on hardware, software and the entire system. When an error occurs, it will be indicated in many ways to help identify it and determine the possible causes and solutions. Errors can be indicated by alarm tone, alarm message, color, alarm message box, result flag and error log, through which you will obtain detailed information about errors and find the relevant solutions.

Alarm tone

When an error occurs, the buzzer gives alarm tone reminding you to notice the error and take corrective actions. Alarm tone can be adjusted manually or silenced.

Perform the following steps to adjust the alarm tone:

- 1 Select **Utility-System Setup**.
- 2 Adjust the alarm tone in the **Alarm Volume** field.
- 3 Test the alarm tone until it is satisfied.
- 4 To silence the alarm tone, drag the slider to the leftmost position of the scale.
- 5 Select **Save F8** to save the adjustment.

Alarm message

When an error occurs, the system gives an alarm and displays the alarm message in the second line of the prompt message area. For details of troubleshooting, refer to 17.5 Error Messages and Corrective Actions (page 17-29).

Color highlight

An error will be indicated by highlighting relevant buttons and screen texts with different colors. Yellow indicates a warning, and red indicates a serious warning or error.

- **Reagent** button
- **Utility** button
- **Alarm** button

Select a button to access relevant function page, check for abnormalities and take corrective actions. When the problem is solved, the alarm indication disappears.

Alarm message box

An error can also be shown in an alarm message box, which contains the date/time, event ID, time(s) and help icon.

Errors that are indicated through an alarm message box are divided into the following types:

- Common error: including those that are indicated by warning the user, and by invalidating tests, reagents and samples. When such error occurs, the alarm message box shows with the title bar highlighted in yellow.
- Serious error: including those except for the common error. When such error occurs, the alarm message box shows with the title bar highlighted in red, and you are only allowed to reboot or exit the system.

When an alarm message box appears, select the **Alarm** button to view the new error logs, analyze the possible causes and determine relevant corrective actions.

Flag

Flag is also called data alarm. When calibration error or failure, or sample result error occurs due to the sample, reagent or system failure, a flag will appear near the corresponding calibration result or sample results.

Error log

All alarms are recorded in the error logs. By recalling the error logs you are enabled to master the current status of the system and troubleshoot errors.

17.3.3 Identifying Errors

To identify errors, understand the error indication thoroughly, check the error logs and system status, and then determine relevant solutions.

The table below shows the error types that may occur on the system. Find relevant corrective actions according to the description.

Table 17.6 Error types

Error Type	Description
Instrument failure and error	Instrument failure and error may be detected on all subsystems and processed in different ways. Such errors are shown in the Error messages and corrective actions table, and can be identified through the event ID.
Data alarm	Data alarm is a flag indicating biochemistry or ISE chemistry result error. The flags are included in the Result flags table, and can be identified through the flag symbol.

17.4 Data Alarm

17.4.1 Introduction

Data alarm is a result flag indicating that an error or abnormality occurs to a result. By identifying results flags can evaluate if the results are reliable and acceptable. Data alarm is not necessarily an error but will definitely influence the result and should be considered carefully.

The system provides monitoring of biochemistry results and ISE chemistry results. When calibration error or failure, or sample result error occurs due to the sample, reagent or system failure, a flag will appear near the corresponding calibration result or sample results. The following pages summary the result flags of the system.

17.4.2 Result Flags

Table 17.7 Result flags and corrective actions

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
<	Result related	Exceeds linearity range low	The result exceeds the low limit of the linearity range.	Take no actions, or rerun the test for confirmation.
>	Result related	Exceeds linearity range high	The result exceeds the high limit of the linearity range.	Rerun the test with sample diluted or decreased.
▲	Result related	Sample volume is increased one	Sample volume is Increased one	No actions are required.
▼	Result related	Sample volume is decreased one	Sample volume is decreased one	No actions are required.
∧	Result related	Exceeds reference range high	The result exceeds the high limit of the reference range.	No actions are required.
∧!	Result related	Exceeds critical range high	The result exceeds the high limit of the critical range.	No actions are required.
∨	Result related	Exceeds reference range low	The result exceeds the low limit of the reference range.	No actions are required.
∨!	Result related	Exceeds critical range low	The result exceeds the low limit of the critical range.	No actions are required.
10-x	Result related	10-x	Results of five runs (10 results), or 10 continuous results of a control are on the same side.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.
1-2s	Result related	1-2s	The current QC result is between ± 2 and ± 3 standard deviations from the assigned mean concentration.	No actions are required.
1-3s	Result related	1-3s	The current QC result is greater than ± 3 standard deviations from the assigned mean concentration.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.

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Flag	Alarm Type	Description	Probable Causes	Corrective Actions
2-2s	Result related	2-2s	Results of two controls in the same run or two continuous results of a control are on the same side and greater than ± 2 standard deviations from the assigned mean concentration.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.
4-1s	Result related	4-1s	Results of two runs (4 results), or 4 continuous results of a control are on the same side and greater than ± 1 standard deviation from the assigned mean concentration.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.
ABS	Result related	Absorbance out of range	The absorbance of primary or secondary wavelength used for calculating results is greater than 3.4A.	Check the sample for foreign matters or interferences; check if the reagent is qualified and placed in the correct position; check the cuvette is clean; check if the photometric system is working normally.
BLK	Calibration related	Blank response out of range	The reagent goes wrong; insufficient reagent is dispensed; the cuvette contains air bubbles; the light drifts; or the cuvette is overflowed.	Check if the cuvette is not overflowed, the reagent is sufficient without air bubbles, the light does not drift and the chemistry parameters are reasonable. If yes, replace the reagent and then rerun the test.
BOE	Result related	Substrate depletion	The sample concentration is too high, and substrate depletion occurs during fixed-time measurements.	Check the reaction curve and the substrate depletion limit. Rerun the test with diluted sample.
CALE	Result related	Edited calibration factor	The calibration factors are edited.	No actions are required.
CALF	Result related	Calibration failed.(for	The calibration is failed.	Recalibrate.

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
		biochemistries)		
CALF	Result related	No fluid in tubing	1. Waste pump tube is aging, blocked, or broken; 2. Sample injection port and fluidic path are blocked or leaking. 3. Air bubble detector failed.	1. Replace the reagent pack with a new one 2. Perform purge B to remove bubbles 3. /4. Replace the pump tube 5. Clean the sample injection port and reinstall electrodes. 6. Replace the bubble detector. 1. Place sufficient ISE wash solution. 2. Replace the pump tube 3. Clean the sample injection port and reinstall electrodes. 4. Replace the bubble detector.
CALF	Calibration related	No fluid in tubing	1. Waste pump tube is aging, blocked, or broken; 2. Sample injection port and fluidic path are blocked or leaking. 3. Air bubble detector failed.	1. Replace the reagent pack with a new one 2. Perform purge B to remove bubbles 3. /4. Replace the pump tube 5. Clean the sample injection port and reinstall electrodes. 6. Replace the bubble detector.

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
				1. Place sufficient ISE wash solution. 2. Replace the pump tube 3. Clean the sample injection port and reinstall electrodes. 4. Replace the bubble detector.
CALJ	Calibration related	Rejected calibration factor	The calibration factors are rejected	No actions are required.
CALM	Result related	Air in segment	1. Waste pump tube is aging, blocked, or broken; 2. Sample injection port and fluidic path are blocked or leaking. 3. Air bubble detector failed.	1. Replace the pump tube 2. Clean the sample injection port and reinstall electrodes. 3. Replace the bubble detector.
CALM	Calibration related	Air in segment	1. Waste pump tube is aging, blocked, or broken; 2. Sample injection port and fluidic path are blocked or leaking. 3. Air bubble detector failed.	1. Replace the pump tube 2. Clean the sample injection port and reinstall electrodes. 3. Replace the bubble detector.
CALR	Result related	Recalculated calibration factor	The calibration factors are recalculated.	No actions are required.
COV	Calibration related	Calibration curve not convergent	For nonlinear calibration, a satisfying base cannot be calculated and no calibration curve is drawn.	Check that the reagent and calibrator are normal, and then recalibrate. If the error remains, contact our customer service department.

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
CSD	Calibration related	Calibration curve standard deviation out of range	The calculated standard deviation of the calibration curve exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
DEL	Calibration related	Deleted QC result	The QC result has been deleted.	No actions are required.
DET	Calibration related	Calibration determination coefficient out of range	The calculated determination coefficient of the calibration curve exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
DEP	Calibration related	Saving calibration result error	1. ISE communication cable failure. 2. Communication interface or pins failure 3. Main control board of the ISE module goes wrong. 4. Software error.	1. Replace the ISE communication cable. 2. Replace the interface or pins. 3. Replace the main control board of the ISE module. 4. Upgrade the operating software or reinstall it.
DTGL	Result related	Insufficient concentrated wash solution	The concentrated wash solution is insufficient during measurement.	Fill more concentrated wash solution.
DUP	Calibration related	Calibration repeatability error	The difference between the maximum and minimum response of the calibrator exceeds the specified limit.	Check if the acceptance limit is reasonable, troubleshoot the error, and then recalibrate.
EDT	Result related	Edited result	The result has been edited.	No actions are required.
EDT	Calibration related	Edited calibration factor	The calibration factors have been edited.	No actions are required.
ENC	Result related	No calculation interval	The sample concentration is too high, and	Check the reaction curve

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
			substrate depletion occurs within the lag time of rate check measurements.	and the substrate depletion limit. Rerun the test with diluted sample.
EXP	Result related	Enzyme linearity range extension	The high-concentration sample leads to substrate depletion during the reaction time, and the result is calculated by using measuring points within the lag time.	Rerun the test with diluted sample.
EXT	Result related	Extended calibration factor	The result is obtained by extending the calibration time.	Take no actions, or recalibrate.
FAC	Calibration related	Calibration slope difference out of range	The slope difference is applicable to linear calibration only and refers to the K factor (slope) difference between two adjacent calibrations. It exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
ICA	Result related	The response is normal, but results cannot be calculated.	The chemistry has not been calibrated.	Rerun it after calibration.
LI	Result related	Water blank fluctuation is out of range.	1. The cuvette is overflowing. 2. The lamp has been replaced incorrectly. 3. Cuvette check is not performed after maintenance. 4. The cable connectors are not tightened. 5. The retaining screw is not tightened. 6. Cleaning liquid inside the cuvette is little. 7. The lamp is aged. 8. The photometer goes wrong.	1. Check if the cuvette is overflowing. 2. Check if the Replace Lamp command is executed during lamp replacement. 3. Check if the cable connectors and retaining screw of the lamp have been tightened. 4. Check if the cleaning liquid inside the cuvette is no less than half of the cuvette. 5. Check if the reaction curve fluctuates irregularly.

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
				If yes, replace the lamp. 6. If the error remains, contact our customer service department.
LIN	Result related	Non-linear	The measuring points for result calculation are nonlinear, because the sample concentration is too high, or the substrate depletion limit is not specified or unreasonable. The lamp is aged.	Check the reaction curve and the substrate depletion limit. Rerun the test with diluted sample. If the alarm occurs for more than one chemistry, and the reaction curve fluctuates irregularly, replace the lamp.
LOW	Result related	Response less than that of the minimum-concentration calibrator	The sample concentration is lower than the sensitivity indicated on the reagent pack, making response less than that of the lowest-concentration calibrator.	For ascending calibration curve, rerun the test with standard or increased sample volume; for descending calibration curve, rerun the test with diluted sample.
MBK	Calibration related	Mixed blank absorbance out of range	The reagent goes wrong; the cuvette is not clear; the reaction cuvette is overflowed; or insufficient reagent is dispensed.	Check if the cuvette is clear and not overflowed, the reagent is sufficient without air bubbles, and the chemistry parameters are reasonable. If yes, replace the reagent and then rerun the test.
MON	Calibration related	Calibration curve not monotonic	The calibration data and calibration curve are not monotonic.	Check if the calibrator is defined and placed correctly, and then recalibrate.
NLN	Result related	No linear interval	The high-concentration sample leads to less than 3 valid measuring points within the	Rerun the test with diluted sample.

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
			reaction time of rate check measurements.	
NOIS	Result related	Electrode voltage noise	1. Electrode failure. 2. Environment interference. 3. ISE main control board failure. 4. Salt buildup around electrodes or tubes due to fluidic leaks.	1. Replace the electrode. 2. Relocate the instrument. 3. Replace the main control board of the ISE module. 4. Clean the tubes and electrodes.
NOIS	Calibration related	Electrode voltage noise	1. Electrode failure. 2. Environment interference. 3. ISE main control board failure. 4. Salt buildup around electrodes or tubes due to fluidic leaks.	1. Replace the electrode. 2. Relocate the instrument. 3. Replace the main control board of the ISE module. 4. Clean the tubes and electrodes.
OVE	Result related	Overridden calibration factor	The result is obtained by overriding a failed calibration.	Take no actions, or recalibrate.
PUGA	Result related	Air in calibrator A	1. Calibrator A is exhausted. 2. Bubbles exist in calibrator tube A 3. Pump tube A is aging, blocked, or broken. 4. Waste pump tube is aging, blocked, or broken; 5. Sample injection port and fluidic path are blocked or leaking. 6. Air bubble detector failed.	1. Replace the reagent pack with a new one 2. Perform purge B to remove bubbles 3. /4. Replace the pump tube 5. Clean the sample injection port and reinstall electrodes. 6. Replace the bubble detector.
PUGA	Calibration	Air in calibrator A	1. Calibrator A is exhausted.	1. Replace the reagent

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
	related		2. Bubbles exist in calibrator tube A 3. Pump tube A is aging, blocked, or broken. 4. Waste pump tube is aging, blocked, or broken; 5. Sample injection port and fluidic path are blocked or leaking. 6. Air bubble detector failed.	pack with a new one 2. Perform purge B to remove bubbles 3. /4. Replace the pump tube 5. Clean the sample injection port and reinstall electrodes. 6. Replace the bubble detector.
PUGB	Result related	Air in calibrator B	1. Calibrator B is exhausted. 2. Bubbles exist in calibrator tube B 3. Pump tube A is aging, blocked, or broken. 4. Waste pump tube is aging, blocked, or broken; 5. Sample injection port and fluidic path are blocked or leaking. 6. Air bubble detector failed.	1. Replace the reagent pack with a new one 2. Perform purge B to remove bubbles 3. /4. Replace the pump tube 5. Clean the sample injection port and reinstall electrodes. 6. Replace the bubble detector.
PUGB	Calibration related	Air in calibrator B	1. Calibrator B is exhausted. 2. Bubbles exist in calibrator tube B 3. Pump tube A is aging, blocked, or broken. 4. Waste pump tube is aging, blocked, or broken; 5. Sample injection port and fluidic path are blocked or leaking. 6. Air bubble detector failed.	1. Replace the reagent pack with a new one 2. Perform purge B to remove bubbles 3. /4. Replace the pump tube 5. Clean the sample injection port and reinstall electrodes. 6. Replace the bubble detector.

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
PRO	Result related	Prozone check error	Antibody excess occurs due to too high sample concentration.	Check the reaction curve and the prozone check parameters. Rerun the test with diluted sample.
R	Result related	Rerun result	The result is obtained by rerunning the test.	No actions are required.
R4S	Result related	R4S	One result of a run is greater than +2 standard deviations from the assigned mean and the other greater than -2SDs.	Check if the reagent is qualified, control sample is normal, and the instrument is working correctly.
RBK	Result related	R1 blank absorbance out of range	The reagent goes wrong; the cuvette is not clear; the reaction cuvette is overflowed; or insufficient reagent is dispensed.	Check if the cuvette is clear and not overflowed, the reagent is sufficient without air bubbles, and the chemistry parameters are reasonable. If yes, replace the reagent and then rerun the test.
RCE	Result related	Response calculation error	Absorbance data for calculation is incomplete, or the dividend is 0.	Rerun the test. If the error remains, contact our customer service department.
REC	Result related	Recalculated result	The sample result is recalculated manually with the latest calibration factors.	/
RESP	Result related	ISE response check code error Command format or execution error	1. ISE communication cable failure. 2. Communication interface or pins failure 3. Main control board of the ISE module goes wrong. 4. Software error.	1. Replace the ISE communication cable 2. Replace the interface or pins. 3. Replace the main control board of the ISE module. 4. Upgrade the operating software or reinstall it.
RESP	Calibration	ISE response check code	1. ISE communication cable failure.	1. Replace the ISE

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
	related	error Command format or execution error	2. Communication interface or pins failure 3. Main control board of the ISE module goes wrong. 4. Software error.	communication cable. 2. Replace the interface or pins. 3. Replace the main control board of the ISE module. 4. Upgrade the operating software or reinstall it
RGTE	Result related	Expired reagent	The result is based on an expired reagent.	Replace the reagent.
RGTL	Result related	Insufficient reagent	The result is based on insufficient reagent.	Replace the reagent.
RGTL	Calibration related	Insufficient reagent	The calibration result is based on insufficient reagent.	Replace the reagent.
RRN	Result related	Response greater than that of the maximum-concentration calibrator	The sample concentration exceeds the high limit of the calibrator concentration.	Rerun the test with diluted sample.
SEN	Calibration related	Calibration sensitivity error	The difference of final response of the maximum and minimum concentration calibrators exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
SJAM	Result related	Sample probe is clogged	Probe clogging is detected during sampling or the sample probe is clogged during sampling.	Sample treatment.
SLDR	Calibration related	Electrode slope drift	1. Electrode or reagent pack is failed. 2. Electrode is unsteady. 3. New reagent pack is unsteady. 4. Reference electrode has been used for over 66 months. 5. ISE main control board failure. 6. Ambient temperature fluctuates drastically	1. Replace the problematic electrode and reagent pack. 2. New electrode will become steady after 15 minutes since installed. 3. Run a couple of calibrations after installing new reagent pack.

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
				4. Replace the reference electrode. 5. Replace the ISE main control board. 6. Control the ambient temperature to make the fluctuation within $\pm 4^{\circ}\text{C}$.
SLEX	Calibration related	Slope out of range	1. Electrode is not installed correctly. 2. Calibrator expired. 3. Electrode degenerated 4. Bubbles in reference electrode 5. Reference electrode has been used for a long time 6. Electrodes interfered. 7. Module or tubing temperature above 32°C .	1. Reinstall the electrode 2. Replace the calibrator. 3. Replace the problematic electrode and rerun. 4. Remove the electrode and clap on it to eliminate bubbles. Reinstall the electrode and run calibration. 5. Replace reference electrode and rerun. 6. Troubleshoot the electrodes by replacing them in different groups. 7. Monitor temperature, if too high, relocate equipment.
SLP	Result related	Corrected result	The result is adjusted with calculation factors.	No actions are required.
SLP	Result related	The results are produced when the calibration factors instead of the default ones are configured for the second time	Calibration factors instead of the default ones are configured for the second time calibration.	No actions are required.

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
		calibration.		
SMPA	Result related	Air in sample	1. Sample is insufficient or contains many bubbles after dispensing. 2. No or insufficient sample has been dispensed into the sample injection port. 3. The electrodes are not properly installed, causing leakage. 4. The waste pump tube is aging or broken.	1. Increase the sample volume. At least 90µl sample should be prepared. 2. Electrode is not installed correctly. Reinstall it. 3. Check the waste tube, and if necessary, replace it.
SMPE	Result related	Expired sample	The sample is expired.	Replace the sample.
SMPL	Result related	Insufficient sample	The sample is insufficient during analysis.	Refill the sample.
SMPL	Calibration related	Insufficient sample	The sample is insufficient during analysis.	Refill the sample.
VDRF	Result related	Electrode voltage drift	1. Electrode or reagent pack is failed. 2. Electrode is unsteady. 3. New reagent pack is unsteady. 4. Reference electrode has been used for over 66 months. 5. ISE main control board failure. 6. Ambient temperature fluctuates drastically.	1. Replace the problematic electrode and reagent pack. 2. New electrode will become steady after 15 minutes since installed. 3. Run a couple of calibrations after installing new reagent pack. 4. Replace the reference electrode. 5. Replace the ISE main control board. 6. Control the ambient temperature to make the fluctuation within +/-4°C
VOUT	Result related	Electrode Voltage Overflow	1. Electrode or reagent pack is failed. 2. Electrode is unsteady. 3. New reagent pack is unsteady. 4. Reference electrode has been used for	1. Replace the problematic electrode and reagent pack. 2. New electrode will become steady after 15

Flag	Alarm Type	Description	Probable Causes	Corrective Actions
			over 66 months. 5. ISE main control board failure.	minutes since installed. 3. Run a couple of calibrations after installing new reagent pack. 4. Replace the reference electrode. 5. Replace the ISE main control board.
VOUT	Calibration related	Electrode Voltage Overflow	1. Electrode or reagent pack is failed. 2. Electrode is unsteady. 3. New reagent pack is unsteady. 4. Reference electrode has been used for over 66 months. 5. ISE main control board failure.	1. Replace the problematic electrode and reagent pack. 2. New electrode will become steady after 15 minutes since installed. 3. Run a couple of calibrations after installing new reagent pack. 4. Replace the reference electrode. 5. Replace the ISE main control board.
T1	Result related	Reaction temperature error disk	1. The ambient temperature is out of range. 2. The temperature sensor goes wrong. (component error and cable error) 3. The temperature protection switch goes wrong. (component error and cable error) 4. The heater goes wrong. (component error and cable error) 5. PCB error 6. Parameters are lost. 7. Electromagnetic interference exists.	1. Check if the error is accidental. 2. If not, contact our customer service department.

17.5 Error Messages and Corrective Actions

Table 17.8 Error messages and corrective actions

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
A00006	Instruction error	Error	Equipment configuration cannot be read or saved Error:	/	E2PROM read/write error	Switch off the analyzing unit power and switch on it again. Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.
A01006	Sample probe unit	Error	Sample probe vertical movement error Position: Error: Or Sample probe horizontal movement error Position: Error: Or	/	Sample probe vertical movement error 1. Sensor status error: The sample probe assembly is probably forced to move vertically. 2. Failed to find the zero position: The sample probe assembly is probably jammed. 3. Collision occurs during operation other than aspirating: The sample probe collides with other object. 4. Collision error: The collision remains. 5. Moving vertically is not allowed in current position:	Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			Sample syringe movement error. Position: Error:		The sample probe moves vertically in an unknown position. Sample probe horizontal movement error 1. Sensor status error: The sample probe assembly is probably forced to move horizontally. 2. Failed to find the zero position: The sample probe assembly is obstructed when rotating. 3. Collision occurs during horizontal movement: The sample probe assembly is obstructed when rotating. 4. Moving horizontally is not allowed in current position: The sample probe assembly is probably forced to move vertically. Sample syringe movement error. 1. Sensor status error: The syringe assembly is probably forced to move. 2. Failed to find the zero position: The syringe assembly is probably jammed.	
A01007	Sample probe unit	Warning	Sample probe collides with an obstacle when aspirating Position:	/	Sample probe vertical movement error 1. Collision occurs during aspirating: The sample probe collides with other object.	Sample probe vertical movement error 1. Collision occurs during aspirating: Remove the obstacle, and then recover failure by performing the Home maintenance procedure.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
A01021	Sample probe unit	Error	Clog detection board communication error.	/	Clog detection board communication error.	Recover the failure. If this message appears for 3 times, contact our customer service department or your local distributor.
A01022	Sample probe unit	Warning	Sample syringe aspirates too much Sample ID/bar code: Position:	/	The aspirate volume is beyond the range of the syringe.	Define the aspirate volume correctly.
A01023	Sample probe unit	Warning	Sample syringe dispenses too much Cuvette No.: Sample ID/bar code: Chemistry:	/	The dispense volume is beyond the range of the syringe.	Define the dispense volume correctly.
A01024	Sample probe unit	Warning	Insufficient sample Sample position: Sample ID/bar code: or Sample probe level detection failed.	/	There is no sample or insufficient sample on the designated position.	1. Check if the sample is sufficient, and then try again. 2. If the error remains, contact our customer service department or your local distributor.
A01027	Sample probe unit	Error	Sample is insufficient or contains air bubbles Position: Sample ID/bar code: or Sample probe level detection failed	/	There is no sample or insufficient sample on the designated position.	1. Check if the sample is sufficient, and then try again. 2. If the error remains, contact our customer service department or your local distributor.
A01028	Sample probe unit	Error	Sample probe fails to detect liquid level	/	There is no deionized water, or the deionized water is not supplied normally.	1. Check if the water supply is normal.

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Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			during cleaning			2. Recover the failure for 3 times. If the error remains, contact our Customer Service Department or your local distributor.
A01029	Sample probe unit	Warning	Sample is insufficient or contains fibrins and clots Position: Sample ID/bar code: or Sample probe clog detection failed.	/	1. The sample contains clots, or is too thick or insufficient. 2. The sample probe is clogged.	1. Check that the sample is preprocessed correctly; or check if the sample contains foreign matters such as clot. If it does, change the sample. 2. Check if the sample is sufficient. 3. Clean the sample probe with wash solution. If the problem remains, remove the sample probe and unclog it, and then continue with the measurement.
A01030	Sample probe unit	Error	Sample probe is clogged during cleaning Sample ID/bar code: Position: or Sample probe clog detection failed.	/	The sample probe is clogged.	1. Clean the sample probe with wash solution. Remove the sample probe and unclog it. 2. If the problem remains, contact the manufacturer.
A01033	Sample probe unit	Warning	Sample probe fails to detect liquid level on reaction carousel when dispensing. Cuvette No.: sample ID/bar code: chemistry: or	/	There is no reagent or insufficient reagent in the reaction cuvette.	1. Check if R1 volume is sufficient and the reagent bottle is free of air bubbles, and then try again. 2. If the problem remains, contact the manufacturer.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			Sample probe level detection failed.			
A01036	Sample probe unit	Error	Sample probe level detection board communication error	/	Level detection board communication error	Recover the failure. If this message appears for 3 times, contact our customer service department or your local distributor.
A01037	Sample probe unit	Error	Sample probe level detection board self-calibrating failed	/	1. The sample probe is not installed correctly or goes wrong. 2. Level detection board communication error	1. Check if the sample is installed correctly or damaged. 2. Recover the failure. If your attempt is failed, contact our customer service department or your local distributor.
A01038	Sample probe unit	Error	Sample probe interior wash is abnormal.	/	Sample probe interior wash is abnormal.	Contact our customer service department or your local distributor.
A02006	Reagent probe unit	Error	Reagent probe vertical movement error Position: Error: Or Reagent probe horizontal movement error Position: Error: Or	/	Reagent probe vertical movement error 1. Sensor status error: The reagent probe assembly is probably forced to move vertically. 2. Failed to find the zero position: The reagent probe assembly is probably jammed. 3. Collision occurs during operation other than aspirating: The reagent probe collides with other object. 4. Collision error: The collision remains. 5. Moving vertically is not allowed in current position:	Switch off the analyzing unit power and switch on it again. Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			Reagent syringe movement error Position: Error:		The reagent probe moves vertically in an unknown position. Reagent probe horizontal movement error 1. Sensor status error: The reagent probe assembly is probably forced to move horizontally. 2. Failed to find the zero position: The reagent probe assembly is obstructed when rotating. 3. Collision occurs during horizontal movement: The reagent probe assembly is obstructed when rotating. 4. Moving horizontally is not allowed in current position: The reagent probe assembly is probably forced to move vertically. Reagent syringe movement error. 1. Sensor status error: The syringe assembly is probably forced to move. 2. Failed to find the zero position: The syringe assembly is probably jammed.	
A02007	Reagent probe unit	Warning	Reagent probe collides with an obstacle when aspirating Position:	/	Reagent probe vertical movement error 1. Collision occurs during aspirating: The reagent probe collides with other object.	Reagent probe vertical movement error 1. Collision occurs during aspirating: Remove the obstacle and then recover the failure.
A02021	Reagent probe	Warning	Reagent syringe	/	The aspirate volume is beyond the range of	Define the aspirate volume

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
	unit		aspirates too much Chemistry: Position:		the syringe.	correctly.
A02022	Reagent probe unit	Warning	Reagent syringe dispenses too much Cuvette No.: Sample ID/bar code: Chemistry:	/	The dispense volume is beyond the range of the syringe.	Define the dispense volume correctly.
A02023	Reagent probe unit	Warning	Insufficient reagent Chemistry: Position on outer ring: Or reagent probe level detection failed.	/	There is no reagent or insufficient reagent on the designated position.	1. Check if the reagent is sufficient, and then try again. 2. If the error remains, contact our customer service department or your local distributor.
A02025	Reagent probe unit	Warning	Reagent probe dispenses insufficient reagent Cuvette No.: Sample ID/bar code: Chemistry:	/	1. The reagent probe aspirates nothing.	1. Check if the reagent satisfies the requirement and is sufficient in volume, and then try again. 2. Recover the failure. If this message appears for 3 times, contact our customer service department or your local distributor.
A02026	Reagent probe unit	Error	Reagent probe fails to detect liquid level during cleaning.	/	There is no deionized water, or the deionized water is not supplied normally.	1. Check if the water supply is normal. 2. If the error remains, contact our customer service department or your local distributor.
A02027	Reagent probe unit	Warning	Water residues exist in the cuvette	/	There is deionized water left in the reaction cuvette.	Recover the failure. If this message appears for 3 times, contact our

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			Cuvette No.: Sample ID/bar code: Chemistry: Or Reagent probe level detection failed.			customer service department or your local distributor.
A02030	Reagent probe unit	Error	Reagent probe level detection board communication error	/	Level detection board communication error	Recover the failure. If this message appears for 3 times, contact our customer service department or your local distributor.
A02031	Reagent probe unit	Error	Reagent probe level detection board self-calibrating failed	/	Level detection board communication error	1. Check if the reagent probe is installed correctly and intact. 2. Recover the failure. If this message appears for 3 times, contact our customer service department or your local distributor.
A02032	Reagent probe unit	Warning	Reagent is insufficient or contains air bubbles Chemistry: Position on outer ring:	/	1. Air bubbles exist in the reagent bottle. 2. The reagent bottle does not meet the requirements.	1. Check if the reagent bottle contains air bubbles, and then try again. 2. Check if the reagent bottle meets the requirements. 3. If the error remains, contact our customer service department or your local distributor.
A02033	Reagent probe unit	Warning	Insufficient reagent is dispensed or air bubbles exist Cuvette No.: Sample ID/bar code:	/	The reagent is insufficient, or air bubbles exist in the reagent bottle.	1. Check if the reagent is sufficient and the reagent bottle contains air bubbles, and then try again. 2. If the problem remains, contact the manufacturer.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			Chemistry: Position: Or Reagent probe level detection failed.			
A04006	Sample mixer unit	Error	Sample mixer vertical movement error Position: Error: Or Sample mixer horizontal movement error Position: Error:	/	Sample mixer vertical movement error 1. Sensor status error The sample mixer assembly is probably forced to move vertically. 2. Failed to find the zero position The sample mixer assembly is probably jammed. 3. Moving vertically is not allowed in current position The sample mixer moves vertically in an unknown position. Sample mixer horizontal movement error 1. Sensor status error The sample mixer assembly is probably forced to move horizontally. 2. Failed to find the zero position The sample mixer assembly is obstructed when rotating. 3. Moving horizontally is not allowed in current position The sample mixer moves vertically in an unknown position.	Switch off the analyzing unit power and switch on it again. Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.
A04015	Sample mixer unit	Error	Sample mixer rotation error Rotation speed:	/	The mixer is obstructed by other object or interfered by the reaction cuvette.	Recover failure by performing the Home maintenance procedure. If this message appears for 3 times,

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
						contact our customer service department or your local distributor.
A05006	Reagent mixer unit	Error	Reagent mixer vertical movement error Position: Error: Or Reagent mixer horizontal movement error Position: Error:	/	Reagent mixer vertical movement error 1. Sensor status error The reagent mixer assembly is probably forced to move vertically. 2. Failed to find the zero position The reagent mixer assembly is probably jammed. 3. Moving vertically is not allowed in current position The reagent mixer moves vertically in an unknown position. Reagent mixer horizontal movement error 1. Sensor status error The reagent mixer assembly is probably forced to move vertically. 2. Failed to find the zero position The reagent mixer assembly is obstructed when rotating. 3. Moving horizontally is not allowed in current position The reagent mixer moves vertically in an unknown position.	Switch off the analyzing unit power and switch on it again. Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.
A05016	Reagent mixer unit	Error	Reagent mixer rotation error Rotation speed:	/	The mixer is obstructed by other object or interfered by the reaction cuvette.	Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
A06006	Reaction carousel unit	Error	Reaction carousel movement error Error:	/	Reaction carousel movement error 1. Failed to find the home position The reaction carousel is obstructed or blocked. 2. The coder missed steps The reaction carousel is obstructed or blocked. 3. The reaction carousel missed steps when moving to the home position. The reaction carousel is obstructed or blocked.	Switch off the analyzing unit power and switch on it again. Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.
A07006	Sample carousel unit	Error	Sample carousel outer ring movement error Error:	/	Sample carousel outer ring movement error 1. Failed to find the home position The sample carousel outer ring is obstructed or blocked. 2. The coder missed steps The sample carousel outer ring is obstructed or blocked. 3. The sample carousel outer ring missed steps when moving to the home position. The sample carousel outer ring is obstructed or blocked.	Recover the failure. If this message appears for 3 times, contact our customer service department or your local distributor.
A07009	Sample carousel unit	Error	Sample bar code reader error	/	The sample bar coder reader goes wrong due to system failure.	Recover the failure. If the error remains, initialize the sample bar code reader. If the error still remains, contact our Customer Service Department or your local distributor.
A07010	Sample carousel unit	Warning	Sample bar code error	/	Sample bar coder reader does not work normally due to communication error.	Initialize the sample bar code reader and try again. If your attempt is failed, contact our

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Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			Position:			customer service department or your local distributor.
A07011	Sample carousel unit	Error	Sample bar code sending buffer is full	/	Sample bar coder sending buffer is full due to communication error.	Recover the failure or reboot the analyzing unit.
A09006	Reagent carousel unit	Error	Reagent carousel outer ring movement error Error:	/	Reagent carousel outer ring movement error 1. Failed to find the home position The reagent carousel outer ring is obstructed or blocked. 2. The coder missed steps The reagent carousel outer ring is obstructed or blocked. 3. The reagent carousel outer ring missed steps when moving to the home position. The reagent carousel outer ring is obstructed or blocked.	Recover the failure. If this message appears for 3 times, contact our customer service department or your local distributor.
A09011	Reagent carousel unit	Error	Reagent bar code reader does not work normally	/	The reagent bar coder reader goes wrong due to system failure.	Recover the failure. If the problem remains, initialize the sample bar code reader. If the error still remains, contact our Customer Service Department or your local distributor.
A09012	Reagent carousel unit	Warning	Reagent bar code error Position:	/	Reagent bar coder sending buffer is full due to communication error.	Initialize the sample bar code reader and try again. If your attempt is failed, contact our customer service department or your local distributor.
A09014	Reagent carousel unit	Error	Reagent bar code sending buffer is full Position:	/	Reagent bar coder reader does not work normally due to communication error.	Recover the failure or reboot the analyzing unit.
A11005	Wash station	Error	Wash station	/	Wash station movement error	Switch off the analyzing unit power

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			movement error Error:		1. Sensor status error The wash station assembly is probably forced to move. 2. Failed to find the home position The wash station assembly is obstructed by other object. 3. The wash station collides with an obstacle when moving. The wash station collides with other object, or the wash probes then collide with the reaction carousel.	and switch on it again. Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.
A11010	Wash station	Error	Releasing vacuum failed	/	1. Solenoid valves V23-V27 go wrong. 2. The vacuum pump goes wrong. 3. The vacuum sensor goes wrong.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A11012	Wash station	Warning	Water supplying is too slow	/	1. The water unit goes wrong. 2. The water supply valve goes wrong. 3. The water supply ball valve is not opened. 4. The water supply ball valve goes wrong. 5. The low-level floater of the water tank goes wrong. 6. The water supply tube is bent. 7. The outlet filter of the water supply tube is clogged.	1. Check the water unit. 2. Check if the water supply ball valve is opened and the handle is level. 3. Check if the water supply tube is smooth. 4. Check if the water level inside the water tank is low (at the scale of 5L). 5. Check if the error is accidental. 6. If the error is not accidental, contact our customer service department or your local distributor.
A11013	Wash station	Error	Water tank is empty	/	1. The water unit goes wrong.	1. Check the water unit.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
					2. The water supply valve goes wrong. 3. The water supply ball valve is not opened. 4. The water supply ball valve goes wrong. 5. The low-level floater of the water tank goes wrong. 6. The water supply tube is bent. 7. The outlet filter of the water supply tube is clogged.	2. Check if the water supply ball valve is opened and the handle is level. 3. Check if the water supply tube is smooth. 4. Check if the water level inside the water tank is low (at the scale of 5L). 4. Check if the error is accidental. 5. If the error is not accidental, contact our customer service department or your local distributor.
A11014	Wash station	Warning	Priming diluted wash solution is slow	/	1. The solenoid valve V06 goes wrong. 2. The restrictor ring is clogged. 3. The inlet filter at the front panel is clogged. 4. The deionized water circulating pump goes wrong. 5. The water tank is empty. 6. The concentrated wash solution tank is empty. 7. The concentrated wash solution pump goes wrong. 8. The low-level floater of the diluted wash solution tank goes wrong.	1. Check if the deionized water pump is opened and the pressure gauge reads between 40kPa-50kPa. 2. Check the floater of the deionized water tank. 3. Check the floater of the diluted wash solution tank. 4. Check the floater of the concentrated wash solution tank. 5. Check if the error is accidental. 6. If the error is not accidental, contact our customer service department or your local distributor.
A11015	Wash station	Error	Insufficient diluted wash solution	/	1. The solenoid valve V06 goes wrong. 2. The restrictor ring is clogged. 3. The inlet filter at the front panel is clogged.	1. Check if the deionized water pump is opened and the pressure gauge reads between 40kPa-50kPa. 2. Check the floater of the

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
					4. The deionized water circulating pump goes wrong. 5. The water tank is empty. 6. The concentrated wash solution tank is empty. 7. The concentrated wash solution pump P05 goes wrong. 8. The low-level floater of the diluted wash solution tank goes wrong.	deionized water tank. 3. Check the floater of the diluted wash solution tank. 4. Check the floater of the concentrated wash solution tank. 5. Check if the error is accidental. 6. If the error is not accidental, contact our customer service department or your local distributor.
A11016	Wash station	Warning	Insufficient concentrated wash solution	/	1. Concentrated wash solution is exhausted and the floater status is Empty. 2. The low-level floater of the concentrated wash solution tank goes wrong.	1. Check if the concentrated wash solution is exhausted and the floater status is Empty. If yes, fill more concentrated wash solution. 2. Check if the error is accidental. 3. If the error is not accidental, contact our customer service department or your local distributor.
A11017	Wash station	Error	Liquid accumulates in primary vacuum container	/	1. The solenoid valve V27 goes wrong. 2. The waste pump P07 goes wrong. 3. The low-concentration waste drain tube is bent.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A11018	Wash station	Error	High concentration waste collector is full	/	1. The waste pump P07 goes wrong. 2. The high-concentration waste drain tube is bent.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A11019	Wash station	Error	Low concentration waste collector is full	/	1. The waste pump P07 goes wrong. 2. The low-concentration waste drain tube	1. Check if the error is accidental. 2. If the error is not accidental,

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Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
					is bent.	contact our customer service department or your local distributor.
A11020	Wash station	Error	High concentration waste tank is full	/	1. The high concentration waste tank is full 2. The floater of the high concentration waste tank goes wrong.	1. Check the high-concentration waste tank. If it is full, replace the waste tank, close the full tank and dispose of the waste properly. 2. Check if the error is accidental. 3. If the error is not accidental, contact our customer service department or your local distributor.
A11027	Wash station	Error	Insufficient vacuum. Cleaning failed	/	1. The primary vacuum pump P08 goes wrong. 2. The primary vacuum sensor goes wrong. 3. The connectors and tubes go wrong. 4. The primary vacuum container goes wrong. 5. Solenoid valves V23-V26 go wrong.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A11028	Wash station	Error	Water tank floater logic error	/	1. Water tank high level and low level floater go wrong	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A11029	Wash station	Error	Diluted wash solution tank floater logic error	/	1. High level and low level floater of diluted wash solution container go wrong.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A11030	Wash station	Error	External Drain	/	1. The drainage module is not turned on.	Check if the drainage module is

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			Canister Overflow		2. The pump of the drainage module goes wrong. 3. High level and low level floater of the drainage module go wrong. 4. The board of the drainage module go wrong. 5. Leakage occurs in waste pipes or the pipes are twisted.	turned on. 2. Check if the error is accidental. 3. If the error is not accidental, contact our customer service department or your local distributor.
A11031	Wash station	Error	Insufficient vacuum degree of the primary vacuum container	/	1. The primary vacuum pump P08 goes wrong. 2. The primary vacuum sensor goes wrong. 3. The connectors and tubes go wrong. 4. The primary vacuum container goes wrong. 5. Solenoid valves V21,V22,V23,V25 or V27 go wrong.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A11034	Wash station	Error	Wash syringe sensor is in incorrect status	/	The assembly is jammed or damaged.	Switch off the analyzing unit power and switch on it again. Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.
A12005	Temperature unit	Warning	Reaction carousel temperature is out of range TDISP temperature: TS01: TS02: TS03:(Adjusted	T1	1. The ambient temperature is out of range. 2. The temperature sensor goes wrong. (component error and cable error) 3. The temperature protection switch goes wrong. (component error and cable error) 4. The heater goes wrong. (component error and cable error)	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			temperature ΔT for 3 Pt1000 sensors)		5. PCB error 6. Parameters are lost. 7. Electromagnetic interference exists.	
A12006	Temperature unit	Warning	Temperature of wash solution for cleaning cuvettes is out of range Temperature:	/	1. The ambient temperature is out of range. 2. The temperature sensor goes wrong. (component error and cable error) 3. The temperature protection switch goes wrong. (component error and cable error) 4. The heater goes wrong. (component error and cable error) 5. PCB error 6. Parameters are lost. 7. Electromagnetic interference exists.	1. Check the temperature of the deionized water for cleaning the whole unit. 2. Check if the water supply is normal and has the temperature between 15°C-30°C. 3. Check if the error is accidental. 4. If the error is not accidental, contact our customer service department or your local distributor.
A12007	Temperature unit	Warning	Temperature of deionized water for cleaning cuvettes is out of range Temperature:	/	1. The ambient temperature is out of range. 2. The temperature sensor goes wrong. (component error and cable error) 3. The temperature protection switch goes wrong. (component error and cable error) 4. The heater goes wrong. (component error and cable error) 5. PCB error 6. Parameters are lost. 7. Electromagnetic interference exists.	1. Check the temperature of the deionized water for cleaning the whole unit. 2. Check if the water supply is normal and has the temperature between 15°C-30°C. 3. Check if the error is accidental. 4. If the error is not accidental, contact our customer service department or your local distributor.
A12009	Temperature unit	Warning	Internal temperature of the whole unit is out of range Temperature:	/	1. The ambient temperature is out of range. 2. The cooling fan goes wrong. 3. The dust screen is blocked. 4. The air vent is blocked in the specified range.	1. Check if the air vent is blocked. Clean the dust screen if it is blocked. 2. Check if enough space is reserved between the air vent and the wall. If not, reallocate the

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
						instrument. 3. Check if the error is accidental. 4. If the error is not accidental, contact our customer service department or your local distributor.
A14006	Reagent refrigeration unit	Warning	Reagent refrigeration temperature is out of range	/	1. The ambient temperature is out of range. 2. The temperature sensor goes wrong. (component error and cable error) 3. The temperature protection switch goes wrong. (component error and cable error) 4. The cooler goes wrong. (component error and cable error) 5. The fan goes wrong. (component error and cable error) 6. The recycle pump goes wrong. (component error and cable error) 7. The refrigerant goes wrong. 8. PCB error 9. Parameters are lost. 10. Electromagnetic interference exists.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A14011	Temperature unit	Warning	Refrigeration Radiating Fan1 is abnormal	/	1. The fan is blocked. 2. The fan is damaged. 3. The power supply goes wrong.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A14012	Reagent refrigeration unit	Warning	Reagent refrigerating fan 2 is abnormal	/	1. The fan is blocked. 2. The fan is damaged. 3. The power supply goes wrong.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.

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Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
						distributor.
A14013	Reagent refrigeration unit	Error	Light source fan is abnormal	/	The fan is blocked. The fan is damaged. The power supply goes wrong.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A14014	Temperature unit	Warning	Vacuum pump fan is abnormal	/	1. The fan is blocked. 2. The fan is damaged. 3. The power supply goes wrong.	1. Check if the error is accidental. 2. If the error is not accidental, contact our customer service department or your local distributor.
A21001	Probe Interior Wash Unit	Error	Sensor of probe interior wash syringe is in incorrect status.	/	The assembly is jammed or damaged.	Switch off the analyzing unit power and switch on it again. Recover failure by performing the Home maintenance procedure. If this message appears for 3 times, contact our customer service department or your local distributor.
A22001	ISE unit	Error	Slope out of range, electrode:%s	SLEX	1. Electrode installation incorrect. 2. Calibrator expired. 3. Electrode degenerated. 4. Bubbles in reference electrode. 5. Reference electrode has been used for a long time. 6. Electrodes interfered. 7. Module or tubing temperature above 32°C.	1. Reinstall the electrode. 2. Replace the calibrator. 3. Replace the problematic electrode and rerun. 4. Remove the electrode and clap on it to eliminate bubbles. Reinstall the electrode and run calibration. 5. Replace reference electrode and rerun. 6. Troubleshoot the electrodes by replacing them in different groups. 7. Monitor temperature, if too

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
						high, relocate equipment.
A22002	ISE unit	Error	Air in sample	SMPA	1. Sample is insufficient or contain much bubbles after dispensing. 2. No or insufficient sample has been dispensed into the sample injection port. 3. Liquid leakage due to that the electrodes are not properly installed. 4. The waste pump tube is aging or broken.	1.&2. Increase the sample volume. At least 90μl sample should be prepared. 3. Electrode is not installed correctly. Reinstall it. 4. Check the waste tube, and if necessary, replace it.
A22004	ISE unit	Error	ISE unit cannot be connected.	/	1. ISE power supply failure. 2. ISE communication cable failure. 3. Communication interface or pins failure. 4. ISE main control board failure.	1. Replace the 24V power supply board. 2. Replace the ISE communication cable. 3. Replace the interface or pins. 4. Replace the ISE main control board.
A22005	ISE unit	Error	ISE unit response error	/	1. ISE communication cable failure. 2. Communication interface or pins failure. 3. ISE main control board failure. 4. Software failure.	1. Replace the ISE communication cable. 2. Replace the interface or pins. 3. Replace the ISE main control board. 4. Upgrade the operating software or reinstall it.
A22006	ISE unit	Error	Purge A and B failed.	/	1. Leaks exist due to improperly-installed electrode or missing O ring. 2. Sample injection port or electrode inside is clogged. 3. Calibrator is exhausted. 4. Prime combinations are not enough. 5. Pump tube is aging, blocked, or broken. 6. Calibrator cannot be dispensed normally due to clogged reagent pack tube.	1. Reinstall the electrode and check for O ring. 2. Use warm water to clean and unclog the sample injection port with fresh water and unclog the electrode tube. Check the reference electrode for crystallized salt. 3. Replace the reagent pack. 4. Increase the prime cycle.

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Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
						5. Replace the pump tube. 6. Unclog the reagent pack tube with warm water.
A22007	ISE unit	Warning	ISE reagent is going to be exhausted.	/	Calibrator is exhausted.	Replace the reagent pack with a new one.
A22008	ISE unit	Error	Na ⁺ electrode voltage overflow, electrode: %s	VOUT	1. Electrode or reagent pack failed. 2. Electrode is unsteady. 3. New reagent pack is unsteady. 4. Reference electrode has been used for over 6 months. 5. ISE main control board failure.	1. Replace the problematic electrode and reagent pack. 2. New electrode will become steady after 15 minutes since installed. 3. Run a couple of calibrations after installing new reagent pack. 4. Replace the reference electrode. 5. Replace the ISE main control board.
A22009	ISE unit	Error	Electrode slope drift. Electrode voltage drift. Electrode:	VDRF/SLDR	1. Electrode or reagent pack failed. 2. Electrode is unsteady. 3. New reagent pack is unsteady. 4. Reference electrode has been used for over 6 months. 5. ISE main control board failure. 6. Ambient temperature fluctuates drastically.	1. Replace the problematic electrode and reagent pack. 2. New electrode will become steady after 15 minutes since installed. 3. Run a couple of calibrations after installing new reagent pack. 4. Replace the reference electrode. 5. Replace the ISE main control board. 6. Control the ambient temperature to make the fluctuation within +/-4°C.
A22010	ISE unit	Error	Voltage noise, electrode: %s	NOIS	1. Electrode failure. 2. Environment interference.	1. Replace the electrode. 2. Relocate the instrument.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
					3. ISE main control board failure. 4. Salt buildup around electrodes or tubes due to fluidic leaks.	3. Replace the ISE main control board. 4. Clean the tubes and electrodes.
A22011	ISE unit	Error	Air in calibrator B	PUGB	1. Calibrator B is exhausted. 2. Bubbles exist in calibrator tube B. 3. Pump tube B is aging, blocked, or broken. 4. Waste pump tube B is aging, blocked, or broken. 5. Sample injection port and fluidic path are blocked or leaking. 6. Air bubble detector is failed.	1. Replace the reagent pack with a new one. 2. Perform purge B to remove bubbles. 3.&4. Replace the pump tube. 5. Clean the sample injection port and reinstall electrodes. 6. Replace the air bubble detector.
A22012	ISE unit	Error	Air in calibrator A	PUGA	1. Calibrator A is exhausted. 2. Bubbles exist in calibrator tube A. 3. Pump tube B is aging, blocked, or broken. 4. Waste pump tube B is aging, blocked, or broken. 5. Sample injection port and fluidic path are blocked or leaking. 6. Air bubble detector is failed.	1. Replace the reagent pack with a new one. 2. Perform purge A to remove bubbles. 3.&4. Replace the pump tube. 5. Clean the sample injection port and reinstall electrodes. 6. Replace the air bubble detector.
A22013	ISE unit	Error	Pump calibration result error	/	1. Pump tube is aging. 2. Sample probe aspiration/dispensing failure.	1. Replace the pump tube. 2. Replace the sample probe.
A22014	ISE unit	Error	Air bubble detector failure	/	1. Air bubble detector board is eroded due to the leaks at the joint of sample injection port and bubble detector. 2. Air bubble detector is failed.	Replace the bubble detector.
A22015	ISE unit	Error	Reading reagent pack chip error	/	1. Reagent pack is not installed. 2. Reagent pack wand is failed.	1. Install reagent pack. 2. Replace the wand.
A22016	ISE unit	Error	Writing reagent pack chip error	/	1. Reagent pack is not installed. 3. Reagent pack wand is failed.	1. Install reagent pack. 3. Replace the wand.
A22017	ISE unit	Error	Air in ISE wash	/	1. ISE wash solution is insufficient.	1. Place sufficient ISE wash

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Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			solution		2. Waste pump tube B is aging, blocked, or broken. 3. Sample injection port and fluidic path are blocked or leaking. 4. Air bubble detector is failed.	solution. 2. Replace the pump tube. 3. Clean the sample injection port and reinstall electrodes. 4. Replace the air bubble detector.
A22018	ISE unit	Error	No fluid in tubing	CALF	1. Waste pump tube B is aging, blocked, or broken. 2. Sample injection port and fluidic path are blocked or leaking. 3. Air bubble detector is failed.	1. Replace the pump tube. 2. Clean the sample injection port and reinstall electrodes. 3. Replace the air bubble detector.
A22019	ISE unit	Error	Saving calibration result error	DEP	1. ISE communication cable failure. 2. Communication interface or pins failure. 3. ISE main control board failure. 4. Software failure.	1. Replace the ISE communication cable. 2. Replace the interface or pins. 3. Replace the ISE main control board. 4. Upgrade the operating software or reinstall it.
A22021	ISE unit	Error	Command format or execution error	RESP	1. ISE communication cable failure. 2. Communication interface or pins failure. 3. ISE main control board failure. 4. Software failure.	1. Replace the ISE communication cable. 2. Replace the interface or pins. 3. Replace the ISE main control board. 5. Upgrade the operating software or reinstall it.
A22022	ISE unit	Error	Air in segment	CALM	1. Waste pump tube B is aging, blocked, or broken. 2. Sample injection port and fluidic path are blocked or leaking. 3. Air bubble detector is failed.	1. Replace the pump tube. 2. Clean the sample injection port and reinstall electrodes. 3. Replace the air bubble detector.
A22023	ISE unit	Error	No reagent module has been loaded.	/	1. Reagent pack is not installed. 2. Reagent pack wand is failed.	1. Install reagent pack. 2. Replace the wand.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
A22024	ISE unit	Error	ISE response check code error	RESP	1. The communication wire between ISE and the middle-layer unit goes wrong. 2. Communication interface or pin error. 3. Main control board does not function. 4. Software error	1. Replace the communication wire 2. Change the interface or the pin. 3. Change the main control board. 4. Upgrade the software or reinstall the software.
A22025	/	Error	Equipment cannot be connected Unit:	/	1. The serial port wire is not connected 2. The power supply of the analyzer is off.	1. Check the connection of the serial port wire. 2. Remove and replug in the serial port wire. 3. Check whether the power supply of the analyzer is on. 4. Perform the Home procedure. 5. Reboot the PC and the analyzer. 6. If the error still remains for continuous three times after the above procedures have been performed, please contact our customer service or your local distributor.
A22026	/	Error	Configuring key parameters failed. Unit: %s	/	Key parameters are not configured	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
A22027	/	Error	Fluidic prime failed.	/	Fluidic is not primed.	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
A22028	/	Error	Downloading key parameters failed. Unit:	/	1.Downloading key parameters failed. 2. Reading the parameters from E2ROM failed. 3.Configuring the parameters of the smart module failed.	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
A22029	/	Error	Collecting dark current failed.	/	Collecting dark current failed.	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
A22030	/	Error	Discharging primary vacuum waste failed.	/	Discharging primary vacuum waste failed.	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
A22031	/	Error	Establishing vacuum failed.	/	Establishing vacuum failed.	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
A22032	/	Error	Floater status error: Floater:	/	1. Low level floater status of the water tank is full. 2. The floater status of the diluted wash solution container is full. 3. The floater status of the concentrated wash solution container is full. 4. The floater status of the low	1. Check the water unit and the water supply tubes. 2. Check the floater status of the water tank, diluted wash solution container, concentrated wash solution container, low concentration waste container and

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
					concentration waste container is empty. 5. The status of the primary vacuum container is empty. 6. The status of the high concentration waste container is empty.	the high concentration waste container 3. Check the status of the primary vacuum container. 4. Check if the error is accidental. 5. If the error is not accidental, contact our customer service department or your local distributor.
A22033	/	Error	Resetting probe interior wash syringe failed.	/	Resetting probe interior wash syringe failed.	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
A22034	/	Error	Resetting auto wash syringe failed.	/	Resetting auto wash syringe failed.	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
A22035	/	Error	Releasing vacuum failed.	/	1. Electromagnetic Valve V23, V24, V25, V26 or V27 failure 2. Vacuum pump failure 3. Vacuum sensor failure	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
A22036	/	Error	Initializing sample bar code reader failed.	/	Sample bar code reader failed due to system error.	1. Recover failure by performing the Home maintenance procedure. 2. If this error remains contact our customer service

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
						department or your local distributor.
A22037	/	Error	Initializing reagent bar code reader failed.	/	Reagent bar code reader failed due to system error.	1. Recover failure by performing the Home maintenance procedure. 2. If this error remains contact our customer service department or your local distributor.
A22038	/	Error	Scanning reagent bar code failed.	/	Scanning reagent bar code failed.	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
A22039	/	Error	Unmatched software version.	/	1.Version inquiry instruction failed. 2. The version information of the control software does not match the one stored in the operating software.	1. Turn off the analyzing unit power and reswitch it on. 2. If the error occurs for continuous three times, please contact our customer service or your local distributor.
C00007	Operating system	Error	CPU performance low	/	The CPU is too busy.	Reboot the computer and operating software. If this message appears for 3 times, contact our customer service department or your local distributor.
C00011	Operating system	Error	The last abnormal exit may cause carryover not handled. Execute the Special Wash maintenance	/	The operating software is abnormal, or the instrument power is cut off unexpectedly.	Restart the operating software, and execute the Special Wash maintenance command before starting analysis.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			command before starting analysis to ensure accurate results.			
C00012	Operating system	Warning	Sound card failure	/	No sound card is installed. Sound card failure. Incorrect sound card driver.	Reinstall the sound card or the sound card driver.
C01001	Instrument connection	Error	Equipment cannot be connected	/	The serial cable is not connected; or the analyzing unit power is switched off.	Check the serial port connection. Replug the cable. Check if the analyzing unit is powered on. Start the initialization again. Restart the computer and analyzing unit. If three continuous attempts are failed, contact our customer service department or your local distributor.
C02001	Database	Error	Database initialing failed	/	The database file is damaged or lost.	Reboot the computer and analyzing unit. If three continuous attempts are failed, contact our customer service department or your local distributor.
C02002	Database	Error	Database upgrade failed	/	The database file is damaged or lost.	Reboot the computer and analyzing unit. If three continuous attempts are failed, contact our customer service department or your local distributor.
C02004	Database	Warning	Database backup failed	/	The database file is damaged or lost.	Reboot the computer and analyzing unit. If three continuous attempts are failed, contact our customer service department or your local distributor.
C02005	Database	Warning	Reading/Writing	/	The database does not work normally.	Reboot the computer and analyzing unit. If three continuous attempts

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			database failed			are failed, contact our customer service department or your local distributor.
C03001	Result calculation	Warning	Result cannot be calculated Sample ID/bar code: Position: Chemistry:	RCE	Absorbance data for calculation is incomplete, or the dividend is 0.	Rerun the test. If the error remains, contact our customer service department or your local distributor.
C03002	Result calculation	Warning	Absorbance out of range Sample ID/bar code: Position: Chemistry:	ABS	The absorbance measured at the primary and secondary wavelength is greater than 3.4A.	Check the sample for foreign matters or interferents; check if the reagent is qualified and placed in the correct position; check the cuvette is clean; check if the photometric system is working normally. If the problem remains, contact our customer service department or your local distributor.
C03003	Result calculation	Warning	R1 blank absorbance out of range Sample ID/bar code: Position: Chemistry:	RBK	The reagent goes wrong; the cuvette is not clear; the reaction cuvette is overflowed; or insufficient reagent is dispensed.	Check if the reagent is sufficient without air bubbles and the chemistry parameters are reasonable. If yes, replace the reagent and then rerun the test. Check if the cuvette is normal. If the error remains, contact our customer service department or your local distributor.
C03004	Result calculation	Warning	Substrate depletion Sample ID/bar code: Position: Chemistry:	BOE	The sample concentration is too high, and substrate depletion occurs during fixed-time measurements.	Check the reaction curve and the substrate depletion limit. Rerun the test with diluted sample.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
C03005	Result calculation	Warning	Result cannot be calculated Sample ID/bar code: Position: Chemistry:	ENC	The sample concentration is too high, and substrate depletion occurs within the lag time of rate check measurements.	Check the reaction curve and the substrate depletion limit. Rerun the test with diluted sample.
C03006	Result calculation	Warning	Linearity limit out of range Sample ID/bar code: Position: Chemistry:	LIN	The measuring points for result calculation are nonlinear, because the sample concentration is too high, or the substrate depletion limit is not specified or unreasonable.	Check the reaction curve and the substrate depletion limit. Rerun the test with diluted sample.
C03007	Result calculation	Warning	Prozone check error Sample ID/bar code: Position: Chemistry:	PRO	Antibody excess occurs due to too high sample concentration.	Check the reaction curve and the prozone check parameters. Rerun the test with diluted sample.
C03008	Result calculation	Warning	Sample concentration is higher than that of the highest-level calibrator Sample ID/bar code: Position: Chemistry:	RRN	The sample concentration exceeds the high limit of the calibrator concentration.	Rerun the test with diluted sample.
C03009	Result calculation	Warning	Mixed blank absorbance out of range Chemistry: Calibrator: Position:	MBK	The reagent goes wrong; the cuvette is not clear; the reaction cuvette is overflowed; or insufficient reagent is dispensed.	Check if the reagent is sufficient without air bubbles and the chemistry parameters are reasonable. Check if the cuvette is normal. Replace the reagent and then rerun the test. If the error remains, contact our customer service department or your local distributor.

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Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
C03010	Result calculation	Warning	Blank response out of range Chemistry: Calibrator: Position:	BLK	The reagent goes wrong; insufficient reagent is dispensed; the cuvette contains air bubbles; the light drifts; or the cuvette is overflowed.	Check if the reagent is sufficient without air bubbles and the chemistry parameters are reasonable. Check if the cuvette is normal. Replace the reagent and then rerun the test. If the error remains, contact our customer service department or your local distributor.
C03011	Result calculation	Warning	Calibration repeatability exceeds limit. Chem:	DUP	The difference between the maximum and minimum response of the calibrator exceeds the specified limit.	Check if the acceptance limit is reasonable, troubleshoot the error, and then recalibrate.
C03012	Result calculation	Warning	Calibration sensitivity exceeds limit. Chem:	SEN	The difference of final response of the maximum and minimum concentration calibrators exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
C03013	Result calculation	Warning	Calibration curve standard deviation out of range Chemistry: Calibrator: Position:	CSD	The calculated standard deviation of the calibration curve exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
C03014	Result calculation	Warning	Calibration determination coefficient out of range Chemistry: Calibrator: Position:	DET	The calculated determination coefficient of the calibration curve exceeds the specified limit.	Check if the acceptance limit is reasonable and the reagent and calibrator are normal, and then recalibrate.
C03015	Result calculation	Warning	Calibration slope difference out of	FAC	The slope difference is applicable to linear calibration only and refers to the K factor	Check if the acceptance limit is reasonable and the reagent and

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			range Chemistry: Calibrator: Position:		(slope) difference between two adjacent calibrations. It exceeds the specified limit.	calibrator are normal, and then recalibrate.
C03016	Result calculation	Warning	Calibration curve not monotonic Chemistry: Calibrator: Position:	MON	The calibration data and calibration curve are not monotonic.	Check if the calibrator is defined and placed correctly, and then recalibrate.
C03017	Result calculation	Warning	Calibration curve not convergent Chemistry: Calibrator: Position:	COV	For nonlinear calibration, a satisfying base cannot be calculated and no calibration curve is drawn.	Check that the reagent and calibrator are normal, and then recalibrate. If the error remains, contact our customer service department or your local distributor.
C03018	Result calculation	Warning	Chemistry: Control: 1-2s warning	1-2s	The QC result is between ± 2 and ± 3 standard deviations from the assigned mean concentration.	No actions are required.
C03019	Result calculation	Warning	Chemistry: Control: 1-3s out of control	1-3s	The QC result is greater than ± 3 standard deviations from the assigned mean concentration.	Check if the reagent is qualified and control is normal. If the error remains, contact our customer service department or your local distributor.
C03020	Result calculation	Warning	Chemistry: Control: 2-2s out of control	2-2s	Results of two controls or two results of one control within a run are simultaneously greater than +2 or -2 standard deviations from the assigned mean.	Check if the reagent is qualified and control is normal. If the error remains, contact our customer service department or your local distributor.
C03021	Result calculation	Warning	Chemistry: Control: R-4s out of	R-4s	One result of a run is greater than +2 standard deviations from the assigned mean	Check if the reagent is qualified and control is normal. If the error remains, contact our customer

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Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			control		and the other greater than -2SDs.	service department or your local distributor.
C03022	Result calculation	Warning	Chemistry: Control: 4-1s out of control	4-1s	Results of two runs in two-control evaluation or four continuous results of a control are greater than +1 or -1 standard deviation from the assigned mean concentration.	Check if the reagent is qualified and control is normal. If the error remains, contact our customer service department or your local distributor.
C03023	Result calculation	Warning	Chemistry: Control: 10-x out of control	10-x	Results of five runs in two-control evaluation or ten continuous results of a control that are being compared are on the same side.	Check if the reagent is qualified and control is normal. If the error remains, contact our customer service department or your local distributor.
C03024	Result calculation	Error	Biochemistry test period time out. Cannot continue	/	1. Software error 2. Operating system error	Rerun the test. Reboot the operating software, analyzing unit and computer. If the error remains, contact our customer service department or your local distributor.
C03026	Result calculation	Warning	Photoelectric data is lost	/	Communication error.	If the error persists, contact our customer service department or your local distributor.
C03030	Result calculation	Error	Photoelectric measurement period is out of range Sample ID/bar code: Position: Chemistry:	/	1. Software error	1. Rerun the operating software. 2. Reboot the operation unit. 3. If the error remains, contact our customer service department or your local distributor.
C03031	Result calculation	Error	Multiple photoelectric measurements are time out	/	1. Software error	1. Rerun the operating software. 2. Reboot the operation unit. 3. If the error remains, contact our customer service department or

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			Sample ID/bar code: Position: Chemistry:			your local distributor.
C04001	Sample bar code	Warning	Duplicate sample bar code. Sample ID/bar code: Position 1: Position 2:	/	Duplicate bar code is used.	Replace the duplicate sample bar code label.
C04002	Sample bar code	Warning	Bar code has no corresponding programming. Sample ID bar code: Position:	/	The sample of the bar code has not been programmed.	Program the sample of the bar code.
C04006	Sample bar code	Warning	Sample is expired Sample ID/bar code: Position:	/	The sample is loaded after its shelf life is exceeded.	The sample is expired. Replace the sample and program it again. Reject the expired sample. If the sample shelf life is too short, change it to a reasonable one.
C04008	Sample bar code	Warning	Sample bar code too long. Position:	/	The bar code length is greater than the maximum value of 27 digits.	Redefine the bar code with no more than 27 digits.
C04009	Sample bar code	Warning	Sample bar code is less than 3 digits. position:	/	The sample bar code is too short, less than the minimum range of 3 digits.	Reprint the bar code and ensure it is no less than 3 digits.
C04012	Sample bar code	Warning	Sample bar code analysis error Sample bar code: Position:	/	Barcode information does not conform with the barcode format	Reset the barcode format or reprint the barcode and scan it.
C05001	Reagent bar code	Warning	Duplicate reagent bar code Reagent:	/	Incorrect reagent or reagent bar code is being used, or an invalid reagent bar code is being used. Bar code is aligned with	Reprint the reagent bar code, or replace the reagent bottle with an invalid bar code.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
			Position 1: Position 2:		reagents, and cannot be used again for new reagent when a reagent is exhausted.	
C05002	Reagent bar code	Warning	Reagent bar code information error. Position:	/	Incorrect reagent bar code is being used, or reagent bar code is not configured reasonably. The reagent bar code contains incomplete or incorrect reagent information, such as expiration date, reagent volume, etc.	Print the new reagent bar code with correct settings and check the bar code against the settings. Replace the reagent bottle, or contact the reagent supplier.
C05003	Reagent bar code	Warning	Reagent bar code analysis error Position:	/	Incorrect reagent bar code is being used, or reagent bar code settings are incorrect. The system is failed to extract reagent information from the bar code.	Check the reagent bar code settings, or reprint the reagent bar code against the settings. Contact the reagent supplier.
C05006	Reagent bar code	Error	Wash solution position on reagent carousel is occupied by another reagent Position:	/	Reagent rather than wash solution is placed in the fixed wash solution position (D) on reagent carousel.	Reposition the reagent, or remove it from the fixed reagent position.
C05008	Reagent bar code	Error	Physiological saline position on reagent carousel is occupied by another reagent Position:	/	Reagent rather than physiological saline is placed in the fixed physiological saline position (W) on reagent carousel.	Reposition the reagent, or remove it from the fixed wash solution position.
C06001	Host communication	Error	LIS initialization error	/	Host file is damaged or does not exist.	Reinstall the operating software.
C06002	Host communication	Error	LIS communication parameter error	/	Host parameters error	Re-set or modify the host communication parameters.
C06003	Host communication	Error	LIS communication error	/	Communication error	If the error occurs accidentally, send or receive the instruction again. If the error still remains, contact our customer service department or your local

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
						distributor.
C06004	Host communication	Error	LIS host cannot be connected	/	Abnormal network connection or the LIS host is not started.	Check LIS connection and network cable. Check if LIS host and LIS station can start normally.
C06005	Host communication	Warning	Sending sample results failed. Sample ID/bar code: Position:	/	Communication error	If the error occurs accidentally, send or receive the instruction again. If the error still remains, contact our customer service department or your local distributor.
C06006	Host communication	Warning	Sending sample information failed. Sample ID/bar code: Position:	/	Communication error	If the error occurs accidentally, send or receive the instruction again. If the error still remains, contact our customer service department or your local distributor.
C06007	Host communication	Warning	Inquiring sample information failed. Sample ID/bar code: Position:	/	LIS host failure.	If the error occurs accidentally, neglect it. If the error occurs frequently, contact the manufacturer of LIS or contact our customer service department or your local distributor.
C06008	Host communication	Warning	Downloading sample failed. Sample ID/bar code: Position:	/	Incorrect channel settings or insufficient or redundant chemistries on the LIS host.	Check and re-set the chemistry correspondence between the operating software and the LIS host.
C07003	Light source	Error	Light intensity is too weak	/	1. The lamp is not installed correctly. 2. The cuvette is contaminated. 3. The lamp is aging. 4. The wash station dispenses liquid incorrectly.	1. Check if the lamp is installed correctly. 2. Perform the diluted wash procedure and then the lamp check procedure.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
						3. Replace the lamp. 4. Check if the wash station dispenses liquid with correct volume to reaction cuvettes. 5. If your attempt is failed, contact our customer service department or your local distributor.
C07004	Light source	Warning	Cuvette blank out of range Cuvette No.:	/	1. The cuvette is contaminated. 2. The lamp is aging. 3. The lamp is not installed correctly. 4. The wash station dispenses liquid incorrectly. 5. The photoelectric collection board goes wrong.	1. Open the reaction carousel and check if the lamp is turned on. If it is not, rerun the operating software. 2. Check if the lamp is installed correctly. 3. Perform the diluted wash procedure and then the cuvette check procedure. 4. Replace or clean the failed cuvette. 5. Replace the lamp. 6. Check if the wash station dispenses liquid with correct volume to reaction cuvettes. 7. If your attempt is failed, contact our customer service department or your local distributor.
C07005	Light source	Error	Lamp is not turned on	/	1. The lamp is damaged. 2. The lamp cable is not connected properly. 3. The power board of the lamp is not connected properly. 4. The power supply of the analyzing unit is disconnected.	1. Open the reaction carousel and check if the lamp is turned on. If it is not, rerun the operating software. 2. Check if the lamp cable is tightened.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
					5. The photoelectric collection board goes wrong.	3. Replace the lamp. 4. Check if the connect of the lamp power board is loose, and if necessary, reinsert the connector. 5. If your attempt is failed, contact our customer service department or your local distributor.
C07006	Light source	Error	Light intensity is too strong	/	1. A cuvette position has no cuvette installed. 2. The circuit gain is too high and beyond the measurement range.	1. Check if all cuvette positions have cuvettes installed. 2. Contact our customer service department or your local distributor to adjust the gain.
C07007	Light source	Error	Dark current is too high Channel: AD:	/	1. The circuit gain is too high and beyond the measurement range. 2. The power board of the lamp is not connected properly. 3. The photoelectric collection board goes wrong.	3. If three continuous attempts are failed, contact our customer service department or your local distributor.
C07008	Light source	Warning	Lamp has exceeded its life span. Replace it immediately.	/	1. The lamp has been used for over 2000 hours. 2. The lamp has been replaced incorrectly.	1. Replace the lamp. 2. Perform the Replace Lamp maintenance procedure again.
C07009	Light source	Error	Water blank out of range (10X)	/	1. The cuvette is overflowing. 2. The lamp has been replaced incorrectly. 3. Cuvette check is not performed after maintenance. 4. The cable connectors are not tightened. 5. The retaining screw is not tightened. 6. Cleaning liquid inside the cuvette is little. 7. The lamp is aged. 8. The photometer goes wrong.	1. Check if the cuvette is overflowing. 2. Check if the Replace Lamp command is executed during lamp replacement. 3. Check if the Cuvette Check command is executed after maintenance. 4. Check if the cleaning liquid inside the cuvette is no less than

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
						half of the cuvette. 5. Check if the cable connectors and retaining screw of the lamp have been tightened. 6. Check if the reaction curve fluctuates irregularly. If yes, replace the lamp. 7. If the error remains, contact our customer service department.
C07012	Other error of operation unit	Warning	Storage device error. Cannot import data	/	No U disk is inserted. No file is found in the U disk, or file error, or file is damaged. The U disk is locked or damaged.	Check if a U disk is inserted or full. Check if the storage device is damaged.
C07013	Other error of operation unit	Warning	Storage device error. Cannot export data	/	No U disk is inserted. Insufficient disk space. The U disk is locked or damaged.	Check if a U disk is inserted or full. Check if the storage device is damaged.
C07014	Other error of operation unit	Warning	Reagent exhausted Chemistry: Position:	/	All reagents of the reagent type for the chemistry are less than the minimum limit. All reagents of the type are too little to be detected.	Refill or replace the reagent.
C07016	Other error of operation unit	Warning	Insufficient wash solution Position:	/	Insufficient wash solution on the reagent carousel.	Refill the wash solution on the reagent carousel.
C07017	Other error of operation unit	Warning	Wash solution is exhausted Position:	/	The wash solution on the reagent carousel is exhausted.	Refill the wash solution on the reagent carousel.
C07022	Other error of operation unit	Warning	Less than X tests are left in biochemistry reagent Chemistry: Position:	/	All reagents of the reagent type for the chemistry are less than the minimum limit. All reagents of the type are too little to be detected.	Refill or replace the reagent.

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
C07023	Other error of operation unit	Warning	Chemistry: %s, 30 minutes left for next calibration.	/	The calibration factors will be expired.	Recalibrate the chemistries.
C07027	Other error of operation unit	Warning	Calibrator %s has been expired	/	The calibrator is expired.	Replace the calibrator.
C07028	Other error of operation unit	Warning	Chemistry: %s, lot No.: %s, position: %s, has been expired	/	The reagent is expired.	Replace the reagent.
C07029	Other error of operation unit	Warning	Chemistry: %s, lot No.: %s, position: %s, has exceeded the uncapping time	/	The uncapping time of the reagent pack is too long.	Replace the reagent.
C07034	Other error of operation unit	Warning	Insufficient physiological saline Position:	/	Insufficient physiological saline.	Refill the physiological saline on the reagent carousel.
C07035	Other error of operation unit	Warning	Physiological saline is exhausted Position:	/	Physiological saline is exhausted.	Refill the physiological saline on the reagent carousel.
C07036	Other	Warning	Chemistry: %s. Calibration factors are expired	/	The calibration factors have been expired.	Recalibrate the chemistry.
C07037	Other	Warning	Reagent bottle number of %s chemistry is changed. Please recalibrate	/	Serial number of the reagent is changed.	Recalibrate the chemistry.
C07038	Other	Warning	Reagent lot number of %s chemistry is changed. Please recalibrate	/	Lot number of the reagent is changed.	Recalibrate the chemistry.

17 Alarms and Troubleshooting

Event ID	Component	Event class	Error Message and Event Log	Flag	Probable Causes	Corrective Actions
C07039	Other	Warning	Calibration factors of %s chemistry are expired. Recalibrate	/	The calibration factors are expired.	Recalibrate the chemistry.
C07040	Other	Warning	Reagent exhausted Chemistry:	/	The reagent is running out. The reagent is too little to be detected.	Refill or replace the reagent.
C07041	Other	Error	ISE reagent is less than %s	/	ISE reagent inventory is below the alarm limit	Check the inventory. If the reagent is insufficient, load the reagent.
C07042	Other	Warning	%s, lot number: %s, position: %s, has been expired	/	Solutions and reagents other than biochemical reagents are expired.	Replace the expired reagents or solutions.

18

Template Modifying Software

The Template Modifying Software is affiliated with the Operating Software and used to create or edit print templates, which illustrate the contents and format of patient reports.

The Template Modifying Software can be started separately or together with the Operating Software. To start the Template Modifying Software, select the **Edit** button on the **Print** page of the operating software.

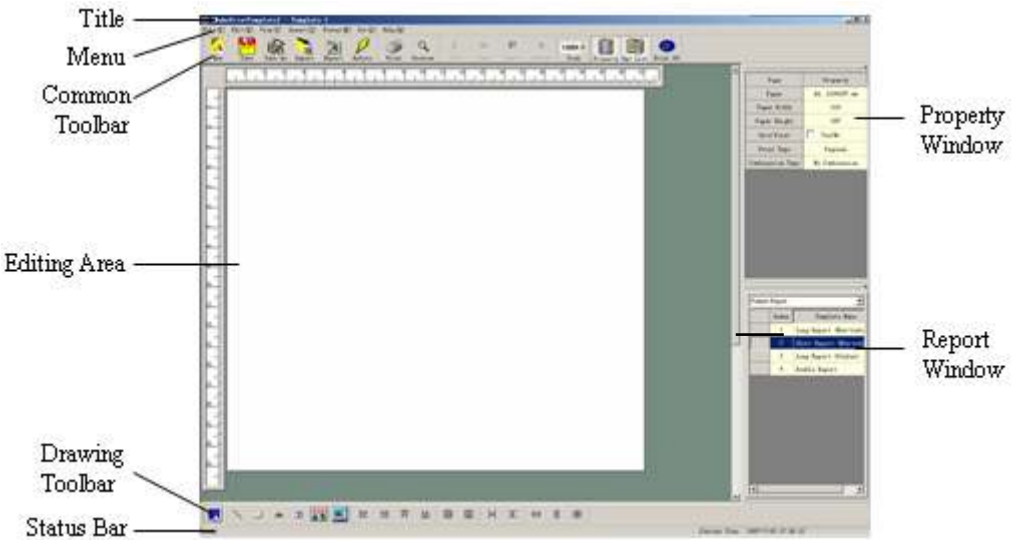
The following sections introduce the Template Modifying Software by menus and toolbars.

18.1 Main Screen

18.1.1 Main Screen

The following figure shows the main screen of the Template Modifying Software.

Figure 18.1 Main screen



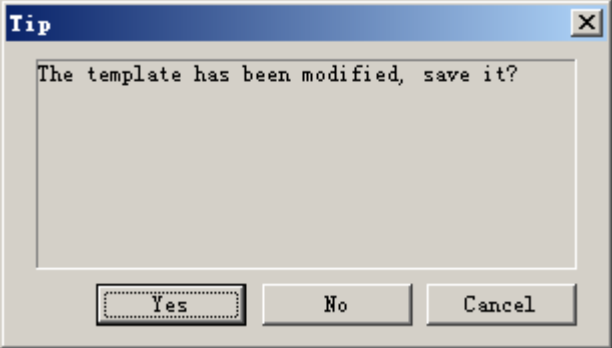
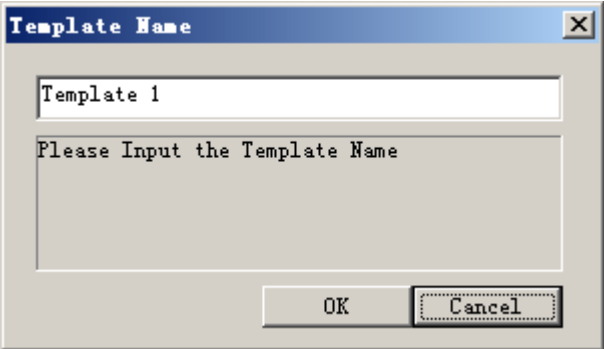
18.1.2 File (F)

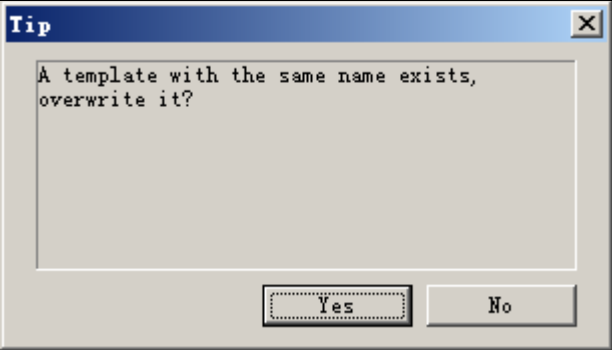
The **File** menu is used to create, save, import/export and print out the templates. Select **File** on upper-left corner of the main screen. The **File** menu is displayed.

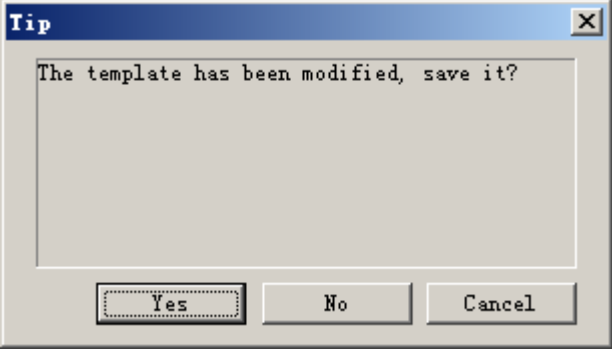
New (N)	Ctrl+N
Save (S)	Ctrl+S
Save As (A)...	
Delete (D)	
Import (I)...	
Export (E)...	
Print (P)...	Ctrl+P
Preview (V)...	
Exit (X)	Alt+F4

The following table explains the menu in detail.

Option	Description
New	Select New to create a template. The type of the template is determined by the report that is currently selected on the report

Option	Description
	window. You can also use the shortcut key Ctrl+N instead. After changing the currently-displayed template, select New to display the following dialog box.  The dialog box is titled 'Tip' and contains the text 'The template has been modified, save it?'. It has three buttons: 'Yes', 'No', and 'Cancel'. • Select Yes to save the changes and create a template. • Select No to cancel the changes and create a template. • Select Cancel to abort creating and return to the previous screen.
Save	Select Save to save the newly-created template or the changes to a template. You can also use the shortcut key Ctrl+S instead. To save a new template, you should define the template name:  The dialog box is titled 'Template Name' and contains a text box with 'Template 1' and a label 'Please Input the Template Name'. It has two buttons: 'OK' and 'Cancel'. • Enter the name in the edit box. • Select OK to save the template and add the name to the template list on the report window. • Select Cancel to abort saving and return to the previous screen. If a template with the same name already exists, the following dialog box pops up.

Option	Description
	 • Select Yes to overwrite the template. • Select No to cancel saving and return to the previous screen.
Save As	Save the current template with another name. If a template with the same name already exists, a dialog box appears to ask for your confirmation.
Preview	Select this option to view the template exactly as it will be printed out. The main screen will be hidden when you preview a template. The tool bar on the Preview window is as follows.  • If the template has more than one page,  and  are available. • : Go to the first page. • : Go to the previous page. • : Go to the specified page. • : Go to the next page. • : Go to the last page. • : Select to expand the template view among 25%, 50%, 75% and 100%. The default is 100%. • : Select to shrink the template view. • : Print out the template. It is equivalent to the Print

Option	Description
	option in the File menu. : Select to exit the preview window and return to the template.
Exit	Select this option to close the Template Modifying Software. You can use the shortcut key Alt+F4 instead. If the template is changed, the following dialog box pops up.  - Select Yes to save the changes and exit the software. - Select No to exit the software without saving the changes. - Select Cancel to abort exiting and return to the previous screen.

18.1.3 Edit (E)

The **Edit** menu provides the functions like cut, copy, paste and delete. Select **Edit** on the menu bar of the main screen. The **Edit** menu is displayed.

Cut (T)	Ctrl+X
Copy (C)	Ctrl+C
Paste (P)	Ctrl+V
Delete (D)	Ctrl+D



NOTE

The control(s) you have cut or copied can only be pasted on the current Template Modifying Software rather than another one or other software.

The following table explains the menu in detail.

Option	Description
Cut	Select this option to copy and delete single or multiple controls. You can use the shortcut key Ctrl+X instead.

Option	Description
	This option is available only when a control(s) is selected.
Copy	Select this option to copy single or multiple controls. You can use the shortcut key Ctrl+C instead. This option is available only when a control(s) is selected.
Paste	Select this option to paste the controls that are previously cut or copied at the same place as where the controls are from. You can use the shortcut key Ctrl+P instead. This option is available only when a control(s) is cut or copied.
Delete	Select this option to delete single or multiple controls You can use the shortcut key Ctrl+D instead. This option is available only when a control(s) is selected.

18.1.4 View (V)

The **View** menu is used to enable or disable the toolbars and to set up the displaying proportion. Select **View** on the menu bar of the main screen. The **View** menu is displayed.

✓ Common Tool (C)
✓ Draw Tool (D)
✓ Property Window (P)
✓ Report Window (R)
✓ Status Bar (S)
25%
50%
75%
✓ 100%
125%
150%
175%
200%

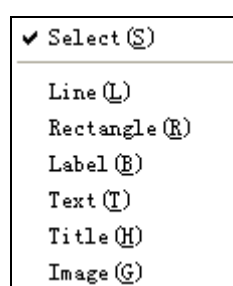
The following table explains the menu in detail.

Option	Description
Common Tool	Enable or disable the common toolbar.
Draw Tool	Enable or disable the draw toolbar.
Property Window	Enable or disable the property window.

Option	Description
Report Window	Enable or disable the report window.
Status Bar	Enable or disable the status bar.
25%-200%	Select a proportion to display the template. The default is 100%.  NOTE You are recommended to select 100% when saving a template.

18.1.5 Insert (I)

The **Insert** menu is used to create controls in the editing page. Select **Insert** on the menu bar of the main screen. The **Insert** menu is displayed.



Only one option on the **Insert** menu can be selected simultaneously.

Option	Description
Select	Select this option to change the mouse pointer to a  . When the mouse pointer changes to a  , you can select single or multiple controls in the editing area.  NOTE Selecting a control while holding the Ctrl key copies the control.
Line	Select this option to insert a line in the editing area. The mouse pointer changes into a +. Click once in the editing area and drag the mouse to draw a line.
Rectangle	Select this option to insert a rectangle in the editing area. The mouse pointer changes into a +. Click once in the editing area and drag the mouse to draw a rectangle.

Option	Description
Label	Select this option to insert a label in the editing area. The mouse pointer changes into a +. Click once in the editing area and drag the mouse to draw a label. Label is a type of text control and the contents on a label will not change when printed.
Text	Select this option to insert a text control in the editing area. The mouse pointer changes into a +. Click once in the editing area and drag the mouse to create a text. The contents in a text control will be replaced by the actual test data when printed.
Title	Select this option to insert a title in the editing area. The mouse pointer changes into a +. Click once in the editing area and drag the mouse to create a title. Title is a type of text control. The “%s” will be replaced by a hospital name when printed. Please note “%s” is added by user and not produced automatically.
Image	Select this option to insert an image in the editing area. The mouse pointer changes into a +. Click once in the editing area and drag the mouse to create an image. The image on the template is for illustration only and will be replaced by corresponding curve graph when printed.

18.1.6 Format (M)

The **Format** menu is used to arrange the controls on a template. Select **Format** on the menu bar of the main screen. The **Format** menu is displayed.

Left (A)
Right (B)
Top (C)
Bottom (D)
Center H (E)
Center V (F)
Even Space H (G)
Even Space V (H)
Same Width (I)
Same Height (J)
Same Size (K)

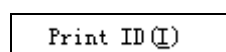
The following table explains the menu in detail.

Option	Description
Left	Align the specified controls with the left of the

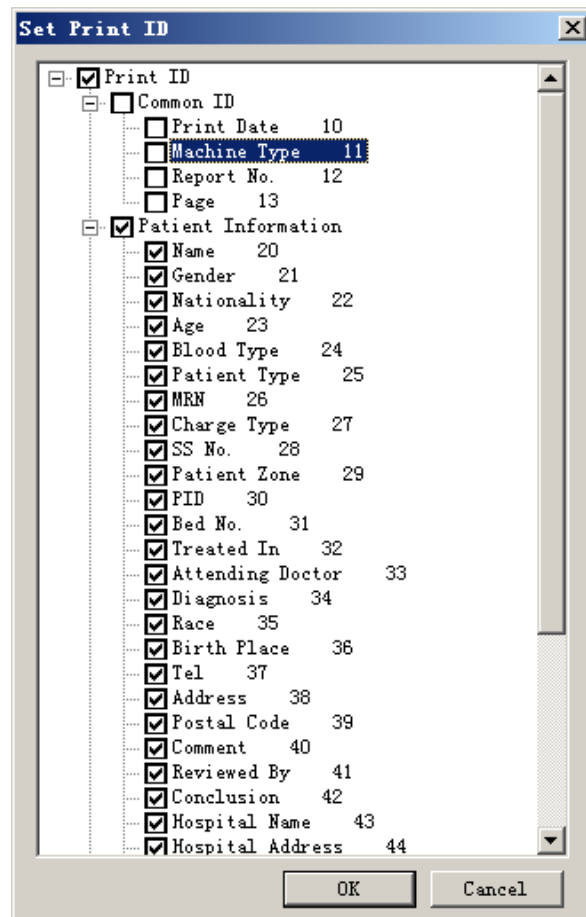
Option	Description
	lastly-selected control.
Right	Align the specified controls with the right of the lastly-selected control.
Top	Align the specified controls with the top side of the lastly-selected control.
Bottom	Align the specified controls with the bottom side of the lastly-selected control.
Center H	Align one or multiple controls to the horizontal center of current template.
Center V	Align one or multiple controls to the vertical center of current template.
Even Space H	Arrange three or more controls with same space horizontally.
Even Space V	Arrange three or more controls with same space vertically.
Same Width	Adjust the specified controls to the same width as the lastly-selected control.
Same Height	Adjust the specified controls to the same height as the lastly-selected control.
Same Size	Adjust the specified controls to the same width and height as the lastly-selected control.

18.1.7 Set(S)

The **Set** menu only includes an option, **Print ID**. See the figure below.

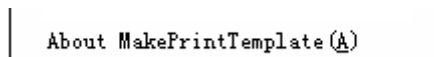


Select **Print ID**. The **Set Print ID** dialog box is displayed. You can enable or disable the print fields and view the corresponding ID of each field.



18.1.8 Help (H)

Select **Help** on the menu bar of the main screen. The **Help** menu is displayed.



The following table explains the menu in detail.

Option	Description
About MakePrintTemplate	Select this option to view the version information of the template modifying software.

18.2 Common Tools

The common toolbar provides the shortcut buttons that enables you to perform an operation quickly.

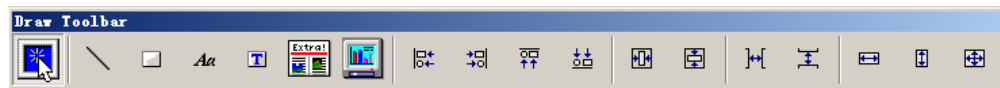


The following table shows the correspondence between the shortcut buttons and menu options.

Button	Menu Option
New	File/New
Save	File/Save (not enabled)
Save As	File/Save As
Import	File/Import
Export	File/Export
Delete	File/Delete
Print	File/Print
Preview	File/Preview
Cut	Edit/Cut
Copy	Edit/Copy
Paste	Edit/Paste
Delete	Edit/Delete
Zoom	View/25%-200%
Property	View/Property Window
Rpt List	View/Report Window
Print ID	Set/Print ID

18.3 Draw Tools

The draw toolbar provides the shortcut buttons that enables you to create and draw controls quickly.



The following table shows the correspondence between the shortcut buttons and menu options.

Button	Menu Option
	Insert/Select
	Insert/Line
	Insert/Rectangle
	Insert/Label
	Insert/Text
	Insert/Title
	Insert/Image
	Format/Left
	Format/Right
	Format/Top
	Format/Bottom
	Format/Center H
	Format/Center V

Button	Menu Option
	Format/Even Space H
	Format/Even Space V
	Format/Same Width
	Format/Same Height
	Format/Same Size

18.4 Property Window

The property window enables you to view and edit the properties of the selected control. If no control is selected in the window, the properties of the current template are displayed.

18.4.1 Page

When no control is selected, the property window shows the properties of the current template, such as paper, print type, etc.

Page	Property
Paper	A4, 210*297 mm
Paper Width	210
Paper Height	297
Grid Point	☐ Yes/No
Print Type	Paginal
Combination Type	No Combination

The following table explains the template properties in detail.

Parameter	Description
Paper	Define the paper type of the template. There are 9 common types available. If the paper width and height you defined are beyond the specified range, Custom is displayed in the Paper field.
Paper Width	Define the width of the template.
Paper Height	Define the height of the template.
Grid Point	Enable or disable grid points on the template.
Print Type	Includes Paginal and Serial.
Combination Type	Whether to print two reports on one piece of paper. Merging reports is now not permitted.

18.4.2 Line

When a line control is selected, the property window shows the properties of the line.

Line	Property
ID	2
Start X	36
Start Y	66
End X	140
End Y	139
Line Width	1
Group No.	0
Line Color	Line Color
Print	☒ Yes/No

The following table explains the line properties in detail.

Parameter	Description
ID	Print ID of the line. The ID is 2.
Start X	Set the X-coordinate value of the start point.  NOTE The control coordinate originates from the upper-left corner of the editing area, from which the X axis (positive) is extended horizontally to the right and the Y axis (positive) vertically to the bottom. The unit is mm.
Start Y	Set the Y-coordinate value of the start point.
End X	Set the X-coordinate value of the end point.
End Y	Set the Y-coordinate value of the end point.
Line Width	Set the width of the line. The unit is mm.
Group No.	A group gathers multiple controls that will be used frequently on the template. e.g. a line of controls constitutes a group. The group No. is 0 if not defined.
Line Color	Set the color of the line.
Print	Enable or disable printing the line on actual reports.

18.4.3 Rectangle

When a rectangle control is selected, the property window shows the properties of the rectangle.

Frame	Property
ID	1
Start X	54
Start Y	54
Width	96
Height	59
Frame Width	1
Group No.	0
Frame Color	Frame Color
Print	☒ Yes/No

The following table explains the rectangle properties in detail.

Parameter	Description
ID	Print ID of the line. The ID is 1.
Start X	Set the X-coordinate value of the start point (upper-left corner).
Start Y	Set the Y-coordinate value of the start point (upper-left corner).
Width	Set the width of the rectangle.
Height	Set the height of the rectangle.
Frame Width	Set the frame width of the rectangle.
Group No.	A group gathers multiple controls that will be used frequently on the template. e.g. a line of controls constitutes a group. The group No. is 0 if not defined.
Frame Color	Set the color of the frame.
Print	Enable or disable printing the rectangle on actual reports.

18.4.4 Label

When a label control is selected, the property window shows the properties of the label.

Label	Property
ID	4
Text	Test
Start X	15
Start Y	48
Width	25
Height	6
Group No.	0
Bk Color	Bk Color
Font	Tahoma
Text Place	Left
Show Frame	☐ Yes/No
Frame Width	1
Frame Color	Frame Color
Print	☒ Yes/No

The following table explains the label properties in detail.

Parameter	Description
ID	Print ID of the label. The ID is 4.
Text	Set the text on the label. It will be printed unchanged on actual reports.
Start X	Set the X-coordinate value of the start point (upper-left corner).
Start Y	Set the Y-coordinate value of the start point.
Width	Set the width of the label.
Height	Set the height of the label.
Group No.	A group gathers multiple controls that will be used frequently on the template. e.g. a line of controls constitutes a group. The group No. is 0 if not defined.
Bk Color	Set the background color of the label.
Font	Set the font of the label text.
Text Place	Set the aligning mode of label text. It includes Left, Center and Right.
Show Frame	Enable and disable printing frame.

Parameter	Description
Frame Width	Set the width of the label frame.
Frame Color	Set the color of the label frame.
Print	Enable or disable printing the label on actual reports.
Text ID	The ID of the label text represented by number. When it is edited, the template is refreshed.
Replace text	Choose whether to use the defined text ID to replace the text of the control.

18.4.5 Text

When a text control is selected, the property window shows the properties of the text.

Text	Property
ID	96
Name	Reference Range
Text	3-42
Show Detail	☒ Yes/No
Start X	161
Start Y	56
Width	31
Height	6
Group No.	1
Text Type	0
Bk Color	Bk Color
Font	Tahoma
Text Place	Left
Show Frame	☐ Yes/No
Frame Width	1
Frame Color	Frame Color
Print	☒ Yes/No

The following table explains the text properties in detail.

Parameter	Description
ID	Print ID of the text. The default is 0 and means unknown ID. Print ID indicates the meaning of the text. Correct printout can be ensured only when print ID is set properly.

Parameter	Description
Name	Set the contents to be displayed on the text control. It varies from different IDs.
Text	Set the contents displayed on the text control. It will be replaced by actual data when printed.
Show Detail	Set the text as table data.  NOTE Only the text that not only is enabled in Show Details field but also has a group No. is table data.
Start X	Set the X-coordinate value of the start point (upper-left corner).
Start Y	Set the Y-coordinate value of the start point.
Width	Set the width of the text.
Height	Set the height of the text.
Group No.	A group gathers multiple controls that will be used frequently on the template. e.g. a line of controls constitutes a group. The group No. is 0 if not defined.
Text Type	Reserved parameter. The default is 0.
Bk Color	Set the background color of the text.
Font	Set the font of the text.
Text Place	Set the aligning mode of the text. It includes Left, Center and Right.
Show Frame	Enable and disable printing frame.
Frame Width	Set the width of the text frame.
Frame Color	Set the color of the label frame.
Print	Enable or disable printing the text on actual reports.

18.4.6 Title

When a title control is selected, the property window shows the properties of the title.

Title	Property
ID	5
Text	%sPatient Report
Start X	12
Start Y	6
Width	190
Height	12
Bk Color	Bk Color
Font	Tahoma
Text Place	Center
Show Frame	☐ Yes/No
Frame Width	1
Frame Color	Frame Color
Print	☒ Yes/No

The following table explains the title properties in detail.

Parameter	Description
ID	Print ID of the title. The ID is 5.
Text	Set the contents to be displayed on the title. “%s” will be replaced by a hospital name and can be displayed in any place of the title.
Start X	Set the X-coordinate value of the start point (upper-left corner).
Start Y	Set the Y-coordinate value of the start point.
Width	Set the width of the title.
Height	Set the height of the title.
Bk Color	Set the background color of the title.
Font	Set the font of the title text.
Text Place	Set the aligning mode of title text. It includes Left, Center and Right.
Show Frame	Enable and disable printing frame.
Frame Width	Set the width of the title frame.
Frame Color	Set the color of the title frame.

Parameter	Description
Print	Enable or disable printing the title on actual reports.
Text ID	The ID of the label text represented by number. When it is edited, the template is refreshed.
Replace text	Choose whether to use the defined text ID to replace the text of the control.

18.4.7 Image

When an image control is selected, the property window shows the properties of the image.

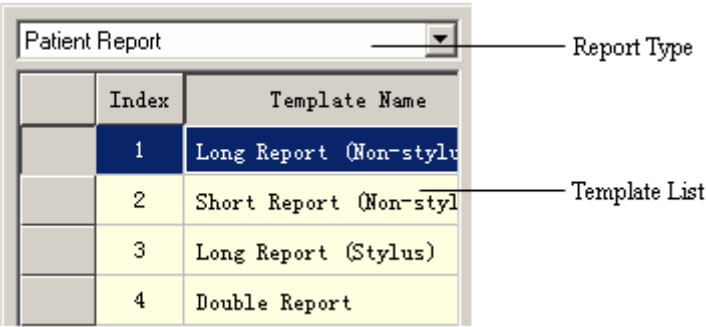
Image	Property
ID	3
Start X	73
Start Y	104
Width	105
Height	54
Group No.	0
Print	☒ Yes/No

The following table explains the image properties in detail.

Parameter	Description
ID	Print ID of the image. The ID is 3.
Start X	Set the X-coordinate value of the start point (upper-left corner).
Start Y	Set the Y-coordinate value of the start point.
Width	Set the width of the image.
Height	Set the height of the image.
Group No.	A group gathers multiple controls that will be used frequently on the template. e.g. a line of controls constitutes a group. The group No. is 0 if not defined.
Print	Enable or disable printing the image on actual reports.

18.5 Report Window

The report window locates on the lower-right corner of the main screen and shows all the templates of a selected report type.



Vocabulary

Absorbance

The difference between the amount of light entering a solution (incident light) and the amount of light passing through the solution (transmitted light) without being absorbed, to determine the concentration of the substance in the solution.

Analyzing unit

The analyzing unit, the analyzer, determines various clinical chemistries in samples and displays the test results. It consists of the sample handling system, reagent handling system, reaction system, cuvette wash station, photometric system, and mixer assembly.

Auto rerun

When a result is beyond the defined range or satisfies the defined conditions, the chemistry will be run again.

Auto serum index

When the Auto Serum Index function is enabled, the system will select the SI chemistry automatically for serum or plasma samples. The SI chemistry will also be requested automatically when you program routine samples manually or by using the LIS host, or program STAT samples, or program routine samples with the default panels.

Bar code reader

Fixed laser beam scanner. It scans the bar code label on sample tubes to identify samples and match the obtained programming information with the scanned samples.

Batch program

Batch program is to program a group of samples with identical programming information, with the exception of the sample ID.

Blank time

Blank time refers to the period between dispensing of the second reactant (reagent or sample) in reversed order and of the last reactant (reagent or sample).

Bottle type

Volume of the reagent bottle.

Calibration curve

A calibration curve reflects the mathematical relation between calibrator concentration and response. It is drawn based on the obtained response and the multiple values between the minimum and maximum concentrations of the calibrator.

Calibration factor

Calibration factor is obtained based on the equation of calibrator concentration (known) and response (calibration math model).

Calibration math model

Calibration math model is used to calculate calibration factors and create calibration curves. It includes single-point K factor, two-point linear, multi-point linear, Logit-Log4P, Logit-Log5P, Exponential5P, Polynomial5P, Parabola and Spline.

Calibration trend

Calibration trend summarizes a chemistry's calibrations during a period of time and reflect the trends of the calibrations.

Carryover

Carryover is the interference of certain substance contained in a reagent. It can influence measurement of another chemistry or the reaction of other mixture, resulting in inaccurate results.

Chemistry configuration

Chemistry configuration is applicable to all chemistries other than ISE chemistry and SI, and used to enable or disable chemistries that have been defined correctly.

Closed-reagent chemistry

Closed-reagent chemistry is run by using the reagents provided by the analyzer manufacturer. Closed-reagent chemistries cannot be modified or deleted.

Concentrated wash

Concentrated wash is an additional cleaning procedure performed on the sample probe, reagent probe, mixers and reaction cuvettes with the aim of eliminating carryover and preventing stains from leaving on exterior and interior of the probes, mixers and cuvettes.

Concentrated wash solution

Concentrated wash solution is used to clean the reaction cuvettes with the aim of

keeping the reagents stable and analyzing samples with increased volume.

Critical range

An allowable result range from the perspective of clinical diagnosis. If the test result is beyond the critical range, the patient may need immediate treatment. You may enable the auto rerun function for a chemistry, which will be rerun automatically once the test result is beyond the critical range.

Current results

Current results include those that are in Incomplete status until the current system time and those programmed and analyzed on the current day.

Cuvette wash station

The cuvette wash station consists of the wash probes, elevating motor and related tubing, and is used to clean the reaction cuvettes with the eight wash probes when a test is finished.

Database

A collection of data arranged for quick search and retrieval.

Decreased

Decreased indicates the sample volume required for analysis and can be defined on the **Define/Edit Chemistries** window.

Diluent

Liquid used to dilute other liquids.

Dilution factor

User-defined dilution ratio, to be multiplied with sample result to obtain the final result.

Download

To obtain sample programming information from the LIS host and match it with the scanned samples. The system supports real-time and manual downloading of sample programming information.

EMF

EMF stands for Electromotive Force. The ISE module determines the concentration of ion by measuring the electromotive force of ion with ion selective electrodes. A calibrator with constant concentration should have electromotive force within certain range.

Endpoint

The endpoint method, also called equilibrium method, is most ideal for measurements. In endpoint measurements, the reaction reaches equilibrium after a period of time. Since the equilibrium constant is quite high, it can be considered that

all substrates (analytes) have changed into products, and the absorbance of the reactant will not change any more. The absorbance change is directly proportional to the analytes' concentration.

Fixed-time

In fixed-time measurements, namely, rate measurements, the reaction velocity (v) is directly proportional to the substrate concentration $[S]$ within a specific period, that is, $v=k[S]$.

Flag

Flag is a manufacturer-defined symbol, which appears on patient reports or result list when a result is beyond the user-defined reference range or exceeds the defined limits.

High-concentration waste

High-concentration waste is produced during the 1st-3rd phase of cuvette cleaning and includes the ISE waste. It can be drained in a waste tank or to the sewer according to your local or national regulations on waste liquid disposal.

History results

Stored results are those programmed and analyzed before the current day.

Increased

Increased indicates the sample volume required for analysis and can be defined on the **Define/Edit Chemistries** window.

Initialization

Initialization is a series of operations automatically performed by the system during the startup procedure. It includes parameters check, reset, testing, cleaning and priming.

Inventory check

Used to check the remaining volume of the biochemistry reagents, sample probe wash solution and reagent probe wash solution and refresh the tests left and wash solution volume on the **Reagent/Calibration** screen.

ISE

ISE is the abbreviation of Ion Selective Electrode. It consists of the Na electrode, K electrode, Cl electrode, reference electrode, sampling and measuring channel, syringe, heat stabilizer, degassing unit and waste discharger. The ISE module measures the concentration of Na, K and Cl in serum, plasma and diluted urine.

K factor

K factor is manually input for single-point linear calibration formula $C = K \times (R - R_0)$ and used to calculate results.

Lamp

Lamp is located on the photometer assembly and used to measure the absorbance of mixture in a reaction cuvette. It should be replaced regularly.

Linearity

Degree of linearity for a reaction curve or calibration curve. Reaction curve linearity is available in fixed-time measurements, while calibration curve linearity specifies the allowable concentration range for result calculation.

LIS

LIS stands for Laboratory Information System. It is a host computer and communicates with chemistry analyzers through the internet interface.

L-J chart

A Levey-Jennings (L-J) chart, drawn based on the QC date (X) and test results (Y), shows the QC result trend of a chemistry during the specified period. The graphical trends of up to 3 controls can be displayed on one L-J chart and distinguished with different colors.

Lot number

Lot number is assigned to controls, calibrators or wash solutions of the same lot for identifying manufacture date, quality, expiration date and other related information.

Low-concentration waste

Low-concentration waste is produced during the 4th-8th phase of cuvette cleaning. It can be drained to the sewer of your laboratory.

Mask/Unmask chemistries

Used when a chemistry needs to be disabled temporarily due to abnormal result or reagent exhaustion. The masked chemistry will have a  symbol appearing on its upper-left corner, and will still be displayed on the **Sample, Quality Control** and **Reagent/Calibration** screens but not run for sample analysis. Masked chemistries cannot be requested until they are unmasked.

Mixer

The system provides sample mixer and reagent mixer, which stir the mixture inside a reaction cuvette when sample/R3 and R2/R4 are respectively dispensed.

Multi-sample report

Containing the results of multiple samples, and can be printed out on the **Current Results** and **History Results** screens.

Off-line dilution

Prior to analysis, samples are diluted manually based on specific ratio.

Offset

Offset is a value added or subtracted to compensate a result. It is often used along with the slope in the equation $y=kx+b$, in which k is the slope and b is the offset.

Off-system chemistry

All the chemistries that are not run by the analyzer are referred to as the off-system chemistries.

Online help

Online help provides you with help information about the screens. If you do not understand a parameter or an operation on a screen, you can go to the online help for relevant information. Access the online help from the following screens:

- Select the  icon on the upper right corner to display the help topic related to the current screen.
- Select the  button in front of each maintenance instruction or item to display the relevant operating instructions.
- Select the  button in front of each error log to display the corresponding topic.
- Click the  button on a warning message window to display the corresponding descriptions and solutions.
- Press the shortcut combination key Alt+F1 to display the topics related to the current screen or window.

Open-reagent chemistry

Open-reagent chemistry, an opposite of the closed-reagent chemistry, can be measured by using the reagents provided by other manufacturers. It can be user-defined, edited and deleted.

Operation unit

The operation unit, a computer configured with the operating software, controls the analyzing unit to finish tests and produce test results.

Output unit

A printer used to print out test results and other data.

Panel

Consists of a couple of chemistries combined together for certain clinical purposes, such as liver function, kidney function, etc. Panels can help fast programming of samples.

Patient demographics

Patient demographics contain information related to the patient and sample, such as patient name, age, gender, collection date/time, etc.

Physiological saline

0.9% sodium chloride solution, used for reagent blank and sample dilution.

Predilution

Prior to analysis, samples are diluted automatically based on the defined dilution factor.

Primary wavelength

The primary wavelength is chosen based on the light absorption features of the reactant and used to measure the absorbed light intensity. Options for primary wavelength include: 340nm, 380nm, 412nm, 450nm, 505nm, 546nm, 570nm, 605nm, 660nm, 700nm, 740nm and 800nm

Prime

Prime is an action to replace the reagents in tubing of the ISE module. A prime is required to replace the reagents in tubing with new ones during the startup procedure or when a reagent is changed.

Print name

Print name appears on a patient report representing a chemistry, and if left blank, will be replaced by the short name of the chemistry.

Prozone check

Prozone check is intended to checking samples with quite different concentrations, which may generate the equivalent amount of insoluble antigen/antibody compound and can have the same test results. The Prozone check can be performed in two ways: rate check and antigen addition.

Pull-down list

A control of the software screen or window. Select the down-triangle button on the right of a pull-down list to show multiple options.

QC panel

Used for analysis of control samples.

QC rule

A set of rules to evaluate if the QC results are under control and the analyzing system is stable. Examples of QC rule are 1-2s, 1-3s, etc.

QC summary

Contains the mean values and standard deviations of controls analyzed within the specified period, as well as the set mean and SD value. The obtained results are compared with the set values to judge if the system is working normally.

Qualitative analysis

Qualitative analysis is used to analyze every sample for the detection of lipemia,

hemolysis and icterus and calculate the numeric values of the index. If the volume of the interferences contained in a sample is beyond the set range, a flag will be added to the patient report.

Random error

An alarm of quality control monitoring. A random error may occur when the lowest and highest values of QC results respectively exceed $-2SD/-3SD$ and $+2SD/+3SD$.

Reaction carousel

Reaction carousel is a turntable, and used to hold reaction cuvettes and transmit each of them to the photometric position for signal detecting and absorbance calculation.

Reaction curve

A reaction curve reflects the relationship of the absorbance measured at the primary wavelength, secondary wavelength and primary-secondary wavelength. It is drawn based on the absorbance of the sample-reagent mixture measured within the reaction period. The system provides 4 types of reaction curves: calibration reaction curve, QC reaction curve, sample blank reaction curve, and sample reaction curve.

Reaction cuvette

Reaction cuvette is a carrier in which reagents and samples react with each other and then carried to the photoelectric position for signal detecting and response calculation.

Reaction direction

Reaction direction refers to the change trend of absorbance during the reaction process. It includes positive and negative.

Reaction time

For endpoint analysis, the reaction time refers to the time span from the start point of the reaction to the end point; for fixed-time and Kinetic analysis, it refers to the period from reaction equilibrium to the end of monitoring.

Reagent blank

In the reagent blank test, the reagents react with the physiological saline, and the blank absorbance is calculated to correct the calibration factors. Only the reagents that are in Calibrated, Cal Time Out or Cal Required status can be requested for reagent blank.

Reagent carousel

The reagent carousel is located on left side of the analyzer panel. It holds reagent bottles and carries each of them to the reagent aspirate position for aspirating.

Reagent carryover

Cross contamination between the reagent probe and the mixers. When the number of tests between the contaminating chemistry and the contaminated is less than or

equal to the defined number (N), and no concentrated wash is inserted between the two chemistries, it indicates that the reagents underlie the risk of carryover.

Reagent inventory alarm limit

Alarm limit of reagents and wash solutions. When the reagent inventory is lower than the alarm limits during or before the analysis, the system will give an alarm and display the reagent or wash solution name in yellow on the **Reagent/Calibration** screen.

Reagent probe

The reagent probe aspirates the specified amount of reagent from a reagent bottle and then dispenses it into a cuvette for reaction and analysis. The system has one reagent probe.

Reagent probe wash solution

Used for cleaning the reagent probe.

Reference range

Reference range is a user-defined range consisting of low limit and high limit. When a result is beyond the reference range, a flag will appear near the result.

Release

Used to clear the specified sample position or all positions on the current sample carousel. When a sample is released, its results and programming information can be still recalled. The released position can be used for programming of new samples.

Replicates

Number of times to run a test, to ensure accurate results.

Result statistics

Result statistics option can summarize the total chemistries and the distribution trend of its results and provide the test data and graph.

Sample blank

Sample blank is similar to sample analysis except for use of equivalent amount of physiological saline. Sample blank is used for removal of non-chromogenesis reaction, such as influence of sample interference (Hemolysis, icterus and lipemia) on absorbance readings.

Sample carousel

The sample carousel is located on right side of the analyzer panel. It holds sample tubes and carries each of them to the sample aspirate position for aspirating.

Sample comments

Remarks for some special samples, such as, ** sample has hemolysis; ** sample needs to be analyzed immediately, etc.

Sample log

Contains the controls and patient samples that are not complete within the recent 24 hours due to certain reasons. Based on the sample log you are allowed to rerun the samples or take other actions for the controls and samples.

Sample panel

Used for analysis of patient samples.

Sample probe

The sample probe aspirates the specified amount of sample from a sample tube and then dispenses it into a cuvette for reaction and analysis.

Sample probe wash solution

Used to clean the sample probe and located in position D3 of the analyzer's front panel.

Sample type

Type of sample. The sample type options include serum, plasma, urine, CSF and other.

Screen

Screen is a part of the software interface. It is rectangular and contains various controls, such as edit box, function button, etc.

Secondary wavelength

The secondary wavelength is used to remove the interference in primary wavelength values and eliminate the influence of noise, such as light flash and drift, and scratches on cuvettes, etc. It cannot be the same as the primary wavelength.

Serial number

Sequence number of the reagent bottle.

Slope

Multiplied with the test result to make it consistent with that obtained on other instruments. It is often used along with the offset in the equation $y=kx+b$, in which k is the slope and b is the offset.

Special calculation

Special calculation is derived from calculation of certain chemistries and has specific clinical purposes, such as A/G, TBil-DBil, etc.

Standard deviation (SD)

Standard deviation is the mean of deviations from the mean value. It is an index to judge the measurement accuracy under specific conditions. In this manual, SD refers to the standard deviation of control concentration.

Standby

Standby is one of the system statuses. When the system status is Standby, it indicates that all tests are finished and all actions of the system have stopped.

STAT

STAT means emergent, including common STAT and quick STAT program. STAT sample program allows emergent samples to be programmed and analyzed with high priority. Common STAT program is used in daytime to run emergent samples with higher priority than routine samples. Quick STAT program is mainly used in nighttime and weekends to program emergent samples quickly with higher priority than routine and common STAT samples.

Symbology

Symbology is a set of rules for encoding and decoding information contained in a bar code label. The system provides a couple of symbologies, such as Codabar, ITF, code128, code39, UPC/EAN, and Code93.

Systematic error

An alarm of quality control monitoring. A systematic error may occur when both the lowest value and highest value of a QC result are on the same side.

Transmit

Transmit is an action sending specified sample results or QC results to the LIS host.

Twin chemistries

Twin chemistries are run with the same reagents and calculated through the same test. For two twin chemistries, the sample volume, volume of shared reagent, calibration replicates, and auto calibration conditions should be the same. When either of the two chemistries is requested for calibration, quality control or sample analysis, the other chemistry will be automatically requested, and finally results of both chemistries will be calculated.

Twin-Plot chart

A twin-plot chart, drawn based on the results of control X and control Y in the same run, is used to detect systematic errors and random errors. It shows the recent 10 QC results of a chemistry and excludes those that have been deleted.

Two-control evaluation

In two-control evaluation, two results are obtained: X_n and Y_n , which are used to define a point on the Twin-plot chart. In this way, a complete twin-plot chart is drawn based on all the QC results and used for detecting systematic errors and random errors.

Unpositioned samples

Samples without positions assigned or with positions not assigned successfully, including those:

Vocabulary

- downloaded from the LIS host and not positioned yet.
- that are in Incomplete status when their positions are assigned for new samples.
- that are incomplete when their positions are released.

Westgard rule

Westgard rule is used for monitoring of quality control. In the Westgard rule, single rules such as 12S, 13S, 22S and 41S are combined to evaluate results of single or multiple controls.

Workload statistics

On the Workload screen, you can view test requests and reagent application for each chemistry during a period, and sample requests and the quantity of its chemistries

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